

# How to fit an animal model

An ecologist guide

Julien Martin & Mathieu Videlier



# Table of contents

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>Preface</b>                                                  | <b>iv</b> |
| Citing the book . . . . .                                       | iv        |
| License . . . . .                                               | iv        |
| Image credit . . . . .                                          | iv        |
| Who are we ? . . . . .                                          | v         |
| Hex Sticker . . . . .                                           | v         |
| <b>1. Introduction</b>                                          | <b>1</b>  |
| 1.1. Data . . . . .                                             | 1         |
| 1.1.1. Data files . . . . .                                     | 1         |
| 1.1.2. Notes on data and pedigree . . . . .                     | 2         |
| 1.2. R . . . . .                                                | 2         |
| 1.2.1. R packages . . . . .                                     | 2         |
| <b>1. Univariate animal model</b>                               | <b>4</b>  |
| Scenario and data . . . . .                                     | 5         |
| Scenario . . . . .                                              | 5         |
| Data files . . . . .                                            | 5         |
| <b>2. Asreml-R</b>                                              | <b>7</b>  |
| 2.0.1. Running the model . . . . .                              | 7         |
| 2.0.2. Estimating heritability . . . . .                        | 9         |
| 2.0.3. Adding fixed effects . . . . .                           | 10        |
| 2.0.4. Adding random effects . . . . .                          | 12        |
| 2.0.5. Testing significance of random effects . . . . .         | 15        |
| 2.0.6. Further partitioning the variance . . . . .              | 16        |
| 2.0.7. Modification of the varaince matrix parameters . . . . . | 19        |
| 2.0.8. Covariance between two random effects . . . . .          | 21        |
| <b>3. MCMCglmm</b>                                              | <b>23</b> |
| 3.0.1. Running the model . . . . .                              | 23        |
| 3.0.2. Change in iteration and sampling . . . . .               | 27        |
| 3.0.3. Change priors parameters . . . . .                       | 29        |
| 3.0.4. Estimating heritability . . . . .                        | 30        |
| 3.0.5. Adding fixed effects . . . . .                           | 31        |
| 3.0.6. Adding random effects . . . . .                          | 34        |
| 3.0.7. Testing significance of variance components . . . . .    | 36        |
| 3.0.8. Further partitioning variance . . . . .                  | 36        |
| 3.0.9. Modification of model parameter . . . . .                | 39        |
| 3.0.10. Covariance between two random effects . . . . .         | 39        |



|                                                                             |                |
|-----------------------------------------------------------------------------|----------------|
| <b>4. brms</b>                                                              | <b>41</b>      |
| 4.0.1. Running the model . . . . .                                          | 41             |
| 4.0.2. Adding fixed effects . . . . .                                       | 47             |
| 4.0.3. Adding random effects . . . . .                                      | 51             |
| 4.0.4. Testing significance of variance components . . . . .                | 57             |
| 4.0.5. Further partitioning of the variance . . . . .                       | 57             |
| 4.0.6. Modification of model parameter . . . . .                            | 64             |
| 4.0.7. Covariance between two random effects . . . . .                      | 64             |
| <br><b>II. Multivariate animal model</b>                                    | <br><b>65</b>  |
| Scenario and data . . . . .                                                 | 66             |
| Scenario . . . . .                                                          | 66             |
| gryphon files . . . . .                                                     | 66             |
| <br><b>5. Asreml-R</b>                                                      | <br><b>68</b>  |
| 5.0.1. Running the model . . . . .                                          | 68             |
| 5.0.2. Adding fixed and random effects . . . . .                            | 72             |
| 5.0.3. Significance testing . . . . .                                       | 76             |
| 5.0.4. Estimate directly the genetic correlation within the model . . . . . | 79             |
| 5.0.5. Visualisation of the correlation (aka BLUP extraction) . . . . .     | 82             |
| 5.0.6. Partitioning (co)variance between groups . . . . .                   | 87             |
| 5.0.7. Between groups (co)variances and the B-matrix . . . . .              | 99             |
| <br><b>6. MCMCglmm</b>                                                      | <br><b>108</b> |
| 6.0.1. Fitting the model . . . . .                                          | 109            |
| 6.0.2. Adding fixed and random effects . . . . .                            | 113            |
| 6.0.3. Partitioning (co)variances . . . . .                                 | 119            |
| 6.0.4. Between groups (co)variances and the B-matrix . . . . .              | 191            |
| <br><b>7. brms</b>                                                          | <br><b>357</b> |
| 7.0.1. Fitting the model . . . . .                                          | 357            |
| 7.0.2. Adding fixed and random effects . . . . .                            | 368            |
| 7.0.3. Partitioning (co)variances . . . . .                                 | 387            |
| 7.0.4. Between groups (co)variances and the B-matrix . . . . .              | 411            |
| <br><b>III. A repeated measures animal model</b>                            | <br><b>431</b> |
| Scenario and data . . . . .                                                 | 432            |
| scenario . . . . .                                                          | 432            |
| Data files . . . . .                                                        | 432            |
| <br><b>8. Asreml-R</b>                                                      | <br><b>434</b> |
| 8.0.1. Estimating repeatability . . . . .                                   | 434            |
| 8.0.2. Partitioning additive and permanent environment effects . . . . .    | 438            |
| 8.0.3. Adding additional effects and testing significance . . . . .         | 441            |
| <br><b>9. MCMCglmm</b>                                                      | <br><b>446</b> |
| 9.0.1. Estimating repeatability . . . . .                                   | 446            |
| 9.0.2. Partitioning additive and permanent environment effects . . . . .    | 451            |
| 9.0.3. Adding additional effects and testing significance . . . . .         | 453            |

|                                                         |            |
|---------------------------------------------------------|------------|
| <b>10. brms</b>                                         | <b>457</b> |
| <b>11. Quick comparison of codes</b>                    | <b>458</b> |
| 11.1. Univariate model with repeated measures . . . . . | 458        |
| 11.1.1. Asreml-R . . . . .                              | 458        |
| 11.1.2. gremlin . . . . .                               | 458        |
| 11.1.3. MCMCglmm . . . . .                              | 458        |
| 11.1.4. brms . . . . .                                  | 458        |
| 11.2. bivariate model . . . . .                         | 458        |
| 11.2.1. Asreml-R . . . . .                              | 458        |
| 11.2.2. gremlin . . . . .                               | 458        |
| 11.2.3. MCMCglmm . . . . .                              | 458        |
| 11.2.4. brms . . . . .                                  | 458        |
| <b>References</b>                                       | <b>459</b> |
| R packages . . . . .                                    | 459        |
| Bibliography . . . . .                                  | 460        |
| <b>Appendices</b>                                       | <b>463</b> |
| <b>A. Data used in this book</b>                        | <b>463</b> |
| A.1. All in one zip file . . . . .                      | 463        |
| A.2. Data files . . . . .                               | 463        |
| A.3. R objects . . . . .                                | 463        |
| A.4. R code & functions . . . . .                       | 464        |

# Preface

This book is a collection of tutorial from the excellent paper by (Wilson et al. 2010). The tutorials have been updated to work with the latest version of the softwares, extended with extra information and other softwares have been added. Eric Postma kindly provided an updated version of the original asreml-r tutorial from asreml-r v3 to asreml-r v4.

**This is a living document and information is updated/added regularly**

## Citing the book

Julien Martin & Mathieu Videlier (2024). How to fit an Animal model? A guide for evolutionary ecologist using multiple R packages. Version: 1.0.0 (2025-01-22). <https://doi.org/10.5281/zenodo.10891227>

## License

We provide this book under the license [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International](https://creativecommons.org/licenses/by-nc-sa/4.0/).



Feel free to use some or all of the content in this book. The only thing we ask is that you acknowledge the original source and authors as per the license and use the same license if you share that work publicly. If you find this book useful or have any comments or suggestions we would love to hear from you (contact info).

## Image credit

The cover image was generated by Julien Martin using the Nightcafe AI image generator using the call “Family tree of legendary beast”.

## Who are we ?



**Julien Martin** is a Professor at the University of Ottawa working on Evolutionary Ecology and has discovered R with version 1.8.1 and teaches R since v2.4.0. He developed the first version of the book in February 2021 and now continues to develop the book and maintains the site for all languages (currently only english but french and spanish comin soon).

- 📖: [uOttawa page](#), [lab page](#)
- 🐦: [jgamartin](#)
- 🌐: [juliengamartin](#)



**Mathieu Videlier** is currently a postdoctoral fellow collaborating with [Pr. Jacqueline Sztepanacz](#) at University of Toronto. In joined the project in May 2021 . He added and continues to add more details in the document. He is also spearing the translation fo French and Spanish.

- 📖: [personal page](#)
- 🐦: [BioBipbip](#)
- 🌐: [MathieuVID](#)

## Hex Sticker



# Chapter 1

## Introduction

The book provides a series of tutorials (and accompanying data files) to fit animal model in R using different packages (ASReml-R, gremlin, MCMCglmm and brms/stan) . You will need to carefully follow the instructions below to first download the data files and second install the R packages. Before beginning the tutorial, we assume the reader has successfully installed the chosen R package on their computer and has saved the required data files to an appropriate directory from which they will be read. Full instructions for how to do this are provided with software distributions.

To work through the different tutorial I would recommend to create a folder where you will save your different R scripts for the tutorials.

In addition, the tutorial is here to help researchers in their coding and understanding of models and outputs, but it is required that you read and understand the literature in quantitative genetics and animal model.

### 1.1. Data

#### 1.1.1. Data files

You will need to download 3 data files for the tutorial in R:

- gryphon.csv: data on gryphon birth weight and morphology
- gryphonRM.csv: data on gryphon repeated measurement of lay date.
- gryphonped.csv: data on the associated pedigree of the data gryphon

In addition, some models presented in the tutorials can take a while to run (sometimes > 1 hour), thus we are also providing the model outputs to allow you continue the tutorial without waiting for the model to run. (But you are free to run models)

The files are available [here](#). We recommend to save the data and Rdata files in a subfolder data in the folder you will use as your working directory for R and where you will save your R scripts. It should be noted that the tutorial are using this structure to read or save data.

### 1.1.2. Notes on data and pedigree

It is always important to take time to think carefully about the strengths and potential limitations of your pedigree information before embarking on quantitative genetic analyses. Pedigree Viewer, written by Brian Kinghorn, is an extremely useful application for visualizing pedigrees, and can be downloaded from: <http://www-personal.une.edu.au/~bkinghor/pedigree.htm>. `pedtricks` an R package developed by Martin et al. (2024) and distributed through [CRAN](#) can also be used for this and offers some nice additional features for visualizing pedigree structures and generating associated statistics. Before you begin running through the tutorials, we advise taking a moment to look at the pedigree files provided with them using Pedigree Viewer or `pedtricks`.

## 1.2. R

You should check that you have the most current version of R and R packages. You can check the number of the current version on CRAN. If you need to update (or install) R packages, use `install.packages()` and follow the prompted instructions.

### 1.2.1. R packages

#### 1.2.1.1. `asreml-r`

ASReml-R is commercial software published by VSN international (<http://www.vsn.co.uk/software/asreml/>). This package is not free and requires a key access. Additional information and guide can be found in the [Asreml-R manual](#). VSNi has provided a 12 months free Asreml-R licence to support the development of this guide. In addition, VSNi also features a blog with data analytics guides and opinions (<https://vsni.co.uk/blogs/>) as well as private courses.



### 1.2.1.2. **gremlin**

`gremlin` is a little monster appearing if you feed a mugwai after midnight. It is also a great and promising software written by Matthew E. Wolak to fit mixed models using a frequentist approach .

### 1.2.1.3. **MCMCglmm**

`MCMCglmm` is an R package for Bayesian mixed model analysis written by Jarrod Hadfield. It is a freeware distributed through CRAN (<http://cran.r-project.org/>). Information and guide about the package can be find in the user manual and vignettes (<http://cran.r-project.org/web/packages/MCMCglmm/index.html>). Reference: (Hadfield 2010).

This module provides some information that applies to `MCMCglmm`-based analyses in general, but that will not be included in other tutorials. Most importantly, this applies to some of the simplest ways of determining the performance of a run using `MCMCglmm`, i.e., verification of the validity of of the posterior distribution. This tutorial is not a substitute for working through the `MCMCglmm` course notes, which is available from CRAN (the Comprehensive R ArchiveNetwork, <http://cran.r-project.org/>, or can be accessed in R using the command `vignette("CourseNotes","MCMCglmm")`). These tutorials do not introduce one of the main advantages of using `MCMCglmm` for analyses of data from natural populations -the ability to properly model non-normal responses. These capabilities are introduced in the documentation that is distributed with `MCMCglmm`, and available from CRAN. Another specific animal guide for `MCMCglmm` can be find ([https://devillemereuil.legitux.org/wp-content/uploads/2021/09/tuto\\_en.pdf](https://devillemereuil.legitux.org/wp-content/uploads/2021/09/tuto_en.pdf)). Pr. Pierre de Villemereuil provide more information in Bayesian concept and focus more on non-gaussian variable.

### 1.2.1.4. **brms**

`brms` provides an interface to fit Bayesian generalized multivariate (non-)linear multilevel models using `Stan`, which is a C++ package for obtaining full Bayesian inference (see <https://mc-stan.org/>). The formula syntax is an extended version of the syntax applied in the ‘`lme4`’ package to provide a familiar and simple interface for performing regression analyses.

It should be noted that if `brms` is able to fit animal model the parametrization used is not the most efficient and can take quite longer than using a different parametrization directly in `stan`.

## **Part I.**

# **Univariate animal model**

This tutorial will demonstrate how to run a univariate animal model to estimate genetic variance in birth weight in the mighty gryphons.

## Scenario and data

### Scenario

In a population of gryphons there is strong positive selection on birth weight with heavier born individuals having, on average higher fitness. To find out whether increased birth weight will evolve in response to the selection, and if so how quickly, we want to estimate the heritability of birth weight.

### Data files

Open `gryphonped.csv` and `gryphon.csv` in your text editor. The structure and contents of these files is fairly self-explanatory. The pedigree file `gryphonped.csv` contains three columns containing unique IDs that correspond to each animal, its father, and its mother. Note that this is a multigenerational pedigree, with the earliest generation (for which parentage information is necessarily missing) at the beginning of the file. For later-born individuals maternal identities are all known but paternity information is incomplete (a common situation in real world applications).

The phenotype data, as well as additional factors and covariates that we may wish to include in our model are contained in `gryphon.csv`. Columns correspond to individual identity (`animal`), maternal identity (`mother`), year of birth (`byear`), sex (`sex`, where 1 is female and 2 is male), birth weight (`bwt`), and tarsus length (`tarsus`). Each row of the data file contains a record for a different offspring individual. Note that all individuals included in the data file must be included as offspring in the pedigree file.

We can read the data file, using `read.csv()` which consider by default that NA is the symbol for missing values and that the first line of the file contains the column headers.

It is a good idea to make sure that all variables are correctly assigned as numeric or factors:

```
gryphon$animal <- as.factor(gryphon$animal)
gryphon$mother <- as.factor(gryphon$mother)
gryphon$byear <- as.factor(gryphon$byear)
gryphon$sex <- as.factor(gryphon$sex)
gryphon$bwt <- as.numeric(gryphon$bwt)
```

---

```
gryphon$tarsus <- as.numeric(gryphon$tarsus)
str(gryphon)
```

```
'data.frame':  1084 obs. of  6 variables:
 $ animal: Factor w/ 1084 levels "1","2","3","5",...: 864 1076 549 989 1030 751 987 490 906 591 ..
 $ mother: Factor w/ 429 levels "1","2","3","8",...: 362 268 216 375 396 289 328 255 347 240 ...
 $ byear  : Factor w/ 34 levels "968","970","971",...: 1 1 2 2 2 2 3 3 3 3 ...
 $ sex    : Factor w/ 2 levels "1","2": 1 1 2 1 2 1 2 1 1 1 ...
 $ bwt    : num  10.77 9.3 3.98 5.39 12.12 ...
 $ tarsus: num  24.8 22.5 12.9 20.5 NA ...
```

Similarly we can read in the pedigree file, using `read.csv()` which consider by default that NA is the symbol for missing values and that the first line of the file contains the column headers.

```
'data.frame':  1309 obs. of  3 variables:
 $ id      : int  1306 1304 1298 1293 1290 1288 1284 1283 1282 1278 ...
 $ father: int   NA NA NA NA NA NA NA NA NA NA ...
 $ mother: int   NA NA NA NA NA NA NA NA NA NA ...
```

```
gryphonped$id <- as.factor(gryphonped$id)
gryphonped$father <- as.factor(gryphonped$father)
gryphonped$mother <- as.factor(gryphonped$mother)
str(gryphonped)
```

```
'data.frame':  1309 obs. of  3 variables:
 $ id      : Factor w/ 1309 levels "1","2","3","4",...: 1306 1304 1298 1293 1290 1288 1284 1283 1282
 $ father: Factor w/ 158 levels "4","13","18",...: NA NA NA NA NA NA NA NA NA NA ...
 $ mother: Factor w/ 429 levels "1","2","3","8",...: NA NA NA NA NA NA NA NA NA NA ...
```

Now that we have imported the data and the pedigree file, we are ready to fit an animal model.

# Chapter 2

## Asreml-R

### 2.0.1. Running the model

First we need to load the asreml library:

```
library(asreml)
```

Loading required package: Matrix

Online License checked out Tue Oct 8 11:02:02 2024

Loading ASReml-R version 4.2

To be able to fit an animal model, Asreml-r needs (the inverse of) the relationship matrix using the ainverse function:

```
ainv <- ainverse(gryphonped)
```

We are now ready to specify our first model:

```
model1 <- asreml(  
  fixed = bwt ~ 1, random = ~ vm(animal, ainv),  
  residual = ~ idv(units),  
  data = gryphon,  
  na.action = na.method(x = "omit", y = "omit")  
)
```



ASReml Version 4.2 08/10/2024 11:02:03

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -4128.454 | 1.0    | 853 | 11:02:03 |
| 2 | -3284.272 | 1.0    | 853 | 11:02:03 |
| 3 | -2354.992 | 1.0    | 853 | 11:02:03 |
| 4 | -1710.357 | 1.0    | 853 | 11:02:03 |
| 5 | -1363.555 | 1.0    | 853 | 11:02:03 |
| 6 | -1263.516 | 1.0    | 853 | 11:02:03 |
| 7 | -1247.854 | 1.0    | 853 | 11:02:03 |
| 8 | -1247.185 | 1.0    | 853 | 11:02:03 |
| 9 | -1247.183 | 1.0    | 853 | 11:02:03 |

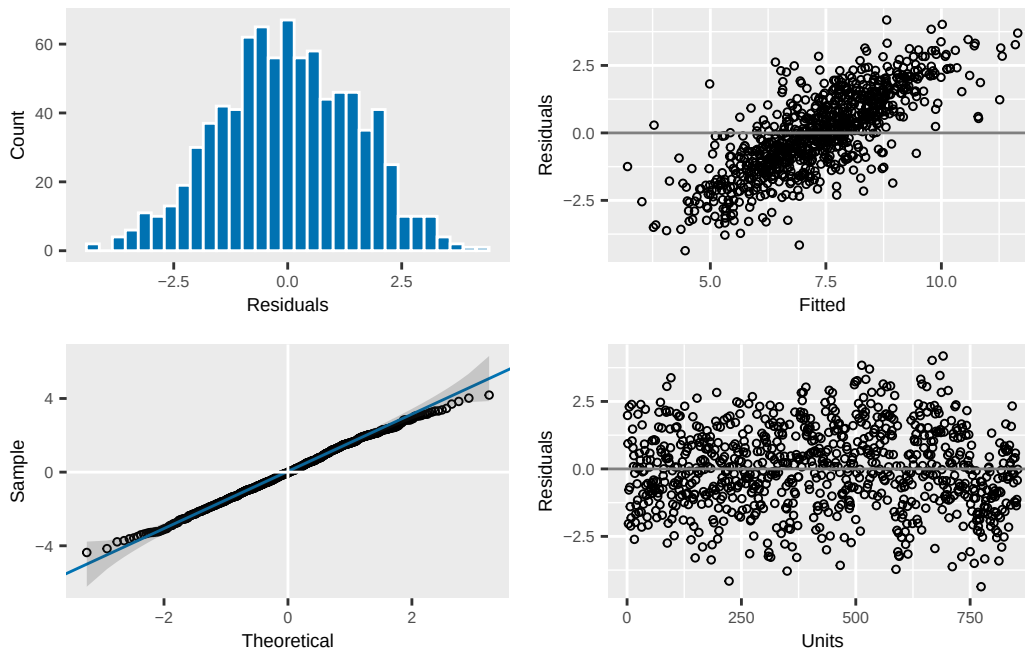
In this model, `bwt` is the response variable and the only fixed effect is the intercept, denoted as 1. The only random effect we have fitted is `animal`, which will provide an estimate of  $V_A$ . Our random `animal` effect is connected to the inverse related matrix `ainv` which integrate the relativeness or pedigree information.

`data=` specifies the name of the dataframe that contains our variables. Finally, we inform `asreml()` what to when it encounters NAs in either the dependent or predictor variables (in this case we choose to remove the records). If you use the argument “include” instead of “omit”, model will keep the NA. With `x=“include”`, the model will exchange NA with 0. Be careful you need to standardize your trait so the mean will be equal to 0, if not estimates (including covariance in multivariate models) could be strongly biased due to the the missing values considered as 0. `y=“include”` will exchange NA with a factor labeled `mv` which will be included in the sparse equation. For more details see Asreml-R manual.

A note of the specification of the structure of the residuals: This simple univariate model will run fine without `residual=~idv(units)`. However, if you are going to use `vpredict()` to calculate the heritability (see below), without specifying the residuals in this way will result in a standard error for the heritability that is incorrect.

Any model has assumption which need to be checked. The model can be plot which help visualizing the distribution of the model residual and check the different assumptions.

```
plot(model1)
```



To see the estimates for the variance components, we run:

```
summary(model1)$varcomp
```

|                  | component | std.error | z.ratio  | bound | %ch |
|------------------|-----------|-----------|----------|-------|-----|
| vm(animal, ainv) | 3.395398  | 0.6349915 | 5.347154 | P     | 0   |
| units!units      | 3.828602  | 0.5185919 | 7.382687 | P     | 0   |
| units!R          | 1.000000  | NA        | NA       | F     | 0   |

We fitted a single random effect so we partitioned the phenotypic variance into two components. The `vm(animal, ainv)` variance component is  $V_A$  and is estimated as 3.4. Given that the ratio of  $V_A$  to its standard error (`z.ratio`) is considerably larger than 2 (*i.e.* the parameter estimate is more than 2 SEs from zero), this looks likely to be significant. The `units!units` component refers to the residual variance  $V_R$ , and `units$R` should be ignored. If you don't include `residual=~idv(units)` in your model specification, `units$R` will provide you with the residual variance.

## 2.0.2. Estimating heritability

We can calculate the  $h^2$  of birth weight from the components above since  $h^2 = V_A/V_P = V_A/(V_A + V_R)$ . Thus according to this model,  $h^2 = 3.4 / (3.4 + 3.83) = 0.47$ .

Alternatively we can use the `vpredict()` function to calculate  $h^2$  and its standard error. `vpredict()` function has two structures, first the model used (here `model1`) and then the estimate name with its associated equation. The equation used different  $V$  and their associated numbers depend of the order of the different random and residual effects included in the model.

```
vpredict(model1, h2.bwt ~ V1 / (V1 + V2))
```

|        | Estimate  | SE        |
|--------|-----------|-----------|
| h2.bwt | 0.4700163 | 0.0765088 |

### 2.0.3. Adding fixed effects

To add fixed effects to a univariate model, we simply modify the model statement. For example, we might know (or suspect) that birth weight is a sexually dimorphic trait and therefore fit in the model.

```
model2 <- asreml(
  fixed = bwt ~ 1 + sex,
  random = ~ vm(animal, ainv),
  residual = ~ idv(units),
  data = gryphon,
  na.action = na.method(x = "omit", y = "omit")
)
```

ASReml Version 4.2 08/10/2024 11:02:04

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -3364.126 | 1.0    | 852 | 11:02:04 |
| 2 | -2702.117 | 1.0    | 852 | 11:02:04 |
| 3 | -1978.916 | 1.0    | 852 | 11:02:04 |
| 4 | -1487.834 | 1.0    | 852 | 11:02:04 |
| 5 | -1236.350 | 1.0    | 852 | 11:02:04 |
| 6 | -1172.771 | 1.0    | 852 | 11:02:04 |
| 7 | -1165.270 | 1.0    | 852 | 11:02:04 |
| 8 | -1165.093 | 1.0    | 852 | 11:02:04 |
| 9 | -1165.093 | 1.0    | 852 | 11:02:04 |

Now we can look at the fixed effects parameters and assess their significance with a conditional Wald F-test:

```
          solution std error  z.ratio
(Intercept) 6.058669 0.1718244 35.26082
sex_1        0.000000          NA      NA
sex_2        2.206996 0.1619974 13.62365
```

ASReml Version 4.2 08/10/2024 11:02:05

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -1165.093 | 1.0    | 852 | 11:02:05 |
| 2 | -1165.093 | 1.0    | 852 | 11:02:05 |

\$Wald

|             | Df | denDF | F.inc  | F.con  | Margin | Pr |
|-------------|----|-------|--------|--------|--------|----|
| (Intercept) | 1  | 251   | 3491.0 | 3491.0 |        | 0  |
| sex         | 1  | 831   | 185.6  | 185.6  | A      | 0  |

\$stratumVariances

|                  | df        | Variance | vm(animal, ainv) | units!units |
|------------------|-----------|----------|------------------|-------------|
| vm(animal, ainv) | 752.28476 | 5.957254 | 0.9864077        | 1           |
| units!units      | 99.71524  | 2.938413 | 0.0000000        | 1           |

The very small probability (Pr) in the Wald test above shows that `sex` is a highly significant fixed effect, and from the parameter estimates (`summary(model2,coef=T)$coef.fixed`) we can see that the average male (sex 2) is 2.2 kg ( $\pm 0.16$  SE) heavier than the average female (sex 1). However, when we look at the variance components in the model including `sex` as a fixed effect, we see that they have changed slightly from the previous model:

```
summary(model2)$varcomp
```

|                  | component | std.error | z.ratio  | bound | %ch |
|------------------|-----------|-----------|----------|-------|-----|
| vm(animal, ainv) | 3.060441  | 0.5243571 | 5.836558 | P     | 0   |
| units!units      | 2.938412  | 0.4161473 | 7.060991 | P     | 0   |
| units!R          | 1.000000  | NA        | NA       | F     | 0   |

| component | std.error | z.ratio | bound | %ch |
|-----------|-----------|---------|-------|-----|
|-----------|-----------|---------|-------|-----|

In fact since sex effects were previously contributing to the residual variance of the model, our estimate of  $V_R$  (denoted units!R in the output) is now slightly lower than before. This has an important consequence for estimating heritability since if we calculate  $V_P$  as  $V_A + V_R$  then as we include fixed effects we will soak up more residual variance driving  $V_P$ . Assuming that  $V_A$  is more or less unaffected by the fixed effects fitted then as  $V_P$  goes down we expect our estimate of  $h^2$  will go up:

```
(h2.1 <- vpredict(model1, h2.bwt ~ V1 / (V1 + V2)))
```

|        | Estimate  | SE        |
|--------|-----------|-----------|
| h2.bwt | 0.4700163 | 0.0765088 |

```
(h2.2 <- vpredict(model2, h2.bwt ~ V1 / (V1 + V2)))
```

|        | Estimate | SE        |
|--------|----------|-----------|
| h2.bwt | 0.510171 | 0.0743239 |

Here  $h^2$  has increased slightly from  $|| \text{Estimate} || : \text{h2.bwt} | 0.47 |$  to  $|| \text{Estimate} || : \text{h2.bwt} | 0.51 |$ . Which is the better estimate? It depends on what your question is. The first is an estimate of the proportion of variance in birth weight explained by additive effects, the latter is an estimate of the proportion of variance in birth weight *after conditioning on sex* that is explained by additive effects.

An important piece of advice, each researcher should be consistent in how they name their estimates and always correctly describe which estimates they are using conditional or not (to avoid any confusion).

#### 2.0.4. Adding random effects

This is done by simply modifying the model statement in the same way. For instance fitting:

```

model3 <- asreml(
  fixed = bwt ~ 1 + sex,
  random = ~ vm(animal, ainv) + byear,
  residual = ~ idv(units),
  data = gryphon,
  na.action = na.method(x = "omit", y = "omit")
)

```

ASReml Version 4.2 08/10/2024 11:02:05

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -2742.658 | 1.0    | 852 | 11:02:05 |
| 2 | -2237.268 | 1.0    | 852 | 11:02:05 |
| 3 | -1690.453 | 1.0    | 852 | 11:02:05 |
| 4 | -1328.910 | 1.0    | 852 | 11:02:05 |
| 5 | -1154.597 | 1.0    | 852 | 11:02:05 |
| 6 | -1116.992 | 1.0    | 852 | 11:02:05 |
| 7 | -1113.809 | 1.0    | 852 | 11:02:05 |
| 8 | -1113.772 | 1.0    | 852 | 11:02:05 |
| 9 | -1113.772 | 1.0    | 852 | 11:02:05 |

```
summary(model3)$varcomp
```

|                  | component | std.error | z.ratio  | bound | %ch |
|------------------|-----------|-----------|----------|-------|-----|
| byear            | 0.8862604 | 0.2695918 | 3.287415 | P     | 0   |
| vm(animal, ainv) | 2.7068665 | 0.4422140 | 6.121169 | P     | 0   |
| units!units      | 2.3092415 | 0.3451025 | 6.691466 | P     | 0   |
| units!R          | 1.0000000 | NA        | NA       | F     | 0   |

```
(h2.3 <- vpredict(model3, h2.bwt ~ V2 / (V1 + V2 + V3)))
```

|        | Estimate  | SE        |
|--------|-----------|-----------|
| h2.bwt | 0.4586068 | 0.0674036 |

Here the variance in bwt explained by byear is 0.89 and, based on the `z.ratio`, appears to be significant ( $>2$ ). Thus we would conclude that year-to-year variation (*e.g.*, in weather, resource abundance) contributes to  $V_P$ . Note that although  $V_A$  has changed somewhat, as most of what is now partitioned as a birth year effect was previously partitioned as  $V_R$ . Thus what we have really done here is to partition environmental effects into those arising from year-to-year differences versus everything else, and we do not really expect much change in  $h^2$  (since now  $h^2 = V_A/(V_A + V_{BY} + V_R)$ ).

However, we get a somewhat different result if we also add a random effect of mother to test for maternal effects:

```
model4 <- asreml(
  fixed = bwt ~ 1 + sex,
  random = ~ vm(animal, ainv) + byear + mother,
  residual = ~ idv(units),
  data = gryphon,
  na.action = na.method(x = "omit", y = "omit")
)
```

ASReml Version 4.2 08/10/2024 11:02:05

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -2033.178 | 1.0    | 852 | 11:02:05 |
| 2 | -1723.734 | 1.0    | 852 | 11:02:05 |
| 3 | -1396.354 | 1.0    | 852 | 11:02:05 |
| 4 | -1193.012 | 1.0    | 852 | 11:02:05 |
| 5 | -1107.946 | 1.0    | 852 | 11:02:05 |
| 6 | -1095.327 | 1.0    | 852 | 11:02:05 |
| 7 | -1094.816 | 1.0    | 852 | 11:02:05 |
| 8 | -1094.815 | 1.0    | 852 | 11:02:05 |

```
summary(model4)$varcomp
```

|                  | component | std.error | z.ratio  | bound | %ch |
|------------------|-----------|-----------|----------|-------|-----|
| byear            | 0.8820313 | 0.2632455 | 3.350604 | P     | 0   |
| mother           | 1.1184698 | 0.2386239 | 4.687167 | P     | 0   |
| vm(animal, ainv) | 2.2985320 | 0.4962496 | 4.631806 | P     | 0   |



---

|             | component | std.error | z.ratio  | bound | %ch |
|-------------|-----------|-----------|----------|-------|-----|
| units!units | 1.6290034 | 0.3714154 | 4.385934 | P     | 0   |
| units!R     | 1.0000000 | NA        | NA       | F     | 0   |

---

```
(h2.4 <- vpredict(model4, h2.bwt ~ V1 / (V1 + V2 + V3 + V4)))
```

---

|        | Estimate  | SE        |
|--------|-----------|-----------|
| h2.bwt | 0.1487898 | 0.0386155 |

---

Here partitioning of significant maternal variance has resulted in a further decrease in  $V_R$  but also a decrease in  $V_A$ . The latter is because maternal effects of the sort we simulated (fixed differences between mothers) will have the consequence of increasing similarity among maternal siblings. Consequently they can look very much like additive genetic effects and if present, but unmodelled, represent a type of “common environment effect” that can - and will - cause upward bias in  $V_A$  and so  $h^2$ . The “common environment” can be conceived as the inextricable sum of the maternal additive genetic effect (such as maternal loci) and the maternal environment or permanent environment (such as litter or nest environment created or modified by the mother).

### 2.0.5. Testing significance of random effects

An important point to note in this tutorial is that while the `z.ratio` (`component/std.error`) reported is a good indicator of likely statistical significance ( $>1.96?$ ), the standard errors are approximate and are not recommended for formal hypothesis testing. A better approach is to use likelihood-ratio tests (LRT).

For example, to test the significance of maternal effects we could compare models with and without the inclusion of maternal identity as a random effect and compare the final log-likelihoods of these models.

```
model4$loglik
```

```
[1] -1094.815
```

shows that the model including maternal identity has a log-likelihood of -1094.815, and

```
model3$loglik
```

```
[1] -1113.772
```

shows that the model excluding maternal identity has a log-likelihood of -1113.772.

A test statistic equal to twice the absolute difference in these log-likelihoods is assumed to be distributed as Chi square with one degree of freedom (one term of difference between the two models). In this case we would conclude that the maternal effects are highly significant since:  $2 \times (-1094.8145793 - -1113.7719147)$  equals 37.9146708, and the p-value that comes with this is:

```
1 - pchisq(2 * (model4$loglik - model3$loglik), 1)
```

```
[1] 7.390738e-10
```

As  $P < 0.0001$  we would therefore conclude that the additional of maternal identity as a random effect significantly improves the fit of the model, given an increase in log-likelihood of approximately 19.

### 2.0.6. Further partitioning the variance

A population can be further fragmented into different groups or categories (such as females and males, juveniles and adults or treated and untreated). Some scientific questions require further and deeper analysis of the variance. To avoid multiple model (one for each group), we can directly partition the variance between groups in a unique model. In addition, by doing so, we can also test if the variance are different between groups.

As example, we decide to take the model4 and partition its additive genetic variance and residual variance by sex. It is possible to further partition the other random effects but it will complexity the animal model and requires sufficient sample size.

First, it required to order the dataset by group (here sex).

```
gryphon <- gryphon[order(gryphon$sex), ]
```

To partition variances between sex, two distinct functions are require `at()` for the random level, and `dsum()` for the residual level:

```

model_SEX <- asreml(
  fixed = bwt ~ 1 + sex,
  random = ~ at(sex):vm(animal, ainv) + byear + mother,
  residual = ~ dsum(~ units | sex),
  data = gryphon,
  na.action = na.method(x = "omit", y = "omit")
)

```

ASReml Version 4.2 08/10/2024 11:02:05

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -1142.164 | 1.0    | 852 | 11:02:05 |
| 2 | -1126.308 | 1.0    | 852 | 11:02:06 |
| 3 | -1111.536 | 1.0    | 852 | 11:02:06 |
| 4 | -1105.383 | 1.0    | 852 | 11:02:06 |
| 5 | -1104.375 | 1.0    | 852 | 11:02:06 |
| 6 | -1104.364 | 1.0    | 852 | 11:02:06 |

```
summary(model_SEX)$varcomp
```

|                               | component | std.error | z.ratio  | bound | %ch |
|-------------------------------|-----------|-----------|----------|-------|-----|
| byear                         | 0.9001595 | 0.2690012 | 3.346303 | P     | 0.0 |
| mother                        | 1.3396184 | 0.2663118 | 5.030263 | P     | 0.0 |
| at(sex, '1'):vm(animal, ainv) | 1.4372390 | 0.6514306 | 2.206281 | P     | 0.1 |
| at(sex, '2'):vm(animal, ainv) | 1.9861434 | 0.9974302 | 1.991261 | P     | 0.3 |
| sex_1!R                       | 2.1706213 | 0.5542492 | 3.916327 | P     | 0.0 |
| sex_2!R                       | 1.7112948 | 0.8246188 | 2.075256 | P     | 0.3 |

By partitioning the additive genetic variance and the residual variance, the model estimates the  $V_A$  and  $V_R$  for each group (sex). Doing so, we can calculate the  $h^2$  for each group of sex. Here, it's important to know in which order the variances are estimated to extract the correct variance in the heritability equation.

```
(h2.F <- vpredict(model_SEX, h2.bwt ~ V3 / (V1 + V2 + V3 + V5)))
```

|        | Estimate  | SE        |
|--------|-----------|-----------|
| h2.bwt | 0.2457811 | 0.1070794 |

```
(h2.M <- vpredict(model_SEX, h2.bwt ~ V4 / (V1 + V2 + V4 + V6)))
```

|        | Estimate  | SE        |
|--------|-----------|-----------|
| h2.bwt | 0.3345244 | 0.1619218 |

To test if the variances are different between sexes, we can compare the model partitioned `model_SEX` and the previous model without the partitioning `model4` in a likelihood ratio test (LRT) with 2 degrees of freedom since models have two components of variance of difference.

```
model_SEX$loglik
```

```
[1] -1104.364
```

```
model4$loglik
```

```
[1] -1094.815
```

```
1 - pchisq(2 * (model_SEX$loglik - model4$loglik), 2)
```

```
[1] 1
```

Here, we can see the point estimates of  $h^2$  seems to differ between sexes ( | Estimate | :—|—: | h2.bwt | 0.25 | and | Estimate | :—|—: | h2.bwt | 0.33 | ), but their SE overlaps. LRT give more information and showed that partitioning the variance and the residual between sexes did not improved the fit of the model and so their variance are not significantly different.

```
h2.sex <- rbind(h2.F, h2.M)
```

```
plot(c(0.95, 1.05) ~ h2.sex[, 1], xlim = c(0, 0.8), ylim = c(0.5, 1.5), , xlab = "", ylab = "", c
arrows(y0 = 0.95, x0 = h2.sex[1, 1] - h2.sex[1, 2], y1 = 0.95, x1 = h2.sex[1, 1] + h2.sex[1, 2],
arrows(y0 = 1.05, x0 = h2.sex[2, 1] - h2.sex[2, 2], y1 = 1.05, x1 = h2.sex[2, 1] + h2.sex[2, 2],
mtext("Narrow-sense heritability ( $\pm$ se)", side = 1, las = 1, adj = 0.4, line = 3, cex = 1.6)
axis(2, at = 1, labels = c("birth weight"), las = 3, cex.axis = 1.6)
```

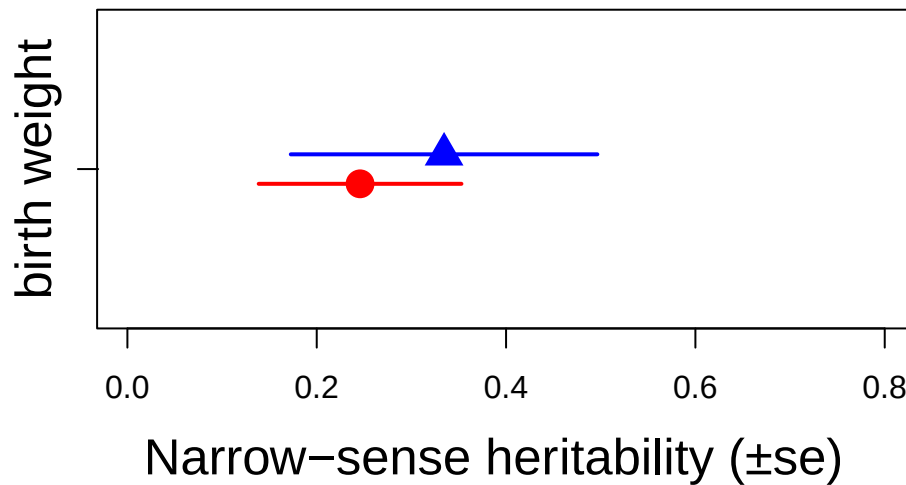


Figure 2.1.: Female and male heritability of birth weight

### 2.0.7. Modification of the variance matrix parameters

Variance represents the deviation of the distribution and it expected to be a positive values. Due to a lack of power, a structural problem in the dataset or a very low variance, Asreml-r often fixes the variance to a boundary B instead of a positive value P. When it is happen, it is generally a good idea to examine it.

To examine the boundary effect, we can explore an alternative model where the model allowed a unstructured parameter for the variance of interest or the entire variance matrix. For this example: we allowed the model to estimate any values (so allowing possible negative values of estimates) for the random and residual matrix.

First, we create a temporary model `model.temp` with the exact structure to modify.

```
model.temp <- asreml(
  fixed = bwt ~ 1,
  random = ~ vm(animal, ainv) + byear + mother,
  residual = ~ idv(units),
```

```
data = gryphon,
na.action = na.method(x = "omit", y = "omit"),
start.values = T
)
G.temp <- model.temp$vpparameters[(1:3), ]
G.temp$Constraint <- "U"
R.temp <- model.temp$vpparameters[-(1:3), ]
R.temp$Constraint[2] <- "U"
```

The argument `start.values=T` allowed the `model.temp` to change its random parameters. We can create the two different matrices and specify which parameters will be modified. For this example we modified the G and the R matrix to fit all variance to be U unstructured. it is important to note for the R matrix the line `units!R` has to be fix to 1, so it will never change.

The object `G.temp` and `R.temp` can be implemented in the following model as new parameters using the argument `R.param` and `G.param`.

```
model5 <- asreml(
  fixed = bwt ~ 1 + sex,
  random = ~ vm(animal, ainv) + byear + mother,
  residual = ~ idv(units),
  data = gryphon,
  na.action = na.method(x = "omit", y = "omit"),
  R.param = R.temp, G.param = G.temp
)
```

ASReml Version 4.2 08/10/2024 11:02:06

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -2033.178 | 1.0    | 852 | 11:02:06 |
| 2 | -1723.734 | 1.0    | 852 | 11:02:06 |
| 3 | -1396.354 | 1.0    | 852 | 11:02:06 |
| 4 | -1193.012 | 1.0    | 852 | 11:02:06 |
| 5 | -1107.946 | 1.0    | 852 | 11:02:06 |
| 6 | -1095.327 | 1.0    | 852 | 11:02:06 |

---

|   |           |     |     |          |
|---|-----------|-----|-----|----------|
| 7 | -1094.816 | 1.0 | 852 | 11:02:06 |
| 8 | -1094.815 | 1.0 | 852 | 11:02:06 |

```
summary(model5)$varcomp
```

|                  | component | std.error | z.ratio  | bound | %ch |
|------------------|-----------|-----------|----------|-------|-----|
| byear            | 0.8820313 | 0.2632455 | 3.350604 | U     | 0   |
| mother           | 1.1184698 | 0.2386239 | 4.687167 | U     | 0   |
| vm(animal, ainv) | 2.2985320 | 0.4962496 | 4.631806 | U     | 0   |
| units!units      | 1.6290034 | 0.3714154 | 4.385934 | U     | 0   |
| units!R          | 1.0000000 | NA        | NA       | F     | 0   |

Since model4 did not showed boundary, the model5 is very similar.

## 2.0.8. Covariance between two random effects

Some research questions require to estimate the covariance between two random effects within a univariate model. To do so, we can use the argument `str`. As an example, we fit a model which estimate the covariance between the additive genetic variance and the mother variance. Both variances require to operate on the same level, thus `animal` and `mother` require to be associated to the pedigree information.

The argument `str` has two components: first the equation term with the two random effects `~vm(animal, Ainv)+vm(mother, ainv)` and second the structural term `~us(2):id(number)`. Here within the structural term, we fit a 2x2 unstructured matrix `us(2)` which estimated the variance and the covariance between the random effects in the equation term. To successfully work, the structural term also requires the number of level identified within `id()`. Here a small tip, if you don't know the number of level identified within `id()`, run the model with a random number. The model will not converge and a error message will appear like this one: `Size of direct product (4) does not conform with total size of included terms (2618)`. The error message can help you determine the required level within the `str` function, as here 2618 divide by 2. In addition, it is necessary the random effects

```
model.temp2 <- asreml(
  fixed = bwt ~ 1,
  random = ~ str(~ vm(animal, ainv) + vm(mother, ainv), ~ us(2):id(1309)) + byear,
  residual = ~ idv(units),
```

```
data = gryphon,
na.action = na.method(x = "omit", y = "omit"),
start.values = T
)

G.temp2 <- model.temp2$vpparameters[(1:4), ]
G.temp2$Constraint <- "U"
model6 <- asreml(
  fixed = bwt ~ 1 + sex,
  random = ~ str(~ vm(animal, ainv) + vm(mother, ainv), ~ us(2):id(1309)) + byear,
  residual = ~ idv(units),
  data = gryphon,
  na.action = na.method(x = "omit", y = "omit"),
  # equate.levels = c("animal", "mother"),
  , G.param = G.temp2
)
summary(model6)$varcomp
```

We have successfully produced a code to estimate the covariance between two random effects. However for this example, the dataset is not sufficient to properly estimate it and the model did not converge but you have the idea of how to use the function `str`.

Additional and final tip: It is happen that Asreml will estimate negative variance if you allow the variance matrix to be unstructured. A negative variance is counter-intuitive meaning statistically the mean within the random effect is less similar than expected by chance. However a possible biological reason can be hypothesized such as a sibling competition within the nest creating a negative among-individual covariance within the nest. Thus to test this hypotheses, it is required to estimate the covariance between two random effects.



# Chapter 3

## MCMCglmm

### 3.0.1. Running the model

First load MCMCglmm:

```
library(MCMCglmm)
```

Loading required package: Matrix

Loading required package: coda

Loading required package: ape

The first model we will fit is a simple animal model with no fixed effects, and only an ‘animal’ random effect relating individuals to their additive genetic values through the pedigree.

First we are going to define the priors. In a way we might want to avoid using priors, because we would like all of the information in our analysis to come from our data. By default MCMCglmm uses improper priors, but this can cause inferential and numerical problems. We will specify priors for the animal effect and the residual variance using the following code:

```
prior1.1 <- list(  
  G = list(G1 = list(V = 1, nu = 0.002)),  
  R = list(V = 1, nu = 0.002)  
)
```

A prior allowed the model to fit different variance structures. With the unique random effect “animal”, we partitioned the phenotypic variance into two distinct variances matrices  $G$  (additive genetic) and  $R$  (residual). This prior specification is the simplistic one and often used because it was believed to be relatively uninformative, and is equivalent to an inverse-gamma prior with shape and scale equal to 0.001. In many cases it is relatively uninformative but when the posterior distribution for the variances has support close to zero it can behave poorly. Parameter expanded priors (See Chapter 8 of the MCMCglmm CourseNotes, available from CRAN) are gaining in popularity due to their better behaviour but for the purposes of this tutorial we will stick with the inverse-gamma prior.

We have told MCMCglmm to pay little heed to our prior expectation ( $V$ ) by specifying a small degree of belief parameter ( $\nu$ ) of 0.002. Since this is a univariate analysis, the priors are matrix of order 1 and thus  $\nu > 0$  is the smallest degree of belief that provides what is known as a ‘proper’ prior, avoiding numerical problems. In fact, there is a lot of information in the data regarding the marginal distributions of the parameters, and MCMCglmm will run most of the models that we suggest in these tutorials without priors. However, this is poor practice, but we will therefore use this simple priors throughout these tutorials. We can now fit an animal model. The model to decompose variation in birth weight into genetic and residual effects is as follows:

The lower case “animal” is a can be a **special** word for MCMCglmm. If a pedigree argument is provided then MCMCglmm will recognize the term `animal` as the term to use to estimate additive genetic variance. When the argument `pedigree` is not provided then the word `animal` is not different than any other variable. However, instead of providing a pedigree argument to the call to MCMCglmm function, it is much more flexible to use the `ginv` argument to specify the random effect that must be linked to the pedigree (with the inverse relatedness matrix). We thus first estimate the inverse relatedness matrix using `inverseA()` then fit the animal model.

```
Ainv <- inverseA(gryphonped)$Ainv
model1.1 <- MCMCglmm(bwt ~ 1,
  random = ~animal, ginv = list(animal = Ainv),
  data = gryphon, prior = prior1.1
)
```

MCMC iteration = 0

MCMC iteration = 1000

MCMC iteration = 2000

---

```
MCMC iteration = 3000  
  
MCMC iteration = 4000  
  
MCMC iteration = 5000  
  
MCMC iteration = 6000  
  
MCMC iteration = 7000  
  
MCMC iteration = 8000  
  
MCMC iteration = 9000  
  
MCMC iteration = 10000  
  
MCMC iteration = 11000  
  
MCMC iteration = 12000  
  
MCMC iteration = 13000
```

After typing this code, MCMCglmm will run, taking about 20 seconds on a modern desktop computer. The progress of the run will be printed to the screen. Also, note the warning message will be printed at the end of the run. This is natural too. In order for the MCMC algorithm to work, MCMCglmm must keep track of effects associated with unmeasured individuals appearing in the pedigree. This will not affect the answers, but when many unmeasured individuals exist, it can hinder the ability of the algorithm to explore the parameter space (more on this, and a solution, later). Lets have a look at the MCMCglmm outputs. First we will evaluate how confident we can be that MCMCglmm found good answers. By entering

```
plot(model1.1$Sol)
```

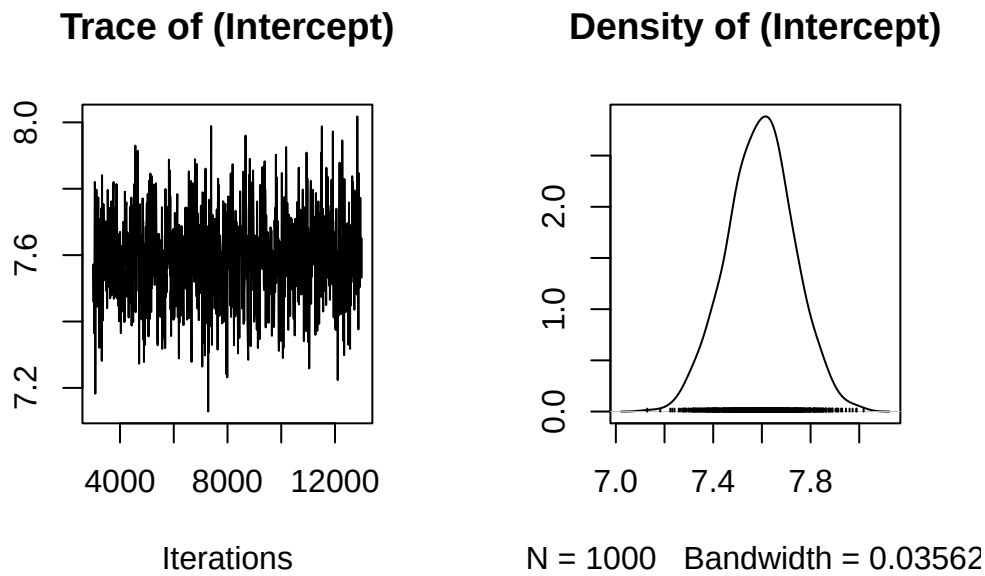


Figure 3.1.: The posterior distribution of the fixed effect (the intercept, or mean) in model 1.1

in the console, we get Figure 2.2. The plot on the left shows a time series of the values of 1000 samples of the posterior distribution of the the model intercept (mean birth weight). The plot on the right shows the same data as a distribution. Complicated statistical methods for estimating population means are of course of little interest; rather, we are examining these outputs to check that MCMCglmm's algorithms worked well for our data and for this model. The important point here is that a consistent amount of variation around a largely unchanging mean value of the intercept was obtained (which give this fluctuating trace concentrated around the mean), and the posterior distribution of the intercept appears to be valid. More rigorous means of evaluation the independence of the samples in the posterior distribution (evaluating autocorrelation) are discussed in the MCMCglmm CourseNotes, available from CRAN. Note that your output for model 1.1 may not be identical to this due to Monte Carlo (random number) error. So every times, you run the model, you will get similar but slightly different results.

The posterior distributions of the the variance components are generally of more interest to animal model users. We can view plots of the posterior distribution for the variance components for model 1.1 by

```
plot(model1.1$VCV)
```

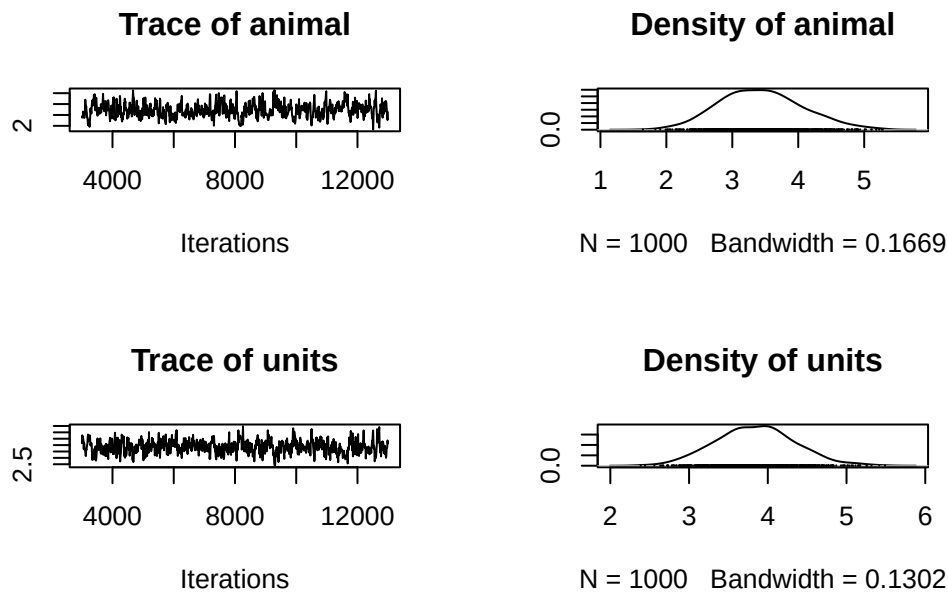


Figure 3.2.: The posterior distributions of the variance components of model 1.1, based on an analysis with the default values for nitt, burnin, and thin in MCMCglmm

which generates Figure 2.3. Here we see distributions of the estimates of the additive genetic (animal) and residual (units) effects. These samples contain some autocorrelation, i.e., trends are apparent in the left-hand plot. We can deal with this easily.

### 3.0.2. Change in iteration and sampling

We will simply re-run the model for a longer number of iterations, and sample the chain less frequently. So far we have been running MCMCglmm with its default values. These defaults are a total run length of 13000 iterations, the first 3000 of which are discarded as a ‘burn-in’ period to make sure that the converges to the part of the parameter space where the maximum likelihood exists. The remaining 10000 iterations are sampled (estimates retained) every 10 iterations (the thinning interval). Because the values in the left-hand plots in figure 2.2 to appear to have different values at the beginning of the run, we might suspect that a longer burn-in period might be required. We can reduce the autocorrelation by lengthening the rest of the run and sampling the chain less frequently. The following code runs the same model 1.1, but is likely to produce better samples of the posterior distributions. This model should take about two minutes to analyze.

```
model1.1 <- MCMCglmm(bwt ~ 1,
  random = ~animal, ginv = list(animal = Ainv),
  data = gryphon, nitt = 65000, thin = 50, burnin = 15000,
```

```
prior = prior1.1, verbose = FALSE
)
```

Notes that we have now included the argument `verbose=FALSE` in the `MCMCglmm` call. We will continue this throughout the tutorial so that more complete screen outputs can be included in this document without using too much space. Note that the autocorrelation is much reduced. A more compact way to evaluate the validity of the posterior distributions is to calculate autocorrelation among samples, as follows:

```
autocorr.diag(model1.1$VCV)
```

|          | animal      | units       |
|----------|-------------|-------------|
| Lag 0    | 1.00000000  | 1.00000000  |
| Lag 50   | 0.19000363  | 0.19720861  |
| Lag 250  | 0.03080980  | 0.01546578  |
| Lag 500  | -0.01505963 | -0.05286184 |
| Lag 2500 | -0.01755035 | -0.02538836 |

We will consider these levels of autocorrelation acceptable, at least for the purposes of this tutorial. Ideally, all samples of the posterior distribution should be independent, and the autocorrelation for all lag values greater than zero should be near zero. However, in practice this will not strictly be achievable for all analytic scenarios. Certainly the levels of autocorrelation observed here should not be tolerated in any formal analysis. Note that the validity of posterior distributions of any analysis should always be checked; however, for brevity we will not continue to be so consistently diligent throughout the rest of these tutorials. We can now proceed with confidence to recover some more information from these samples. We can obtain estimates of the additive genetic and residual variance by calculating the modes of the posterior distributions:

```
posterior.mode(model1.1$VCV)
```

| animal   | units    |
|----------|----------|
| 3.324804 | 3.789779 |

We can obtain the Bayesian equivalent of confidence intervals by calculating the the values of the estimates that bound 95% (or any other proportion) of the posterior distributions:

---

```
HPDinterval(model1.1$VCV)
```

```
      lower    upper
animal 2.249203 4.764617
units  2.929122 4.923796
attr(,"Probability")
[1] 0.95
```

### 3.0.3. Change priors parameters

We specified weak priors in this analyses. Now we will check whether or not proper priors would have influenced the results that we obtained. The simplest way to do this is to re-run the model with different priors. In the previous model we specified a prior where the size of genetic and residual variance were similar. Here we construct priors with a larger degree of belief parameter ( $\nu$ ), and we will specify that a large proportion (95%) of the variation is under genetic control (V). Thus, the residual variance contains 05% of the phenotypic variance.

```
p.var <- var(gryphon$bwt, na.rm = TRUE)
prior1.1.2 <- list(
  G = list(G1 = list(V = matrix(p.var * 0.95), nu = 1)),
  R = list(V = matrix(p.var * 0.05), nu = 1)
)

model1.1.2 <- MCMCglmm(bwt ~ 1,
  random = ~animal, ginv = list(animal = Ainv),
  data = gryphon, prior = prior1.1.2, nitt = 65000, thin = 50,
  burnin = 15000, verbose = FALSE
)
posterior.mode(model1.1$VCV)
```

```
      animal    units
3.324804 3.789779
```

```
posterior.mode(model1.1.2$VCV)
```

```
      animal      units  
3.769996 3.812074
```

and we can therefore conclude that the difference in the priors has little effect on the outcome of the analysis. This is typical for an analysis where lots of data are available relative to the complexity of the model, but is often not the case. In all cases, it is important to check the effect of priors on conclusions drawn from a model. In addition, you can also specify the prior with previous knowledge or expectation for the variance.

### 3.0.4. Estimating heritability

A useful property of Bayesian posterior distributions is that we can apply almost any transformation to these distributions and they will remain valid. This applies to the calculation of heritability. We can obtain an estimate of the heritability by applying the basic formula  $h^2 = V_A/V_P$  to each sample of the posterior distribution:

```
posterior.heritability1.1 <- model1.1$VCV[, "animal"] /  
  (model1.1$VCV[, "animal"] + model1.1$VCV[, "units"])  
  
posterior.mode(posterior.heritability1.1)
```

```
      var1  
0.4674414
```

```
HPDinterval(posterior.heritability1.1, 0.95)
```

```
      lower      upper  
var1 0.3287511 0.6268801  
attr(,"Probability")  
[1] 0.95
```

Generate a plot of the posterior distribution of this heritability estimate:



```
plot(posterior.heritability1.1)
```

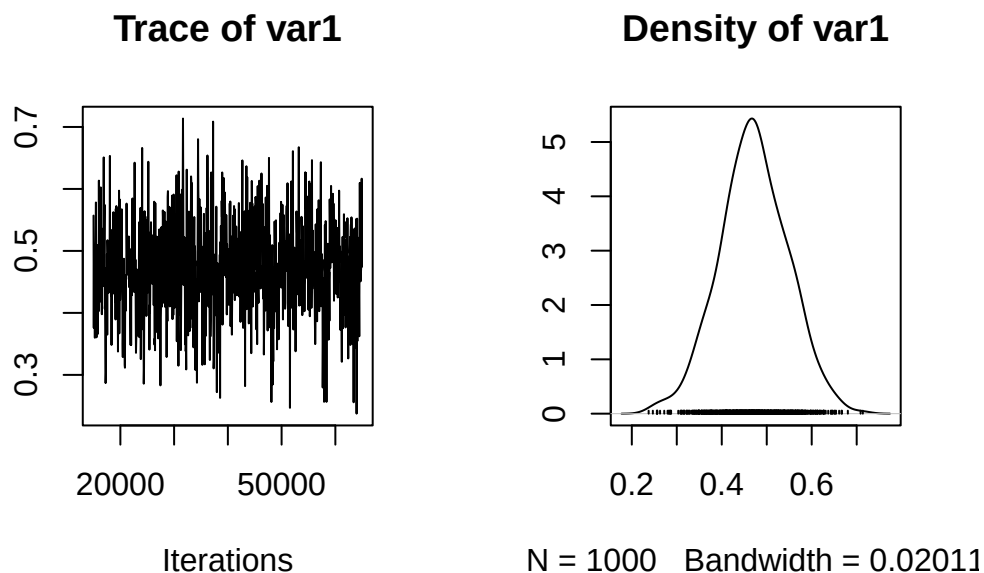


Figure 3.3.: The posterior distributions the heritability from model 1.1

### 3.0.5. Adding fixed effects

To add effects to a univariate model, we simply modify the fixed effect part of the model specification:

```
model1.2 <- MCMCglmm(bwt ~ sex,  
  random = ~animal, ginv = list(animal = Ainv),  
  data = gryphon, prior = prior1.1,  
  nitt = 65000, thin = 50, burnin = 15000, verbose = FALSE  
)  
summary(model1.2)
```

Iterations = 15001:64951

Thinning interval = 50

Sample size = 1000

DIC: 3717.393

G-structure: ~animal

```
      post.mean l-95% CI u-95% CI eff.samp
animal      3.083      2.05      4.016      755.1
```

```
R-structure: ~units
```

```
      post.mean l-95% CI u-95% CI eff.samp
units      2.944      2.189      3.779      805.6
```

```
Location effects: bwt ~ sex
```

```
      post.mean l-95% CI u-95% CI eff.samp pMCMC
(Intercept)      6.065      5.760      6.394      1000 <0.001 ***
sex2              2.207      1.849      2.492      1000 <0.001 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can assess the significance of `sex` as a fixed effect by examining its posterior distribution. Important notes here, it is important to know how the model names their fixed effect level to call them properly.

```
posterior.mode(model1.2$Sol[, "sex2"])
```

```
var1
2.22336
```

```
HPDinterval(model1.2$Sol[, "sex2"], 0.95)
```

```
      lower      upper
var1 1.849157 2.491577
attr(,"Probability")
[1] 0.95
```

The posterior distribution of the `sex2` term does not overlap zero. Thus, we can infer that sex has an effect on birth weight (presence of a sexual dimorphism) in this model and is a useful addition to the model, for most purposes. It is also worth noting that the variance components have changed slightly:

---

```
posterior.mode(model1.2$VCV)
```

```
      animal      units  
3.356471 2.875270
```

In fact since sex effects were previously contributing to the residual variance of the model our estimate of  $V_R$  (denoted 'units' in the output) is now slightly lower than before. This has an important consequence for estimating heritability since if we calculate  $V_P$  as  $V_A + V_R$  then as we include fixed effects we will soak up more residual variance driving  $V_P$ . Assuming that  $V_A$  is more or less unaffected by the fixed effects fitted then as  $V_P$  goes down we expect our estimate of  $h^2$  will go up.

```
posterior.heritability1.2 <- model1.2$VCV[, "animal"] /  
  (model1.2$VCV[, "animal"] + model1.2$VCV[, "units"])  
posterior.mode(posterior.heritability1.2)
```

```
      var1  
0.5494853
```

```
HPDinterval(posterior.heritability1.2, 0.95)
```

```
      lower      upper  
var1 0.3719122 0.6471908  
attr(,"Probability")  
[1] 0.95
```

Here  $h^2$  has increased slightly from 0.4829 to 0.5079 (again, your values may differ slightly due to Monte Carlo error). Which is the better estimate? It depends on what your question is. The first is an estimate of the proportion of variance in birth weight explained by additive effects, the latter is an estimate of the proportion of variance in birth weight after conditioning on sex that is explained by additive effects. An important piece of advice, each researcher should be consistent in how they name their estimates and always correctly describe which estimates they are using conditional or not (to avoid any confusion).

### 3.0.6. Adding random effects

This is done by simply modifying the model statement in the same way, but requires addition of a prior for the new random effect. For instance, we can fit an effect of birth year:

```
prior1.3 <- list(
  G = list(G1 = list(V = 1, nu = 0.002), G2 = list(V = 1, nu = 0.002)),
  R = list(V = 1, nu = 0.002)
)

model1.3 <- MCMCglmm(bwt ~ sex,
  random = ~ animal + byear, ginv = list(animal = Ainv),
  data = gryphon,
  nitt = 65000, thin = 50, burnin = 15000,
  prior = prior1.3, verbose = FALSE
)

posterior.mode(model1.3$VCV)
```

```
      animal      byear      units
2.7101798 0.8393404 2.3517527
```

Here the variance in birth weight explained by birth year is 0.84. Note that although  $V_A$  has changed somewhat, most of what is now partitioned as a birth year effect was previously partitioned as  $V_R$ . Thus what we have really done here is to partition environmental effects into those arising from year to year differences versus everything else, and we do not really expect much change in  $h^2$  (since now  $h^2 = V_A / (V_A + V_{BY} + V_R)$ ). However, we get a somewhat different result if we also add a random effect of mother to test for maternal effects:

```
prior1.4 <- list(
  G = list(
    G1 = list(V = 1, nu = 0.002),
    G2 = list(V = 1, nu = 0.002),
    G3 = list(V = 1, nu = 0.002)
  ),
```

---

```

R = list(V = 1, nu = 0.002)
)

model1.4 <- MCMCglmm(bwt ~ sex,
  random = ~ animal + byear + mother,
  ginv = list(animal = Ainv), data = gryphon,
  nitt = 65000, thin = 50, burnin = 15000,
  prior = prior1.4, verbose = FALSE
)

posterior.mode(model1.4$VCV)

```

```

      animal      byear      mother      units
2.2100541 0.8649762 1.0710032 1.5762728

```

Here partitioning of significant maternal variance has resulted in a further decrease in  $V_R$  but also a decrease in  $V_A$ . The latter is because maternal effects of the sort we simulated (fixed differences between mothers) will have the consequence of increasing similarity among maternal siblings. Consequently they can look very much like an additive genetic effects and if present, but unmodelled, represent a type of ‘common environment effect’ that can - and will- cause upward bias in  $V_A$  and so  $h^2$ . Let’s compare the estimates of heritability from each of models 1.2, 1.3 and 1.4:

```

posterior.heritability1.3 <- model1.3$VCV[, "animal"] /
  (model1.3$VCV[, "animal"] + model1.3$VCV[, "byear"] + model1.3$VCV[, "units"])
posterior.heritability1.4 <- model1.4$VCV[, "animal"] /
  (model1.4$VCV[, "animal"] + model1.4$VCV[, "byear"] + model1.4$VCV[, "mother"] + model1.4$VCV[, "units"])
posterior.mode(posterior.heritability1.2)

```

```

      var1
0.5494853

```

```

posterior.mode(posterior.heritability1.3)

```

```

      var1
0.4500106

```

```
posterior.mode(posterior.heritability1.4)
```

```
var1  
0.3812813
```

### 3.0.7. Testing significance of variance components

While testing the significance of fixed effects by evaluating whether or not their posterior distributions overlap zero was simple and valid, this approach does not work for variance components. Variance components are bounded to be positive (given a proper prior), and thus even when a random effect is not meaningful, its posterior distribution will never overlap zero. Model comparisons can be performed using the deviance information criterion (DIC), although it should be noted that the properties of DIC are not well understood and that the DIC may be focused at the wrong level for most people's intended level of inference - particularly with non-Gaussian responses. The implementation of DIC in MCMCglmm is further described in the reference manual. DIC values are calculated by MCMCglmm by default. Briefly, DIC like other information criteria balance model fit and model complexity simultaneously, and small values of DIC are preferred. We can compare models 1.4 and 1.3, i.e., models with and without the mother term:

```
model1.3$DIC
```

```
[1] 3547.575
```

```
model1.4$DIC
```

```
[1] 3297.787
```

model 1.4 has a much lower DIC value. Since the maternal effect term is the only difference between the models, we can consider the inclusion of this term statistically justifiable. We should note however that DIC has a large sampling variance and should probably only be calculated based on much longer MCMC runs.

### 3.0.8. Further partitioning variance

A population can be further fragmented into different groups or categories (such as females and males, juveniles and adults or treated and untreated). Some scientific questions require further and deeper analysis of the variance. To

avoid multiple model (one for each group), we can directly partition the variance between groups in a unique model. In addition, by doing so, we can also test if the variance are different between groups.

As example, we can partition the additive genetic variance and residual variance by sex. It is impossible to further partition the other variances but complexity an animal model requires sufficient sample size.

```
prior1.4.SEX <- list(
  G = list(G1 = list(V = diag(2), nu = 1.002), G2 = list(V = 1, nu = 0.002), G3 = list(V = 1, nu
  R = list(V = diag(2), nu = 1.002)
)

model1.4.SEX <- MCMCglmm(bwt ~ sex,
  random = ~ idh(sex):animal + byear + mother,
  rcov = ~ idh(sex):units,
  ginv = list(animal = Ainv), data = gryphon, nitt = 65000, thin = 50, burnin = 15000,
  prior = prior1.4.SEX, verbose = FALSE
)

posterior.mode(model1.4.SEX$VCV)
```

| sex1.animal | sex2.animal | byear     | mother    | sex1.units | sex2.units |
|-------------|-------------|-----------|-----------|------------|------------|
| 0.6470916   | 2.5074849   | 0.8829385 | 1.4621951 | 2.3933947  | 1.6630896  |

```
posterior.heritability1.4.FEM <- model1.4.SEX$VCV[, "sex1.animal"] /
  (model1.4.SEX$VCV[, "sex1.animal"] + model1.4.SEX$VCV[, "byear"] +
    model1.4.SEX$VCV[, "mother"] + model1.4.SEX$VCV[, "sex1.units"])
posterior.heritability1.4.MAL <- model1.4.SEX$VCV[, "sex2.animal"] /
  (model1.4.SEX$VCV[, "sex2.animal"] + model1.4.SEX$VCV[, "byear"] +
    model1.4.SEX$VCV[, "mother"] + model1.4.SEX$VCV[, "sex2.units"])

posterior.mode(posterior.heritability1.4.FEM)
```

var1

0.1742903

```
HPDinterval(posterior.heritability1.4.FEM, 0.95)
```

```

      lower      upper
var1 0.04975175 0.4445928
attr(,"Probability")
[1] 0.95
```

```
posterior.mode(posterior.heritability1.4.MAL)
```

```

      var1
0.2857213
```

```
HPDinterval(posterior.heritability1.4.MAL, 0.95)
```

```

      lower      upper
var1 0.08249098 0.6198509
attr(,"Probability")
[1] 0.95
```

Here, we can estimate the heritability for each sex. Both doesn't overlap with zero, so we can conclude both sexes have significant heritability. However due to their overlaps CIs, we can not conclude the heritability is not significantly different between sexes. An important quote to remember is "A difference in significance is not a significant difference"

```

h2.sex <- rbind(
  cbind(posterior.mode(posterior.heritability1.4.FEM), HPDinterval(posterior.heritability1.4.FEM,
  cbind(posterior.mode(posterior.heritability1.4.MAL), HPDinterval(posterior.heritability1.4.MAL,
)

plot(c(0.95, 1.05) ~ h2.sex[, 1], xlim = c(0, 0.8), ylim = c(0.5, 1.5), , xlab = "", ylab = "", c
arrows(y0 = 0.95, x0 = h2.sex[1, 2], y1 = 0.95, x1 = h2.sex[1, 3], code = 3, angle = 90, length =
arrows(y0 = 1.05, x0 = h2.sex[2, 2], y1 = 1.05, x1 = h2.sex[2, 3], code = 3, angle = 90, length =
mtext("Narrow-sense heritability ( $\pm$ CI)", side = 1, las = 1, adj = 0.4, line = 3, cex = 1.6)
axis(2, at = 1, labels = c("birth weight"), las = 3, cex.axis = 1.6)
```



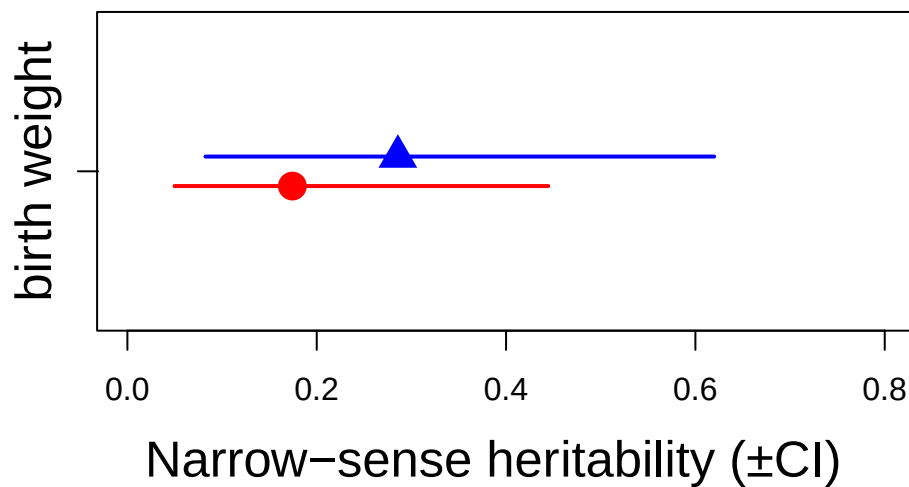


Figure 3.4.: Female and male heritability of birth weight

### 3.0.9. Modification of model parameter

Unfortunately (to our knowledge), it is not possible to alter the variance matrices and refit them within the model.

### 3.0.10. Covariance between two random effects

Some research questions require to estimate the covariance between two random effects within a univariate model. To do so, we can use the argument `str`. A similar argument or linking function `mm` can be used but it will force the variance of `animal` and `mother` to be equal and the covariance to 1. As an example, we fit a model which estimates the covariance between the additive genetic variance and the mother variance. Both variances require to operate on the same level, thus `animal` and `mother` require to be associated to the pedigree information. The `ginverse` list name has to correspond to the first term in the argument or linking function.

```
prior1.5 <- list(
  G = list(G1 = list(V = diag(2), nu = 0.002)),
  R = list(V = 1, nu = 0.002)
)

model1.5 <- MCMCglmm(bwt ~ sex,
  random = ~ str(animal + mother), ginv = list(animal = Ainv),
  rcov = ~ idh(1):units,
```

```
data = gryphon, nitt = 65000, thin = 50, burnin = 15000,  
prior = prior1.5, verbose = FALSE  
)  
  
posterior.mode(model1.5$VCV)
```

# Chapter 4

## brms

### 4.0.1. Running the model

First we need to load the brms library:

```
library(brms)
```

```
Loading required package: Rcpp
```

```
Loading 'brms' package (version 2.21.0). Useful instructions  
can be found by typing help('brms'). A more detailed introduction  
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':
```

```
ar
```

To be able to fit an animal model, brms needs the relativeness (relationship) matrix of the pedigree and not its inverse (as in other softwares). This can be estimated using the nadiv package created by Pr. Matthew Wolak (<https://cran.r-project.org/web/packages/nadiv/index.html>).

```
Amat <- as.matrix(nadiv::makeA(gryphonped))
```

We are now ready to specify our first model: The structure of a `brms` model is similar to `lme4`, thus the random effect is added to the model with the term `(1 | gr(animal, cov = Amat))` which associate the id `animal` to the matrix of relativeness. In addition to the synthase of `lme4`, we includes other features or parameters within the models such as `chain` which represent the number of Markov chains (defaults to 4), `core` which represents the number of cores to use when executing the chains in parallel and `iter` which represents the number of total iterations per chain. For more parameters such as `thin` or `warmup/burnin`, you can read the Cran R page of the package (<https://cran.r-project.org/web/packages/brms/brms.pdf>)

`brms` is a Bayesian Multilevel Models using Stan, doing so we can apply a prior to the model to better shape the distribution of the different variances estimated by the model. Given that `brms` fit the model using a Bayesian approach via the software `stan`, we need to specify priors for the model. Default priors in `brms` work relatively well, however we strongly suggest to carefully select an adequate prior for your analysis. In this tutorial we will use the default priors. To get the prior used by default, we can use the `get_prior()` function.

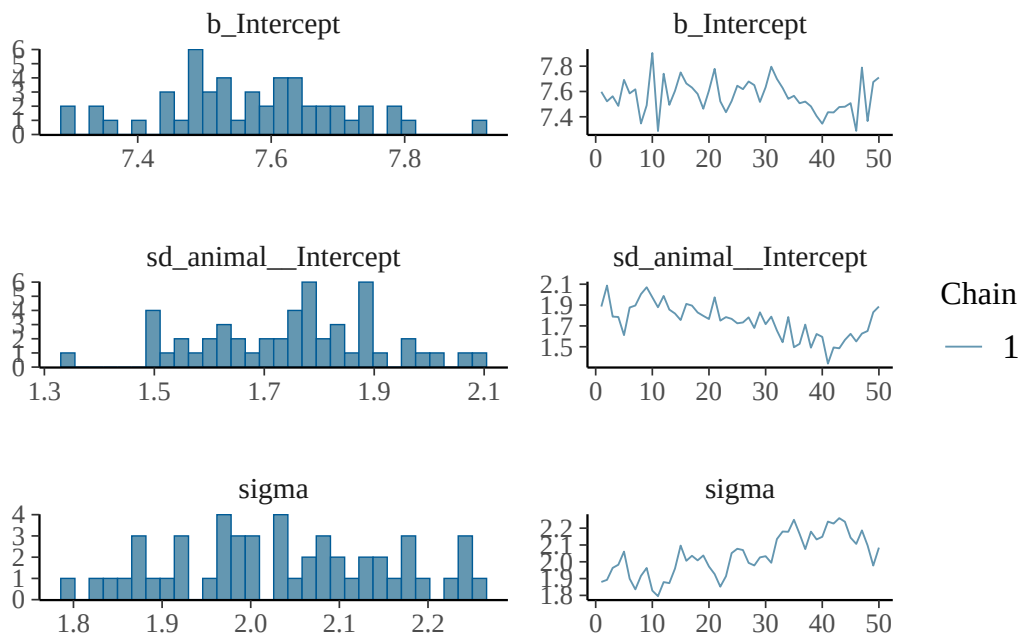
```
brms_m1.1 <- brm(
  bwt ~ 1 + (1 | gr(animal, cov = Amat)),
  data = gryphon,
  data2 = list(Amat = Amat),
  family = gaussian(),
  chains = 1, cores = 1, iter = 100
)

save(brms_m1.1, file = "r-obj/brms_m1_1.rda")
```

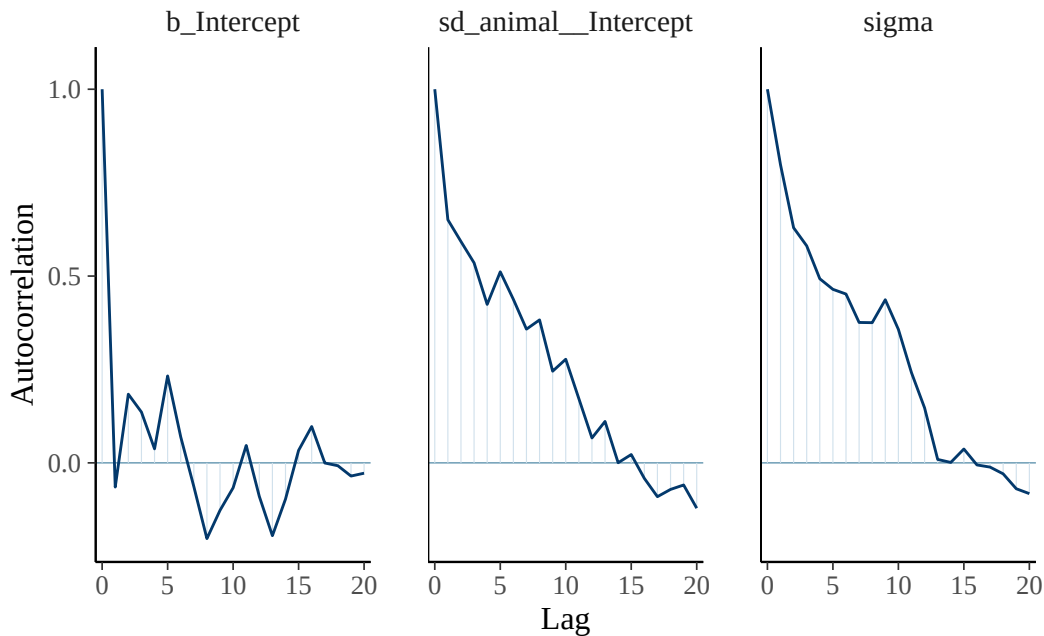
The result of the long model calculation is save in a spare file `brms_m1_1.rda`". To help readers, we can directly reloading it. Two distinct plot can be produce to produce some diagnostics graphs `mcmc_plot`.Note, that `sigma` represents the residual standard deviation.

Next,we examine (or directly using the model) the variance estimate and their distributions (via `summary` or `plot`).

```
load("r-obj/brms_m1_1.rda")
plot(brms_m1.1)
```



```
mcmc_plot(brms_m1.1, type = "acf")
```



```
summary(brms_m1.1)
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

Family: gaussian

```
Links: mu = identity; sigma = identity
Formula: bwt ~ 1 + (1 | gr(animal, cov = Amat))
Data: gryphon (Number of observations: 854)
Draws: 1 chains, each with iter = 100; warmup = 50; thin = 1;
      total post-warmup draws = 50
```

#### Multilevel Hyperparameters:

```
~animal (Number of levels: 1084)
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)      1.75      0.17      1.49      2.06 1.55          2          20
```

#### Regression Coefficients:

```
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept      7.57      0.13      7.30      7.79 1.00          27          63
```

#### Further Distributional Parameters:

```
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma      2.04      0.12      1.83      2.25 1.68          2          20
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

The plot of variance showed that the different variances have an normal distribution, the autocorrelation plot or ‘acf’ show that the autocorrelation is close to 0. The summary exposes the mean (Estimate) of each variance or fixed effect (here just the intercept) associated to their posterior distribution with standard deviation (Est.Error) and two-sided 95% Credible intervals. Rhat provides information on the estimate convergence. If it’s greater than 1, the chains have not yet converged and it will be require to run more iterations and/or set stronger priors. ESS represents the Effective sample values as the number of independent samples from the posterior distribution. However, for the purpose of this guide, the Rhat values are acceptable.

It is also possible to calculate the heritability using the function ‘as.mcmc’

```
v_animal <- (VarCorr(brms_m1.1, summary = FALSE)$animal$sd)^2
v_r <- (VarCorr(brms_m1.1, summary = FALSE)$residual$sd)^2
h.bwt.1 <- as.mcmc(v_animal / (v_animal + v_r))
summary(h.bwt.1)
```

Iterations = 1:50

Thinning interval = 1

Number of chains = 1

Sample size per chain = 50

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

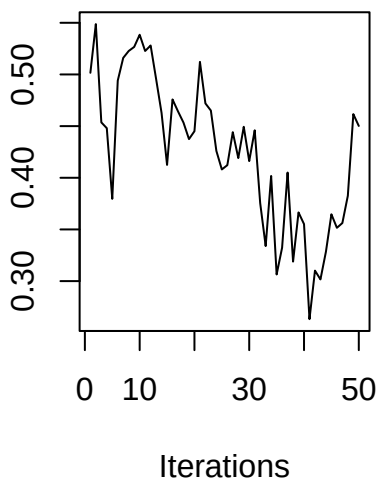
| Mean    | SD      | Naive SE | Time-series SE |
|---------|---------|----------|----------------|
| 0.42526 | 0.07162 | 0.01013  | 0.02854        |

2. Quantiles for each variable:

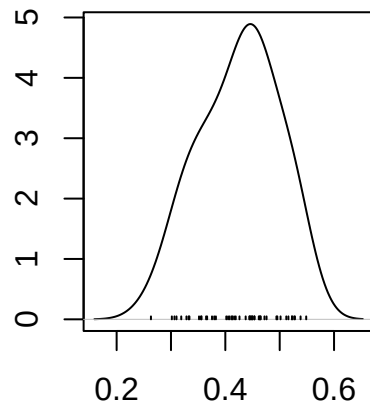
| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.3027 | 0.3687 | 0.4408 | 0.4702 | 0.5361 |

```
plot(h.bwt.1)
```

**Trace of Intercept**



**Density of Intercept**



```
# or
Var.table <- as_draws_df(brms_m1.1)
Var.table$h.bwt.1 <- as.mcmc((Var.table$sd_animal__Intercept)^2 / ((Var.table$sd_animal__Intercept)
summary(Var.table$h.bwt.1)
```

Iterations = 1:50

Thinning interval = 1

Number of chains = 1

Sample size per chain = 50

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

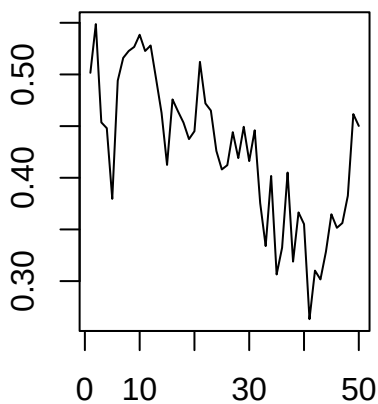
| Mean    | SD      | Naive SE | Time-series SE |
|---------|---------|----------|----------------|
| 0.42526 | 0.07162 | 0.01013  | 0.02854        |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.3027 | 0.3687 | 0.4408 | 0.4702 | 0.5361 |

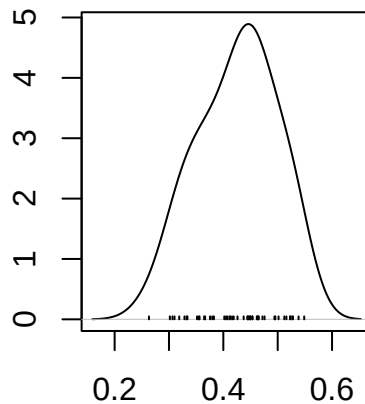
```
plot(Var.table$h.bwt.1)
```

**Trace of var1**



Iterations

**Density of var1**



N = 50 Bandwidth = 0.03472



---

## 4.0.2. Adding fixed effects

To add effects to a univariate model, we simply modify the priors and the fixed effect portion of the model specification:

```
brms_m1.2 <- brm(
  bwt ~ 1 + sex + (1 | gr(animal, cov = Amat)),
  data = gryphon,
  data2 = list(Amat = Amat),
  family = gaussian(),
  chains = 2, cores = 2, iter = 1000
)

save(brms_m1.2, file = "r-obj/brms_m1_2.rda")
```

To save time, the results of the calculation is stored in the spare file `brms_m1_2.rda`". We can assess the significance of sex as a fixed effect by examining its posterior distribution.

```
load("r-obj/brms_m1_2.rda")
summary(brms_m1.2)
```

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: bwt ~ 1 + sex + (1 | gr(animal, cov = Amat))
Data: gryphon (Number of observations: 854)
Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;
       total post-warmup draws = 1000

Multilevel Hyperparameters:
~animal (Number of levels: 1084)

      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)      1.67      0.14    1.39    1.97 1.02      93      120

Regression Coefficients:
```

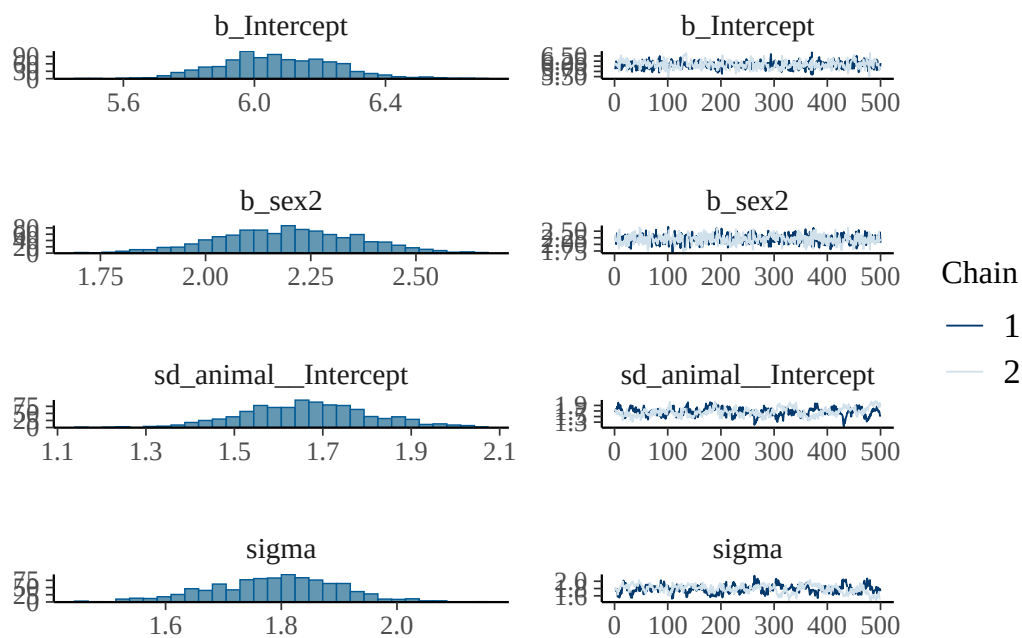
|           | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------|----------|-----------|----------|----------|------|----------|----------|
| Intercept | 6.07     | 0.18      | 5.75     | 6.43     | 1.00 | 616      | 451      |
| sex2      | 2.20     | 0.17      | 1.87     | 2.52     | 1.00 | 749      | 502      |

Further Distributional Parameters:

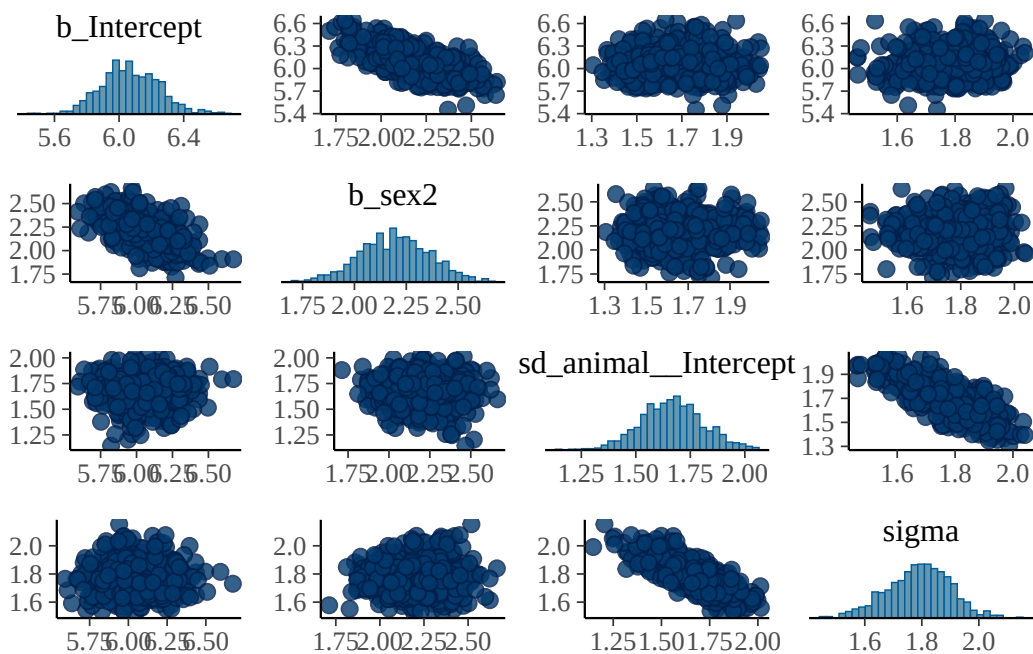
|       | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------|----------|-----------|----------|----------|------|----------|----------|
| sigma | 1.79     | 0.11      | 1.56     | 1.99     | 1.03 | 96       | 119      |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(brms_m1.2)
```



```
mcmc_plot(brms_m1.2, type = "pairs")
```



```
summary(brms_m1.2)$fixed
```

|           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat     | Bulk_ESS | Tail_ESS |
|-----------|----------|-----------|----------|----------|----------|----------|----------|
| Intercept | 6.071245 | 0.1774787 | 5.748606 | 6.428933 | 1.001657 | 615.7726 | 451.3620 |
| sex2      | 2.195386 | 0.1667400 | 1.872125 | 2.518738 | 1.002985 | 749.4558 | 502.4438 |

```
summary(brms_m1.2)$random
```

```
$animal
```

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat     | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|----------|----------|----------|
| sd(Intercept) | 1.667668 | 0.1442392 | 1.393627 | 1.967308 | 1.023006 | 93.37621 | 119.8231 |

The posterior distribution of the sex2 term does not overlap zero. Thus, we can infer that sex has an effect on birth weight (presence of a sexual dimorphism) in this model and is a useful addition to the model, for most purposes. It is also worth noting that the variance components have changed slightly:

```
summary(brms_m1.2)$random
```

```
$animal
```

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat     | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|----------|----------|----------|
| sd(Intercept) | 1.667668 | 0.1442392 | 1.393627 | 1.967308 | 1.023006 | 93.37621 | 119.8231 |

In fact since sex effects were previously contributing to the residual variance of the model our estimate of  $V_R$  (denoted 'units' in the output) is now slightly lower than before. This has an important consequence for estimating heritability since if we calculate  $V_P$  as  $V_A + V_R$  then as we include fixed effects we will soak up more residual variance driving  $V_P$ . Assuming that  $V_A$  is more or less unaffected by the fixed effects fitted then as  $V_P$  goes down we expect our estimate of  $h^2$  will go up.

```
v_animal <- (VarCorr(brms_m1.2, summary = FALSE)$animal$sd)^2
v_r <- (VarCorr(brms_m1.2, summary = FALSE)$residual$sd)^2
h.bwt.2 <- as.mcmc(v_animal / (v_animal + v_r))

summary(h.bwt.2)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.464637 | 0.068645 | 0.002171 | 0.007561       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.3375 | 0.4183 | 0.4620 | 0.5089 | 0.6030 |

```
summary(h.bwt.1)
```

```
Iterations = 1:50
```

```
Thinning interval = 1
```

---

Number of chains = 1

Sample size per chain = 50

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean    | SD      | Naive SE | Time-series SE |
|---------|---------|----------|----------------|
| 0.42526 | 0.07162 | 0.01013  | 0.02854        |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.3027 | 0.3687 | 0.4408 | 0.4702 | 0.5361 |

Here  $h^2$  has increased slightly from 0.5010 to 0.4192 (again, your values may differ slightly due to Monte Carlo error). Which is the better estimate? It depends on what your question is. The first is an estimate of the proportion of variance in birth weight explained by additive effects, the latter is an estimate of the proportion of variance in birth weight after conditioning on sex that is explained by additive effects. An important piece of advice, each researcher should be consistent in how they name their estimates and always correctly describe which estimates they are using conditional or not (to avoid any confusion).

### 4.0.3. Adding random effects

This is done by simply modifying the model statement in the same way, but requires addition of a prior for the new random effect. For instance, we can fit an effect of birth year:

```
brms_m1.3 <- brm(  
  bwt ~ 1 + sex + (1 | gr(animal, cov = Amat)) + (1 | byear) + (1 | mother),  
  data = gryphon,  
  data2 = list(Amat = Amat),  
  family = gaussian(),  
  chains = 2, cores = 2, iter = 1000  
)
```

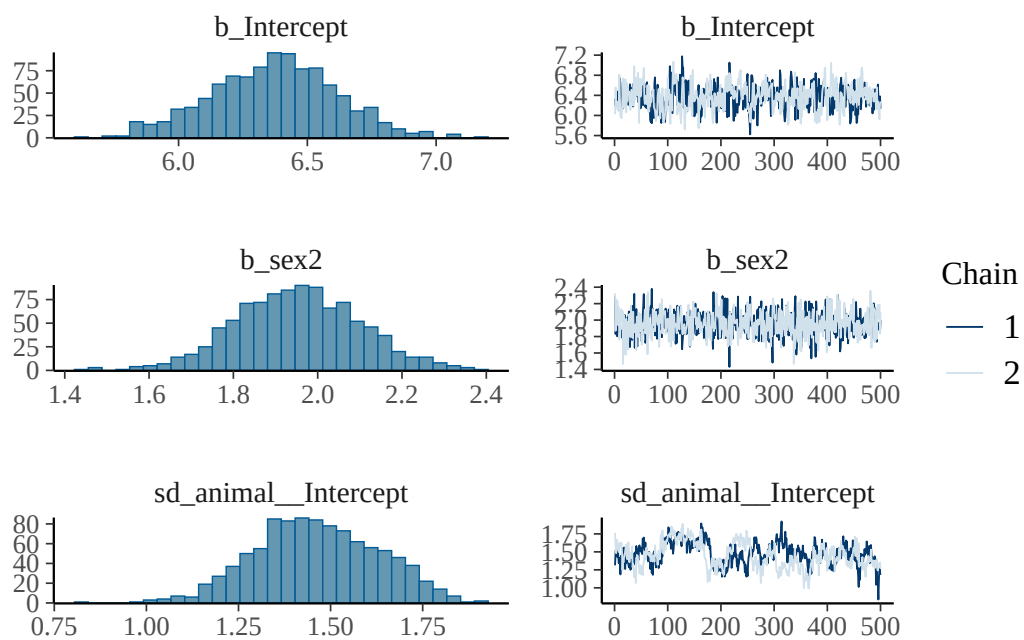
```
save(brms_m1.3, file = "r-obj/brms_m1_3.rda")
```

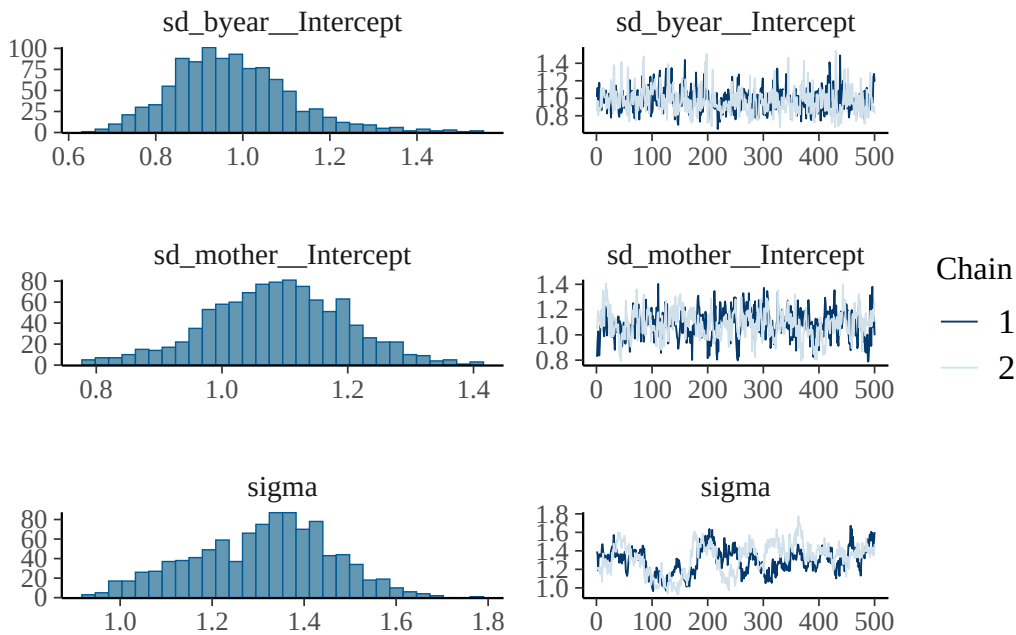
To save time, the results of the calculation is stored in the spare file `brms_m1_3.rda`". We can assess the significance of `sex` as a fixed effect by examining its posterior distribution.

```
load("r-obj/brms_m1_3.rda")
```

```
plot(brms_m1.3, ask = FALSE, N = 3)
```

Warning: Argument 'N' is deprecated. Please use argument 'nvariables' instead.





```
summary(brms_m1.3)$random
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

\$animal

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat     | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|----------|----------|----------|
| sd(Intercept) | 1.462757 | 0.1673822 | 1.153225 | 1.768881 | 1.079817 | 32.11486 | 156.8095 |

\$byear

|               | Estimate  | Est.Error | l-95% CI  | u-95% CI | Rhat     | Bulk_ESS | Tail_ESS |
|---------------|-----------|-----------|-----------|----------|----------|----------|----------|
| sd(Intercept) | 0.9767418 | 0.140156  | 0.7415211 | 1.306467 | 1.007016 | 469.316  | 407.5334 |

\$mother

|               | Estimate | Est.Error | l-95% CI  | u-95% CI | Rhat     | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|-----------|----------|----------|----------|----------|
| sd(Intercept) | 1.086973 | 0.1107873 | 0.8507715 | 1.296683 | 1.001921 | 228.2474 | 380.5216 |

Here partitioning of significant birth year and maternal variance has resulted in a further decrease in  $V_R$  but also a decrease in  $V_A$ . The latter is because maternal effects of the sort we simulated (fixed differences between mothers) will have the consequence of increasing similarity among maternal siblings. Consequently they can look very much

like an additive genetic effects and if present, but unmodelled, represent a type of ‘common environment effect’ that can - and will- cause upward bias in  $V_A$  and so  $h^2$ . Let’s compare the estimates of heritability from each of models 1.2, 1.3 and 1.4:

```
v_animal <- (VarCorr(brms_m1.3, summary = FALSE)$animal$sd)^2
v_byear <- (VarCorr(brms_m1.3, summary = FALSE)$byear$sd)^2
v_mother <- (VarCorr(brms_m1.3, summary = FALSE)$mother$sd)^2
v_r <- (VarCorr(brms_m1.3, summary = FALSE)$residual$sd)^2
h.bwt.3 <- as.mcmc(v_animal / (v_animal + v_byear + v_mother + v_r))
summary(h.bwt.3)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.356562 | 0.073890 | 0.002337 | 0.010196       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.2208 | 0.3042 | 0.3517 | 0.4111 | 0.5045 |

```
summary(h.bwt.2)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```



---

Sample size per chain = 1000

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.464637 | 0.068645 | 0.002171 | 0.007561       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.3375 | 0.4183 | 0.4620 | 0.5089 | 0.6030 |

```
summary(h.bwt.1)
```

Iterations = 1:50

Thinning interval = 1

Number of chains = 1

Sample size per chain = 50

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean    | SD      | Naive SE | Time-series SE |
|---------|---------|----------|----------------|
| 0.42526 | 0.07162 | 0.01013  | 0.02854        |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.3027 | 0.3687 | 0.4408 | 0.4702 | 0.5361 |

```
# or
Var.table <- as_draws_df(brms_m1.3)
Var.table$h.bwt.3 <- as.mcmc((Var.table$sd_animal__Intercept)^2 / ((Var.table$sd_animal__Intercept)
summary(Var.table$h.bwt.3)
```

Iterations = 1:1000

Thinning interval = 1

Number of chains = 1

Sample size per chain = 1000

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

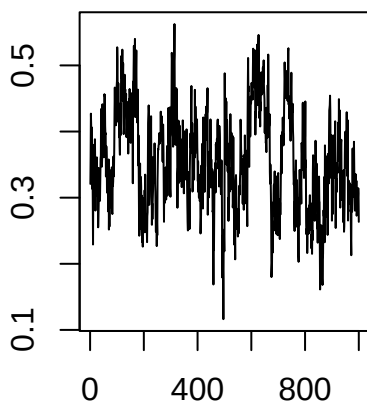
| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.356562 | 0.073890 | 0.002337 | 0.010196       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.2208 | 0.3042 | 0.3517 | 0.4111 | 0.5045 |

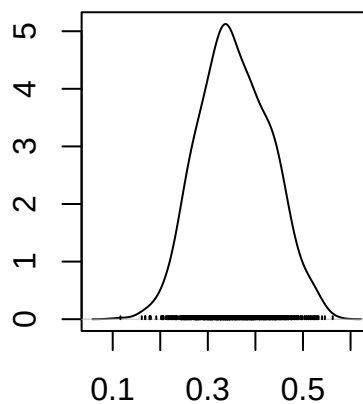
```
plot(Var.table$h.bwt.3)
```

**Trace of var1**



Iterations

**Density of var1**



N = 1000 Bandwidth = 0.01967

---

#### 4.0.4. Testing significance of variance components

While testing the significance of fixed effects by evaluating whether or not their posterior distributions overlap zero was simple and valid, this approach does not work for variance components. Variance components are bounded to be positive (given a proper prior), and thus even when a random effect is not meaningful, its posterior distribution will never overlap zero.

Model comparisons can be performed using the function `loo_compare` using `waic` or weighted AIC.

```
brms_m1.3 <- add_criterion(brms_m1.3, "loo")
```

```
Warning: Found 364 observations with a pareto_k > 0.67 in model 'brms_m1.3'. We
recommend to run more iterations to get at least about 2200 posterior draws to
improve LOO-CV approximation accuracy.
```

```
brms_m1.1 <- add_criterion(brms_m1.1, "loo")
```

```
Warning: Found 476 observations with a pareto_k > 0.41 in model 'brms_m1.1'. We
recommend to run more iterations to get at least about 2200 posterior draws to
improve LOO-CV approximation accuracy.
```

```
loo_compare(brms_m1.3, brms_m1.1, criterion = "loo")
```

|           | elpd_diff | se_diff |
|-----------|-----------|---------|
| brms_m1.3 | 0.0       | 0.0     |
| brms_m1.1 | -208.3    | 15.7    |

#### 4.0.5. Further partitioning of the variance

Depending of the research question and the presence of different group within the dataset, `brms` allowed to partition the variance at different groups. Two distinct approach can be done to partition the different random effect: using an extra argument `by=sex` or by adding `(0+sex|)` before the `|`. Notes, here we used `||` which not estimate a possible covariance between groups (female and male) for the random effect.

```
brms_m1.4 <- brm(
  # bwt ~ 1 + sex + (1 | gr(animal, cov = Amat, by = sex)) + (1 | gr(byear, by = sex)) + (1 | gr(mother)),
  bwt ~ 1 + sex + (0 + sex || gr(animal, cov = Amat)) + (0 + sex || byear) + (0 + sex || mother),
  data = gryphon,
  data2 = list(Amat = Amat),
  family = gaussian(),
  chains = 2, cores = 2, iter = 1000
)

save(brms_m1.4, file = "r-obj/brms_m1_4.rda")
```

To save time, the results of the calculation is stored in the spare file `brms_m1_4.rda`.

```
load("r-obj/brms_m1_4.rda")
summary(brms_m1.4)
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

Family: gaussian

Links: mu = identity; sigma = identity

Formula: `bwt ~ 1 + sex + (0 + sex || gr(animal, cov = Amat)) + (0 + sex || byear) + (0 + sex || mother)`

Data: gryphon (Number of observations: 854)

Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;  
total post-warmup draws = 1000

Multilevel Hyperparameters:

~animal (Number of levels: 1084)

|          | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|----------|----------|-----------|----------|----------|------|----------|----------|
| sd(sex1) | 1.32     | 0.24      | 0.81     | 1.75     | 1.02 | 40       | 124      |
| sd(sex2) | 0.91     | 0.38      | 0.12     | 1.53     | 1.07 | 22       | 70       |

~byear (Number of levels: 34)

---

|          | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|----------|----------|-----------|----------|----------|------|----------|----------|
| sd(sex1) | 0.91     | 0.17      | 0.62     | 1.26     | 1.00 | 632      | 604      |
| sd(sex2) | 1.08     | 0.20      | 0.76     | 1.50     | 1.00 | 429      | 614      |

~mother (Number of levels: 429)

|          | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|----------|----------|-----------|----------|----------|------|----------|----------|
| sd(sex1) | 0.91     | 0.24      | 0.33     | 1.33     | 1.03 | 93       | 93       |
| sd(sex2) | 1.39     | 0.16      | 1.09     | 1.69     | 1.02 | 210      | 262      |

Regression Coefficients:

|           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------|----------|-----------|----------|----------|------|----------|----------|
| Intercept | 6.28     | 0.23      | 5.85     | 6.74     | 1.00 | 635      | 490      |
| sex2      | 2.05     | 0.34      | 1.39     | 2.67     | 1.00 | 636      | 698      |

Further Distributional Parameters:

|       | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------|----------|-----------|----------|----------|------|----------|----------|
| sigma | 1.49     | 0.17      | 1.15     | 1.76     | 1.05 | 22       | 53       |

Draws were sampled using `sampling(NUTS)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat = 1`).

We can see the model estimate variance for both sexes. However, the residual level or `sigma` is not splitted by sexes. A futher and more complex code need to be performed, thus we can estimate the sex-specific heritability.

```
bf_m1.5 <- bf(
  bwt ~ 1 + sex + (0 + sex || gr(animal, cov = Amat)) + (0 + sex || mother) + (0 + sex || byear),
  sigma ~ sex - 1
)

brms_m1.5 <- brm(bf_m1.5,
  data = gryphon,
  data2 = list(Amat = Amat),
  family = gaussian(),
```

```
chains = 1, cores = 1, iter = 1000
)

save(brms_m1.5, file = "r-obj/brms_m1_5.rda")
```

To save time, the results of the calculation is stored in the spare file `brms_m1_4.rda`.

```
load("r-obj/brms_m1_5.rda")
summary(brms_m1.5)
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

```
Family: gaussian
Links: mu = identity; sigma = log
Formula: bwt ~ 1 + sex + (0 + sex || gr(animal, cov = Amat)) + (0 + sex || mother) + (0 + sex ||
          sigma ~ sex - 1
Data: gryphon (Number of observations: 854)
Draws: 1 chains, each with iter = 1000; warmup = 500; thin = 1;
       total post-warmup draws = 500
```

Multilevel Hyperparameters:

~animal (Number of levels: 854)

|          | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|----------|----------|-----------|----------|----------|------|----------|----------|
| sd(sex1) | 1.56     | 0.29      | 1.02     | 2.09     | 1.17 | 4        | 30       |
| sd(sex2) | 1.61     | 0.41      | 0.52     | 2.08     | 1.36 | 2        | 21       |

~byear (Number of levels: 34)

|          | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|----------|----------|-----------|----------|----------|------|----------|----------|
| sd(sex1) | 0.91     | 0.18      | 0.59     | 1.36     | 1.01 | 153      | 229      |
| sd(sex2) | 1.06     | 0.20      | 0.75     | 1.49     | 1.00 | 170      | 143      |

~mother (Number of levels: 394)

---

|          | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|----------|----------|-----------|----------|----------|------|----------|----------|
| sd(sex1) | 0.88     | 0.21      | 0.41     | 1.25     | 1.01 | 73       | 134      |
| sd(sex2) | 1.27     | 0.18      | 0.88     | 1.59     | 1.01 | 31       | 64       |

Regression Coefficients:

|            | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|------------|----------|-----------|----------|----------|------|----------|----------|
| Intercept  | 6.29     | 0.23      | 5.88     | 6.75     | 1.00 | 209      | 313      |
| sex2       | 2.02     | 0.31      | 1.49     | 2.66     | 1.00 | 127      | 296      |
| sigma_sex1 | 0.22     | 0.21      | -0.25    | 0.54     | 1.15 | 5        | 12       |
| sigma_sex2 | -0.20    | 0.40      | -0.82    | 0.54     | 1.59 | 2        | 15       |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
#
```

```
Var.table <- as_draws_df(brms_m1.5)
```

```
Var.table$h.bwt.f <- as.mcmc((Var.table$sd_animal__sex1)^2 / ((Var.table$sd_animal__sex1)^2 + (Var.table$sd_animal__sex1)^2))
```

```
Var.table$h.bwt.m <- as.mcmc((Var.table$sd_animal__sex2)^2 / ((Var.table$sd_animal__sex2)^2 + (Var.table$sd_animal__sex2)^2))
```

```
summary(Var.table$h.bwt.f)
```

```
Iterations = 1:500
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 500
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.575443 | 0.126621 | 0.005663 | 0.031251       |

2. Quantiles for each variable:

|  | 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--|--------|--------|--------|--------|--------|
|  | 0.3075 | 0.4863 | 0.5811 | 0.6741 | 0.7800 |

```
summary(Var.table$h.bwt.m)
```

```
Iterations = 1:500
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 500
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

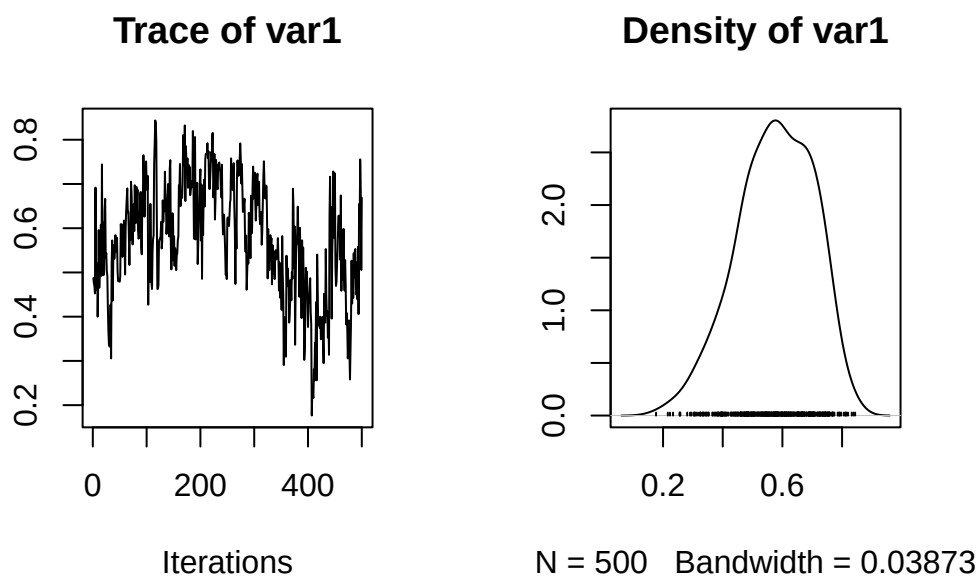
| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.463879 | 0.155395 | 0.006949 | 0.078323       |

2. Quantiles for each variable:

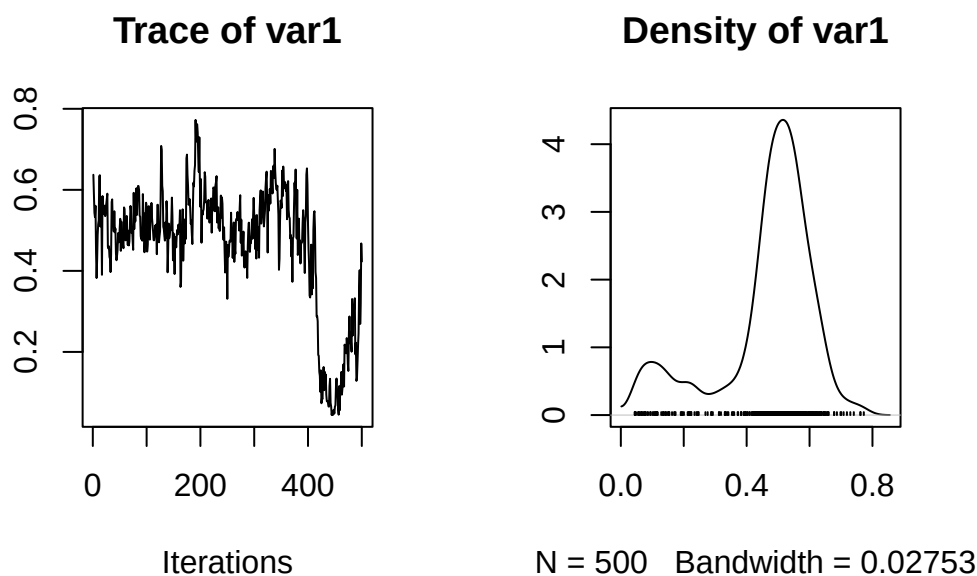
|  | 2.5%    | 25%     | 50%     | 75%     | 97.5%   |
|--|---------|---------|---------|---------|---------|
|  | 0.06693 | 0.43668 | 0.50150 | 0.55729 | 0.66016 |

```
plot(Var.table$h.bwt.f)
```





```
plot(Var.table$h.bwt.m)
```



Here, we can plot the point estimates of the  $h^2$  which seems to differ between sexes, but their CI overlaps.

```
h2.sex <- rbind(
  cbind(summary(Var.table$h.bwt.f)$statistics[1], summary(Var.table$h.bwt.f)$quantiles[1], summary(Var.table$h.bwt.f)$quantiles[2]),
  cbind(summary(Var.table$h.bwt.m)$statistics[1], summary(Var.table$h.bwt.m)$quantiles[1], summary(Var.table$h.bwt.m)$quantiles[2])
)

plot(c(0.95, 1.05) ~ h2.sex[, 1], xlim = c(0, 0.8), ylim = c(0.5, 1.5), , xlab = "", ylab = "", cex.lab = 1.2,
     arrows(y0 = 0.95, x0 = h2.sex[1, 2], y1 = 0.95, x1 = h2.sex[1, 3], code = 3, angle = 90, length = 10),
     arrows(y0 = 1.05, x0 = h2.sex[2, 2], y1 = 1.05, x1 = h2.sex[2, 3], code = 3, angle = 90, length = 10))
```

```
mtext("Narrow-sense heritability ( $\pm$ CI)", side = 1, las = 1, adj = 0.4, line = 3, cex = 1.6)
axis(2, at = 1, labels = c("birth weight"), las = 3, cex.axis = 1.6)
```

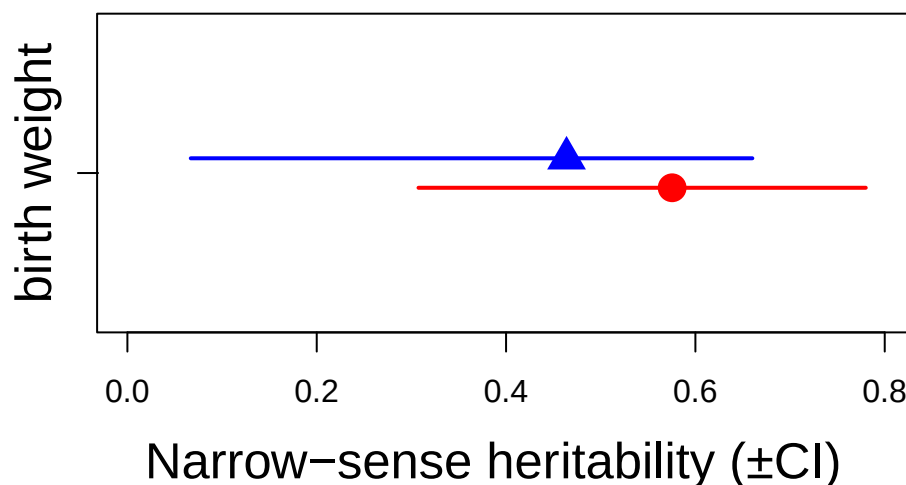


Figure 4.1.: Female and male heritability of birth weight

#### 4.0.6. Modification of model parameter

Unfortunately (to our knowledge), it is not possible to alter the variance matrices and refit them within the model.

#### 4.0.7. Covariance between two random effects

Some research questions require to estimate the covariance between two random effects within a univariate model. Unfortunately (to our knowledge), it is not possible to create a covariance between distinct random effects (<https://github.com/paul-buerkner/brms/issues/502>). However, a multi-membership model can be fit using the linking function `mm`, thus forcing the variance of two variables to be equal and the covariance to 1.

## **Part II.**

# **Multivariate animal model**

---

This tutorial will demonstrate how to run a multivariate animal model looking at birth weight and tarsus length of the phenomenal gryphons.

## Scenario and data

### Scenario

Since natural selection rarely acts on single traits, to understand how birth weight might evolve in our population of gryphons, we may also want to think about possible covariance with other traits. If tarsus length at fledging is also under positive selection, what implications does it have for birth weight and vice versa? If the two traits are positively genetically correlated then this will facilitate evolution of larger size (since response of one trait will induce a positively correlated response in the other). If there is negative genetic covariance then this could act as an evolutionary constraint.

Using multivariate models allows the estimation of parameters relating to each trait alone (*i.e.*  $V_A$ ,  $h^2$ , etc), but also yields estimates of covariance components between traits. These include the (additive) genetic covariance  $COV_A$  which is often rescaled to give the additive genetic correlation  $r_A$ . However, covariance can also arise through other random effects (*e.g.* maternal covariance) and these sources can also be explicitly modeled in a bivariate analysis.

### **gryphon files**

gryphonpedigree and phenotypic data files are the same as those used in tutorial 1 (*i.e.* gryphonped.csv and gryphon.csv respectively).

Reading the data

```
gryphon <- read.csv("data/gryphon.csv")
gryphon$animal <- as.factor(gryphon$animal)
gryphon$mother <- as.factor(gryphon$mother)
gryphon$byear <- as.factor(gryphon$byear)
gryphon$sex <- as.factor(gryphon$sex)
gryphon$bwt <- as.numeric(gryphon$bwt)
gryphon$tarsus <- as.numeric(gryphon$tarsus)
```

Reading the pedigree

```
gryphonped <- read.csv("data/gryphonped.csv")
gryphonped$id <- as.factor(gryphonped$id)
gryphonped$father <- as.factor(gryphonped$father)
gryphonped$mother <- as.factor(gryphonped$mother)
```

# Chapter 5

## Asreml-R

### 5.0.1. Running the model

First we need to load the `asreml` library:

```
library(asreml)
```

```
Loading required package: Matrix
```

```
Attaching package: 'Matrix'
```

```
The following objects are masked from 'package:tidyr':
```

```
expand, pack, unpack
```

```
Online License checked out Tue Oct 8 11:28:58 2024
```

```
Loading ASReml-R version 4.2
```

For running multivariate analyses in ASReml-R, the code is slightly more complex than for the univariate case. This is because ASReml-R allows us to make different assumptions about the way in which traits might be related. We need to explicitly specify a covariance structure with difference covariance functions `us()`, `idh()` or `corgh()` which for example would estimate an unconstrained (co)variance matrix, an identity matrix and a variance and correlation matrix respectively. We can also specify some starting values for the variance matrices. These can be

very approximate *guestimates* or not at all, but having reasonable starting values can help convergence. It is also possible to let the model running without specifying starting values. Finally, we have increased the default maximum number of iterations (`maxiter`) which can help to achieve convergence for more complicated models. Another way to increase the number of iteration will be to use the update function. Notes that if the `LogLik` is not stabilized after several iterations, it is good indication of the model require more iteration.

```
ainv <- ainverse(gryphonped)

modela <- asreml(
  fixed = cbind(bwt, tarsus) ~ trait,
  random = ~ us(trait, init = c(1, 0.1, 1)):vm(animal, ainv),
  residual = ~ id(units):us(trait, init = c(1, 0.1, 1)),
  data = gryphon,
  na.action = na.method(x = "include", y = "include"),
  maxit = 20
)
```

ASReml Version 4.2 08/10/2024 11:28:59

|    | LogLik    | Sigma2 | DF   | wall     |
|----|-----------|--------|------|----------|
| 1  | -7108.741 | 1.0    | 1535 | 11:28:59 |
| 2  | -5837.803 | 1.0    | 1535 | 11:28:59 |
| 3  | -4437.495 | 1.0    | 1535 | 11:28:59 |
| 4  | -3459.378 | 1.0    | 1535 | 11:28:59 |
| 5  | -2914.034 | 1.0    | 1535 | 11:28:59 |
| 6  | -2729.131 | 1.0    | 1535 | 11:28:59 |
| 7  | -2684.659 | 1.0    | 1535 | 11:28:59 |
| 8  | -2679.838 | 1.0    | 1535 | 11:28:59 |
| 9  | -2679.742 | 1.0    | 1535 | 11:28:59 |
| 10 | -2679.741 | 1.0    | 1535 | 11:28:59 |

```
modela <- update(modela)
```

ASReml Version 4.2 08/10/2024 11:28:59

| LogLik | Sigma2 | DF | wall |
|--------|--------|----|------|
|--------|--------|----|------|

|   |           |     |      |          |
|---|-----------|-----|------|----------|
| 1 | -2679.741 | 1.0 | 1535 | 11:28:59 |
| 2 | -2679.741 | 1.0 | 1535 | 11:28:59 |

modela has fitted a bivariate model of bwt and tarsus, with the mean for each of the traits as a fixed effect (trait). The additive genetic variance-covariance matrix (**G**) is unstructured (us; *i.e.* all elements are free to vary) and the starting values for  $V_A$  for bwt,  $COV_A$  between bwt and tarsus, and  $V_A$  for tarsus are set to 1, 0.1 and 1, respectively. Similarly, the residual matrix is unstructured and uses the same starting values.

Note that the argument `na.action = na.method(x = "include", y = "include")` can be added to the model. In a bivariate model, it will help calculate the covariance between two traits with different missing information NA and so help imbalance phenotypage and save sample size. However, it is important to scale (mean = 0, var = 1) the two traits to correctly adjust the model (see Asreml-R manual for more information).

Let's have a look at the variance components, and notice that there are now seven (co)variance components reported in the table:

```
summary(modela)$varcomp
```

|                                            | component | std.error | z.ratio  | bound | %ch |
|--------------------------------------------|-----------|-----------|----------|-------|-----|
| trait:vm(animal, ainv)!trait_bwt:bwt       | 3.368397  | 0.6348307 | 5.305977 | P     | 0   |
| trait:vm(animal, ainv)!trait_tarsus:bwt    | 2.459809  | 1.0732644 | 2.291895 | P     | 0   |
| trait:vm(animal, ainv)!trait_tarsus:tarsus | 12.345792 | 3.0744285 | 4.015638 | P     | 0   |
| units:trait!R                              | 1.000000  | NA        | NA       | F     | 0   |
| units:trait!trait_bwt:bwt                  | 3.849916  | 0.5200101 | 7.403542 | P     | 0   |
| units:trait!trait_tarsus:bwt               | 3.313282  | 0.9129234 | 3.629310 | P     | 0   |
| units:trait!trait_tarsus:tarsus            | 17.646432 | 2.6670380 | 6.616491 | P     | 0   |

The first three terms are related to the genetic matrix and, in order are  $V_{A,bwt}$ ,  $COV_A$ ,  $V_{A,tarsus}$ . Below is again a line where the `units:trait!R` component equals to 1, which again can be ignored. The final three terms relate to the residual matrix and correspond to  $V_{R,bwt}$ ,  $COV_R$ ,  $V_{R,tarsus}$ . Based on our quick and dirty check (is `z.ratio` > 1.96?) all components look to be statistically significant.

We can calculate the genetic correlation as  $COV_A / \sqrt{V_{A,bwt} \cdot V_{A,tarsus}}$ . Thus this model gives an estimate of  $r_A = 0.38$ . It is also possible to estimate the residual correlation  $r_{res} = 0.4$ .



---

Both correlations are distinct in nature. The genetic correlation reflects how much the traits are linked by genetic via polygenic effect or linkage disequilibrium, whereas the residual correlation reflects the environmental correlation or errors measurement correlation.

Although we can calculate this by hand, we can also use `vpredict()`, which also provides an (approximate) standard error:

```
vpredict(modela, r_A ~ V2 / sqrt(V1 * V3))
```

|     | Estimate  | SE        |
|-----|-----------|-----------|
| r_A | 0.3814436 | 0.1299759 |

```
vpredict(modela, r_res ~ V6 / sqrt(V5 * V7))
```

|       | Estimate  | SE       |
|-------|-----------|----------|
| r_res | 0.4019799 | 0.086071 |

Of course we can also calculate the heritability of bwt and tarsus from this model:

```
vpredict(modela, h2.bwt ~ V1 / (V1 + V5))
```

|        | Estimate | SE        |
|--------|----------|-----------|
| h2.bwt | 0.466646 | 0.0767153 |

```
vpredict(modela, h2.tarsus ~ V3 / (V3 + V7))
```

|           | Estimate  | SE        |
|-----------|-----------|-----------|
| h2.tarsus | 0.4116331 | 0.0930586 |

### 5.0.2. Adding fixed and random effects

Fixed and random effects can be added just as for the univariate case. Given that our full model of bwt from tutorial 1 had sex as a fixed effect as well as birth year and mother as random effects, we could specify a bivariate formulation with the same complexity:

```
modelb <- asreml(
  fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex,
  random = ~ us(trait, init = c(1, 0.1, 1)):vm(animal, ainv) +
    us(trait, init = c(1, 0.1, 1)):byear +
    us(trait, init = c(1, 0.1, 1)):mother,
  residual = ~ id(units):us(trait, init = c(1, 0.1, 1)),
  data = gryphon,
  na.action = na.method(x = "include", y = "include"),
  maxit = 20
)
```

ASReml Version 4.2 08/10/2024 11:28:59

|   | LogLik    | Sigma2 | DF   | wall                     |
|---|-----------|--------|------|--------------------------|
| 1 | -4672.301 | 1.0    | 1533 | 11:28:59                 |
| 2 | -4005.616 | 1.0    | 1533 | 11:28:59                 |
| 3 | -3271.484 | 1.0    | 1533 | 11:28:59 ( 1 restrained) |
| 4 | -2761.414 | 1.0    | 1533 | 11:29:00 ( 1 restrained) |
| 5 | -2481.355 | 1.0    | 1533 | 11:29:00                 |
| 6 | -2395.858 | 1.0    | 1533 | 11:29:00                 |
| 7 | -2381.050 | 1.0    | 1533 | 11:29:00                 |
| 8 | -2380.251 | 1.0    | 1533 | 11:29:00                 |
| 9 | -2380.246 | 1.0    | 1533 | 11:29:00                 |

```
modelb <- update(modelb)
```

ASReml Version 4.2 08/10/2024 11:29:00

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2380.246 | 1.0    | 1533 | 11:29:00 |
| 2 | -2380.246 | 1.0    | 1533 | 11:29:00 |

Note that we have specified a covariance structure for each random effect and an estimate of the effect of sex on both birth weight and tarsus length.

There will now be thirteen (co)variance components reported after running the code:

```
summary(modelb)$varcomp
```

|                                            | component  | std.error | z.ratio    | bound | %ch |
|--------------------------------------------|------------|-----------|------------|-------|-----|
| trait:byear!trait_bwt:bwt                  | 0.9746385  | 0.2825727 | 3.4491602  | P     | 0   |
| trait:byear!trait_tarsus:bwt               | 0.1624076  | 0.4185079 | 0.3880635  | P     | 0   |
| trait:byear!trait_tarsus:tarsus            | 3.7383721  | 1.2065992 | 3.0982716  | P     | 0   |
| trait:mother!trait_bwt:bwt                 | 1.1445184  | 0.2302182 | 4.9714512  | P     | 0   |
| trait:mother!trait_tarsus:bwt              | -1.5567306 | 0.4051848 | -3.8420260 | P     | 0   |
| trait:mother!trait_tarsus:tarsus           | 4.8206132  | 1.3201300 | 3.6516202  | P     | 0   |
| trait:vm(animal, ainv)!trait_bwt:bwt       | 1.9893546  | 0.4410246 | 4.5107569  | P     | 0   |
| trait:vm(animal, ainv)!trait_tarsus:bwt    | 3.3170404  | 0.9032323 | 3.6724110  | P     | 0   |
| trait:vm(animal, ainv)!trait_tarsus:tarsus | 10.2294887 | 2.8077066 | 3.6433610  | P     | 0   |
| units:trait!R                              | 1.0000000  | NA        | NA         | F     | 0   |
| units:trait!trait_bwt:bwt                  | 1.8443110  | 0.3443178 | 5.3564203  | P     | 0   |
| units:trait!trait_tarsus:bwt               | 4.0142841  | 0.7412540 | 5.4155308  | P     | 0   |
| units:trait!trait_tarsus:tarsus            | 12.4845955 | 2.2893363 | 5.4533690  | P     | 0   |

we can estimate the different correlations using `vpredict`:

```
vpredict(modelb, r_byear ~ V2 / sqrt(V1 * V3))
```

|         | Estimate  | SE        |
|---------|-----------|-----------|
| r_byear | 0.0850831 | 0.2134209 |

```
vpredict(modelb, r_M ~ V5 / sqrt(V4 * V6))
```

|     | Estimate   | SE        |
|-----|------------|-----------|
| r_M | -0.6627518 | 0.2487963 |

```
vpredict(modelb, r_A ~ V8 / sqrt(V7 * V9))
```

|     | Estimate  | SE        |
|-----|-----------|-----------|
| r_A | 0.7353053 | 0.1094747 |

```
vpredict(modelb, r_res ~ V12 / sqrt(V11 * V13))
```

|       | Estimate  | SE        |
|-------|-----------|-----------|
| r_res | 0.8365729 | 0.0736676 |

Now we can look at the fixed effects parameters and assess their significance with a conditional Wald F-test:

|                           | solution   | std error | z.ratio    |
|---------------------------|------------|-----------|------------|
| trait_bwt                 | 6.3844483  | 0.2328210 | 27.4221324 |
| trait_tarsus              | 20.5936436 | 0.5098944 | 40.3880569 |
| at(trait, 'bwt'):sex_1    | 0.0000000  | NA        | NA         |
| at(trait, 'bwt'):sex_2    | 1.9502053  | 0.1480467 | 13.1729086 |
| at(trait, 'tarsus'):sex_1 | 0.0000000  | NA        | NA         |
| at(trait, 'tarsus'):sex_2 | -0.0684413 | 0.3823448 | -0.1790041 |

ASReml Version 4.2 08/10/2024 11:29:00

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2380.246 | 1.0    | 1533 | 11:29:01 |
| 2 | -2380.246 | 1.0    | 1533 | 11:29:01 |

|                      | Df | denDF | F.inc     | F.con     | Margin | Pr       |
|----------------------|----|-------|-----------|-----------|--------|----------|
| trait                | 2  | 52.6  | 1.396e+03 | 1.396e+03 |        | 0.000000 |
| at(trait, 'bwt'):sex | 1  | 812.8 | 2.984e+02 | 1.735e+02 | B      | 0.000000 |

|                         | Df | denDF | F.inc     | F.con     | Margin | Pr       |
|-------------------------|----|-------|-----------|-----------|--------|----------|
| at(trait, 'tarsus'):sex | 1  | 747.9 | 3.204e-02 | 3.204e-02 | B      | 0.857984 |

Note that it is possible to specify a fixed effect to a specific trait by adding the number of order within `cbind` inside the argument `at(trait, x)`. For example, here we apply the fixed effect `sex` only to the response variable `tarsus`.

```
modelb_2 <- asreml(
  fixed = cbind(bwt, tarsus) ~ trait + at(trait, 2):sex,
  random = ~ us(trait, init = c(1, 0.1, 1)):vm(animal, ainv) +
    us(trait, init = c(1, 0.1, 1)):byear +
    us(trait, init = c(1, 0.1, 1)):mother,
  residual = ~ id(units):us(trait, init = c(1, 0.1, 1)),
  data = gryphon,
  na.action = na.method(x = "include", y = "include"),
  maxit = 20
)
```

ASReml Version 4.2 08/10/2024 11:29:01

|    | LogLik    | Sigma2 | DF   | wall     |                 |
|----|-----------|--------|------|----------|-----------------|
| 1  | -4810.563 | 1.0    | 1534 | 11:29:01 |                 |
| 2  | -4129.799 | 1.0    | 1534 | 11:29:01 |                 |
| 3  | -3382.529 | 1.0    | 1534 | 11:29:01 | ( 1 restrained) |
| 4  | -2864.076 | 1.0    | 1534 | 11:29:01 |                 |
| 5  | -2574.891 | 1.0    | 1534 | 11:29:01 |                 |
| 6  | -2478.879 | 1.0    | 1534 | 11:29:01 |                 |
| 7  | -2458.305 | 1.0    | 1534 | 11:29:01 |                 |
| 8  | -2456.425 | 1.0    | 1534 | 11:29:01 |                 |
| 9  | -2456.377 | 1.0    | 1534 | 11:29:01 |                 |
| 10 | -2456.376 | 1.0    | 1534 | 11:29:01 |                 |

|              | solution  | std error | z.ratio  |
|--------------|-----------|-----------|----------|
| trait_bwt    | 7.636226  | 0.2389515 | 31.95722 |
| trait_tarsus | 22.703658 | 0.4827348 | 47.03133 |

```
at(trait, 'tarsus'):sex_1 0.000000      NA      NA
at(trait, 'tarsus'):sex_2 -3.267042 0.2953279 -11.06242
```

```
ASReml Version 4.2 08/10/2024 11:29:02
```

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2456.376 | 1.0    | 1534 | 11:29:02 |
| 2 | -2456.376 | 1.0    | 1534 | 11:29:02 |

|                         | Df | denDF | F.inc  | F.con  | Margin | Pr |
|-------------------------|----|-------|--------|--------|--------|----|
| trait                   | 2  | 50.7  | 1233.0 | 1233.0 |        | 0  |
| at(trait, 'tarsus'):sex | 1  | 522.9 | 122.4  | 122.4  | B      | 0  |

### 5.0.3. Significance testing

Under the model above  $r_M$  is estimated as -0.66 and the `z.ratio` associated with the corresponding covariance ( $COV_M$ ) is >2 (in absolute terms). We might therefore infer that there is evidence for a strong negative correlation between the traits with respect to the mother and that while maternal identity explains variance in both traits those mothers that tend to produce heavier offspring actually tend to produce offspring with shorter tarsus lengths.

To formally test if  $COV_M$  is significantly different from zero, we can compare the log-likelihood for this model:

```
modelb$loglik
```

```
[1] -2380.246
```

to a model in which we specify that  $COV_M=0$ . Since this constraint reduces the number of parameters to be estimated by one, we can use a likelihood ratio test (LRT) with one degree of freedom. To run the constrained model, we modify the G structure defined for the mother random effect to diagonal (`diag`), which means we only estimate the variances (the diagonal of the matrix) but not the covariance (the covariance are fixed to 0):

```
modelc <- asreml(
  fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex,
  random = ~ us(trait, init = c(1, 0.1, 1)):vm(animal, ainv) +
    us(trait, init = c(1, 0.1, 1)):byear +
```

```

    diag(trait, init = c(1, 1)):mother,
  residual = ~ id(units):us(trait, init = c(1, 0.1, 1)),
  data = gryphon,
  na.action = na.method(x = "include", y = "include"),
  maxit = 20
)

```

ASReml Version 4.2 08/10/2024 11:29:02

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -4677.820 | 1.0    | 1533 | 11:29:02 |
| 2 | -4010.442 | 1.0    | 1533 | 11:29:02 |
| 3 | -3275.409 | 1.0    | 1533 | 11:29:02 |
| 4 | -2763.519 | 1.0    | 1533 | 11:29:02 |
| 5 | -2483.732 | 1.0    | 1533 | 11:29:02 |
| 6 | -2400.242 | 1.0    | 1533 | 11:29:02 |
| 7 | -2386.663 | 1.0    | 1533 | 11:29:02 |
| 8 | -2386.049 | 1.0    | 1533 | 11:29:02 |
| 9 | -2386.045 | 1.0    | 1533 | 11:29:02 |

You can run `summary(modelc)$varcomp` to confirm this worked. We can now obtain the log-likelihood of this model and compare this to that of `modelb` using a likelihood ratio test:

```
modelc$loglik
```

```
[1] -2386.045
```

We can see that the model log-likelihood is now -2386.05. And comparing the models using a likelihood ratio test:

```
2 * (modelb$loglik - modelc$loglik)
```

```
[1] 11.59835
```

So our chi-square test statistic is  $\chi^2_1 = 11.6$ . The p-value that goes with this is obtained by:

```
1 - pchisq(2 * (modelb$loglik - modelc$loglik), 1)
```

```
[1] 0.0006601037
```

We would therefore conclude that the maternal covariance is significantly different from zero.

We could apply the same procedure to show that the residual (environmental) covariance and the genetic covariance estimates are significantly greater than zero (*i.e.*, heavier individuals tend to have longer tarsus lengths). In contrast, we should find that the byear covariance between the two traits is non-significant.

```
modeld <- asreml(  
  fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex,  
  random = ~ us(trait, init = c(1, 0.1, 1)):vm(animal, ainv) +  
    diag(trait, init = c(1, 1)):byear +  
    us(trait, init = c(1, 0.1, 1)):mother,  
  residual = ~ id(units):us(trait, init = c(1, 0.1, 1)),  
  data = gryphon,  
  na.action = na.method(x = "include", y = "include"),  
  maxit = 20  
)
```

ASReml Version 4.2 08/10/2024 11:29:03

|   | LogLik    | Sigma2 | DF   | wall                     |
|---|-----------|--------|------|--------------------------|
| 1 | -4672.708 | 1.0    | 1533 | 11:29:03                 |
| 2 | -4005.954 | 1.0    | 1533 | 11:29:03                 |
| 3 | -3271.738 | 1.0    | 1533 | 11:29:03 ( 1 restrained) |
| 4 | -2761.626 | 1.0    | 1533 | 11:29:03 ( 1 restrained) |
| 5 | -2481.647 | 1.0    | 1533 | 11:29:03                 |
| 6 | -2395.992 | 1.0    | 1533 | 11:29:04                 |
| 7 | -2381.136 | 1.0    | 1533 | 11:29:04                 |
| 8 | -2380.331 | 1.0    | 1533 | 11:29:04                 |
| 9 | -2380.326 | 1.0    | 1533 | 11:29:04                 |



---

```
2 * (modelb$loglik - modeld$loglik)
```

```
[1] 0.1600641
```

```
1 - pchisq(2 * (modelb$loglik - modeld$loglik), 1)
```

```
[1] 0.6890975
```

#### 5.0.4. Estimate directly the genetic correlation within the model

Within Asreml-r, different matrix structure can be specify such as us,corg, diag, etc (cf see the Asreml-r guide). Instead of the fitting an unstructured matrix with the argument us or a reduced model with no covariance with the argument diag, we can also directly estimate the genetic correlation between the bwt and tarsus with corgh.

Here we decide to estimate directly the additive genetic correlation.

```
modele <- asreml(  
  fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex,  
  random = ~ corgh(trait, init = c(0.1, 1, 1)):vm(animal, ainv) +  
    us(trait, init = c(1, 0.1, 1)):byear +  
    us(trait, init = c(1, 0.1, 1)):mother,  
  residual = ~ id(units):us(trait, init = c(1, 0.1, 1)),  
  data = gryphon,  
  na.action = na.method(x = "include", y = "include"),  
  maxit = 20  
)
```

ASReml Version 4.2 08/10/2024 11:29:04

|   | LogLik    | Sigma2 | DF   | wall     |                 |
|---|-----------|--------|------|----------|-----------------|
| 1 | -4672.301 | 1.0    | 1533 | 11:29:04 |                 |
| 2 | -4003.183 | 1.0    | 1533 | 11:29:05 |                 |
| 3 | -3266.521 | 1.0    | 1533 | 11:29:05 | ( 1 restrained) |
| 4 | -2757.188 | 1.0    | 1533 | 11:29:05 | ( 1 restrained) |
| 5 | -2479.291 | 1.0    | 1533 | 11:29:05 |                 |

|   |           |     |      |          |
|---|-----------|-----|------|----------|
| 6 | -2395.476 | 1.0 | 1533 | 11:29:05 |
| 7 | -2381.026 | 1.0 | 1533 | 11:29:05 |
| 8 | -2380.251 | 1.0 | 1533 | 11:29:05 |
| 9 | -2380.246 | 1.0 | 1533 | 11:29:05 |

```
modele <- update(modele)
```

```
ASReml Version 4.2 08/10/2024 11:29:05
```

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2380.246 | 1.0    | 1533 | 11:29:06 |
| 2 | -2380.246 | 1.0    | 1533 | 11:29:06 |

```
summary(modele)$varcomp
```

|                                                    | component  | std.error | z.ratio   | bound | %ch |
|----------------------------------------------------|------------|-----------|-----------|-------|-----|
| trait:byear!trait_bwt:bwt                          | 0.9746386  | 0.2825728 | 3.449159  | P     | 0   |
| trait:byear!trait_tarsus:bwt                       | 0.1624071  | 0.4185082 | 0.388062  | P     | 0   |
| trait:byear!trait_tarsus:tarsus                    | 3.7383734  | 1.2066018 | 3.098266  | P     | 0   |
| trait:mother!trait_bwt:bwt                         | 1.1445186  | 0.2302183 | 4.971450  | P     | 0   |
| trait:mother!trait_tarsus:bwt                      | -1.5567316 | 0.4051850 | -3.842027 | P     | 0   |
| trait:mother!trait_tarsus:tarsus                   | 4.8206154  | 1.3201324 | 3.651615  | P     | 0   |
| trait:vm(animal, ainv)!trait!tarsus:!trait!bwt.cor | 0.7353061  | 0.1094807 | 6.716310  | U     | 0   |
| trait:vm(animal, ainv)!trait_bwt                   | 1.9893543  | 0.4410243 | 4.510758  | P     | 0   |
| trait:vm(animal, ainv)!trait_tarsus                | 10.2294850 | 2.8077055 | 3.643361  | P     | 0   |
| units:trait!R                                      | 1.0000000  | NA        | NA        | F     | 0   |
| units:trait!trait_bwt:bwt                          | 1.8443112  | 0.3443178 | 5.356422  | P     | 0   |
| units:trait!trait_tarsus:bwt                       | 4.0142825  | 0.7412540 | 5.415529  | P     | 0   |
| units:trait!trait_tarsus:tarsus                    | 12.4845977 | 2.2893355 | 5.453372  | P     | 0   |

It is important to note that using `corgh` change the order of the estimate (co)variance/correlation. Thus, the initial values need to be reorder and all different calculation need to be adjust in consequence. It is also important to check the difference between the model with `us` and `corgh` to make sure any mistake are made.

---

```
summary(modelb)$loglik
```

```
[1] -2380.246
```

```
summary(modele)$loglik
```

```
[1] -2380.246
```

There two main advantages to use `corgh`: first, a direct estimation of correlation within the G matrix can avoid mistake in the `vpredict` calculation; second, it is possible to test if the correlation is significantly different than 0 (similar result as LRT with the covariance) but also to -1 and 1 which correspond of the correlation boundaries. The following code showed how to create a reduced model with the correlation close to 1 and compared to the initial model. Since we compared the correlation to its boundary, the degree of freedom is only half as a one tail LTR.

```
MODEL_MODIF <- update.asreml(modele, start.values = T)
G_MOD <- MODEL_MODIF$vpparameters.table[(1:9), ]
G_MOD[1, 2] <- 0.99999
G_MOD[1, 3] <- "F"
modele.red <- asreml(
  fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex,
  random = ~ corgh(trait, init = c(0.1, 1, 1)):vm(animal, ainv) +
    us(trait, init = c(1, 0.1, 1)):byear +
    us(trait, init = c(1, 0.1, 1)):mother,
  residual = ~ id(units):us(trait, init = c(1, 0.1, 1)),
  data = gryphon,
  na.action = na.method(x = "include", y = "include"),
  maxit = 20,
  G.param = G_MOD
)
```

ASReml Version 4.2 08/10/2024 11:29:06

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2545.233 | 1.0    | 1533 | 11:29:06 |
| 2 | -2483.883 | 1.0    | 1533 | 11:29:06 |

|    |           |     |      |          |
|----|-----------|-----|------|----------|
| 3  | -2423.504 | 1.0 | 1533 | 11:29:06 |
| 4  | -2392.509 | 1.0 | 1533 | 11:29:07 |
| 5  | -2383.661 | 1.0 | 1533 | 11:29:07 |
| 6  | -2383.084 | 1.0 | 1533 | 11:29:07 |
| 7  | -2383.033 | 1.0 | 1533 | 11:29:07 |
| 8  | -2383.022 | 1.0 | 1533 | 11:29:07 |
| 9  | -2383.019 | 1.0 | 1533 | 11:29:07 |
| 10 | -2383.019 | 1.0 | 1533 | 11:29:07 |

```
2 * (modele$loglik - modele.red$loglik)
```

```
[1] 5.544679
```

```
1 - pchisq(2 * (modele$loglik - modele.red$loglik), df = 0.5)
```

```
[1] 0.006598676
```

Here, the correlation is significantly different than 1 (~0.99999).

### 5.0.5. Visualisation of the correlation (aka BLUP extraction)

When estimating correlation between traits, having a visualization of it can help the interpretation. In addition, visualizing the correlation can spot outliers in the dataset. Thanks to mixed model, each breeding values is stored within the model and can be extract as BLUP (Best Linear Unbiased Predictor). BLUP should be normally distributed, if not you need to check the assumption of your animal model.

To simplify the following code, we rename the variable T1 and T2.

```
gryphon$T1 <- gryphon$bwt
gryphon$T2 <- gryphon$tarsus
#####
modele <- asreml(
  fixed = cbind(T1, T2) ~ trait + at(trait):sex,
  random = ~ corgh(trait, init = c(0.1, 1, 1)):vm(animal, ainv) +
    us(trait, init = c(1, 0.1, 1)):byear +
```

```

us(trait, init = c(1, 0.1, 1)):mother,
residual = ~ id(units):us(trait, init = c(1, 0.1, 1)),
data = gryphon,
na.action = na.method(x = "include", y = "include"),
maxit = 20
)

```

ASReml Version 4.2 08/10/2024 11:29:07

|   | LogLik    | Sigma2 | DF   | wall                     |
|---|-----------|--------|------|--------------------------|
| 1 | -4672.301 | 1.0    | 1533 | 11:29:07                 |
| 2 | -4003.183 | 1.0    | 1533 | 11:29:07                 |
| 3 | -3266.521 | 1.0    | 1533 | 11:29:07 ( 1 restrained) |
| 4 | -2757.188 | 1.0    | 1533 | 11:29:07 ( 1 restrained) |
| 5 | -2479.291 | 1.0    | 1533 | 11:29:07                 |
| 6 | -2395.476 | 1.0    | 1533 | 11:29:07                 |
| 7 | -2381.026 | 1.0    | 1533 | 11:29:07                 |
| 8 | -2380.251 | 1.0    | 1533 | 11:29:07                 |
| 9 | -2380.246 | 1.0    | 1533 | 11:29:08                 |

```
modele <- update(modele)
```

ASReml Version 4.2 08/10/2024 11:29:08

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2380.246 | 1.0    | 1533 | 11:29:08 |
| 2 | -2380.246 | 1.0    | 1533 | 11:29:08 |

```
summary(modele)$varcomp
```

|                          | component | std.error | z.ratio  | bound | %ch |
|--------------------------|-----------|-----------|----------|-------|-----|
| trait:byear!trait_T1:T1  | 0.9746386 | 0.2825728 | 3.449159 | P     | 0   |
| trait:byear!trait_T2:T1  | 0.1624071 | 0.4185082 | 0.388062 | P     | 0   |
| trait:byear!trait_T2:T2  | 3.7383734 | 1.2066018 | 3.098266 | P     | 0   |
| trait:mother!trait_T1:T1 | 1.1445186 | 0.2302183 | 4.971450 | P     | 0   |

|                                               | component  | std.error | z.ratio   | bound | %ch |
|-----------------------------------------------|------------|-----------|-----------|-------|-----|
| trait:mother!trait_T2:T1                      | -1.5567316 | 0.4051850 | -3.842027 | P     | 0   |
| trait:mother!trait_T2:T2                      | 4.8206154  | 1.3201324 | 3.651615  | P     | 0   |
| trait:vm(animal, ainv)!trait!T2:!trait!T1.cor | 0.7353061  | 0.1094807 | 6.716310  | U     | 0   |
| trait:vm(animal, ainv)!trait_T1               | 1.9893543  | 0.4410243 | 4.510758  | P     | 0   |
| trait:vm(animal, ainv)!trait_T2               | 10.2294850 | 2.8077055 | 3.643361  | P     | 0   |
| units:trait!R                                 | 1.0000000  | NA        | NA        | F     | 0   |
| units:trait!trait_T1:T1                       | 1.8443112  | 0.3443178 | 5.356422  | P     | 0   |
| units:trait!trait_T2:T1                       | 4.0142825  | 0.7412540 | 5.415529  | P     | 0   |
| units:trait!trait_T2:T2                       | 12.4845977 | 2.2893355 | 5.453372  | P     | 0   |

```
#####
DvsS <- data.frame(
  Trait = rownames(modele$coefficients$random),
  BLUP = modele$coefficients$random,
  SE = sqrt(modele$vccoeff$random * modele$sigma2)
)
DvsS$ID <- substr(DvsS$Trait, 27, 30)
DvsS$TRAIT <- substr(DvsS$Trait, 7, 8)
DvsS <- DvsS[927:3544, ] # keep only row associated to animal
summary(factor(DvsS$TRAIT)) # 1309 each
```

```
T1  T2
846 1772
```

```
#
```

```
DvsS$Trait <- NULL
colnames(DvsS)[1] <- "BLUP"
BLUPS <- reshape(DvsS, v.names = c("BLUP", "SE"), idvar = "ID", timevar = "TRAIT", direction = "w")
```

```
Warning in reshapeWide(data, idvar = idvar, timevar = timevar, varying =
varying, : multiple rows match for TRAIT=T1: first taken
```

---

Warning in reshapeWide(data, idvar = idvar, timevar = timevar, varying =  
varying, : multiple rows match for TRAIT=T2: first taken

```
nrow(BLUPS)
```

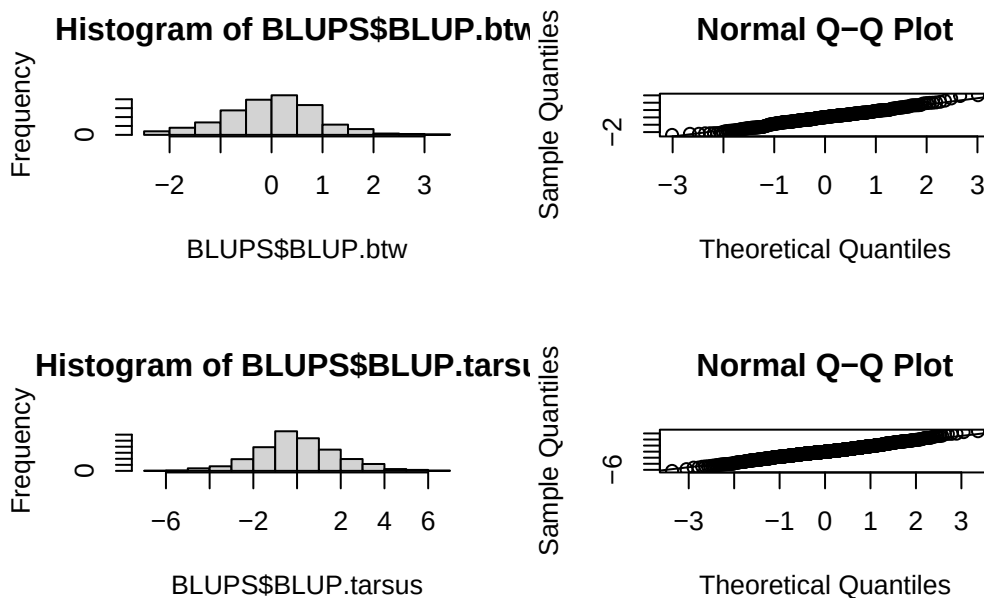
```
[1] 1310
```

```
rownames(BLUPS) <- c()  
colnames(BLUPS) <- c("ID", "BLUP.btw", "SE.btw", "BLUP.tarsus", "SE.tarsus")  
summary(BLUPS)
```

| ID               | BLUP.btw         | SE.btw          | BLUP.tarsus       |
|------------------|------------------|-----------------|-------------------|
| Length:1310      | Min. : -2.3778   | Min. : 0.7494   | Min. : -6.34104   |
| Class :character | 1st Qu.: -0.5797 | 1st Qu.: 0.9993 | 1st Qu.: -1.14403 |
| Mode :character  | Median : 0.0350  | Median : 1.0223 | Median : -0.02524 |
|                  | Mean : -0.0082   | Mean : 1.0640   | Mean : 0.02189    |
|                  | 3rd Qu.: 0.5911  | 3rd Qu.: 1.0552 | 3rd Qu.: 1.17735  |
|                  | Max. : 3.0123    | Max. : 1.4377   | Max. : 6.71502    |
|                  | NA's : 926       | NA's : 926      |                   |
| SE.tarsus        |                  |                 |                   |
| Min. : 1.616     |                  |                 |                   |
| 1st Qu.: 2.371   |                  |                 |                   |
| Median : 2.451   |                  |                 |                   |
| Mean : 2.576     |                  |                 |                   |
| 3rd Qu.: 2.810   |                  |                 |                   |
| Max. : 3.287     |                  |                 |                   |

```
# write.csv(BLUPS,file="BLUPS_6x6.csv",row.names=F)  
#####  
par(mfrow = c(2, 2))  
hist(BLUPS$BLUP.btw)  
qqnorm(BLUPS$BLUP.btw)  
qqline(BLUPS$BLUP.btw)
```

```
hist(BLUPS$BLUP.tarsus)
qqnorm(BLUPS$BLUP.tarsus)
qqline(BLUPS$BLUP.tarsus)
```

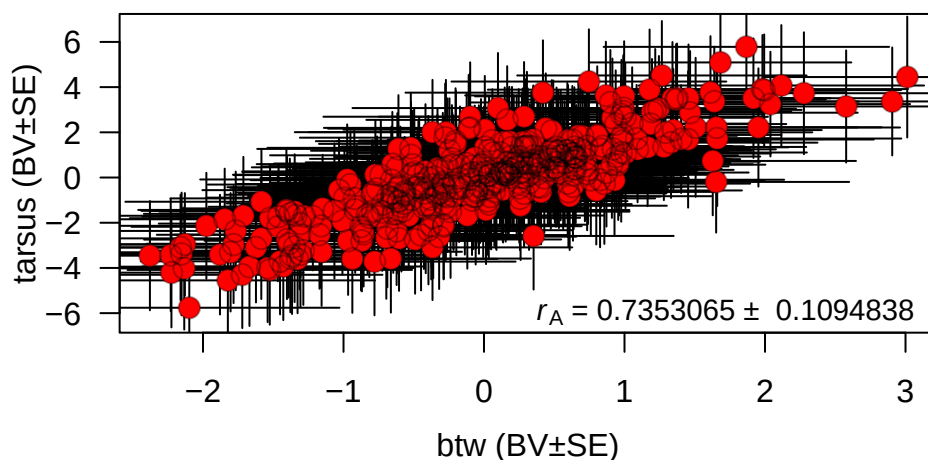


#

Here, some simple code to plot the genetic correlation.

```
plot(BLUP.tarsus ~ BLUP.btw, BLUPS, xlab = "", ylab = "", las = 1.2, bty = "n", col = "white")
arrows(x0 = BLUPS$BLUP.btw, y0 = BLUPS$BLUP.tarsus - BLUPS$SE.tarsus, x1 = BLUPS$BLUP.btw, y1 = BLUPS$BLUP.tarsus + BLUPS$SE.tarsus)
arrows(x0 = BLUPS$BLUP.btw - BLUPS$SE.btw, y0 = BLUPS$BLUP.tarsus, x1 = BLUPS$BLUP.btw + BLUPS$SE.btw, y1 = BLUPS$BLUP.tarsus)
points(BLUP.tarsus ~ BLUP.btw, BLUPS, pch = 16, col = "red", cex = 1.5)
points(BLUP.tarsus ~ BLUP.btw, BLUPS, pch = 1, col = rgb(0, 0, 0, 0.3), cex = c(1.5))
mtext("btw (BV±SE)", side = 1, line = 2.4)
mtext("tarsus (BV±SE)", side = 2, line = 2, las = 3)
mtext(expression(paste(italic(r)[A], " = 0.7353065 ± 0.1094838")), side = 1, line = -1, adj = 0.)
```





### 5.0.6. Partitionning (co)variance between groups

Similar to the univariate model, it is possible to partition the variance and also the covariance between different groups within the dataset. Here, we can estimate sex-specific genetic correlation. Note, to partition a correlation, it is require to have important sample size within each group. For this example, we simplify the model !

```
gryphon <- gryphon[order(gryphon$sex), ]
model_sex <- asreml(
  fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex,
  random = ~ at(sex):us(trait, init = c(1, 0.1, 1)):vm(animal, ainvs) +
    us(trait, init = c(1, 0.1, 1)):byear +
    us(trait, init = c(1, 0.1, 1)):mother,
  residual = ~ dsum(~ id(units):us(trait) | sex),
  data = gryphon,
  na.action = na.method(x = "include", y = "include"),
  maxit = 20
)
```

ASReml Version 4.2 08/10/2024 11:29:09

|   | LogLik    | Sigma2 | DF   | wall     |                 |
|---|-----------|--------|------|----------|-----------------|
| 1 | -2522.729 | 1.0    | 1807 | 11:29:10 | ( 1 restrained) |
| 2 | -2459.512 | 1.0    | 1807 | 11:29:10 | ( 3 restrained) |
| 3 | -2408.940 | 1.0    | 1807 | 11:29:10 |                 |
| 4 | -2392.691 | 1.0    | 1807 | 11:29:10 |                 |

```

5      -2388.962          1.0   1807   11:29:10
6      -2388.743          1.0   1807   11:29:10
7      -2388.736          1.0   1807   11:29:10
8      -2388.736          1.0   1807   11:29:10

```

```

Warning in asreml(fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex, random =
~at(sex):us(trait, : Warning : US updates modified 1 times in iteration 2 to
remain positive definite.

```

```
model_sex <- update(model_sex)
```

```
ASReml Version 4.2 08/10/2024 11:29:10
```

```

      LogLik      Sigma2      DF      wall
1    -2388.736        1.0   1807   11:29:11
2    -2388.736        1.0   1807   11:29:11

```

```
summary(model_sex)$varcomp
```

|                                                         | component  | std.error | z.ratio    | bound | %ch |
|---------------------------------------------------------|------------|-----------|------------|-------|-----|
| trait:byear!trait_bwt:bwt                               | 0.9858478  | 0.2863878 | 3.4423530  | P     | 0   |
| trait:byear!trait_tarsus:bwt                            | 0.1525063  | 0.4334263 | 0.3518622  | P     | 0   |
| trait:byear!trait_tarsus:tarsus                         | 3.9981983  | 1.2798747 | 3.1238982  | P     | 0   |
| trait:mother!trait_bwt:bwt                              | 1.3312734  | 0.2484444 | 5.3584371  | P     | 0   |
| trait:mother!trait_tarsus:bwt                           | -1.6174228 | 0.4283851 | -3.7756279 | P     | 0   |
| trait:mother!trait_tarsus:tarsus                        | 4.7542338  | 1.3546517 | 3.5095618  | P     | 0   |
| at(sex, '1'):trait:vm(animal, ainv)!trait_bwt:bwt       | 1.3402853  | 0.5670773 | 2.3634965  | P     | 0   |
| at(sex, '1'):trait:vm(animal, ainv)!trait_tarsus:bwt    | 2.3608392  | 1.1348473 | 2.0803144  | P     | 0   |
| at(sex, '1'):trait:vm(animal, ainv)!trait_tarsus:tarsus | 6.0625993  | 3.1304394 | 1.9366608  | P     | 0   |
| at(sex, '2'):trait:vm(animal, ainv)!trait_bwt:bwt       | 1.8645998  | 0.8888206 | 2.0978361  | P     | 0   |
| at(sex, '2'):trait:vm(animal, ainv)!trait_tarsus:bwt    | 5.0954811  | 2.0684729 | 2.4634024  | P     | 0   |
| at(sex, '2'):trait:vm(animal, ainv)!trait_tarsus:tarsus | 14.9771870 | 6.4479787 | 2.3227724  | P     | 0   |
| sex_1!R                                                 | 1.0000000  | NA        | NA         | F     | 0   |
| sex_1!trait_bwt:bwt                                     | 2.3079850  | 0.5015651 | 4.6015657  | P     | 0   |

|                           | component  | std.error | z.ratio   | bound | %ch |
|---------------------------|------------|-----------|-----------|-------|-----|
| sex_1!trait_tarsus:bwt    | 4.4287898  | 1.0376370 | 4.2681493 | P     | 0   |
| sex_1!trait_tarsus:tarsus | 13.4857819 | 2.9284922 | 4.6050257 | P     | 0   |
| sex_2!R                   | 1.0000000  | NA        | NA        | F     | 0   |
| sex_2!trait_bwt:bwt       | 1.7956612  | 0.7549779 | 2.3784288 | P     | 0   |
| sex_2!trait_tarsus:bwt    | 2.6340448  | 1.7685804 | 1.4893554 | P     | 0   |
| sex_2!trait_tarsus:tarsus | 9.6094528  | 5.4917853 | 1.7497867 | P     | 0   |

we can estimate the different correlations using `vpredict`:

```
vpredict(model_sex, r_byear ~ V2 / sqrt(V1 * V3))
```

|         | Estimate  | SE       |
|---------|-----------|----------|
| r_byear | 0.0768158 | 0.213141 |

```
vpredict(model_sex, r_M ~ V5 / sqrt(V4 * V6))
```

|     | Estimate   | SE       |
|-----|------------|----------|
| r_M | -0.6429092 | 0.248944 |

```
vpredict(model_sex, r_A.1 ~ V8 / sqrt(V7 * V9))
```

|       | Estimate  | SE        |
|-------|-----------|-----------|
| r_A.1 | 0.8282059 | 0.1723596 |

```
vpredict(model_sex, r_A.2 ~ V11 / sqrt(V10 * V12))
```

|       | Estimate  | SE        |
|-------|-----------|-----------|
| r_A.2 | 0.9642225 | 0.1241668 |

```
vpredict(model_sex, r_res.1 ~ V15 / sqrt(V14 * V16))
```

|         | Estimate  | SE        |
|---------|-----------|-----------|
| r_res.1 | 0.7938355 | 0.0789263 |

```
vpredict(model_sex, r_res.2 ~ V19 / sqrt(V18 * V20))
```

|         | Estimate  | SE        |
|---------|-----------|-----------|
| r_res.2 | 0.6341057 | 0.1894837 |

and the heritability too:

```
vpredict(model_sex, h2.bwt.1 ~ V7 / (V1 + V4 + V7 + V14))
```

|          | Estimate  | SE        |
|----------|-----------|-----------|
| h2.bwt.1 | 0.2246768 | 0.0917683 |

```
vpredict(model_sex, h2.bwt.2 ~ V10 / (V1 + V4 + V10 + V18))
```

|          | Estimate  | SE        |
|----------|-----------|-----------|
| h2.bwt.2 | 0.3119425 | 0.1442547 |

```
vpredict(model_sex, h2.tarsus.1 ~ V9 / (V3 + V6 + V9 + V16))
```

|             | Estimate | SE        |
|-------------|----------|-----------|
| h2.tarsus.1 | 0.21422  | 0.1070464 |

```
vpredict(model_sex, h2.tarsus.2 ~ V12 / (V3 + V6 + V12 + V20))
```

|             | Estimate  | SE        |
|-------------|-----------|-----------|
| h2.tarsus.2 | 0.4492383 | 0.1833858 |

Now we can look at the fixed effects parameters and assess their significance with a conditional Wald F-test:

|                           | solution   | std error | z.ratio    |
|---------------------------|------------|-----------|------------|
| trait_bwt                 | 6.3779149  | 0.2311766 | 27.5889321 |
| trait_tarsus              | 20.5838787 | 0.4942649 | 41.6454395 |
| at(trait, 'bwt'):sex_1    | 0.0000000  | NA        | NA         |
| at(trait, 'bwt'):sex_2    | 1.9393688  | 0.1903239 | 10.1898321 |
| at(trait, 'tarsus'):sex_1 | 0.0000000  | NA        | NA         |
| at(trait, 'tarsus'):sex_2 | -0.0554799 | 0.4758708 | -0.1165861 |

ASReml Version 4.2 08/10/2024 11:29:12

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2388.736 | 1.0    | 1807 | 11:29:12 |
| 2 | -2388.736 | 1.0    | 1807 | 11:29:12 |

|                         | Df | denDF | F.inc     | F.con     | Margin | Pr        |
|-------------------------|----|-------|-----------|-----------|--------|-----------|
| trait                   | 2  | 44.8  | 1.522e+03 | 1.522e+03 |        | 0.0000000 |
| at(trait, 'bwt'):sex    | 1  | 137.5 | 2.209e+02 | 1.038e+02 | B      | 0.0000000 |
| at(trait, 'tarsus'):sex | 1  | 138.6 | 1.359e-02 | 1.359e-02 | B      | 0.9073748 |

To assess the significant of the covariance, a LTR test can be done with a reduced model where a specific covariance can be fixed to 0 (for example the female covariance, following code).

```
model_modif <- update.asreml(model_sex, start.values = T)
G <- model_modif$vpparameters[(1:12), ]
G$Constraint[(2)] <- "F"
G$Value[(2)] <- 0
#
reduc.model_sex <- asreml(
```

```

fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex,
random = ~ at(sex):us(trait, init = c(1, 0.1, 1)):vm(animal, ainv) +
  us(trait, init = c(1, 0.1, 1)):byear +
  us(trait, init = c(1, 0.1, 1)):mother,
residual = ~ dsum(~ id(units):us(trait) | sex),
data = gryphon,
na.action = na.method(x = "include", y = "include"),
maxit = 20,
G.param = G
)

```

ASReml Version 4.2 08/10/2024 11:29:13

|   | LogLik    | Sigma2 | DF   | wall     |                 |
|---|-----------|--------|------|----------|-----------------|
| 1 | -2474.972 | 1.0    | 1807 | 11:29:13 | ( 3 restrained) |
| 2 | -2406.283 | 1.0    | 1807 | 11:29:13 |                 |
| 3 | -2394.010 | 1.0    | 1807 | 11:29:14 |                 |
| 4 | -2391.718 | 1.0    | 1807 | 11:29:14 |                 |
| 5 | -2391.480 | 1.0    | 1807 | 11:29:15 |                 |
| 6 | -2391.477 | 1.0    | 1807 | 11:29:15 |                 |

Warning in asreml(fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex, random = ~at(sex):us(trait, : Warning : US updates modified 1 times in iteration 1 to remain positive definite.

```

reduc.model_sex <- update(reduc.model_sex)

```

ASReml Version 4.2 08/10/2024 11:29:15

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2391.476 | 1.0    | 1807 | 11:29:15 |
| 2 | -2391.476 | 1.0    | 1807 | 11:29:15 |

```

summary(reduc.model_sex)$varcomp

```

|                                                         | component  | std.error | z.ratio    | bound | %ch |
|---------------------------------------------------------|------------|-----------|------------|-------|-----|
| trait:byear!trait_bwt:bwt                               | 0.9794331  | 0.2848997 | 3.4378175  | P     | 0   |
| trait:byear!trait_tarsus:bwt                            | 0.1428995  | 0.4322719 | 0.3305778  | P     | 0   |
| trait:byear!trait_tarsus:tarsus                         | 4.0021595  | 1.2818624 | 3.1221444  | P     | 0   |
| trait:mother!trait_bwt:bwt                              | 1.4956509  | 0.2568074 | 5.8240170  | P     | 0   |
| trait:mother!trait_tarsus:bwt                           | -1.2460057 | 0.4438357 | -2.8073580 | P     | 0   |
| trait:mother!trait_tarsus:tarsus                        | 5.3945609  | 1.4035705 | 3.8434556  | P     | 0   |
| at(sex, '1'):trait:vm(animal, ainv)!trait_bwt:bwt       | 0.5265716  | 0.3579555 | 1.4710530  | P     | 0   |
| at(sex, '1'):trait:vm(animal, ainv)!trait_tarsus:bwt    | 0.0000000  | NA        | NA         | F     | NA  |
| at(sex, '1'):trait:vm(animal, ainv)!trait_tarsus:tarsus | 1.4223969  | 1.9103795 | 0.7445625  | P     | 0   |
| at(sex, '2'):trait:vm(animal, ainv)!trait_bwt:bwt       | 1.5835813  | 0.8671365 | 1.8262193  | P     | 0   |
| at(sex, '2'):trait:vm(animal, ainv)!trait_tarsus:bwt    | 4.4288714  | 2.0173971 | 2.1953395  | P     | 0   |
| at(sex, '2'):trait:vm(animal, ainv)!trait_tarsus:tarsus | 12.9349047 | 6.2946996 | 2.0548883  | P     | 0   |
| sex_1!R                                                 | 1.0000000  | NA        | NA         | F     | 0   |
| sex_1!trait_bwt:bwt                                     | 2.9539767  | 0.4196755 | 7.0387165  | P     | 0   |
| sex_1!trait_tarsus:bwt                                  | 6.3138301  | 0.6802598 | 9.2814981  | P     | 0   |
| sex_1!trait_tarsus:tarsus                               | 17.3577089 | 2.4730547 | 7.0187323  | P     | 0   |
| sex_2!R                                                 | 1.0000000  | NA        | NA         | F     | 0   |
| sex_2!trait_bwt:bwt                                     | 1.9341439  | 0.7416691 | 2.6078261  | P     | 0   |
| sex_2!trait_tarsus:bwt                                  | 2.9467290  | 1.7370018 | 1.6964455  | P     | 0   |
| sex_2!trait_tarsus:tarsus                               | 10.7245912 | 5.4025888 | 1.9850837  | P     | 0   |

```
2 * (model_sex$loglik - reduc.model_sex$loglik)
```

```
[1] 5.481033
```

```
1 - pchisq(2 * (model_sex$loglik - reduc.model_sex$loglik), df = 1)
```

```
[1] 0.0192239
```

In addition, it is also possible to test the sexes if sexes has significant differences with another reduced model where both covariance are fixed to their average values.

```
# code provided as an example for the moment since the model cannot run on this data
model_modif <- update.asreml(model_sex, start.values = T)
G <- model_modif$vparameters[(1:12), ]
G$fac <- factor(
  c(
    1, 2, 3, 4, 2, 6, # Additive genetic matrix 2 =5
    7, 8, 9, # byear matrix
    10, 11, 12 # mother matrix
  )
)
Modif <- vcm.lm(~fac, data = G)
attr(Modif, "assign") <- NULL
attr(Modif, "contrasts") <- NULL
#
reduc.model_sex_2 <- asreml(
  fixed = cbind(bwt, tarsus) ~ trait + at(trait):sex,
  random = ~ at(sex):us(trait, init = c(1, 0.1, 1)):vm(animal, ainv) +
    us(trait, init = c(1, 0.1, 1)):byear +
    us(trait, init = c(1, 0.1, 1)):mother,
  residual = ~ dsum(~ id(units):us(trait) | sex),
  data = gryphon,
  na.action = na.method(x = "include", y = "include"),
  maxit = 20,
  G.param = G, vcm = Modif
)
reduc.model_sex_2 <- update(reduc.model_sex_2)
summary(reduc.model_sex_2)$varcomp

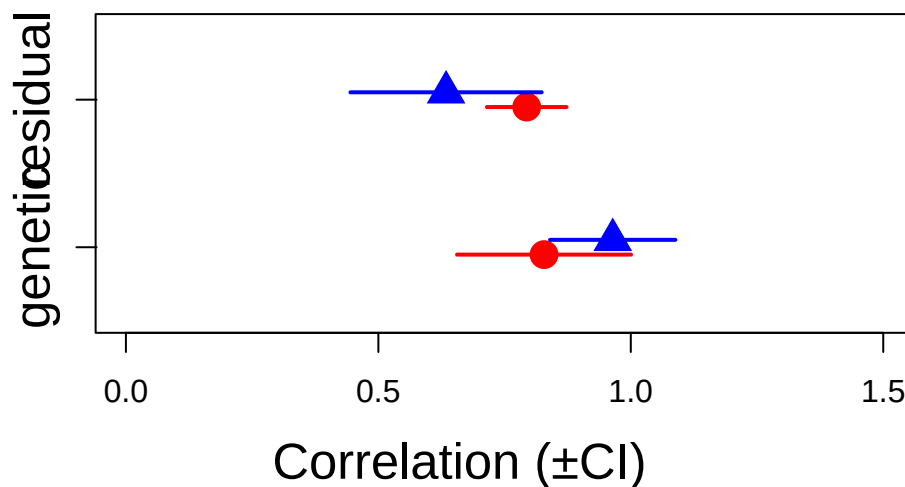
2 * (model_sex$loglik - reduc.model_sex_2$loglik)
1 - pchisq(2 * (model_sex$loglik - reduc.model_sex_2$loglik), df = 2)
```



Here a plot to visualize the overlaps of covariances.

```
genetic.correlation.F <- vpredict(model_sex, r_A.1 ~ V8 / sqrt(V7 * V9))
genetic.correlation.M <- vpredict(model_sex, r_A.2 ~ V11 / sqrt(V10 * V12))
residual.correlation.F <- vpredict(model_sex, r_res.1 ~ V15 / sqrt(V14 * V16))
residual.correlation.M <- vpredict(model_sex, r_res.2 ~ V19 / sqrt(V18 * V20))
cor.est <- rbind(genetic.correlation.F, genetic.correlation.M, residual.correlation.F, residual.c

plot(c(0.95, 1.05, 1.95, 2.05) ~ cor.est[, 1], xlim = c(0, 1.5), ylim = c(0.5, 2.5), xlab = "", y
arrows(y0 = 0.95, x0 = cor.est[1, 1] - cor.est[1, 2], y1 = 0.95, x1 = cor.est[1, 1] + cor.est[1,
arrows(y0 = 1.05, x0 = cor.est[2, 1] - cor.est[2, 2], y1 = 1.05, x1 = cor.est[2, 1] + cor.est[2,
arrows(y0 = 1.95, x0 = cor.est[3, 1] - cor.est[3, 2], y1 = 1.95, x1 = cor.est[3, 1] + cor.est[3,
arrows(y0 = 2.05, x0 = cor.est[4, 1] - cor.est[4, 2], y1 = 2.05, x1 = cor.est[4, 1] + cor.est[4,
mtext("Correlation ( $\pm$ CI)", side = 1, las = 1, adj = 0.4, line = 3, cex = 1.6)
axis(2, at = 1, labels = c("genetic"), las = 3, cex.axis = 1.6)
axis(2, at = 2, labels = c("residual"), las = 3, cex.axis = 1.6)
```



By using `corgh`, we can extract the BLUPs and plot the sex-specific correlation.

```
gryphon$T1 <- gryphon$bwt
gryphon$T2 <- gryphon$tarsus
###
model_sex <- asreml(
  fixed = cbind(T1, T2) ~ trait + at(trait):sex,
  random = ~ at(sex):corgh(trait, init = c(0.1, 1, 1)):vm(animal, ainv) +
```

```

us(trait, init = c(1, 0.1, 1)):byear +
us(trait, init = c(1, 0.1, 1)):mother,
residual = ~ dsum(~ id(units):us(trait) | sex),
data = gryphon,
na.action = na.method(x = "include", y = "include"),
maxit = 20
)

```

ASReml Version 4.2 08/10/2024 11:29:16

|   | LogLik    | Sigma2 | DF   | wall     |                 |
|---|-----------|--------|------|----------|-----------------|
| 1 | -2522.729 | 1.0    | 1807 | 11:29:16 | ( 2 restrained) |
| 2 | -2457.755 | 1.0    | 1807 | 11:29:17 | ( 2 restrained) |
| 3 | -2407.462 | 1.0    | 1807 | 11:29:17 | ( 2 restrained) |
| 4 | -2394.143 | 1.0    | 1807 | 11:29:18 | ( 1 restrained) |
| 5 | -2389.368 | 1.0    | 1807 | 11:29:18 |                 |
| 6 | -2388.741 | 1.0    | 1807 | 11:29:18 |                 |
| 7 | -2388.736 | 1.0    | 1807 | 11:29:18 |                 |

```
model_sex <- update(model_sex)
```

ASReml Version 4.2 08/10/2024 11:29:18

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2388.736 | 1.0    | 1807 | 11:29:19 |
| 2 | -2388.736 | 1.0    | 1807 | 11:29:19 |

```

DvsS <- data.frame(
  Trait = rownames(model_sex$coefficients$random),
  BLUP = model_sex$coefficients$random,
  SE = sqrt(model_sex$vccoeff$random * model_sex$sigma2)
) %>%
  filter(grepl("at\\(sex", Trait)) %>%
  mutate(
    ID = substr(Trait, 40, 44),

```

```

    TRAIT = substr(Trait, 20, 21),
    SEX = substr(Trait, 10, 10)
  ) %>%
  rename(
    BLUP = "effect"
  ) %>%
  select(BLUP:SEX)
summary(factor(DvsS$TRAIT)) # 1309 each

```

```

T1    T2
2618 2618

```

```
#
```

```

BLUPS <- reshape(DvsS, v.names = c("BLUP", "SE"), idvar = c("ID", "SEX"), timevar = "TRAIT", direction = "long")
nrow(BLUPS)

```

```
[1] 2618
```

```

rownames(BLUPS) <- c()
colnames(BLUPS) <- c("ID", "SEX", "BLUP.btw", "SE.btw", "BLUP.tarsus", "SE.tarsus")
summary(BLUPS)

```

| ID                | SEX              | BLUP.btw           | SE.btw          |
|-------------------|------------------|--------------------|-----------------|
| Length:2618       | Length:2618      | Min. : -2.669649   | Min. : 0.8383   |
| Class :character  | Class :character | 1st Qu.: -0.281979 | 1st Qu.: 0.9366 |
| Mode :character   | Mode :character  | Median : 0.000000  | Median : 1.1001 |
|                   |                  | Mean : 0.009574    | Mean : 1.0913   |
|                   |                  | 3rd Qu.: 0.295795  | 3rd Qu.: 1.1780 |
|                   |                  | Max. : 2.895393    | Max. : 1.4276   |
| BLUP.tarsus       | SE.tarsus        |                    |                 |
| Min. : -7.81574   | Min. : 1.829     |                    |                 |
| 1st Qu.: -0.64388 | 1st Qu.: 2.342   |                    |                 |

```

Median : 0.00000   Median :2.462
Mean   : 0.03319   Mean    :2.728
3rd Qu.: 0.74473   3rd Qu.:3.329
Max.    : 8.77778   Max.    :4.038

```

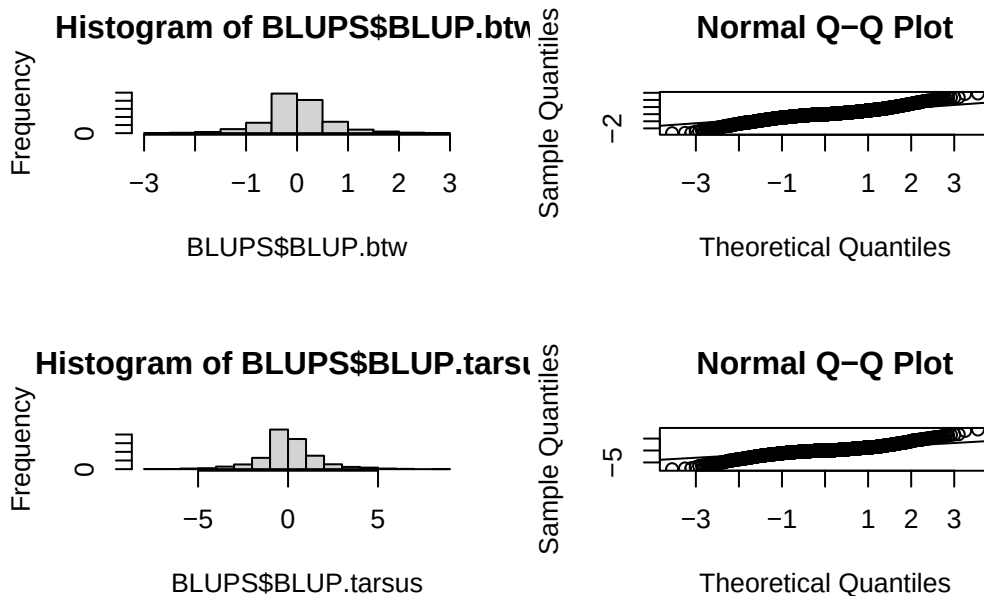
```

# write.csv(BLUPS,file="BLUPS_6x6_SEX.csv",row.names=F)

#####

par(mfrow = c(2, 2))
hist(BLUPS$BLUP.btw)
qqnorm(BLUPS$BLUP.btw)
qqline(BLUPS$BLUP.btw)
hist(BLUPS$BLUP.tarsus)
qqnorm(BLUPS$BLUP.tarsus)
qqline(BLUPS$BLUP.tarsus)

```



Here, some simple codes to plot the genetic correlation.

```

FEM <- subset(BLUPS, SEX == "1")
MAL <- subset(BLUPS, SEX == "2")

#

par(mfrow = c(1, 2))

#

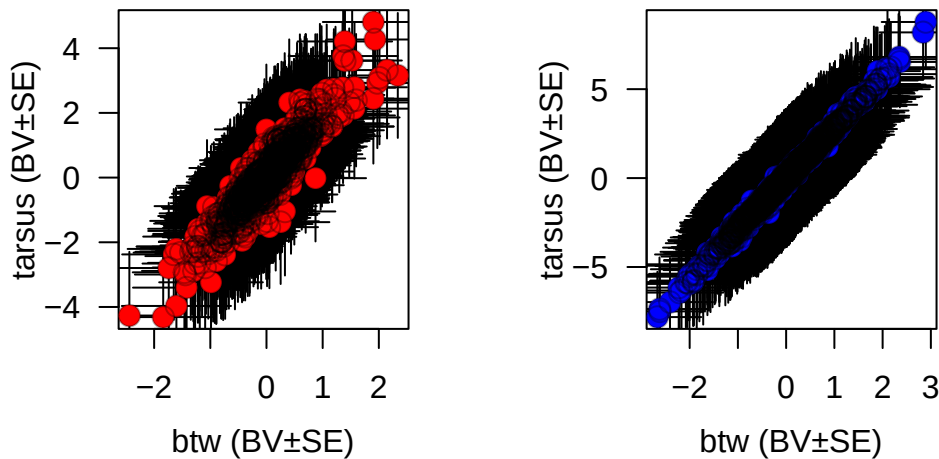
plot(BLUP.tarsus ~ BLUP.btw, FEM, xlab = "", ylab = "", las = 1.2, bty = "o", col = "white")

```

```

arrows(x0 = FEM$BLUP.btw, y0 = FEM$BLUP.tarsus - FEM$SE.tarsus, x1 = FEM$BLUP.btw, y1 = FEM$BLUP.
arrows(x0 = FEM$BLUP.btw - FEM$SE.btw, y0 = FEM$BLUP.tarsus, x1 = FEM$BLUP.btw + FEM$SE.btw, y1 =
points(BLUP.tarsus ~ BLUP.btw, FEM, pch = 16, col = "red", cex = 1.5)
points(BLUP.tarsus ~ BLUP.btw, FEM, pch = 1, col = rgb(0, 0, 0, 0.3), cex = c(1.5))
mtext("btw (BV±SE)", side = 1, line = 2.4)
mtext("tarsus (BV±SE)", side = 2, line = 2, las = 3)
#
plot(BLUP.tarsus ~ BLUP.btw, MAL, xlab = "", ylab = "", las = 1.2, bty = "o", col = "white")
arrows(x0 = MAL$BLUP.btw, y0 = MAL$BLUP.tarsus - MAL$SE.tarsus, x1 = MAL$BLUP.btw, y1 = MAL$BLUP.
arrows(x0 = MAL$BLUP.btw - MAL$SE.btw, y0 = MAL$BLUP.tarsus, x1 = MAL$BLUP.btw + MAL$SE.btw, y1 =
points(BLUP.tarsus ~ BLUP.btw, MAL, pch = 16, col = "blue", cex = 1.5)
points(BLUP.tarsus ~ BLUP.btw, MAL, pch = 1, col = rgb(0, 0, 0, 0.3), cex = c(1.5))
mtext("btw (BV±SE)", side = 1, line = 2.4)
mtext("tarsus (BV±SE)", side = 2, line = 2, las = 3)

```



### 5.0.7. Between groups (co)variances and the B-matrix

Animal models are amazing model. With different group within a population, it is also possible to estimate how much the different groups shared the same genetic via the cross-group genetic covariance. This covariance is essential to understand ontogenic or sexual conflict, which can constraint or enhanced response to evolution. As an example, we estimate the cross-sex genetic correlation  $r_{\{fm\}}$

First, we need to dissociate the trait values for females and males into distinct variables. Then, we use a bivariate model (for one trait: tarsus) and a multivariate model (for various traits: tarsus and bwt). With a multivariate model, the cross-sex-cross trait covariance matrix is also named B matrix.

The coding is a bit complex but pretty straightforward. It is important to modify the covariance matrix at the residual level to avoid the calculation of a cross-sex residual covariance (no individual switched sex during the experiment).

```
gryphon$bwt.1 <- NA
gryphon$tarsus.1 <- NA
animal <- gryphon[gryphon$sex == "1", ]$animal
for (i in unique(animal)) {
  gryphon$bwt.1[which(gryphon$animal == i)] <- gryphon$bwt[which(gryphon$animal == i)]
  gryphon$tarsus.1[which(gryphon$animal == i)] <- gryphon$tarsus[which(gryphon$animal == i)]
}
#
gryphon$bwt.2 <- NA
gryphon$tarsus.2 <- NA
animal <- gryphon[gryphon$sex == "2", ]$animal
for (i in unique(animal)) {
  gryphon$bwt.2[which(gryphon$animal == i)] <- gryphon$bwt[which(gryphon$animal == i)]
  gryphon$tarsus.2[which(gryphon$animal == i)] <- gryphon$tarsus[which(gryphon$animal == i)]
}

#####
temp <- asreml(cbind(tarsus.1, tarsus.2) ~ trait,
  random = ~ us(trait):vm(animal, ainv) +
    diag(trait):byear + diag(trait):mother,
  residual = ~ units:us(trait),
  data = gryphon, na.action = na.method(y = "include", x = "include"), maxiter = 20,
  start.values = T
)
G <- temp$vparameters[(1:7), ]
R <- temp$vparameters[-(1:7), ]
#
G$Constraint <- "U"
R$Value[3] <- 0
R$Constraint[3] <- "F"
#
```

```

model.BiV_Sex <- asreml(cbind(tarsus.1, tarsus.2) ~ trait,
  random = ~ us(trait):vm(animal, ainv) +
    diag(trait):byear + diag(trait):mother,
  residual = ~ units:us(trait),
  data = gryphon, na.action = na.method(y = "include", x = "include"), maxiter = 20,
  G.param = G, R.param = R
)

```

ASReml Version 4.2 08/10/2024 11:29:22

|   | LogLik    | Sigma2 | DF  | wall     |                 |
|---|-----------|--------|-----|----------|-----------------|
| 1 | -1494.807 | 1.0    | 681 | 11:29:22 | ( 1 restrained) |
| 2 | -1484.793 | 1.0    | 681 | 11:29:22 | ( 1 restrained) |
| 3 | -1475.726 | 1.0    | 681 | 11:29:22 | ( 1 restrained) |
| 4 | -1471.905 | 1.0    | 681 | 11:29:22 | ( 1 restrained) |
| 5 | -1470.716 | 1.0    | 681 | 11:29:22 |                 |
| 6 | -1468.154 | 1.0    | 681 | 11:29:22 |                 |
| 7 | -1467.969 | 1.0    | 681 | 11:29:22 |                 |
| 8 | -1467.967 | 1.0    | 681 | 11:29:22 |                 |

```

model.BiV_Sex <- update.asreml(model.BiV_Sex)

```

ASReml Version 4.2 08/10/2024 11:29:22

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -1467.967 | 1.0    | 681 | 11:29:22 |
| 2 | -1467.967 | 1.0    | 681 | 11:29:22 |

#

```

summary(model.BiV_Sex)$varcomp

```

|                             | component | std.error | z.ratio   | bound | %ch |
|-----------------------------|-----------|-----------|-----------|-------|-----|
| trait:byear!trait_tarsus.1  | 3.280319  | 1.532909  | 2.1399299 | U     | 0   |
| trait:byear!trait_tarsus.2  | 4.743134  | 1.891252  | 2.5079332 | U     | 0   |
| trait:mother!trait_tarsus.1 | 1.875131  | 2.424092  | 0.7735398 | U     | 0   |

|                                                | component | std.error | z.ratio   | bound | %ch |
|------------------------------------------------|-----------|-----------|-----------|-------|-----|
| trait:mother!trait_tarsus.2                    | 4.314158  | 2.785254  | 1.5489283 | U     | 0   |
| trait:vm(animal, ainv)!trait_tarsus.1:tarsus.1 | 6.582654  | 3.636467  | 1.8101781 | U     | 0   |
| trait:vm(animal, ainv)!trait_tarsus.2:tarsus.1 | 8.396245  | 3.278591  | 2.5609306 | U     | 0   |
| trait:vm(animal, ainv)!trait_tarsus.2:tarsus.2 | 12.898424 | 8.038362  | 1.6046084 | U     | 0   |
| units:trait!R                                  | 1.000000  | NA        | NA        | F     | 0   |
| units:trait!trait_tarsus.1:tarsus.1            | 14.872757 | 3.637545  | 4.0886803 | P     | 0   |
| units:trait!trait_tarsus.2:tarsus.1            | 0.000000  | NA        | NA        | F     | NA  |
| units:trait!trait_tarsus.2:tarsus.2            | 10.760849 | 6.294585  | 1.7095406 | P     | 0   |

The cross-sex genetic correlation can estimate from the output of the model. For tarsus length at fledging, sexes shared a lot of genetic variance which is common for a trait with low sexual dimorphism. If the selection is antagonistic between males and females, sexes can not evolve freely from the other sexes and a sexual conflict appears.

```
vpredict(model.BiV_Sex, r_fm ~ V6 / sqrt(V5 * V7))
```

|      | Estimate  | SE        |
|------|-----------|-----------|
| r_fm | 0.9112054 | 0.4229764 |

We can estimate directly the correlation and plot the cross-sex genetic correlation

```
temp <- asreml(cbind(tarsus.1, tarsus.2) ~ trait,
  random = ~ corgh(trait):vm(animal, ainv) +
    diag(trait):byear + diag(trait):mother,
  residual = ~ units:corgh(trait),
  data = gryphon, na.action = na.method(y = "include", x = "include"), maxiter = 20,
  start.values = T
)
G <- temp$vpparameters[(1:7), ]
R <- temp$vpparameters[-(1:7), ]
#
G$Constraint <- "U"
```



```

R$Value[2] <- 0
R$Constraint[2] <- "F"
#
model.BiV_Sex <- asreml(cbind(tarsus.1, tarsus.2) ~ trait,
  random = ~ corgh(trait):vm(animal, ainv) +
    diag(trait):byear + diag(trait):mother,
  residual = ~ units:corgh(trait),
  data = gryphon, na.action = na.method(y = "include", x = "include"), maxiter = 20,
  G.param = G, R.param = R
)

```

ASReml Version 4.2 08/10/2024 11:29:23

|   | LogLik    | Sigma2 | DF  | wall     |                 |
|---|-----------|--------|-----|----------|-----------------|
| 1 | -1494.323 | 1.0    | 681 | 11:29:23 | ( 1 restrained) |
| 2 | -1482.996 | 1.0    | 681 | 11:29:23 | ( 1 restrained) |
| 3 | -1472.827 | 1.0    | 681 | 11:29:23 | ( 1 restrained) |
| 4 | -1468.707 | 1.0    | 681 | 11:29:23 |                 |
| 5 | -1467.984 | 1.0    | 681 | 11:29:23 |                 |
| 6 | -1467.968 | 1.0    | 681 | 11:29:23 |                 |
| 7 | -1467.967 | 1.0    | 681 | 11:29:23 |                 |

```

model.BiV_Sex <- update.asreml(model.BiV_Sex)

```

ASReml Version 4.2 08/10/2024 11:29:23

|   | LogLik    | Sigma2 | DF  | wall     |
|---|-----------|--------|-----|----------|
| 1 | -1467.967 | 1.0    | 681 | 11:29:24 |
| 2 | -1467.967 | 1.0    | 681 | 11:29:24 |

```

#
summary(model.BiV_Sex)$varcomp

```

|                                                           | component  | std.error | z.ratio   | bound | %ch |
|-----------------------------------------------------------|------------|-----------|-----------|-------|-----|
| trait:byear!trait_tarsus.1                                | 3.2803263  | 1.5329224 | 2.1399167 | U     | 0   |
| trait:byear!trait_tarsus.2                                | 4.7431679  | 1.8913244 | 2.5078553 | U     | 0   |
| trait:mother!trait_tarsus.1                               | 1.8751274  | 2.4240942 | 0.7735373 | U     | 0   |
| trait:mother!trait_tarsus.2                               | 4.3141262  | 2.7852550 | 1.5489160 | U     | 0   |
| trait:vm(animal, ainv)!trait!tarsus.2:!trait!tarsus.1.cor | 0.9111864  | 0.4230261 | 2.1539720 | U     | 0   |
| trait:vm(animal, ainv)!trait_tarsus.1                     | 6.5826478  | 3.6364929 | 1.8101638 | U     | 0   |
| trait:vm(animal, ainv)!trait_tarsus.2                     | 12.8988848 | 8.0388517 | 1.6045681 | U     | 0   |
| units:trait!R                                             | 1.0000000  | NA        | NA        | F     | 0   |
| units:trait!trait!tarsus.2:!trait!tarsus.1.cor            | 0.0000000  | NA        | NA        | F     | NA  |
| units:trait!trait_tarsus.1                                | 14.8727602 | 3.6375549 | 4.0886696 | P     | 0   |
| units:trait!trait_tarsus.2                                | 10.7604420 | 6.2948051 | 1.7094162 | P     | 0   |

```
#####
DvsS <- data.frame(
  Trait = rownames(model.BiV_Sex$coefficients$random),
  BLUP = model.BiV_Sex$coefficients$random,
  SE = sqrt(model.BiV_Sex$vccoeff$random * model.BiV_Sex$sigma2)
) %>%
  filter(grepl("vm\\(animal", Trait)) %>%
  mutate(
    ID = substr(Trait, 33, 36),
    TRAIT = substr(Trait, 7, 14)
  ) %>%
  rename(
    BLUP = "effect"
  ) %>%
  select(BLUP:TRAIT)

summary(factor(DvsS$TRAIT))
```

```
tarsus.1 tarsus.2
      1309      1309
```

---

```
#
```

```
BLUPS <- reshape(DvsS, v.names = c("BLUP", "SE"), idvar = "ID", timevar = "TRAIT", direction = "w",
nrow(BLUPS)
```

```
[1] 1309
```

```
rownames(BLUPS) <- c()
colnames(BLUPS) <- c("ID", "BLUP.1", "SE.1", "BLUP.2", "SE.2")
summary(BLUPS)
```

| ID               | BLUP.1           | SE.1           | BLUP.2            |
|------------------|------------------|----------------|-------------------|
| Length:1309      | Min. : -4.2702   | Min. : 1.724   | Min. : -6.10276   |
| Class :character | 1st Qu.: -0.7149 | 1st Qu.: 2.010 | 1st Qu.: -0.99945 |
| Mode :character  | Median : 0.0000  | Median : 2.127 | Median : 0.00000  |
|                  | Mean : 0.0718    | Mean : 2.198   | Mean : 0.09409    |
|                  | 3rd Qu.: 0.8386  | 3rd Qu.: 2.421 | 3rd Qu.: 1.15952  |
|                  | Max. : 4.9297    | Max. : 2.677   | Max. : 7.57246    |

SE.2

Min. :2.375  
1st Qu.:2.679  
Median :3.051  
Mean :3.041  
3rd Qu.:3.375  
Max. :3.732

```
#####
```

```
Y <- BLUPS$BLUP.1
```

```
X <- BLUPS$BLUP.2
```

```
se.Y <- BLUPS$SE.1
```

```
se.X <- BLUPS$SE.2
```

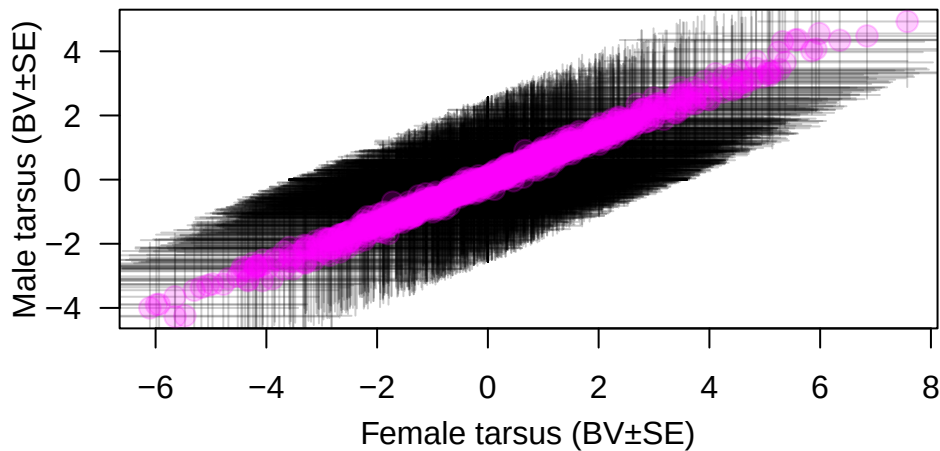
```
plot(X, Y, xlab = "", ylab = "", las = 1.2, bty = "o", col = "white")
```

```
arrows(x0 = X, y0 = Y - se.Y, x1 = X, y1 = Y + se.Y, col = rgb(0, 0, 0, 0.2), code = 3, angle = 90)
```

```

arrows(x0 = X - se.X, y0 = Y, x1 = X + se.X, y1 = Y, col = rgb(0, 0, 0, 0.2), code = 3, angle = 90)
points(X, Y, pch = 1, col = rgb(1, 0, 1, 0.2), cex = 1.5)
points(X, Y, pch = 16, col = rgb(1, 0, 1, 0.2), cex = 1.5)
# abline(v=0,lty=3);abline(h=0,lty=3)
mtext("Male tarsus (BV±SE)", side = 2, line = 2, las = 3)
mtext("Female tarsus (BV±SE)", side = 1, line = 2.2)

```



The B matrix used the same code but in a multivariate animal model framework. Here some example code, however due to the nature of the dataset, the cross-sex genetic covariance for birth weight is hard to estimate making difficulty to fit this multivariate animal model.

```

temp <- asreml(cbind(tarsus.1, bwt.1, tarsus.2, bwt.2) ~ trait,
  random = ~ us(trait):vm(animal, ainv) +
    diag(trait):byear + diag(trait):mother,
  residual = ~ units:us(trait),
  data = gryphon, na.action = na.method(y = "include", x = "include"), maxiter = 20,
  start.values = T
)
G <- temp$vparameters[(1:18), ]
R <- temp$vparameters[-(1:18), ]
#
G$Constraint <- "U"
R$Value[5:6] <- 0
R$Constraint[5:6] <- "F"
R$Value[8:9] <- 0

```

---

```
R$Constraint[8:9] <- "F"
#
# model.MultV_Sex<-asreml(cbind(tarsus.1,bwt.1,tarsus.2,bwt.2)~trait,
#       random=~us(trait):vm(animal,ainv)+
#       diag(trait):byear +   diag(trait):mother,
#       residual = ~units:us(trait),
#       data=gryphon,na.action=na.method(y="include",x="include"),maxiter=20,
#       G.param=G,R.param=R)
# model.MultV_Sex<-update.asreml(model.MultV_Sex)
#
# summary(model.MultV_Sex)$varcomp
```

# Chapter 6

## MCMCglmm

MCMCglmm has the advantage to keep automatically keep the lines with missing data and will try to fit the model use latent variables for missing data. We will remove the missing values from the data before fitting the model.

```
gryphon2 <- subset(gryphon, !is.na(bwt) & !is.na(tarsus))
```

First load MCMCglmm:

```
library(MCMCglmm)
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Loading required package: coda

Loading required package: ape

Attaching package: 'ape'

---

The following object is masked from 'package:dplyr':

where

```
Ainv <- inverseA(gryphonped)$Ainv
```

### 6.0.1. Fitting the model

Fitting a multivariate model in MCMCglmm involves several new consideration above those for fitting univariate models. First, we have to fit multivariate priors; second, we have to specify the ways in which effects on different traits may covary, including the nature of residual (co)variation; and third, we will have to be a little more specific when specifying to MCMCglmm what type of distributions from which we assume our data are drawn. Our most basic model can be specified as:

```
prior2.1 <- list(  
  G = list(G1 = list(V = diag(2), nu = 1.002)),  
  R = list(V = diag(2), nu = 1.002)  
)  
  
model2.1 <- MCMCglmm(cbind(bwt, tarsus) ~ trait - 1,  
  random = ~ us(trait):animal,  
  rcov = ~ us(trait):units,  
  family = c("gaussian", "gaussian"),  
  ginv = list(animal = Ainv),  
  data = gryphon2, prior = prior2.1, verbose = FALSE  
)  
summary(model2.1)
```

```
Iterations = 3001:12991
```

```
Thinning interval = 10
```

```
Sample size = 1000
```

```
DIC: 7156.733
```

G-structure: ~us(trait):animal

|                                | post.mean | l-95% CI | u-95% CI | eff.samp |
|--------------------------------|-----------|----------|----------|----------|
| traitbwt:traitbwt.animal       | 3.234     | 1.7827   | 4.429    | 127.34   |
| traittarsus:traitbwt.animal    | 2.022     | -0.5907  | 4.541    | 59.82    |
| traitbwt:traittarsus.animal    | 2.022     | -0.5907  | 4.541    | 59.82    |
| traittarsus:traittarsus.animal | 11.156    | 5.2966   | 18.249   | 70.53    |

R-structure: ~us(trait):units

|                               | post.mean | l-95% CI | u-95% CI | eff.samp |
|-------------------------------|-----------|----------|----------|----------|
| traitbwt:traitbwt.units       | 3.982     | 2.917    | 5.209    | 148.37   |
| traittarsus:traitbwt.units    | 3.687     | 1.580    | 5.983    | 64.13    |
| traitbwt:traittarsus.units    | 3.687     | 1.580    | 5.983    | 64.13    |
| traittarsus:traittarsus.units | 18.818    | 12.772   | 24.582   | 83.82    |

Location effects: cbind(bwt, tarsus) ~ trait - 1

|             | post.mean | l-95% CI | u-95% CI | eff.samp | pMCMC      |
|-------------|-----------|----------|----------|----------|------------|
| traitbwt    | 7.475     | 7.165    | 7.746    | 923.9    | <0.001 *** |
| traittarsus | 20.430    | 19.882   | 21.035   | 819.7    | <0.001 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
plot(model2.1$VCV[, "traittarsus:traittarsus.animal"])
```



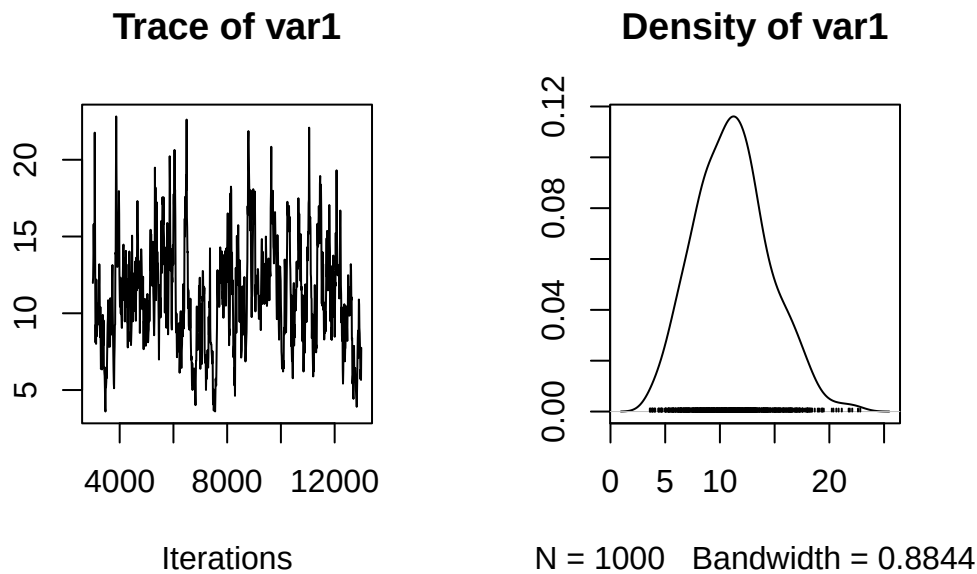


Figure 6.1.: The posterior distribution of the additive genetic effect for tarsus length in a MCMCglmm run with default values

```
autocorr.diag(model2.1$VCV)[, "traittarsus:traittarsus.animal"][2]
```

Lag 10

0.8551806

We have constructed the prior similarly to the those in the univariate models in tutorial 1, only we are specifying a 2x2 covariance matrix rather than a single variance. In order to provide proper priors, we have set the degree of belief parameter to greater than 1 (1.002). Those priors are not necessarily weak or uninformative in all circumstances. We will consider them adequate nonetheless for this tutorial. Please the vignette of the MCMCglmm packages (Hadfield 2010) for more information on priors. In tutorial 1, we used full autocorrelation tables to evaluate the validity of the posterior distribution. Note that we have not done this here.

For a bivariate model this table can become very complex. Nonetheless, it is worth evaluating, rather it is simply to large to include here. It can be viewed in the console as before. Here we have displayed only the autocorrelation for estimates of additive genetic effects for tarsus length with a lag of one samples (10 iterations given this MCMCglmm run with default values). This lag of 0.8551806 is clearly unacceptable. The posterior distribution of the additive genetic effect on tarsus length is shown in Figure 4, note the autocorrelation evident in the left-hand plot.

We will opt to run the analysis for longer. This longer run could be run using the following code (including a line to save the output):

```
model2.1 <- MCMCglmm(cbind(bwt, tarsus) ~ trait - 1,
  random = ~ us(trait):animal,
  rcov = ~ us(trait):units,
  family = c("gaussian", "gaussian"),
  ginv = list(animal = Ainv),
  data = gryphon2,
  nitt = 130000, thin = 100, burnin = 30000,
  prior = prior2.1, verbose = FALSE
)
save(model2.1, file = "r-obj/MCMCglmm_model2_1_LongRun.rda")
```

However, this run might take as long as an hour. For the purpose of this tutorial we have provided an output for such a run. It can be obtained and manipulated as follows, assuming that the file `MCMCglmm_model2_1_LongRun.rda` is available at the specified location:

```
load(file = "r-obj/MCMCglmm_model2_1_LongRun.rda")
autocorr.diag(model2.1$VCV)[, "traittarsus:traittarsus.animal"] [2]
```

```
Lag 100
0.3501744
```

This level of autocorrelation is more acceptable, at least for the purpose of demonstration in this tutorial. We can recover variance components, heritabilities, and genetic correlations from the posterior distribution of this model:

```
posterior.mode(model2.1$VCV)
```

|                             |                                |
|-----------------------------|--------------------------------|
| traitbwt:traitbwt.animal    | traittarsus:traitbwt.animal    |
| 3.471888                    | 2.949939                       |
| traitbwt:traittarsus.animal | traittarsus:traittarsus.animal |
| 2.949939                    | 12.012766                      |
| traitbwt:traitbwt.units     | traittarsus:traitbwt.units     |
| 3.953930                    | 3.876887                       |
| traitbwt:traittarsus.units  | traittarsus:traittarsus.units  |
| 3.876887                    | 19.604460                      |

---

```
heritability.bwt2.1 <- model2.1$VCV[, "traitbwt:traitbwt.animal"] / (model2.1$VCV[, "traitbwt:tra  
posterior.mode(heritability.bwt2.1)
```

```
var1  
0.4999336
```

```
heritability.tarsus2.1 <- model2.1$VCV[, "traittarsus:traittarsus.animal"] / (model2.1$VCV[, "tra  
posterior.mode(heritability.tarsus2.1)
```

```
var1  
0.3719549
```

```
genetic.correlation2.1 <- model2.1$VCV[, "traitbwt:traittarsus.animal"] / sqrt(model2.1$VCV[, "tr  
posterior.mode(genetic.correlation2.1)
```

```
var1  
0.4041642
```

## 6.0.2. Adding fixed and random effects

Fixed and random effects can be added just as for the univariate case. Given that our full model of bwt from tutorial 1 had sex as a fixed effect as well as random effects of byear and mother, we could specify a bivariate formulation of this using the following code (including a line to save the output):

```
prior2.2 <- list(  
  G = list(  
    G1 = list(V = diag(2), nu = 1.002),  
    G2 = list(V = diag(2), nu = 1.002),  
    G3 = list(V = diag(2), nu = 1.002)  
  ),  
  R = list(V = diag(2), nu = 1.002)  
)  
model2.2 <- MCMCglmm(cbind(bwt, tarsus) ~ trait - 1 + trait:sex,  
  random = ~ us(trait):animal + us(trait):byear + us(trait):mother,
```

```
rcov = ~ us(trait):units,
family = c("gaussian", "gaussian"),
ginv = list(animal = Ainv), data = gryphon2,
nitt = 130000, thin = 100, burnin = 30000,
prior = prior2.2, verbose = FALSE
)
save(model2.2, file = "r-obj/MCMCglmm_model2_2_LongRun.rda")
```

Again we have provided the data from one such run. It can be accessed using the code:

```
load(file = "r-obj/MCMCglmm_model2_2_LongRun.rda")
summary(model2.2)
```

Iterations = 30001:129901

Thinning interval = 100

Sample size = 1000

DIC: 5832.952

G-structure: ~us(trait):animal

|                                | post.mean | l-95% CI | u-95% CI | eff.samp |
|--------------------------------|-----------|----------|----------|----------|
| traitbwt:traitbwt.animal       | 1.558     | 0.5616   | 2.488    | 230.8    |
| traittarsus:traitbwt.animal    | 2.290     | 0.3241   | 4.264    | 274.8    |
| traitbwt:traittarsus.animal    | 2.290     | 0.3241   | 4.264    | 274.8    |
| traittarsus:traittarsus.animal | 8.083     | 0.9063   | 13.599   | 228.1    |

~us(trait):byear

|                            | post.mean | l-95% CI | u-95% CI | eff.samp |
|----------------------------|-----------|----------|----------|----------|
| traitbwt:traitbwt.byear    | 0.96775   | 0.4124   | 1.5053   | 1000     |
| traittarsus:traitbwt.byear | 0.07332   | -0.8100  | 0.9791   | 1000     |
| traitbwt:traittarsus.byear | 0.07332   | -0.8100  | 0.9791   | 1000     |

---

```
traittarsus:traittarsus.byear    3.80720    1.6291    6.3986    1000
```

```
~us(trait):mother
```

|                                | post.mean | l-95% CI | u-95% CI | eff.samp |
|--------------------------------|-----------|----------|----------|----------|
| traitbwt:traitbwt.mother       | 1.335     | 0.8564   | 1.8090   | 871.2    |
| traittarsus:traitbwt.mother    | -1.508    | -2.1667  | -0.8288  | 648.6    |
| traitbwt:traittarsus.mother    | -1.508    | -2.1667  | -0.8288  | 648.6    |
| traittarsus:traittarsus.mother | 4.292     | 2.2380   | 6.6336   | 796.0    |

```
R-structure: ~us(trait):units
```

|                               | post.mean | l-95% CI | u-95% CI | eff.samp |
|-------------------------------|-----------|----------|----------|----------|
| traitbwt:traitbwt.units       | 2.13      | 1.304    | 2.939    | 469.2    |
| traittarsus:traitbwt.units    | 4.81      | 3.111    | 6.568    | 414.7    |
| traitbwt:traittarsus.units    | 4.81      | 3.111    | 6.568    | 414.7    |
| traittarsus:traittarsus.units | 14.51     | 9.419    | 19.892   | 261.3    |

```
Location effects: cbind(bwt, tarsus) ~ trait - 1 + trait:sex
```

|                  | post.mean | l-95% CI | u-95% CI | eff.samp | pMCMC      |
|------------------|-----------|----------|----------|----------|------------|
| traitbwt         | 6.2734    | 5.8152   | 6.7272   | 1205     | <0.001 *** |
| traittarsus      | 20.3985   | 19.4021  | 21.4106  | 1000     | <0.001 *** |
| traitbwt:sex2    | 2.0354    | 1.7347   | 2.3529   | 1000     | <0.001 *** |
| traittarsus:sex2 | 0.0705    | -0.6949  | 0.7686   | 1000     | 0.868      |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
autocorr(model2.2$VCV)[, , "traittarsus:traittarsus.animal"][3, 4]
```

```
[1] 0.1026744
```

We can evaluate the fixed effect, their Ci evaluate their significance.

```
posterior.mode(model2.2$Sol)
```

| traitbwt   | traittarsus | traitbwt:sex2 | traittarsus:sex2 |
|------------|-------------|---------------|------------------|
| 6.26902047 | 20.35816977 | 2.06048779    | -0.06501522      |

```
HPDinterval(model2.2$Sol, 0.95)
```

|                  | lower      | upper      |
|------------------|------------|------------|
| traitbwt         | 5.8151983  | 6.7272503  |
| traittarsus      | 19.4021008 | 21.4106029 |
| traitbwt:sex2    | 1.7347121  | 2.3528879  |
| traittarsus:sex2 | -0.6948574 | 0.7686074  |

```
attr(,"Probability")  
[1] 0.95
```

```
plot(model2.2$Sol)
```

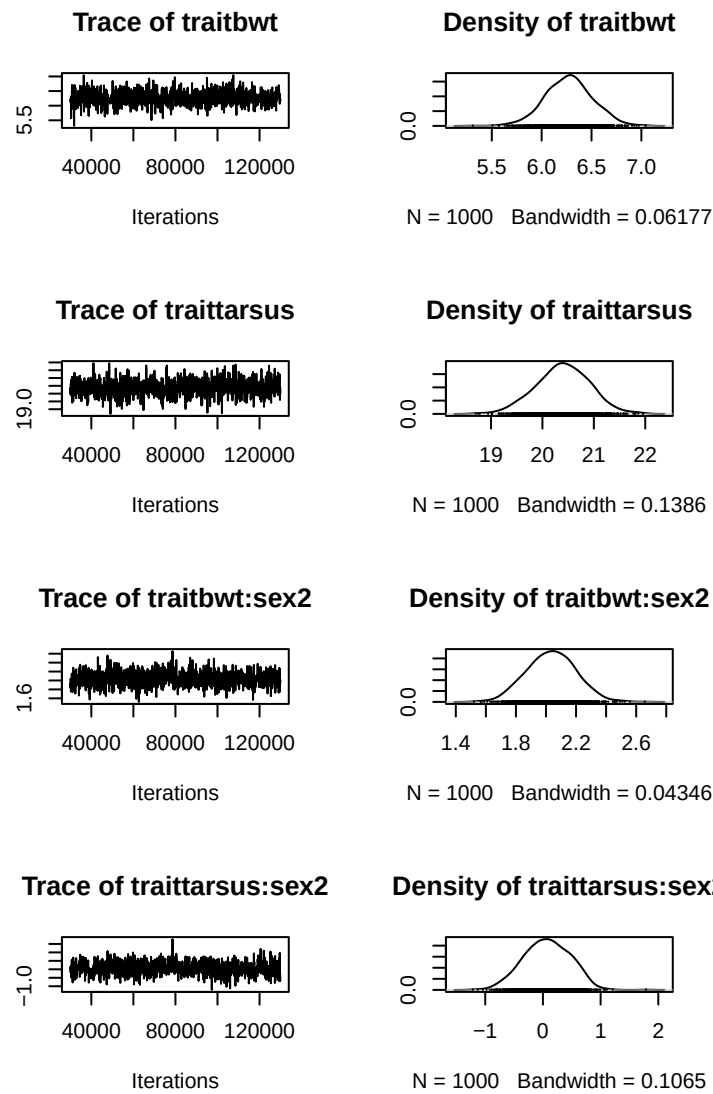


Figure 6.2.: Posterior trace and distribution for the fixed effects in model 2.2

As before we can obtain the raw variance component estimates and genetic correlations for the random effects:

```
posterior.mode(model2.2$VCV)
```

```

traitbwt:traitbwt.animal    traittarsus:traitbwt.animal
      1.3294950              2.0622374
traitbwt:traittarsus.animal traittarsus:traittarsus.animal
      2.0622374              8.3900676
traitbwt:traitbwt.byear     traittarsus:traitbwt.byear
      0.8118565              0.2327381
traitbwt:traittarsus.byear  traittarsus:traittarsus.byear

```

|                             |            |                                |
|-----------------------------|------------|--------------------------------|
|                             | 0.2327381  | 3.7375906                      |
| traitbwt:traitbwt.mother    |            | traittarsus:traitbwt.mother    |
|                             | 1.4089440  | -1.4963686                     |
| traitbwt:traittarsus.mother |            | traittarsus:traittarsus.mother |
|                             | -1.4963686 | 3.9386669                      |
| traitbwt:traitbwt.units     |            | traittarsus:traitbwt.units     |
|                             | 2.2353960  | 4.3432849                      |
| traitbwt:traittarsus.units  |            | traittarsus:traittarsus.units  |
|                             | 4.3432849  | 15.0853981                     |

```
genetic.correlation2.2 <- model2.2$VCV[, "traitbwt:traittarsus.animal"] / sqrt(model2.2$VCV[, "tr  
maternal.correlation2.2 <- model2.2$VCV[, "traitbwt:traittarsus.mother"] / sqrt(model2.2$VCV[, "t  
posterior.mode(genetic.correlation2.2)
```

```
var1  
0.6932486
```

```
posterior.mode(maternal.correlation2.2)
```

```
var1  
-0.7431221
```

Evaluation of the statistical support for these genetic and maternal correlations is straightforward. Because we imposed no constraint on their estimation, we can evaluate the extent to which the posterior distributions overlap zero:

```
HPDinterval(genetic.correlation2.2, 0.95)
```

```
lower upper  
var1 0.3062932 0.9197543  
attr(,"Probability")  
[1] 0.95
```



---

```
HPDinterval(maternal.correlation2.2, 0.95)
```

```
      lower      upper
var1 -0.9432297 -0.3210149
attr(,"Probability")
[1] 0.95
```

Neither of these posterior distributions overlaps zero, so we can consider them both statistically supported.

### 6.0.3. Partitioning (co)variances

As in the tutorial 1, it is possible to partition the variance-covariance matrix between groups (here sex) Note: the model is simplified without sex-specific covariance for the byear and mother random effect.

```
gryphon2 <- gryphon2[order(gryphon2$sex), ]

prior2.3 <- list(
  G = list(
    G1 = list(V = diag(2), nu = 1.002),
    G2 = list(V = diag(2), nu = 1.002),
    G3 = list(V = diag(2), nu = 1.002),
    G4 = list(V = diag(2), nu = 1.002)
  ),
  R = list(
    V1 = list(V = diag(2), nu = 1.002),
    V2 = list(V = diag(2), nu = 1.002)
  )
)

model2.4 <- MCMCglmm(cbind(bwt, tarsus) ~ trait - 1 + trait:sex,
  random = ~ us(at.level(sex, "1"):trait):animal + us(at.level(sex, "2"):trait):animal + idh(trait),
  rcov = ~ us(at.level(sex, "1"):trait):units + us(at.level(sex, "2"):trait):units,
  family = c("gaussian", "gaussian"),
```

```
ginv = list(animal = Ainv), data = gryphon2,
nitt = 130000, thin = 100, burnin = 30000,
prior = prior2.3, verbose = FALSE, pr = TRUE,
)
save(model2.4, file = "r-obj/MCMCglmm_model2_4_LongRun.rda")
```

Again we have provided the data from one such run. It can be accessed using the code:

```
load(file = "r-obj/MCMCglmm_model2_4_LongRun.rda")
summary(model2.4)
```

Iterations = 30001:129901

Thinning interval = 100

Sample size = 1000

DIC: 5576.328

G-structure: ~us(at.level(sex, "1"):trait):animal

|                                                                      | post.mean |
|----------------------------------------------------------------------|-----------|
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal       | 1.122     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal    | 1.127     |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    | 1.127     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal | 3.379     |
|                                                                      | 1-95% CI  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal       | 0.1602    |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal    | -0.6531   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    | -0.6531   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal | 0.1844    |
|                                                                      | u-95% CI  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal       | 2.359     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal    | 3.496     |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    | 3.496     |

---

```

at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal      8.918
                                                                    eff.samp
at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal          167.5
at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal        119.3
at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal        119.3
at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal     102.6

~us(at.level(sex, "2"):trait):animal

                                                                    post.mean
at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal          1.598
at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal        3.099
at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal        3.099
at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal     10.218
                                                                    1-95% CI
at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal          0.1895
at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal       -0.5506
at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal       -0.5506
at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal     0.2127
                                                                    u-95% CI
at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal          3.305
at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal        7.864
at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal        7.864
at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal     24.230
                                                                    eff.samp
at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal          57.28
at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal        42.01
at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal        42.01
at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal     37.21

~idh(trait):byear

post.mean 1-95% CI u-95% CI eff.samp

```

---

|                   |        |        |       |      |
|-------------------|--------|--------|-------|------|
| traitbwt.byear    | 0.9309 | 0.4614 | 1.463 | 1000 |
| traittarsus.byear | 4.0310 | 1.9268 | 6.724 | 1000 |

~idh(trait):mother

|                    | post.mean | l-95% CI | u-95% CI | eff.samp |
|--------------------|-----------|----------|----------|----------|
| traitbwt.mother    | 1.924     | 1.406    | 2.398    | 667.6    |
| traittarsus.mother | 7.093     | 4.626    | 9.681    | 698.5    |

R-structure: ~us(at.level(sex, "1"):trait):units

|                                                                     | post.mean |
|---------------------------------------------------------------------|-----------|
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       | 2.090     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    | 4.533     |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    | 4.533     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units | 14.113    |
|                                                                     | l-95% CI  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       | 0.9958    |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    | 2.4185    |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    | 2.4185    |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units | 8.1848    |
|                                                                     | u-95% CI  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       | 3.128     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    | 6.875     |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    | 6.875     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units | 18.935    |
|                                                                     | eff.samp  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       | 207.2     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    | 168.4     |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    | 168.4     |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units | 185.3     |

~us(at.level(sex, "2"):trait):units

---

|                                                                     | post.mean |
|---------------------------------------------------------------------|-----------|
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       | 1.782     |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    | 3.697     |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    | 3.697     |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units | 12.437    |
|                                                                     | 1-95% CI  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       | 0.2776    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    | -0.3141   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    | -0.3141   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units | 0.1776    |
|                                                                     | u-95% CI  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       | 3.115     |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    | 7.218     |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    | 7.218     |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units | 21.903    |
|                                                                     | eff.samp  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       | 52.55     |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    | 51.90     |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    | 51.90     |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units | 39.20     |

Location effects: cbind(bwt, tarsus) ~ trait - 1 + trait:sex

|                  | post.mean | 1-95% CI | u-95% CI | eff.samp | pMCMC      |
|------------------|-----------|----------|----------|----------|------------|
| traitbwt         | 6.30098   | 5.89218  | 6.78834  | 1000.0   | <0.001 *** |
| traittarsus      | 20.45577  | 19.53577 | 21.34719 | 1129.8   | <0.001 *** |
| traitbwt:sex2    | 2.01306   | 1.63662  | 2.38011  | 887.4    | <0.001 *** |
| traittarsus:sex2 | 0.05817   | -0.86635 | 0.89119  | 1016.6   | 0.896      |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
autocorr(model2.4$VCV)
```

```
, , at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0                                     1.00000000
Lag 100                                  0.64694479
Lag 500                                  0.18648179
Lag 1000                                 0.10392219
Lag 5000                                 -0.04275072

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0                                     0.84704874
Lag 100                                  0.60161240
Lag 500                                  0.20180692
Lag 1000                                 0.10068129
Lag 5000                                 -0.03878312

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal
Lag 0                                     0.84704874
Lag 100                                  0.60161240
Lag 500                                  0.20180692
Lag 1000                                 0.10068129
Lag 5000                                 -0.03878312

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal
Lag 0                                     0.53041407
Lag 100                                  0.39413485
Lag 500                                  0.16964194
Lag 1000                                 0.11264314
Lag 5000                                 -0.01013697

      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal
Lag 0                                     -0.07132000
Lag 100                                  -0.09608251
Lag 500                                  -0.05360431
Lag 1000                                 -0.02600414
```

---

|                                                                      |              |            |             |              |
|----------------------------------------------------------------------|--------------|------------|-------------|--------------|
| Lag 5000                                                             | -0.02326421  |            |             |              |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |              |            |             |              |
| Lag 0                                                                | -0.07404287  |            |             |              |
| Lag 100                                                              | -0.08742103  |            |             |              |
| Lag 500                                                              | -0.05376905  |            |             |              |
| Lag 1000                                                             | -0.03219125  |            |             |              |
| Lag 5000                                                             | 0.02771727   |            |             |              |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |              |            |             |              |
| Lag 0                                                                | -0.07404287  |            |             |              |
| Lag 100                                                              | -0.08742103  |            |             |              |
| Lag 500                                                              | -0.05376905  |            |             |              |
| Lag 1000                                                             | -0.03219125  |            |             |              |
| Lag 5000                                                             | 0.02771727   |            |             |              |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |              |            |             |              |
| Lag 0                                                                | -0.06663301  |            |             |              |
| Lag 100                                                              | -0.07398282  |            |             |              |
| Lag 500                                                              | -0.03873715  |            |             |              |
| Lag 1000                                                             | -0.03346102  |            |             |              |
| Lag 5000                                                             | 0.06535632   |            |             |              |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |              |            |             |              |
| Lag 0                                                                | -0.002044905 | 0.06061428 | -0.13681757 | 0.063034744  |
| Lag 100                                                              | -0.029101625 | 0.04741082 | -0.09232454 | 0.062553003  |
| Lag 500                                                              | -0.025891155 | 0.04101237 | -0.01510511 | -0.026837884 |
| Lag 1000                                                             | 0.029398462  | 0.02792539 | -0.02514900 | 0.009578198  |
| Lag 5000                                                             | -0.016122661 | 0.03081539 | 0.04189460  | -0.039791141 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |              |            |             |              |
| Lag 0                                                                | -0.842319278 |            |             |              |
| Lag 100                                                              | -0.569203867 |            |             |              |
| Lag 500                                                              | -0.167844469 |            |             |              |
| Lag 1000                                                             | -0.114647645 |            |             |              |
| Lag 5000                                                             | -0.002132053 |            |             |              |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units     |              |            |             |              |
| Lag 0                                                                | -0.708901550 |            |             |              |

---

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Lag 100                                                             | -0.517998161 |
| Lag 500                                                             | -0.167589741 |
| Lag 1000                                                            | -0.110500558 |
| Lag 5000                                                            | 0.002914291  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |              |
| Lag 0                                                               | -0.708901550 |
| Lag 100                                                             | -0.517998161 |
| Lag 500                                                             | -0.167589741 |
| Lag 1000                                                            | -0.110500558 |
| Lag 5000                                                            | 0.002914291  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |              |
| Lag 0                                                               | -0.438123204 |
| Lag 100                                                             | -0.337083166 |
| Lag 500                                                             | -0.129292647 |
| Lag 1000                                                            | -0.103678560 |
| Lag 5000                                                            | -0.001926232 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |              |
| Lag 0                                                               | 0.07807105   |
| Lag 100                                                             | 0.10707885   |
| Lag 500                                                             | 0.05568856   |
| Lag 1000                                                            | 0.02521629   |
| Lag 5000                                                            | 0.01402475   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |              |
| Lag 0                                                               | 0.06679340   |
| Lag 100                                                             | 0.08704308   |
| Lag 500                                                             | 0.05892190   |
| Lag 1000                                                            | 0.02676188   |
| Lag 5000                                                            | -0.03056683  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    |              |
| Lag 0                                                               | 0.06679340   |
| Lag 100                                                             | 0.08704308   |
| Lag 500                                                             | 0.05892190   |
| Lag 1000                                                            | 0.02676188   |



---

|                                                                       |             |
|-----------------------------------------------------------------------|-------------|
| Lag 5000                                                              | -0.03056683 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units   |             |
| Lag 0                                                                 | 0.04797898  |
| Lag 100                                                               | 0.05730717  |
| Lag 500                                                               | 0.04727555  |
| Lag 1000                                                              | 0.02677473  |
| Lag 5000                                                              | -0.06608227 |
| , , at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal |             |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal        |             |
| Lag 0                                                                 | 0.847048735 |
| Lag 100                                                               | 0.596411029 |
| Lag 500                                                               | 0.228550625 |
| Lag 1000                                                              | 0.137616124 |
| Lag 5000                                                              | 0.009429906 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal     |             |
| Lag 0                                                                 | 1.00000000  |
| Lag 100                                                               | 0.71730236  |
| Lag 500                                                               | 0.27616079  |
| Lag 1000                                                              | 0.13795063  |
| Lag 5000                                                              | 0.01144749  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal     |             |
| Lag 0                                                                 | 1.00000000  |
| Lag 100                                                               | 0.71730236  |
| Lag 500                                                               | 0.27616079  |
| Lag 1000                                                              | 0.13795063  |
| Lag 5000                                                              | 0.01144749  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal  |             |
| Lag 0                                                                 | 0.7989970   |
| Lag 100                                                               | 0.6014134   |
| Lag 500                                                               | 0.2515262   |
| Lag 1000                                                              | 0.1354306   |

---

```
Lag 5000                                0.0136445
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal
Lag 0                                  -0.069644149
Lag 100                               -0.094348331
Lag 500                               -0.069174874
Lag 1000                             -0.030980734
Lag 5000                             -0.001770693
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal
Lag 0                                  -0.09266557
Lag 100                               -0.10417316
Lag 500                               -0.06908668
Lag 1000                             -0.04934221
Lag 5000                             0.03968797
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal
Lag 0                                  -0.09266557
Lag 100                               -0.10417316
Lag 500                               -0.06908668
Lag 1000                             -0.04934221
Lag 5000                             0.03968797
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal
Lag 0                                  -0.10219567
Lag 100                               -0.10740690
Lag 500                               -0.05829130
Lag 1000                             -0.05667648
Lag 5000                             0.08311412
      traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother
Lag 0      -0.03731153      0.05572330      -0.12626725      0.06865980
Lag 100     -0.04492620      0.05076637      -0.08142219      0.05404288
Lag 500     -0.03460527      0.03246607      -0.03107773     -0.02899561
Lag 1000     0.01459594      0.01717445      -0.05078674      0.01110690
Lag 5000    -0.01688700      0.03883380      0.02698184     -0.03307579
      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units
Lag 0                                  -0.73143141
```

---

---

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Lag 100                                                             | -0.52689086  |
| Lag 500                                                             | -0.22551523  |
| Lag 1000                                                            | -0.12616708  |
| Lag 5000                                                            | -0.04647821  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |              |
| Lag 0                                                               | -0.82495927  |
| Lag 100                                                             | -0.60990914  |
| Lag 500                                                             | -0.24464022  |
| Lag 1000                                                            | -0.12721295  |
| Lag 5000                                                            | -0.03841367  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |              |
| Lag 0                                                               | -0.82495927  |
| Lag 100                                                             | -0.60990914  |
| Lag 500                                                             | -0.24464022  |
| Lag 1000                                                            | -0.12721295  |
| Lag 5000                                                            | -0.03841367  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |              |
| Lag 0                                                               | -0.64394327  |
| Lag 100                                                             | -0.48995337  |
| Lag 500                                                             | -0.19725633  |
| Lag 1000                                                            | -0.10852446  |
| Lag 5000                                                            | -0.02105523  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |              |
| Lag 0                                                               | 0.082481767  |
| Lag 100                                                             | 0.105417000  |
| Lag 500                                                             | 0.073280263  |
| Lag 1000                                                            | 0.028355398  |
| Lag 5000                                                            | -0.006019758 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |              |
| Lag 0                                                               | 0.09308370   |
| Lag 100                                                             | 0.10680733   |
| Lag 500                                                             | 0.07810010   |
| Lag 1000                                                            | 0.04359553   |

---

```
Lag 5000                                -0.04726853
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
Lag 0                                   0.09308370
Lag 100                                0.10680733
Lag 500                                0.07810010
Lag 1000                               0.04359553
Lag 5000                                -0.04726853
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0                                   0.08843633
Lag 100                                0.09343376
Lag 500                                0.06886426
Lag 1000                               0.05338682
Lag 5000                                -0.09185034

, , at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0                                   0.847048735
Lag 100                                0.596411029
Lag 500                                0.228550625
Lag 1000                               0.137616124
Lag 5000                                0.009429906
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0                                   1.000000000
Lag 100                                0.71730236
Lag 500                                0.27616079
Lag 1000                               0.13795063
Lag 5000                                0.01144749
      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal
Lag 0                                   1.000000000
Lag 100                                0.71730236
Lag 500                                0.27616079
Lag 1000                               0.13795063
```

---

|          |                                                                      |            |             |              |
|----------|----------------------------------------------------------------------|------------|-------------|--------------|
| Lag 5000 |                                                                      |            |             | 0.01144749   |
|          | at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |            |             |              |
| Lag 0    |                                                                      |            |             | 0.7989970    |
| Lag 100  |                                                                      |            |             | 0.6014134    |
| Lag 500  |                                                                      |            |             | 0.2515262    |
| Lag 1000 |                                                                      |            |             | 0.1354306    |
| Lag 5000 |                                                                      |            |             | 0.0136445    |
|          | at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |            |             |              |
| Lag 0    |                                                                      |            |             | -0.069644149 |
| Lag 100  |                                                                      |            |             | -0.094348331 |
| Lag 500  |                                                                      |            |             | -0.069174874 |
| Lag 1000 |                                                                      |            |             | -0.030980734 |
| Lag 5000 |                                                                      |            |             | -0.001770693 |
|          | at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |            |             |              |
| Lag 0    |                                                                      |            |             | -0.09266557  |
| Lag 100  |                                                                      |            |             | -0.10417316  |
| Lag 500  |                                                                      |            |             | -0.06908668  |
| Lag 1000 |                                                                      |            |             | -0.04934221  |
| Lag 5000 |                                                                      |            |             | 0.03968797   |
|          | at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |            |             |              |
| Lag 0    |                                                                      |            |             | -0.09266557  |
| Lag 100  |                                                                      |            |             | -0.10417316  |
| Lag 500  |                                                                      |            |             | -0.06908668  |
| Lag 1000 |                                                                      |            |             | -0.04934221  |
| Lag 5000 |                                                                      |            |             | 0.03968797   |
|          | at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |            |             |              |
| Lag 0    |                                                                      |            |             | -0.10219567  |
| Lag 100  |                                                                      |            |             | -0.10740690  |
| Lag 500  |                                                                      |            |             | -0.05829130  |
| Lag 1000 |                                                                      |            |             | -0.05667648  |
| Lag 5000 |                                                                      |            |             | 0.08311412   |
|          | traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |            |             |              |
| Lag 0    | -0.03731153                                                          | 0.05572330 | -0.12626725 | 0.06865980   |

---

|                                                                     |             |            |             |             |
|---------------------------------------------------------------------|-------------|------------|-------------|-------------|
| Lag 100                                                             | -0.04492620 | 0.05076637 | -0.08142219 | 0.05404288  |
| Lag 500                                                             | -0.03460527 | 0.03246607 | -0.03107773 | -0.02899561 |
| Lag 1000                                                            | 0.01459594  | 0.01717445 | -0.05078674 | 0.01110690  |
| Lag 5000                                                            | -0.01688700 | 0.03883380 | 0.02698184  | -0.03307579 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       |             |            |             |             |
| Lag 0                                                               |             |            |             | -0.73143141 |
| Lag 100                                                             |             |            |             | -0.52689086 |
| Lag 500                                                             |             |            |             | -0.22551523 |
| Lag 1000                                                            |             |            |             | -0.12616708 |
| Lag 5000                                                            |             |            |             | -0.04647821 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |             |            |             |             |
| Lag 0                                                               |             |            |             | -0.82495927 |
| Lag 100                                                             |             |            |             | -0.60990914 |
| Lag 500                                                             |             |            |             | -0.24464022 |
| Lag 1000                                                            |             |            |             | -0.12721295 |
| Lag 5000                                                            |             |            |             | -0.03841367 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |             |            |             |             |
| Lag 0                                                               |             |            |             | -0.82495927 |
| Lag 100                                                             |             |            |             | -0.60990914 |
| Lag 500                                                             |             |            |             | -0.24464022 |
| Lag 1000                                                            |             |            |             | -0.12721295 |
| Lag 5000                                                            |             |            |             | -0.03841367 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |             |            |             |             |
| Lag 0                                                               |             |            |             | -0.64394327 |
| Lag 100                                                             |             |            |             | -0.48995337 |
| Lag 500                                                             |             |            |             | -0.19725633 |
| Lag 1000                                                            |             |            |             | -0.10852446 |
| Lag 5000                                                            |             |            |             | -0.02105523 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |             |            |             |             |
| Lag 0                                                               |             |            |             | 0.082481767 |
| Lag 100                                                             |             |            |             | 0.105417000 |
| Lag 500                                                             |             |            |             | 0.073280263 |
| Lag 1000                                                            |             |            |             | 0.028355398 |

---

|                                                                          |              |
|--------------------------------------------------------------------------|--------------|
| Lag 5000                                                                 | -0.006019758 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units         |              |
| Lag 0                                                                    | 0.09308370   |
| Lag 100                                                                  | 0.10680733   |
| Lag 500                                                                  | 0.07810010   |
| Lag 1000                                                                 | 0.04359553   |
| Lag 5000                                                                 | -0.04726853  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units         |              |
| Lag 0                                                                    | 0.09308370   |
| Lag 100                                                                  | 0.10680733   |
| Lag 500                                                                  | 0.07810010   |
| Lag 1000                                                                 | 0.04359553   |
| Lag 5000                                                                 | -0.04726853  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units      |              |
| Lag 0                                                                    | 0.08843633   |
| Lag 100                                                                  | 0.09343376   |
| Lag 500                                                                  | 0.06886426   |
| Lag 1000                                                                 | 0.05338682   |
| Lag 5000                                                                 | -0.09185034  |
| , , at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |              |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal           |              |
| Lag 0                                                                    | 0.5304141    |
| Lag 100                                                                  | 0.3737195    |
| Lag 500                                                                  | 0.1441203    |
| Lag 1000                                                                 | 0.1503417    |
| Lag 5000                                                                 | 0.1187940    |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal        |              |
| Lag 0                                                                    | 0.7989970    |
| Lag 100                                                                  | 0.5706521    |
| Lag 500                                                                  | 0.2430451    |
| Lag 1000                                                                 | 0.1680830    |

---

|                                                                      |              |
|----------------------------------------------------------------------|--------------|
| Lag 5000                                                             | 0.1259980    |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    |              |
| Lag 0                                                                | 0.7989970    |
| Lag 100                                                              | 0.5706521    |
| Lag 500                                                              | 0.2430451    |
| Lag 1000                                                             | 0.1680830    |
| Lag 5000                                                             | 0.1259980    |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |              |
| Lag 0                                                                | 1.00000000   |
| Lag 100                                                              | 0.73196692   |
| Lag 500                                                              | 0.31335783   |
| Lag 1000                                                             | 0.18501263   |
| Lag 5000                                                             | 0.08438218   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |              |
| Lag 0                                                                | -0.01785209  |
| Lag 100                                                              | -0.03508025  |
| Lag 500                                                              | -0.04733762  |
| Lag 1000                                                             | -0.01709422  |
| Lag 5000                                                             | -0.01586047  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |              |
| Lag 0                                                                | -0.038020441 |
| Lag 100                                                              | -0.045171003 |
| Lag 500                                                              | -0.050004069 |
| Lag 1000                                                             | -0.054183547 |
| Lag 5000                                                             | -0.004955516 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |              |
| Lag 0                                                                | -0.038020441 |
| Lag 100                                                              | -0.045171003 |
| Lag 500                                                              | -0.050004069 |
| Lag 1000                                                             | -0.054183547 |
| Lag 5000                                                             | -0.004955516 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |              |
| Lag 0                                                                | -0.05447207  |



---

|          |                                 |                                      |                 |                    |
|----------|---------------------------------|--------------------------------------|-----------------|--------------------|
| Lag 100  |                                 |                                      |                 | -0.05998184        |
| Lag 500  |                                 |                                      |                 | -0.06158778        |
| Lag 1000 |                                 |                                      |                 | -0.08267333        |
| Lag 5000 |                                 |                                      |                 | 0.02065741         |
|          | traitbwt.byear                  | traittarsus.byear                    | traitbwt.mother | traittarsus.mother |
| Lag 0    | -0.060159939                    | 0.06450755                           | -0.0973321863   | -0.009350685       |
| Lag 100  | -0.043720033                    | 0.03483594                           | -0.0765923141   | -0.006212912       |
| Lag 500  | -0.052466217                    | 0.02987272                           | -0.0662772868   | -0.030465249       |
| Lag 1000 | -0.001034192                    | 0.03110963                           | -0.0728720391   | 0.009855596        |
| Lag 5000 | -0.034160786                    | 0.05472996                           | 0.0008533055    | 0.003426058        |
|          | at.level(sex, "1"):traitbwt:    | at.level(sex, "1"):traitbwt.units    |                 |                    |
| Lag 0    |                                 |                                      |                 | -0.4637838         |
| Lag 100  |                                 |                                      |                 | -0.3337342         |
| Lag 500  |                                 |                                      |                 | -0.1618186         |
| Lag 1000 |                                 |                                      |                 | -0.1329387         |
| Lag 5000 |                                 |                                      |                 | -0.1239580         |
|          | at.level(sex, "1"):traittarsus: | at.level(sex, "1"):traitbwt.units    |                 |                    |
| Lag 0    |                                 |                                      |                 | -0.6583252         |
| Lag 100  |                                 |                                      |                 | -0.4881849         |
| Lag 500  |                                 |                                      |                 | -0.2287167         |
| Lag 1000 |                                 |                                      |                 | -0.1381960         |
| Lag 5000 |                                 |                                      |                 | -0.1179873         |
|          | at.level(sex, "1"):traitbwt:    | at.level(sex, "1"):traittarsus.units |                 |                    |
| Lag 0    |                                 |                                      |                 | -0.6583252         |
| Lag 100  |                                 |                                      |                 | -0.4881849         |
| Lag 500  |                                 |                                      |                 | -0.2287167         |
| Lag 1000 |                                 |                                      |                 | -0.1381960         |
| Lag 5000 |                                 |                                      |                 | -0.1179873         |
|          | at.level(sex, "1"):traittarsus: | at.level(sex, "1"):traittarsus.units |                 |                    |
| Lag 0    |                                 |                                      |                 | -0.76001059        |
| Lag 100  |                                 |                                      |                 | -0.57358014        |
| Lag 500  |                                 |                                      |                 | -0.25179771        |
| Lag 1000 |                                 |                                      |                 | -0.12123408        |

---

```
Lag 5000                                -0.07986147
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units
Lag 0                                   0.02870904
Lag 100                                0.03399695
Lag 500                                0.05485675
Lag 1000                               0.01651664
Lag 5000                               0.01516504
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units
Lag 0                                   0.039633904
Lag 100                                0.041502118
Lag 500                                0.059018043
Lag 1000                               0.052958967
Lag 5000                               -0.002274568
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
Lag 0                                   0.039633904
Lag 100                                0.041502118
Lag 500                                0.059018043
Lag 1000                               0.052958967
Lag 5000                               -0.002274568
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0                                   0.04584246
Lag 100                                0.04763329
Lag 500                                0.06852725
Lag 1000                               0.08362165
Lag 5000                               -0.03430204

, , at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0                                   -0.07132000
Lag 100                                -0.06108550
Lag 500                                -0.06344456
Lag 1000                               -0.02628413
```

---

---

|                                                                      |               |
|----------------------------------------------------------------------|---------------|
| Lag 5000                                                             | 0.10351490    |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal    |               |
| Lag 0                                                                | -0.0696441487 |
| Lag 100                                                              | -0.0685711479 |
| Lag 500                                                              | -0.0543839240 |
| Lag 1000                                                             | 0.0004950661  |
| Lag 5000                                                             | 0.1221823016  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    |               |
| Lag 0                                                                | -0.0696441487 |
| Lag 100                                                              | -0.0685711479 |
| Lag 500                                                              | -0.0543839240 |
| Lag 1000                                                             | 0.0004950661  |
| Lag 5000                                                             | 0.1221823016  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |               |
| Lag 0                                                                | -0.0178520882 |
| Lag 100                                                              | -0.0173974776 |
| Lag 500                                                              | -0.0002494694 |
| Lag 1000                                                             | 0.0551913450  |
| Lag 5000                                                             | 0.1333840825  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |               |
| Lag 0                                                                | 1.0000000     |
| Lag 100                                                              | 0.8242352     |
| Lag 500                                                              | 0.5296802     |
| Lag 1000                                                             | 0.3040607     |
| Lag 5000                                                             | -0.1276161    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |               |
| Lag 0                                                                | 0.9099634     |
| Lag 100                                                              | 0.8047694     |
| Lag 500                                                              | 0.5857973     |
| Lag 1000                                                             | 0.3552775     |
| Lag 5000                                                             | -0.1485103    |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |               |
| Lag 0                                                                | 0.9099634     |

---

|                                                                      |             |            |              |              |
|----------------------------------------------------------------------|-------------|------------|--------------|--------------|
| Lag 100                                                              |             |            |              | 0.8047694    |
| Lag 500                                                              |             |            |              | 0.5857973    |
| Lag 1000                                                             |             |            |              | 0.3552775    |
| Lag 5000                                                             |             |            |              | -0.1485103   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |             |            |              |              |
| Lag 0                                                                |             |            |              | 0.7704756    |
| Lag 100                                                              |             |            |              | 0.7082472    |
| Lag 500                                                              |             |            |              | 0.5569812    |
| Lag 1000                                                             |             |            |              | 0.3517296    |
| Lag 5000                                                             |             |            |              | -0.1453072   |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |             |            |              |              |
| Lag 0                                                                | -0.03784246 | 0.04916122 | -0.025145260 | -0.13999847  |
| Lag 100                                                              | -0.01888261 | 0.04551933 | -0.009748633 | -0.14616483  |
| Lag 500                                                              | -0.01864811 | 0.07395050 | 0.029035276  | -0.12958636  |
| Lag 1000                                                             | -0.02117775 | 0.06164183 | 0.068666314  | -0.09577992  |
| Lag 5000                                                             | 0.01769136  | 0.04869291 | 0.037573009  | 0.01686724   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |             |            |              |              |
| Lag 0                                                                |             |            |              | 0.048426202  |
| Lag 100                                                              |             |            |              | 0.039795097  |
| Lag 500                                                              |             |            |              | 0.046311373  |
| Lag 1000                                                             |             |            |              | 0.005469282  |
| Lag 5000                                                             |             |            |              | -0.104125437 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units     |             |            |              |              |
| Lag 0                                                                |             |            |              | 0.043207572  |
| Lag 100                                                              |             |            |              | 0.043210916  |
| Lag 500                                                              |             |            |              | 0.033044478  |
| Lag 1000                                                             |             |            |              | -0.004411742 |
| Lag 5000                                                             |             |            |              | -0.110707718 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units     |             |            |              |              |
| Lag 0                                                                |             |            |              | 0.043207572  |
| Lag 100                                                              |             |            |              | 0.043210916  |
| Lag 500                                                              |             |            |              | 0.033044478  |
| Lag 1000                                                             |             |            |              | -0.004411742 |

---

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Lag 5000                                                            | -0.110707718 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |              |
| Lag 0                                                               | 0.01554818   |
| Lag 100                                                             | 0.02352457   |
| Lag 500                                                             | 0.01217491   |
| Lag 1000                                                            | -0.02394172  |
| Lag 5000                                                            | -0.13812594  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |              |
| Lag 0                                                               | -0.9369474   |
| Lag 100                                                             | -0.8092241   |
| Lag 500                                                             | -0.5186132   |
| Lag 1000                                                            | -0.2947735   |
| Lag 5000                                                            | 0.1226249    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |              |
| Lag 0                                                               | -0.8725969   |
| Lag 100                                                             | -0.7954903   |
| Lag 500                                                             | -0.5688119   |
| Lag 1000                                                            | -0.3390939   |
| Lag 5000                                                            | 0.1455303    |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    |              |
| Lag 0                                                               | -0.8725969   |
| Lag 100                                                             | -0.7954903   |
| Lag 500                                                             | -0.5688119   |
| Lag 1000                                                            | -0.3390939   |
| Lag 5000                                                            | 0.1455303    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units |              |
| Lag 0                                                               | -0.7525481   |
| Lag 100                                                             | -0.7086874   |
| Lag 500                                                             | -0.5421136   |
| Lag 1000                                                            | -0.3389906   |
| Lag 5000                                                            | 0.1485387    |

, , at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal

```
      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0                                     -0.074042865
Lag 100                                  -0.072737049
Lag 500                                  -0.064855516
Lag 1000                                 -0.004245299
Lag 5000                                 0.126495395

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0                                     -0.092665568
Lag 100                                  -0.096939661
Lag 500                                  -0.070837135
Lag 1000                                 0.006501962
Lag 5000                                 0.148898005

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal
Lag 0                                     -0.092665568
Lag 100                                  -0.096939661
Lag 500                                  -0.070837135
Lag 1000                                 0.006501962
Lag 5000                                 0.148898005

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal
Lag 0                                     -0.038020441
Lag 100                                  -0.039681669
Lag 500                                  -0.006820427
Lag 1000                                 0.063529955
Lag 5000                                 0.163665055

      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal
Lag 0                                     0.9099634
Lag 100                                  0.7863387
Lag 500                                  0.5413307
Lag 1000                                 0.3118422
Lag 5000                                 -0.1191809

      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal
Lag 0                                     1.0000000
```

---

|                                                                      |                                               |
|----------------------------------------------------------------------|-----------------------------------------------|
| Lag 100                                                              | 0.8933098                                     |
| Lag 500                                                              | 0.6382613                                     |
| Lag 1000                                                             | 0.3875538                                     |
| Lag 5000                                                             | -0.1480316                                    |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |                                               |
| Lag 0                                                                | 1.0000000                                     |
| Lag 100                                                              | 0.8933098                                     |
| Lag 500                                                              | 0.6382613                                     |
| Lag 1000                                                             | 0.3875538                                     |
| Lag 5000                                                             | -0.1480316                                    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |                                               |
| Lag 0                                                                | 0.9445430                                     |
| Lag 100                                                              | 0.8642926                                     |
| Lag 500                                                              | 0.6377101                                     |
| Lag 1000                                                             | 0.3971583                                     |
| Lag 5000                                                             | -0.1545401                                    |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |                                               |
| Lag 0                                                                | -0.04691870 0.05505699 0.03372293 -0.18311492 |
| Lag 100                                                              | -0.03261563 0.04790144 0.03175029 -0.18180192 |
| Lag 500                                                              | -0.02904559 0.05050843 0.04377317 -0.16642684 |
| Lag 1000                                                             | -0.03811545 0.05361475 0.07019878 -0.12467546 |
| Lag 5000                                                             | 0.04062218 0.04654678 0.03310770 0.01019974   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |                                               |
| Lag 0                                                                | 0.04443896                                    |
| Lag 100                                                              | 0.04551916                                    |
| Lag 500                                                              | 0.04357037                                    |
| Lag 1000                                                             | -0.01792034                                   |
| Lag 5000                                                             | -0.12752563                                   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units     |                                               |
| Lag 0                                                                | 0.04900783                                    |
| Lag 100                                                              | 0.06050152                                    |
| Lag 500                                                              | 0.04114675                                    |
| Lag 1000                                                             | -0.01839006                                   |

---

```
Lag 5000                                -0.13689966
      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units
Lag 0                                   0.04900783
Lag 100                                0.06050152
Lag 500                                0.04114675
Lag 1000                               -0.01839006
Lag 5000                               -0.13689966
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units
Lag 0                                   0.02477085
Lag 100                                0.03939172
Lag 500                                0.02069295
Lag 1000                               -0.04020316
Lag 5000                               -0.16427556
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units
Lag 0                                   -0.8748666
Lag 100                                -0.7865075
Lag 500                                -0.5347377
Lag 1000                               -0.3099292
Lag 5000                               0.1242901
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units
Lag 0                                   -0.9629330
Lag 100                                -0.8851422
Lag 500                                -0.6211970
Lag 1000                               -0.3754777
Lag 5000                               0.1511004
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
Lag 0                                   -0.9629330
Lag 100                                -0.8851422
Lag 500                                -0.6211970
Lag 1000                               -0.3754777
Lag 5000                               0.1511004
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0                                   -0.9191068
```



---

|          |            |
|----------|------------|
| Lag 100  | -0.8609263 |
| Lag 500  | -0.6233180 |
| Lag 1000 | -0.3879589 |
| Lag 5000 | 0.1606066  |

, , at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal

at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal

|          |              |
|----------|--------------|
| Lag 0    | -0.074042865 |
| Lag 100  | -0.072737049 |
| Lag 500  | -0.064855516 |
| Lag 1000 | -0.004245299 |
| Lag 5000 | 0.126495395  |

at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal

|          |              |
|----------|--------------|
| Lag 0    | -0.092665568 |
| Lag 100  | -0.096939661 |
| Lag 500  | -0.070837135 |
| Lag 1000 | 0.006501962  |
| Lag 5000 | 0.148898005  |

at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal

|          |              |
|----------|--------------|
| Lag 0    | -0.092665568 |
| Lag 100  | -0.096939661 |
| Lag 500  | -0.070837135 |
| Lag 1000 | 0.006501962  |
| Lag 5000 | 0.148898005  |

at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal

|          |              |
|----------|--------------|
| Lag 0    | -0.038020441 |
| Lag 100  | -0.039681669 |
| Lag 500  | -0.006820427 |
| Lag 1000 | 0.063529955  |
| Lag 5000 | 0.163665055  |

at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal

|       |           |
|-------|-----------|
| Lag 0 | 0.9099634 |
|-------|-----------|

|                                                                      |             |            |            |             |
|----------------------------------------------------------------------|-------------|------------|------------|-------------|
| Lag 100                                                              | 0.7863387   |            |            |             |
| Lag 500                                                              | 0.5413307   |            |            |             |
| Lag 1000                                                             | 0.3118422   |            |            |             |
| Lag 5000                                                             | -0.1191809  |            |            |             |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |             |            |            |             |
| Lag 0                                                                | 1.0000000   |            |            |             |
| Lag 100                                                              | 0.8933098   |            |            |             |
| Lag 500                                                              | 0.6382613   |            |            |             |
| Lag 1000                                                             | 0.3875538   |            |            |             |
| Lag 5000                                                             | -0.1480316  |            |            |             |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |             |            |            |             |
| Lag 0                                                                | 1.0000000   |            |            |             |
| Lag 100                                                              | 0.8933098   |            |            |             |
| Lag 500                                                              | 0.6382613   |            |            |             |
| Lag 1000                                                             | 0.3875538   |            |            |             |
| Lag 5000                                                             | -0.1480316  |            |            |             |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |             |            |            |             |
| Lag 0                                                                | 0.9445430   |            |            |             |
| Lag 100                                                              | 0.8642926   |            |            |             |
| Lag 500                                                              | 0.6377101   |            |            |             |
| Lag 1000                                                             | 0.3971583   |            |            |             |
| Lag 5000                                                             | -0.1545401  |            |            |             |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |             |            |            |             |
| Lag 0                                                                | -0.04691870 | 0.05505699 | 0.03372293 | -0.18311492 |
| Lag 100                                                              | -0.03261563 | 0.04790144 | 0.03175029 | -0.18180192 |
| Lag 500                                                              | -0.02904559 | 0.05050843 | 0.04377317 | -0.16642684 |
| Lag 1000                                                             | -0.03811545 | 0.05361475 | 0.07019878 | -0.12467546 |
| Lag 5000                                                             | 0.04062218  | 0.04654678 | 0.03310770 | 0.01019974  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |             |            |            |             |
| Lag 0                                                                | 0.04443896  |            |            |             |
| Lag 100                                                              | 0.04551916  |            |            |             |
| Lag 500                                                              | 0.04357037  |            |            |             |
| Lag 1000                                                             | -0.01792034 |            |            |             |

---

|                                                                     |             |
|---------------------------------------------------------------------|-------------|
| Lag 5000                                                            | -0.12752563 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |             |
| Lag 0                                                               | 0.04900783  |
| Lag 100                                                             | 0.06050152  |
| Lag 500                                                             | 0.04114675  |
| Lag 1000                                                            | -0.01839006 |
| Lag 5000                                                            | -0.13689966 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |             |
| Lag 0                                                               | 0.04900783  |
| Lag 100                                                             | 0.06050152  |
| Lag 500                                                             | 0.04114675  |
| Lag 1000                                                            | -0.01839006 |
| Lag 5000                                                            | -0.13689966 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |             |
| Lag 0                                                               | 0.02477085  |
| Lag 100                                                             | 0.03939172  |
| Lag 500                                                             | 0.02069295  |
| Lag 1000                                                            | -0.04020316 |
| Lag 5000                                                            | -0.16427556 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |             |
| Lag 0                                                               | -0.8748666  |
| Lag 100                                                             | -0.7865075  |
| Lag 500                                                             | -0.5347377  |
| Lag 1000                                                            | -0.3099292  |
| Lag 5000                                                            | 0.1242901   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |             |
| Lag 0                                                               | -0.9629330  |
| Lag 100                                                             | -0.8851422  |
| Lag 500                                                             | -0.6211970  |
| Lag 1000                                                            | -0.3754777  |
| Lag 5000                                                            | 0.1511004   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    |             |
| Lag 0                                                               | -0.9629330  |

---

```
Lag 100 -0.8851422
Lag 500 -0.6211970
Lag 1000 -0.3754777
Lag 5000 0.1511004

      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0 -0.9191068
Lag 100 -0.8609263
Lag 500 -0.6233180
Lag 1000 -0.3879589
Lag 5000 0.1606066

, , at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0 -0.066633008
Lag 100 -0.069354252
Lag 500 -0.053416684
Lag 1000 0.001180564
Lag 5000 0.142470162

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0 -0.102195672
Lag 100 -0.107130141
Lag 500 -0.073576929
Lag 1000 -0.004189061
Lag 5000 0.167339055

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal
Lag 0 -0.102195672
Lag 100 -0.107130141
Lag 500 -0.073576929
Lag 1000 -0.004189061
Lag 5000 0.167339055

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal
Lag 0 -0.05447207
```

---

|                                                                      |             |            |            |             |
|----------------------------------------------------------------------|-------------|------------|------------|-------------|
| Lag 100                                                              |             |            |            | -0.05707224 |
| Lag 500                                                              |             |            |            | -0.01853426 |
| Lag 1000                                                             |             |            |            | 0.04684883  |
| Lag 5000                                                             |             |            |            | 0.17837197  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |             |            |            |             |
| Lag 0                                                                |             |            |            | 0.7704756   |
| Lag 100                                                              |             |            |            | 0.6843951   |
| Lag 500                                                              |             |            |            | 0.5050605   |
| Lag 1000                                                             |             |            |            | 0.2977972   |
| Lag 5000                                                             |             |            |            | -0.1058223  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |             |            |            |             |
| Lag 0                                                                |             |            |            | 0.9445430   |
| Lag 100                                                              |             |            |            | 0.8556548   |
| Lag 500                                                              |             |            |            | 0.6303848   |
| Lag 1000                                                             |             |            |            | 0.3904095   |
| Lag 5000                                                             |             |            |            | -0.1424423  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |             |            |            |             |
| Lag 0                                                                |             |            |            | 0.9445430   |
| Lag 100                                                              |             |            |            | 0.8556548   |
| Lag 500                                                              |             |            |            | 0.6303848   |
| Lag 1000                                                             |             |            |            | 0.3904095   |
| Lag 5000                                                             |             |            |            | -0.1424423  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |             |            |            |             |
| Lag 0                                                                |             |            |            | 1.0000000   |
| Lag 100                                                              |             |            |            | 0.9100529   |
| Lag 500                                                              |             |            |            | 0.6616827   |
| Lag 1000                                                             |             |            |            | 0.4121439   |
| Lag 5000                                                             |             |            |            | -0.1590568  |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |             |            |            |             |
| Lag 0                                                                | -0.03974818 | 0.04854354 | 0.04924110 | -0.22289117 |
| Lag 100                                                              | -0.03767078 | 0.04176415 | 0.03930577 | -0.20950998 |
| Lag 500                                                              | -0.03440434 | 0.03555315 | 0.05305906 | -0.18185253 |
| Lag 1000                                                             | -0.02822560 | 0.04567963 | 0.06510782 | -0.12092658 |

---

|                                                                     |            |            |             |            |
|---------------------------------------------------------------------|------------|------------|-------------|------------|
| Lag 5000                                                            | 0.05272181 | 0.04246380 | 0.02211597  | 0.02036647 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       |            |            |             |            |
| Lag 0                                                               |            |            | 0.04061110  |            |
| Lag 100                                                             |            |            | 0.04206692  |            |
| Lag 500                                                             |            |            | 0.03373343  |            |
| Lag 1000                                                            |            |            | -0.01881550 |            |
| Lag 5000                                                            |            |            | -0.13859641 |            |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |            |            |             |            |
| Lag 0                                                               |            |            | 0.05879880  |            |
| Lag 100                                                             |            |            | 0.06872711  |            |
| Lag 500                                                             |            |            | 0.04634425  |            |
| Lag 1000                                                            |            |            | -0.01579800 |            |
| Lag 5000                                                            |            |            | -0.14960797 |            |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |            |            |             |            |
| Lag 0                                                               |            |            | 0.05879880  |            |
| Lag 100                                                             |            |            | 0.06872711  |            |
| Lag 500                                                             |            |            | 0.04634425  |            |
| Lag 1000                                                            |            |            | -0.01579800 |            |
| Lag 5000                                                            |            |            | -0.14960797 |            |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |            |            |             |            |
| Lag 0                                                               |            |            | 0.04254878  |            |
| Lag 100                                                             |            |            | 0.05461840  |            |
| Lag 500                                                             |            |            | 0.03646094  |            |
| Lag 1000                                                            |            |            | -0.04023461 |            |
| Lag 5000                                                            |            |            | -0.17207003 |            |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |            |            |             |            |
| Lag 0                                                               |            |            | -0.7532567  |            |
| Lag 100                                                             |            |            | -0.6923793  |            |
| Lag 500                                                             |            |            | -0.5032427  |            |
| Lag 1000                                                            |            |            | -0.2991945  |            |
| Lag 5000                                                            |            |            | 0.1154818   |            |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |            |            |             |            |
| Lag 0                                                               |            |            | -0.9140715  |            |

---

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Lag 100                                                             | -0.8493221   |
| Lag 500                                                             | -0.6164356   |
| Lag 1000                                                            | -0.3808768   |
| Lag 5000                                                            | 0.1473936    |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    |              |
| Lag 0                                                               | -0.9140715   |
| Lag 100                                                             | -0.8493221   |
| Lag 500                                                             | -0.6164356   |
| Lag 1000                                                            | -0.3808768   |
| Lag 5000                                                            | 0.1473936    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units |              |
| Lag 0                                                               | -0.9650519   |
| Lag 100                                                             | -0.9008410   |
| Lag 500                                                             | -0.6496396   |
| Lag 1000                                                            | -0.4073019   |
| Lag 5000                                                            | 0.1644354    |
| , , traitbwt.byear                                                  |              |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal      |              |
| Lag 0                                                               | -0.002044905 |
| Lag 100                                                             | 0.018082206  |
| Lag 500                                                             | -0.019694583 |
| Lag 1000                                                            | -0.033624772 |
| Lag 5000                                                            | -0.025949000 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal   |              |
| Lag 0                                                               | -0.03731153  |
| Lag 100                                                             | -0.01355344  |
| Lag 500                                                             | -0.01864081  |
| Lag 1000                                                            | -0.05745850  |
| Lag 5000                                                            | -0.01235998  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal   |              |
| Lag 0                                                               | -0.03731153  |

---

|                                                                      |              |
|----------------------------------------------------------------------|--------------|
| Lag 100                                                              | -0.01355344  |
| Lag 500                                                              | -0.01864081  |
| Lag 1000                                                             | -0.05745850  |
| Lag 5000                                                             | -0.01235998  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |              |
| Lag 0                                                                | -0.06015994  |
| Lag 100                                                              | -0.03212487  |
| Lag 500                                                              | -0.02412236  |
| Lag 1000                                                             | -0.05846861  |
| Lag 5000                                                             | -0.02882580  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |              |
| Lag 0                                                                | -0.03784246  |
| Lag 100                                                              | -0.04180932  |
| Lag 500                                                              | -0.04438042  |
| Lag 1000                                                             | -0.01257459  |
| Lag 5000                                                             | 0.00630995   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |              |
| Lag 0                                                                | -0.04691870  |
| Lag 100                                                              | -0.04133222  |
| Lag 500                                                              | -0.05230682  |
| Lag 1000                                                             | -0.02870414  |
| Lag 5000                                                             | -0.00469889  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |              |
| Lag 0                                                                | -0.04691870  |
| Lag 100                                                              | -0.04133222  |
| Lag 500                                                              | -0.05230682  |
| Lag 1000                                                             | -0.02870414  |
| Lag 5000                                                             | -0.00469889  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |              |
| Lag 0                                                                | -0.039748177 |
| Lag 100                                                              | -0.032934501 |
| Lag 500                                                              | -0.055295362 |
| Lag 1000                                                             | -0.027884156 |



---

|          |                                                                     |                   |                 |                    |
|----------|---------------------------------------------------------------------|-------------------|-----------------|--------------------|
| Lag 5000 |                                                                     |                   |                 | -0.007044631       |
|          | traitbwt.byear                                                      | traittarsus.byear | traitbwt.mother | traittarsus.mother |
| Lag 0    | 1.00000000                                                          | -0.0251146296     | 0.03365469      | 0.03928862         |
| Lag 100  | 0.03109454                                                          | 0.0004436899      | -0.05764761     | -0.01264335        |
| Lag 500  | 0.03937305                                                          | 0.0006604187      | -0.00457655     | 0.02746272         |
| Lag 1000 | 0.01680424                                                          | -0.0194711518     | 0.03737600      | -0.04627035        |
| Lag 5000 | 0.03318792                                                          | 0.0155533971      | -0.02558374     | 0.05305580         |
|          | at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       |                   |                 |                    |
| Lag 0    |                                                                     |                   |                 | 0.005983125        |
| Lag 100  |                                                                     |                   |                 | -0.001889062       |
| Lag 500  |                                                                     |                   |                 | -0.018793288       |
| Lag 1000 |                                                                     |                   |                 | 0.027363658        |
| Lag 5000 |                                                                     |                   |                 | 0.010334637        |
|          | at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |                   |                 |                    |
| Lag 0    |                                                                     |                   |                 | 0.052806759        |
| Lag 100  |                                                                     |                   |                 | 0.014657374        |
| Lag 500  |                                                                     |                   |                 | -0.020921457       |
| Lag 1000 |                                                                     |                   |                 | 0.041519184        |
| Lag 5000 |                                                                     |                   |                 | -0.000172048       |
|          | at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |                   |                 |                    |
| Lag 0    |                                                                     |                   |                 | 0.052806759        |
| Lag 100  |                                                                     |                   |                 | 0.014657374        |
| Lag 500  |                                                                     |                   |                 | -0.020921457       |
| Lag 1000 |                                                                     |                   |                 | 0.041519184        |
| Lag 5000 |                                                                     |                   |                 | -0.000172048       |
|          | at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |                   |                 |                    |
| Lag 0    |                                                                     |                   |                 | 0.072105699        |
| Lag 100  |                                                                     |                   |                 | 0.023891187        |
| Lag 500  |                                                                     |                   |                 | -0.022626087       |
| Lag 1000 |                                                                     |                   |                 | 0.038071084        |
| Lag 5000 |                                                                     |                   |                 | 0.008022532        |
|          | at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |                   |                 |                    |
| Lag 0    |                                                                     |                   |                 | 0.02472261         |

---

```
Lag 100                                0.04922524
Lag 500                                0.05684465
Lag 1000                               0.02213746
Lag 5000                               -0.02587314
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units
Lag 0                                  0.035150019
Lag 100                                0.039823504
Lag 500                                0.058615425
Lag 1000                               0.039863168
Lag 5000                               -0.002768445
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
Lag 0                                  0.035150019
Lag 100                                0.039823504
Lag 500                                0.058615425
Lag 1000                               0.039863168
Lag 5000                               -0.002768445
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0                                  0.032332494
Lag 100                                0.036826480
Lag 500                                0.056371336
Lag 1000                               0.037506421
Lag 5000                               -0.002071877

, , traittarsus.byear

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0                                  0.06061428
Lag 100                                0.06276970
Lag 500                                -0.02842127
Lag 1000                               0.01799228
Lag 5000                               0.02740499
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0                                  0.05572330
```

---

---

|                                                                      |              |
|----------------------------------------------------------------------|--------------|
| Lag 100                                                              | 0.06655805   |
| Lag 500                                                              | -0.02673025  |
| Lag 1000                                                             | 0.04345968   |
| Lag 5000                                                             | 0.05112113   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    |              |
| Lag 0                                                                | 0.05572330   |
| Lag 100                                                              | 0.06655805   |
| Lag 500                                                              | -0.02673025  |
| Lag 1000                                                             | 0.04345968   |
| Lag 5000                                                             | 0.05112113   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |              |
| Lag 0                                                                | 0.064507548  |
| Lag 100                                                              | 0.074840509  |
| Lag 500                                                              | -0.003777881 |
| Lag 1000                                                             | 0.058609933  |
| Lag 5000                                                             | 0.063485567  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |              |
| Lag 0                                                                | 0.049161224  |
| Lag 100                                                              | 0.009105861  |
| Lag 500                                                              | 0.005065210  |
| Lag 1000                                                             | 0.016389664  |
| Lag 5000                                                             | -0.029590445 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |              |
| Lag 0                                                                | 0.055056994  |
| Lag 100                                                              | 0.035711495  |
| Lag 500                                                              | 0.012368434  |
| Lag 1000                                                             | 0.004770290  |
| Lag 5000                                                             | -0.009144398 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |              |
| Lag 0                                                                | 0.055056994  |
| Lag 100                                                              | 0.035711495  |
| Lag 500                                                              | 0.012368434  |
| Lag 1000                                                             | 0.004770290  |

---

```
Lag 5000                                -0.009144398
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal
Lag 0                                    0.048543542
Lag 100                                 0.034488675
Lag 500                                 0.009703880
Lag 1000                                -0.001685047
Lag 5000                                0.005011858

      traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother
Lag 0      -0.025114630      1.000000000      0.03708995      0.07084541
Lag 100     -0.033801997      0.041927040     -0.06653239      0.04503853
Lag 500      0.009533405     -0.020053055      0.01042960     -0.03755216
Lag 1000    -0.003946143      0.011455578     -0.01588844      0.01986940
Lag 5000     0.027020776      0.002689451     -0.02585871      0.02687208

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units
Lag 0                                -0.10335662
Lag 100                               -0.05376108
Lag 500                               0.01626001
Lag 1000                              -0.01344940
Lag 5000                              -0.02833156

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units
Lag 0                                -0.09746412
Lag 100                               -0.05607997
Lag 500                               0.01829969
Lag 1000                              -0.04558284
Lag 5000                              -0.04241014

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units
Lag 0                                -0.09746412
Lag 100                               -0.05607997
Lag 500                               0.01829969
Lag 1000                              -0.04558284
Lag 5000                              -0.04241014

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units
Lag 0                                -0.108393481
```

---

|          |              |
|----------|--------------|
| Lag 100  | -0.049448255 |
| Lag 500  | 0.002404817  |
| Lag 1000 | -0.086421792 |
| Lag 5000 | -0.055605953 |

at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units

|          |              |
|----------|--------------|
| Lag 0    | -0.025767223 |
| Lag 100  | -0.006184423 |
| Lag 500  | -0.002201914 |
| Lag 1000 | -0.011174601 |
| Lag 5000 | 0.029950491  |

at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units

|          |              |
|----------|--------------|
| Lag 0    | -0.049170830 |
| Lag 100  | -0.040276502 |
| Lag 500  | 0.003528012  |
| Lag 1000 | -0.010662330 |
| Lag 5000 | 0.002523808  |

at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units

|          |              |
|----------|--------------|
| Lag 0    | -0.049170830 |
| Lag 100  | -0.040276502 |
| Lag 500  | 0.003528012  |
| Lag 1000 | -0.010662330 |
| Lag 5000 | 0.002523808  |

at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units

|          |              |
|----------|--------------|
| Lag 0    | -0.053929470 |
| Lag 100  | -0.043281273 |
| Lag 500  | 0.009317392  |
| Lag 1000 | -0.010594624 |
| Lag 5000 | -0.013754908 |

, , traitbwt.mother

at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal

|       |             |
|-------|-------------|
| Lag 0 | -0.13681757 |
|-------|-------------|

|                                                                      |               |
|----------------------------------------------------------------------|---------------|
| Lag 100                                                              | -0.09694549   |
| Lag 500                                                              | -0.06857367   |
| Lag 1000                                                             | -0.04540954   |
| Lag 5000                                                             | -0.01652050   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal    |               |
| Lag 0                                                                | -0.12626725   |
| Lag 100                                                              | -0.12449687   |
| Lag 500                                                              | -0.05181080   |
| Lag 1000                                                             | -0.03932960   |
| Lag 5000                                                             | -0.01141931   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    |               |
| Lag 0                                                                | -0.12626725   |
| Lag 100                                                              | -0.12449687   |
| Lag 500                                                              | -0.05181080   |
| Lag 1000                                                             | -0.03932960   |
| Lag 5000                                                             | -0.01141931   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |               |
| Lag 0                                                                | -0.09733219   |
| Lag 100                                                              | -0.10049386   |
| Lag 500                                                              | -0.04634235   |
| Lag 1000                                                             | -0.01749975   |
| Lag 5000                                                             | -0.03285757   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |               |
| Lag 0                                                                | -0.02514526   |
| Lag 100                                                              | 0.01494004    |
| Lag 500                                                              | -0.05400749   |
| Lag 1000                                                             | -0.02644804   |
| Lag 5000                                                             | -0.02759428   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |               |
| Lag 0                                                                | 0.0337229276  |
| Lag 100                                                              | 0.0355284011  |
| Lag 500                                                              | -0.0008562576 |
| Lag 1000                                                             | -0.0192570169 |

---

|          |                                                                      |                   |                 |                    |
|----------|----------------------------------------------------------------------|-------------------|-----------------|--------------------|
| Lag 5000 |                                                                      |                   |                 | -0.0314028551      |
|          | at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |                   |                 |                    |
| Lag 0    |                                                                      |                   |                 | 0.0337229276       |
| Lag 100  |                                                                      |                   |                 | 0.0355284011       |
| Lag 500  |                                                                      |                   |                 | -0.0008562576      |
| Lag 1000 |                                                                      |                   |                 | -0.0192570169      |
| Lag 5000 |                                                                      |                   |                 | -0.0314028551      |
|          | at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |                   |                 |                    |
| Lag 0    |                                                                      |                   |                 | 0.04924110         |
| Lag 100  |                                                                      |                   |                 | 0.04492689         |
| Lag 500  |                                                                      |                   |                 | 0.02422838         |
| Lag 1000 |                                                                      |                   |                 | -0.04034312        |
| Lag 5000 |                                                                      |                   |                 | -0.03784851        |
|          | traitbwt.byear                                                       | traittarsus.byear | traitbwt.mother | traittarsus.mother |
| Lag 0    | 0.033654686                                                          | 0.037089946       | 1.000000000     | -0.267715213       |
| Lag 100  | 0.020365368                                                          | 0.031597781       | 0.039464037     | 0.006711605        |
| Lag 500  | 0.007110008                                                          | 0.046188516       | 0.095615498     | -0.023010721       |
| Lag 1000 | -0.019597442                                                         | 0.001266059       | 0.065362608     | 0.041196297        |
| Lag 5000 | 0.019704700                                                          | -0.034265234      | -0.005121853    | 0.041919494        |
|          | at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |                   |                 |                    |
| Lag 0    |                                                                      |                   |                 | 0.093898173        |
| Lag 100  |                                                                      |                   |                 | 0.109444195        |
| Lag 500  |                                                                      |                   |                 | 0.055322096        |
| Lag 1000 |                                                                      |                   |                 | 0.036648121        |
| Lag 5000 |                                                                      |                   |                 | 0.003492676        |
|          | at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units     |                   |                 |                    |
| Lag 0    |                                                                      |                   |                 | 0.11659923         |
| Lag 100  |                                                                      |                   |                 | 0.12608289         |
| Lag 500  |                                                                      |                   |                 | 0.03369955         |
| Lag 1000 |                                                                      |                   |                 | 0.04312267         |
| Lag 5000 |                                                                      |                   |                 | -0.01418292        |
|          | at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units     |                   |                 |                    |
| Lag 0    |                                                                      |                   |                 | 0.11659923         |

---

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Lag 100                                                             | 0.12608289   |
| Lag 500                                                             | 0.03369955   |
| Lag 1000                                                            | 0.04312267   |
| Lag 5000                                                            | -0.01418292  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |              |
| Lag 0                                                               | 0.1007331132 |
| Lag 100                                                             | 0.0846271381 |
| Lag 500                                                             | 0.0314559531 |
| Lag 1000                                                            | 0.0313378649 |
| Lag 5000                                                            | 0.0006030047 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |              |
| Lag 0                                                               | -0.02385685  |
| Lag 100                                                             | -0.01929710  |
| Lag 500                                                             | 0.05442789   |
| Lag 1000                                                            | 0.02999688   |
| Lag 5000                                                            | 0.02354946   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |              |
| Lag 0                                                               | -0.037784916 |
| Lag 100                                                             | -0.040619404 |
| Lag 500                                                             | 0.009246757  |
| Lag 1000                                                            | 0.021585046  |
| Lag 5000                                                            | 0.029657103  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    |              |
| Lag 0                                                               | -0.037784916 |
| Lag 100                                                             | -0.040619404 |
| Lag 500                                                             | 0.009246757  |
| Lag 1000                                                            | 0.021585046  |
| Lag 5000                                                            | 0.029657103  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units |              |
| Lag 0                                                               | -0.04048151  |
| Lag 100                                                             | -0.04664306  |
| Lag 500                                                             | -0.02420607  |
| Lag 1000                                                            | 0.03414246   |



---

|          |            |
|----------|------------|
| Lag 5000 | 0.04075949 |
|----------|------------|

, , traittarsus.mother

at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal

|          |             |
|----------|-------------|
| Lag 0    | 0.06303474  |
| Lag 100  | 0.09137304  |
| Lag 500  | 0.05137407  |
| Lag 1000 | 0.02569160  |
| Lag 5000 | -0.08057411 |

at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal

|          |             |
|----------|-------------|
| Lag 0    | 0.06865980  |
| Lag 100  | 0.10336350  |
| Lag 500  | 0.01711371  |
| Lag 1000 | 0.03032742  |
| Lag 5000 | -0.08530728 |

at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal

|          |             |
|----------|-------------|
| Lag 0    | 0.06865980  |
| Lag 100  | 0.10336350  |
| Lag 500  | 0.01711371  |
| Lag 1000 | 0.03032742  |
| Lag 5000 | -0.08530728 |

at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal

|          |              |
|----------|--------------|
| Lag 0    | -0.009350685 |
| Lag 100  | 0.024288467  |
| Lag 500  | -0.016438414 |
| Lag 1000 | -0.005767054 |
| Lag 5000 | -0.063956859 |

at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal

|          |             |
|----------|-------------|
| Lag 0    | -0.13999847 |
| Lag 100  | -0.14330129 |
| Lag 500  | -0.12944310 |
| Lag 1000 | -0.08446537 |

```
Lag 5000                                0.04104776
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal
Lag 0                                   -0.18311492
Lag 100                                -0.16614099
Lag 500                                -0.14634974
Lag 1000                               -0.09164415
Lag 5000                               0.05689178
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal
Lag 0                                   -0.18311492
Lag 100                                -0.16614099
Lag 500                                -0.14634974
Lag 1000                               -0.09164415
Lag 5000                               0.05689178
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal
Lag 0                                   -0.22289117
Lag 100                                -0.19084285
Lag 500                                -0.14966648
Lag 1000                               -0.07718413
Lag 5000                               0.07429663
      traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother
Lag 0      0.039288617      0.07084541   -0.267715213      1.000000000
Lag 100     0.047364166     -0.01932534   -0.036245609      0.088363955
Lag 500     0.005475011     -0.04374386   -0.028017777      0.043041568
Lag 1000    -0.022031785      0.01752292    0.005084865     -0.003286219
Lag 5000    -0.004605383     -0.01801176   -0.022372822     -0.055614496
      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units
Lag 0                                   -0.04283736
Lag 100                                -0.05247358
Lag 500                                -0.03520027
Lag 1000                               -0.02685262
Lag 5000                               0.08204082
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units
Lag 0                                   -0.080407800
```

---

---

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Lag 100                                                             | -0.079283374 |
| Lag 500                                                             | 0.007896803  |
| Lag 1000                                                            | -0.024892006 |
| Lag 5000                                                            | 0.090297411  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |              |
| Lag 0                                                               | -0.080407800 |
| Lag 100                                                             | -0.079283374 |
| Lag 500                                                             | 0.007896803  |
| Lag 1000                                                            | -0.024892006 |
| Lag 5000                                                            | 0.090297411  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |              |
| Lag 0                                                               | -0.09075088  |
| Lag 100                                                             | -0.04605175  |
| Lag 500                                                             | 0.02816248   |
| Lag 1000                                                            | 0.01663048   |
| Lag 5000                                                            | 0.08246697   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |              |
| Lag 0                                                               | 0.14299876   |
| Lag 100                                                             | 0.14159603   |
| Lag 500                                                             | 0.13583757   |
| Lag 1000                                                            | 0.06878998   |
| Lag 5000                                                            | -0.03559959  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |              |
| Lag 0                                                               | 0.16291072   |
| Lag 100                                                             | 0.16523003   |
| Lag 500                                                             | 0.14972084   |
| Lag 1000                                                            | 0.08594345   |
| Lag 5000                                                            | -0.05664141  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    |              |
| Lag 0                                                               | 0.16291072   |
| Lag 100                                                             | 0.16523003   |
| Lag 500                                                             | 0.14972084   |
| Lag 1000                                                            | 0.08594345   |

---

```
Lag 5000                                -0.05664141
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0                                   0.17073426
Lag 100                                0.18983154
Lag 500                                0.15067362
Lag 1000                               0.07940280
Lag 5000                               -0.07670042

, , at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0                                   -0.84231928
Lag 100                                -0.57945611
Lag 500                                -0.19062716
Lag 1000                               -0.08668794
Lag 5000                               0.01749515

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0                                   -0.73143141
Lag 100                                -0.54471632
Lag 500                                -0.20876950
Lag 1000                               -0.08873424
Lag 5000                               0.01590906

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal
Lag 0                                   -0.73143141
Lag 100                                -0.54471632
Lag 500                                -0.20876950
Lag 1000                               -0.08873424
Lag 5000                               0.01590906

      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal
Lag 0                                   -0.463783799
Lag 100                                -0.353636210
Lag 500                                -0.169902631
Lag 1000                               -0.092155921
```

---

|                                                                      |              |              |               |               |
|----------------------------------------------------------------------|--------------|--------------|---------------|---------------|
| Lag 5000                                                             | -0.009813848 |              |               |               |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |              |              |               |               |
| Lag 0                                                                | 0.048426202  |              |               |               |
| Lag 100                                                              | 0.082944312  |              |               |               |
| Lag 500                                                              | 0.033604930  |              |               |               |
| Lag 1000                                                             | 0.005353498  |              |               |               |
| Lag 5000                                                             | 0.014843101  |              |               |               |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |              |              |               |               |
| Lag 0                                                                | 0.04443896   |              |               |               |
| Lag 100                                                              | 0.05903744   |              |               |               |
| Lag 500                                                              | 0.02909782   |              |               |               |
| Lag 1000                                                             | 0.01112292   |              |               |               |
| Lag 5000                                                             | -0.02822596  |              |               |               |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |              |              |               |               |
| Lag 0                                                                | 0.04443896   |              |               |               |
| Lag 100                                                              | 0.05903744   |              |               |               |
| Lag 500                                                              | 0.02909782   |              |               |               |
| Lag 1000                                                             | 0.01112292   |              |               |               |
| Lag 5000                                                             | -0.02822596  |              |               |               |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |              |              |               |               |
| Lag 0                                                                | 0.04061110   |              |               |               |
| Lag 100                                                              | 0.04147746   |              |               |               |
| Lag 500                                                              | 0.01338370   |              |               |               |
| Lag 1000                                                             | 0.01709317   |              |               |               |
| Lag 5000                                                             | -0.05401662  |              |               |               |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |              |              |               |               |
| Lag 0                                                                | 0.005983125  | -0.103356615 | 0.0938981727  | -0.0428373611 |
| Lag 100                                                              | 0.042918382  | -0.047853012 | 0.0762514968  | -0.0699317580 |
| Lag 500                                                              | -0.002564868 | -0.017832691 | 0.0157985450  | 0.0235550018  |
| Lag 1000                                                             | 0.014867426  | -0.027311021 | 0.0003829822  | -0.0006460503 |
| Lag 5000                                                             | 0.009101393  | 0.004430949  | -0.0233572527 | 0.0174497074  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |              |              |               |               |
| Lag 0                                                                | 1.000000000  |              |               |               |

---

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Lag 100                                                             | 0.503164974  |
| Lag 500                                                             | 0.176766919  |
| Lag 1000                                                            | 0.100125592  |
| Lag 5000                                                            | 0.003858174  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |              |
| Lag 0                                                               | 0.862131977  |
| Lag 100                                                             | 0.467009268  |
| Lag 500                                                             | 0.173071923  |
| Lag 1000                                                            | 0.091405415  |
| Lag 5000                                                            | 0.001214249  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |              |
| Lag 0                                                               | 0.862131977  |
| Lag 100                                                             | 0.467009268  |
| Lag 500                                                             | 0.173071923  |
| Lag 1000                                                            | 0.091405415  |
| Lag 5000                                                            | 0.001214249  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |              |
| Lag 0                                                               | 0.556754885  |
| Lag 100                                                             | 0.300025056  |
| Lag 500                                                             | 0.130914950  |
| Lag 1000                                                            | 0.084742808  |
| Lag 5000                                                            | 0.008877657  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |              |
| Lag 0                                                               | -0.066893803 |
| Lag 100                                                             | -0.088991169 |
| Lag 500                                                             | -0.041740101 |
| Lag 1000                                                            | -0.013627370 |
| Lag 5000                                                            | -0.004864431 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |              |
| Lag 0                                                               | -0.041695516 |
| Lag 100                                                             | -0.058375215 |
| Lag 500                                                             | -0.038590340 |
| Lag 1000                                                            | -0.008465406 |

---

|                                                                      |              |
|----------------------------------------------------------------------|--------------|
| Lag 5000                                                             | 0.033653455  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units     |              |
| Lag 0                                                                | -0.041695516 |
| Lag 100                                                              | -0.058375215 |
| Lag 500                                                              | -0.038590340 |
| Lag 1000                                                             | -0.008465406 |
| Lag 5000                                                             | 0.033653455  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units  |              |
| Lag 0                                                                | -0.020866307 |
| Lag 100                                                              | -0.024784750 |
| Lag 500                                                              | -0.024026474 |
| Lag 1000                                                             | -0.008875445 |
| Lag 5000                                                             | 0.060413047  |
| , , at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units |              |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal       |              |
| Lag 0                                                                | -0.70890155  |
| Lag 100                                                              | -0.51875562  |
| Lag 500                                                              | -0.20600428  |
| Lag 1000                                                             | -0.10994315  |
| Lag 5000                                                             | -0.01039121  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal    |              |
| Lag 0                                                                | -0.82495927  |
| Lag 100                                                              | -0.61316330  |
| Lag 500                                                              | -0.24079957  |
| Lag 1000                                                             | -0.11403500  |
| Lag 5000                                                             | -0.01904085  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    |              |
| Lag 0                                                                | -0.82495927  |
| Lag 100                                                              | -0.61316330  |
| Lag 500                                                              | -0.24079957  |
| Lag 1000                                                             | -0.11403500  |

---

```
Lag 5000                                -0.01904085
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal
Lag 0                                   -0.65832516
Lag 100                                -0.49644020
Lag 500                                -0.22156354
Lag 1000                               -0.11606788
Lag 5000                               -0.03431864
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal
Lag 0                                   0.0432075717
Lag 100                                0.0731729081
Lag 500                                0.0351321242
Lag 1000                               0.0008078044
Lag 5000                               0.0060255376
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal
Lag 0                                   0.04900783
Lag 100                                0.06034422
Lag 500                                0.03147880
Lag 1000                               0.01663910
Lag 5000                               -0.02708842
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal
Lag 0                                   0.04900783
Lag 100                                0.06034422
Lag 500                                0.03147880
Lag 1000                               0.01663910
Lag 5000                               -0.02708842
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal
Lag 0                                   0.05879880
Lag 100                                0.06066830
Lag 500                                0.02143188
Lag 1000                               0.02664768
Lag 5000                               -0.05794619
      traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother
Lag 0                0.05280676      -0.097464115      0.11659923      -0.080407800
```

---



---

|                                                                     |             |              |             |              |
|---------------------------------------------------------------------|-------------|--------------|-------------|--------------|
| Lag 100                                                             | 0.06749027  | -0.048884392 | 0.05889480  | -0.042771302 |
| Lag 500                                                             | 0.01581190  | -0.013402510 | 0.01353140  | 0.023637489  |
| Lag 1000                                                            | 0.02427770  | -0.012851631 | 0.01997733  | 0.008272035  |
| Lag 5000                                                            | -0.00118757 | 0.004909443  | -0.02042686 | 0.001970307  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       |             |              |             |              |
| Lag 0                                                               |             |              |             | 0.8621320    |
| Lag 100                                                             |             |              |             | 0.4520274    |
| Lag 500                                                             |             |              |             | 0.1932988    |
| Lag 1000                                                            |             |              |             | 0.1091092    |
| Lag 5000                                                            |             |              |             | 0.0303842    |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |             |              |             |              |
| Lag 0                                                               |             |              |             | 1.00000000   |
| Lag 100                                                             |             |              |             | 0.52253994   |
| Lag 500                                                             |             |              |             | 0.19894517   |
| Lag 1000                                                            |             |              |             | 0.10703411   |
| Lag 5000                                                            |             |              |             | 0.03011013   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |             |              |             |              |
| Lag 0                                                               |             |              |             | 1.00000000   |
| Lag 100                                                             |             |              |             | 0.52253994   |
| Lag 500                                                             |             |              |             | 0.19894517   |
| Lag 1000                                                            |             |              |             | 0.10703411   |
| Lag 5000                                                            |             |              |             | 0.03011013   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |             |              |             |              |
| Lag 0                                                               |             |              |             | 0.83204286   |
| Lag 100                                                             |             |              |             | 0.40931694   |
| Lag 500                                                             |             |              |             | 0.16394862   |
| Lag 1000                                                            |             |              |             | 0.09392225   |
| Lag 5000                                                            |             |              |             | 0.02432038   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |             |              |             |              |
| Lag 0                                                               |             |              |             | -0.066443086 |
| Lag 100                                                             |             |              |             | -0.077647262 |
| Lag 500                                                             |             |              |             | -0.036441782 |
| Lag 1000                                                            |             |              |             | -0.007304102 |

---

```
Lag 5000                                -0.002379788
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units
Lag 0                                    -0.05557720
Lag 100                                 -0.05970375
Lag 500                                 -0.03617882
Lag 1000                                -0.01614682
Lag 5000                                0.03234714
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
Lag 0                                    -0.05557720
Lag 100                                 -0.05970375
Lag 500                                 -0.03617882
Lag 1000                                -0.01614682
Lag 5000                                0.03234714
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0                                    -0.04717349
Lag 100                                 -0.04356349
Lag 500                                 -0.02677549
Lag 1000                                -0.02460366
Lag 5000                                0.06745537

, , at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0                                    -0.70890155
Lag 100                                 -0.51875562
Lag 500                                 -0.20600428
Lag 1000                                -0.10994315
Lag 5000                                -0.01039121
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0                                    -0.82495927
Lag 100                                 -0.61316330
Lag 500                                 -0.24079957
Lag 1000                                -0.11403500
```

---

|                                                                      |              |
|----------------------------------------------------------------------|--------------|
| Lag 5000                                                             | -0.01904085  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    |              |
| Lag 0                                                                | -0.82495927  |
| Lag 100                                                              | -0.61316330  |
| Lag 500                                                              | -0.24079957  |
| Lag 1000                                                             | -0.11403500  |
| Lag 5000                                                             | -0.01904085  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |              |
| Lag 0                                                                | -0.65832516  |
| Lag 100                                                              | -0.49644020  |
| Lag 500                                                              | -0.22156354  |
| Lag 1000                                                             | -0.11606788  |
| Lag 5000                                                             | -0.03431864  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |              |
| Lag 0                                                                | 0.0432075717 |
| Lag 100                                                              | 0.0731729081 |
| Lag 500                                                              | 0.0351321242 |
| Lag 1000                                                             | 0.0008078044 |
| Lag 5000                                                             | 0.0060255376 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |              |
| Lag 0                                                                | 0.04900783   |
| Lag 100                                                              | 0.06034422   |
| Lag 500                                                              | 0.03147880   |
| Lag 1000                                                             | 0.01663910   |
| Lag 5000                                                             | -0.02708842  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |              |
| Lag 0                                                                | 0.04900783   |
| Lag 100                                                              | 0.06034422   |
| Lag 500                                                              | 0.03147880   |
| Lag 1000                                                             | 0.01663910   |
| Lag 5000                                                             | -0.02708842  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |              |
| Lag 0                                                                | 0.05879880   |

---

|                                                                     |             |              |             |              |
|---------------------------------------------------------------------|-------------|--------------|-------------|--------------|
| Lag 100                                                             |             |              |             | 0.06066830   |
| Lag 500                                                             |             |              |             | 0.02143188   |
| Lag 1000                                                            |             |              |             | 0.02664768   |
| Lag 5000                                                            |             |              |             | -0.05794619  |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother |             |              |             |              |
| Lag 0                                                               | 0.05280676  | -0.097464115 | 0.11659923  | -0.080407800 |
| Lag 100                                                             | 0.06749027  | -0.048884392 | 0.05889480  | -0.042771302 |
| Lag 500                                                             | 0.01581190  | -0.013402510 | 0.01353140  | 0.023637489  |
| Lag 1000                                                            | 0.02427770  | -0.012851631 | 0.01997733  | 0.008272035  |
| Lag 5000                                                            | -0.00118757 | 0.004909443  | -0.02042686 | 0.001970307  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       |             |              |             |              |
| Lag 0                                                               |             |              |             | 0.8621320    |
| Lag 100                                                             |             |              |             | 0.4520274    |
| Lag 500                                                             |             |              |             | 0.1932988    |
| Lag 1000                                                            |             |              |             | 0.1091092    |
| Lag 5000                                                            |             |              |             | 0.0303842    |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |             |              |             |              |
| Lag 0                                                               |             |              |             | 1.00000000   |
| Lag 100                                                             |             |              |             | 0.52253994   |
| Lag 500                                                             |             |              |             | 0.19894517   |
| Lag 1000                                                            |             |              |             | 0.10703411   |
| Lag 5000                                                            |             |              |             | 0.03011013   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |             |              |             |              |
| Lag 0                                                               |             |              |             | 1.00000000   |
| Lag 100                                                             |             |              |             | 0.52253994   |
| Lag 500                                                             |             |              |             | 0.19894517   |
| Lag 1000                                                            |             |              |             | 0.10703411   |
| Lag 5000                                                            |             |              |             | 0.03011013   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |             |              |             |              |
| Lag 0                                                               |             |              |             | 0.83204286   |
| Lag 100                                                             |             |              |             | 0.40931694   |
| Lag 500                                                             |             |              |             | 0.16394862   |
| Lag 1000                                                            |             |              |             | 0.09392225   |

---

|                                                                         |              |
|-------------------------------------------------------------------------|--------------|
| Lag 5000                                                                | 0.02432038   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units           |              |
| Lag 0                                                                   | -0.066443086 |
| Lag 100                                                                 | -0.077647262 |
| Lag 500                                                                 | -0.036441782 |
| Lag 1000                                                                | -0.007304102 |
| Lag 5000                                                                | -0.002379788 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units        |              |
| Lag 0                                                                   | -0.05557720  |
| Lag 100                                                                 | -0.05970375  |
| Lag 500                                                                 | -0.03617882  |
| Lag 1000                                                                | -0.01614682  |
| Lag 5000                                                                | 0.03234714   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units        |              |
| Lag 0                                                                   | -0.05557720  |
| Lag 100                                                                 | -0.05970375  |
| Lag 500                                                                 | -0.03617882  |
| Lag 1000                                                                | -0.01614682  |
| Lag 5000                                                                | 0.03234714   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units     |              |
| Lag 0                                                                   | -0.04717349  |
| Lag 100                                                                 | -0.04356349  |
| Lag 500                                                                 | -0.02677549  |
| Lag 1000                                                                | -0.02460366  |
| Lag 5000                                                                | 0.06745537   |
| , , at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |              |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal          |              |
| Lag 0                                                                   | -0.43812320  |
| Lag 100                                                                 | -0.32924526  |
| Lag 500                                                                 | -0.13020212  |
| Lag 1000                                                                | -0.13254862  |

---

```
Lag 5000 -0.07309476
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0 -0.64394327
Lag 100 -0.46780888
Lag 500 -0.20079012
Lag 1000 -0.14480996
Lag 5000 -0.09575862
      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal
Lag 0 -0.64394327
Lag 100 -0.46780888
Lag 500 -0.20079012
Lag 1000 -0.14480996
Lag 5000 -0.09575862
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal
Lag 0 -0.76001059
Lag 100 -0.54663983
Lag 500 -0.24713058
Lag 1000 -0.16089016
Lag 5000 -0.08907569
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal
Lag 0 0.01554818
Lag 100 0.03347226
Lag 500 0.03054674
Lag 1000 0.01335762
Lag 5000 0.01644186
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal
Lag 0 0.0247708510
Lag 100 0.0298811053
Lag 500 0.0393319911
Lag 1000 0.0410794684
Lag 5000 0.0003106287
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal
Lag 0 0.0247708510
```

---

|                                                                      |              |               |              |              |
|----------------------------------------------------------------------|--------------|---------------|--------------|--------------|
| Lag 100                                                              |              |               |              | 0.0298811053 |
| Lag 500                                                              |              |               |              | 0.0393319911 |
| Lag 1000                                                             |              |               |              | 0.0410794684 |
| Lag 5000                                                             |              |               |              | 0.0003106287 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |              |               |              |              |
| Lag 0                                                                |              |               |              | 0.04254878   |
| Lag 100                                                              |              |               |              | 0.04748420   |
| Lag 500                                                              |              |               |              | 0.04789053   |
| Lag 1000                                                             |              |               |              | 0.06796255   |
| Lag 5000                                                             |              |               |              | -0.01851941  |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |              |               |              |              |
| Lag 0                                                                | 0.072105699  | -0.1083934806 | 0.1007331132 | -0.090750879 |
| Lag 100                                                              | 0.069867202  | -0.0486077910 | 0.0545104394 | -0.013859645 |
| Lag 500                                                              | 0.026575267  | -0.0133212485 | 0.0286661585 | 0.008613454  |
| Lag 1000                                                             | 0.015663649  | -0.0195256514 | 0.0413886079 | 0.020780006  |
| Lag 5000                                                             | -0.002018726 | -0.0001505633 | 0.0007733641 | -0.036366883 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |              |               |              |              |
| Lag 0                                                                |              |               |              | 0.55675488   |
| Lag 100                                                              |              |               |              | 0.28216537   |
| Lag 500                                                              |              |               |              | 0.12989226   |
| Lag 1000                                                             |              |               |              | 0.12353001   |
| Lag 5000                                                             |              |               |              | 0.07787577   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units     |              |               |              |              |
| Lag 0                                                                |              |               |              | 0.83204286   |
| Lag 100                                                              |              |               |              | 0.39760050   |
| Lag 500                                                              |              |               |              | 0.17265831   |
| Lag 1000                                                             |              |               |              | 0.12489795   |
| Lag 5000                                                             |              |               |              | 0.08926567   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units     |              |               |              |              |
| Lag 0                                                                |              |               |              | 0.83204286   |
| Lag 100                                                              |              |               |              | 0.39760050   |
| Lag 500                                                              |              |               |              | 0.17265831   |
| Lag 1000                                                             |              |               |              | 0.12489795   |

---

```
Lag 5000                                0.08926567
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units
Lag 0                                  1.00000000
Lag 100                               0.43985525
Lag 500                               0.18705457
Lag 1000                              0.11273003
Lag 5000                              0.07297707
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units
Lag 0                                 -0.03083664
Lag 100                               -0.03304267
Lag 500                               -0.03123886
Lag 1000                              -0.02337670
Lag 5000                              -0.02222725
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units
Lag 0                                 -0.0304234955
Lag 100                               -0.0241051816
Lag 500                               -0.0412351901
Lag 1000                              -0.0471997761
Lag 5000                              0.0005985541
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
Lag 0                                 -0.0304234955
Lag 100                               -0.0241051816
Lag 500                               -0.0412351901
Lag 1000                              -0.0471997761
Lag 5000                              0.0005985541
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0                                 -0.03748483
Lag 100                               -0.02874080
Lag 500                               -0.04666191
Lag 1000                              -0.07162962
Lag 5000                              0.02693394

, , at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units
```



---

|          |                                                                      |
|----------|----------------------------------------------------------------------|
|          | at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal       |
| Lag 0    | 0.07807105                                                           |
| Lag 100  | 0.04809322                                                           |
| Lag 500  | 0.07342298                                                           |
| Lag 1000 | 0.02747559                                                           |
| Lag 5000 | -0.08865706                                                          |
|          | at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal    |
| Lag 0    | 0.082481767                                                          |
| Lag 100  | 0.069884466                                                          |
| Lag 500  | 0.053669242                                                          |
| Lag 1000 | -0.003247808                                                         |
| Lag 5000 | -0.105675290                                                         |
|          | at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    |
| Lag 0    | 0.082481767                                                          |
| Lag 100  | 0.069884466                                                          |
| Lag 500  | 0.053669242                                                          |
| Lag 1000 | -0.003247808                                                         |
| Lag 5000 | -0.105675290                                                         |
|          | at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |
| Lag 0    | 0.0287090438                                                         |
| Lag 100  | 0.0207457037                                                         |
| Lag 500  | 0.0006081912                                                         |
| Lag 1000 | -0.0572950737                                                        |
| Lag 5000 | -0.1156100914                                                        |
|          | at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |
| Lag 0    | -0.9369474                                                           |
| Lag 100  | -0.8074872                                                           |
| Lag 500  | -0.5046449                                                           |
| Lag 1000 | -0.2917207                                                           |
| Lag 5000 | 0.1083680                                                            |
|          | at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |
| Lag 0    | -0.8748666                                                           |

---

|                                                                      |             |             |             |             |              |
|----------------------------------------------------------------------|-------------|-------------|-------------|-------------|--------------|
| Lag 100                                                              |             |             |             |             | -0.8019088   |
| Lag 500                                                              |             |             |             |             | -0.5689707   |
| Lag 1000                                                             |             |             |             |             | -0.3490005   |
| Lag 5000                                                             |             |             |             |             | 0.1337402    |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |             |             |             |             |              |
| Lag 0                                                                |             |             |             |             | -0.8748666   |
| Lag 100                                                              |             |             |             |             | -0.8019088   |
| Lag 500                                                              |             |             |             |             | -0.5689707   |
| Lag 1000                                                             |             |             |             |             | -0.3490005   |
| Lag 5000                                                             |             |             |             |             | 0.1337402    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |             |             |             |             |              |
| Lag 0                                                                |             |             |             |             | -0.7532567   |
| Lag 100                                                              |             |             |             |             | -0.7106727   |
| Lag 500                                                              |             |             |             |             | -0.5497772   |
| Lag 1000                                                             |             |             |             |             | -0.3502307   |
| Lag 5000                                                             |             |             |             |             | 0.1338020    |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |             |             |             |             |              |
| Lag 0                                                                | 0.024722609 | -0.02576722 | -0.02385685 | 0.14299876  |              |
| Lag 100                                                              | 0.017186912 | -0.04974888 | -0.00291607 | 0.12584167  |              |
| Lag 500                                                              | 0.027788923 | -0.07216401 | -0.03437633 | 0.12502432  |              |
| Lag 1000                                                             | 0.030599400 | -0.05970350 | -0.07019603 | 0.09866987  |              |
| Lag 5000                                                             | 0.003688852 | -0.03623879 | -0.01956357 | -0.02929408 |              |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |             |             |             |             |              |
| Lag 0                                                                |             |             |             |             | -0.066893803 |
| Lag 100                                                              |             |             |             |             | -0.023868973 |
| Lag 500                                                              |             |             |             |             | -0.055699109 |
| Lag 1000                                                             |             |             |             |             | -0.006728342 |
| Lag 5000                                                             |             |             |             |             | 0.089063097  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units     |             |             |             |             |              |
| Lag 0                                                                |             |             |             |             | -0.06644309  |
| Lag 100                                                              |             |             |             |             | -0.03252225  |
| Lag 500                                                              |             |             |             |             | -0.03013542  |
| Lag 1000                                                             |             |             |             |             | 0.00413424   |

---

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Lag 5000                                                            | 0.09523719   |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |              |
| Lag 0                                                               | -0.06644309  |
| Lag 100                                                             | -0.03252225  |
| Lag 500                                                             | -0.03013542  |
| Lag 1000                                                            | 0.00413424   |
| Lag 5000                                                            | 0.09523719   |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |              |
| Lag 0                                                               | -0.030836644 |
| Lag 100                                                             | -0.012261003 |
| Lag 500                                                             | -0.009828189 |
| Lag 1000                                                            | 0.015985541  |
| Lag 5000                                                            | 0.120961240  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |              |
| Lag 0                                                               | 1.0000000    |
| Lag 100                                                             | 0.7895424    |
| Lag 500                                                             | 0.4966907    |
| Lag 1000                                                            | 0.2824837    |
| Lag 5000                                                            | -0.1086586   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |              |
| Lag 0                                                               | 0.9162648    |
| Lag 100                                                             | 0.7897342    |
| Lag 500                                                             | 0.5546655    |
| Lag 1000                                                            | 0.3340592    |
| Lag 5000                                                            | -0.1357395   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    |              |
| Lag 0                                                               | 0.9162648    |
| Lag 100                                                             | 0.7897342    |
| Lag 500                                                             | 0.5546655    |
| Lag 1000                                                            | 0.3340592    |
| Lag 5000                                                            | -0.1357395   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units |              |
| Lag 0                                                               | 0.7818908    |

---

|          |            |
|----------|------------|
| Lag 100  | 0.7088815  |
| Lag 500  | 0.5356249  |
| Lag 1000 | 0.3410568  |
| Lag 5000 | -0.1407239 |

, , at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units

at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal

|          |              |
|----------|--------------|
| Lag 0    | 0.066793402  |
| Lag 100  | 0.050378446  |
| Lag 500  | 0.077515159  |
| Lag 1000 | 0.008812794  |
| Lag 5000 | -0.121465553 |

at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal

|          |              |
|----------|--------------|
| Lag 0    | 0.093083697  |
| Lag 100  | 0.084384646  |
| Lag 500  | 0.074812950  |
| Lag 1000 | -0.007598499 |
| Lag 5000 | -0.145198558 |

at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal

|          |              |
|----------|--------------|
| Lag 0    | 0.093083697  |
| Lag 100  | 0.084384646  |
| Lag 500  | 0.074812950  |
| Lag 1000 | -0.007598499 |
| Lag 5000 | -0.145198558 |

at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal

|          |              |
|----------|--------------|
| Lag 0    | 0.039633904  |
| Lag 100  | 0.030631751  |
| Lag 500  | 0.008216661  |
| Lag 1000 | -0.067574178 |
| Lag 5000 | -0.158681853 |

at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal

|       |            |
|-------|------------|
| Lag 0 | -0.8725969 |
|-------|------------|

---

|                                                                      |             |             |             |             |
|----------------------------------------------------------------------|-------------|-------------|-------------|-------------|
| Lag 100                                                              | -0.7772224  |             |             |             |
| Lag 500                                                              | -0.5293889  |             |             |             |
| Lag 1000                                                             | -0.3047292  |             |             |             |
| Lag 5000                                                             | 0.1032909   |             |             |             |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |             |             |             |             |
| Lag 0                                                                | -0.9629330  |             |             |             |
| Lag 100                                                              | -0.8857980  |             |             |             |
| Lag 500                                                              | -0.6314814  |             |             |             |
| Lag 1000                                                             | -0.3830396  |             |             |             |
| Lag 5000                                                             | 0.1372333   |             |             |             |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |             |             |             |             |
| Lag 0                                                                | -0.9629330  |             |             |             |
| Lag 100                                                              | -0.8857980  |             |             |             |
| Lag 500                                                              | -0.6314814  |             |             |             |
| Lag 1000                                                             | -0.3830396  |             |             |             |
| Lag 5000                                                             | 0.1372333   |             |             |             |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |             |             |             |             |
| Lag 0                                                                | -0.9140715  |             |             |             |
| Lag 100                                                              | -0.8566927  |             |             |             |
| Lag 500                                                              | -0.6364708  |             |             |             |
| Lag 1000                                                             | -0.3975119  |             |             |             |
| Lag 5000                                                             | 0.1452190   |             |             |             |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |             |             |             |             |
| Lag 0                                                                | 0.03515002  | -0.04917083 | -0.03778492 | 0.16291072  |
| Lag 100                                                              | 0.03091121  | -0.05304287 | -0.03754739 | 0.16477625  |
| Lag 500                                                              | 0.02594763  | -0.04411058 | -0.04091176 | 0.15960207  |
| Lag 1000                                                             | 0.03943697  | -0.05730659 | -0.07505811 | 0.12634662  |
| Lag 5000                                                             | -0.02494957 | -0.03580137 | -0.02083442 | -0.01955372 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units        |             |             |             |             |
| Lag 0                                                                | -0.04169552 |             |             |             |
| Lag 100                                                              | -0.02400995 |             |             |             |
| Lag 500                                                              | -0.05483846 |             |             |             |
| Lag 1000                                                             | 0.01413904  |             |             |             |

---

```
Lag 5000                                0.12137475
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units
Lag 0                                   -0.05557720
Lag 100                                -0.04292481
Lag 500                                -0.04373833
Lag 1000                               0.01820822
Lag 5000                               0.13061028
      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units
Lag 0                                   -0.05557720
Lag 100                                -0.04292481
Lag 500                                -0.04373833
Lag 1000                               0.01820822
Lag 5000                               0.13061028
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units
Lag 0                                   -0.03042350
Lag 100                                -0.02364770
Lag 500                                -0.02090983
Lag 1000                               0.03823823
Lag 5000                               0.15598999
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units
Lag 0                                   0.9162648
Lag 100                                0.7742948
Lag 500                                0.5235576
Lag 1000                               0.3032634
Lag 5000                               -0.1115658
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units
Lag 0                                   1.0000000
Lag 100                                0.8752177
Lag 500                                0.6156893
Lag 1000                               0.3715418
Lag 5000                               -0.1435973
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
Lag 0                                   1.0000000
```

---

|                                                                      |              |
|----------------------------------------------------------------------|--------------|
| Lag 100                                                              | 0.8752177    |
| Lag 500                                                              | 0.6156893    |
| Lag 1000                                                             | 0.3715418    |
| Lag 5000                                                             | -0.1435973   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units  |              |
| Lag 0                                                                | 0.9475981    |
| Lag 100                                                              | 0.8522663    |
| Lag 500                                                              | 0.6217954    |
| Lag 1000                                                             | 0.3896947    |
| Lag 5000                                                             | -0.1540929   |
| , , at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units |              |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal       |              |
| Lag 0                                                                | 0.066793402  |
| Lag 100                                                              | 0.050378446  |
| Lag 500                                                              | 0.077515159  |
| Lag 1000                                                             | 0.008812794  |
| Lag 5000                                                             | -0.121465553 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal    |              |
| Lag 0                                                                | 0.093083697  |
| Lag 100                                                              | 0.084384646  |
| Lag 500                                                              | 0.074812950  |
| Lag 1000                                                             | -0.007598499 |
| Lag 5000                                                             | -0.145198558 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal    |              |
| Lag 0                                                                | 0.093083697  |
| Lag 100                                                              | 0.084384646  |
| Lag 500                                                              | 0.074812950  |
| Lag 1000                                                             | -0.007598499 |
| Lag 5000                                                             | -0.145198558 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |              |
| Lag 0                                                                | 0.039633904  |

---

|                                                                      |            |             |             |              |
|----------------------------------------------------------------------|------------|-------------|-------------|--------------|
| Lag 100                                                              |            |             |             | 0.030631751  |
| Lag 500                                                              |            |             |             | 0.008216661  |
| Lag 1000                                                             |            |             |             | -0.067574178 |
| Lag 5000                                                             |            |             |             | -0.158681853 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |            |             |             |              |
| Lag 0                                                                |            |             |             | -0.8725969   |
| Lag 100                                                              |            |             |             | -0.7772224   |
| Lag 500                                                              |            |             |             | -0.5293889   |
| Lag 1000                                                             |            |             |             | -0.3047292   |
| Lag 5000                                                             |            |             |             | 0.1032909    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |            |             |             |              |
| Lag 0                                                                |            |             |             | -0.9629330   |
| Lag 100                                                              |            |             |             | -0.8857980   |
| Lag 500                                                              |            |             |             | -0.6314814   |
| Lag 1000                                                             |            |             |             | -0.3830396   |
| Lag 5000                                                             |            |             |             | 0.1372333    |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |            |             |             |              |
| Lag 0                                                                |            |             |             | -0.9629330   |
| Lag 100                                                              |            |             |             | -0.8857980   |
| Lag 500                                                              |            |             |             | -0.6314814   |
| Lag 1000                                                             |            |             |             | -0.3830396   |
| Lag 5000                                                             |            |             |             | 0.1372333    |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |            |             |             |              |
| Lag 0                                                                |            |             |             | -0.9140715   |
| Lag 100                                                              |            |             |             | -0.8566927   |
| Lag 500                                                              |            |             |             | -0.6364708   |
| Lag 1000                                                             |            |             |             | -0.3975119   |
| Lag 5000                                                             |            |             |             | 0.1452190    |
| traitbwt.byear traittarsus.byear traitbwt.mother traittarsus.mother  |            |             |             |              |
| Lag 0                                                                | 0.03515002 | -0.04917083 | -0.03778492 | 0.16291072   |
| Lag 100                                                              | 0.03091121 | -0.05304287 | -0.03754739 | 0.16477625   |
| Lag 500                                                              | 0.02594763 | -0.04411058 | -0.04091176 | 0.15960207   |
| Lag 1000                                                             | 0.03943697 | -0.05730659 | -0.07505811 | 0.12634662   |



---

|                                                                     |             |             |             |             |
|---------------------------------------------------------------------|-------------|-------------|-------------|-------------|
| Lag 5000                                                            | -0.02494957 | -0.03580137 | -0.02083442 | -0.01955372 |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units       |             |             |             |             |
| Lag 0                                                               |             |             |             | -0.04169552 |
| Lag 100                                                             |             |             |             | -0.02400995 |
| Lag 500                                                             |             |             |             | -0.05483846 |
| Lag 1000                                                            |             |             |             | 0.01413904  |
| Lag 5000                                                            |             |             |             | 0.12137475  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units    |             |             |             |             |
| Lag 0                                                               |             |             |             | -0.05557720 |
| Lag 100                                                             |             |             |             | -0.04292481 |
| Lag 500                                                             |             |             |             | -0.04373833 |
| Lag 1000                                                            |             |             |             | 0.01820822  |
| Lag 5000                                                            |             |             |             | 0.13061028  |
| at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units    |             |             |             |             |
| Lag 0                                                               |             |             |             | -0.05557720 |
| Lag 100                                                             |             |             |             | -0.04292481 |
| Lag 500                                                             |             |             |             | -0.04373833 |
| Lag 1000                                                            |             |             |             | 0.01820822  |
| Lag 5000                                                            |             |             |             | 0.13061028  |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units |             |             |             |             |
| Lag 0                                                               |             |             |             | -0.03042350 |
| Lag 100                                                             |             |             |             | -0.02364770 |
| Lag 500                                                             |             |             |             | -0.02090983 |
| Lag 1000                                                            |             |             |             | 0.03823823  |
| Lag 5000                                                            |             |             |             | 0.15598999  |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units       |             |             |             |             |
| Lag 0                                                               |             |             |             | 0.9162648   |
| Lag 100                                                             |             |             |             | 0.7742948   |
| Lag 500                                                             |             |             |             | 0.5235576   |
| Lag 1000                                                            |             |             |             | 0.3032634   |
| Lag 5000                                                            |             |             |             | -0.1115658  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |             |             |             |             |
| Lag 0                                                               |             |             |             | 1.0000000   |

---

```
Lag 100                                0.8752177
Lag 500                                0.6156893
Lag 1000                               0.3715418
Lag 5000                               -0.1435973
      at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
Lag 0                                  1.0000000
Lag 100                                0.8752177
Lag 500                                0.6156893
Lag 1000                               0.3715418
Lag 5000                               -0.1435973
      at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
Lag 0                                  0.9475981
Lag 100                                0.8522663
Lag 500                                0.6217954
Lag 1000                               0.3896947
Lag 5000                               -0.1540929

, , at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units

      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
Lag 0                                  0.047978981
Lag 100                                0.045079451
Lag 500                                0.067970538
Lag 1000                               -0.006909171
Lag 5000                               -0.132371879
      at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
Lag 0                                  0.08843633
Lag 100                                0.08879180
Lag 500                                0.08319912
Lag 1000                               -0.00586568
Lag 5000                               -0.15675135
      at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal
Lag 0                                  0.08843633
```

---

|                                                                      |             |
|----------------------------------------------------------------------|-------------|
| Lag 100                                                              | 0.08879180  |
| Lag 500                                                              | 0.08319912  |
| Lag 1000                                                             | -0.00586568 |
| Lag 5000                                                             | -0.15675135 |
| at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal |             |
| Lag 0                                                                | 0.04584246  |
| Lag 100                                                              | 0.04189200  |
| Lag 500                                                              | 0.02195919  |
| Lag 1000                                                             | -0.05747491 |
| Lag 5000                                                             | -0.16717974 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal       |             |
| Lag 0                                                                | -0.75254810 |
| Lag 100                                                              | -0.68626575 |
| Lag 500                                                              | -0.50089775 |
| Lag 1000                                                             | -0.29031256 |
| Lag 5000                                                             | 0.09258748  |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal    |             |
| Lag 0                                                                | -0.9191068  |
| Lag 100                                                              | -0.8549977  |
| Lag 500                                                              | -0.6281177  |
| Lag 1000                                                             | -0.3837276  |
| Lag 5000                                                             | 0.1299735   |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal    |             |
| Lag 0                                                                | -0.9191068  |
| Lag 100                                                              | -0.8549977  |
| Lag 500                                                              | -0.6281177  |
| Lag 1000                                                             | -0.3837276  |
| Lag 5000                                                             | 0.1299735   |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal |             |
| Lag 0                                                                | -0.9650519  |
| Lag 100                                                              | -0.9020388  |
| Lag 500                                                              | -0.6603730  |
| Lag 1000                                                             | -0.4085407  |

---

|          |                                 |                                 |                 |                    |
|----------|---------------------------------|---------------------------------|-----------------|--------------------|
| Lag 5000 |                                 |                                 |                 | 0.1451290          |
|          | traitbwt.byear                  | traittarsus.byear               | traitbwt.mother | traittarsus.mother |
| Lag 0    | 0.03233249                      | -0.05392947                     | -0.04048151     | 0.17073426         |
| Lag 100  | 0.03971497                      | -0.04863212                     | -0.04603016     | 0.19100707         |
| Lag 500  | 0.02833531                      | -0.02831174                     | -0.04356222     | 0.17180604         |
| Lag 1000 | 0.03018475                      | -0.05344380                     | -0.07230573     | 0.12082866         |
| Lag 5000 | -0.04000580                     | -0.02623878                     | -0.01322038     | -0.01980137        |
|          | at.level(sex, "1"):traitbwt:    | at.level(sex, "1"):traitbwt.    | units           |                    |
| Lag 0    |                                 |                                 |                 | -0.02086631        |
| Lag 100  |                                 |                                 |                 | -0.02091564        |
| Lag 500  |                                 |                                 |                 | -0.04738890        |
| Lag 1000 |                                 |                                 |                 | 0.02308918         |
| Lag 5000 |                                 |                                 |                 | 0.12589958         |
|          | at.level(sex, "1"):traittarsus: | at.level(sex, "1"):traitbwt.    | units           |                    |
| Lag 0    |                                 |                                 |                 | -0.04717349        |
| Lag 100  |                                 |                                 |                 | -0.04819145        |
| Lag 500  |                                 |                                 |                 | -0.05369332        |
| Lag 1000 |                                 |                                 |                 | 0.02184183         |
| Lag 5000 |                                 |                                 |                 | 0.13760532         |
|          | at.level(sex, "1"):traitbwt:    | at.level(sex, "1"):traittarsus. | units           |                    |
| Lag 0    |                                 |                                 |                 | -0.04717349        |
| Lag 100  |                                 |                                 |                 | -0.04819145        |
| Lag 500  |                                 |                                 |                 | -0.05369332        |
| Lag 1000 |                                 |                                 |                 | 0.02184183         |
| Lag 5000 |                                 |                                 |                 | 0.13760532         |
|          | at.level(sex, "1"):traittarsus: | at.level(sex, "1"):traittarsus. | units           |                    |
| Lag 0    |                                 |                                 |                 | -0.03748483        |
| Lag 100  |                                 |                                 |                 | -0.03257654        |
| Lag 500  |                                 |                                 |                 | -0.03463265        |
| Lag 1000 |                                 |                                 |                 | 0.04295442         |
| Lag 5000 |                                 |                                 |                 | 0.15951258         |
|          | at.level(sex, "2"):traitbwt:    | at.level(sex, "2"):traitbwt.    | units           |                    |
| Lag 0    |                                 |                                 |                 | 0.7818908          |

---

|                                                                     |            |
|---------------------------------------------------------------------|------------|
| Lag 100                                                             | 0.6906357  |
| Lag 500                                                             | 0.4994770  |
| Lag 1000                                                            | 0.2932880  |
| Lag 5000                                                            | -0.1017144 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units    |            |
| Lag 0                                                               | 0.9475981  |
| Lag 100                                                             | 0.8462154  |
| Lag 500                                                             | 0.6150082  |
| Lag 1000                                                            | 0.3748486  |
| Lag 5000                                                            | -0.1355925 |
| at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units    |            |
| Lag 0                                                               | 0.9475981  |
| Lag 100                                                             | 0.8462154  |
| Lag 500                                                             | 0.6150082  |
| Lag 1000                                                            | 0.3748486  |
| Lag 5000                                                            | -0.1355925 |
| at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units |            |
| Lag 0                                                               | 1.0000000  |
| Lag 100                                                             | 0.8919817  |
| Lag 500                                                             | 0.6480975  |
| Lag 1000                                                            | 0.4037684  |
| Lag 5000                                                            | -0.1516651 |

As before we can obtain the raw variance component estimates and genetic correlations for the random effects:

```
posterior.mode(model2.4$VCV)
```

```

at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.animal
0.9669729
at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.animal
0.1859324
at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.animal
0.1859324
at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.animal

```

```
1.9958915
  at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.animal
0.8390295
  at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.animal
-0.1015747
  at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.animal
-0.1015747
  at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.animal
0.6444978
  traitbwt.byear
0.8339224
  traittarsus.byear
2.9032045
  traitbwt.mother
1.8371233
  traittarsus.mother
6.4835733
  at.level(sex, "1"):traitbwt:at.level(sex, "1"):traitbwt.units
2.1384066
  at.level(sex, "1"):traittarsus:at.level(sex, "1"):traitbwt.units
4.8421570
  at.level(sex, "1"):traitbwt:at.level(sex, "1"):traittarsus.units
4.8421570
  at.level(sex, "1"):traittarsus:at.level(sex, "1"):traittarsus.units
13.0119702
  at.level(sex, "2"):traitbwt:at.level(sex, "2"):traitbwt.units
2.1428481
  at.level(sex, "2"):traittarsus:at.level(sex, "2"):traitbwt.units
6.0868703
  at.level(sex, "2"):traitbwt:at.level(sex, "2"):traittarsus.units
6.0868703
  at.level(sex, "2"):traittarsus:at.level(sex, "2"):traittarsus.units
18.3836767
```

---

```
#
```

```
genetic.correlation2.4.F <- model2.4$VCV[, "at.level(sex, \"1\")":traittarsus:at.level(sex, \"1\")]
```

```
residual.correlation2.4.F <- model2.4$VCV[, "at.level(sex, \"1\")":traittarsus:at.level(sex, \"1\")]
```

```
genetic.correlation2.4.M <- model2.4$VCV[, "at.level(sex, \"2\")":traittarsus:at.level(sex, \"2\")]
```

```
residual.correlation2.4.M <- model2.4$VCV[, "at.level(sex, \"2\")":traittarsus:at.level(sex, \"2\")]
```

```
#
```

```
posterior.mode(genetic.correlation2.4.F)
```

```
var1
```

```
0.7852385
```

```
posterior.mode(residual.correlation2.4.F)
```

```
var1
```

```
0.8738073
```

```
posterior.mode(genetic.correlation2.4.M)
```

```
var1
```

```
0.9166822
```

```
posterior.mode(residual.correlation2.4.M)
```

```
var1
```

```
0.8658358
```

Evaluation of the statistical support for these sex-specific correlations is straightforward. Because we imposed no constraint on their estimation, we can evaluate the extent to which the posterior distributions overlap zero or overlap each other:

```
HPDinterval(genetic.correlation2.4.F, 0.95)
```

```
      lower      upper
var1 -0.3742433 0.963575
attr(,"Probability")
[1] 0.95
```

```
HPDinterval(genetic.correlation2.4.M, 0.95)
```

```
      lower      upper
var1 -0.3275444 0.9801895
attr(,"Probability")
[1] 0.95
```

```
HPDinterval(residual.correlation2.4.F, 0.95)
```

```
      lower      upper
var1 0.698292 0.962472
attr(,"Probability")
[1] 0.95
```

```
HPDinterval(residual.correlation2.4.M, 0.95)
```

```
      lower      upper
var1 -0.1744922 0.9796736
attr(,"Probability")
[1] 0.95
```

Here a plot to visualize the overlaps of covariances.

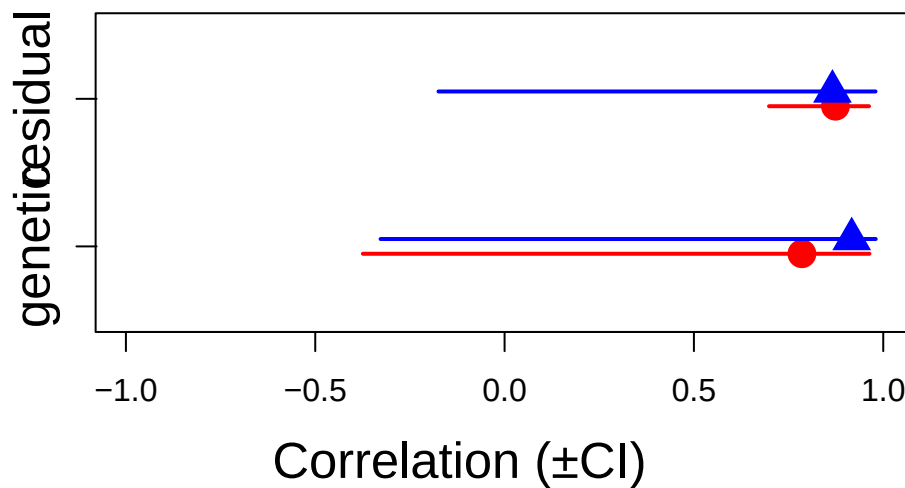
```
cor.est <- rbind(
  cbind(posterior.mode(genetic.correlation2.4.F, ), HPDinterval(genetic.correlation2.4.F, , 0.95)),
  cbind(posterior.mode(genetic.correlation2.4.M), HPDinterval(genetic.correlation2.4.M, 0.95)),
  cbind(posterior.mode(residual.correlation2.4.F, ), HPDinterval(residual.correlation2.4.F, , 0.95)),
  cbind(posterior.mode(residual.correlation2.4.M), HPDinterval(residual.correlation2.4.M, 0.95))
)
```



```

plot(c(0.95, 1.05, 1.95, 2.05) ~ cor.est[, 1], xlim = c(-1, 1), ylim = c(0.5, 2.5), xlab = "", ylab = "genetic",
arrows(y0 = 0.95, x0 = cor.est[1, 2], y1 = 0.95, x1 = cor.est[1, 3], code = 3, angle = 90, length = 10),
arrows(y0 = 1.05, x0 = cor.est[2, 2], y1 = 1.05, x1 = cor.est[2, 3], code = 3, angle = 90, length = 10),
arrows(y0 = 1.95, x0 = cor.est[3, 2], y1 = 1.95, x1 = cor.est[3, 3], code = 3, angle = 90, length = 10),
arrows(y0 = 2.05, x0 = cor.est[4, 2], y1 = 2.05, x1 = cor.est[4, 3], code = 3, angle = 90, length = 10),
mtext("Correlation ( $\pm$ CI)", side = 1, las = 1, adj = 0.4, line = 3, cex = 1.6)
axis(2, at = 1, labels = c("genetic"), las = 3, cex.axis = 1.6)
axis(2, at = 2, labels = c("residual"), las = 3, cex.axis = 1.6)

```



These posterior distributions overlap between each other, which suggested the correlation were not significantly different between sexes.

#### 6.0.4. Between groups (co)variances and the B-matrix

Animal models are amazing model. With different group within a population, it is also possible to estimate how much the different groups shared the same genetic via the cross-group genetic covariance. This covariance is essential to understand ontogenic or sexual conflict, which can constraint or enhanced response to evolution. As an example, we estimate the cross-sex genetic correlation  $r_{fm}$

First, we need to dissociate the trait values for females and males into distinct variables. Then, we use a bivariate model (for one trait: tarsus) and a multivariate model (for various traits: tarsus and bwt). With a multivariate model, the cross-sex-cross trait covariance matrix is also named B matrix.

The coding is a bit complain but pretty straightforward. It is important to modify the covariance matrix at the residual level to avoid the calculation of a cross-sex residual covariance (no individual switched sex during the experiment).

```
gryphon2$bwt.1 <- NA
gryphon2$tarsus.1 <- NA
animal <- gryphon2[gryphon2$sex == "1", ]$animal
for (i in unique(animal)) {
  gryphon2$bwt.1[which(gryphon2$animal == i)] <- gryphon2$bwt[which(gryphon2$animal == i)]
  gryphon2$tarsus.1[which(gryphon2$animal == i)] <- gryphon2$tarsus[which(gryphon2$animal == i)]
}
#
gryphon2$bwt.2 <- NA
gryphon2$tarsus.2 <- NA
animal <- gryphon2[gryphon2$sex == "2", ]$animal
for (i in unique(animal)) {
  gryphon2$bwt.2[which(gryphon2$animal == i)] <- gryphon2$bwt[which(gryphon2$animal == i)]
  gryphon2$tarsus.2[which(gryphon2$animal == i)] <- gryphon2$tarsus[which(gryphon2$animal == i)]
}
#
prior2.4 <- list(
  G = list(
    G1 = list(V = diag(2), nu = 1.002),
    G2 = list(V = diag(2), nu = 1.002),
    G3 = list(V = diag(2), nu = 1.002)
  ),
  R = list(
    V1 = list(V = diag(2), nu = 1.002)
  )
)
#
model.BivSex <- MCMCglmm(cbind(tarsus.1, tarsus.2) ~ trait - 1,
  random = ~ us(trait):animal + idh(trait):byear + idh(trait):mother,
  rcov = ~ us(trait):units,
```

---

```

family = c("gaussian", "gaussian"),
ginv = list(animal = Ainv), data = gryphon2,
nitt = 130000, thin = 100, burnin = 30000,
prior = prior2.4, verbose = FALSE, pr = TRUE
)

save(model.BivSex, file = "r-obj/MCMCglmm_model_BivSex_LongRun.rda")

```

Again we have provided the data from one such run. It can be accessed using the code:

```

load(file = "r-obj/MCMCglmm_model_BivSex_LongRun.rda")
summary(model.BivSex)

```

Iterations = 30001:129901

Thinning interval = 100

Sample size = 1000

DIC: 1670.599

G-structure: ~us(trait):animal

|                                    | post.mean | l-95% CI | u-95% CI | eff.samp |
|------------------------------------|-----------|----------|----------|----------|
| traittarsus.1:traittarsus.1.animal | 6.632     | 2.136    | 12.69    | 85.74    |
| traittarsus.2:traittarsus.1.animal | 8.043     | 2.389    | 13.54    | 117.04   |
| traittarsus.1:traittarsus.2.animal | 8.043     | 2.389    | 13.54    | 117.04   |
| traittarsus.2:traittarsus.2.animal | 16.145    | 3.128    | 28.93    | 21.81    |

~idh(trait):byear

|                     | post.mean | l-95% CI | u-95% CI | eff.samp |
|---------------------|-----------|----------|----------|----------|
| traittarsus.1.byear | 3.184     | 0.505    | 6.515    | 357.4    |
| traittarsus.2.byear | 4.576     | 1.346    | 8.476    | 442.5    |

```
~idh(trait):mother
```

|                      | post.mean | l-95% CI | u-95% CI | eff.samp |
|----------------------|-----------|----------|----------|----------|
| traittarsus.1.mother | 1.777     | 0.07858  | 4.714    | 299.68   |
| traittarsus.2.mother | 2.980     | 0.12204  | 7.328    | 70.26    |

```
R-structure: ~us(trait):units
```

|                                   | post.mean | l-95% CI | u-95% CI | eff.samp |
|-----------------------------------|-----------|----------|----------|----------|
| traittarsus.1:traittarsus.1.units | 15.455    | 8.998    | 21.84    | 104.923  |
| traittarsus.2:traittarsus.1.units | -1.497    | -15.500  | 15.53    | 8.767    |
| traittarsus.1:traittarsus.2.units | -1.497    | -15.500  | 15.53    | 8.767    |
| traittarsus.2:traittarsus.2.units | 9.356     | 0.239    | 19.10    | 21.548   |

```
Location effects: cbind(tarsus.1, tarsus.2) ~ trait - 1
```

|               | post.mean | l-95% CI | u-95% CI | eff.samp | pMCMC      |
|---------------|-----------|----------|----------|----------|------------|
| traittarsus.1 | 20.48     | 19.62    | 21.48    | 703.9    | <0.001 *** |
| traittarsus.2 | 20.46     | 19.42    | 21.40    | 846.8    | <0.001 *** |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
autocorr(model.BivSex$VCV)
```

```
, , traittarsus.1:traittarsus.1.animal
```

|          | traittarsus.1:traittarsus.1.animal | traittarsus.2:traittarsus.1.animal |
|----------|------------------------------------|------------------------------------|
| Lag 0    | 1.00000000                         | 0.48422763                         |
| Lag 100  | 0.73797990                         | 0.27701934                         |
| Lag 500  | 0.43151596                         | 0.08460564                         |
| Lag 1000 | 0.27709357                         | 0.03138071                         |
| Lag 5000 | 0.09623473                         | 0.02508175                         |

|       | traittarsus.1:traittarsus.2.animal | traittarsus.2:traittarsus.2.animal |
|-------|------------------------------------|------------------------------------|
| Lag 0 | 0.48422763                         | -0.20835012                        |

---

|          |            |             |
|----------|------------|-------------|
| Lag 100  | 0.27701934 | -0.26518087 |
| Lag 500  | 0.08460564 | -0.27929151 |
| Lag 1000 | 0.03138071 | -0.23408954 |
| Lag 5000 | 0.02508175 | -0.02431482 |

traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother

|          |            |              |              |
|----------|------------|--------------|--------------|
| Lag 0    | 0.05650214 | -0.059474225 | -0.038396947 |
| Lag 100  | 0.03805347 | -0.063250151 | -0.012779432 |
| Lag 500  | 0.03353186 | -0.031583736 | 0.059794328  |
| Lag 1000 | 0.02971111 | 0.032191172  | -0.004310584 |
| Lag 5000 | 0.04021796 | -0.001198619 | -0.065221991 |

traittarsus.2.mother traittarsus.1:traittarsus.1.units

|          |           |            |
|----------|-----------|------------|
| Lag 0    | 0.1285643 | -0.7394524 |
| Lag 100  | 0.1416089 | -0.5874113 |
| Lag 500  | 0.1227812 | -0.3892610 |
| Lag 1000 | 0.1054744 | -0.2272794 |
| Lag 5000 | 0.1046093 | -0.0490914 |

traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units

|          |              |              |
|----------|--------------|--------------|
| Lag 0    | -0.030661923 | -0.030661923 |
| Lag 100  | -0.009066847 | -0.009066847 |
| Lag 500  | -0.014021640 | -0.014021640 |
| Lag 1000 | -0.038727365 | -0.038727365 |
| Lag 5000 | -0.053406989 | -0.053406989 |

traittarsus.2:traittarsus.2.units

|          |             |
|----------|-------------|
| Lag 0    | 0.24444681  |
| Lag 100  | 0.28630940  |
| Lag 500  | 0.28676382  |
| Lag 1000 | 0.25016902  |
| Lag 5000 | -0.01889626 |

, , traittarsus.2:traittarsus.1.animal

traittarsus.1:traittarsus.1.animal traittarsus.2:traittarsus.1.animal

|       |            |            |
|-------|------------|------------|
| Lag 0 | 0.48422763 | 1.00000000 |
|-------|------------|------------|

---

|                                                                       |              |             |              |
|-----------------------------------------------------------------------|--------------|-------------|--------------|
| Lag 100                                                               | 0.25316393   | 0.68722653  |              |
| Lag 500                                                               | -0.01525426  | 0.30994286  |              |
| Lag 1000                                                              | -0.13117432  | 0.12750201  |              |
| Lag 5000                                                              | -0.03481149  | 0.01967858  |              |
| traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal |              |             |              |
| Lag 0                                                                 | 1.00000000   | 0.4372811   |              |
| Lag 100                                                               | 0.68722653   | 0.3171927   |              |
| Lag 500                                                               | 0.30994286   | 0.2173200   |              |
| Lag 1000                                                              | 0.12750201   | 0.2034517   |              |
| Lag 5000                                                              | 0.01967858   | 0.1155543   |              |
| traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother          |              |             |              |
| Lag 0                                                                 | -0.028123268 | -0.10117282 | 0.068035501  |
| Lag 100                                                               | -0.036441213 | -0.06326083 | 0.058376180  |
| Lag 500                                                               | -0.017206400 | -0.06159010 | -0.002869166 |
| Lag 1000                                                              | -0.004346341 | 0.06173108  | -0.017320527 |
| Lag 5000                                                              | -0.044798021 | 0.07734401  | -0.083350187 |
| traittarsus.2.mother traittarsus.1:traittarsus.1.units                |              |             |              |
| Lag 0                                                                 | -0.261520775 | -0.32691256 |              |
| Lag 100                                                               | -0.219989670 | -0.19037150 |              |
| Lag 500                                                               | -0.171008994 | 0.03568009  |              |
| Lag 1000                                                              | -0.073253782 | 0.15052971  |              |
| Lag 5000                                                              | -0.002628408 | 0.06147675  |              |
| traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units   |              |             |              |
| Lag 0                                                                 | -0.15561461  | -0.15561461 |              |
| Lag 100                                                               | -0.11940169  | -0.11940169 |              |
| Lag 500                                                               | -0.10722180  | -0.10722180 |              |
| Lag 1000                                                              | -0.11888700  | -0.11888700 |              |
| Lag 5000                                                              | -0.07749379  | -0.07749379 |              |
| traittarsus.2:traittarsus.2.units                                     |              |             |              |
| Lag 0                                                                 | -0.3304177   |             |              |
| Lag 100                                                               | -0.2624116   |             |              |
| Lag 500                                                               | -0.1893048   |             |              |
| Lag 1000                                                              | -0.2045441   |             |              |

---

Lag 5000 -0.1580773

, , traittarsus.1:traittarsus.2.animal

traittarsus.1:traittarsus.1.animal traittarsus.2:traittarsus.1.animal

|          |             |            |
|----------|-------------|------------|
| Lag 0    | 0.48422763  | 1.00000000 |
| Lag 100  | 0.25316393  | 0.68722653 |
| Lag 500  | -0.01525426 | 0.30994286 |
| Lag 1000 | -0.13117432 | 0.12750201 |
| Lag 5000 | -0.03481149 | 0.01967858 |

traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal

|          |            |           |
|----------|------------|-----------|
| Lag 0    | 1.00000000 | 0.4372811 |
| Lag 100  | 0.68722653 | 0.3171927 |
| Lag 500  | 0.30994286 | 0.2173200 |
| Lag 1000 | 0.12750201 | 0.2034517 |
| Lag 5000 | 0.01967858 | 0.1155543 |

traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother

|          |              |             |              |
|----------|--------------|-------------|--------------|
| Lag 0    | -0.028123268 | -0.10117282 | 0.068035501  |
| Lag 100  | -0.036441213 | -0.06326083 | 0.058376180  |
| Lag 500  | -0.017206400 | -0.06159010 | -0.002869166 |
| Lag 1000 | -0.004346341 | 0.06173108  | -0.017320527 |
| Lag 5000 | -0.044798021 | 0.07734401  | -0.083350187 |

traittarsus.2.mother traittarsus.1:traittarsus.1.units

|          |              |             |
|----------|--------------|-------------|
| Lag 0    | -0.261520775 | -0.32691256 |
| Lag 100  | -0.219989670 | -0.19037150 |
| Lag 500  | -0.171008994 | 0.03568009  |
| Lag 1000 | -0.073253782 | 0.15052971  |
| Lag 5000 | -0.002628408 | 0.06147675  |

traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.15561461 | -0.15561461 |
| Lag 100  | -0.11940169 | -0.11940169 |
| Lag 500  | -0.10722180 | -0.10722180 |
| Lag 1000 | -0.11888700 | -0.11888700 |

```
Lag 5000                -0.07749379                -0.07749379
      traittarsus.2:traittarsus.2.units
Lag 0                   -0.3304177
Lag 100                 -0.2624116
Lag 500                 -0.1893048
Lag 1000                -0.2045441
Lag 5000                -0.1580773

, , traittarsus.2:traittarsus.2.animal

      traittarsus.1:traittarsus.1.animal traittarsus.2:traittarsus.1.animal
Lag 0                   -0.2083501                0.43728107
Lag 100                 -0.2828015                0.29272735
Lag 500                 -0.3313152                0.14658444
Lag 1000                -0.3483078                0.06012732
Lag 5000                -0.1542996                -0.06471744

      traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal
Lag 0                   0.43728107                1.00000000
Lag 100                 0.29272735                0.87775419
Lag 500                 0.14658444                0.72227084
Lag 1000                0.06012732                0.61062886
Lag 5000                -0.06471744                0.08074184

      traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother
Lag 0                   -0.020084226            -0.013645105            0.007263224
Lag 100                 -0.015303798            0.015235251            0.013922907
Lag 500                 -0.009604004            -0.004276257            -0.022794079
Lag 1000                -0.036443873            0.045216146            0.038406389
Lag 5000                -0.046131641            0.012581835            -0.012187074

      traittarsus.2.mother traittarsus.1:traittarsus.1.units
Lag 0                   -0.50677816                0.23024103
Lag 100                 -0.46549011                0.26517948
Lag 500                 -0.35999881                0.31894466
Lag 1000                -0.26148886                0.30787546
```

---



---

|                                                                       |             |              |              |
|-----------------------------------------------------------------------|-------------|--------------|--------------|
| Lag 5000                                                              | -0.09601646 | 0.09998448   |              |
| traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units   |             |              |              |
| Lag 0                                                                 | 0.03215520  | 0.03215520   |              |
| Lag 100                                                               | 0.04234930  | 0.04234930   |              |
| Lag 500                                                               | 0.01485912  | 0.01485912   |              |
| Lag 1000                                                              | -0.02777646 | -0.02777646  |              |
| Lag 5000                                                              | -0.10000281 | -0.10000281  |              |
| traittarsus.2:traittarsus.2.units                                     |             |              |              |
| Lag 0                                                                 | -0.90778147 |              |              |
| Lag 100                                                               | -0.84400702 |              |              |
| Lag 500                                                               | -0.70957533 |              |              |
| Lag 1000                                                              | -0.62450980 |              |              |
| Lag 5000                                                              | -0.06262314 |              |              |
| , , traittarsus.1.byear                                               |             |              |              |
| traittarsus.1:traittarsus.1.animal traittarsus.2:traittarsus.1.animal |             |              |              |
| Lag 0                                                                 | 0.05650214  | -0.02812327  |              |
| Lag 100                                                               | 0.06068975  | -0.02687949  |              |
| Lag 500                                                               | 0.02342116  | -0.03285780  |              |
| Lag 1000                                                              | 0.03532201  | -0.03419221  |              |
| Lag 5000                                                              | 0.02786557  | -0.01481811  |              |
| traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal |             |              |              |
| Lag 0                                                                 | -0.02812327 | -0.02008423  |              |
| Lag 100                                                               | -0.02687949 | -0.02175035  |              |
| Lag 500                                                               | -0.03285780 | -0.01196604  |              |
| Lag 1000                                                              | -0.03419221 | -0.03085934  |              |
| Lag 5000                                                              | -0.01481811 | 0.01496886   |              |
| traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother          |             |              |              |
| Lag 0                                                                 | 1.00000000  | -0.008840935 | -0.034026465 |
| Lag 100                                                               | 0.23201455  | -0.040849216 | 0.006952432  |
| Lag 500                                                               | 0.10173419  | 0.018677216  | 0.021915445  |
| Lag 1000                                                              | 0.05656314  | 0.045347703  | -0.069023041 |

---

```
Lag 5000          -0.02674433          0.003879773          -0.031727649
      traittarsus.2.mother traittarsus.1:traittarsus.1.units
Lag 0              0.041524616              -0.131561587
Lag 100            0.022654125              -0.091920671
Lag 500            -0.003137523              -0.023174815
Lag 1000           0.093233105              0.017723781
Lag 5000           -0.001576843              0.003971118
      traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units
Lag 0              0.06264243              0.06264243
Lag 100            0.06240060              0.06240060
Lag 500            0.03789653              0.03789653
Lag 1000           0.03352306              0.03352306
Lag 5000           0.05768588              0.05768588
      traittarsus.2:traittarsus.2.units
Lag 0              0.013826338
Lag 100            0.015690413
Lag 500            0.020546446
Lag 1000           -0.006052117
Lag 5000           -0.020241900

, , traittarsus.2.byear

      traittarsus.1:traittarsus.1.animal traittarsus.2:traittarsus.1.animal
Lag 0              -0.059474225              -0.10117282
Lag 100            -0.029542673              -0.06571088
Lag 500            0.019127206              0.02553144
Lag 1000           -0.006675716              0.03933732
Lag 5000           -0.020363086              -0.01801919
      traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal
Lag 0              -0.10117282              -0.013645105
Lag 100            -0.06571088              -0.001743299
Lag 500            0.02553144              0.024080673
Lag 1000           0.03933732              0.020998801
```

---

---

|          |             |              |
|----------|-------------|--------------|
| Lag 5000 | -0.01801919 | -0.036422613 |
|----------|-------------|--------------|

|          |                     |                     |                      |
|----------|---------------------|---------------------|----------------------|
|          | traittarsus.1.byear | traittarsus.2.byear | traittarsus.1.mother |
| Lag 0    | -0.008840935        | 1.00000000          | 0.019871390          |
| Lag 100  | 0.003895756         | 0.15813110          | 0.000200299          |
| Lag 500  | -0.028206511        | 0.02861588          | 0.028238837          |
| Lag 1000 | -0.022327614        | 0.02577769          | 0.069995240          |
| Lag 5000 | 0.054519302         | -0.01863939         | 0.021348106          |

|          |                      |                                   |
|----------|----------------------|-----------------------------------|
|          | traittarsus.2.mother | traittarsus.1:traittarsus.1.units |
| Lag 0    | 0.02790731           | 0.04582166                        |
| Lag 100  | -0.02533496          | 0.01584625                        |
| Lag 500  | -0.04228406          | -0.01965332                       |
| Lag 1000 | -0.03776489          | -0.05592538                       |
| Lag 5000 | 0.04026220           | 0.02293994                        |

|          |                                   |                                   |
|----------|-----------------------------------|-----------------------------------|
|          | traittarsus.2:traittarsus.1.units | traittarsus.1:traittarsus.2.units |
| Lag 0    | -0.01692229                       | -0.01692229                       |
| Lag 100  | -0.03014201                       | -0.03014201                       |
| Lag 500  | -0.05685394                       | -0.05685394                       |
| Lag 1000 | -0.05775898                       | -0.05775898                       |
| Lag 5000 | -0.04474406                       | -0.04474406                       |

|          |                                   |
|----------|-----------------------------------|
|          | traittarsus.2:traittarsus.2.units |
| Lag 0    | -0.05647151                       |
| Lag 100  | -0.01120227                       |
| Lag 500  | -0.02826105                       |
| Lag 1000 | -0.01902149                       |
| Lag 5000 | 0.04653422                        |

, , traittarsus.1.mother

|          |                                    |                                    |
|----------|------------------------------------|------------------------------------|
|          | traittarsus.1:traittarsus.1.animal | traittarsus.2:traittarsus.1.animal |
| Lag 0    | -0.03839695                        | 0.068035501                        |
| Lag 100  | -0.01125448                        | 0.090872487                        |
| Lag 500  | -0.01822303                        | 0.074904451                        |
| Lag 1000 | -0.06684368                        | -0.009049799                       |

---

```
Lag 5000                                0.01882807                    -0.038704215
      traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal
Lag 0                                0.068035501                    0.007263224
Lag 100                             0.090872487                    0.009427782
Lag 500                             0.074904451                    -0.008966636
Lag 1000                           -0.009049799                    -0.062208885
Lag 5000                           -0.038704215                    -0.090802212
      traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother
Lag 0          -0.034026465            0.019871390            1.000000000
Lag 100         -0.044009625           -0.022642909            0.53848292
Lag 500          0.033149894           -0.058758981            0.05060514
Lag 1000        -0.034811076           -0.041691079           -0.01263351
Lag 5000         0.003520887           -0.007417182           -0.01561154
      traittarsus.2.mother traittarsus.1:traittarsus.1.units
Lag 0          -0.053312892                        -0.274551627
Lag 100         -0.021399368                        -0.183188795
Lag 500          0.001127949                        -0.004741630
Lag 1000         0.017008055                        0.049168722
Lag 5000        -0.013865361                        -0.003898312
      traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units
Lag 0                                0.021389916                    0.021389916
Lag 100                             0.010504830                    0.010504830
Lag 500                             0.005067944                    0.005067944
Lag 1000                            0.024425517                    0.024425517
Lag 5000                           -0.004626176                    -0.004626176
      traittarsus.2:traittarsus.2.units
Lag 0                                0.02429791
Lag 100                             0.01285473
Lag 500                             0.02017728
Lag 1000                            0.06093906
Lag 5000                            0.10720949

, , traittarsus.2.mother
```

---

|          |                                    |                     |                                    |
|----------|------------------------------------|---------------------|------------------------------------|
|          | traittarsus.1:traittarsus.1.animal |                     | traittarsus.2:traittarsus.1.animal |
| Lag 0    | 0.12856434                         |                     | -0.2615208                         |
| Lag 100  | 0.14693139                         |                     | -0.2325836                         |
| Lag 500  | 0.21342005                         |                     | -0.1621874                         |
| Lag 1000 | 0.23353394                         |                     | -0.1119149                         |
| Lag 5000 | 0.02771044                         |                     | 0.0396288                          |
|          | traittarsus.1:traittarsus.2.animal |                     | traittarsus.2:traittarsus.2.animal |
| Lag 0    | -0.2615208                         |                     | -0.50677816                        |
| Lag 100  | -0.2325836                         |                     | -0.48725853                        |
| Lag 500  | -0.1621874                         |                     | -0.40888208                        |
| Lag 1000 | -0.1119149                         |                     | -0.34699264                        |
| Lag 5000 | 0.0396288                          |                     | 0.02549716                         |
|          | traittarsus.1.byear                | traittarsus.2.byear | traittarsus.1.mother               |
| Lag 0    | 0.04152462                         | 0.027907314         | -0.05331289                        |
| Lag 100  | 0.02194445                         | 0.054336046         | -0.04284508                        |
| Lag 500  | -0.03327513                        | -0.020857444        | -0.02968159                        |
| Lag 1000 | 0.02573536                         | -0.040238713        | -0.03336616                        |
| Lag 5000 | 0.01374507                         | 0.005808512         | 0.07687084                         |
|          | traittarsus.2.mother               |                     | traittarsus.1:traittarsus.1.units  |
| Lag 0    | 1.00000000                         |                     | -0.13910701                        |
| Lag 100  | 0.70222369                         |                     | -0.13691118                        |
| Lag 500  | 0.35625516                         |                     | -0.18735521                        |
| Lag 1000 | 0.19638031                         |                     | -0.21082439                        |
| Lag 5000 | -0.04114072                        |                     | -0.04387536                        |
|          | traittarsus.2:traittarsus.1.units  |                     | traittarsus.1:traittarsus.2.units  |
| Lag 0    | 0.04614450                         |                     | 0.04614450                         |
| Lag 100  | 0.04486226                         |                     | 0.04486226                         |
| Lag 500  | 0.04022585                         |                     | 0.04022585                         |
| Lag 1000 | 0.05277963                         |                     | 0.05277963                         |
| Lag 5000 | 0.04252673                         |                     | 0.04252673                         |
|          | traittarsus.2:traittarsus.2.units  |                     |                                    |
| Lag 0    | 0.268960307                        |                     |                                    |

---

|          |              |
|----------|--------------|
| Lag 100  | 0.289158808  |
| Lag 500  | 0.345252707  |
| Lag 1000 | 0.323960441  |
| Lag 5000 | -0.002082367 |

, , traittarsus.1:traittarsus.1.units

|          | traittarsus.1:traittarsus.1.animal | traittarsus.2:traittarsus.1.animal |
|----------|------------------------------------|------------------------------------|
| Lag 0    | -0.7394524                         | -0.32691256                        |
| Lag 100  | -0.5906993                         | -0.20009588                        |
| Lag 500  | -0.3828210                         | -0.08127245                        |
| Lag 1000 | -0.2031251                         | 0.02298348                         |
| Lag 5000 | -0.0979703                         | -0.01920779                        |

|          | traittarsus.1:traittarsus.2.animal | traittarsus.2:traittarsus.2.animal |
|----------|------------------------------------|------------------------------------|
| Lag 0    | -0.32691256                        | 0.23024103                         |
| Lag 100  | -0.20009588                        | 0.28084172                         |
| Lag 500  | -0.08127245                        | 0.28272296                         |
| Lag 1000 | 0.02298348                         | 0.28283381                         |
| Lag 5000 | -0.01920779                        | 0.05177093                         |

|          | traittarsus.1.byear | traittarsus.2.byear | traittarsus.1.mother |
|----------|---------------------|---------------------|----------------------|
| Lag 0    | -0.13156159         | 0.04582166          | -0.27455163          |
| Lag 100  | -0.05719958         | 0.04337991          | -0.16825519          |
| Lag 500  | -0.06997182         | 0.07140946          | -0.07087194          |
| Lag 1000 | -0.02088140         | 0.01426241          | 0.03955072           |
| Lag 5000 | -0.01887279         | 0.04431921          | 0.06767122           |

|          | traittarsus.2.mother | traittarsus.1:traittarsus.1.units |
|----------|----------------------|-----------------------------------|
| Lag 0    | -0.1391070           | 1.00000000                        |
| Lag 100  | -0.1379410           | 0.56579808                        |
| Lag 500  | -0.1301093           | 0.35250381                        |
| Lag 1000 | -0.1253506           | 0.15585076                        |
| Lag 5000 | -0.0735286           | 0.05909602                        |

|       | traittarsus.2:traittarsus.1.units | traittarsus.1:traittarsus.2.units |
|-------|-----------------------------------|-----------------------------------|
| Lag 0 | 0.04896537                        | 0.04896537                        |

---

|                                                                       |              |              |            |
|-----------------------------------------------------------------------|--------------|--------------|------------|
| Lag 100                                                               | 0.02746645   | 0.02746645   |            |
| Lag 500                                                               | 0.03218528   | 0.03218528   |            |
| Lag 1000                                                              | 0.05596211   | 0.05596211   |            |
| Lag 5000                                                              | 0.05413531   | 0.05413531   |            |
| traittarsus.2:traittarsus.2.units                                     |              |              |            |
| Lag 0                                                                 | -0.2610614   |              |            |
| Lag 100                                                               | -0.3079509   |              |            |
| Lag 500                                                               | -0.2969894   |              |            |
| Lag 1000                                                              | -0.2847495   |              |            |
| Lag 5000                                                              | -0.0427216   |              |            |
| , , traittarsus.2:traittarsus.1.units                                 |              |              |            |
| traittarsus.1:traittarsus.1.animal traittarsus.2:traittarsus.1.animal |              |              |            |
| Lag 0                                                                 | -0.030661923 | -0.15561461  |            |
| Lag 100                                                               | 0.003884354  | -0.11920858  |            |
| Lag 500                                                               | 0.042099074  | -0.08915914  |            |
| Lag 1000                                                              | 0.042025296  | -0.07981143  |            |
| Lag 5000                                                              | -0.018911064 | -0.09774452  |            |
| traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal |              |              |            |
| Lag 0                                                                 | -0.15561461  | 0.03215520   |            |
| Lag 100                                                               | -0.11920858  | 0.04834806   |            |
| Lag 500                                                               | -0.08915914  | 0.05802195   |            |
| Lag 1000                                                              | -0.07981143  | 0.02656239   |            |
| Lag 5000                                                              | -0.09774452  | -0.04245498  |            |
| traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother          |              |              |            |
| Lag 0                                                                 | 0.06264243   | -0.016922288 | 0.02138992 |
| Lag 100                                                               | 0.06751797   | -0.004991909 | 0.02085557 |
| Lag 500                                                               | 0.06731777   | -0.030284706 | 0.05500122 |
| Lag 1000                                                              | 0.06629827   | -0.044495833 | 0.10333723 |
| Lag 5000                                                              | 0.08927471   | -0.038637660 | 0.06016204 |
| traittarsus.2.mother traittarsus.1:traittarsus.1.units                |              |              |            |
| Lag 0                                                                 | 0.04614450   | 0.048965370  |            |

---

|          |            |              |
|----------|------------|--------------|
| Lag 100  | 0.04880037 | 0.017340477  |
| Lag 500  | 0.04329157 | -0.013844496 |
| Lag 1000 | 0.06724925 | -0.039591736 |
| Lag 5000 | 0.04795409 | -0.003568665 |

traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 1.0000000 | 1.0000000 |
| Lag 100  | 0.9675526 | 0.9675526 |
| Lag 500  | 0.9029928 | 0.9029928 |
| Lag 1000 | 0.8401123 | 0.8401123 |
| Lag 5000 | 0.4663345 | 0.4663345 |

traittarsus.2:traittarsus.2.units

|          |             |
|----------|-------------|
| Lag 0    | -0.04275237 |
| Lag 100  | -0.06227874 |
| Lag 500  | -0.06672641 |
| Lag 1000 | -0.04382724 |
| Lag 5000 | 0.03699475  |

, , traittarsus.1:traittarsus.2.units

traittarsus.1:traittarsus.1.animal traittarsus.2:traittarsus.1.animal

|          |              |             |
|----------|--------------|-------------|
| Lag 0    | -0.030661923 | -0.15561461 |
| Lag 100  | 0.003884354  | -0.11920858 |
| Lag 500  | 0.042099074  | -0.08915914 |
| Lag 1000 | 0.042025296  | -0.07981143 |
| Lag 5000 | -0.018911064 | -0.09774452 |

traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.15561461 | 0.03215520  |
| Lag 100  | -0.11920858 | 0.04834806  |
| Lag 500  | -0.08915914 | 0.05802195  |
| Lag 1000 | -0.07981143 | 0.02656239  |
| Lag 5000 | -0.09774452 | -0.04245498 |

traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother

|       |            |              |            |
|-------|------------|--------------|------------|
| Lag 0 | 0.06264243 | -0.016922288 | 0.02138992 |
|-------|------------|--------------|------------|

---



---

|          |            |              |            |
|----------|------------|--------------|------------|
| Lag 100  | 0.06751797 | -0.004991909 | 0.02085557 |
| Lag 500  | 0.06731777 | -0.030284706 | 0.05500122 |
| Lag 1000 | 0.06629827 | -0.044495833 | 0.10333723 |
| Lag 5000 | 0.08927471 | -0.038637660 | 0.06016204 |

traittarsus.2.mother traittarsus.1:traittarsus.1.units

|          |            |              |
|----------|------------|--------------|
| Lag 0    | 0.04614450 | 0.048965370  |
| Lag 100  | 0.04880037 | 0.017340477  |
| Lag 500  | 0.04329157 | -0.013844496 |
| Lag 1000 | 0.06724925 | -0.039591736 |
| Lag 5000 | 0.04795409 | -0.003568665 |

traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 1.0000000 | 1.0000000 |
| Lag 100  | 0.9675526 | 0.9675526 |
| Lag 500  | 0.9029928 | 0.9029928 |
| Lag 1000 | 0.8401123 | 0.8401123 |
| Lag 5000 | 0.4663345 | 0.4663345 |

traittarsus.2:traittarsus.2.units

|          |             |
|----------|-------------|
| Lag 0    | -0.04275237 |
| Lag 100  | -0.06227874 |
| Lag 500  | -0.06672641 |
| Lag 1000 | -0.04382724 |
| Lag 5000 | 0.03699475  |

, , traittarsus.2:traittarsus.2.units

traittarsus.1:traittarsus.1.animal traittarsus.2:traittarsus.1.animal

|          |           |             |
|----------|-----------|-------------|
| Lag 0    | 0.2444468 | -0.33041768 |
| Lag 100  | 0.2953169 | -0.24524259 |
| Lag 500  | 0.3257997 | -0.12021844 |
| Lag 1000 | 0.3204576 | -0.05040170 |
| Lag 5000 | 0.1966910 | 0.07386444  |

traittarsus.1:traittarsus.2.animal traittarsus.2:traittarsus.2.animal

|       |             |            |
|-------|-------------|------------|
| Lag 0 | -0.33041768 | -0.9077815 |
|-------|-------------|------------|

---

|                                                                     |            |              |              |
|---------------------------------------------------------------------|------------|--------------|--------------|
| Lag 100                                                             |            | -0.24524259  | -0.8410651   |
| Lag 500                                                             |            | -0.12021844  | -0.6969304   |
| Lag 1000                                                            |            | -0.05040170  | -0.5893775   |
| Lag 5000                                                            |            | 0.07386444   | -0.1004775   |
| traittarsus.1.byear traittarsus.2.byear traittarsus.1.mother        |            |              |              |
| Lag 0                                                               | 0.01382634 | -0.056471508 | 0.024297909  |
| Lag 100                                                             | 0.01730830 | -0.030701856 | 0.017739447  |
| Lag 500                                                             | 0.03968933 | 0.004065694  | 0.010636299  |
| Lag 1000                                                            | 0.03341332 | -0.048898719 | -0.028816725 |
| Lag 5000                                                            | 0.05219060 | -0.013503372 | -0.005167407 |
| traittarsus.2.mother traittarsus.1:traittarsus.1.units              |            |              |              |
| Lag 0                                                               | 0.2689603  |              | -0.2610614   |
| Lag 100                                                             | 0.3013532  |              | -0.3031510   |
| Lag 500                                                             | 0.3094825  |              | -0.3179085   |
| Lag 1000                                                            | 0.2400190  |              | -0.2838200   |
| Lag 5000                                                            | 0.1233946  |              | -0.1252443   |
| traittarsus.2:traittarsus.1.units traittarsus.1:traittarsus.2.units |            |              |              |
| Lag 0                                                               |            | -0.04275237  | -0.04275237  |
| Lag 100                                                             |            | -0.05037125  | -0.05037125  |
| Lag 500                                                             |            | -0.01879502  | -0.01879502  |
| Lag 1000                                                            |            | 0.02100080   | 0.02100080   |
| Lag 5000                                                            |            | 0.10654662   | 0.10654662   |
| traittarsus.2:traittarsus.2.units                                   |            |              |              |
| Lag 0                                                               |            | 1.00000000   |              |
| Lag 100                                                             |            | 0.87986649   |              |
| Lag 500                                                             |            | 0.70214749   |              |
| Lag 1000                                                            |            | 0.61035106   |              |
| Lag 5000                                                            |            | 0.07522425   |              |

The cross-sex genetic correlation can estimate from the output of the model. For tarsus length at fledging, sexes shared a lot of genetic variance which is common for a trait with low sexual dimorphism. If the selection is antagonistic between males and females, sexes can not evolve freely from the other sexes and a sexual conflict appears.

---

```
rfm <- model.BivSex$VCV[, "traittarsus.1:traittarsus.2.animal"] / sqrt(model.BivSex$VCV[, "traittarsus.1:traittarsus.2.animal"])
posterior.mode(rfm)
```

```
var1
0.9664439
```

```
HPDinterval(rfm, 0.95)
```

```
      lower    upper
var1 0.4630817 0.992376
attr(,"Probability")
[1] 0.95
```

We can estimate directly the correlation and plot the cross-sex genetic correlation

```
DvsS <- data.frame(
  Trait = colnames(model.BivSex$Sol),
  BLUP = posterior.mode(model.BivSex$Sol),
  CI = HPDinterval((model.BivSex$Sol))
) %>%
  filter(grepl("animal", Trait))

DvsS$ID <- substr(DvsS$Trait, 22, 26)
DvsS$TRAIT <- substr(DvsS$Trait, 6, 13)
summary(factor(DvsS$TRAIT))
```

```
tarsus.1 tarsus.2
1309      1309
```

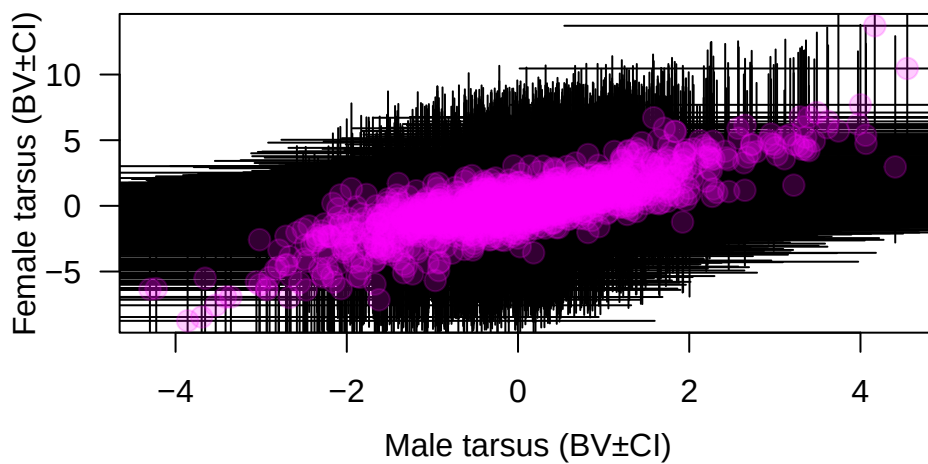
```
DvsS$Trait <- NULL
BLUPS <- reshape(DvsS, v.names = c("BLUP", "CI.lower", "CI.upper"), idvar = "ID", timevar = "TRAIT",
  nrow(BLUPS)
```

```
[1] 1309
```

```
rownames(BLUPS) <- c()
colnames(BLUPS) <- c("ID", "BLUP.btw", "CI.L.btw", "CI.U.btw", "BLUP.tarsus", "CI.L.tarsus", "CI.U.tarsus")
summary(BLUPS)
```

| ID                | BLUP.btw           | CI.L.btw         | CI.U.btw        |
|-------------------|--------------------|------------------|-----------------|
| Length:1309       | Min. : -4.299559   | Min. : -9.4393   | Min. : 0.5871   |
| Class :character  | 1st Qu.: -0.743429 | 1st Qu.: -5.2018 | 1st Qu.: 3.5625 |
| Mode :character   | Median : -0.000879 | Median : -4.3976 | Median : 4.5287 |
|                   | Mean : 0.024560    | Mean : -4.3251   | Mean : 4.4753   |
|                   | 3rd Qu.: 0.762532  | 3rd Qu.: -3.4818 | 3rd Qu.: 5.3499 |
|                   | Max. : 4.546380    | Max. : 0.5408    | Max. : 10.9441  |
| BLUP.tarsus       | CI.L.tarsus        | CI.U.tarsus      |                 |
| Min. : -8.75836   | Min. : -14.320     | Min. : -0.3279   |                 |
| 1st Qu.: -1.07251 | 1st Qu.: -8.156    | 1st Qu.: 4.9947  |                 |
| Median : 0.07087  | Median : -6.720    | Median : 6.8749  |                 |
| Mean : 0.07139    | Mean : -6.464      | Mean : 6.6127    |                 |
| 3rd Qu.: 1.17195  | 3rd Qu.: -4.869    | 3rd Qu.: 8.3021  |                 |
| Max. : 13.71503   | Max. : 1.264       | Max. : 16.7611   |                 |

```
plot(BLUP.tarsus ~ BLUP.btw, BLUPS, xlab = "", ylab = "", las = 1.2, bty = "o", col = "white")
arrows(x0 = BLUPS$BLUP.btw, y0 = BLUPS$CI.L.tarsus, x1 = BLUPS$BLUP.btw, y1 = BLUPS$CI.U.tarsus,
arrows(x0 = BLUPS$CI.L.btw, y0 = BLUPS$BLUP.tarsus, x1 = BLUPS$CI.U.btw, y1 = BLUPS$BLUP.tarsus,
points(BLUP.tarsus ~ BLUP.btw, BLUPS, pch = 16, col = rgb(1, 0, 1, 0.2), cex = 1.5)
points(BLUP.tarsus ~ BLUP.btw, BLUPS, pch = 1, col = rgb(1, 0, 1, 0.2), cex = c(1.5))
mtext("Male tarsus (BV±CI)", side = 1, line = 2.4)
mtext("Female tarsus (BV±CI)", side = 2, line = 2, las = 3)
```



#

The B matrix used the same code but in a multivariate animal model framework. Here some example code, however due to the nature of the dataset, the cross-sex genetic covariance for birth weight is hard to estimate making difficulty to fit this multivariate animal model.

```
prior2.5 <- list(
  G = list(
    G1 = list(V = diag(4), nu = 1.002),
    G2 = list(V = diag(4), nu = 1.002),
    G3 = list(V = diag(4), nu = 1.002)
  ),
  R = list(
    V1 = list(V = diag(4), nu = 1.002)
  )
)
#
model.MultivSex <- MCMCglmm(cbind(tarsus.1, bwt.1, tarsus.2, bwt.2) ~ trait - 1,
  random = ~ us(trait):animal + idh(trait):byear + idh(trait):mother,
  rcov = ~ us(trait):units,
  family = c("gaussian", "gaussian", "gaussian", "gaussian"),
  ginv = list(animal = Ainv), data = gryphon2,
  nitt = 130000, thin = 100, burnin = 30000,
  prior = prior2.5, verbose = FALSE, pr = TRUE)
```

```
)  
save(model.MultivSex, file = "r-obj/MCMCglmm_model_MultivSex_LongRun.rda")
```

Again we have provided the data from one such run. It can be accessed using the code:

```
load(file = "r-obj/MCMCglmm_model_MultivSex_LongRun.rda")  
summary(model.MultivSex)
```

Iterations = 30001:129901

Thinning interval = 100

Sample size = 1000

DIC: 2590.513

G-structure: ~us(trait):animal

|                                    | post.mean | 1-95% CI   | u-95% CI | eff.samp |
|------------------------------------|-----------|------------|----------|----------|
| traittarsus.1:traittarsus.1.animal | 5.2542    | 0.6145315  | 11.414   | 28.46    |
| traitbwt.1:traittarsus.1.animal    | 1.2682    | -0.7026690 | 3.298    | 63.97    |
| traittarsus.2:traittarsus.1.animal | 5.9945    | 0.3482679  | 13.405   | 21.59    |
| traitbwt.2:traittarsus.1.animal    | 1.4467    | -0.3853824 | 4.110    | 35.84    |
| traittarsus.1:traitbwt.1.animal    | 1.2682    | -0.7026690 | 3.298    | 63.97    |
| traitbwt.1:traitbwt.1.animal       | 1.7891    | 0.5620761  | 3.005    | 97.87    |
| traittarsus.2:traitbwt.1.animal    | 0.7719    | -1.8038433 | 4.190    | 22.64    |
| traitbwt.2:traitbwt.1.animal       | 0.9939    | 0.0009052  | 2.069    | 50.24    |
| traittarsus.1:traittarsus.2.animal | 5.9945    | 0.3482679  | 13.405   | 21.59    |
| traitbwt.1:traittarsus.2.animal    | 0.7719    | -1.8038433 | 4.190    | 22.64    |
| traittarsus.2:traittarsus.2.animal | 12.7341   | 1.4093961  | 23.690   | 28.81    |
| traitbwt.2:traittarsus.2.animal    | 2.7675    | -0.5938976 | 6.927    | 20.64    |
| traittarsus.1:traitbwt.2.animal    | 1.4467    | -0.3853824 | 4.110    | 35.84    |
| traitbwt.1:traitbwt.2.animal       | 0.9939    | 0.0009052  | 2.069    | 50.24    |
| traittarsus.2:traitbwt.2.animal    | 2.7675    | -0.5938976 | 6.927    | 20.64    |
| traitbwt.2:traitbwt.2.animal       | 1.5560    | 0.2002403  | 3.046    | 27.21    |

---

~idh(trait):byear

|                     | post.mean | l-95% CI | u-95% CI | eff.samp |
|---------------------|-----------|----------|----------|----------|
| traittarsus.1.byear | 3.3123    | 0.9318   | 6.450    | 391.3    |
| traitbwt.1.byear    | 0.6822    | 0.2403   | 1.253    | 522.1    |
| traittarsus.2.byear | 4.2198    | 1.3966   | 7.713    | 245.2    |
| traitbwt.2.byear    | 1.1743    | 0.5405   | 1.992    | 669.6    |

~idh(trait):mother

|                      | post.mean | l-95% CI | u-95% CI | eff.samp |
|----------------------|-----------|----------|----------|----------|
| traittarsus.1.mother | 4.858     | 0.5149   | 8.841    | 122.5    |
| traitbwt.1.mother    | 1.307     | 0.5752   | 2.041    | 369.0    |
| traittarsus.2.mother | 5.389     | 0.7457   | 9.557    | 140.4    |
| traitbwt.2.mother    | 2.003     | 1.2844   | 2.770    | 409.1    |

R-structure: ~us(trait):units

|                                   | post.mean | l-95% CI | u-95% CI | eff.samp |
|-----------------------------------|-----------|----------|----------|----------|
| traittarsus.1:traittarsus.1.units | 14.0783   | 8.6213   | 20.698   | 76.808   |
| traitbwt.1:traittarsus.1.units    | 4.0764    | 2.2358   | 6.357    | 47.762   |
| traittarsus.2:traittarsus.1.units | -3.6471   | -16.9514 | 14.938   | 4.746    |
| traitbwt.2:traittarsus.1.units    | -1.3655   | -6.7308  | 4.970    | 7.185    |
| traittarsus.1:traitbwt.1.units    | 4.0764    | 2.2358   | 6.357    | 47.762   |
| traitbwt.1:traitbwt.1.units       | 1.7295    | 0.7344   | 2.785    | 57.968   |
| traittarsus.2:traitbwt.1.units    | -1.1455   | -5.8008  | 4.672    | 6.837    |
| traitbwt.2:traitbwt.1.units       | -0.4245   | -2.3300  | 1.630    | 7.646    |
| traittarsus.1:traittarsus.2.units | -3.6471   | -16.9514 | 14.938   | 4.746    |
| traitbwt.1:traittarsus.2.units    | -1.1455   | -5.8008  | 4.672    | 6.837    |
| traittarsus.2:traittarsus.2.units | 10.8365   | 0.5947   | 19.576   | 26.795   |
| traitbwt.2:traittarsus.2.units    | 3.7358    | -0.1168  | 6.848    | 25.426   |
| traittarsus.1:traitbwt.2.units    | -1.3655   | -6.7308  | 4.970    | 7.185    |

```

traitbwt.1:traitbwt.2.units      -0.4245  -2.3300    1.630    7.646
traittarsus.2:traitbwt.2.units    3.7358  -0.1168    6.848   25.426
traitbwt.2:traitbwt.2.units      1.7825   0.2691    2.916   28.817

```

```
Location effects: cbind(tarsus.1, bwt.1, tarsus.2, bwt.2) ~ trait - 1
```

```

          post.mean 1-95% CI u-95% CI eff.samp  pMCMC
traittarsus.1   20.424   19.488   21.324    484.6 <0.001 ***
traitbwt.1       6.143    5.686    6.550    596.7 <0.001 ***
traittarsus.2   20.487   19.421   21.461    587.3 <0.001 ***
traitbwt.2       8.247    7.744    8.741    876.7 <0.001 ***

```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
autocorr(model.MultivSex$VCV)
```

```
, , traittarsus.1:traittarsus.1.animal
```

```

          traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal
Lag 0                      1.0000000                      0.6872795
Lag 100                    0.8646238                      0.6023267
Lag 500                    0.6217623                      0.4701016
Lag 1000                   0.4759845                      0.3306117
Lag 5000                   0.1189988                      0.1665003

```

```

          traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal
Lag 0                      0.6821642                      0.5157628
Lag 100                    0.5837197                      0.4557449
Lag 500                    0.4393984                      0.3819538
Lag 1000                   0.3626317                      0.3277211
Lag 5000                   0.1659171                      0.2490413

```

```

          traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal
Lag 0                      0.6872795                      0.2316436
Lag 100                    0.6023267                      0.1917562
Lag 500                    0.4701016                      0.1506507

```



---

|                                                                    |            |            |
|--------------------------------------------------------------------|------------|------------|
| Lag 1000                                                           | 0.3306117  | 0.1026122  |
| Lag 5000                                                           | 0.1665003  | 0.1840457  |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |            |            |
| Lag 0                                                              | 0.4371184  | 0.2363148  |
| Lag 100                                                            | 0.3790517  | 0.1969987  |
| Lag 500                                                            | 0.3316844  | 0.1943316  |
| Lag 1000                                                           | 0.2706112  | 0.1653149  |
| Lag 5000                                                           | 0.2226417  | 0.2722600  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |            |            |
| Lag 0                                                              | 0.6821642  | 0.4371184  |
| Lag 100                                                            | 0.5837197  | 0.3790517  |
| Lag 500                                                            | 0.4393984  | 0.3316844  |
| Lag 1000                                                           | 0.3626317  | 0.2706112  |
| Lag 5000                                                           | 0.1659171  | 0.2226417  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |            |            |
| Lag 0                                                              | 0.10780644 | 0.12397199 |
| Lag 100                                                            | 0.07023035 | 0.10315317 |
| Lag 500                                                            | 0.03867618 | 0.09153994 |
| Lag 1000                                                           | 0.03903023 | 0.09577409 |
| Lag 5000                                                           | 0.21464371 | 0.32154133 |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |            |            |
| Lag 0                                                              | 0.5157628  | 0.2363148  |
| Lag 100                                                            | 0.4557449  | 0.1969987  |
| Lag 500                                                            | 0.3819538  | 0.1943316  |
| Lag 1000                                                           | 0.3277211  | 0.1653149  |
| Lag 5000                                                           | 0.2490413  | 0.2722600  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |            |            |
| Lag 0                                                              | 0.12397199 | 0.07790198 |
| Lag 100                                                            | 0.10315317 | 0.07045248 |
| Lag 500                                                            | 0.09153994 | 0.07781235 |
| Lag 1000                                                           | 0.09577409 | 0.07872349 |
| Lag 5000                                                           | 0.32154133 | 0.36408510 |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |            |            |

---

|                                                                  |               |              |              |
|------------------------------------------------------------------|---------------|--------------|--------------|
| Lag 0                                                            | -0.0001991343 | 0.02533600   | 0.027261984  |
| Lag 100                                                          | 0.0100119397  | 0.03699313   | 0.023145145  |
| Lag 500                                                          | 0.0417069693  | 0.01039048   | 0.001545709  |
| Lag 1000                                                         | 0.0504304342  | 0.06238542   | -0.035646379 |
| Lag 5000                                                         | 0.0318527577  | 0.07041675   | 0.069731639  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |               |              |              |
| Lag 0                                                            | -0.01231264   | -0.150788349 | 0.10750858   |
| Lag 100                                                          | -0.02806645   | -0.118379074 | 0.11004526   |
| Lag 500                                                          | 0.01040024    | -0.006641147 | 0.07412673   |
| Lag 1000                                                         | -0.02300762   | 0.007943789  | 0.04372918   |
| Lag 5000                                                         | -0.01416454   | 0.056559933  | -0.06110098  |
| traittarsus.2.mother traitbwt.2.mother                           |               |              |              |
| Lag 0                                                            | 0.01502226    | 0.050820670  |              |
| Lag 100                                                          | 0.01411095    | 0.039228813  |              |
| Lag 500                                                          | 0.05290822    | 0.008673539  |              |
| Lag 1000                                                         | 0.06760176    | -0.017960145 |              |
| Lag 5000                                                         | 0.02399980    | -0.029552126 |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |              |              |
| Lag 0                                                            |               | -0.6964793   | -0.5806661   |
| Lag 100                                                          |               | -0.6332557   | -0.5256332   |
| Lag 500                                                          |               | -0.5006482   | -0.4211229   |
| Lag 1000                                                         |               | -0.3965468   | -0.3341455   |
| Lag 5000                                                         |               | -0.1551518   | -0.1703888   |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |               |              |              |
| Lag 0                                                            |               | 0.1834123    | 0.1843936    |
| Lag 100                                                          |               | 0.1989318    | 0.1972224    |
| Lag 500                                                          |               | 0.2183849    | 0.2244201    |
| Lag 1000                                                         |               | 0.2393546    | 0.2533883    |
| Lag 5000                                                         |               | 0.1874294    | 0.2352808    |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |               |              |              |
| Lag 0                                                            |               | -0.5806661   | -0.2972330   |
| Lag 100                                                          |               | -0.5256332   | -0.2627074   |
| Lag 500                                                          |               | -0.4211229   | -0.1975278   |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                         | -0.3341455  | -0.1864674  |
| Lag 5000                                                         | -0.1703888  | -0.1779074  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.1148525   | 0.1045202   |
| Lag 100                                                          | 0.1322561   | 0.1205853   |
| Lag 500                                                          | 0.1661494   | 0.1609281   |
| Lag 1000                                                         | 0.2078063   | 0.2097516   |
| Lag 5000                                                         | 0.2120391   | 0.2411848   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.1834123   | 0.1148525   |
| Lag 100                                                          | 0.1989318   | 0.1322561   |
| Lag 500                                                          | 0.2183849   | 0.1661494   |
| Lag 1000                                                         | 0.2393546   | 0.2078063   |
| Lag 5000                                                         | 0.1874294   | 0.2120391   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.09279221 | -0.09066377 |
| Lag 100                                                          | -0.06552157 | -0.08137057 |
| Lag 500                                                          | -0.03120806 | -0.06842888 |
| Lag 1000                                                         | -0.04688856 | -0.07652447 |
| Lag 5000                                                         | -0.22692538 | -0.31476665 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | 0.1843936   | 0.1045202   |
| Lag 100                                                          | 0.1972224   | 0.1205853   |
| Lag 500                                                          | 0.2244201   | 0.1609281   |
| Lag 1000                                                         | 0.2533883   | 0.2097516   |
| Lag 5000                                                         | 0.2352808   | 0.2411848   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |             |
| Lag 0                                                            | -0.09066377 | -0.04305989 |
| Lag 100                                                          | -0.08137057 | -0.04967401 |
| Lag 500                                                          | -0.06842888 | -0.06533316 |
| Lag 1000                                                         | -0.07652447 | -0.05918369 |
| Lag 5000                                                         | -0.31476665 | -0.34290345 |

---

, , traitbwt.1:traittarsus.1.animal

|          | traittarsus.1:traittarsus.1.animal | traitbwt.1:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.6872795                          | 1.0000000                       |
| Lag 100  | 0.5870266                          | 0.8045048                       |
| Lag 500  | 0.4080911                          | 0.4908098                       |
| Lag 1000 | 0.3227808                          | 0.3613161                       |
| Lag 5000 | 0.1778345                          | 0.1860991                       |

|          | traittarsus.2:traittarsus.1.animal | traitbwt.2:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.3863431                          | 0.5234222                       |
| Lag 100  | 0.3311212                          | 0.4545868                       |
| Lag 500  | 0.2390588                          | 0.3427345                       |
| Lag 1000 | 0.2351271                          | 0.3004179                       |
| Lag 5000 | 0.1671396                          | 0.1589172                       |

|          | traittarsus.1:traitbwt.1.animal | traitbwt.1:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | 1.0000000                       | 0.7001005                    |
| Lag 100  | 0.8045048                       | 0.5457846                    |
| Lag 500  | 0.4908098                       | 0.3331394                    |
| Lag 1000 | 0.3613161                       | 0.2645656                    |
| Lag 5000 | 0.1860991                       | 0.1481314                    |

|          | traittarsus.2:traitbwt.1.animal | traitbwt.2:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | 0.4680554                       | 0.4293517                    |
| Lag 100  | 0.3667985                       | 0.3303034                    |
| Lag 500  | 0.2878690                       | 0.2612182                    |
| Lag 1000 | 0.2588194                       | 0.2248089                    |
| Lag 5000 | 0.2468277                       | 0.2268437                    |

|          | traittarsus.1:traittarsus.2.animal | traitbwt.1:traittarsus.2.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.3863431                          | 0.4680554                       |
| Lag 100  | 0.3311212                          | 0.3667985                       |
| Lag 500  | 0.2390588                          | 0.2878690                       |
| Lag 1000 | 0.2351271                          | 0.2588194                       |
| Lag 5000 | 0.1671396                          | 0.2468277                       |

|  | traittarsus.2:traittarsus.2.animal | traitbwt.2:traittarsus.2.animal |
|--|------------------------------------|---------------------------------|
|--|------------------------------------|---------------------------------|

---

|                                                              |              |              |              |
|--------------------------------------------------------------|--------------|--------------|--------------|
| Lag 0                                                        | 0.096601163  | 0.14997594   |              |
| Lag 100                                                      | 0.063514030  | 0.11517198   |              |
| Lag 500                                                      | 0.007343656  | 0.08445112   |              |
| Lag 1000                                                     | 0.062005867  | 0.13299248   |              |
| Lag 5000                                                     | 0.138711493  | 0.22091811   |              |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal |              |              |              |
| Lag 0                                                        | 0.5234222    | 0.4293517    |              |
| Lag 100                                                      | 0.4545868    | 0.3303034    |              |
| Lag 500                                                      | 0.3427345    | 0.2612182    |              |
| Lag 1000                                                     | 0.3004179    | 0.2248089    |              |
| Lag 5000                                                     | 0.1589172    | 0.2268437    |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal |              |              |              |
| Lag 0                                                        | 0.14997594   | 0.14180302   |              |
| Lag 100                                                      | 0.11517198   | 0.11177118   |              |
| Lag 500                                                      | 0.08445112   | 0.09651159   |              |
| Lag 1000                                                     | 0.13299248   | 0.13501835   |              |
| Lag 5000                                                     | 0.22091811   | 0.25836873   |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear     |              |              |              |
| Lag 0                                                        | 0.06781216   | -0.04857154  | -0.001115860 |
| Lag 100                                                      | 0.05744185   | -0.04365919  | -0.003028122 |
| Lag 500                                                      | 0.04182580   | -0.01683610  | -0.013871766 |
| Lag 1000                                                     | 0.08409670   | 0.04075709   | -0.088936973 |
| Lag 5000                                                     | 0.07211397   | 0.04755605   | 0.061879136  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother      |              |              |              |
| Lag 0                                                        | -0.026101857 | 0.015326642  | -0.015001711 |
| Lag 100                                                      | -0.026229006 | 0.004791955  | -0.002553299 |
| Lag 500                                                      | -0.004795139 | 0.018197593  | -0.001734987 |
| Lag 1000                                                     | -0.025567829 | -0.014419690 | -0.054320480 |
| Lag 5000                                                     | -0.011401634 | -0.003782365 | -0.012255190 |
| traittarsus.2.mother traitbwt.2.mother                       |              |              |              |
| Lag 0                                                        | 0.05171415   | -0.01155255  |              |
| Lag 100                                                      | 0.04549989   | -0.01648838  |              |
| Lag 500                                                      | 0.09888411   | -0.02579001  |              |

---

|                                                                  |            |             |
|------------------------------------------------------------------|------------|-------------|
| Lag 1000                                                         | 0.07526847 | -0.03519968 |
| Lag 5000                                                         | 0.07495551 | -0.03929467 |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |            |             |
| Lag 0                                                            | -0.5374156 | -0.7822168  |
| Lag 100                                                          | -0.4631107 | -0.6561709  |
| Lag 500                                                          | -0.3115622 | -0.4297448  |
| Lag 1000                                                         | -0.2502951 | -0.3378405  |
| Lag 5000                                                         | -0.1757000 | -0.1812141  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |            |             |
| Lag 0                                                            | 0.1247101  | 0.1399193   |
| Lag 100                                                          | 0.1426212  | 0.1604774   |
| Lag 500                                                          | 0.1604094  | 0.2063094   |
| Lag 1000                                                         | 0.1703454  | 0.2249688   |
| Lag 5000                                                         | 0.1859265  | 0.1911673   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |            |             |
| Lag 0                                                            | -0.7822168 | -0.6444049  |
| Lag 100                                                          | -0.6561709 | -0.5391921  |
| Lag 500                                                          | -0.4297448 | -0.3429426  |
| Lag 1000                                                         | -0.3378405 | -0.2744658  |
| Lag 5000                                                         | -0.1812141 | -0.1591715  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |            |             |
| Lag 0                                                            | 0.1003793  | 0.1067549   |
| Lag 100                                                          | 0.1314503  | 0.1404067   |
| Lag 500                                                          | 0.1658186  | 0.1981668   |
| Lag 1000                                                         | 0.1899850  | 0.2354607   |
| Lag 5000                                                         | 0.2090708  | 0.1964048   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |            |             |
| Lag 0                                                            | 0.1247101  | 0.1003793   |
| Lag 100                                                          | 0.1426212  | 0.1314503   |
| Lag 500                                                          | 0.1604094  | 0.1658186   |
| Lag 1000                                                         | 0.1703454  | 0.1899850   |
| Lag 5000                                                         | 0.1859265  | 0.2090708   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |            |             |

---

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.09677793 | -0.12847847 |
| Lag 100  | -0.06212434 | -0.10051402 |
| Lag 500  | -0.01671120 | -0.07130994 |
| Lag 1000 | -0.07460059 | -0.11790351 |
| Lag 5000 | -0.16171357 | -0.21125583 |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.1399193 | 0.1067549 |
| Lag 100  | 0.1604774 | 0.1404067 |
| Lag 500  | 0.2063094 | 0.1981668 |
| Lag 1000 | 0.2249688 | 0.2354607 |
| Lag 5000 | 0.1911673 | 0.1964048 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.12847847 | -0.10947233 |
| Lag 100  | -0.10051402 | -0.09309607 |
| Lag 500  | -0.07130994 | -0.08545450 |
| Lag 1000 | -0.11790351 | -0.11149356 |
| Lag 5000 | -0.21125583 | -0.23203853 |

, , traittarsus.2:traittarsus.1.animal

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |             |            |
|----------|-------------|------------|
| Lag 0    | 0.682164176 | 0.38634307 |
| Lag 100  | 0.591700877 | 0.34935277 |
| Lag 500  | 0.451559860 | 0.32828629 |
| Lag 1000 | 0.378373002 | 0.27249926 |
| Lag 5000 | 0.004289439 | 0.08293975 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |             |            |
|----------|-------------|------------|
| Lag 0    | 1.00000000  | 0.78206164 |
| Lag 100  | 0.87865696  | 0.70110541 |
| Lag 500  | 0.69161215  | 0.56910648 |
| Lag 1000 | 0.56265051  | 0.44347532 |
| Lag 5000 | -0.04003226 | 0.01712343 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

---

|                                                                    |             |            |
|--------------------------------------------------------------------|-------------|------------|
| Lag 0                                                              | 0.38634307  | 0.09679049 |
| Lag 100                                                            | 0.34935277  | 0.10017661 |
| Lag 500                                                            | 0.32828629  | 0.12775760 |
| Lag 1000                                                           | 0.27249926  | 0.11433423 |
| Lag 5000                                                           | 0.08293975  | 0.09000388 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |            |
| Lag 0                                                              | 0.71265304  | 0.46511587 |
| Lag 100                                                            | 0.64354820  | 0.41849123 |
| Lag 500                                                            | 0.54967619  | 0.35510793 |
| Lag 1000                                                           | 0.45077032  | 0.27846473 |
| Lag 5000                                                           | 0.02763651  | 0.08826768 |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |            |
| Lag 0                                                              | 1.00000000  | 0.71265304 |
| Lag 100                                                            | 0.87865696  | 0.64354820 |
| Lag 500                                                            | 0.69161215  | 0.54967619 |
| Lag 1000                                                           | 0.56265051  | 0.45077032 |
| Lag 5000                                                           | -0.04003226 | 0.02763651 |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |            |
| Lag 0                                                              | 0.5942621   | 0.5613010  |
| Lag 100                                                            | 0.5307170   | 0.5141865  |
| Lag 500                                                            | 0.4678297   | 0.4619296  |
| Lag 1000                                                           | 0.4001045   | 0.3816713  |
| Lag 5000                                                           | 0.1478270   | 0.1489578  |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |            |
| Lag 0                                                              | 0.78206164  | 0.46511587 |
| Lag 100                                                            | 0.70110541  | 0.41849123 |
| Lag 500                                                            | 0.56910648  | 0.35510793 |
| Lag 1000                                                           | 0.44347532  | 0.27846473 |
| Lag 5000                                                           | 0.01712343  | 0.08826768 |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |            |
| Lag 0                                                              | 0.5613010   | 0.4107260  |
| Lag 100                                                            | 0.5141865   | 0.3793894  |
| Lag 500                                                            | 0.4619296   | 0.3449678  |



---

|                                                                  |             |              |             |
|------------------------------------------------------------------|-------------|--------------|-------------|
| Lag 1000                                                         |             | 0.3816713    | 0.2731469   |
| Lag 5000                                                         |             | 0.1489578    | 0.1431489   |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |             |              |             |
| Lag 0                                                            | 0.01620495  | 0.04295234   | 0.052223393 |
| Lag 100                                                          | 0.02029711  | 0.05648775   | 0.034069858 |
| Lag 500                                                          | 0.04765917  | 0.03285861   | 0.007554307 |
| Lag 1000                                                         | 0.03767470  | 0.07587887   | 0.003191847 |
| Lag 5000                                                         | -0.02249363 | 0.05599353   | 0.040862718 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |             |              |             |
| Lag 0                                                            | -0.04161269 | -0.102952660 | 0.04055573  |
| Lag 100                                                          | -0.05474586 | -0.077407539 | 0.03752540  |
| Lag 500                                                          | -0.01536913 | -0.009985327 | 0.02041393  |
| Lag 1000                                                         | -0.02242552 | -0.016880479 | 0.02095844  |
| Lag 5000                                                         | 0.05382429  | 0.093039192  | -0.03853343 |
| traittarsus.2.mother traitbwt.2.mother                           |             |              |             |
| Lag 0                                                            | -0.15988076 | 0.11172969   |             |
| Lag 100                                                          | -0.14920373 | 0.08235912   |             |
| Lag 500                                                          | -0.10669450 | 0.07860308   |             |
| Lag 1000                                                         | -0.09563279 | 0.04882603   |             |
| Lag 5000                                                         | -0.03477033 | 0.03882608   |             |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |             |
| Lag 0                                                            |             | -0.48510213  | -0.35513667 |
| Lag 100                                                          |             | -0.43884380  | -0.32128424 |
| Lag 500                                                          |             | -0.37005382  | -0.29419691 |
| Lag 1000                                                         |             | -0.29875897  | -0.25541065 |
| Lag 5000                                                         |             | -0.06651357  | -0.08518695 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |             |
| Lag 0                                                            |             | 0.1906441    | 0.2159001   |
| Lag 100                                                          |             | 0.2109187    | 0.2292953   |
| Lag 500                                                          |             | 0.2433902    | 0.2525070   |
| Lag 1000                                                         |             | 0.2670207    | 0.2801991   |
| Lag 5000                                                         |             | 0.2055428    | 0.2202544   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |              |             |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | -0.35513667 | -0.16353731 |
| Lag 100                                                          | -0.32128424 | -0.14737332 |
| Lag 500                                                          | -0.29419691 | -0.14406290 |
| Lag 1000                                                         | -0.25541065 | -0.15449858 |
| Lag 5000                                                         | -0.08518695 | -0.09726346 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.1588923   | 0.1751169   |
| Lag 100                                                          | 0.1763493   | 0.1863279   |
| Lag 500                                                          | 0.2035612   | 0.2019259   |
| Lag 1000                                                         | 0.2408050   | 0.2434815   |
| Lag 5000                                                         | 0.2188271   | 0.2190020   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.1906441   | 0.1588923   |
| Lag 100                                                          | 0.2109187   | 0.1763493   |
| Lag 500                                                          | 0.2433902   | 0.2035612   |
| Lag 1000                                                         | 0.2670207   | 0.2408050   |
| Lag 5000                                                         | 0.2055428   | 0.2188271   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.5600992  | -0.5208193  |
| Lag 100                                                          | -0.5263626  | -0.5010452  |
| Lag 500                                                          | -0.4620756  | -0.4434018  |
| Lag 1000                                                         | -0.4037773  | -0.3729971  |
| Lag 5000                                                         | -0.1493219  | -0.1529356  |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | 0.2159001   | 0.1751169   |
| Lag 100                                                          | 0.2292953   | 0.1863279   |
| Lag 500                                                          | 0.2525070   | 0.2019259   |
| Lag 1000                                                         | 0.2801991   | 0.2434815   |
| Lag 5000                                                         | 0.2202544   | 0.2190020   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |             |
| Lag 0                                                            | -0.5208193  | -0.3939711  |
| Lag 100                                                          | -0.5010452  | -0.3851102  |
| Lag 500                                                          | -0.4434018  | -0.3487254  |

---

|          |            |            |
|----------|------------|------------|
| Lag 1000 | -0.3729971 | -0.2846600 |
| Lag 5000 | -0.1529356 | -0.1449719 |

, , traitbwt.2:traittarsus.1.animal

|          |                                    |                                 |
|----------|------------------------------------|---------------------------------|
|          | traittarsus.1:traittarsus.1.animal | traitbwt.1:traittarsus.1.animal |
| Lag 0    | 0.51576277                         | 0.52342220                      |
| Lag 100  | 0.44159056                         | 0.43580048                      |
| Lag 500  | 0.30785687                         | 0.33037079                      |
| Lag 1000 | 0.24390874                         | 0.26926704                      |
| Lag 5000 | -0.05005481                        | 0.04687804                      |

|          |                                    |                                 |
|----------|------------------------------------|---------------------------------|
|          | traittarsus.2:traittarsus.1.animal | traitbwt.2:traittarsus.1.animal |
| Lag 0    | 0.782061640                        | 1.00000000                      |
| Lag 100  | 0.696299215                        | 0.87453824                      |
| Lag 500  | 0.538889846                        | 0.65219311                      |
| Lag 1000 | 0.445389577                        | 0.48957859                      |
| Lag 5000 | -0.006519724                       | 0.01842864                      |

|          |                                 |                              |
|----------|---------------------------------|------------------------------|
|          | traittarsus.1:traitbwt.1.animal | traitbwt.1:traitbwt.1.animal |
| Lag 0    | 0.52342220                      | 0.32999449                   |
| Lag 100  | 0.43580048                      | 0.28148773                   |
| Lag 500  | 0.33037079                      | 0.26589289                   |
| Lag 1000 | 0.26926704                      | 0.24323871                   |
| Lag 5000 | 0.04687804                      | 0.09890983                   |

|          |                                 |                              |
|----------|---------------------------------|------------------------------|
|          | traittarsus.2:traitbwt.1.animal | traitbwt.2:traitbwt.1.animal |
| Lag 0    | 0.59763628                      | 0.6478170                    |
| Lag 100  | 0.52643466                      | 0.5573055                    |
| Lag 500  | 0.44551868                      | 0.4418750                    |
| Lag 1000 | 0.37736602                      | 0.3442054                    |
| Lag 5000 | 0.08285167                      | 0.1125165                    |

|         |                                    |                                 |
|---------|------------------------------------|---------------------------------|
|         | traittarsus.1:traittarsus.2.animal | traitbwt.1:traittarsus.2.animal |
| Lag 0   | 0.782061640                        | 0.59763628                      |
| Lag 100 | 0.696299215                        | 0.52643466                      |
| Lag 500 | 0.538889846                        | 0.44551868                      |

---

|                                                                    |              |            |              |
|--------------------------------------------------------------------|--------------|------------|--------------|
| Lag 1000                                                           | 0.445389577  | 0.37736602 |              |
| Lag 5000                                                           | -0.006519724 | 0.08285167 |              |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |            |              |
| Lag 0                                                              | 0.5635407    | 0.6729793  |              |
| Lag 100                                                            | 0.5125673    | 0.6052812  |              |
| Lag 500                                                            | 0.4465910    | 0.5277439  |              |
| Lag 1000                                                           | 0.4070664    | 0.4538221  |              |
| Lag 5000                                                           | 0.1850370    | 0.1792542  |              |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |            |              |
| Lag 0                                                              | 1.00000000   | 0.6478170  |              |
| Lag 100                                                            | 0.87453824   | 0.5573055  |              |
| Lag 500                                                            | 0.65219311   | 0.4418750  |              |
| Lag 1000                                                           | 0.48957859   | 0.3442054  |              |
| Lag 5000                                                           | 0.01842864   | 0.1125165  |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |            |              |
| Lag 0                                                              | 0.6729793    | 0.6191252  |              |
| Lag 100                                                            | 0.6052812    | 0.5529718  |              |
| Lag 500                                                            | 0.5277439    | 0.4729419  |              |
| Lag 1000                                                           | 0.4538221    | 0.3884032  |              |
| Lag 5000                                                           | 0.1792542    | 0.1542326  |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |            |              |
| Lag 0                                                              | 0.063463728  | 0.02730871 | 0.051964042  |
| Lag 100                                                            | 0.066558472  | 0.03961464 | 0.040967359  |
| Lag 500                                                            | 0.076728186  | 0.05374642 | -0.012930939 |
| Lag 1000                                                           | 0.100842393  | 0.09752083 | -0.062954038 |
| Lag 5000                                                           | -0.006751012 | 0.04984406 | 0.002434151  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |              |            |              |
| Lag 0                                                              | -0.09337461  | 0.02006907 | -0.06942502  |
| Lag 100                                                            | -0.09001277  | 0.02763705 | -0.07262374  |
| Lag 500                                                            | -0.04869538  | 0.02000750 | -0.09505843  |
| Lag 1000                                                           | -0.05831890  | 0.02560894 | -0.08617648  |
| Lag 5000                                                           | 0.04246982   | 0.06392166 | -0.03722137  |
| traittarsus.2.mother traitbwt.2.mother                             |              |            |              |

---

|                                                                  |               |             |
|------------------------------------------------------------------|---------------|-------------|
| Lag 0                                                            | -0.026436938  | 0.04826105  |
| Lag 100                                                          | -0.041329867  | 0.04078039  |
| Lag 500                                                          | -0.028290001  | 0.03056152  |
| Lag 1000                                                         | -0.036326123  | 0.04300492  |
| Lag 5000                                                         | -0.008439906  | 0.00653221  |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |             |
| Lag 0                                                            | -0.3969629895 | -0.44235441 |
| Lag 100                                                          | -0.3457725771 | -0.37788435 |
| Lag 500                                                          | -0.2409128696 | -0.27612777 |
| Lag 1000                                                         | -0.2064627528 | -0.23939733 |
| Lag 5000                                                         | -0.0008284304 | -0.05932951 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |               |             |
| Lag 0                                                            | 0.1332860     | 0.1894193   |
| Lag 100                                                          | 0.1542105     | 0.2129929   |
| Lag 500                                                          | 0.1772766     | 0.2426247   |
| Lag 1000                                                         | 0.1956287     | 0.2577964   |
| Lag 5000                                                         | 0.1561845     | 0.1623293   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |               |             |
| Lag 0                                                            | -0.44235441   | -0.3300209  |
| Lag 100                                                          | -0.37788435   | -0.2868342  |
| Lag 500                                                          | -0.27612777   | -0.2200427  |
| Lag 1000                                                         | -0.23939733   | -0.2196364  |
| Lag 5000                                                         | -0.05932951   | -0.1105188  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |               |             |
| Lag 0                                                            | 0.1392793     | 0.1845588   |
| Lag 100                                                          | 0.1618837     | 0.2068320   |
| Lag 500                                                          | 0.1824833     | 0.2322375   |
| Lag 1000                                                         | 0.2062851     | 0.2545405   |
| Lag 5000                                                         | 0.1708879     | 0.1679111   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |               |             |
| Lag 0                                                            | 0.1332860     | 0.1392793   |
| Lag 100                                                          | 0.1542105     | 0.1618837   |
| Lag 500                                                          | 0.1772766     | 0.1824833   |

---

|                                                                    |            |            |
|--------------------------------------------------------------------|------------|------------|
| Lag 1000                                                           | 0.1956287  | 0.2062851  |
| Lag 5000                                                           | 0.1561845  | 0.1708879  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |            |            |
| Lag 0                                                              | -0.5541670 | -0.6200400 |
| Lag 100                                                            | -0.5163721 | -0.5823652 |
| Lag 500                                                            | -0.4591717 | -0.5082023 |
| Lag 1000                                                           | -0.4081394 | -0.4277328 |
| Lag 5000                                                           | -0.1850899 | -0.1717978 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |            |            |
| Lag 0                                                              | 0.1894193  | 0.1845588  |
| Lag 100                                                            | 0.2129929  | 0.2068320  |
| Lag 500                                                            | 0.2426247  | 0.2322375  |
| Lag 1000                                                           | 0.2577964  | 0.2545405  |
| Lag 5000                                                           | 0.1623293  | 0.1679111  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |            |            |
| Lag 0                                                              | -0.6200400 | -0.5649761 |
| Lag 100                                                            | -0.5823652 | -0.5353605 |
| Lag 500                                                            | -0.5082023 | -0.4627844 |
| Lag 1000                                                           | -0.4277328 | -0.3777429 |
| Lag 5000                                                           | -0.1717978 | -0.1507443 |
| , , traittarsus.1:traitbwt.1.animal                                |            |            |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |            |            |
| Lag 0                                                              | 0.6872795  | 1.0000000  |
| Lag 100                                                            | 0.5870266  | 0.8045048  |
| Lag 500                                                            | 0.4080911  | 0.4908098  |
| Lag 1000                                                           | 0.3227808  | 0.3613161  |
| Lag 5000                                                           | 0.1778345  | 0.1860991  |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |            |            |
| Lag 0                                                              | 0.3863431  | 0.5234222  |
| Lag 100                                                            | 0.3311212  | 0.4545868  |
| Lag 500                                                            | 0.2390588  | 0.3427345  |

---

|                                                                    |             |            |
|--------------------------------------------------------------------|-------------|------------|
| Lag 1000                                                           | 0.2351271   | 0.3004179  |
| Lag 5000                                                           | 0.1671396   | 0.1589172  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |            |
| Lag 0                                                              | 1.0000000   | 0.7001005  |
| Lag 100                                                            | 0.8045048   | 0.5457846  |
| Lag 500                                                            | 0.4908098   | 0.3331394  |
| Lag 1000                                                           | 0.3613161   | 0.2645656  |
| Lag 5000                                                           | 0.1860991   | 0.1481314  |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |            |
| Lag 0                                                              | 0.4680554   | 0.4293517  |
| Lag 100                                                            | 0.3667985   | 0.3303034  |
| Lag 500                                                            | 0.2878690   | 0.2612182  |
| Lag 1000                                                           | 0.2588194   | 0.2248089  |
| Lag 5000                                                           | 0.2468277   | 0.2268437  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |            |
| Lag 0                                                              | 0.3863431   | 0.4680554  |
| Lag 100                                                            | 0.3311212   | 0.3667985  |
| Lag 500                                                            | 0.2390588   | 0.2878690  |
| Lag 1000                                                           | 0.2351271   | 0.2588194  |
| Lag 5000                                                           | 0.1671396   | 0.2468277  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |            |
| Lag 0                                                              | 0.096601163 | 0.14997594 |
| Lag 100                                                            | 0.063514030 | 0.11517198 |
| Lag 500                                                            | 0.007343656 | 0.08445112 |
| Lag 1000                                                           | 0.062005867 | 0.13299248 |
| Lag 5000                                                           | 0.138711493 | 0.22091811 |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |            |
| Lag 0                                                              | 0.5234222   | 0.4293517  |
| Lag 100                                                            | 0.4545868   | 0.3303034  |
| Lag 500                                                            | 0.3427345   | 0.2612182  |
| Lag 1000                                                           | 0.3004179   | 0.2248089  |
| Lag 5000                                                           | 0.1589172   | 0.2268437  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |            |

---

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.14997594 | 0.14180302 |
| Lag 100  | 0.11517198 | 0.11177118 |
| Lag 500  | 0.08445112 | 0.09651159 |
| Lag 1000 | 0.13299248 | 0.13501835 |
| Lag 5000 | 0.22091811 | 0.25836873 |

traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear

|          |            |             |              |
|----------|------------|-------------|--------------|
| Lag 0    | 0.06781216 | -0.04857154 | -0.001115860 |
| Lag 100  | 0.05744185 | -0.04365919 | -0.003028122 |
| Lag 500  | 0.04182580 | -0.01683610 | -0.013871766 |
| Lag 1000 | 0.08409670 | 0.04075709  | -0.088936973 |
| Lag 5000 | 0.07211397 | 0.04755605  | 0.061879136  |

traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother

|          |              |              |              |
|----------|--------------|--------------|--------------|
| Lag 0    | -0.026101857 | 0.015326642  | -0.015001711 |
| Lag 100  | -0.026229006 | 0.004791955  | -0.002553299 |
| Lag 500  | -0.004795139 | 0.018197593  | -0.001734987 |
| Lag 1000 | -0.025567829 | -0.014419690 | -0.054320480 |
| Lag 5000 | -0.011401634 | -0.003782365 | -0.012255190 |

traittarsus.2.mother traitbwt.2.mother

|          |            |             |
|----------|------------|-------------|
| Lag 0    | 0.05171415 | -0.01155255 |
| Lag 100  | 0.04549989 | -0.01648838 |
| Lag 500  | 0.09888411 | -0.02579001 |
| Lag 1000 | 0.07526847 | -0.03519968 |
| Lag 5000 | 0.07495551 | -0.03929467 |

traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.5374156 | -0.7822168 |
| Lag 100  | -0.4631107 | -0.6561709 |
| Lag 500  | -0.3115622 | -0.4297448 |
| Lag 1000 | -0.2502951 | -0.3378405 |
| Lag 5000 | -0.1757000 | -0.1812141 |

traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units

|         |           |           |
|---------|-----------|-----------|
| Lag 0   | 0.1247101 | 0.1399193 |
| Lag 100 | 0.1426212 | 0.1604774 |
| Lag 500 | 0.1604094 | 0.2063094 |



---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                         | 0.1703454   | 0.2249688   |
| Lag 5000                                                         | 0.1859265   | 0.1911673   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.7822168  | -0.6444049  |
| Lag 100                                                          | -0.6561709  | -0.5391921  |
| Lag 500                                                          | -0.4297448  | -0.3429426  |
| Lag 1000                                                         | -0.3378405  | -0.2744658  |
| Lag 5000                                                         | -0.1812141  | -0.1591715  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.1003793   | 0.1067549   |
| Lag 100                                                          | 0.1314503   | 0.1404067   |
| Lag 500                                                          | 0.1658186   | 0.1981668   |
| Lag 1000                                                         | 0.1899850   | 0.2354607   |
| Lag 5000                                                         | 0.2090708   | 0.1964048   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.1247101   | 0.1003793   |
| Lag 100                                                          | 0.1426212   | 0.1314503   |
| Lag 500                                                          | 0.1604094   | 0.1658186   |
| Lag 1000                                                         | 0.1703454   | 0.1899850   |
| Lag 5000                                                         | 0.1859265   | 0.2090708   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.09677793 | -0.12847847 |
| Lag 100                                                          | -0.06212434 | -0.10051402 |
| Lag 500                                                          | -0.01671120 | -0.07130994 |
| Lag 1000                                                         | -0.07460059 | -0.11790351 |
| Lag 5000                                                         | -0.16171357 | -0.21125583 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | 0.1399193   | 0.1067549   |
| Lag 100                                                          | 0.1604774   | 0.1404067   |
| Lag 500                                                          | 0.2063094   | 0.1981668   |
| Lag 1000                                                         | 0.2249688   | 0.2354607   |
| Lag 5000                                                         | 0.1911673   | 0.1964048   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |             |

---

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.12847847 | -0.10947233 |
| Lag 100  | -0.10051402 | -0.09309607 |
| Lag 500  | -0.07130994 | -0.08545450 |
| Lag 1000 | -0.11790351 | -0.11149356 |
| Lag 5000 | -0.21125583 | -0.23203853 |

, , traitbwt.1:traitbwt.1.animal

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.23164362 | 0.70010050 |
| Lag 100  | 0.18054608 | 0.54456316 |
| Lag 500  | 0.08781934 | 0.30301892 |
| Lag 1000 | 0.03769032 | 0.17531595 |
| Lag 5000 | 0.09013727 | 0.08812775 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |            |             |
|----------|------------|-------------|
| Lag 0    | 0.09679049 | 0.329994492 |
| Lag 100  | 0.07358032 | 0.280145064 |
| Lag 500  | 0.03161374 | 0.197610680 |
| Lag 1000 | 0.07044623 | 0.170452855 |
| Lag 5000 | 0.09611101 | 0.004847697 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|          |            |           |
|----------|------------|-----------|
| Lag 0    | 0.70010050 | 1.0000000 |
| Lag 100  | 0.54456316 | 0.6856952 |
| Lag 500  | 0.30301892 | 0.3598203 |
| Lag 1000 | 0.17531595 | 0.2634115 |
| Lag 5000 | 0.08812775 | 0.0581774 |

traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal

|          |           |            |
|----------|-----------|------------|
| Lag 0    | 0.2789729 | 0.50673141 |
| Lag 100  | 0.2383906 | 0.37724913 |
| Lag 500  | 0.1850306 | 0.26913927 |
| Lag 1000 | 0.1760392 | 0.21403804 |
| Lag 5000 | 0.1959228 | 0.08822723 |

traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal

|                                                                    |              |              |               |
|--------------------------------------------------------------------|--------------|--------------|---------------|
| Lag 0                                                              | 0.09679049   | 0.2789729    |               |
| Lag 100                                                            | 0.07358032   | 0.2383906    |               |
| Lag 500                                                            | 0.03161374   | 0.1850306    |               |
| Lag 1000                                                           | 0.07044623   | 0.1760392    |               |
| Lag 5000                                                           | 0.09611101   | 0.1959228    |               |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |              |               |
| Lag 0                                                              | 0.10548333   | 0.17131088   |               |
| Lag 100                                                            | 0.08666667   | 0.15141502   |               |
| Lag 500                                                            | 0.03855034   | 0.12592562   |               |
| Lag 1000                                                           | 0.09753802   | 0.15461525   |               |
| Lag 5000                                                           | 0.05868589   | 0.07881607   |               |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |              |               |
| Lag 0                                                              | 0.329994492  | 0.50673141   |               |
| Lag 100                                                            | 0.280145064  | 0.37724913   |               |
| Lag 500                                                            | 0.197610680  | 0.26913927   |               |
| Lag 1000                                                           | 0.170452855  | 0.21403804   |               |
| Lag 5000                                                           | 0.004847697  | 0.08822723   |               |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |              |               |
| Lag 0                                                              | 0.17131088   | 0.21046169   |               |
| Lag 100                                                            | 0.15141502   | 0.17525516   |               |
| Lag 500                                                            | 0.12592562   | 0.17738787   |               |
| Lag 1000                                                           | 0.15461525   | 0.17597147   |               |
| Lag 5000                                                           | 0.07881607   | 0.08328384   |               |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |              |               |
| Lag 0                                                              | 0.065188114  | -0.054239888 | -0.0315843380 |
| Lag 100                                                            | 0.038096586  | -0.064724456 | -0.0570202816 |
| Lag 500                                                            | -0.002027739 | 0.020824115  | 0.0009269064  |
| Lag 1000                                                           | 0.057546907  | 0.008429484  | -0.0456402525 |
| Lag 5000                                                           | 0.105389029  | 0.003598701  | 0.0158576632  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |              |              |               |
| Lag 0                                                              | -0.02489988  | 0.063535708  | -0.301057181  |
| Lag 100                                                            | -0.02027214  | 0.044670327  | -0.205303237  |
| Lag 500                                                            | -0.01893478  | 0.000885945  | -0.083381850  |

|                                                                  |             |              |              |
|------------------------------------------------------------------|-------------|--------------|--------------|
| Lag 1000                                                         | -0.01593842 | -0.053672083 | -0.098869582 |
| Lag 5000                                                         | -0.06322669 | 0.022199101  | 0.007755258  |
| traittarsus.2.mother traitbwt.2.mother                           |             |              |              |
| Lag 0                                                            | 0.02830176  | -0.06829043  |              |
| Lag 100                                                          | 0.01508424  | -0.03965583  |              |
| Lag 500                                                          | 0.07970255  | -0.07134929  |              |
| Lag 1000                                                         | 0.03715760  | -0.03341572  |              |
| Lag 5000                                                         | 0.05894613  | -0.03420041  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |              |
| Lag 0                                                            |             | -0.224526532 | -0.57857216  |
| Lag 100                                                          |             | -0.156949752 | -0.45706691  |
| Lag 500                                                          |             | -0.038181523 | -0.25766540  |
| Lag 1000                                                         |             | -0.006974297 | -0.17163469  |
| Lag 5000                                                         |             | -0.082791880 | -0.06775113  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |              |
| Lag 0                                                            |             | 0.08095300   | 0.11864383   |
| Lag 100                                                          |             | 0.08975567   | 0.13271891   |
| Lag 500                                                          |             | 0.10744809   | 0.16596526   |
| Lag 1000                                                         |             | 0.10586708   | 0.15987797   |
| Lag 5000                                                         |             | 0.11020681   | 0.08507017   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |              |              |
| Lag 0                                                            |             | -0.57857216  | -0.72140390  |
| Lag 100                                                          |             | -0.45706691  | -0.56413893  |
| Lag 500                                                          |             | -0.25766540  | -0.34043397  |
| Lag 1000                                                         |             | -0.17163469  | -0.22135712  |
| Lag 5000                                                         |             | -0.06775113  | -0.03117589  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |              |              |
| Lag 0                                                            |             | 0.1206260    | 0.15466547   |
| Lag 100                                                          |             | 0.1358748    | 0.17988396   |
| Lag 500                                                          |             | 0.1545988    | 0.21095632   |
| Lag 1000                                                         |             | 0.1778949    | 0.23064318   |
| Lag 5000                                                         |             | 0.1189483    | 0.08708938   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |              |              |

---

|          |            |           |
|----------|------------|-----------|
| Lag 0    | 0.08095300 | 0.1206260 |
| Lag 100  | 0.08975567 | 0.1358748 |
| Lag 500  | 0.10744809 | 0.1545988 |
| Lag 1000 | 0.10586708 | 0.1778949 |
| Lag 5000 | 0.11020681 | 0.1189483 |

traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units

|          |             |            |
|----------|-------------|------------|
| Lag 0    | -0.11346055 | -0.1672807 |
| Lag 100  | -0.08013985 | -0.1375357 |
| Lag 500  | -0.05850422 | -0.1306059 |
| Lag 1000 | -0.11497643 | -0.1592872 |
| Lag 5000 | -0.07297426 | -0.0658664 |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.11864383 | 0.15466547 |
| Lag 100  | 0.13271891 | 0.17988396 |
| Lag 500  | 0.16596526 | 0.21095632 |
| Lag 1000 | 0.15987797 | 0.23064318 |
| Lag 5000 | 0.08507017 | 0.08708938 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.1672807 | -0.1878114 |
| Lag 100  | -0.1375357 | -0.1655553 |
| Lag 500  | -0.1306059 | -0.1703498 |
| Lag 1000 | -0.1592872 | -0.1664878 |
| Lag 5000 | -0.0658664 | -0.0614103 |

, , traittarsus.2:traitbwt.1.animal

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |            |           |
|----------|------------|-----------|
| Lag 0    | 0.43711841 | 0.4680554 |
| Lag 100  | 0.38614287 | 0.3822935 |
| Lag 500  | 0.32973015 | 0.2828059 |
| Lag 1000 | 0.30898433 | 0.3053279 |
| Lag 5000 | 0.06068506 | 0.1038801 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                              | 0.71265304  | 0.59763628  |
| Lag 100                                                            | 0.64495665  | 0.53517485  |
| Lag 500                                                            | 0.56187680  | 0.46435370  |
| Lag 1000                                                           | 0.52770068  | 0.43493174  |
| Lag 5000                                                           | -0.04596197 | -0.09919957 |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.4680554   | 0.27897287  |
| Lag 100                                                            | 0.3822935   | 0.24804717  |
| Lag 500                                                            | 0.2828059   | 0.18131034  |
| Lag 1000                                                           | 0.3053279   | 0.20383689  |
| Lag 5000                                                           | 0.1038801   | 0.07466278  |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 1.00000000  | 0.75320258  |
| Lag 100                                                            | 0.84412396  | 0.63847339  |
| Lag 500                                                            | 0.63408467  | 0.47179404  |
| Lag 1000                                                           | 0.55041345  | 0.40200369  |
| Lag 5000                                                           | 0.07989653  | 0.02863099  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.71265304  | 1.00000000  |
| Lag 100                                                            | 0.64495665  | 0.84412396  |
| Lag 500                                                            | 0.56187680  | 0.63408467  |
| Lag 1000                                                           | 0.52770068  | 0.55041345  |
| Lag 5000                                                           | -0.04596197 | 0.07989653  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.39609257  | 0.50085427  |
| Lag 100                                                            | 0.37249703  | 0.45415166  |
| Lag 500                                                            | 0.36147616  | 0.41592880  |
| Lag 1000                                                           | 0.38702161  | 0.40262387  |
| Lag 5000                                                           | 0.05902071  | 0.05027704  |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |             |
| Lag 0                                                              | 0.59763628  | 0.75320258  |
| Lag 100                                                            | 0.53517485  | 0.63847339  |
| Lag 500                                                            | 0.46435370  | 0.47179404  |

---

|                                                                  |              |              |              |
|------------------------------------------------------------------|--------------|--------------|--------------|
| Lag 1000                                                         |              | 0.43493174   | 0.40200369   |
| Lag 5000                                                         |              | -0.09919957  | 0.02863099   |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |              |              |              |
| Lag 0                                                            |              | 0.50085427   | 0.47574281   |
| Lag 100                                                          |              | 0.45415166   | 0.42373764   |
| Lag 500                                                          |              | 0.41592880   | 0.36577921   |
| Lag 1000                                                         |              | 0.40262387   | 0.31479935   |
| Lag 5000                                                         |              | 0.05027704   | 0.04228229   |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |              |              |              |
| Lag 0                                                            | 0.029421346  | 0.01370326   | 0.08861601   |
| Lag 100                                                          | 0.024425366  | 0.01320865   | 0.06508810   |
| Lag 500                                                          | 0.050837464  | 0.03060238   | 0.05880699   |
| Lag 1000                                                         | 0.065685010  | 0.04013922   | 0.02624809   |
| Lag 5000                                                         | -0.006279369 | 0.01390888   | 0.06953734   |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |              |              |
| Lag 0                                                            | -0.06566102  | 0.010266822  | -0.002921727 |
| Lag 100                                                          | -0.05957647  | 0.001133316  | -0.025849735 |
| Lag 500                                                          | -0.03310752  | -0.035302602 | -0.011049599 |
| Lag 1000                                                         | -0.01325750  | -0.005132729 | -0.030094999 |
| Lag 5000                                                         | 0.07737072   | 0.051936326  | 0.028222378  |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |              |
| Lag 0                                                            | -0.01173970  | 0.004891297  |              |
| Lag 100                                                          | -0.01984404  | 0.001495609  |              |
| Lag 500                                                          | -0.01047992  | 0.019305497  |              |
| Lag 1000                                                         | -0.06166849  | 0.035361493  |              |
| Lag 5000                                                         | -0.03572076  | 0.037357859  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |              |
| Lag 0                                                            |              | -0.3729388   | -0.3850991   |
| Lag 100                                                          |              | -0.3393916   | -0.3381615   |
| Lag 500                                                          |              | -0.2662669   | -0.2603097   |
| Lag 1000                                                         |              | -0.2581177   | -0.2862286   |
| Lag 5000                                                         |              | -0.1138844   | -0.1255711   |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |              |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | 0.1304165   | 0.1778597   |
| Lag 100                                                          | 0.1523428   | 0.1937314   |
| Lag 500                                                          | 0.1726064   | 0.2144724   |
| Lag 1000                                                         | 0.1844905   | 0.2275146   |
| Lag 5000                                                         | 0.1876933   | 0.1730132   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.3850991  | -0.2793636  |
| Lag 100                                                          | -0.3381615  | -0.2451230  |
| Lag 500                                                          | -0.2603097  | -0.1889420  |
| Lag 1000                                                         | -0.2862286  | -0.2310740  |
| Lag 5000                                                         | -0.1255711  | -0.1098246  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.1142625   | 0.1618783   |
| Lag 100                                                          | 0.1516538   | 0.1886145   |
| Lag 500                                                          | 0.1942499   | 0.2259939   |
| Lag 1000                                                         | 0.1947749   | 0.2314168   |
| Lag 5000                                                         | 0.1938069   | 0.1610221   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.1304165   | 0.1142625   |
| Lag 100                                                          | 0.1523428   | 0.1516538   |
| Lag 500                                                          | 0.1726064   | 0.1942499   |
| Lag 1000                                                         | 0.1844905   | 0.1947749   |
| Lag 5000                                                         | 0.1876933   | 0.1938069   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.41858410 | -0.47291409 |
| Lag 100                                                          | -0.40815749 | -0.45309828 |
| Lag 500                                                          | -0.38612479 | -0.40783101 |
| Lag 1000                                                         | -0.40349965 | -0.40278702 |
| Lag 5000                                                         | -0.06617356 | -0.06076514 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | 0.1778597   | 0.1618783   |
| Lag 100                                                          | 0.1937314   | 0.1886145   |
| Lag 500                                                          | 0.2144724   | 0.2259939   |



---

|                                                            |             |            |
|------------------------------------------------------------|-------------|------------|
| Lag 1000                                                   | 0.2275146   | 0.2314168  |
| Lag 5000                                                   | 0.1730132   | 0.1610221  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units |             |            |
| Lag 0                                                      | -0.47291409 | -0.4396042 |
| Lag 100                                                    | -0.45309828 | -0.4158969 |
| Lag 500                                                    | -0.40783101 | -0.3577962 |
| Lag 1000                                                   | -0.40278702 | -0.3301300 |
| Lag 5000                                                   | -0.06076514 | -0.0490055 |

, , traitbwt.2:traitbwt.1.animal

|                                                                    |             |            |
|--------------------------------------------------------------------|-------------|------------|
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |             |            |
| Lag 0                                                              | 0.23631479  | 0.42935174 |
| Lag 100                                                            | 0.19631829  | 0.33111043 |
| Lag 500                                                            | 0.13140023  | 0.20166811 |
| Lag 1000                                                           | 0.13084790  | 0.23051272 |
| Lag 5000                                                           | -0.05046358 | 0.04857134 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |             |            |
| Lag 0                                                              | 0.4651159   | 0.6478170  |
| Lag 100                                                            | 0.4250027   | 0.5751741  |
| Lag 500                                                            | 0.3644232   | 0.4605468  |
| Lag 1000                                                           | 0.3695356   | 0.4130944  |
| Lag 5000                                                           | -0.0748654  | -0.1705451 |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |            |
| Lag 0                                                              | 0.42935174  | 0.50673141 |
| Lag 100                                                            | 0.33111043  | 0.36623376 |
| Lag 500                                                            | 0.20166811  | 0.24599792 |
| Lag 1000                                                           | 0.23051272  | 0.26598499 |
| Lag 5000                                                           | 0.04857134  | 0.06879203 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |            |
| Lag 0                                                              | 0.75320258  | 1.00000000 |
| Lag 100                                                            | 0.64643730  | 0.80429897 |
| Lag 500                                                            | 0.48359147  | 0.55454840 |

---

|                                                                    |              |              |             |
|--------------------------------------------------------------------|--------------|--------------|-------------|
| Lag 1000                                                           | 0.43668798   | 0.44468678   |             |
| Lag 5000                                                           | 0.08606661   | -0.01807979  |             |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |              |             |
| Lag 0                                                              | 0.4651159    | 0.75320258   |             |
| Lag 100                                                            | 0.4250027    | 0.64643730   |             |
| Lag 500                                                            | 0.3644232    | 0.48359147   |             |
| Lag 1000                                                           | 0.3695356    | 0.43668798   |             |
| Lag 5000                                                           | -0.0748654   | 0.08606661   |             |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |              |             |
| Lag 0                                                              | 0.35637772   | 0.518702070  |             |
| Lag 100                                                            | 0.34579510   | 0.481108274  |             |
| Lag 500                                                            | 0.32649959   | 0.434843612  |             |
| Lag 1000                                                           | 0.37931051   | 0.436821292  |             |
| Lag 5000                                                           | 0.03852246   | 0.006399124  |             |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |              |             |
| Lag 0                                                              | 0.6478170    | 1.00000000   |             |
| Lag 100                                                            | 0.5751741    | 0.80429897   |             |
| Lag 500                                                            | 0.4605468    | 0.55454840   |             |
| Lag 1000                                                           | 0.4130944    | 0.44468678   |             |
| Lag 5000                                                           | -0.1705451   | -0.01807979  |             |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |              |             |
| Lag 0                                                              | 0.518702070  | 0.64907876   |             |
| Lag 100                                                            | 0.481108274  | 0.56161598   |             |
| Lag 500                                                            | 0.434843612  | 0.47984513   |             |
| Lag 1000                                                           | 0.436821292  | 0.40938531   |             |
| Lag 5000                                                           | 0.006399124  | -0.01977366  |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |              |             |
| Lag 0                                                              | 0.034800266  | 0.007431509  | 0.10446126  |
| Lag 100                                                            | 0.051936557  | -0.001493245 | 0.08007660  |
| Lag 500                                                            | 0.071129943  | 0.067797207  | 0.05478615  |
| Lag 1000                                                           | 0.103624443  | 0.063225351  | 0.01489821  |
| Lag 5000                                                           | -0.000397122 | 0.008471637  | -0.01350094 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |              |              |             |

---

|                                                                  |              |              |             |
|------------------------------------------------------------------|--------------|--------------|-------------|
| Lag 0                                                            | -0.08333568  | 0.100888337  | -0.21005355 |
| Lag 100                                                          | -0.07762909  | 0.079323546  | -0.17806133 |
| Lag 500                                                          | -0.03825641  | -0.007478373 | -0.11051022 |
| Lag 1000                                                         | -0.02784742  | 0.017077993  | -0.10296388 |
| Lag 5000                                                         | 0.04558001   | 0.022598106  | 0.02483781  |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |             |
| Lag 0                                                            | 0.018470325  | -0.095372303 |             |
| Lag 100                                                          | -0.002048636 | -0.029630368 |             |
| Lag 500                                                          | 0.014260425  | -0.034300090 |             |
| Lag 1000                                                         | -0.048567932 | 0.067004499  |             |
| Lag 5000                                                         | -0.025983993 | 0.005631837  |             |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |             |
| Lag 0                                                            |              | -0.25474138  | -0.36301593 |
| Lag 100                                                          |              | -0.21067390  | -0.29399524 |
| Lag 500                                                          |              | -0.11042093  | -0.17823668 |
| Lag 1000                                                         |              | -0.13264168  | -0.21582496 |
| Lag 5000                                                         |              | 0.02716341   | -0.03930422 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |             |
| Lag 0                                                            |              | 0.06808542   | 0.1473359   |
| Lag 100                                                          |              | 0.08333997   | 0.1644163   |
| Lag 500                                                          |              | 0.11004808   | 0.1967466   |
| Lag 1000                                                         |              | 0.13239890   | 0.2087490   |
| Lag 5000                                                         |              | 0.14121855   | 0.1213850   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |              |             |
| Lag 0                                                            |              | -0.36301593  | -0.39024104 |
| Lag 100                                                          |              | -0.29399524  | -0.30803165 |
| Lag 500                                                          |              | -0.17823668  | -0.22428734 |
| Lag 1000                                                         |              | -0.21582496  | -0.24800908 |
| Lag 5000                                                         |              | -0.03930422  | -0.07900553 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |              |             |
| Lag 0                                                            |              | 0.09800038   | 0.1786089   |
| Lag 100                                                          |              | 0.12840120   | 0.2150929   |
| Lag 500                                                          |              | 0.17479984   | 0.2654006   |

---

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                           | 0.19253125  | 0.2644751   |
| Lag 5000                                                           | 0.13671344  | 0.1087463   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units   |             |             |
| Lag 0                                                              | 0.06808542  | 0.09800038  |
| Lag 100                                                            | 0.08333997  | 0.12840120  |
| Lag 500                                                            | 0.11004808  | 0.17479984  |
| Lag 1000                                                           | 0.13239890  | 0.19253125  |
| Lag 5000                                                           | 0.14121855  | 0.13671344  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |             |             |
| Lag 0                                                              | -0.38430674 | -0.49584021 |
| Lag 100                                                            | -0.36892511 | -0.47162690 |
| Lag 500                                                            | -0.35875583 | -0.43506761 |
| Lag 1000                                                           | -0.38190122 | -0.42199356 |
| Lag 5000                                                           | -0.04674851 | -0.01287201 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |             |             |
| Lag 0                                                              | 0.1473359   | 0.1786089   |
| Lag 100                                                            | 0.1644163   | 0.2150929   |
| Lag 500                                                            | 0.1967466   | 0.2654006   |
| Lag 1000                                                           | 0.2087490   | 0.2644751   |
| Lag 5000                                                           | 0.1213850   | 0.1087463   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |             |             |
| Lag 0                                                              | -0.49584021 | -0.56988609 |
| Lag 100                                                            | -0.47162690 | -0.53124887 |
| Lag 500                                                            | -0.43506761 | -0.45896948 |
| Lag 1000                                                           | -0.42199356 | -0.40139643 |
| Lag 5000                                                           | -0.01287201 | 0.01366036  |
| , , traittarsus.1:traittarsus.2.animal                             |             |             |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |             |             |
| Lag 0                                                              | 0.682164176 | 0.38634307  |
| Lag 100                                                            | 0.591700877 | 0.34935277  |
| Lag 500                                                            | 0.451559860 | 0.32828629  |

---

|                                                                    |             |            |
|--------------------------------------------------------------------|-------------|------------|
| Lag 1000                                                           | 0.378373002 | 0.27249926 |
| Lag 5000                                                           | 0.004289439 | 0.08293975 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |             |            |
| Lag 0                                                              | 1.00000000  | 0.78206164 |
| Lag 100                                                            | 0.87865696  | 0.70110541 |
| Lag 500                                                            | 0.69161215  | 0.56910648 |
| Lag 1000                                                           | 0.56265051  | 0.44347532 |
| Lag 5000                                                           | -0.04003226 | 0.01712343 |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |            |
| Lag 0                                                              | 0.38634307  | 0.09679049 |
| Lag 100                                                            | 0.34935277  | 0.10017661 |
| Lag 500                                                            | 0.32828629  | 0.12775760 |
| Lag 1000                                                           | 0.27249926  | 0.11433423 |
| Lag 5000                                                           | 0.08293975  | 0.09000388 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |            |
| Lag 0                                                              | 0.71265304  | 0.46511587 |
| Lag 100                                                            | 0.64354820  | 0.41849123 |
| Lag 500                                                            | 0.54967619  | 0.35510793 |
| Lag 1000                                                           | 0.45077032  | 0.27846473 |
| Lag 5000                                                           | 0.02763651  | 0.08826768 |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |            |
| Lag 0                                                              | 1.00000000  | 0.71265304 |
| Lag 100                                                            | 0.87865696  | 0.64354820 |
| Lag 500                                                            | 0.69161215  | 0.54967619 |
| Lag 1000                                                           | 0.56265051  | 0.45077032 |
| Lag 5000                                                           | -0.04003226 | 0.02763651 |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |            |
| Lag 0                                                              | 0.5942621   | 0.5613010  |
| Lag 100                                                            | 0.5307170   | 0.5141865  |
| Lag 500                                                            | 0.4678297   | 0.4619296  |
| Lag 1000                                                           | 0.4001045   | 0.3816713  |
| Lag 5000                                                           | 0.1478270   | 0.1489578  |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |            |

---

|                                                                  |             |              |             |
|------------------------------------------------------------------|-------------|--------------|-------------|
| Lag 0                                                            | 0.78206164  | 0.46511587   |             |
| Lag 100                                                          | 0.70110541  | 0.41849123   |             |
| Lag 500                                                          | 0.56910648  | 0.35510793   |             |
| Lag 1000                                                         | 0.44347532  | 0.27846473   |             |
| Lag 5000                                                         | 0.01712343  | 0.08826768   |             |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |             |              |             |
| Lag 0                                                            | 0.5613010   | 0.4107260    |             |
| Lag 100                                                          | 0.5141865   | 0.3793894    |             |
| Lag 500                                                          | 0.4619296   | 0.3449678    |             |
| Lag 1000                                                         | 0.3816713   | 0.2731469    |             |
| Lag 5000                                                         | 0.1489578   | 0.1431489    |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |             |              |             |
| Lag 0                                                            | 0.01620495  | 0.04295234   | 0.052223393 |
| Lag 100                                                          | 0.02029711  | 0.05648775   | 0.034069858 |
| Lag 500                                                          | 0.04765917  | 0.03285861   | 0.007554307 |
| Lag 1000                                                         | 0.03767470  | 0.07587887   | 0.003191847 |
| Lag 5000                                                         | -0.02249363 | 0.05599353   | 0.040862718 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |             |              |             |
| Lag 0                                                            | -0.04161269 | -0.102952660 | 0.04055573  |
| Lag 100                                                          | -0.05474586 | -0.077407539 | 0.03752540  |
| Lag 500                                                          | -0.01536913 | -0.009985327 | 0.02041393  |
| Lag 1000                                                         | -0.02242552 | -0.016880479 | 0.02095844  |
| Lag 5000                                                         | 0.05382429  | 0.093039192  | -0.03853343 |
| traittarsus.2.mother traitbwt.2.mother                           |             |              |             |
| Lag 0                                                            | -0.15988076 | 0.11172969   |             |
| Lag 100                                                          | -0.14920373 | 0.08235912   |             |
| Lag 500                                                          | -0.10669450 | 0.07860308   |             |
| Lag 1000                                                         | -0.09563279 | 0.04882603   |             |
| Lag 5000                                                         | -0.03477033 | 0.03882608   |             |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |             |
| Lag 0                                                            | -0.48510213 | -0.35513667  |             |
| Lag 100                                                          | -0.43884380 | -0.32128424  |             |
| Lag 500                                                          | -0.37005382 | -0.29419691  |             |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                         | -0.29875897 | -0.25541065 |
| Lag 5000                                                         | -0.06651357 | -0.08518695 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |             |
| Lag 0                                                            | 0.1906441   | 0.2159001   |
| Lag 100                                                          | 0.2109187   | 0.2292953   |
| Lag 500                                                          | 0.2433902   | 0.2525070   |
| Lag 1000                                                         | 0.2670207   | 0.2801991   |
| Lag 5000                                                         | 0.2055428   | 0.2202544   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.35513667 | -0.16353731 |
| Lag 100                                                          | -0.32128424 | -0.14737332 |
| Lag 500                                                          | -0.29419691 | -0.14406290 |
| Lag 1000                                                         | -0.25541065 | -0.15449858 |
| Lag 5000                                                         | -0.08518695 | -0.09726346 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.1588923   | 0.1751169   |
| Lag 100                                                          | 0.1763493   | 0.1863279   |
| Lag 500                                                          | 0.2035612   | 0.2019259   |
| Lag 1000                                                         | 0.2408050   | 0.2434815   |
| Lag 5000                                                         | 0.2188271   | 0.2190020   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.1906441   | 0.1588923   |
| Lag 100                                                          | 0.2109187   | 0.1763493   |
| Lag 500                                                          | 0.2433902   | 0.2035612   |
| Lag 1000                                                         | 0.2670207   | 0.2408050   |
| Lag 5000                                                         | 0.2055428   | 0.2188271   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.5600992  | -0.5208193  |
| Lag 100                                                          | -0.5263626  | -0.5010452  |
| Lag 500                                                          | -0.4620756  | -0.4434018  |
| Lag 1000                                                         | -0.4037773  | -0.3729971  |
| Lag 5000                                                         | -0.1493219  | -0.1529356  |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |

---

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.2159001 | 0.1751169 |
| Lag 100  | 0.2292953 | 0.1863279 |
| Lag 500  | 0.2525070 | 0.2019259 |
| Lag 1000 | 0.2801991 | 0.2434815 |
| Lag 5000 | 0.2202544 | 0.2190020 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.5208193 | -0.3939711 |
| Lag 100  | -0.5010452 | -0.3851102 |
| Lag 500  | -0.4434018 | -0.3487254 |
| Lag 1000 | -0.3729971 | -0.2846600 |
| Lag 5000 | -0.1529356 | -0.1449719 |

, , traitbwt.1:traittarsus.2.animal

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |            |           |
|----------|------------|-----------|
| Lag 0    | 0.43711841 | 0.4680554 |
| Lag 100  | 0.38614287 | 0.3822935 |
| Lag 500  | 0.32973015 | 0.2828059 |
| Lag 1000 | 0.30898433 | 0.3053279 |
| Lag 5000 | 0.06068506 | 0.1038801 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | 0.71265304  | 0.59763628  |
| Lag 100  | 0.64495665  | 0.53517485  |
| Lag 500  | 0.56187680  | 0.46435370  |
| Lag 1000 | 0.52770068  | 0.43493174  |
| Lag 5000 | -0.04596197 | -0.09919957 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|          |           |            |
|----------|-----------|------------|
| Lag 0    | 0.4680554 | 0.27897287 |
| Lag 100  | 0.3822935 | 0.24804717 |
| Lag 500  | 0.2828059 | 0.18131034 |
| Lag 1000 | 0.3053279 | 0.20383689 |
| Lag 5000 | 0.1038801 | 0.07466278 |

traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal



---

|                                                                    |             |            |            |
|--------------------------------------------------------------------|-------------|------------|------------|
| Lag 0                                                              | 1.00000000  | 0.75320258 |            |
| Lag 100                                                            | 0.84412396  | 0.63847339 |            |
| Lag 500                                                            | 0.63408467  | 0.47179404 |            |
| Lag 1000                                                           | 0.55041345  | 0.40200369 |            |
| Lag 5000                                                           | 0.07989653  | 0.02863099 |            |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |            |            |
| Lag 0                                                              | 0.71265304  | 1.00000000 |            |
| Lag 100                                                            | 0.64495665  | 0.84412396 |            |
| Lag 500                                                            | 0.56187680  | 0.63408467 |            |
| Lag 1000                                                           | 0.52770068  | 0.55041345 |            |
| Lag 5000                                                           | -0.04596197 | 0.07989653 |            |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |            |            |
| Lag 0                                                              | 0.39609257  | 0.50085427 |            |
| Lag 100                                                            | 0.37249703  | 0.45415166 |            |
| Lag 500                                                            | 0.36147616  | 0.41592880 |            |
| Lag 1000                                                           | 0.38702161  | 0.40262387 |            |
| Lag 5000                                                           | 0.05902071  | 0.05027704 |            |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |            |            |
| Lag 0                                                              | 0.59763628  | 0.75320258 |            |
| Lag 100                                                            | 0.53517485  | 0.63847339 |            |
| Lag 500                                                            | 0.46435370  | 0.47179404 |            |
| Lag 1000                                                           | 0.43493174  | 0.40200369 |            |
| Lag 5000                                                           | -0.09919957 | 0.02863099 |            |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |            |            |
| Lag 0                                                              | 0.50085427  | 0.47574281 |            |
| Lag 100                                                            | 0.45415166  | 0.42373764 |            |
| Lag 500                                                            | 0.41592880  | 0.36577921 |            |
| Lag 1000                                                           | 0.40262387  | 0.31479935 |            |
| Lag 5000                                                           | 0.05027704  | 0.04228229 |            |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |             |            |            |
| Lag 0                                                              | 0.029421346 | 0.01370326 | 0.08861601 |
| Lag 100                                                            | 0.024425366 | 0.01320865 | 0.06508810 |
| Lag 500                                                            | 0.050837464 | 0.03060238 | 0.05880699 |

---

|                                                                  |              |              |              |
|------------------------------------------------------------------|--------------|--------------|--------------|
| Lag 1000                                                         | 0.065685010  | 0.04013922   | 0.02624809   |
| Lag 5000                                                         | -0.006279369 | 0.01390888   | 0.06953734   |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |              |              |
| Lag 0                                                            | -0.06566102  | 0.010266822  | -0.002921727 |
| Lag 100                                                          | -0.05957647  | 0.001133316  | -0.025849735 |
| Lag 500                                                          | -0.03310752  | -0.035302602 | -0.011049599 |
| Lag 1000                                                         | -0.01325750  | -0.005132729 | -0.030094999 |
| Lag 5000                                                         | 0.07737072   | 0.051936326  | 0.028222378  |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |              |
| Lag 0                                                            | -0.01173970  | 0.004891297  |              |
| Lag 100                                                          | -0.01984404  | 0.001495609  |              |
| Lag 500                                                          | -0.01047992  | 0.019305497  |              |
| Lag 1000                                                         | -0.06166849  | 0.035361493  |              |
| Lag 5000                                                         | -0.03572076  | 0.037357859  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |              |
| Lag 0                                                            |              | -0.3729388   | -0.3850991   |
| Lag 100                                                          |              | -0.3393916   | -0.3381615   |
| Lag 500                                                          |              | -0.2662669   | -0.2603097   |
| Lag 1000                                                         |              | -0.2581177   | -0.2862286   |
| Lag 5000                                                         |              | -0.1138844   | -0.1255711   |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |              |
| Lag 0                                                            |              | 0.1304165    | 0.1778597    |
| Lag 100                                                          |              | 0.1523428    | 0.1937314    |
| Lag 500                                                          |              | 0.1726064    | 0.2144724    |
| Lag 1000                                                         |              | 0.1844905    | 0.2275146    |
| Lag 5000                                                         |              | 0.1876933    | 0.1730132    |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |              |              |
| Lag 0                                                            |              | -0.3850991   | -0.2793636   |
| Lag 100                                                          |              | -0.3381615   | -0.2451230   |
| Lag 500                                                          |              | -0.2603097   | -0.1889420   |
| Lag 1000                                                         |              | -0.2862286   | -0.2310740   |
| Lag 5000                                                         |              | -0.1255711   | -0.1098246   |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |              |              |

---

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.1142625 | 0.1618783 |
| Lag 100  | 0.1516538 | 0.1886145 |
| Lag 500  | 0.1942499 | 0.2259939 |
| Lag 1000 | 0.1947749 | 0.2314168 |
| Lag 5000 | 0.1938069 | 0.1610221 |

traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.1304165 | 0.1142625 |
| Lag 100  | 0.1523428 | 0.1516538 |
| Lag 500  | 0.1726064 | 0.1942499 |
| Lag 1000 | 0.1844905 | 0.1947749 |
| Lag 5000 | 0.1876933 | 0.1938069 |

traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.41858410 | -0.47291409 |
| Lag 100  | -0.40815749 | -0.45309828 |
| Lag 500  | -0.38612479 | -0.40783101 |
| Lag 1000 | -0.40349965 | -0.40278702 |
| Lag 5000 | -0.06617356 | -0.06076514 |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.1778597 | 0.1618783 |
| Lag 100  | 0.1937314 | 0.1886145 |
| Lag 500  | 0.2144724 | 0.2259939 |
| Lag 1000 | 0.2275146 | 0.2314168 |
| Lag 5000 | 0.1730132 | 0.1610221 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |             |            |
|----------|-------------|------------|
| Lag 0    | -0.47291409 | -0.4396042 |
| Lag 100  | -0.45309828 | -0.4158969 |
| Lag 500  | -0.40783101 | -0.3577962 |
| Lag 1000 | -0.40278702 | -0.3301300 |
| Lag 5000 | -0.06076514 | -0.0490055 |

, , traittarsus.2:traittarsus.2.animal

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|                                                                    |              |             |
|--------------------------------------------------------------------|--------------|-------------|
| Lag 0                                                              | 0.107806440  | 0.09660116  |
| Lag 100                                                            | 0.063335645  | 0.07671143  |
| Lag 500                                                            | 0.015199121  | 0.08067887  |
| Lag 1000                                                           | -0.004286261 | 0.05833877  |
| Lag 5000                                                           | -0.070723679 | -0.04217947 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |              |             |
| Lag 0                                                              | 0.5942621    | 0.5635407   |
| Lag 100                                                            | 0.5173844    | 0.5077019   |
| Lag 500                                                            | 0.4100415    | 0.4261910   |
| Lag 1000                                                           | 0.3053499    | 0.3088610   |
| Lag 5000                                                           | -0.2062339   | -0.1851005  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |              |             |
| Lag 0                                                              | 0.09660116   | 0.10548333  |
| Lag 100                                                            | 0.07671143   | 0.09974202  |
| Lag 500                                                            | 0.08067887   | 0.11354731  |
| Lag 1000                                                           | 0.05833877   | 0.12159525  |
| Lag 5000                                                           | -0.04217947  | -0.06338610 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |              |             |
| Lag 0                                                              | 0.3960926    | 0.3563777   |
| Lag 100                                                            | 0.3697695    | 0.3333969   |
| Lag 500                                                            | 0.3480333    | 0.3176505   |
| Lag 1000                                                           | 0.2823012    | 0.2658600   |
| Lag 5000                                                           | -0.2062348   | -0.1460308  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |             |
| Lag 0                                                              | 0.5942621    | 0.3960926   |
| Lag 100                                                            | 0.5173844    | 0.3697695   |
| Lag 500                                                            | 0.4100415    | 0.3480333   |
| Lag 1000                                                           | 0.3053499    | 0.2823012   |
| Lag 5000                                                           | -0.2062339   | -0.2062348  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |             |
| Lag 0                                                              | 1.00000000   | 0.89913093  |
| Lag 100                                                            | 0.90510783   | 0.83322028  |
| Lag 500                                                            | 0.73781155   | 0.70412544  |

---

|                                                                  |              |              |              |
|------------------------------------------------------------------|--------------|--------------|--------------|
| Lag 1000                                                         | 0.57269345   | 0.54745411   |              |
| Lag 5000                                                         | -0.03830606  | -0.09250669  |              |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal     |              |              |              |
| Lag 0                                                            | 0.5635407    | 0.3563777    |              |
| Lag 100                                                          | 0.5077019    | 0.3333969    |              |
| Lag 500                                                          | 0.4261910    | 0.3176505    |              |
| Lag 1000                                                         | 0.3088610    | 0.2658600    |              |
| Lag 5000                                                         | -0.1851005   | -0.1460308   |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |              |              |              |
| Lag 0                                                            | 0.89913093   | 0.6844205    |              |
| Lag 100                                                          | 0.83322028   | 0.6434324    |              |
| Lag 500                                                          | 0.70412544   | 0.5640206    |              |
| Lag 1000                                                         | 0.54745411   | 0.4452587    |              |
| Lag 5000                                                         | -0.09250669  | -0.0996112   |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |              |              |              |
| Lag 0                                                            | 0.035961574  | -0.017469135 | 0.007171364  |
| Lag 100                                                          | 0.026817555  | -0.002104493 | 0.011633070  |
| Lag 500                                                          | 0.008446146  | -0.011130916 | 0.013837207  |
| Lag 1000                                                         | 0.029054326  | 0.040567400  | 0.023793648  |
| Lag 5000                                                         | -0.032575030 | 0.020439004  | -0.049218434 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |              |              |
| Lag 0                                                            | -0.08333086  | 0.04651404   | -0.06375729  |
| Lag 100                                                          | -0.08053153  | 0.06577446   | -0.05659105  |
| Lag 500                                                          | -0.03847864  | 0.11200938   | -0.07300512  |
| Lag 1000                                                         | -0.04249559  | 0.08134977   | -0.07101148  |
| Lag 5000                                                         | 0.07016400   | 0.02226530   | 0.03188352   |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |              |
| Lag 0                                                            | -0.34477064  | 0.15871038   |              |
| Lag 100                                                          | -0.31220276  | 0.12040861   |              |
| Lag 500                                                          | -0.20777383  | 0.08824805   |              |
| Lag 1000                                                         | -0.12481403  | 0.05105578   |              |
| Lag 5000                                                         | 0.01048624   | 0.01454861   |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |              |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | -0.10376953 | -0.10398125 |
| Lag 100                                                          | -0.07294547 | -0.07699566 |
| Lag 500                                                          | -0.06977230 | -0.08979287 |
| Lag 1000                                                         | -0.03903802 | -0.06150263 |
| Lag 5000                                                         | 0.09531506  | 0.08613972  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |             |
| Lag 0                                                            | 0.18812698  | 0.21480017  |
| Lag 100                                                          | 0.19528250  | 0.21914991  |
| Lag 500                                                          | 0.19556551  | 0.21256430  |
| Lag 1000                                                         | 0.18892755  | 0.20764610  |
| Lag 5000                                                         | 0.06665558  | 0.07468684  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.10398125 | -0.10182068 |
| Lag 100                                                          | -0.07699566 | -0.08777589 |
| Lag 500                                                          | -0.08979287 | -0.09511315 |
| Lag 1000                                                         | -0.06150263 | -0.09765668 |
| Lag 5000                                                         | 0.08613972  | 0.05634093  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.22270138  | 0.23393547  |
| Lag 100                                                          | 0.22287169  | 0.23382912  |
| Lag 500                                                          | 0.20351827  | 0.21079497  |
| Lag 1000                                                         | 0.20537971  | 0.21383439  |
| Lag 5000                                                         | 0.03798515  | 0.03641653  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.18812698  | 0.22270138  |
| Lag 100                                                          | 0.19528250  | 0.22287169  |
| Lag 500                                                          | 0.19556551  | 0.20351827  |
| Lag 1000                                                         | 0.18892755  | 0.20537971  |
| Lag 5000                                                         | 0.06665558  | 0.03798515  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.9107369  | -0.84107771 |
| Lag 100                                                          | -0.8633544  | -0.80717702 |
| Lag 500                                                          | -0.7252004  | -0.68217338 |

---

|          |            |             |
|----------|------------|-------------|
| Lag 1000 | -0.5917870 | -0.54946518 |
| Lag 5000 | 0.0476902  | 0.09830838  |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.21480017 | 0.23393547 |
| Lag 100  | 0.21914991 | 0.23382912 |
| Lag 500  | 0.21256430 | 0.21079497 |
| Lag 1000 | 0.20764610 | 0.21383439 |
| Lag 5000 | 0.07468684 | 0.03641653 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |             |            |
|----------|-------------|------------|
| Lag 0    | -0.84107771 | -0.6791085 |
| Lag 100  | -0.80717702 | -0.6539764 |
| Lag 500  | -0.68217338 | -0.5583969 |
| Lag 1000 | -0.54946518 | -0.4464232 |
| Lag 5000 | 0.09830838  | 0.1028304  |

, , traitbwt.2:traittarsus.2.animal

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | 0.12397199  | 0.14997594  |
| Lag 100  | 0.08643519  | 0.11314881  |
| Lag 500  | 0.03605225  | 0.09323271  |
| Lag 1000 | 0.03084094  | 0.10586940  |
| Lag 5000 | -0.09632022 | -0.03936224 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.5613010  | 0.6729793  |
| Lag 100  | 0.5033508  | 0.6010903  |
| Lag 500  | 0.4274531  | 0.5133373  |
| Lag 1000 | 0.3517565  | 0.4086869  |
| Lag 5000 | -0.1677216 | -0.1581302 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|         |            |            |
|---------|------------|------------|
| Lag 0   | 0.14997594 | 0.17131088 |
| Lag 100 | 0.11314881 | 0.15721379 |
| Lag 500 | 0.09323271 | 0.15933093 |

---

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                           | 0.10586940  | 0.18501406  |
| Lag 5000                                                           | -0.03936224 | -0.01791672 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.5008543   | 0.51870207  |
| Lag 100                                                            | 0.4453244   | 0.46669360  |
| Lag 500                                                            | 0.3907627   | 0.41453930  |
| Lag 1000                                                           | 0.3366895   | 0.35501264  |
| Lag 5000                                                           | -0.1140445  | -0.07692479 |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.5613010   | 0.5008543   |
| Lag 100                                                            | 0.5033508   | 0.4453244   |
| Lag 500                                                            | 0.4274531   | 0.3907627   |
| Lag 1000                                                           | 0.3517565   | 0.3366895   |
| Lag 5000                                                           | -0.1677216  | -0.1140445  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.8991309   | 1.00000000  |
| Lag 100                                                            | 0.8359176   | 0.91646465  |
| Lag 500                                                            | 0.7187715   | 0.77900191  |
| Lag 1000                                                           | 0.5982337   | 0.63306113  |
| Lag 5000                                                           | 0.0176654   | -0.03136451 |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |             |
| Lag 0                                                              | 0.6729793   | 0.51870207  |
| Lag 100                                                            | 0.6010903   | 0.46669360  |
| Lag 500                                                            | 0.5133373   | 0.41453930  |
| Lag 1000                                                           | 0.4086869   | 0.35501264  |
| Lag 5000                                                           | -0.1581302  | -0.07692479 |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |             |
| Lag 0                                                              | 1.00000000  | 0.89818763  |
| Lag 100                                                            | 0.91646465  | 0.82112793  |
| Lag 500                                                            | 0.77900191  | 0.69592684  |
| Lag 1000                                                           | 0.63306113  | 0.55065244  |
| Lag 5000                                                           | -0.03136451 | -0.04949222 |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |             |             |



---

|                                                                  |             |             |             |
|------------------------------------------------------------------|-------------|-------------|-------------|
| Lag 0                                                            | 0.05124741  | 0.001646162 | 0.04068399  |
| Lag 100                                                          | 0.04921037  | 0.008930910 | 0.03759685  |
| Lag 500                                                          | 0.03899327  | 0.015514352 | 0.02580930  |
| Lag 1000                                                         | 0.06406637  | 0.051682789 | 0.02147033  |
| Lag 5000                                                         | -0.01137788 | 0.015052216 | -0.04729885 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |             |             |             |
| Lag 0                                                            | -0.12402916 | 0.08639137  | -0.09704390 |
| Lag 100                                                          | -0.10713277 | 0.09204331  | -0.10427670 |
| Lag 500                                                          | -0.07124572 | 0.09145340  | -0.11605857 |
| Lag 1000                                                         | -0.06554488 | 0.09437629  | -0.10182520 |
| Lag 5000                                                         | 0.06888625  | 0.02727995  | 0.01776003  |
| traittarsus.2.mother traitbwt.2.mother                           |             |             |             |
| Lag 0                                                            | -0.15610253 | 0.09891371  |             |
| Lag 100                                                          | -0.16408367 | 0.08304541  |             |
| Lag 500                                                          | -0.12517718 | 0.04261766  |             |
| Lag 1000                                                         | -0.09948987 | 0.04824881  |             |
| Lag 5000                                                         | -0.01179365 | -0.01278893 |             |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |             |             |
| Lag 0                                                            |             | -0.14515627 | -0.1467921  |
| Lag 100                                                          |             | -0.11636785 | -0.1156256  |
| Lag 500                                                          |             | -0.08159624 | -0.1042741  |
| Lag 1000                                                         |             | -0.07861755 | -0.1076860  |
| Lag 5000                                                         |             | 0.10186075  | 0.0712461   |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |             |             |
| Lag 0                                                            |             | 0.18051752  | 0.23807593  |
| Lag 100                                                          |             | 0.18953377  | 0.24534120  |
| Lag 500                                                          |             | 0.17920764  | 0.22900261  |
| Lag 1000                                                         |             | 0.16794260  | 0.21104881  |
| Lag 5000                                                         |             | 0.04389169  | 0.05181362  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |             |
| Lag 0                                                            |             | -0.1467921  | -0.14376430 |
| Lag 100                                                          |             | -0.1156256  | -0.12366939 |
| Lag 500                                                          |             | -0.1042741  | -0.12078005 |

---

|                                                                  |               |             |
|------------------------------------------------------------------|---------------|-------------|
| Lag 1000                                                         | -0.1076860    | -0.15700540 |
| Lag 5000                                                         | 0.0712461     | 0.01799705  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |               |             |
| Lag 0                                                            | 0.21778240    | 0.25816273  |
| Lag 100                                                          | 0.22684710    | 0.26502282  |
| Lag 500                                                          | 0.20506975    | 0.23751874  |
| Lag 1000                                                         | 0.19012136    | 0.21960899  |
| Lag 5000                                                         | 0.02816484    | 0.03038383  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |               |             |
| Lag 0                                                            | 0.18051752    | 0.21778240  |
| Lag 100                                                          | 0.18953377    | 0.22684710  |
| Lag 500                                                          | 0.17920764    | 0.20506975  |
| Lag 1000                                                         | 0.16794260    | 0.19012136  |
| Lag 5000                                                         | 0.04389169    | 0.02816484  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |               |             |
| Lag 0                                                            | -0.8750125593 | -0.93853010 |
| Lag 100                                                          | -0.8368578282 | -0.89227332 |
| Lag 500                                                          | -0.7287281819 | -0.75764556 |
| Lag 1000                                                         | -0.6175608959 | -0.62307537 |
| Lag 5000                                                         | -0.0008665459 | 0.04348967  |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |               |             |
| Lag 0                                                            | 0.23807593    | 0.25816273  |
| Lag 100                                                          | 0.24534120    | 0.26502282  |
| Lag 500                                                          | 0.22900261    | 0.23751874  |
| Lag 1000                                                         | 0.21104881    | 0.21960899  |
| Lag 5000                                                         | 0.05181362    | 0.03038383  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |               |             |
| Lag 0                                                            | -0.93853010   | -0.85888343 |
| Lag 100                                                          | -0.89227332   | -0.81357341 |
| Lag 500                                                          | -0.75764556   | -0.67868373 |
| Lag 1000                                                         | -0.62307537   | -0.54366192 |
| Lag 5000                                                         | 0.04348967    | 0.05718606  |

---

, , traittarsus.1:traitbwt.2.animal

|          | traittarsus.1:traittarsus.1.animal | traitbwt.1:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.51576277                         | 0.52342220                      |
| Lag 100  | 0.44159056                         | 0.43580048                      |
| Lag 500  | 0.30785687                         | 0.33037079                      |
| Lag 1000 | 0.24390874                         | 0.26926704                      |
| Lag 5000 | -0.05005481                        | 0.04687804                      |

|          | traittarsus.2:traittarsus.1.animal | traitbwt.2:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.782061640                        | 1.00000000                      |
| Lag 100  | 0.696299215                        | 0.87453824                      |
| Lag 500  | 0.538889846                        | 0.65219311                      |
| Lag 1000 | 0.445389577                        | 0.48957859                      |
| Lag 5000 | -0.006519724                       | 0.01842864                      |

|          | traittarsus.1:traitbwt.1.animal | traitbwt.1:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | 0.52342220                      | 0.32999449                   |
| Lag 100  | 0.43580048                      | 0.28148773                   |
| Lag 500  | 0.33037079                      | 0.26589289                   |
| Lag 1000 | 0.26926704                      | 0.24323871                   |
| Lag 5000 | 0.04687804                      | 0.09890983                   |

|          | traittarsus.2:traitbwt.1.animal | traitbwt.2:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | 0.59763628                      | 0.6478170                    |
| Lag 100  | 0.52643466                      | 0.5573055                    |
| Lag 500  | 0.44551868                      | 0.4418750                    |
| Lag 1000 | 0.37736602                      | 0.3442054                    |
| Lag 5000 | 0.08285167                      | 0.1125165                    |

|          | traittarsus.1:traittarsus.2.animal | traitbwt.1:traittarsus.2.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.782061640                        | 0.59763628                      |
| Lag 100  | 0.696299215                        | 0.52643466                      |
| Lag 500  | 0.538889846                        | 0.44551868                      |
| Lag 1000 | 0.445389577                        | 0.37736602                      |
| Lag 5000 | -0.006519724                       | 0.08285167                      |

|  | traittarsus.2:traittarsus.2.animal | traitbwt.2:traittarsus.2.animal |
|--|------------------------------------|---------------------------------|
|--|------------------------------------|---------------------------------|

---

|                                                              |              |            |              |
|--------------------------------------------------------------|--------------|------------|--------------|
| Lag 0                                                        | 0.5635407    | 0.6729793  |              |
| Lag 100                                                      | 0.5125673    | 0.6052812  |              |
| Lag 500                                                      | 0.4465910    | 0.5277439  |              |
| Lag 1000                                                     | 0.4070664    | 0.4538221  |              |
| Lag 5000                                                     | 0.1850370    | 0.1792542  |              |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal |              |            |              |
| Lag 0                                                        | 1.00000000   | 0.6478170  |              |
| Lag 100                                                      | 0.87453824   | 0.5573055  |              |
| Lag 500                                                      | 0.65219311   | 0.4418750  |              |
| Lag 1000                                                     | 0.48957859   | 0.3442054  |              |
| Lag 5000                                                     | 0.01842864   | 0.1125165  |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal |              |            |              |
| Lag 0                                                        | 0.6729793    | 0.6191252  |              |
| Lag 100                                                      | 0.6052812    | 0.5529718  |              |
| Lag 500                                                      | 0.5277439    | 0.4729419  |              |
| Lag 1000                                                     | 0.4538221    | 0.3884032  |              |
| Lag 5000                                                     | 0.1792542    | 0.1542326  |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear     |              |            |              |
| Lag 0                                                        | 0.063463728  | 0.02730871 | 0.051964042  |
| Lag 100                                                      | 0.066558472  | 0.03961464 | 0.040967359  |
| Lag 500                                                      | 0.076728186  | 0.05374642 | -0.012930939 |
| Lag 1000                                                     | 0.100842393  | 0.09752083 | -0.062954038 |
| Lag 5000                                                     | -0.006751012 | 0.04984406 | 0.002434151  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother      |              |            |              |
| Lag 0                                                        | -0.09337461  | 0.02006907 | -0.06942502  |
| Lag 100                                                      | -0.09001277  | 0.02763705 | -0.07262374  |
| Lag 500                                                      | -0.04869538  | 0.02000750 | -0.09505843  |
| Lag 1000                                                     | -0.05831890  | 0.02560894 | -0.08617648  |
| Lag 5000                                                     | 0.04246982   | 0.06392166 | -0.03722137  |
| traittarsus.2.mother traitbwt.2.mother                       |              |            |              |
| Lag 0                                                        | -0.026436938 | 0.04826105 |              |
| Lag 100                                                      | -0.041329867 | 0.04078039 |              |
| Lag 500                                                      | -0.028290001 | 0.03056152 |              |

---

|                                                                  |               |             |
|------------------------------------------------------------------|---------------|-------------|
| Lag 1000                                                         | -0.036326123  | 0.04300492  |
| Lag 5000                                                         | -0.008439906  | 0.00653221  |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |             |
| Lag 0                                                            | -0.3969629895 | -0.44235441 |
| Lag 100                                                          | -0.3457725771 | -0.37788435 |
| Lag 500                                                          | -0.2409128696 | -0.27612777 |
| Lag 1000                                                         | -0.2064627528 | -0.23939733 |
| Lag 5000                                                         | -0.0008284304 | -0.05932951 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |               |             |
| Lag 0                                                            | 0.1332860     | 0.1894193   |
| Lag 100                                                          | 0.1542105     | 0.2129929   |
| Lag 500                                                          | 0.1772766     | 0.2426247   |
| Lag 1000                                                         | 0.1956287     | 0.2577964   |
| Lag 5000                                                         | 0.1561845     | 0.1623293   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |               |             |
| Lag 0                                                            | -0.44235441   | -0.3300209  |
| Lag 100                                                          | -0.37788435   | -0.2868342  |
| Lag 500                                                          | -0.27612777   | -0.2200427  |
| Lag 1000                                                         | -0.23939733   | -0.2196364  |
| Lag 5000                                                         | -0.05932951   | -0.1105188  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |               |             |
| Lag 0                                                            | 0.1392793     | 0.1845588   |
| Lag 100                                                          | 0.1618837     | 0.2068320   |
| Lag 500                                                          | 0.1824833     | 0.2322375   |
| Lag 1000                                                         | 0.2062851     | 0.2545405   |
| Lag 5000                                                         | 0.1708879     | 0.1679111   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |               |             |
| Lag 0                                                            | 0.1332860     | 0.1392793   |
| Lag 100                                                          | 0.1542105     | 0.1618837   |
| Lag 500                                                          | 0.1772766     | 0.1824833   |
| Lag 1000                                                         | 0.1956287     | 0.2062851   |
| Lag 5000                                                         | 0.1561845     | 0.1708879   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |               |             |

---

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.5541670 | -0.6200400 |
| Lag 100  | -0.5163721 | -0.5823652 |
| Lag 500  | -0.4591717 | -0.5082023 |
| Lag 1000 | -0.4081394 | -0.4277328 |
| Lag 5000 | -0.1850899 | -0.1717978 |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.1894193 | 0.1845588 |
| Lag 100  | 0.2129929 | 0.2068320 |
| Lag 500  | 0.2426247 | 0.2322375 |
| Lag 1000 | 0.2577964 | 0.2545405 |
| Lag 5000 | 0.1623293 | 0.1679111 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.6200400 | -0.5649761 |
| Lag 100  | -0.5823652 | -0.5353605 |
| Lag 500  | -0.5082023 | -0.4627844 |
| Lag 1000 | -0.4277328 | -0.3777429 |
| Lag 5000 | -0.1717978 | -0.1507443 |

, , traitbwt.1:traitbwt.2.animal

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |             |            |
|----------|-------------|------------|
| Lag 0    | 0.23631479  | 0.42935174 |
| Lag 100  | 0.19631829  | 0.33111043 |
| Lag 500  | 0.13140023  | 0.20166811 |
| Lag 1000 | 0.13084790  | 0.23051272 |
| Lag 5000 | -0.05046358 | 0.04857134 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.4651159  | 0.6478170  |
| Lag 100  | 0.4250027  | 0.5751741  |
| Lag 500  | 0.3644232  | 0.4605468  |
| Lag 1000 | 0.3695356  | 0.4130944  |
| Lag 5000 | -0.0748654 | -0.1705451 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

---

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                              | 0.42935174  | 0.50673141  |
| Lag 100                                                            | 0.33111043  | 0.36623376  |
| Lag 500                                                            | 0.20166811  | 0.24599792  |
| Lag 1000                                                           | 0.23051272  | 0.26598499  |
| Lag 5000                                                           | 0.04857134  | 0.06879203  |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.75320258  | 1.00000000  |
| Lag 100                                                            | 0.64643730  | 0.80429897  |
| Lag 500                                                            | 0.48359147  | 0.55454840  |
| Lag 1000                                                           | 0.43668798  | 0.44468678  |
| Lag 5000                                                           | 0.08606661  | -0.01807979 |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.4651159   | 0.75320258  |
| Lag 100                                                            | 0.4250027   | 0.64643730  |
| Lag 500                                                            | 0.3644232   | 0.48359147  |
| Lag 1000                                                           | 0.3695356   | 0.43668798  |
| Lag 5000                                                           | -0.0748654  | 0.08606661  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.35637772  | 0.518702070 |
| Lag 100                                                            | 0.34579510  | 0.481108274 |
| Lag 500                                                            | 0.32649959  | 0.434843612 |
| Lag 1000                                                           | 0.37931051  | 0.436821292 |
| Lag 5000                                                           | 0.03852246  | 0.006399124 |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |             |
| Lag 0                                                              | 0.6478170   | 1.00000000  |
| Lag 100                                                            | 0.5751741   | 0.80429897  |
| Lag 500                                                            | 0.4605468   | 0.55454840  |
| Lag 1000                                                           | 0.4130944   | 0.44468678  |
| Lag 5000                                                           | -0.1705451  | -0.01807979 |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |             |
| Lag 0                                                              | 0.518702070 | 0.64907876  |
| Lag 100                                                            | 0.481108274 | 0.56161598  |
| Lag 500                                                            | 0.434843612 | 0.47984513  |

---

|                                                                  |              |              |             |
|------------------------------------------------------------------|--------------|--------------|-------------|
| Lag 1000                                                         | 0.436821292  | 0.40938531   |             |
| Lag 5000                                                         | 0.006399124  | -0.01977366  |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |              |              |             |
| Lag 0                                                            | 0.034800266  | 0.007431509  | 0.10446126  |
| Lag 100                                                          | 0.051936557  | -0.001493245 | 0.08007660  |
| Lag 500                                                          | 0.071129943  | 0.067797207  | 0.05478615  |
| Lag 1000                                                         | 0.103624443  | 0.063225351  | 0.01489821  |
| Lag 5000                                                         | -0.000397122 | 0.008471637  | -0.01350094 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |              |             |
| Lag 0                                                            | -0.08333568  | 0.100888337  | -0.21005355 |
| Lag 100                                                          | -0.07762909  | 0.079323546  | -0.17806133 |
| Lag 500                                                          | -0.03825641  | -0.007478373 | -0.11051022 |
| Lag 1000                                                         | -0.02784742  | 0.017077993  | -0.10296388 |
| Lag 5000                                                         | 0.04558001   | 0.022598106  | 0.02483781  |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |             |
| Lag 0                                                            | 0.018470325  | -0.095372303 |             |
| Lag 100                                                          | -0.002048636 | -0.029630368 |             |
| Lag 500                                                          | 0.014260425  | -0.034300090 |             |
| Lag 1000                                                         | -0.048567932 | 0.067004499  |             |
| Lag 5000                                                         | -0.025983993 | 0.005631837  |             |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |             |
| Lag 0                                                            | -0.25474138  | -0.36301593  |             |
| Lag 100                                                          | -0.21067390  | -0.29399524  |             |
| Lag 500                                                          | -0.11042093  | -0.17823668  |             |
| Lag 1000                                                         | -0.13264168  | -0.21582496  |             |
| Lag 5000                                                         | 0.02716341   | -0.03930422  |             |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |             |
| Lag 0                                                            | 0.06808542   | 0.1473359    |             |
| Lag 100                                                          | 0.08333997   | 0.1644163    |             |
| Lag 500                                                          | 0.11004808   | 0.1967466    |             |
| Lag 1000                                                         | 0.13239890   | 0.2087490    |             |
| Lag 5000                                                         | 0.14121855   | 0.1213850    |             |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |              |             |



---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | -0.36301593 | -0.39024104 |
| Lag 100                                                          | -0.29399524 | -0.30803165 |
| Lag 500                                                          | -0.17823668 | -0.22428734 |
| Lag 1000                                                         | -0.21582496 | -0.24800908 |
| Lag 5000                                                         | -0.03930422 | -0.07900553 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.09800038  | 0.1786089   |
| Lag 100                                                          | 0.12840120  | 0.2150929   |
| Lag 500                                                          | 0.17479984  | 0.2654006   |
| Lag 1000                                                         | 0.19253125  | 0.2644751   |
| Lag 5000                                                         | 0.13671344  | 0.1087463   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.06808542  | 0.09800038  |
| Lag 100                                                          | 0.08333997  | 0.12840120  |
| Lag 500                                                          | 0.11004808  | 0.17479984  |
| Lag 1000                                                         | 0.13239890  | 0.19253125  |
| Lag 5000                                                         | 0.14121855  | 0.13671344  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.38430674 | -0.49584021 |
| Lag 100                                                          | -0.36892511 | -0.47162690 |
| Lag 500                                                          | -0.35875583 | -0.43506761 |
| Lag 1000                                                         | -0.38190122 | -0.42199356 |
| Lag 5000                                                         | -0.04674851 | -0.01287201 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | 0.1473359   | 0.1786089   |
| Lag 100                                                          | 0.1644163   | 0.2150929   |
| Lag 500                                                          | 0.1967466   | 0.2654006   |
| Lag 1000                                                         | 0.2087490   | 0.2644751   |
| Lag 5000                                                         | 0.1213850   | 0.1087463   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |             |
| Lag 0                                                            | -0.49584021 | -0.56988609 |
| Lag 100                                                          | -0.47162690 | -0.53124887 |
| Lag 500                                                          | -0.43506761 | -0.45896948 |

---

|          |             |             |
|----------|-------------|-------------|
| Lag 1000 | -0.42199356 | -0.40139643 |
| Lag 5000 | -0.01287201 | 0.01366036  |

, , traittarsus.2:traitbwt.2.animal

    traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | 0.12397199  | 0.14997594  |
| Lag 100  | 0.08643519  | 0.11314881  |
| Lag 500  | 0.03605225  | 0.09323271  |
| Lag 1000 | 0.03084094  | 0.10586940  |
| Lag 5000 | -0.09632022 | -0.03936224 |

    traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.5613010  | 0.6729793  |
| Lag 100  | 0.5033508  | 0.6010903  |
| Lag 500  | 0.4274531  | 0.5133373  |
| Lag 1000 | 0.3517565  | 0.4086869  |
| Lag 5000 | -0.1677216 | -0.1581302 |

    traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | 0.14997594  | 0.17131088  |
| Lag 100  | 0.11314881  | 0.15721379  |
| Lag 500  | 0.09323271  | 0.15933093  |
| Lag 1000 | 0.10586940  | 0.18501406  |
| Lag 5000 | -0.03936224 | -0.01791672 |

    traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal

|          |            |             |
|----------|------------|-------------|
| Lag 0    | 0.5008543  | 0.51870207  |
| Lag 100  | 0.4453244  | 0.46669360  |
| Lag 500  | 0.3907627  | 0.41453930  |
| Lag 1000 | 0.3366895  | 0.35501264  |
| Lag 5000 | -0.1140445 | -0.07692479 |

    traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal

|         |           |           |
|---------|-----------|-----------|
| Lag 0   | 0.5613010 | 0.5008543 |
| Lag 100 | 0.5033508 | 0.4453244 |
| Lag 500 | 0.4274531 | 0.3907627 |

---

|                                                                    |             |             |             |
|--------------------------------------------------------------------|-------------|-------------|-------------|
| Lag 1000                                                           | 0.3517565   | 0.3366895   |             |
| Lag 5000                                                           | -0.1677216  | -0.1140445  |             |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |             |             |
| Lag 0                                                              | 0.8991309   | 1.00000000  |             |
| Lag 100                                                            | 0.8359176   | 0.91646465  |             |
| Lag 500                                                            | 0.7187715   | 0.77900191  |             |
| Lag 1000                                                           | 0.5982337   | 0.63306113  |             |
| Lag 5000                                                           | 0.0176654   | -0.03136451 |             |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |             |             |
| Lag 0                                                              | 0.6729793   | 0.51870207  |             |
| Lag 100                                                            | 0.6010903   | 0.46669360  |             |
| Lag 500                                                            | 0.5133373   | 0.41453930  |             |
| Lag 1000                                                           | 0.4086869   | 0.35501264  |             |
| Lag 5000                                                           | -0.1581302  | -0.07692479 |             |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |             |             |
| Lag 0                                                              | 1.00000000  | 0.89818763  |             |
| Lag 100                                                            | 0.91646465  | 0.82112793  |             |
| Lag 500                                                            | 0.77900191  | 0.69592684  |             |
| Lag 1000                                                           | 0.63306113  | 0.55065244  |             |
| Lag 5000                                                           | -0.03136451 | -0.04949222 |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |             |             |             |
| Lag 0                                                              | 0.05124741  | 0.001646162 | 0.04068399  |
| Lag 100                                                            | 0.04921037  | 0.008930910 | 0.03759685  |
| Lag 500                                                            | 0.03899327  | 0.015514352 | 0.02580930  |
| Lag 1000                                                           | 0.06406637  | 0.051682789 | 0.02147033  |
| Lag 5000                                                           | -0.01137788 | 0.015052216 | -0.04729885 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |             |             |             |
| Lag 0                                                              | -0.12402916 | 0.08639137  | -0.09704390 |
| Lag 100                                                            | -0.10713277 | 0.09204331  | -0.10427670 |
| Lag 500                                                            | -0.07124572 | 0.09145340  | -0.11605857 |
| Lag 1000                                                           | -0.06554488 | 0.09437629  | -0.10182520 |
| Lag 5000                                                           | 0.06888625  | 0.02727995  | 0.01776003  |
| traittarsus.2.mother traitbwt.2.mother                             |             |             |             |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | -0.15610253 | 0.09891371  |
| Lag 100                                                          | -0.16408367 | 0.08304541  |
| Lag 500                                                          | -0.12517718 | 0.04261766  |
| Lag 1000                                                         | -0.09948987 | 0.04824881  |
| Lag 5000                                                         | -0.01179365 | -0.01278893 |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |             |
| Lag 0                                                            | -0.14515627 | -0.1467921  |
| Lag 100                                                          | -0.11636785 | -0.1156256  |
| Lag 500                                                          | -0.08159624 | -0.1042741  |
| Lag 1000                                                         | -0.07861755 | -0.1076860  |
| Lag 5000                                                         | 0.10186075  | 0.0712461   |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |             |
| Lag 0                                                            | 0.18051752  | 0.23807593  |
| Lag 100                                                          | 0.18953377  | 0.24534120  |
| Lag 500                                                          | 0.17920764  | 0.22900261  |
| Lag 1000                                                         | 0.16794260  | 0.21104881  |
| Lag 5000                                                         | 0.04389169  | 0.05181362  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.1467921  | -0.14376430 |
| Lag 100                                                          | -0.1156256  | -0.12366939 |
| Lag 500                                                          | -0.1042741  | -0.12078005 |
| Lag 1000                                                         | -0.1076860  | -0.15700540 |
| Lag 5000                                                         | 0.0712461   | 0.01799705  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.21778240  | 0.25816273  |
| Lag 100                                                          | 0.22684710  | 0.26502282  |
| Lag 500                                                          | 0.20506975  | 0.23751874  |
| Lag 1000                                                         | 0.19012136  | 0.21960899  |
| Lag 5000                                                         | 0.02816484  | 0.03038383  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.18051752  | 0.21778240  |
| Lag 100                                                          | 0.18953377  | 0.22684710  |
| Lag 500                                                          | 0.17920764  | 0.20506975  |

---

|                                                                    |               |             |
|--------------------------------------------------------------------|---------------|-------------|
| Lag 1000                                                           | 0.16794260    | 0.19012136  |
| Lag 5000                                                           | 0.04389169    | 0.02816484  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |               |             |
| Lag 0                                                              | -0.8750125593 | -0.93853010 |
| Lag 100                                                            | -0.8368578282 | -0.89227332 |
| Lag 500                                                            | -0.7287281819 | -0.75764556 |
| Lag 1000                                                           | -0.6175608959 | -0.62307537 |
| Lag 5000                                                           | -0.0008665459 | 0.04348967  |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |               |             |
| Lag 0                                                              | 0.23807593    | 0.25816273  |
| Lag 100                                                            | 0.24534120    | 0.26502282  |
| Lag 500                                                            | 0.22900261    | 0.23751874  |
| Lag 1000                                                           | 0.21104881    | 0.21960899  |
| Lag 5000                                                           | 0.05181362    | 0.03038383  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |               |             |
| Lag 0                                                              | -0.93853010   | -0.85888343 |
| Lag 100                                                            | -0.89227332   | -0.81357341 |
| Lag 500                                                            | -0.75764556   | -0.67868373 |
| Lag 1000                                                           | -0.62307537   | -0.54366192 |
| Lag 5000                                                           | 0.04348967    | 0.05718606  |
| , , traitbwt.2:traitbwt.2.animal                                   |               |             |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |               |             |
| Lag 0                                                              | 0.07790198    | 0.14180302  |
| Lag 100                                                            | 0.04618642    | 0.09666633  |
| Lag 500                                                            | 0.01084756    | 0.06603768  |
| Lag 1000                                                           | 0.02267992    | 0.11225922  |
| Lag 5000                                                           | -0.11897702   | -0.02765292 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |               |             |
| Lag 0                                                              | 0.4107260     | 0.6191252   |
| Lag 100                                                            | 0.3762287     | 0.5533278   |
| Lag 500                                                            | 0.3541228     | 0.4906565   |

---

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                           | 0.3161080   | 0.4110850   |
| Lag 5000                                                           | -0.1301409  | -0.1457415  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.14180302  | 0.21046169  |
| Lag 100                                                            | 0.09666633  | 0.16894953  |
| Lag 500                                                            | 0.06603768  | 0.16336058  |
| Lag 1000                                                           | 0.11225922  | 0.19977861  |
| Lag 5000                                                           | -0.02765292 | 0.02292947  |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.47574281  | 0.64907876  |
| Lag 100                                                            | 0.42046513  | 0.55150494  |
| Lag 500                                                            | 0.36650261  | 0.46465819  |
| Lag 1000                                                           | 0.33756450  | 0.39404623  |
| Lag 5000                                                           | -0.04573143 | -0.05021868 |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.4107260   | 0.47574281  |
| Lag 100                                                            | 0.3762287   | 0.42046513  |
| Lag 500                                                            | 0.3541228   | 0.36650261  |
| Lag 1000                                                           | 0.3161080   | 0.33756450  |
| Lag 5000                                                           | -0.1301409  | -0.04573143 |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.6844205   | 0.89818763  |
| Lag 100                                                            | 0.6509791   | 0.82577566  |
| Lag 500                                                            | 0.5935139   | 0.71300098  |
| Lag 1000                                                           | 0.5428655   | 0.62246323  |
| Lag 5000                                                           | 0.0305572   | -0.02000725 |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |             |
| Lag 0                                                              | 0.6191252   | 0.64907876  |
| Lag 100                                                            | 0.5533278   | 0.55150494  |
| Lag 500                                                            | 0.4906565   | 0.46465819  |
| Lag 1000                                                           | 0.4110850   | 0.39404623  |
| Lag 5000                                                           | -0.1457415  | -0.05021868 |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |             |

|                                                                  |             |              |              |
|------------------------------------------------------------------|-------------|--------------|--------------|
| Lag 0                                                            |             | 0.89818763   | 1.00000000   |
| Lag 100                                                          |             | 0.82577566   | 0.87344161   |
| Lag 500                                                          |             | 0.71300098   | 0.71296418   |
| Lag 1000                                                         |             | 0.62246323   | 0.58027397   |
| Lag 5000                                                         |             | -0.02000725  | -0.05065402  |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |             |              |              |
| Lag 0                                                            | 0.05353374  | 0.007606345  | 0.07295919   |
| Lag 100                                                          | 0.05915538  | 0.016873424  | 0.06649764   |
| Lag 500                                                          | 0.05495199  | 0.034153584  | 0.05087275   |
| Lag 1000                                                         | 0.08867289  | 0.063336260  | 0.03210710   |
| Lag 5000                                                         | 0.01157646  | 0.010648297  | -0.07267030  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |             |              |              |
| Lag 0                                                            | -0.11303991 | 0.128729447  | -0.162810059 |
| Lag 100                                                          | -0.10682241 | 0.132664466  | -0.165762472 |
| Lag 500                                                          | -0.06622430 | 0.082749840  | -0.142795999 |
| Lag 1000                                                         | -0.06796661 | 0.092840268  | -0.120838816 |
| Lag 5000                                                         | 0.07422776  | 0.007625067  | 0.007301082  |
| traittarsus.2.mother traitbwt.2.mother                           |             |              |              |
| Lag 0                                                            | -0.04298505 | -0.033866092 |              |
| Lag 100                                                          | -0.06870971 | 0.016347544  |              |
| Lag 500                                                          | -0.06335390 | 0.003806205  |              |
| Lag 1000                                                         | -0.06335680 | 0.042702082  |              |
| Lag 5000                                                         | -0.03319016 | -0.022982672 |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |              |
| Lag 0                                                            |             | -0.13106867  | -0.13241719  |
| Lag 100                                                          |             | -0.10278270  | -0.09227117  |
| Lag 500                                                          |             | -0.06959318  | -0.07769390  |
| Lag 1000                                                         |             | -0.08442761  | -0.10813364  |
| Lag 5000                                                         |             | 0.12643968   | 0.06443195   |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |              |
| Lag 0                                                            |             | 0.13071885   | 0.21162923   |
| Lag 100                                                          |             | 0.13765665   | 0.22039206   |
| Lag 500                                                          |             | 0.12262833   | 0.19792348   |

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                         | 0.11563474  | 0.17509971  |
| Lag 5000                                                         | 0.01775142  | 0.02534375  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.13241719 | -0.1400636  |
| Lag 100                                                          | -0.09227117 | -0.1032594  |
| Lag 500                                                          | -0.07769390 | -0.1120628  |
| Lag 1000                                                         | -0.10813364 | -0.1695827  |
| Lag 5000                                                         | 0.06443195  | -0.0137381  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.172783869 | 0.24161056  |
| Lag 100                                                          | 0.184368154 | 0.25618384  |
| Lag 500                                                          | 0.169565117 | 0.23188706  |
| Lag 1000                                                         | 0.152615170 | 0.20023901  |
| Lag 5000                                                         | 0.008979599 | 0.01431812  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.13071885  | 0.172783869 |
| Lag 100                                                          | 0.13765665  | 0.184368154 |
| Lag 500                                                          | 0.12262833  | 0.169565117 |
| Lag 1000                                                         | 0.11563474  | 0.152615170 |
| Lag 5000                                                         | 0.01775142  | 0.008979599 |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.70436348 | -0.85484247 |
| Lag 100                                                          | -0.67503813 | -0.80874929 |
| Lag 500                                                          | -0.61496101 | -0.69869786 |
| Lag 1000                                                         | -0.55915090 | -0.60389172 |
| Lag 5000                                                         | -0.01252908 | 0.03149869  |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | 0.21162923  | 0.24161056  |
| Lag 100                                                          | 0.22039206  | 0.25618384  |
| Lag 500                                                          | 0.19792348  | 0.23188706  |
| Lag 1000                                                         | 0.17509971  | 0.20023901  |
| Lag 5000                                                         | 0.02534375  | 0.01431812  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |             |



---

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.85484247 | -0.90703776 |
| Lag 100  | -0.80874929 | -0.84260101 |
| Lag 500  | -0.69869786 | -0.68593227 |
| Lag 1000 | -0.60389172 | -0.56191283 |
| Lag 5000 | 0.03149869  | 0.05571691  |

, , traittarsus.1.byear

    traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |               |             |
|----------|---------------|-------------|
| Lag 0    | -0.0001991343 | 0.06781216  |
| Lag 100  | 0.0149599532  | 0.07212249  |
| Lag 500  | -0.0052702707 | 0.06733748  |
| Lag 1000 | 0.0022177857  | 0.01064963  |
| Lag 5000 | -0.0477855881 | -0.02770227 |

    traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |              |             |
|----------|--------------|-------------|
| Lag 0    | 0.016204947  | 0.06346373  |
| Lag 100  | 0.029428446  | 0.06520358  |
| Lag 500  | 0.003988555  | 0.04831408  |
| Lag 1000 | -0.009687121 | -0.01510108 |
| Lag 5000 | -0.057748697 | -0.07984786 |

    traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | 0.06781216  | 0.06518811  |
| Lag 100  | 0.07212249  | 0.07480376  |
| Lag 500  | 0.06733748  | 0.11082992  |
| Lag 1000 | 0.01064963  | 0.05008650  |
| Lag 5000 | -0.02770227 | -0.01341916 |

    traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal

|          |             |              |
|----------|-------------|--------------|
| Lag 0    | 0.02942135  | 0.034800266  |
| Lag 100  | 0.02240012  | 0.032507546  |
| Lag 500  | 0.02748906  | 0.041726424  |
| Lag 1000 | 0.01673477  | 0.007404418  |
| Lag 5000 | -0.04116453 | -0.064540096 |

    traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal

|                                                                    |               |              |              |
|--------------------------------------------------------------------|---------------|--------------|--------------|
| Lag 0                                                              | 0.016204947   | 0.02942135   |              |
| Lag 100                                                            | 0.029428446   | 0.02240012   |              |
| Lag 500                                                            | 0.003988555   | 0.02748906   |              |
| Lag 1000                                                           | -0.009687121  | 0.01673477   |              |
| Lag 5000                                                           | -0.057748697  | -0.04116453  |              |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |               |              |              |
| Lag 0                                                              | 0.035961574   | 0.05124741   |              |
| Lag 100                                                            | 0.039140531   | 0.04519151   |              |
| Lag 500                                                            | -0.008894598  | 0.03130914   |              |
| Lag 1000                                                           | -0.036681278  | -0.01447981  |              |
| Lag 5000                                                           | -0.088569223  | -0.10213928  |              |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |               |              |              |
| Lag 0                                                              | 0.06346373    | 0.034800266  |              |
| Lag 100                                                            | 0.06520358    | 0.032507546  |              |
| Lag 500                                                            | 0.04831408    | 0.041726424  |              |
| Lag 1000                                                           | -0.01510108   | 0.007404418  |              |
| Lag 5000                                                           | -0.07984786   | -0.064540096 |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |               |              |              |
| Lag 0                                                              | 0.05124741    | 0.05353374   |              |
| Lag 100                                                            | 0.04519151    | 0.03671240   |              |
| Lag 500                                                            | 0.03130914    | 0.04212493   |              |
| Lag 1000                                                           | -0.01447981   | -0.01321894  |              |
| Lag 5000                                                           | -0.10213928   | -0.09784272  |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |               |              |              |
| Lag 0                                                              | 1.00000000    | -0.011500754 | -0.014207130 |
| Lag 100                                                            | 0.29498964    | 0.032775932  | 0.007424037  |
| Lag 500                                                            | 0.01534486    | -0.007063004 | -0.008141387 |
| Lag 1000                                                           | -0.00531817   | 0.074940752  | -0.088872220 |
| Lag 5000                                                           | -0.04617778   | -0.015458692 | -0.020304988 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |               |              |              |
| Lag 0                                                              | -0.0081906518 | 0.107143639  | -0.06029833  |
| Lag 100                                                            | -0.0458427594 | 0.096733446  | -0.08150148  |
| Lag 500                                                            | 0.0004877642  | 0.068025342  | -0.06046456  |

---

|                                                                  |               |              |              |
|------------------------------------------------------------------|---------------|--------------|--------------|
| Lag 1000                                                         | 0.0105349484  | -0.002189929 | -0.01019378  |
| Lag 5000                                                         | -0.0644916207 | -0.020934152 | 0.04368155   |
| traittarsus.2.mother traitbwt.2.mother                           |               |              |              |
| Lag 0                                                            | -0.03378213   | 0.028992058  |              |
| Lag 100                                                          | -0.04096317   | 0.005259162  |              |
| Lag 500                                                          | 0.04192974    | -0.033349969 |              |
| Lag 1000                                                         | -0.02480675   | -0.011058052 |              |
| Lag 5000                                                         | 0.04747283    | -0.001107231 |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |              |              |
| Lag 0                                                            |               | -0.145066040 | -0.104716754 |
| Lag 100                                                          |               | -0.089626633 | -0.078525674 |
| Lag 500                                                          |               | -0.055797909 | -0.064960700 |
| Lag 1000                                                         |               | -0.002415045 | -0.025253406 |
| Lag 5000                                                         |               | 0.031464141  | 0.006356867  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |               |              |              |
| Lag 0                                                            |               | -0.038470154 | -0.03236997  |
| Lag 100                                                          |               | -0.037857054 | -0.03429257  |
| Lag 500                                                          |               | -0.023813333 | -0.02985769  |
| Lag 1000                                                         |               | -0.004466145 | -0.01855463  |
| Lag 5000                                                         |               | 0.031916697  | 0.01037899   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |               |              |              |
| Lag 0                                                            |               | -0.104716754 | -0.071954690 |
| Lag 100                                                          |               | -0.078525674 | -0.071715488 |
| Lag 500                                                          |               | -0.064960700 | -0.078535410 |
| Lag 1000                                                         |               | -0.025253406 | -0.053257563 |
| Lag 5000                                                         |               | 0.006356867  | -0.008584678 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |               |              |              |
| Lag 0                                                            |               | -0.02749973  | -0.02370119  |
| Lag 100                                                          |               | -0.02036580  | -0.01896999  |
| Lag 500                                                          |               | -0.02537270  | -0.03447923  |
| Lag 1000                                                         |               | 0.01650965   | 0.00885959   |
| Lag 5000                                                         |               | 0.03965003   | 0.01362772   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |               |              |              |

---

|                                                                    |              |              |
|--------------------------------------------------------------------|--------------|--------------|
| Lag 0                                                              | -0.038470154 | -0.02749973  |
| Lag 100                                                            | -0.037857054 | -0.02036580  |
| Lag 500                                                            | -0.023813333 | -0.02537270  |
| Lag 1000                                                           | -0.004466145 | 0.01650965   |
| Lag 5000                                                           | 0.031916697  | 0.03965003   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |              |              |
| Lag 0                                                              | -0.029532571 | -0.04533921  |
| Lag 100                                                            | -0.006520508 | -0.01935854  |
| Lag 500                                                            | 0.015074249  | -0.01914032  |
| Lag 1000                                                           | 0.040326869  | 0.01182073   |
| Lag 5000                                                           | 0.077261030  | 0.09163015   |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |              |              |
| Lag 0                                                              | -0.03236997  | -0.02370119  |
| Lag 100                                                            | -0.03429257  | -0.01896999  |
| Lag 500                                                            | -0.02985769  | -0.03447923  |
| Lag 1000                                                           | -0.01855463  | 0.00885959   |
| Lag 5000                                                           | 0.01037899   | 0.01362772   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |              |              |
| Lag 0                                                              | -0.04533921  | -0.044815839 |
| Lag 100                                                            | -0.01935854  | -0.018188105 |
| Lag 500                                                            | -0.01914032  | -0.044430332 |
| Lag 1000                                                           | 0.01182073   | 0.006995154  |
| Lag 5000                                                           | 0.09163015   | 0.087372099  |
| , , traitbwt.1.byyear                                              |              |              |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |              |              |
| Lag 0                                                              | 0.02533600   | -0.048571545 |
| Lag 100                                                            | 0.04296781   | -0.011259967 |
| Lag 500                                                            | 0.04385496   | 0.001433897  |
| Lag 1000                                                           | 0.00311012   | -0.028185947 |
| Lag 5000                                                           | -0.03791093  | -0.035032491 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |              |              |

---

|                                                                    |              |              |
|--------------------------------------------------------------------|--------------|--------------|
| Lag 0                                                              | 0.042952342  | 0.027308707  |
| Lag 100                                                            | 0.053766787  | 0.050479200  |
| Lag 500                                                            | 0.061397390  | 0.045799389  |
| Lag 1000                                                           | 0.026376244  | 0.045575916  |
| Lag 5000                                                           | -0.007467705 | 0.004959177  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |              |              |
| Lag 0                                                              | -0.048571545 | -0.054239888 |
| Lag 100                                                            | -0.011259967 | -0.013618719 |
| Lag 500                                                            | 0.001433897  | -0.023666680 |
| Lag 1000                                                           | -0.028185947 | -0.035975542 |
| Lag 5000                                                           | -0.035032491 | -0.004849703 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |              |              |
| Lag 0                                                              | 0.01370326   | 0.007431509  |
| Lag 100                                                            | 0.02573311   | 0.031124056  |
| Lag 500                                                            | 0.01003935   | -0.020817482 |
| Lag 1000                                                           | -0.01201110  | -0.011831635 |
| Lag 5000                                                           | -0.03020983  | 0.005318089  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |              |
| Lag 0                                                              | 0.042952342  | 0.01370326   |
| Lag 100                                                            | 0.053766787  | 0.02573311   |
| Lag 500                                                            | 0.061397390  | 0.01003935   |
| Lag 1000                                                           | 0.026376244  | -0.01201110  |
| Lag 5000                                                           | -0.007467705 | -0.03020983  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |              |
| Lag 0                                                              | -0.01746914  | 0.001646162  |
| Lag 100                                                            | -0.00918084  | 0.014660207  |
| Lag 500                                                            | 0.03727433   | 0.029738658  |
| Lag 1000                                                           | 0.05233753   | 0.058558627  |
| Lag 5000                                                           | 0.05442837   | 0.036237222  |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |              |
| Lag 0                                                              | 0.027308707  | 0.007431509  |
| Lag 100                                                            | 0.050479200  | 0.031124056  |
| Lag 500                                                            | 0.045799389  | -0.020817482 |

---

|                                                                  |               |              |              |
|------------------------------------------------------------------|---------------|--------------|--------------|
| Lag 1000                                                         |               | 0.045575916  | -0.011831635 |
| Lag 5000                                                         |               | 0.004959177  | 0.005318089  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |               |              |              |
| Lag 0                                                            |               | 0.001646162  | 0.007606345  |
| Lag 100                                                          |               | 0.014660207  | 0.025027029  |
| Lag 500                                                          |               | 0.029738658  | 0.007127126  |
| Lag 1000                                                         |               | 0.058558627  | 0.051997571  |
| Lag 5000                                                         |               | 0.036237222  | 0.030982100  |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |               |              |              |
| Lag 0                                                            | -0.011500754  | 1.000000000  | -0.032565228 |
| Lag 100                                                          | -0.008778968  | 0.12997956   | -0.003228198 |
| Lag 500                                                          | -0.004505343  | 0.03863671   | -0.017827698 |
| Lag 1000                                                         | 0.025463698   | -0.01655625  | 0.013769326  |
| Lag 5000                                                         | 0.001254535   | -0.04239004  | 0.001901775  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |               |              |              |
| Lag 0                                                            | -0.039257246  | -0.02879748  | 0.065880688  |
| Lag 100                                                          | 0.015007558   | -0.03208555  | 0.058933430  |
| Lag 500                                                          | -0.005393351  | 0.03210587   | 0.004743831  |
| Lag 1000                                                         | 0.043913747   | 0.05551756   | 0.012244934  |
| Lag 5000                                                         | -0.026820593  | 0.05843803   | 0.010088100  |
| traittarsus.2.mother traitbwt.2.mother                           |               |              |              |
| Lag 0                                                            | 0.008939011   | 0.059268463  |              |
| Lag 100                                                          | -0.001571382  | 0.086064468  |              |
| Lag 500                                                          | -0.023366713  | 0.008732966  |              |
| Lag 1000                                                         | 0.002165536   | -0.002113612 |              |
| Lag 5000                                                         | -0.038345540  | -0.007683096 |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |              |              |
| Lag 0                                                            | -0.0435586501 | 0.010219509  |              |
| Lag 100                                                          | -0.0505048908 | -0.031324102 |              |
| Lag 500                                                          | -0.0630828149 | -0.004357406 |              |
| Lag 1000                                                         | 0.0199630207  | 0.069957559  |              |
| Lag 5000                                                         | 0.0005522977  | -0.005454216 |              |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |               |              |              |

---

|                                                                  |              |              |
|------------------------------------------------------------------|--------------|--------------|
| Lag 0                                                            | 0.05293519   | 0.05768879   |
| Lag 100                                                          | 0.04858011   | 0.05892110   |
| Lag 500                                                          | 0.05854491   | 0.06372575   |
| Lag 1000                                                         | 0.04969484   | 0.03639502   |
| Lag 5000                                                         | 0.02832954   | 0.02784365   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |              |
| Lag 0                                                            | 0.010219509  | -0.028183637 |
| Lag 100                                                          | -0.031324102 | -0.040106071 |
| Lag 500                                                          | -0.004357406 | 0.006591277  |
| Lag 1000                                                         | 0.069957559  | 0.063937711  |
| Lag 5000                                                         | -0.005454216 | -0.027544320 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |              |
| Lag 0                                                            | 0.04996250   | 0.04700881   |
| Lag 100                                                          | 0.04397513   | 0.04673808   |
| Lag 500                                                          | 0.04695403   | 0.04531904   |
| Lag 1000                                                         | 0.03436998   | 0.02563582   |
| Lag 5000                                                         | 0.02256523   | 0.02398399   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |              |              |
| Lag 0                                                            | 0.05293519   | 0.04996250   |
| Lag 100                                                          | 0.04858011   | 0.04397513   |
| Lag 500                                                          | 0.05854491   | 0.04695403   |
| Lag 1000                                                         | 0.04969484   | 0.03436998   |
| Lag 5000                                                         | 0.02832954   | 0.02256523   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |              |              |
| Lag 0                                                            | 0.01013932   | -0.001638125 |
| Lag 100                                                          | 0.01806400   | 0.001129150  |
| Lag 500                                                          | -0.01420490  | -0.017077159 |
| Lag 1000                                                         | -0.04995245  | -0.056833617 |
| Lag 5000                                                         | -0.04322171  | -0.023913807 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |              |              |
| Lag 0                                                            | 0.05768879   | 0.04700881   |
| Lag 100                                                          | 0.05892110   | 0.04673808   |
| Lag 500                                                          | 0.06372575   | 0.04531904   |

---

|                                                            |              |              |
|------------------------------------------------------------|--------------|--------------|
| Lag 1000                                                   | 0.03639502   | 0.02563582   |
| Lag 5000                                                   | 0.02784365   | 0.02398399   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units |              |              |
| Lag 0                                                      | -0.001638125 | -0.007601342 |
| Lag 100                                                    | 0.001129150  | -0.013307489 |
| Lag 500                                                    | -0.017077159 | -0.008568690 |
| Lag 1000                                                   | -0.056833617 | -0.049677065 |
| Lag 5000                                                   | -0.023913807 | -0.009694337 |

, , traittarsus.2.byear

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |             |             |
| Lag 0                                                              | 0.02726198  | -0.00111586 |
| Lag 100                                                            | 0.05357730  | 0.01828074  |
| Lag 500                                                            | 0.04315378  | 0.02964535  |
| Lag 1000                                                           | 0.08991282  | 0.06959065  |
| Lag 5000                                                           | 0.05518086  | 0.09966305  |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |             |             |
| Lag 0                                                              | 0.05222339  | 0.05196404  |
| Lag 100                                                            | 0.07930026  | 0.07496121  |
| Lag 500                                                            | 0.04558008  | 0.04893690  |
| Lag 1000                                                           | 0.07097163  | 0.08902753  |
| Lag 5000                                                           | 0.06038135  | 0.02474492  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |             |
| Lag 0                                                              | -0.00111586 | -0.03158434 |
| Lag 100                                                            | 0.01828074  | -0.03831807 |
| Lag 500                                                            | 0.02964535  | 0.03186606  |
| Lag 1000                                                           | 0.06959065  | 0.06084561  |
| Lag 5000                                                           | 0.09966305  | 0.04703834  |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.08861601  | 0.10446126  |
| Lag 100                                                            | 0.08291699  | 0.08928353  |
| Lag 500                                                            | 0.07447628  | 0.09304440  |



---

|                                                                    |              |             |             |
|--------------------------------------------------------------------|--------------|-------------|-------------|
| Lag 1000                                                           | 0.05734586   | 0.08937241  |             |
| Lag 5000                                                           | 0.08247995   | 0.05423744  |             |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |             |             |
| Lag 0                                                              | 0.05222339   | 0.08861601  |             |
| Lag 100                                                            | 0.07930026   | 0.08291699  |             |
| Lag 500                                                            | 0.04558008   | 0.07447628  |             |
| Lag 1000                                                           | 0.07097163   | 0.05734586  |             |
| Lag 5000                                                           | 0.06038135   | 0.08247995  |             |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |             |             |
| Lag 0                                                              | 0.007171364  | 0.04068399  |             |
| Lag 100                                                            | 0.043732946  | 0.05987656  |             |
| Lag 500                                                            | 0.015444417  | 0.03370999  |             |
| Lag 1000                                                           | 0.019544081  | 0.04502976  |             |
| Lag 5000                                                           | 0.006685864  | -0.01642810 |             |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |             |             |
| Lag 0                                                              | 0.05196404   | 0.10446126  |             |
| Lag 100                                                            | 0.07496121   | 0.08928353  |             |
| Lag 500                                                            | 0.04893690   | 0.09304440  |             |
| Lag 1000                                                           | 0.08902753   | 0.08937241  |             |
| Lag 5000                                                           | 0.02474492   | 0.05423744  |             |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |             |             |
| Lag 0                                                              | 0.04068399   | 0.07295919  |             |
| Lag 100                                                            | 0.05987656   | 0.07152626  |             |
| Lag 500                                                            | 0.03370999   | 0.07468887  |             |
| Lag 1000                                                           | 0.04502976   | 0.05593692  |             |
| Lag 5000                                                           | -0.01642810  | -0.02210630 |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |             |             |
| Lag 0                                                              | -0.014207130 | -0.03256523 | 1.00000000  |
| Lag 100                                                            | 0.004989327  | 0.01606065  | 0.26418064  |
| Lag 500                                                            | 0.044052804  | 0.03428138  | 0.09537772  |
| Lag 1000                                                           | -0.001440518 | 0.01039861  | 0.10874833  |
| Lag 5000                                                           | 0.022958892  | -0.01931981 | -0.01473189 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |              |             |             |

---

|                                                                  |               |               |               |
|------------------------------------------------------------------|---------------|---------------|---------------|
| Lag 0                                                            | 0.0396353651  | 0.00119313    | 0.009996629   |
| Lag 100                                                          | 0.0105436230  | 0.04327018    | 0.062928368   |
| Lag 500                                                          | -0.0175690949 | 0.04625644    | 0.018026408   |
| Lag 1000                                                         | -0.0005183552 | 0.06289415    | -0.081508956  |
| Lag 5000                                                         | 0.0441472029  | -0.04105115   | 0.035686320   |
| traittarsus.2.mother traitbwt.2.mother                           |               |               |               |
| Lag 0                                                            | 0.04535223    | 0.002495676   |               |
| Lag 100                                                          | 0.02030029    | 0.045447997   |               |
| Lag 500                                                          | 0.02446377    | 0.009649222   |               |
| Lag 1000                                                         | 0.03899831    | 0.024762534   |               |
| Lag 5000                                                         | -0.06415993   | 0.094134035   |               |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |               |               |
| Lag 0                                                            |               | -0.04924174   | 0.0004983836  |
| Lag 100                                                          |               | -0.07260472   | -0.0274666354 |
| Lag 500                                                          |               | -0.08762084   | -0.0545545565 |
| Lag 1000                                                         |               | -0.11600745   | -0.0941075726 |
| Lag 5000                                                         |               | -0.01429942   | -0.0306094192 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |               |               |               |
| Lag 0                                                            |               | 0.06126409    | 0.06054337    |
| Lag 100                                                          |               | 0.06861858    | 0.06794736    |
| Lag 500                                                          |               | 0.06487160    | 0.06219436    |
| Lag 1000                                                         |               | 0.05154526    | 0.06299063    |
| Lag 5000                                                         |               | -0.03961272   | -0.02077759   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |               |               |               |
| Lag 0                                                            |               | 0.0004983836  | 0.045243199   |
| Lag 100                                                          |               | -0.0274666354 | 0.002618325   |
| Lag 500                                                          |               | -0.0545545565 | -0.045086068  |
| Lag 1000                                                         |               | -0.0941075726 | -0.078265909  |
| Lag 5000                                                         |               | -0.0306094192 | -0.039652118  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |               |               |               |
| Lag 0                                                            |               | 0.04569537    | 0.05361674    |
| Lag 100                                                          |               | 0.05790943    | 0.06228633    |
| Lag 500                                                          |               | 0.06629817    | 0.07181673    |

---

|                                                                    |              |              |
|--------------------------------------------------------------------|--------------|--------------|
| Lag 1000                                                           | 0.05677668   | 0.07820270   |
| Lag 5000                                                           | -0.04900870  | -0.01725220  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units   |              |              |
| Lag 0                                                              | 0.06126409   | 0.04569537   |
| Lag 100                                                            | 0.06861858   | 0.05790943   |
| Lag 500                                                            | 0.06487160   | 0.06629817   |
| Lag 1000                                                           | 0.05154526   | 0.05677668   |
| Lag 5000                                                           | -0.03961272  | -0.04900870  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |              |              |
| Lag 0                                                              | -0.05907849  | -0.07410803  |
| Lag 100                                                            | -0.06985906  | -0.08123809  |
| Lag 500                                                            | -0.02440940  | -0.02862324  |
| Lag 1000                                                           | -0.03153841  | -0.04394287  |
| Lag 5000                                                           | 0.01609465   | 0.02558327   |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |              |              |
| Lag 0                                                              | 0.06054337   | 0.05361674   |
| Lag 100                                                            | 0.06794736   | 0.06228633   |
| Lag 500                                                            | 0.06219436   | 0.07181673   |
| Lag 1000                                                           | 0.06299063   | 0.07820270   |
| Lag 5000                                                           | -0.02077759  | -0.01725220  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |              |              |
| Lag 0                                                              | -0.07410803  | -0.09539294  |
| Lag 100                                                            | -0.08123809  | -0.10192651  |
| Lag 500                                                            | -0.02862324  | -0.04787149  |
| Lag 1000                                                           | -0.04394287  | -0.05696449  |
| Lag 5000                                                           | 0.02558327   | 0.02318439   |
| , , traitbwt.2.byear                                               |              |              |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |              |              |
| Lag 0                                                              | -0.012312644 | -0.026101857 |
| Lag 100                                                            | 0.001679568  | -0.022719854 |
| Lag 500                                                            | -0.022095064 | 0.036781391  |

---

|                                                                    |              |              |
|--------------------------------------------------------------------|--------------|--------------|
| Lag 1000                                                           | -0.044649058 | 0.008440477  |
| Lag 5000                                                           | 0.012471911  | 0.017412057  |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |              |              |
| Lag 0                                                              | -0.041612693 | -0.093374609 |
| Lag 100                                                            | -0.044670676 | -0.085938138 |
| Lag 500                                                            | -0.073851310 | -0.057467051 |
| Lag 1000                                                           | -0.112427556 | -0.073729661 |
| Lag 5000                                                           | -0.002302422 | -0.007211525 |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |              |              |
| Lag 0                                                              | -0.026101857 | -0.024899884 |
| Lag 100                                                            | -0.022719854 | -0.008317247 |
| Lag 500                                                            | 0.036781391  | 0.041716499  |
| Lag 1000                                                           | 0.008440477  | -0.015226273 |
| Lag 5000                                                           | 0.017412057  | -0.007915616 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |              |              |
| Lag 0                                                              | -0.06566102  | -0.08333568  |
| Lag 100                                                            | -0.07750349  | -0.08378159  |
| Lag 500                                                            | -0.08329811  | -0.05814502  |
| Lag 1000                                                           | -0.08817705  | -0.08501262  |
| Lag 5000                                                           | 0.01626488   | -0.01063817  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |              |
| Lag 0                                                              | -0.041612693 | -0.06566102  |
| Lag 100                                                            | -0.044670676 | -0.07750349  |
| Lag 500                                                            | -0.073851310 | -0.08329811  |
| Lag 1000                                                           | -0.112427556 | -0.08817705  |
| Lag 5000                                                           | -0.002302422 | 0.01626488   |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |              |
| Lag 0                                                              | -0.08333086  | -0.12402916  |
| Lag 100                                                            | -0.07538028  | -0.10747454  |
| Lag 500                                                            | -0.05024835  | -0.08333129  |
| Lag 1000                                                           | -0.08252216  | -0.09539109  |
| Lag 5000                                                           | -0.05270947  | -0.04030638  |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |              |

---

|                                                                  |              |              |              |
|------------------------------------------------------------------|--------------|--------------|--------------|
| Lag 0                                                            | -0.093374609 | -0.08333568  |              |
| Lag 100                                                          | -0.085938138 | -0.08378159  |              |
| Lag 500                                                          | -0.057467051 | -0.05814502  |              |
| Lag 1000                                                         | -0.073729661 | -0.08501262  |              |
| Lag 5000                                                         | -0.007211525 | -0.01063817  |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |              |              |              |
| Lag 0                                                            | -0.12402916  | -0.11303991  |              |
| Lag 100                                                          | -0.10747454  | -0.09513886  |              |
| Lag 500                                                          | -0.08333129  | -0.08666156  |              |
| Lag 1000                                                         | -0.09539109  | -0.10675958  |              |
| Lag 5000                                                         | -0.04030638  | -0.02380443  |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |              |              |              |
| Lag 0                                                            | -0.008190652 | -0.039257246 | 0.03963537   |
| Lag 100                                                          | -0.031080960 | -0.037426765 | 0.00903900   |
| Lag 500                                                          | 0.006051004  | -0.020419576 | -0.06809153  |
| Lag 1000                                                         | 0.018314138  | -0.002583670 | -0.02323337  |
| Lag 5000                                                         | -0.026798307 | -0.007024901 | -0.01177870  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |              |              |
| Lag 0                                                            | 1.000000000  | -0.03484939  | 0.019595061  |
| Lag 100                                                          | 0.148773685  | -0.05262924  | 0.013805361  |
| Lag 500                                                          | 0.023071337  | 0.01145834   | 0.001042167  |
| Lag 1000                                                         | 0.001235911  | 0.02017891   | 0.034032045  |
| Lag 5000                                                         | -0.035993191 | -0.01462895  | -0.011384188 |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |              |
| Lag 0                                                            | 0.015981233  | -0.007859673 |              |
| Lag 100                                                          | -0.022156260 | -0.004707385 |              |
| Lag 500                                                          | -0.006728665 | -0.029061201 |              |
| Lag 1000                                                         | -0.021805472 | 0.003816387  |              |
| Lag 5000                                                         | -0.009007212 | 0.012478432  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |              |
| Lag 0                                                            | 0.01051817   | 0.02175859   |              |
| Lag 100                                                          | 0.02304329   | 0.01564866   |              |
| Lag 500                                                          | 0.01520723   | -0.00804956  |              |

---

|                                                                  |             |              |
|------------------------------------------------------------------|-------------|--------------|
| Lag 1000                                                         | 0.04140673  | 0.03546143   |
| Lag 5000                                                         | -0.01488195 | -0.01947941  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |
| Lag 0                                                            | -0.04179855 | -0.03235648  |
| Lag 100                                                          | -0.04317226 | -0.03935256  |
| Lag 500                                                          | -0.03459143 | -0.04109499  |
| Lag 1000                                                         | -0.03581518 | -0.03377959  |
| Lag 5000                                                         | -0.04487869 | -0.06427439  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |              |
| Lag 0                                                            | 0.02175859  | 0.008779544  |
| Lag 100                                                          | 0.01564866  | 0.004367987  |
| Lag 500                                                          | -0.00804956 | -0.020584034 |
| Lag 1000                                                         | 0.03546143  | 0.024592354  |
| Lag 5000                                                         | -0.01947941 | -0.008436479 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |              |
| Lag 0                                                            | -0.05587517 | -0.03458419  |
| Lag 100                                                          | -0.04924453 | -0.03546666  |
| Lag 500                                                          | -0.03327777 | -0.03542884  |
| Lag 1000                                                         | -0.02872355 | -0.01384796  |
| Lag 5000                                                         | -0.05472119 | -0.06947879  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |              |
| Lag 0                                                            | -0.04179855 | -0.05587517  |
| Lag 100                                                          | -0.04317226 | -0.04924453  |
| Lag 500                                                          | -0.03459143 | -0.03327777  |
| Lag 1000                                                         | -0.03581518 | -0.02872355  |
| Lag 5000                                                         | -0.04487869 | -0.05472119  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |              |
| Lag 0                                                            | 0.07499456  | 0.09224924   |
| Lag 100                                                          | 0.07208086  | 0.09762890   |
| Lag 500                                                          | 0.07251351  | 0.09224810   |
| Lag 1000                                                         | 0.08656386  | 0.08963227   |
| Lag 5000                                                         | 0.01938132  | 0.01875251   |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |              |

---

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.03235648 | -0.03458419 |
| Lag 100  | -0.03935256 | -0.03546666 |
| Lag 500  | -0.04109499 | -0.03542884 |
| Lag 1000 | -0.03377959 | -0.01384796 |
| Lag 5000 | -0.06427439 | -0.06947879 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.09224924 | 0.07926090 |
| Lag 100  | 0.09762890 | 0.09616451 |
| Lag 500  | 0.09224810 | 0.09549623 |
| Lag 1000 | 0.08963227 | 0.09029398 |
| Lag 5000 | 0.01875251 | 0.01366050 |

, , traittarsus.1.mother

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |             |              |
|----------|-------------|--------------|
| Lag 0    | -0.15078835 | 0.015326642  |
| Lag 100  | -0.14463422 | 0.003365171  |
| Lag 500  | -0.15804338 | -0.076509373 |
| Lag 1000 | -0.09886124 | -0.083743318 |
| Lag 5000 | -0.10523556 | -0.044881123 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |             |              |
|----------|-------------|--------------|
| Lag 0    | -0.10295266 | 0.020069065  |
| Lag 100  | -0.08557369 | 0.024944654  |
| Lag 500  | -0.09200768 | -0.007032787 |
| Lag 1000 | -0.07176616 | -0.035638619 |
| Lag 5000 | -0.04022520 | -0.053311401 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|          |              |              |
|----------|--------------|--------------|
| Lag 0    | 0.015326642  | 0.063535708  |
| Lag 100  | 0.003365171  | 0.050510724  |
| Lag 500  | -0.076509373 | 0.005284771  |
| Lag 1000 | -0.083743318 | -0.009316143 |
| Lag 5000 | -0.044881123 | 0.017585405  |

traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal

|                                                                    |              |              |              |
|--------------------------------------------------------------------|--------------|--------------|--------------|
| Lag 0                                                              | 0.01026682   |              | 0.10088834   |
| Lag 100                                                            | 0.02777302   |              | 0.10524313   |
| Lag 500                                                            | -0.02208151  |              | 0.06189318   |
| Lag 1000                                                           | -0.03691548  |              | 0.02840216   |
| Lag 5000                                                           | -0.05057754  |              | -0.05628831  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |              |              |
| Lag 0                                                              | -0.10295266  |              | 0.01026682   |
| Lag 100                                                            | -0.08557369  |              | 0.02777302   |
| Lag 500                                                            | -0.09200768  |              | -0.02208151  |
| Lag 1000                                                           | -0.07176616  |              | -0.03691548  |
| Lag 5000                                                           | -0.04022520  |              | -0.05057754  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |              |              |
| Lag 0                                                              | 0.046514036  |              | 0.086391373  |
| Lag 100                                                            | 0.041591585  |              | 0.092682345  |
| Lag 500                                                            | 0.014159903  |              | 0.054749349  |
| Lag 1000                                                           | -0.006046443 |              | 0.037148101  |
| Lag 5000                                                           | 0.023770495  |              | -0.002799893 |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |              |              |
| Lag 0                                                              | 0.020069065  |              | 0.10088834   |
| Lag 100                                                            | 0.024944654  |              | 0.10524313   |
| Lag 500                                                            | -0.007032787 |              | 0.06189318   |
| Lag 1000                                                           | -0.035638619 |              | 0.02840216   |
| Lag 5000                                                           | -0.053311401 |              | -0.05628831  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |              |              |
| Lag 0                                                              | 0.086391373  |              | 0.128729447  |
| Lag 100                                                            | 0.092682345  |              | 0.137837082  |
| Lag 500                                                            | 0.054749349  |              | 0.091445074  |
| Lag 1000                                                           | 0.037148101  |              | 0.065510680  |
| Lag 5000                                                           | -0.002799893 |              | -0.009892994 |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |              |              |
| Lag 0                                                              | 0.10714364   | -0.028797482 | 0.001193130  |
| Lag 100                                                            | 0.12475726   | -0.002859105 | 0.017423692  |
| Lag 500                                                            | 0.05635178   | 0.024729770  | 0.044602685  |



---

|                                                                  |             |              |              |
|------------------------------------------------------------------|-------------|--------------|--------------|
| Lag 1000                                                         | -0.01346005 | 0.015026664  | -0.004151145 |
| Lag 5000                                                         | -0.03482857 | -0.062611057 | -0.134069816 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |             |              |              |
| Lag 0                                                            | -0.03484939 | 1.00000000   | -0.271213796 |
| Lag 100                                                          | -0.06696141 | 0.63404978   | -0.170737492 |
| Lag 500                                                          | -0.04137534 | 0.29008219   | -0.136893424 |
| Lag 1000                                                         | 0.02864811  | 0.07345948   | -0.100567641 |
| Lag 5000                                                         | 0.02750431  | -0.03892620  | 0.007641615  |
| traittarsus.2.mother traitbwt.2.mother                           |             |              |              |
| Lag 0                                                            | 0.03387619  | -0.05622667  |              |
| Lag 100                                                          | 0.07299528  | -0.04780327  |              |
| Lag 500                                                          | 0.04150672  | -0.03149897  |              |
| Lag 1000                                                         | 0.02244428  | 0.03981911   |              |
| Lag 5000                                                         | -0.02167543 | 0.03773247   |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |              |
| Lag 0                                                            |             | -0.28511353  | -0.039522955 |
| Lag 100                                                          |             | -0.16904760  | -0.001067247 |
| Lag 500                                                          |             | -0.02585760  | 0.071889419  |
| Lag 1000                                                         |             | 0.03550783   | 0.091662892  |
| Lag 5000                                                         |             | 0.11461631   | 0.071952509  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |              |
| Lag 0                                                            |             | 0.1697424    | 0.17038749   |
| Lag 100                                                          |             | 0.1698287    | 0.16772504   |
| Lag 500                                                          |             | 0.1774711    | 0.18621633   |
| Lag 1000                                                         |             | 0.1539551    | 0.18005891   |
| Lag 5000                                                         |             | 0.1039239    | 0.09992436   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |              |              |
| Lag 0                                                            |             | -0.039522955 | 0.02826569   |
| Lag 100                                                          |             | -0.001067247 | 0.01968611   |
| Lag 500                                                          |             | 0.071889419  | 0.06722240   |
| Lag 1000                                                         |             | 0.091662892  | 0.06515118   |
| Lag 5000                                                         |             | 0.071952509  | -0.00146006  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |              |              |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | 0.1570314   | 0.15014199  |
| Lag 100                                                          | 0.1589658   | 0.15363210  |
| Lag 500                                                          | 0.1753944   | 0.19214621  |
| Lag 1000                                                         | 0.1525229   | 0.17920894  |
| Lag 5000                                                         | 0.0981355   | 0.09929337  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.1697424   | 0.1570314   |
| Lag 100                                                          | 0.1698287   | 0.1589658   |
| Lag 500                                                          | 0.1774711   | 0.1753944   |
| Lag 1000                                                         | 0.1539551   | 0.1525229   |
| Lag 5000                                                         | 0.1039239   | 0.0981355   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.05365944 | -0.08507034 |
| Lag 100                                                          | -0.05114312 | -0.08650307 |
| Lag 500                                                          | -0.03294118 | -0.06365200 |
| Lag 1000                                                         | -0.01342963 | -0.04929539 |
| Lag 5000                                                         | -0.03255967 | -0.00364547 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | 0.17038749  | 0.15014199  |
| Lag 100                                                          | 0.16772504  | 0.15363210  |
| Lag 500                                                          | 0.18621633  | 0.19214621  |
| Lag 1000                                                         | 0.18005891  | 0.17920894  |
| Lag 5000                                                         | 0.09992436  | 0.09929337  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |             |
| Lag 0                                                            | -0.08507034 | -0.11755479 |
| Lag 100                                                          | -0.08650307 | -0.11806828 |
| Lag 500                                                          | -0.06365200 | -0.08269424 |
| Lag 1000                                                         | -0.04929539 | -0.07064952 |
| Lag 5000                                                         | -0.00364547 | 0.01527502  |

, , traitbwt.1.mother

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

---

|                                                                    |              |              |
|--------------------------------------------------------------------|--------------|--------------|
| Lag 0                                                              | 0.10750858   | -0.015001711 |
| Lag 100                                                            | 0.11345621   | -0.018373787 |
| Lag 500                                                            | 0.12529867   | 0.019269248  |
| Lag 1000                                                           | 0.09676891   | 0.063965485  |
| Lag 5000                                                           | 0.03295446   | 0.002551181  |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |              |              |
| Lag 0                                                              | 0.04055573   | -0.06942502  |
| Lag 100                                                            | 0.04246078   | -0.06694784  |
| Lag 500                                                            | 0.05073917   | -0.04628001  |
| Lag 1000                                                           | 0.02033431   | -0.02832344  |
| Lag 5000                                                           | 0.02405793   | 0.06116511   |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |              |              |
| Lag 0                                                              | -0.015001711 | -0.301057181 |
| Lag 100                                                            | -0.018373787 | -0.217367082 |
| Lag 500                                                            | 0.019269248  | -0.084030109 |
| Lag 1000                                                           | 0.063965485  | 0.016600029  |
| Lag 5000                                                           | 0.002551181  | 0.006633359  |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |              |              |
| Lag 0                                                              | -0.002921727 | -0.21005355  |
| Lag 100                                                            | -0.017967358 | -0.17453411  |
| Lag 500                                                            | -0.048665239 | -0.13167639  |
| Lag 1000                                                           | -0.013679253 | -0.06743430  |
| Lag 5000                                                           | -0.015338823 | 0.06036364   |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |              |
| Lag 0                                                              | 0.04055573   | -0.002921727 |
| Lag 100                                                            | 0.04246078   | -0.017967358 |
| Lag 500                                                            | 0.05073917   | -0.048665239 |
| Lag 1000                                                           | 0.02033431   | -0.013679253 |
| Lag 5000                                                           | 0.02405793   | -0.015338823 |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |              |
| Lag 0                                                              | -0.063757286 | -0.09704390  |
| Lag 100                                                            | -0.078555235 | -0.11278771  |
| Lag 500                                                            | -0.075850107 | -0.13102396  |

---

|                                                                  |               |               |              |
|------------------------------------------------------------------|---------------|---------------|--------------|
| Lag 1000                                                         | -0.051714924  | -0.09839159   |              |
| Lag 5000                                                         | 0.004416861   | 0.02809940    |              |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal     |               |               |              |
| Lag 0                                                            | -0.06942502   | -0.21005355   |              |
| Lag 100                                                          | -0.06694784   | -0.17453411   |              |
| Lag 500                                                          | -0.04628001   | -0.13167639   |              |
| Lag 1000                                                         | -0.02832344   | -0.06743430   |              |
| Lag 5000                                                         | 0.06116511    | 0.06036364    |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |               |               |              |
| Lag 0                                                            | -0.09704390   | -0.16281006   |              |
| Lag 100                                                          | -0.11278771   | -0.15402583   |              |
| Lag 500                                                          | -0.13102396   | -0.18417663   |              |
| Lag 1000                                                         | -0.09839159   | -0.13265931   |              |
| Lag 5000                                                         | 0.02809940    | 0.06774942    |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |               |               |              |
| Lag 0                                                            | -0.06029833   | 0.065880688   | 0.009996629  |
| Lag 100                                                          | -0.06060175   | 0.002155759   | 0.013541244  |
| Lag 500                                                          | 0.02852613    | -0.001260087  | 0.019839054  |
| Lag 1000                                                         | -0.02910626   | 0.001290157   | 0.048245645  |
| Lag 5000                                                         | -0.01024266   | 0.050399045   | 0.056737041  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |               |               |              |
| Lag 0                                                            | 0.0195950614  | -0.2712137958 | 1.000000000  |
| Lag 100                                                          | -0.0008933777 | -0.1720110752 | 0.379551682  |
| Lag 500                                                          | 0.0366638614  | -0.0393624856 | 0.041329213  |
| Lag 1000                                                         | -0.0151426446 | 0.0476618317  | 0.002159042  |
| Lag 5000                                                         | 0.0123140414  | 0.0001252644  | -0.018222262 |
| traittarsus.2.mother traitbwt.2.mother                           |               |               |              |
| Lag 0                                                            | 0.031492934   | 0.14519001    |              |
| Lag 100                                                          | 0.002534347   | 0.07587057    |              |
| Lag 500                                                          | -0.013284762  | 0.04165749    |              |
| Lag 1000                                                         | -0.031081556  | -0.02212041   |              |
| Lag 5000                                                         | 0.013850724   | -0.02074787   |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |               |              |

---

|                                                                  |              |              |
|------------------------------------------------------------------|--------------|--------------|
| Lag 0                                                            | 0.01973629   | 0.03665672   |
| Lag 100                                                          | -0.02293101  | 0.01498518   |
| Lag 500                                                          | -0.11066356  | -0.05474960  |
| Lag 1000                                                         | -0.09135360  | -0.03226117  |
| Lag 5000                                                         | -0.07860493  | -0.07560389  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |
| Lag 0                                                            | 0.04294093   | 0.012376673  |
| Lag 100                                                          | 0.03955764   | 0.010683501  |
| Lag 500                                                          | 0.01752126   | -0.006221666 |
| Lag 1000                                                         | 0.02219142   | -0.001297180 |
| Lag 5000                                                         | 0.01239642   | 0.030951874  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |              |
| Lag 0                                                            | 0.03665672   | -0.007047622 |
| Lag 100                                                          | 0.01498518   | 0.037977225  |
| Lag 500                                                          | -0.05474960  | 0.019970710  |
| Lag 1000                                                         | -0.03226117  | -0.007739455 |
| Lag 5000                                                         | -0.07560389  | -0.062919612 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |              |
| Lag 0                                                            | -0.002752353 | -0.03097991  |
| Lag 100                                                          | -0.003093096 | -0.03192449  |
| Lag 500                                                          | -0.009027187 | -0.04102552  |
| Lag 1000                                                         | -0.020527419 | -0.03483856  |
| Lag 5000                                                         | 0.006886321  | 0.02535835   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |              |              |
| Lag 0                                                            | 0.04294093   | -0.002752353 |
| Lag 100                                                          | 0.03955764   | -0.003093096 |
| Lag 500                                                          | 0.01752126   | -0.009027187 |
| Lag 1000                                                         | 0.02219142   | -0.020527419 |
| Lag 5000                                                         | 0.01239642   | 0.006886321  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |              |              |
| Lag 0                                                            | 0.08205572   | 0.11704839   |
| Lag 100                                                          | 0.07607819   | 0.11403531   |
| Lag 500                                                          | 0.10265484   | 0.14762922   |

---

|                                                                    |              |              |
|--------------------------------------------------------------------|--------------|--------------|
| Lag 1000                                                           | 0.06476226   | 0.10125717   |
| Lag 5000                                                           | -0.03096688  | -0.06021103  |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |              |              |
| Lag 0                                                              | 0.012376673  | -0.03097991  |
| Lag 100                                                            | 0.010683501  | -0.03192449  |
| Lag 500                                                            | -0.006221666 | -0.04102552  |
| Lag 1000                                                           | -0.001297180 | -0.03483856  |
| Lag 5000                                                           | 0.030951874  | 0.02535835   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |              |              |
| Lag 0                                                              | 0.11704839   | 0.15336397   |
| Lag 100                                                            | 0.11403531   | 0.15628950   |
| Lag 500                                                            | 0.14762922   | 0.17953508   |
| Lag 1000                                                           | 0.10125717   | 0.12282290   |
| Lag 5000                                                           | -0.06021103  | -0.09027737  |
| , , traittarsus.2.mother                                           |              |              |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |              |              |
| Lag 0                                                              | 0.01502226   | 0.051714153  |
| Lag 100                                                            | 0.02971209   | 0.049905332  |
| Lag 500                                                            | 0.07537838   | 0.055173503  |
| Lag 1000                                                           | 0.12660306   | 0.083736831  |
| Lag 5000                                                           | 0.02592685   | 0.003920255  |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |              |              |
| Lag 0                                                              | -0.15988076  | -0.026436938 |
| Lag 100                                                            | -0.12481132  | -0.008713726 |
| Lag 500                                                            | -0.06365003  | -0.003368880 |
| Lag 1000                                                           | -0.01077423  | 0.048477788  |
| Lag 5000                                                           | 0.17321257   | 0.100587628  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |              |              |
| Lag 0                                                              | 0.051714153  | 0.02830176   |
| Lag 100                                                            | 0.049905332  | 0.03057360   |
| Lag 500                                                            | 0.055173503  | 0.03073470   |

---

|                                                                    |              |               |
|--------------------------------------------------------------------|--------------|---------------|
| Lag 1000                                                           | 0.083736831  | 0.04723145    |
| Lag 5000                                                           | 0.003920255  | 0.01299831    |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |              |               |
| Lag 0                                                              | -0.01173970  | 0.0184703252  |
| Lag 100                                                            | -0.01697136  | 0.0251626879  |
| Lag 500                                                            | -0.05188574  | -0.0005911452 |
| Lag 1000                                                           | -0.04833371  | 0.0163990890  |
| Lag 5000                                                           | 0.24669771   | 0.1785739883  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |               |
| Lag 0                                                              | -0.15988076  | -0.01173970   |
| Lag 100                                                            | -0.12481132  | -0.01697136   |
| Lag 500                                                            | -0.06365003  | -0.05188574   |
| Lag 1000                                                           | -0.01077423  | -0.04833371   |
| Lag 5000                                                           | 0.17321257   | 0.24669771    |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |               |
| Lag 0                                                              | -0.3447706   | -0.1561025    |
| Lag 100                                                            | -0.2981644   | -0.1438613    |
| Lag 500                                                            | -0.2223198   | -0.1486635    |
| Lag 1000                                                           | -0.1932667   | -0.1478947    |
| Lag 5000                                                           | 0.1277913    | 0.1562519     |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |               |
| Lag 0                                                              | -0.026436938 | 0.0184703252  |
| Lag 100                                                            | -0.008713726 | 0.0251626879  |
| Lag 500                                                            | -0.003368880 | -0.0005911452 |
| Lag 1000                                                           | 0.048477788  | 0.0163990890  |
| Lag 5000                                                           | 0.100587628  | 0.1785739883  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |               |
| Lag 0                                                              | -0.1561025   | -0.04298505   |
| Lag 100                                                            | -0.1438613   | -0.03256173   |
| Lag 500                                                            | -0.1486635   | -0.08441125   |
| Lag 1000                                                           | -0.1478947   | -0.11538109   |
| Lag 5000                                                           | 0.1562519    | 0.16485805    |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |               |

---

|                                                                  |              |               |              |
|------------------------------------------------------------------|--------------|---------------|--------------|
| Lag 0                                                            | -0.033782129 | 0.0089390110  | 0.04535223   |
| Lag 100                                                          | 0.009970602  | -0.0185988821 | 0.06005459   |
| Lag 500                                                          | -0.031227653 | 0.0075697533  | -0.02997353  |
| Lag 1000                                                         | 0.011336122  | -0.0006296073 | -0.05122879  |
| Lag 5000                                                         | 0.049341702  | -0.0529101788 | 0.11732582   |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |               |              |
| Lag 0                                                            | 0.015981233  | 0.033876186   | 0.031492934  |
| Lag 100                                                          | -0.001350435 | 0.027458021   | -0.034002906 |
| Lag 500                                                          | -0.067109166 | -0.078872225  | -0.017665845 |
| Lag 1000                                                         | -0.038517473 | -0.030003877  | -0.040451967 |
| Lag 5000                                                         | -0.086683580 | -0.004367187  | 0.002148712  |
| traittarsus.2.mother traitbwt.2.mother                           |              |               |              |
| Lag 0                                                            | 1.00000000   | -0.32918284   |              |
| Lag 100                                                          | 0.57125197   | -0.19561543   |              |
| Lag 500                                                          | 0.26951589   | -0.09671746   |              |
| Lag 1000                                                         | 0.09211216   | -0.01393679   |              |
| Lag 5000                                                         | -0.02238888  | -0.04415446   |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |               |              |
| Lag 0                                                            |              | -0.003303414  | -0.033595784 |
| Lag 100                                                          |              | -0.026601660  | -0.053415941 |
| Lag 500                                                          |              | 0.023989474   | -0.005220881 |
| Lag 1000                                                         |              | -0.059547268  | -0.037570106 |
| Lag 5000                                                         |              | -0.043225022  | -0.018285255 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |               |              |
| Lag 0                                                            |              | -0.032394558  | -0.038463991 |
| Lag 100                                                          |              | -0.025527001  | -0.031296257 |
| Lag 500                                                          |              | -0.026572307  | -0.025766415 |
| Lag 1000                                                         |              | -0.039244262  | -0.042965449 |
| Lag 5000                                                         |              | -0.005888125  | -0.008787816 |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |               |              |
| Lag 0                                                            |              | -0.033595784  | -0.047856148 |
| Lag 100                                                          |              | -0.053415941  | -0.036979690 |
| Lag 500                                                          |              | -0.005220881  | 0.004374636  |



---

|                                                                  |              |              |
|------------------------------------------------------------------|--------------|--------------|
| Lag 1000                                                         | -0.037570106 | -0.014564806 |
| Lag 5000                                                         | -0.018285255 | -0.019587618 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |              |
| Lag 0                                                            | -0.044767110 | -0.050917805 |
| Lag 100                                                          | -0.038505551 | -0.046892201 |
| Lag 500                                                          | -0.026544619 | -0.030053092 |
| Lag 1000                                                         | -0.047964926 | -0.058484565 |
| Lag 5000                                                         | 0.005513663  | 0.008103974  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |              |              |
| Lag 0                                                            | -0.032394558 | -0.044767110 |
| Lag 100                                                          | -0.025527001 | -0.038505551 |
| Lag 500                                                          | -0.026572307 | -0.026544619 |
| Lag 1000                                                         | -0.039244262 | -0.047964926 |
| Lag 5000                                                         | -0.005888125 | 0.005513663  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |              |              |
| Lag 0                                                            | 0.1360740    | 0.1159929    |
| Lag 100                                                          | 0.1741515    | 0.1418886    |
| Lag 500                                                          | 0.1700049    | 0.1445076    |
| Lag 1000                                                         | 0.1777767    | 0.1521923    |
| Lag 5000                                                         | -0.1463666   | -0.1686025   |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |              |              |
| Lag 0                                                            | -0.038463991 | -0.050917805 |
| Lag 100                                                          | -0.031296257 | -0.046892201 |
| Lag 500                                                          | -0.025766415 | -0.030053092 |
| Lag 1000                                                         | -0.042965449 | -0.058484565 |
| Lag 5000                                                         | -0.008787816 | 0.008103974  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |              |              |
| Lag 0                                                            | 0.1159929    | 0.0979877    |
| Lag 100                                                          | 0.1418886    | 0.1021546    |
| Lag 500                                                          | 0.1445076    | 0.1134636    |
| Lag 1000                                                         | 0.1521923    | 0.1278208    |
| Lag 5000                                                         | -0.1686025   | -0.1687524   |

---

, , traitbwt.2.mother

```
      traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal
Lag 0                0.05082067                -0.011552553
Lag 100              0.04450359                -0.023686867
Lag 500              0.04179199                 0.004658431
Lag 1000             0.02883236                -0.006496015
Lag 5000             0.04112469                 0.028334393

      traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal
Lag 0                0.11172969                 0.04826105
Lag 100              0.10951168                 0.03784269
Lag 500              0.08924566                 0.03087286
Lag 1000             0.05576158                 0.03858511
Lag 5000            -0.02612490                 0.02719676

      traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal
Lag 0                -0.011552553                -0.068290432
Lag 100              -0.023686867                -0.061106862
Lag 500              0.004658431                -0.008368994
Lag 1000            -0.006496015                 0.002748701
Lag 5000             0.028334393                 0.008620918

      traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal
Lag 0                0.004891297                -0.095372303
Lag 100              0.018900269                -0.050527725
Lag 500              0.034858608                -0.018316704
Lag 1000             0.021136189                -0.004218270
Lag 5000            -0.083516144                 0.008525728

      traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal
Lag 0                0.11172969                 0.004891297
Lag 100              0.10951168                 0.018900269
Lag 500              0.08924566                 0.034858608
Lag 1000             0.05576158                 0.021136189
Lag 5000            -0.02612490                -0.083516144

      traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal
```

---

|                                                              |              |              |              |
|--------------------------------------------------------------|--------------|--------------|--------------|
| Lag 0                                                        | 0.15871038   | 0.09891371   |              |
| Lag 100                                                      | 0.13316302   | 0.07837814   |              |
| Lag 500                                                      | 0.10976919   | 0.06061434   |              |
| Lag 1000                                                     | 0.08100730   | 0.06929365   |              |
| Lag 5000                                                     | -0.02040836  | -0.01741179  |              |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal |              |              |              |
| Lag 0                                                        | 0.04826105   | -0.095372303 |              |
| Lag 100                                                      | 0.03784269   | -0.050527725 |              |
| Lag 500                                                      | 0.03087286   | -0.018316704 |              |
| Lag 1000                                                     | 0.03858511   | -0.004218270 |              |
| Lag 5000                                                     | 0.02719676   | 0.008525728  |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal |              |              |              |
| Lag 0                                                        | 0.09891371   | -0.033866092 |              |
| Lag 100                                                      | 0.07837814   | -0.011360083 |              |
| Lag 500                                                      | 0.06061434   | 0.012969192  |              |
| Lag 1000                                                     | 0.06929365   | 0.045752648  |              |
| Lag 5000                                                     | -0.01741179  | -0.001193243 |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear     |              |              |              |
| Lag 0                                                        | 0.02899206   | 0.05926846   | 0.0024956756 |
| Lag 100                                                      | -0.01583481  | 0.05520016   | 0.0239528518 |
| Lag 500                                                      | 0.03917478   | 0.01148328   | 0.0002419691 |
| Lag 1000                                                     | -0.03694049  | 0.02504159   | 0.0667961149 |
| Lag 5000                                                     | -0.06196001  | 0.02957462   | 0.0049061407 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother      |              |              |              |
| Lag 0                                                        | -0.007859673 | -0.05622667  | 0.14519001   |
| Lag 100                                                      | 0.010189964  | -0.05585546  | 0.13743423   |
| Lag 500                                                      | 0.007357905  | 0.02124383   | 0.03427438   |
| Lag 1000                                                     | 0.029340288  | 0.06934074   | -0.01797194  |
| Lag 5000                                                     | 0.011164278  | 0.01230641   | 0.01184408   |
| traittarsus.2.mother traitbwt.2.mother                       |              |              |              |
| Lag 0                                                        | -0.329182837 | 1.000000000  |              |
| Lag 100                                                      | -0.235728358 | 0.284175599  |              |
| Lag 500                                                      | -0.148665968 | 0.029162936  |              |

---

|                                                                  |              |              |
|------------------------------------------------------------------|--------------|--------------|
| Lag 1000                                                         | -0.007592145 | -0.008559910 |
| Lag 5000                                                         | 0.006615767  | 0.006917489  |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |
| Lag 0                                                            | -0.04849521  | -0.01247903  |
| Lag 100                                                          | -0.04229095  | -0.01179876  |
| Lag 500                                                          | -0.06190225  | -0.03479262  |
| Lag 1000                                                         | -0.06244621  | -0.01333269  |
| Lag 5000                                                         | -0.04472803  | -0.03405548  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |
| Lag 0                                                            | 0.06299708   | 0.07575889   |
| Lag 100                                                          | 0.05582189   | 0.06465068   |
| Lag 500                                                          | 0.05304268   | 0.04802379   |
| Lag 1000                                                         | 0.07638864   | 0.07282218   |
| Lag 5000                                                         | 0.04062409   | 0.05791562   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |              |
| Lag 0                                                            | -0.01247903  | -0.009223763 |
| Lag 100                                                          | -0.01179876  | -0.004801511 |
| Lag 500                                                          | -0.03479262  | -0.021159851 |
| Lag 1000                                                         | -0.01333269  | -0.005423005 |
| Lag 5000                                                         | -0.03405548  | -0.040128813 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |              |
| Lag 0                                                            | 0.07595500   | 0.08248472   |
| Lag 100                                                          | 0.06741840   | 0.07062568   |
| Lag 500                                                          | 0.04493967   | 0.04348999   |
| Lag 1000                                                         | 0.05963657   | 0.06140170   |
| Lag 5000                                                         | 0.05397711   | 0.06956257   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |              |              |
| Lag 0                                                            | 0.06299708   | 0.07595500   |
| Lag 100                                                          | 0.05582189   | 0.06741840   |
| Lag 500                                                          | 0.05304268   | 0.04493967   |
| Lag 1000                                                         | 0.07638864   | 0.05963657   |
| Lag 5000                                                         | 0.04062409   | 0.05397711   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |              |              |

---

|          |              |              |
|----------|--------------|--------------|
| Lag 0    | -0.063098741 | -0.063844474 |
| Lag 100  | -0.069885333 | -0.060995678 |
| Lag 500  | -0.075115678 | -0.052589081 |
| Lag 1000 | -0.091682781 | -0.072696501 |
| Lag 5000 | 0.004523339  | 0.009847437  |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | 0.07575889 | 0.08248472 |
| Lag 100  | 0.06465068 | 0.07062568 |
| Lag 500  | 0.04802379 | 0.04348999 |
| Lag 1000 | 0.07282218 | 0.06140170 |
| Lag 5000 | 0.05791562 | 0.06956257 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |              |              |
|----------|--------------|--------------|
| Lag 0    | -0.063844474 | -0.079128573 |
| Lag 100  | -0.060995678 | -0.038220443 |
| Lag 500  | -0.052589081 | -0.022953575 |
| Lag 1000 | -0.072696501 | -0.037549862 |
| Lag 5000 | 0.009847437  | -0.001314642 |

, , traittarsus.1:traittarsus.1.units

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.69647928 | -0.53741563 |
| Lag 100  | -0.62623971 | -0.47900505 |
| Lag 500  | -0.45854096 | -0.35990127 |
| Lag 1000 | -0.34465891 | -0.24167598 |
| Lag 5000 | -0.04540565 | -0.08362331 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.4851021 | -0.3969630 |
| Lag 100  | -0.4391087 | -0.3672331 |
| Lag 500  | -0.3514192 | -0.3303341 |
| Lag 1000 | -0.2817756 | -0.2606715 |
| Lag 5000 | -0.1379797 | -0.1850268 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

---

|                                                                    |             |            |
|--------------------------------------------------------------------|-------------|------------|
| Lag 0                                                              | -0.53741563 | -0.2245265 |
| Lag 100                                                            | -0.47900505 | -0.1935129 |
| Lag 500                                                            | -0.35990127 | -0.1387130 |
| Lag 1000                                                           | -0.24167598 | -0.1103208 |
| Lag 5000                                                           | -0.08362331 | -0.1059888 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |            |
| Lag 0                                                              | -0.3729388  | -0.2547414 |
| Lag 100                                                            | -0.3377067  | -0.2223712 |
| Lag 500                                                            | -0.2941086  | -0.2173660 |
| Lag 1000                                                           | -0.2338232  | -0.1759256 |
| Lag 5000                                                           | -0.1733008  | -0.1754188 |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |            |
| Lag 0                                                              | -0.4851021  | -0.3729388 |
| Lag 100                                                            | -0.4391087  | -0.3377067 |
| Lag 500                                                            | -0.3514192  | -0.2941086 |
| Lag 1000                                                           | -0.2817756  | -0.2338232 |
| Lag 5000                                                           | -0.1379797  | -0.1733008 |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |            |
| Lag 0                                                              | -0.10376953 | -0.1451563 |
| Lag 100                                                            | -0.07852837 | -0.1303753 |
| Lag 500                                                            | -0.06730801 | -0.1313359 |
| Lag 1000                                                           | -0.06713014 | -0.1425029 |
| Lag 5000                                                           | -0.15697371 | -0.2346428 |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |            |
| Lag 0                                                              | -0.3969630  | -0.2547414 |
| Lag 100                                                            | -0.3672331  | -0.2223712 |
| Lag 500                                                            | -0.3303341  | -0.2173660 |
| Lag 1000                                                           | -0.2606715  | -0.1759256 |
| Lag 5000                                                           | -0.1850268  | -0.1754188 |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |            |
| Lag 0                                                              | -0.1451563  | -0.1310687 |
| Lag 100                                                            | -0.1303753  | -0.1210976 |
| Lag 500                                                            | -0.1313359  | -0.1454273 |

---

|                                                                  |              |              |             |
|------------------------------------------------------------------|--------------|--------------|-------------|
| Lag 1000                                                         |              | -0.1425029   | -0.1484290  |
| Lag 5000                                                         |              | -0.2346428   | -0.2608162  |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |              |              |             |
| Lag 0                                                            | -0.14506604  | -0.043558650 | -0.04924174 |
| Lag 100                                                          | -0.12498034  | -0.053994536 | -0.03871946 |
| Lag 500                                                          | -0.05225308  | -0.005205432 | -0.05590755 |
| Lag 1000                                                         | -0.05643316  | -0.037921140 | 0.00532337  |
| Lag 5000                                                         | -0.01904738  | -0.070857345 | 0.03761236  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |              |             |
| Lag 0                                                            | 0.01051817   | -0.28511353  | 0.01973629  |
| Lag 100                                                          | 0.03224747   | -0.21849925  | 0.02374680  |
| Lag 500                                                          | 0.03546917   | -0.12897815  | 0.01858679  |
| Lag 1000                                                         | 0.01230618   | -0.03335119  | 0.03274960  |
| Lag 5000                                                         | 0.02825979   | -0.05125552  | 0.04540862  |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |             |
| Lag 0                                                            | -0.003303414 | -0.048495208 |             |
| Lag 100                                                          | -0.027103529 | -0.040346222 |             |
| Lag 500                                                          | -0.025758939 | 0.002369226  |             |
| Lag 1000                                                         | -0.025407178 | -0.024008377 |             |
| Lag 5000                                                         | -0.010658268 | 0.012747460  |             |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |             |
| Lag 0                                                            |              | 1.00000000   | 0.71266479  |
| Lag 100                                                          |              | 0.69601632   | 0.48602800  |
| Lag 500                                                          |              | 0.44391095   | 0.33331074  |
| Lag 1000                                                         |              | 0.32879898   | 0.24626220  |
| Lag 5000                                                         |              | 0.09190517   | 0.09278524  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |             |
| Lag 0                                                            |              | -0.2592850   | -0.2645685  |
| Lag 100                                                          |              | -0.2655767   | -0.2702771  |
| Lag 500                                                          |              | -0.2886811   | -0.2950022  |
| Lag 1000                                                         |              | -0.2858651   | -0.3159051  |
| Lag 5000                                                         |              | -0.2005196   | -0.2351461  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |              |             |

---

|                                                                  |            |            |
|------------------------------------------------------------------|------------|------------|
| Lag 0                                                            | 0.71266479 | 0.3606694  |
| Lag 100                                                          | 0.48602800 | 0.2474550  |
| Lag 500                                                          | 0.33331074 | 0.1595171  |
| Lag 1000                                                         | 0.24626220 | 0.1323001  |
| Lag 5000                                                         | 0.09278524 | 0.1142750  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |            |            |
| Lag 0                                                            | -0.2009925 | -0.1971901 |
| Lag 100                                                          | -0.2133808 | -0.2127295 |
| Lag 500                                                          | -0.2513257 | -0.2534202 |
| Lag 1000                                                         | -0.2644654 | -0.2860502 |
| Lag 5000                                                         | -0.2208762 | -0.2453635 |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |            |            |
| Lag 0                                                            | -0.2592850 | -0.2009925 |
| Lag 100                                                          | -0.2655767 | -0.2133808 |
| Lag 500                                                          | -0.2886811 | -0.2513257 |
| Lag 1000                                                         | -0.2858651 | -0.2644654 |
| Lag 5000                                                         | -0.2005196 | -0.2208762 |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |            |            |
| Lag 0                                                            | 0.09543017 | 0.11602826 |
| Lag 100                                                          | 0.06131639 | 0.09650511 |
| Lag 500                                                          | 0.06137108 | 0.11179013 |
| Lag 1000                                                         | 0.08079826 | 0.13634360 |
| Lag 5000                                                         | 0.18979188 | 0.24893895 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |            |            |
| Lag 0                                                            | -0.2645685 | -0.1971901 |
| Lag 100                                                          | -0.2702771 | -0.2127295 |
| Lag 500                                                          | -0.2950022 | -0.2534202 |
| Lag 1000                                                         | -0.3159051 | -0.2860502 |
| Lag 5000                                                         | -0.2351461 | -0.2453635 |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |            |            |
| Lag 0                                                            | 0.11602826 | 0.10062986 |
| Lag 100                                                          | 0.09650511 | 0.09093566 |
| Lag 500                                                          | 0.11179013 | 0.12421446 |



---

|          |            |            |
|----------|------------|------------|
| Lag 1000 | 0.13634360 | 0.14583460 |
| Lag 5000 | 0.24893895 | 0.25949670 |

, , traitbwt.1:traittarsus.1.units

|          |                                    |                                 |
|----------|------------------------------------|---------------------------------|
|          | traittarsus.1:traittarsus.1.animal | traitbwt.1:traittarsus.1.animal |
| Lag 0    | -0.5806661                         | -0.7822168                      |
| Lag 100  | -0.5195230                         | -0.6845208                      |
| Lag 500  | -0.3880444                         | -0.4391722                      |
| Lag 1000 | -0.2850697                         | -0.3219049                      |
| Lag 5000 | -0.1858774                         | -0.1615545                      |
|          | traittarsus.2:traittarsus.1.animal | traitbwt.2:traittarsus.1.animal |
| Lag 0    | -0.3551367                         | -0.4423544                      |
| Lag 100  | -0.3116917                         | -0.4053917                      |
| Lag 500  | -0.2504438                         | -0.3222535                      |
| Lag 1000 | -0.2164808                         | -0.2714011                      |
| Lag 5000 | -0.1964028                         | -0.1739972                      |
|          | traittarsus.1:traitbwt.1.animal    | traitbwt.1:traitbwt.1.animal    |
| Lag 0    | -0.7822168                         | -0.5785722                      |
| Lag 100  | -0.6845208                         | -0.5036377                      |
| Lag 500  | -0.4391722                         | -0.3105590                      |
| Lag 1000 | -0.3219049                         | -0.2500610                      |
| Lag 5000 | -0.1615545                         | -0.1051180                      |
|          | traittarsus.2:traitbwt.1.animal    | traitbwt.2:traitbwt.1.animal    |
| Lag 0    | -0.3850991                         | -0.3630159                      |
| Lag 100  | -0.3306453                         | -0.3097920                      |
| Lag 500  | -0.2840530                         | -0.2549302                      |
| Lag 1000 | -0.2403468                         | -0.2124423                      |
| Lag 5000 | -0.2587649                         | -0.2036370                      |
|          | traittarsus.1:traittarsus.2.animal | traitbwt.1:traittarsus.2.animal |
| Lag 0    | -0.3551367                         | -0.3850991                      |
| Lag 100  | -0.3116917                         | -0.3306453                      |
| Lag 500  | -0.2504438                         | -0.2840530                      |

---

|                                                                    |             |              |               |
|--------------------------------------------------------------------|-------------|--------------|---------------|
| Lag 1000                                                           | -0.2164808  | -0.2403468   |               |
| Lag 5000                                                           | -0.1964028  | -0.2587649   |               |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |              |               |
| Lag 0                                                              | -0.10398125 | -0.1467921   |               |
| Lag 100                                                            | -0.07656994 | -0.1228344   |               |
| Lag 500                                                            | -0.05666776 | -0.1265197   |               |
| Lag 1000                                                           | -0.08628503 | -0.1605079   |               |
| Lag 5000                                                           | -0.11080574 | -0.1945688   |               |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |              |               |
| Lag 0                                                              | -0.4423544  | -0.3630159   |               |
| Lag 100                                                            | -0.4053917  | -0.3097920   |               |
| Lag 500                                                            | -0.3222535  | -0.2549302   |               |
| Lag 1000                                                           | -0.2714011  | -0.2124423   |               |
| Lag 5000                                                           | -0.1739972  | -0.2036370   |               |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |              |               |
| Lag 0                                                              | -0.1467921  | -0.1324172   |               |
| Lag 100                                                            | -0.1228344  | -0.1123196   |               |
| Lag 500                                                            | -0.1265197  | -0.1443413   |               |
| Lag 1000                                                           | -0.1605079  | -0.1717954   |               |
| Lag 5000                                                           | -0.1945688  | -0.2243561   |               |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |             |              |               |
| Lag 0                                                              | -0.10471675 | 0.01021951   | 0.0004983836  |
| Lag 100                                                            | -0.08194667 | 0.01380504   | -0.0044687157 |
| Lag 500                                                            | -0.01278894 | 0.03290932   | -0.0005650915 |
| Lag 1000                                                           | -0.06407190 | -0.02170329  | 0.0530406940  |
| Lag 5000                                                           | -0.07516029 | -0.06130238  | -0.0224119290 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |             |              |               |
| Lag 0                                                              | 0.02175859  | -0.039522955 | 0.03665672    |
| Lag 100                                                            | 0.03477326  | -0.019056605 | 0.05315654    |
| Lag 500                                                            | -0.01973469 | -0.002626844 | 0.03028858    |
| Lag 1000                                                           | 0.04003712  | 0.003169525  | 0.06449203    |
| Lag 5000                                                           | 0.02550062  | -0.010727363 | 0.04503907    |
| traittarsus.2.mother traitbwt.2.mother                             |             |              |               |

---

|                                                                  |             |              |
|------------------------------------------------------------------|-------------|--------------|
| Lag 0                                                            | -0.03359578 | -0.012479028 |
| Lag 100                                                          | -0.04488434 | -0.002934766 |
| Lag 500                                                          | -0.06697376 | 0.021613573  |
| Lag 1000                                                         | -0.08457909 | 0.019481160  |
| Lag 5000                                                         | -0.08600099 | 0.051485249  |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |
| Lag 0                                                            | 0.7126648   | 1.0000000    |
| Lag 100                                                          | 0.4891102   | 0.6680602    |
| Lag 500                                                          | 0.3020606   | 0.3997173    |
| Lag 1000                                                         | 0.2345067   | 0.2999511    |
| Lag 5000                                                         | 0.1759098   | 0.1561292    |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |
| Lag 0                                                            | -0.1706739  | -0.1801399   |
| Lag 100                                                          | -0.1802375  | -0.1994176   |
| Lag 500                                                          | -0.2103921  | -0.2440747   |
| Lag 1000                                                         | -0.2087782  | -0.2510893   |
| Lag 5000                                                         | -0.1812853  | -0.1905911   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |              |
| Lag 0                                                            | 1.0000000   | 0.8197350    |
| Lag 100                                                          | 0.6680602   | 0.5536026    |
| Lag 500                                                          | 0.3997173   | 0.3326177    |
| Lag 1000                                                         | 0.2999511   | 0.2334663    |
| Lag 5000                                                         | 0.1561292   | 0.1202757    |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |              |
| Lag 0                                                            | -0.1666030  | -0.1700866   |
| Lag 100                                                          | -0.1812316  | -0.1939150   |
| Lag 500                                                          | -0.2194596  | -0.2430469   |
| Lag 1000                                                         | -0.2384353  | -0.2743602   |
| Lag 5000                                                         | -0.2008085  | -0.2014312   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |              |
| Lag 0                                                            | -0.1706739  | -0.1666030   |
| Lag 100                                                          | -0.1802375  | -0.1812316   |
| Lag 500                                                          | -0.2103921  | -0.2194596   |

---

|                                                                    |            |             |
|--------------------------------------------------------------------|------------|-------------|
| Lag 1000                                                           | -0.2087782 | -0.2384353  |
| Lag 5000                                                           | -0.1812853 | -0.2008085  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |            |             |
| Lag 0                                                              | 0.10875399 | 0.13276682  |
| Lag 100                                                            | 0.05904373 | 0.09500984  |
| Lag 500                                                            | 0.06612909 | 0.11639092  |
| Lag 1000                                                           | 0.10572947 | 0.15285062  |
| Lag 5000                                                           | 0.15545551 | 0.20383203  |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |            |             |
| Lag 0                                                              | -0.1801399 | -0.1700866  |
| Lag 100                                                            | -0.1994176 | -0.1939150  |
| Lag 500                                                            | -0.2440747 | -0.2430469  |
| Lag 1000                                                           | -0.2510893 | -0.2743602  |
| Lag 5000                                                           | -0.1905911 | -0.2014312  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |            |             |
| Lag 0                                                              | 0.13276682 | 0.11344391  |
| Lag 100                                                            | 0.09500984 | 0.08651744  |
| Lag 500                                                            | 0.11639092 | 0.13209239  |
| Lag 1000                                                           | 0.15285062 | 0.15588819  |
| Lag 5000                                                           | 0.20383203 | 0.21723655  |
| , , traittarsus.2:traittarsus.1.units                              |            |             |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |            |             |
| Lag 0                                                              | 0.1834123  | 0.124710076 |
| Lag 100                                                            | 0.1968309  | 0.130500544 |
| Lag 500                                                            | 0.2068864  | 0.139328738 |
| Lag 1000                                                           | 0.1939570  | 0.142277480 |
| Lag 5000                                                           | 0.1325680  | 0.009474997 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |            |             |
| Lag 0                                                              | 0.1906441  | 0.13328596  |
| Lag 100                                                            | 0.2063827  | 0.13989222  |
| Lag 500                                                            | 0.2249905  | 0.15953298  |

---

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                           | 0.2243627   | 0.17422740  |
| Lag 5000                                                           | 0.1612708   | 0.09845918  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.124710076 | 0.08095300  |
| Lag 100                                                            | 0.130500544 | 0.07950890  |
| Lag 500                                                            | 0.139328738 | 0.07894049  |
| Lag 1000                                                           | 0.142277480 | 0.08409986  |
| Lag 5000                                                           | 0.009474997 | -0.03476503 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.13041646  | 0.06808542  |
| Lag 100                                                            | 0.13619199  | 0.07439607  |
| Lag 500                                                            | 0.15236746  | 0.10133254  |
| Lag 1000                                                           | 0.16354531  | 0.12513909  |
| Lag 5000                                                           | 0.09727094  | 0.09702098  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.1906441   | 0.13041646  |
| Lag 100                                                            | 0.2063827   | 0.13619199  |
| Lag 500                                                            | 0.2249905   | 0.15236746  |
| Lag 1000                                                           | 0.2243627   | 0.16354531  |
| Lag 5000                                                           | 0.1612708   | 0.09727094  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.1881270   | 0.1805175   |
| Lag 100                                                            | 0.2019192   | 0.1895957   |
| Lag 500                                                            | 0.2143083   | 0.2038416   |
| Lag 1000                                                           | 0.2049762   | 0.2031078   |
| Lag 5000                                                           | 0.2015150   | 0.2236163   |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |             |
| Lag 0                                                              | 0.13328596  | 0.06808542  |
| Lag 100                                                            | 0.13989222  | 0.07439607  |
| Lag 500                                                            | 0.15953298  | 0.10133254  |
| Lag 1000                                                           | 0.17422740  | 0.12513909  |
| Lag 5000                                                           | 0.09845918  | 0.09702098  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |             |

---

|                                                                  |              |             |
|------------------------------------------------------------------|--------------|-------------|
| Lag 0                                                            | 0.1805175    | 0.1307188   |
| Lag 100                                                          | 0.1895957    | 0.1387545   |
| Lag 500                                                          | 0.2038416    | 0.1596754   |
| Lag 1000                                                         | 0.2031078    | 0.1718471   |
| Lag 5000                                                         | 0.2236163    | 0.2388384   |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |              |             |
| Lag 0                                                            | -0.03847015  | 0.05293519  |
| Lag 100                                                          | -0.03144044  | 0.04956734  |
| Lag 500                                                          | -0.01780154  | 0.05591029  |
| Lag 1000                                                         | -0.02643841  | 0.04337011  |
| Lag 5000                                                         | -0.01140358  | 0.03031863  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |             |
| Lag 0                                                            | -0.041798551 | 0.1697424   |
| Lag 100                                                          | -0.034073821 | 0.1688986   |
| Lag 500                                                          | -0.048890409 | 0.1720568   |
| Lag 1000                                                         | -0.040999443 | 0.1790535   |
| Lag 5000                                                         | -0.002784642 | 0.1836021   |
| traittarsus.2.mother traitbwt.2.mother                           |              |             |
| Lag 0                                                            | -0.03239456  | 0.06299708  |
| Lag 100                                                          | -0.03271179  | 0.06442310  |
| Lag 500                                                          | -0.02932124  | 0.06841853  |
| Lag 1000                                                         | -0.01370787  | 0.05547883  |
| Lag 5000                                                         | -0.04156138  | 0.02258640  |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |             |
| Lag 0                                                            | -0.2592850   | -0.17067395 |
| Lag 100                                                          | -0.2666810   | -0.16983312 |
| Lag 500                                                          | -0.2761829   | -0.17499382 |
| Lag 1000                                                         | -0.2748628   | -0.18004636 |
| Lag 5000                                                         | -0.2185399   | -0.04226407 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |             |
| Lag 0                                                            | 1.0000000    | 0.9603400   |
| Lag 100                                                          | 0.9783825    | 0.9444274   |
| Lag 500                                                          | 0.9365031    | 0.9054416   |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                         | 0.8964880   | 0.8617258   |
| Lag 5000                                                         | 0.6039003   | 0.6022291   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.17067395 | -0.12741602 |
| Lag 100                                                          | -0.16983312 | -0.12362277 |
| Lag 500                                                          | -0.17499382 | -0.11972769 |
| Lag 1000                                                         | -0.18004636 | -0.12502460 |
| Lag 5000                                                         | -0.04226407 | 0.01790192  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.9609935   | 0.9034487   |
| Lag 100                                                          | 0.9411829   | 0.8893146   |
| Lag 500                                                          | 0.8990207   | 0.8539291   |
| Lag 1000                                                         | 0.8622161   | 0.8157349   |
| Lag 5000                                                         | 0.5753236   | 0.5773694   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 1.0000000   | 0.9609935   |
| Lag 100                                                          | 0.9783825   | 0.9411829   |
| Lag 500                                                          | 0.9365031   | 0.8990207   |
| Lag 1000                                                         | 0.8964880   | 0.8622161   |
| Lag 5000                                                         | 0.6039003   | 0.5753236   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.1856903  | -0.1801544  |
| Lag 100                                                          | -0.1921763  | -0.1830867  |
| Lag 500                                                          | -0.2110415  | -0.2009371  |
| Lag 1000                                                         | -0.2033807  | -0.1989511  |
| Lag 5000                                                         | -0.1866476  | -0.2075534  |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | 0.9603400   | 0.9034487   |
| Lag 100                                                          | 0.9444274   | 0.8893146   |
| Lag 500                                                          | 0.9054416   | 0.8539291   |
| Lag 1000                                                         | 0.8617258   | 0.8157349   |
| Lag 5000                                                         | 0.6022291   | 0.5773694   |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |             |

---

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.1801544 | -0.1345421 |
| Lag 100  | -0.1830867 | -0.1370653 |
| Lag 500  | -0.2009371 | -0.1566901 |
| Lag 1000 | -0.1989511 | -0.1643648 |
| Lag 5000 | -0.2075534 | -0.2123302 |

, , traitbwt.2:traittarsus.1.units

|          | traittarsus.1:traittarsus.1.animal | traitbwt.1:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.1843936                          | 0.13991929                      |
| Lag 100  | 0.1916063                          | 0.14051238                      |
| Lag 500  | 0.1997384                          | 0.15350375                      |
| Lag 1000 | 0.1779242                          | 0.14333349                      |
| Lag 5000 | 0.1059309                          | 0.02408511                      |

|          | traittarsus.2:traittarsus.1.animal | traitbwt.2:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.2159001                          | 0.18941930                      |
| Lag 100  | 0.2292097                          | 0.19756460                      |
| Lag 500  | 0.2470830                          | 0.21294438                      |
| Lag 1000 | 0.2430281                          | 0.21181509                      |
| Lag 5000 | 0.1564420                          | 0.09549646                      |

|          | traittarsus.1:traitbwt.1.animal | traitbwt.1:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | 0.13991929                      | 0.11864383                   |
| Lag 100  | 0.14051238                      | 0.11475981                   |
| Lag 500  | 0.15350375                      | 0.12191537                   |
| Lag 1000 | 0.14333349                      | 0.12719171                   |
| Lag 5000 | 0.02408511                      | -0.00131242                  |

|          | traittarsus.2:traitbwt.1.animal | traitbwt.2:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | 0.1778597                       | 0.1473359                    |
| Lag 100  | 0.1832967                       | 0.1558541                    |
| Lag 500  | 0.2035349                       | 0.1829320                    |
| Lag 1000 | 0.2085231                       | 0.1947599                    |
| Lag 5000 | 0.1096782                       | 0.1076230                    |

|  | traittarsus.1:traittarsus.2.animal | traitbwt.1:traittarsus.2.animal |
|--|------------------------------------|---------------------------------|
|--|------------------------------------|---------------------------------|



---

|                                                                    |              |            |             |
|--------------------------------------------------------------------|--------------|------------|-------------|
| Lag 0                                                              | 0.2159001    | 0.1778597  |             |
| Lag 100                                                            | 0.2292097    | 0.1832967  |             |
| Lag 500                                                            | 0.2470830    | 0.2035349  |             |
| Lag 1000                                                           | 0.2430281    | 0.2085231  |             |
| Lag 5000                                                           | 0.1564420    | 0.1096782  |             |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |            |             |
| Lag 0                                                              | 0.2148002    | 0.2380759  |             |
| Lag 100                                                            | 0.2260582    | 0.2490150  |             |
| Lag 500                                                            | 0.2434250    | 0.2659527  |             |
| Lag 1000                                                           | 0.2396319    | 0.2621966  |             |
| Lag 5000                                                           | 0.2311987    | 0.2321063  |             |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |            |             |
| Lag 0                                                              | 0.18941930   | 0.1473359  |             |
| Lag 100                                                            | 0.19756460   | 0.1558541  |             |
| Lag 500                                                            | 0.21294438   | 0.1829320  |             |
| Lag 1000                                                           | 0.21181509   | 0.1947599  |             |
| Lag 5000                                                           | 0.09549646   | 0.1076230  |             |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |            |             |
| Lag 0                                                              | 0.2380759    | 0.2116292  |             |
| Lag 100                                                            | 0.2490150    | 0.2231338  |             |
| Lag 500                                                            | 0.2659527    | 0.2429222  |             |
| Lag 1000                                                           | 0.2621966    | 0.2441661  |             |
| Lag 5000                                                           | 0.2321063    | 0.2236298  |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |            |             |
| Lag 0                                                              | -0.032369970 | 0.05768879 | 0.06054337  |
| Lag 100                                                            | -0.031641858 | 0.05864059 | 0.06580478  |
| Lag 500                                                            | -0.001530924 | 0.04503326 | 0.07399288  |
| Lag 1000                                                           | -0.016005519 | 0.03161375 | 0.08585903  |
| Lag 5000                                                           | 0.010619935  | 0.01730446 | -0.02417570 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |              |            |             |
| Lag 0                                                              | -0.032356483 | 0.1703875  | 0.012376673 |
| Lag 100                                                            | -0.034654911 | 0.1647608  | 0.013581085 |
| Lag 500                                                            | -0.052597308 | 0.1600416  | 0.005553001 |

---

|                                                                  |              |             |              |
|------------------------------------------------------------------|--------------|-------------|--------------|
| Lag 1000                                                         | -0.039591608 | 0.1872703   | -0.009042770 |
| Lag 5000                                                         | 0.004455343  | 0.1972374   | -0.027185781 |
| traittarsus.2.mother traitbwt.2.mother                           |              |             |              |
| Lag 0                                                            | -0.03846399  | 0.07575889  |              |
| Lag 100                                                          | -0.03390447  | 0.06853459  |              |
| Lag 500                                                          | -0.02862102  | 0.06274714  |              |
| Lag 1000                                                         | -0.01799649  | 0.06065063  |              |
| Lag 5000                                                         | -0.05555704  | 0.02202828  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |             |              |
| Lag 0                                                            |              | -0.2645685  | -0.18013988  |
| Lag 100                                                          |              | -0.2616234  | -0.17607364  |
| Lag 500                                                          |              | -0.2652457  | -0.18034511  |
| Lag 1000                                                         |              | -0.2671219  | -0.17702256  |
| Lag 5000                                                         |              | -0.2093233  | -0.06028968  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |             |              |
| Lag 0                                                            |              | 0.9603400   | 1.0000000    |
| Lag 100                                                          |              | 0.9432077   | 0.9762869    |
| Lag 500                                                          |              | 0.9124477   | 0.9252632    |
| Lag 1000                                                         |              | 0.8838890   | 0.8792611    |
| Lag 5000                                                         |              | 0.6117319   | 0.6007465    |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |             |              |
| Lag 0                                                            |              | -0.18013988 | -0.15026999  |
| Lag 100                                                          |              | -0.17607364 | -0.14749648  |
| Lag 500                                                          |              | -0.18034511 | -0.15145709  |
| Lag 1000                                                         |              | -0.17702256 | -0.15538707  |
| Lag 5000                                                         |              | -0.06028968 | -0.02621414  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |             |              |
| Lag 0                                                            |              | 0.9303986   | 0.9527160    |
| Lag 100                                                          |              | 0.9147001   | 0.9321247    |
| Lag 500                                                          |              | 0.8850543   | 0.8884276    |
| Lag 1000                                                         |              | 0.8576398   | 0.8431799    |
| Lag 5000                                                         |              | 0.5876791   | 0.5807055    |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |              |             |              |

---

|                                                                    |            |            |
|--------------------------------------------------------------------|------------|------------|
| Lag 0                                                              | 0.9603400  | 0.9303986  |
| Lag 100                                                            | 0.9432077  | 0.9147001  |
| Lag 500                                                            | 0.9124477  | 0.8850543  |
| Lag 1000                                                           | 0.8838890  | 0.8576398  |
| Lag 5000                                                           | 0.6117319  | 0.5876791  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |            |            |
| Lag 0                                                              | -0.2006563 | -0.2276707 |
| Lag 100                                                            | -0.2124295 | -0.2354203 |
| Lag 500                                                            | -0.2379446 | -0.2556157 |
| Lag 1000                                                           | -0.2389714 | -0.2561306 |
| Lag 5000                                                           | -0.2130152 | -0.2137306 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |            |            |
| Lag 0                                                              | 1.0000000  | 0.9527160  |
| Lag 100                                                            | 0.9762869  | 0.9321247  |
| Lag 500                                                            | 0.9252632  | 0.8884276  |
| Lag 1000                                                           | 0.8792611  | 0.8431799  |
| Lag 5000                                                           | 0.6007465  | 0.5807055  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |            |            |
| Lag 0                                                              | -0.2276707 | -0.2128094 |
| Lag 100                                                            | -0.2354203 | -0.2168142 |
| Lag 500                                                            | -0.2556157 | -0.2324758 |
| Lag 1000                                                           | -0.2561306 | -0.2373888 |
| Lag 5000                                                           | -0.2137306 | -0.1974646 |
| , , traittarsus.1:traitbwt.1.units                                 |            |            |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |            |            |
| Lag 0                                                              | -0.5806661 | -0.7822168 |
| Lag 100                                                            | -0.5195230 | -0.6845208 |
| Lag 500                                                            | -0.3880444 | -0.4391722 |
| Lag 1000                                                           | -0.2850697 | -0.3219049 |
| Lag 5000                                                           | -0.1858774 | -0.1615545 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |            |            |

---

|                                                                    |             |            |
|--------------------------------------------------------------------|-------------|------------|
| Lag 0                                                              | -0.3551367  | -0.4423544 |
| Lag 100                                                            | -0.3116917  | -0.4053917 |
| Lag 500                                                            | -0.2504438  | -0.3222535 |
| Lag 1000                                                           | -0.2164808  | -0.2714011 |
| Lag 5000                                                           | -0.1964028  | -0.1739972 |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |            |
| Lag 0                                                              | -0.7822168  | -0.5785722 |
| Lag 100                                                            | -0.6845208  | -0.5036377 |
| Lag 500                                                            | -0.4391722  | -0.3105590 |
| Lag 1000                                                           | -0.3219049  | -0.2500610 |
| Lag 5000                                                           | -0.1615545  | -0.1051180 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |            |
| Lag 0                                                              | -0.3850991  | -0.3630159 |
| Lag 100                                                            | -0.3306453  | -0.3097920 |
| Lag 500                                                            | -0.2840530  | -0.2549302 |
| Lag 1000                                                           | -0.2403468  | -0.2124423 |
| Lag 5000                                                           | -0.2587649  | -0.2036370 |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |            |
| Lag 0                                                              | -0.3551367  | -0.3850991 |
| Lag 100                                                            | -0.3116917  | -0.3306453 |
| Lag 500                                                            | -0.2504438  | -0.2840530 |
| Lag 1000                                                           | -0.2164808  | -0.2403468 |
| Lag 5000                                                           | -0.1964028  | -0.2587649 |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |            |
| Lag 0                                                              | -0.10398125 | -0.1467921 |
| Lag 100                                                            | -0.07656994 | -0.1228344 |
| Lag 500                                                            | -0.05666776 | -0.1265197 |
| Lag 1000                                                           | -0.08628503 | -0.1605079 |
| Lag 5000                                                           | -0.11080574 | -0.1945688 |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |            |
| Lag 0                                                              | -0.4423544  | -0.3630159 |
| Lag 100                                                            | -0.4053917  | -0.3097920 |
| Lag 500                                                            | -0.3222535  | -0.2549302 |

---

|                                                                  |             |              |               |
|------------------------------------------------------------------|-------------|--------------|---------------|
| Lag 1000                                                         |             | -0.2714011   | -0.2124423    |
| Lag 5000                                                         |             | -0.1739972   | -0.2036370    |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |             |              |               |
| Lag 0                                                            |             | -0.1467921   | -0.1324172    |
| Lag 100                                                          |             | -0.1228344   | -0.1123196    |
| Lag 500                                                          |             | -0.1265197   | -0.1443413    |
| Lag 1000                                                         |             | -0.1605079   | -0.1717954    |
| Lag 5000                                                         |             | -0.1945688   | -0.2243561    |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |             |              |               |
| Lag 0                                                            | -0.10471675 | 0.01021951   | 0.0004983836  |
| Lag 100                                                          | -0.08194667 | 0.01380504   | -0.0044687157 |
| Lag 500                                                          | -0.01278894 | 0.03290932   | -0.0005650915 |
| Lag 1000                                                         | -0.06407190 | -0.02170329  | 0.0530406940  |
| Lag 5000                                                         | -0.07516029 | -0.06130238  | -0.0224119290 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |             |              |               |
| Lag 0                                                            | 0.02175859  | -0.039522955 | 0.03665672    |
| Lag 100                                                          | 0.03477326  | -0.019056605 | 0.05315654    |
| Lag 500                                                          | -0.01973469 | -0.002626844 | 0.03028858    |
| Lag 1000                                                         | 0.04003712  | 0.003169525  | 0.06449203    |
| Lag 5000                                                         | 0.02550062  | -0.010727363 | 0.04503907    |
| traittarsus.2.mother traitbwt.2.mother                           |             |              |               |
| Lag 0                                                            | -0.03359578 | -0.012479028 |               |
| Lag 100                                                          | -0.04488434 | -0.002934766 |               |
| Lag 500                                                          | -0.06697376 | 0.021613573  |               |
| Lag 1000                                                         | -0.08457909 | 0.019481160  |               |
| Lag 5000                                                         | -0.08600099 | 0.051485249  |               |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |               |
| Lag 0                                                            |             | 0.7126648    | 1.0000000     |
| Lag 100                                                          |             | 0.4891102    | 0.6680602     |
| Lag 500                                                          |             | 0.3020606    | 0.3997173     |
| Lag 1000                                                         |             | 0.2345067    | 0.2999511     |
| Lag 5000                                                         |             | 0.1759098    | 0.1561292     |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |               |

---

|                                                                  |            |            |
|------------------------------------------------------------------|------------|------------|
| Lag 0                                                            | -0.1706739 | -0.1801399 |
| Lag 100                                                          | -0.1802375 | -0.1994176 |
| Lag 500                                                          | -0.2103921 | -0.2440747 |
| Lag 1000                                                         | -0.2087782 | -0.2510893 |
| Lag 5000                                                         | -0.1812853 | -0.1905911 |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |            |            |
| Lag 0                                                            | 1.0000000  | 0.8197350  |
| Lag 100                                                          | 0.6680602  | 0.5536026  |
| Lag 500                                                          | 0.3997173  | 0.3326177  |
| Lag 1000                                                         | 0.2999511  | 0.2334663  |
| Lag 5000                                                         | 0.1561292  | 0.1202757  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |            |            |
| Lag 0                                                            | -0.1666030 | -0.1700866 |
| Lag 100                                                          | -0.1812316 | -0.1939150 |
| Lag 500                                                          | -0.2194596 | -0.2430469 |
| Lag 1000                                                         | -0.2384353 | -0.2743602 |
| Lag 5000                                                         | -0.2008085 | -0.2014312 |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |            |            |
| Lag 0                                                            | -0.1706739 | -0.1666030 |
| Lag 100                                                          | -0.1802375 | -0.1812316 |
| Lag 500                                                          | -0.2103921 | -0.2194596 |
| Lag 1000                                                         | -0.2087782 | -0.2384353 |
| Lag 5000                                                         | -0.1812853 | -0.2008085 |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |            |            |
| Lag 0                                                            | 0.10875399 | 0.13276682 |
| Lag 100                                                          | 0.05904373 | 0.09500984 |
| Lag 500                                                          | 0.06612909 | 0.11639092 |
| Lag 1000                                                         | 0.10572947 | 0.15285062 |
| Lag 5000                                                         | 0.15545551 | 0.20383203 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |            |            |
| Lag 0                                                            | -0.1801399 | -0.1700866 |
| Lag 100                                                          | -0.1994176 | -0.1939150 |
| Lag 500                                                          | -0.2440747 | -0.2430469 |

---

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                           | -0.2510893  | -0.2743602  |
| Lag 5000                                                           | -0.1905911  | -0.2014312  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |             |             |
| Lag 0                                                              | 0.13276682  | 0.11344391  |
| Lag 100                                                            | 0.09500984  | 0.08651744  |
| Lag 500                                                            | 0.11639092  | 0.13209239  |
| Lag 1000                                                           | 0.15285062  | 0.15588819  |
| Lag 5000                                                           | 0.20383203  | 0.21723655  |
| , , traitbwt.1:traitbwt.1.units                                    |             |             |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |             |             |
| Lag 0                                                              | -0.29723299 | -0.6444049  |
| Lag 100                                                            | -0.26194069 | -0.5583089  |
| Lag 500                                                            | -0.17460653 | -0.3429615  |
| Lag 1000                                                           | -0.09216281 | -0.2402326  |
| Lag 5000                                                           | -0.14907830 | -0.1125298  |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |             |             |
| Lag 0                                                              | -0.16353731 | -0.33002093 |
| Lag 100                                                            | -0.13444098 | -0.30242196 |
| Lag 500                                                            | -0.07860312 | -0.20182430 |
| Lag 1000                                                           | -0.07437860 | -0.16489835 |
| Lag 5000                                                           | -0.15240021 | -0.07855392 |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |             |
| Lag 0                                                              | -0.6444049  | -0.72140390 |
| Lag 100                                                            | -0.5583089  | -0.60150571 |
| Lag 500                                                            | -0.3429615  | -0.35781515 |
| Lag 1000                                                           | -0.2402326  | -0.29870892 |
| Lag 5000                                                           | -0.1125298  | -0.04717998 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | -0.2793636  | -0.3902410  |
| Lag 100                                                            | -0.2441693  | -0.3411243  |
| Lag 500                                                            | -0.1881361  | -0.2431591  |

---

|                                                                    |              |              |             |
|--------------------------------------------------------------------|--------------|--------------|-------------|
| Lag 1000                                                           | -0.1803300   | -0.2025644   |             |
| Lag 5000                                                           | -0.2243905   | -0.1275516   |             |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |              |             |
| Lag 0                                                              | -0.16353731  | -0.2793636   |             |
| Lag 100                                                            | -0.13444098  | -0.2441693   |             |
| Lag 500                                                            | -0.07860312  | -0.1881361   |             |
| Lag 1000                                                           | -0.07437860  | -0.1803300   |             |
| Lag 5000                                                           | -0.15240021  | -0.2243905   |             |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |              |             |
| Lag 0                                                              | -0.10182068  | -0.14376430  |             |
| Lag 100                                                            | -0.07461870  | -0.12129614  |             |
| Lag 500                                                            | -0.03475445  | -0.09690301  |             |
| Lag 1000                                                           | -0.07582375  | -0.13287333  |             |
| Lag 5000                                                           | -0.06172035  | -0.10552070  |             |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |              |             |
| Lag 0                                                              | -0.33002093  | -0.3902410   |             |
| Lag 100                                                            | -0.30242196  | -0.3411243   |             |
| Lag 500                                                            | -0.20182430  | -0.2431591   |             |
| Lag 1000                                                           | -0.16489835  | -0.2025644   |             |
| Lag 5000                                                           | -0.07855392  | -0.1275516   |             |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |              |             |
| Lag 0                                                              | -0.14376430  | -0.1400636   |             |
| Lag 100                                                            | -0.12129614  | -0.1267043   |             |
| Lag 500                                                            | -0.09690301  | -0.1357567   |             |
| Lag 1000                                                           | -0.13287333  | -0.1570152   |             |
| Lag 5000                                                           | -0.10552070  | -0.1203712   |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |              |             |
| Lag 0                                                              | -0.071954690 | -0.028183637 | 0.04524320  |
| Lag 100                                                            | -0.044770596 | 0.002882978  | 0.03767827  |
| Lag 500                                                            | -0.005911696 | 0.003838461  | 0.01619977  |
| Lag 1000                                                           | -0.056263708 | 0.008443638  | 0.05346543  |
| Lag 5000                                                           | -0.087263536 | -0.023015448 | -0.01948411 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |              |              |             |



---

|                                                                  |              |              |              |
|------------------------------------------------------------------|--------------|--------------|--------------|
| Lag 0                                                            | 0.008779544  | 0.028265685  | -0.007047622 |
| Lag 100                                                          | 0.013764785  | 0.026135300  | 0.054993361  |
| Lag 500                                                          | -0.033989612 | 0.002688467  | 0.096589744  |
| Lag 1000                                                         | 0.025220721  | 0.005441149  | 0.099042795  |
| Lag 5000                                                         | 0.047628522  | -0.007718921 | 0.047775710  |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |              |
| Lag 0                                                            | -0.04785615  | -0.009223763 |              |
| Lag 100                                                          | -0.05001172  | 0.003907093  |              |
| Lag 500                                                          | -0.07099090  | 0.044036802  |              |
| Lag 1000                                                         | -0.05777658  | 0.037746309  |              |
| Lag 5000                                                         | -0.09694236  | 0.069793617  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |              |
| Lag 0                                                            |              | 0.36066940   | 0.8197350    |
| Lag 100                                                          |              | 0.25257032   | 0.5640460    |
| Lag 500                                                          |              | 0.11373127   | 0.3085415    |
| Lag 1000                                                         |              | 0.07190253   | 0.2121382    |
| Lag 5000                                                         |              | 0.13152446   | 0.1159014    |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |              |
| Lag 0                                                            |              | -0.1274160   | -0.1502700   |
| Lag 100                                                          |              | -0.1314926   | -0.1663342   |
| Lag 500                                                          |              | -0.1561690   | -0.2058338   |
| Lag 1000                                                         |              | -0.1493199   | -0.1963753   |
| Lag 5000                                                         |              | -0.1353839   | -0.1194476   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |              |              |
| Lag 0                                                            |              | 0.8197350    | 1.00000000   |
| Lag 100                                                          |              | 0.5640460    | 0.66059880   |
| Lag 500                                                          |              | 0.3085415    | 0.35457689   |
| Lag 1000                                                         |              | 0.2121382    | 0.23677816   |
| Lag 5000                                                         |              | 0.1159014    | 0.06278176   |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |              |              |
| Lag 0                                                            |              | -0.1670809   | -0.1902055   |
| Lag 100                                                          |              | -0.1716919   | -0.2107271   |
| Lag 500                                                          |              | -0.1984597   | -0.2462995   |

---

|                                                                    |            |            |
|--------------------------------------------------------------------|------------|------------|
| Lag 1000                                                           | -0.2124167 | -0.2635936 |
| Lag 5000                                                           | -0.1444361 | -0.1252641 |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units   |            |            |
| Lag 0                                                              | -0.1274160 | -0.1670809 |
| Lag 100                                                            | -0.1314926 | -0.1716919 |
| Lag 500                                                            | -0.1561690 | -0.1984597 |
| Lag 1000                                                           | -0.1493199 | -0.2124167 |
| Lag 5000                                                           | -0.1353839 | -0.1444361 |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |            |            |
| Lag 0                                                              | 0.11081955 | 0.13741353 |
| Lag 100                                                            | 0.06249231 | 0.09952814 |
| Lag 500                                                            | 0.04779758 | 0.10118050 |
| Lag 1000                                                           | 0.09619023 | 0.13533567 |
| Lag 5000                                                           | 0.09963826 | 0.10989738 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |            |            |
| Lag 0                                                              | -0.1502700 | -0.1902055 |
| Lag 100                                                            | -0.1663342 | -0.2107271 |
| Lag 500                                                            | -0.2058338 | -0.2462995 |
| Lag 1000                                                           | -0.1963753 | -0.2635936 |
| Lag 5000                                                           | -0.1194476 | -0.1252641 |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |            |            |
| Lag 0                                                              | 0.13741353 | 0.1314508  |
| Lag 100                                                            | 0.09952814 | 0.1030693  |
| Lag 500                                                            | 0.10118050 | 0.1373268  |
| Lag 1000                                                           | 0.13533567 | 0.1500819  |
| Lag 5000                                                           | 0.10989738 | 0.1138216  |
| , , traittarsus.2:traitbwt.1.units                                 |            |            |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |            |            |
| Lag 0                                                              | 0.1148525  | 0.10037928 |
| Lag 100                                                            | 0.1303566  | 0.10703233 |
| Lag 500                                                            | 0.1453432  | 0.12751184 |

---

|                                                                    |            |             |
|--------------------------------------------------------------------|------------|-------------|
| Lag 1000                                                           | 0.1449075  | 0.12663225  |
| Lag 5000                                                           | 0.1652756  | 0.03681818  |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |            |             |
| Lag 0                                                              | 0.1588923  | 0.1392793   |
| Lag 100                                                            | 0.1783256  | 0.1463284   |
| Lag 500                                                            | 0.2040736  | 0.1612874   |
| Lag 1000                                                           | 0.2230315  | 0.1896293   |
| Lag 5000                                                           | 0.1750420  | 0.0827739   |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |            |             |
| Lag 0                                                              | 0.10037928 | 0.12062597  |
| Lag 100                                                            | 0.10703233 | 0.11559443  |
| Lag 500                                                            | 0.12751184 | 0.11837717  |
| Lag 1000                                                           | 0.12663225 | 0.10630852  |
| Lag 5000                                                           | 0.03681818 | -0.04309778 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |            |             |
| Lag 0                                                              | 0.1142625  | 0.09800038  |
| Lag 100                                                            | 0.1363961  | 0.11259128  |
| Lag 500                                                            | 0.1781857  | 0.14358029  |
| Lag 1000                                                           | 0.1973172  | 0.16595148  |
| Lag 5000                                                           | 0.1261037  | 0.08832747  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |            |             |
| Lag 0                                                              | 0.1588923  | 0.1142625   |
| Lag 100                                                            | 0.1783256  | 0.1363961   |
| Lag 500                                                            | 0.2040736  | 0.1781857   |
| Lag 1000                                                           | 0.2230315  | 0.1973172   |
| Lag 5000                                                           | 0.1750420  | 0.1261037   |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |            |             |
| Lag 0                                                              | 0.2227014  | 0.2177824   |
| Lag 100                                                            | 0.2383512  | 0.2331011   |
| Lag 500                                                            | 0.2644369  | 0.2625082   |
| Lag 1000                                                           | 0.2577927  | 0.2670893   |
| Lag 5000                                                           | 0.1772736  | 0.1885761   |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |            |             |

---

|                                                                  |               |             |              |
|------------------------------------------------------------------|---------------|-------------|--------------|
| Lag 0                                                            | 0.1392793     | 0.09800038  |              |
| Lag 100                                                          | 0.1463284     | 0.11259128  |              |
| Lag 500                                                          | 0.1612874     | 0.14358029  |              |
| Lag 1000                                                         | 0.1896293     | 0.16595148  |              |
| Lag 5000                                                         | 0.0827739     | 0.08832747  |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |               |             |              |
| Lag 0                                                            | 0.2177824     | 0.1727839   |              |
| Lag 100                                                          | 0.2331011     | 0.1869982   |              |
| Lag 500                                                          | 0.2625082     | 0.2195418   |              |
| Lag 1000                                                         | 0.2670893     | 0.2381230   |              |
| Lag 5000                                                         | 0.1885761     | 0.1980800   |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |               |             |              |
| Lag 0                                                            | -0.027499725  | 0.04996250  | 0.04569537   |
| Lag 100                                                          | -0.023937020  | 0.05747431  | 0.04863214   |
| Lag 500                                                          | -0.009997799  | 0.05757848  | 0.05649481   |
| Lag 1000                                                         | -0.028336813  | 0.03932991  | 0.08794045   |
| Lag 5000                                                         | -0.028421868  | 0.02850733  | -0.02604903  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |               |             |              |
| Lag 0                                                            | -0.0558751650 | 0.1570314   | -0.002752353 |
| Lag 100                                                          | -0.0390036691 | 0.1577619   | -0.005577721 |
| Lag 500                                                          | -0.0604361094 | 0.1762753   | -0.008689304 |
| Lag 1000                                                         | -0.0544777158 | 0.1711484   | -0.003879605 |
| Lag 5000                                                         | 0.0009703848  | 0.1611550   | 0.001378773  |
| traittarsus.2.mother traitbwt.2.mother                           |               |             |              |
| Lag 0                                                            | -0.04476711   | 0.07595500  |              |
| Lag 100                                                          | -0.04782246   | 0.07837204  |              |
| Lag 500                                                          | -0.04024464   | 0.07773574  |              |
| Lag 1000                                                         | -0.01332984   | 0.04580976  |              |
| Lag 5000                                                         | -0.03731338   | 0.02497852  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |             |              |
| Lag 0                                                            | -0.2009925    | -0.16660301 |              |
| Lag 100                                                          | -0.2135650    | -0.16581994 |              |
| Lag 500                                                          | -0.2387289    | -0.17888627 |              |

---

|                                                                  |             |              |
|------------------------------------------------------------------|-------------|--------------|
| Lag 1000                                                         | -0.2382720  | -0.17175572  |
| Lag 5000                                                         | -0.2290178  | -0.06304922  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |
| Lag 0                                                            | 0.9609935   | 0.9303986    |
| Lag 100                                                          | 0.9414150   | 0.9167945    |
| Lag 500                                                          | 0.9016587   | 0.8805530    |
| Lag 1000                                                         | 0.8646228   | 0.8343108    |
| Lag 5000                                                         | 0.5874768   | 0.5773601    |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |              |
| Lag 0                                                            | -0.16660301 | -0.167080898 |
| Lag 100                                                          | -0.16581994 | -0.156138622 |
| Lag 500                                                          | -0.17888627 | -0.151199166 |
| Lag 1000                                                         | -0.17175572 | -0.139083793 |
| Lag 5000                                                         | -0.06304922 | 0.007994506  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |              |
| Lag 0                                                            | 1.0000000   | 0.9457989    |
| Lag 100                                                          | 0.9647777   | 0.9182192    |
| Lag 500                                                          | 0.8967026   | 0.8587466    |
| Lag 1000                                                         | 0.8499110   | 0.8058180    |
| Lag 5000                                                         | 0.5555747   | 0.5516644    |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |              |
| Lag 0                                                            | 0.9609935   | 1.0000000    |
| Lag 100                                                          | 0.9414150   | 0.9647777    |
| Lag 500                                                          | 0.9016587   | 0.8967026    |
| Lag 1000                                                         | 0.8646228   | 0.8499110    |
| Lag 5000                                                         | 0.5874768   | 0.5555747    |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |              |
| Lag 0                                                            | -0.2146171  | -0.2167857   |
| Lag 100                                                          | -0.2184104  | -0.2221790   |
| Lag 500                                                          | -0.2572470  | -0.2584745   |
| Lag 1000                                                         | -0.2569503  | -0.2596379   |
| Lag 5000                                                         | -0.1645148  | -0.1699811   |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |              |

---

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.9303986 | 0.9457989 |
| Lag 100  | 0.9167945 | 0.9182192 |
| Lag 500  | 0.8805530 | 0.8587466 |
| Lag 1000 | 0.8343108 | 0.8058180 |
| Lag 5000 | 0.5773601 | 0.5516644 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.2167857 | -0.1800172 |
| Lag 100  | -0.2221790 | -0.1849007 |
| Lag 500  | -0.2584745 | -0.2183863 |
| Lag 1000 | -0.2596379 | -0.2235166 |
| Lag 5000 | -0.1699811 | -0.1684397 |

, , traitbwt.2:traitbwt.1.units

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |           |            |
|----------|-----------|------------|
| Lag 0    | 0.1045202 | 0.10675486 |
| Lag 100  | 0.1144301 | 0.10909010 |
| Lag 500  | 0.1244192 | 0.13415361 |
| Lag 1000 | 0.1101292 | 0.11544191 |
| Lag 5000 | 0.1331335 | 0.04367713 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |           |            |
|----------|-----------|------------|
| Lag 0    | 0.1751169 | 0.18455878 |
| Lag 100  | 0.1936998 | 0.19497627 |
| Lag 500  | 0.2199790 | 0.20989089 |
| Lag 1000 | 0.2292872 | 0.21238609 |
| Lag 5000 | 0.1693953 | 0.06254965 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|          |            |             |
|----------|------------|-------------|
| Lag 0    | 0.10675486 | 0.15466547  |
| Lag 100  | 0.10909010 | 0.15753844  |
| Lag 500  | 0.13415361 | 0.17187977  |
| Lag 1000 | 0.11544191 | 0.14889411  |
| Lag 5000 | 0.04367713 | -0.01393939 |

traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal

|                                                                    |              |            |            |
|--------------------------------------------------------------------|--------------|------------|------------|
| Lag 0                                                              | 0.1618783    | 0.17860886 |            |
| Lag 100                                                            | 0.1807130    | 0.20406604 |            |
| Lag 500                                                            | 0.2315550    | 0.24587070 |            |
| Lag 1000                                                           | 0.2433453    | 0.24000705 |            |
| Lag 5000                                                           | 0.1313066    | 0.07959284 |            |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |              |            |            |
| Lag 0                                                              | 0.1751169    | 0.1618783  |            |
| Lag 100                                                            | 0.1936998    | 0.1807130  |            |
| Lag 500                                                            | 0.2199790    | 0.2315550  |            |
| Lag 1000                                                           | 0.2292872    | 0.2433453  |            |
| Lag 5000                                                           | 0.1693953    | 0.1313066  |            |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |            |            |
| Lag 0                                                              | 0.2339355    | 0.2581627  |            |
| Lag 100                                                            | 0.2476485    | 0.2736295  |            |
| Lag 500                                                            | 0.2790724    | 0.3078386  |            |
| Lag 1000                                                           | 0.2842308    | 0.3166841  |            |
| Lag 5000                                                           | 0.2101542    | 0.1923682  |            |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |            |            |
| Lag 0                                                              | 0.18455878   | 0.17860886 |            |
| Lag 100                                                            | 0.19497627   | 0.20406604 |            |
| Lag 500                                                            | 0.20989089   | 0.24587070 |            |
| Lag 1000                                                           | 0.21238609   | 0.24000705 |            |
| Lag 5000                                                           | 0.06254965   | 0.07959284 |            |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |            |            |
| Lag 0                                                              | 0.2581627    | 0.2416106  |            |
| Lag 100                                                            | 0.2736295    | 0.2608676  |            |
| Lag 500                                                            | 0.3078386    | 0.2963880  |            |
| Lag 1000                                                           | 0.3166841    | 0.3068141  |            |
| Lag 5000                                                           | 0.1923682    | 0.1712560  |            |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |            |            |
| Lag 0                                                              | -0.023701193 | 0.04700881 | 0.05361674 |
| Lag 100                                                            | -0.021944414 | 0.05837761 | 0.06114321 |
| Lag 500                                                            | -0.004019018 | 0.04124133 | 0.05472308 |

|                                                                  |              |             |              |
|------------------------------------------------------------------|--------------|-------------|--------------|
| Lag 1000                                                         | -0.014592442 | 0.02897141  | 0.08299624   |
| Lag 5000                                                         | -0.001455363 | 0.01713092  | -0.03330245  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |             |              |
| Lag 0                                                            | -0.03458419  | 0.1501420   | -0.030979909 |
| Lag 100                                                          | -0.02882752  | 0.1495980   | -0.034573917 |
| Lag 500                                                          | -0.05044049  | 0.1542703   | -0.037881298 |
| Lag 1000                                                         | -0.04326520  | 0.1753867   | -0.049179995 |
| Lag 5000                                                         | 0.01976572   | 0.1700994   | -0.002621514 |
| traittarsus.2.mother traitbwt.2.mother                           |              |             |              |
| Lag 0                                                            | -0.05091781  | 0.08248472  |              |
| Lag 100                                                          | -0.04612708  | 0.07643625  |              |
| Lag 500                                                          | -0.04592155  | 0.07211789  |              |
| Lag 1000                                                         | -0.02428388  | 0.04810929  |              |
| Lag 5000                                                         | -0.05454058  | 0.02230047  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |             |              |
| Lag 0                                                            |              | -0.1971901  | -0.17008659  |
| Lag 100                                                          |              | -0.2006748  | -0.16466175  |
| Lag 500                                                          |              | -0.2146434  | -0.17764629  |
| Lag 1000                                                         |              | -0.2152829  | -0.15594317  |
| Lag 5000                                                         |              | -0.2163189  | -0.07951097  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |             |              |
| Lag 0                                                            |              | 0.9034487   | 0.9527160    |
| Lag 100                                                          |              | 0.8867034   | 0.9318204    |
| Lag 500                                                          |              | 0.8558412   | 0.8820862    |
| Lag 1000                                                         |              | 0.8281770   | 0.8332104    |
| Lag 5000                                                         |              | 0.5777198   | 0.5582022    |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |              |             |              |
| Lag 0                                                            |              | -0.17008659 | -0.1902055   |
| Lag 100                                                          |              | -0.16466175 | -0.1881759   |
| Lag 500                                                          |              | -0.17764629 | -0.1918480   |
| Lag 1000                                                         |              | -0.15594317 | -0.1592987   |
| Lag 5000                                                         |              | -0.07951097 | -0.0307547   |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |             |              |



---

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.9457989 | 1.0000000 |
| Lag 100  | 0.9171312 | 0.9591397 |
| Lag 500  | 0.8650556 | 0.8846934 |
| Lag 1000 | 0.8286470 | 0.8252821 |
| Lag 5000 | 0.5520607 | 0.5384118 |

traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.9034487 | 0.9457989 |
| Lag 100  | 0.8867034 | 0.9171312 |
| Lag 500  | 0.8558412 | 0.8650556 |
| Lag 1000 | 0.8281770 | 0.8286470 |
| Lag 5000 | 0.5777198 | 0.5520607 |

traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.2187484 | -0.2507033 |
| Lag 100  | -0.2283894 | -0.2593514 |
| Lag 500  | -0.2686154 | -0.2983604 |
| Lag 1000 | -0.2818961 | -0.3082824 |
| Lag 5000 | -0.1951997 | -0.1732268 |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.9527160 | 1.0000000 |
| Lag 100  | 0.9318204 | 0.9591397 |
| Lag 500  | 0.8820862 | 0.8846934 |
| Lag 1000 | 0.8332104 | 0.8252821 |
| Lag 5000 | 0.5582022 | 0.5384118 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.2507033 | -0.2443613 |
| Lag 100  | -0.2593514 | -0.2551879 |
| Lag 500  | -0.2983604 | -0.2884325 |
| Lag 1000 | -0.3082824 | -0.2942164 |
| Lag 5000 | -0.1732268 | -0.1455134 |

, , traittarsus.1:traittarsus.2.units

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                              | 0.1834123   | 0.124710076 |
| Lag 100                                                            | 0.1968309   | 0.130500544 |
| Lag 500                                                            | 0.2068864   | 0.139328738 |
| Lag 1000                                                           | 0.1939570   | 0.142277480 |
| Lag 5000                                                           | 0.1325680   | 0.009474997 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |             |             |
| Lag 0                                                              | 0.1906441   | 0.13328596  |
| Lag 100                                                            | 0.2063827   | 0.13989222  |
| Lag 500                                                            | 0.2249905   | 0.15953298  |
| Lag 1000                                                           | 0.2243627   | 0.17422740  |
| Lag 5000                                                           | 0.1612708   | 0.09845918  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.124710076 | 0.08095300  |
| Lag 100                                                            | 0.130500544 | 0.07950890  |
| Lag 500                                                            | 0.139328738 | 0.07894049  |
| Lag 1000                                                           | 0.142277480 | 0.08409986  |
| Lag 5000                                                           | 0.009474997 | -0.03476503 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | 0.13041646  | 0.06808542  |
| Lag 100                                                            | 0.13619199  | 0.07439607  |
| Lag 500                                                            | 0.15236746  | 0.10133254  |
| Lag 1000                                                           | 0.16354531  | 0.12513909  |
| Lag 5000                                                           | 0.09727094  | 0.09702098  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.1906441   | 0.13041646  |
| Lag 100                                                            | 0.2063827   | 0.13619199  |
| Lag 500                                                            | 0.2249905   | 0.15236746  |
| Lag 1000                                                           | 0.2243627   | 0.16354531  |
| Lag 5000                                                           | 0.1612708   | 0.09727094  |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |             |
| Lag 0                                                              | 0.1881270   | 0.1805175   |
| Lag 100                                                            | 0.2019192   | 0.1895957   |
| Lag 500                                                            | 0.2143083   | 0.2038416   |

---

|                                                                  |              |            |             |
|------------------------------------------------------------------|--------------|------------|-------------|
| Lag 1000                                                         | 0.2049762    | 0.2031078  |             |
| Lag 5000                                                         | 0.2015150    | 0.2236163  |             |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal     |              |            |             |
| Lag 0                                                            | 0.13328596   | 0.06808542 |             |
| Lag 100                                                          | 0.13989222   | 0.07439607 |             |
| Lag 500                                                          | 0.15953298   | 0.10133254 |             |
| Lag 1000                                                         | 0.17422740   | 0.12513909 |             |
| Lag 5000                                                         | 0.09845918   | 0.09702098 |             |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |              |            |             |
| Lag 0                                                            | 0.1805175    | 0.1307188  |             |
| Lag 100                                                          | 0.1895957    | 0.1387545  |             |
| Lag 500                                                          | 0.2038416    | 0.1596754  |             |
| Lag 1000                                                         | 0.2031078    | 0.1718471  |             |
| Lag 5000                                                         | 0.2236163    | 0.2388384  |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |              |            |             |
| Lag 0                                                            | -0.03847015  | 0.05293519 | 0.06126409  |
| Lag 100                                                          | -0.03144044  | 0.04956734 | 0.06250612  |
| Lag 500                                                          | -0.01780154  | 0.05591029 | 0.07921148  |
| Lag 1000                                                         | -0.02643841  | 0.04337011 | 0.09113494  |
| Lag 5000                                                         | -0.01140358  | 0.03031863 | -0.02594409 |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |            |             |
| Lag 0                                                            | -0.041798551 | 0.1697424  | 0.04294093  |
| Lag 100                                                          | -0.034073821 | 0.1688986  | 0.04145441  |
| Lag 500                                                          | -0.048890409 | 0.1720568  | 0.02794912  |
| Lag 1000                                                         | -0.040999443 | 0.1790535  | 0.02144402  |
| Lag 5000                                                         | -0.002784642 | 0.1836021  | -0.02796491 |
| traittarsus.2.mother traitbwt.2.mother                           |              |            |             |
| Lag 0                                                            | -0.03239456  | 0.06299708 |             |
| Lag 100                                                          | -0.03271179  | 0.06442310 |             |
| Lag 500                                                          | -0.02932124  | 0.06841853 |             |
| Lag 1000                                                         | -0.01370787  | 0.05547883 |             |
| Lag 5000                                                         | -0.04156138  | 0.02258640 |             |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |            |             |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | -0.2592850  | -0.17067395 |
| Lag 100                                                          | -0.2666810  | -0.16983312 |
| Lag 500                                                          | -0.2761829  | -0.17499382 |
| Lag 1000                                                         | -0.2748628  | -0.18004636 |
| Lag 5000                                                         | -0.2185399  | -0.04226407 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |             |
| Lag 0                                                            | 1.0000000   | 0.9603400   |
| Lag 100                                                          | 0.9783825   | 0.9444274   |
| Lag 500                                                          | 0.9365031   | 0.9054416   |
| Lag 1000                                                         | 0.8964880   | 0.8617258   |
| Lag 5000                                                         | 0.6039003   | 0.6022291   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.17067395 | -0.12741602 |
| Lag 100                                                          | -0.16983312 | -0.12362277 |
| Lag 500                                                          | -0.17499382 | -0.11972769 |
| Lag 1000                                                         | -0.18004636 | -0.12502460 |
| Lag 5000                                                         | -0.04226407 | 0.01790192  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.9609935   | 0.9034487   |
| Lag 100                                                          | 0.9411829   | 0.8893146   |
| Lag 500                                                          | 0.8990207   | 0.8539291   |
| Lag 1000                                                         | 0.8622161   | 0.8157349   |
| Lag 5000                                                         | 0.5753236   | 0.5773694   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 1.0000000   | 0.9609935   |
| Lag 100                                                          | 0.9783825   | 0.9411829   |
| Lag 500                                                          | 0.9365031   | 0.8990207   |
| Lag 1000                                                         | 0.8964880   | 0.8622161   |
| Lag 5000                                                         | 0.6039003   | 0.5753236   |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.1856903  | -0.1801544  |
| Lag 100                                                          | -0.1921763  | -0.1830867  |
| Lag 500                                                          | -0.2110415  | -0.2009371  |

---

|          |            |            |
|----------|------------|------------|
| Lag 1000 | -0.2033807 | -0.1989511 |
| Lag 5000 | -0.1866476 | -0.2075534 |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.9603400 | 0.9034487 |
| Lag 100  | 0.9444274 | 0.8893146 |
| Lag 500  | 0.9054416 | 0.8539291 |
| Lag 1000 | 0.8617258 | 0.8157349 |
| Lag 5000 | 0.6022291 | 0.5773694 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.1801544 | -0.1345421 |
| Lag 100  | -0.1830867 | -0.1370653 |
| Lag 500  | -0.2009371 | -0.1566901 |
| Lag 1000 | -0.1989511 | -0.1643648 |
| Lag 5000 | -0.2075534 | -0.2123302 |

, , traitbwt.1:traittarsus.2.units

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |           |            |
|----------|-----------|------------|
| Lag 0    | 0.1148525 | 0.10037928 |
| Lag 100  | 0.1303566 | 0.10703233 |
| Lag 500  | 0.1453432 | 0.12751184 |
| Lag 1000 | 0.1449075 | 0.12663225 |
| Lag 5000 | 0.1652756 | 0.03681818 |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.1588923 | 0.1392793 |
| Lag 100  | 0.1783256 | 0.1463284 |
| Lag 500  | 0.2040736 | 0.1612874 |
| Lag 1000 | 0.2230315 | 0.1896293 |
| Lag 5000 | 0.1750420 | 0.0827739 |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|         |            |            |
|---------|------------|------------|
| Lag 0   | 0.10037928 | 0.12062597 |
| Lag 100 | 0.10703233 | 0.11559443 |
| Lag 500 | 0.12751184 | 0.11837717 |

---

|                                                                    |            |             |
|--------------------------------------------------------------------|------------|-------------|
| Lag 1000                                                           | 0.12663225 | 0.10630852  |
| Lag 5000                                                           | 0.03681818 | -0.04309778 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |            |             |
| Lag 0                                                              | 0.1142625  | 0.09800038  |
| Lag 100                                                            | 0.1363961  | 0.11259128  |
| Lag 500                                                            | 0.1781857  | 0.14358029  |
| Lag 1000                                                           | 0.1973172  | 0.16595148  |
| Lag 5000                                                           | 0.1261037  | 0.08832747  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |            |             |
| Lag 0                                                              | 0.1588923  | 0.1142625   |
| Lag 100                                                            | 0.1783256  | 0.1363961   |
| Lag 500                                                            | 0.2040736  | 0.1781857   |
| Lag 1000                                                           | 0.2230315  | 0.1973172   |
| Lag 5000                                                           | 0.1750420  | 0.1261037   |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |            |             |
| Lag 0                                                              | 0.2227014  | 0.2177824   |
| Lag 100                                                            | 0.2383512  | 0.2331011   |
| Lag 500                                                            | 0.2644369  | 0.2625082   |
| Lag 1000                                                           | 0.2577927  | 0.2670893   |
| Lag 5000                                                           | 0.1772736  | 0.1885761   |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |            |             |
| Lag 0                                                              | 0.1392793  | 0.09800038  |
| Lag 100                                                            | 0.1463284  | 0.11259128  |
| Lag 500                                                            | 0.1612874  | 0.14358029  |
| Lag 1000                                                           | 0.1896293  | 0.16595148  |
| Lag 5000                                                           | 0.0827739  | 0.08832747  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |            |             |
| Lag 0                                                              | 0.2177824  | 0.1727839   |
| Lag 100                                                            | 0.2331011  | 0.1869982   |
| Lag 500                                                            | 0.2625082  | 0.2195418   |
| Lag 1000                                                           | 0.2670893  | 0.2381230   |
| Lag 5000                                                           | 0.1885761  | 0.1980800   |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |            |             |

---

|                                                                  |               |             |              |
|------------------------------------------------------------------|---------------|-------------|--------------|
| Lag 0                                                            | -0.027499725  | 0.04996250  | 0.04569537   |
| Lag 100                                                          | -0.023937020  | 0.05747431  | 0.04863214   |
| Lag 500                                                          | -0.009997799  | 0.05757848  | 0.05649481   |
| Lag 1000                                                         | -0.028336813  | 0.03932991  | 0.08794045   |
| Lag 5000                                                         | -0.028421868  | 0.02850733  | -0.02604903  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |               |             |              |
| Lag 0                                                            | -0.0558751650 | 0.1570314   | -0.002752353 |
| Lag 100                                                          | -0.0390036691 | 0.1577619   | -0.005577721 |
| Lag 500                                                          | -0.0604361094 | 0.1762753   | -0.008689304 |
| Lag 1000                                                         | -0.0544777158 | 0.1711484   | -0.003879605 |
| Lag 5000                                                         | 0.0009703848  | 0.1611550   | 0.001378773  |
| traittarsus.2.mother traitbwt.2.mother                           |               |             |              |
| Lag 0                                                            | -0.04476711   | 0.07595500  |              |
| Lag 100                                                          | -0.04782246   | 0.07837204  |              |
| Lag 500                                                          | -0.04024464   | 0.07773574  |              |
| Lag 1000                                                         | -0.01332984   | 0.04580976  |              |
| Lag 5000                                                         | -0.03731338   | 0.02497852  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |               |             |              |
| Lag 0                                                            |               | -0.2009925  | -0.16660301  |
| Lag 100                                                          |               | -0.2135650  | -0.16581994  |
| Lag 500                                                          |               | -0.2387289  | -0.17888627  |
| Lag 1000                                                         |               | -0.2382720  | -0.17175572  |
| Lag 5000                                                         |               | -0.2290178  | -0.06304922  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |               |             |              |
| Lag 0                                                            |               | 0.9609935   | 0.9303986    |
| Lag 100                                                          |               | 0.9414150   | 0.9167945    |
| Lag 500                                                          |               | 0.9016587   | 0.8805530    |
| Lag 1000                                                         |               | 0.8646228   | 0.8343108    |
| Lag 5000                                                         |               | 0.5874768   | 0.5773601    |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |               |             |              |
| Lag 0                                                            |               | -0.16660301 | -0.167080898 |
| Lag 100                                                          |               | -0.16581994 | -0.156138622 |
| Lag 500                                                          |               | -0.17888627 | -0.151199166 |

---

|                                                                  |             |              |
|------------------------------------------------------------------|-------------|--------------|
| Lag 1000                                                         | -0.17175572 | -0.139083793 |
| Lag 5000                                                         | -0.06304922 | 0.007994506  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |              |
| Lag 0                                                            | 1.0000000   | 0.9457989    |
| Lag 100                                                          | 0.9647777   | 0.9182192    |
| Lag 500                                                          | 0.8967026   | 0.8587466    |
| Lag 1000                                                         | 0.8499110   | 0.8058180    |
| Lag 5000                                                         | 0.5555747   | 0.5516644    |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |              |
| Lag 0                                                            | 0.9609935   | 1.0000000    |
| Lag 100                                                          | 0.9414150   | 0.9647777    |
| Lag 500                                                          | 0.9016587   | 0.8967026    |
| Lag 1000                                                         | 0.8646228   | 0.8499110    |
| Lag 5000                                                         | 0.5874768   | 0.5555747    |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |              |
| Lag 0                                                            | -0.2146171  | -0.2167857   |
| Lag 100                                                          | -0.2184104  | -0.2221790   |
| Lag 500                                                          | -0.2572470  | -0.2584745   |
| Lag 1000                                                         | -0.2569503  | -0.2596379   |
| Lag 5000                                                         | -0.1645148  | -0.1699811   |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |              |
| Lag 0                                                            | 0.9303986   | 0.9457989    |
| Lag 100                                                          | 0.9167945   | 0.9182192    |
| Lag 500                                                          | 0.8805530   | 0.8587466    |
| Lag 1000                                                         | 0.8343108   | 0.8058180    |
| Lag 5000                                                         | 0.5773601   | 0.5516644    |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |              |
| Lag 0                                                            | -0.2167857  | -0.1800172   |
| Lag 100                                                          | -0.2221790  | -0.1849007   |
| Lag 500                                                          | -0.2584745  | -0.2183863   |
| Lag 1000                                                         | -0.2596379  | -0.2235166   |
| Lag 5000                                                         | -0.1699811  | -0.1684397   |



---

, , traittarsus.2:traittarsus.2.units

|          | traittarsus.1:traittarsus.1.animal | traitbwt.1:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | -0.09279221                        | -0.09677793                     |
| Lag 100  | -0.06813005                        | -0.09295582                     |
| Lag 500  | -0.02512501                        | -0.08099239                     |
| Lag 1000 | -0.02287160                        | -0.08741368                     |
| Lag 5000 | 0.07141747                         | 0.05570893                      |

|          | traittarsus.2:traittarsus.1.animal | traitbwt.2:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | -0.5600992                         | -0.5541670                      |
| Lag 100  | -0.5121175                         | -0.5169149                      |
| Lag 500  | -0.4106125                         | -0.4295480                      |
| Lag 1000 | -0.3066655                         | -0.3332778                      |
| Lag 5000 | 0.2026604                          | 0.1966380                       |

|          | traittarsus.1:traitbwt.1.animal | traitbwt.1:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | -0.09677793                     | -0.11346055                  |
| Lag 100  | -0.09295582                     | -0.11622763                  |
| Lag 500  | -0.08099239                     | -0.09958123                  |
| Lag 1000 | -0.08741368                     | -0.12997232                  |
| Lag 5000 | 0.05570893                      | 0.07322980                   |

|          | traittarsus.2:traitbwt.1.animal | traitbwt.2:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | -0.4185841                      | -0.3843067                   |
| Lag 100  | -0.3914286                      | -0.3603585                   |
| Lag 500  | -0.3364209                      | -0.3074805                   |
| Lag 1000 | -0.2650097                      | -0.2595029                   |
| Lag 5000 | 0.1809913                       | 0.1272979                    |

|          | traittarsus.1:traittarsus.2.animal | traitbwt.1:traittarsus.2.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | -0.5600992                         | -0.4185841                      |
| Lag 100  | -0.5121175                         | -0.3914286                      |
| Lag 500  | -0.4106125                         | -0.3364209                      |
| Lag 1000 | -0.3066655                         | -0.2650097                      |
| Lag 5000 | 0.2026604                          | 0.1809913                       |

|  | traittarsus.2:traittarsus.2.animal | traitbwt.2:traittarsus.2.animal |
|--|------------------------------------|---------------------------------|
|--|------------------------------------|---------------------------------|

---

|                                                              |               |              |             |
|--------------------------------------------------------------|---------------|--------------|-------------|
| Lag 0                                                        | -0.91073694   | -0.87501256  |             |
| Lag 100                                                      | -0.86504036   | -0.83310883  |             |
| Lag 500                                                      | -0.70973046   | -0.68729285  |             |
| Lag 1000                                                     | -0.55757696   | -0.54103224  |             |
| Lag 5000                                                     | 0.02356453    | 0.08002102   |             |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal |               |              |             |
| Lag 0                                                        | -0.5541670    | -0.3843067   |             |
| Lag 100                                                      | -0.5169149    | -0.3603585   |             |
| Lag 500                                                      | -0.4295480    | -0.3074805   |             |
| Lag 1000                                                     | -0.3332778    | -0.2595029   |             |
| Lag 5000                                                     | 0.1966380     | 0.1272979    |             |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal |               |              |             |
| Lag 0                                                        | -0.87501256   | -0.70436348  |             |
| Lag 100                                                      | -0.83310883   | -0.67114014  |             |
| Lag 500                                                      | -0.68729285   | -0.55786587  |             |
| Lag 1000                                                     | -0.54103224   | -0.43429408  |             |
| Lag 5000                                                     | 0.08002102    | 0.08588843   |             |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear     |               |              |             |
| Lag 0                                                        | -0.0295325711 | 0.01013932   | -0.05907849 |
| Lag 100                                                      | -0.0258002546 | 0.01023903   | -0.03799824 |
| Lag 500                                                      | -0.0008409219 | 0.01222834   | -0.02632830 |
| Lag 1000                                                     | -0.0645850913 | -0.04476975  | -0.04078215 |
| Lag 5000                                                     | 0.0499883850  | -0.01870262  | 0.03463537  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother      |               |              |             |
| Lag 0                                                        | 0.07499456    | -0.05365944  | 0.08205572  |
| Lag 100                                                      | 0.07846166    | -0.07198121  | 0.07982740  |
| Lag 500                                                      | 0.06122895    | -0.08239796  | 0.07480035  |
| Lag 1000                                                     | 0.04893466    | -0.08171638  | 0.06996265  |
| Lag 5000                                                     | -0.05752959   | -0.01133586  | -0.02071995 |
| traittarsus.2.mother traitbwt.2.mother                       |               |              |             |
| Lag 0                                                        | 0.13607402    | -0.063098741 |             |
| Lag 100                                                      | 0.16280409    | -0.056620767 |             |
| Lag 500                                                      | 0.14111652    | -0.077496866 |             |

---

|                                                                  |             |              |
|------------------------------------------------------------------|-------------|--------------|
| Lag 1000                                                         | 0.12664216  | -0.055395330 |
| Lag 5000                                                         | 0.01649277  | -0.003196996 |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |
| Lag 0                                                            | 0.09543017  | 0.10875399   |
| Lag 100                                                          | 0.06598285  | 0.08293245   |
| Lag 500                                                          | 0.05295587  | 0.08321338   |
| Lag 1000                                                         | 0.06481928  | 0.08697241   |
| Lag 5000                                                         | -0.09911194 | -0.09382365  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |
| Lag 0                                                            | -0.18569026 | -0.20065634  |
| Lag 100                                                          | -0.19134621 | -0.20598290  |
| Lag 500                                                          | -0.19158726 | -0.20439959  |
| Lag 1000                                                         | -0.17346895 | -0.18874412  |
| Lag 5000                                                         | -0.04965337 | -0.05447094  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |              |
| Lag 0                                                            | 0.10875399  | 0.11081955   |
| Lag 100                                                          | 0.08293245  | 0.09377309   |
| Lag 500                                                          | 0.08321338  | 0.07955802   |
| Lag 1000                                                         | 0.08697241  | 0.10737966   |
| Lag 5000                                                         | -0.09382365 | -0.07165472  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |              |
| Lag 0                                                            | -0.21461708 | -0.2187484   |
| Lag 100                                                          | -0.21502402 | -0.2194216   |
| Lag 500                                                          | -0.20000700 | -0.2023328   |
| Lag 1000                                                         | -0.19364240 | -0.1958537   |
| Lag 5000                                                         | -0.02836013 | -0.0258784   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |              |
| Lag 0                                                            | -0.18569026 | -0.21461708  |
| Lag 100                                                          | -0.19134621 | -0.21502402  |
| Lag 500                                                          | -0.19158726 | -0.20000700  |
| Lag 1000                                                         | -0.17346895 | -0.19364240  |
| Lag 5000                                                         | -0.04965337 | -0.02836013  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |              |

---

|          |            |             |
|----------|------------|-------------|
| Lag 0    | 1.0000000  | 0.92685734  |
| Lag 100  | 0.8892164  | 0.83343965  |
| Lag 500  | 0.7197831  | 0.67640907  |
| Lag 1000 | 0.5723349  | 0.53670401  |
| Lag 5000 | -0.0393405 | -0.08858763 |

traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units

|          |             |            |
|----------|-------------|------------|
| Lag 0    | -0.20065634 | -0.2187484 |
| Lag 100  | -0.20598290 | -0.2194216 |
| Lag 500  | -0.20439959 | -0.2023328 |
| Lag 1000 | -0.18874412 | -0.1958537 |
| Lag 5000 | -0.05447094 | -0.0258784 |

traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | 0.92685734  | 0.75421487  |
| Lag 100  | 0.83343965  | 0.68043059  |
| Lag 500  | 0.67640907  | 0.55259827  |
| Lag 1000 | 0.53670401  | 0.43176584  |
| Lag 5000 | -0.08858763 | -0.08998996 |

, , traitbwt.2:traittarsus.2.units

traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.09066377 | -0.12847847 |
| Lag 100  | -0.06791265 | -0.11660390 |
| Lag 500  | -0.02813480 | -0.08793510 |
| Lag 1000 | -0.02464042 | -0.10829057 |
| Lag 5000 | 0.09059977  | 0.04232726  |

traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.5208193 | -0.6200400 |
| Lag 100  | -0.4814739 | -0.5760115 |
| Lag 500  | -0.4160646 | -0.4939845 |
| Lag 1000 | -0.3309504 | -0.3961937 |
| Lag 5000 | 0.1719378  | 0.1704323  |

traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

---

|                                                                    |             |             |
|--------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                              | -0.12847847 | -0.16728073 |
| Lag 100                                                            | -0.11660390 | -0.16431801 |
| Lag 500                                                            | -0.08793510 | -0.14580635 |
| Lag 1000                                                           | -0.10829057 | -0.17627597 |
| Lag 5000                                                           | 0.04232726  | 0.02621324  |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |             |
| Lag 0                                                              | -0.4729141  | -0.49584021 |
| Lag 100                                                            | -0.4400289  | -0.46178772 |
| Lag 500                                                            | -0.3667218  | -0.38855190 |
| Lag 1000                                                           | -0.3075053  | -0.32632756 |
| Lag 5000                                                           | 0.1176403   | 0.08762877  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |             |
| Lag 0                                                              | -0.5208193  | -0.4729141  |
| Lag 100                                                            | -0.4814739  | -0.4400289  |
| Lag 500                                                            | -0.4160646  | -0.3667218  |
| Lag 1000                                                           | -0.3309504  | -0.3075053  |
| Lag 5000                                                           | 0.1719378   | 0.1176403   |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |             |
| Lag 0                                                              | -0.84107771 | -0.93853010 |
| Lag 100                                                            | -0.80537663 | -0.88900373 |
| Lag 500                                                            | -0.70011206 | -0.74957513 |
| Lag 1000                                                           | -0.58965340 | -0.62094832 |
| Lag 5000                                                           | -0.01326034 | 0.04348338  |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |             |
| Lag 0                                                              | -0.6200400  | -0.49584021 |
| Lag 100                                                            | -0.5760115  | -0.46178772 |
| Lag 500                                                            | -0.4939845  | -0.38855190 |
| Lag 1000                                                           | -0.3961937  | -0.32632756 |
| Lag 5000                                                           | 0.1704323   | 0.08762877  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |             |
| Lag 0                                                              | -0.93853010 | -0.85484247 |
| Lag 100                                                            | -0.88900373 | -0.80629890 |
| Lag 500                                                            | -0.74957513 | -0.66699739 |

---

|                                                                  |             |              |              |
|------------------------------------------------------------------|-------------|--------------|--------------|
| Lag 1000                                                         | -0.62094832 |              | -0.53588327  |
| Lag 5000                                                         | 0.04348338  |              | 0.06798066   |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |             |              |              |
| Lag 0                                                            | -0.04533921 | -0.001638125 | -0.07410803  |
| Lag 100                                                          | -0.04526568 | 0.002752216  | -0.05435921  |
| Lag 500                                                          | -0.02361959 | -0.010266769 | -0.04066778  |
| Lag 1000                                                         | -0.08661480 | -0.055915794 | -0.03404716  |
| Lag 5000                                                         | 0.02028860  | -0.015718402 | 0.05000555   |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |             |              |              |
| Lag 0                                                            | 0.09224924  | -0.08507034  | 0.117048388  |
| Lag 100                                                          | 0.09712475  | -0.09090158  | 0.119562799  |
| Lag 500                                                          | 0.06704383  | -0.06950074  | 0.093258667  |
| Lag 1000                                                         | 0.06946025  | -0.09719184  | 0.091569829  |
| Lag 5000                                                         | -0.05957284 | -0.01506617  | -0.004965446 |
| traittarsus.2.mother traitbwt.2.mother                           |             |              |              |
| Lag 0                                                            | 0.11599294  | -0.06384447  |              |
| Lag 100                                                          | 0.12644815  | -0.06061567  |              |
| Lag 500                                                          | 0.12233108  | -0.05933391  |              |
| Lag 1000                                                         | 0.11302925  | -0.05409993  |              |
| Lag 5000                                                         | 0.02623229  | 0.01443371   |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |              |              |
| Lag 0                                                            | 0.11602826  |              | 0.13276682   |
| Lag 100                                                          | 0.08412251  |              | 0.10350782   |
| Lag 500                                                          | 0.06419072  |              | 0.10039070   |
| Lag 1000                                                         | 0.08359941  |              | 0.11689682   |
| Lag 5000                                                         | -0.10317497 |              | -0.07583069  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |              |              |
| Lag 0                                                            | -0.1801544  |              | -0.2276707   |
| Lag 100                                                          | -0.1866591  |              | -0.2347923   |
| Lag 500                                                          | -0.1769783  |              | -0.2196866   |
| Lag 1000                                                         | -0.1617965  |              | -0.1997778   |
| Lag 5000                                                         | -0.0380065  |              | -0.0414611   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |              |              |

---

|                                                                  |              |             |
|------------------------------------------------------------------|--------------|-------------|
| Lag 0                                                            | 0.13276682   | 0.13741353  |
| Lag 100                                                          | 0.10350782   | 0.11641231  |
| Lag 500                                                          | 0.10039070   | 0.12001620  |
| Lag 1000                                                         | 0.11689682   | 0.15480317  |
| Lag 5000                                                         | -0.07583069  | -0.03093518 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |              |             |
| Lag 0                                                            | -0.21678569  | -0.2507033  |
| Lag 100                                                          | -0.22198433  | -0.2562972  |
| Lag 500                                                          | -0.20689452  | -0.2323026  |
| Lag 1000                                                         | -0.19261030  | -0.2149386  |
| Lag 5000                                                         | -0.02955808  | -0.0282441  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |              |             |
| Lag 0                                                            | -0.1801544   | -0.21678569 |
| Lag 100                                                          | -0.1866591   | -0.22198433 |
| Lag 500                                                          | -0.1769783   | -0.20689452 |
| Lag 1000                                                         | -0.1617965   | -0.19261030 |
| Lag 5000                                                         | -0.0380065   | -0.02955808 |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |              |             |
| Lag 0                                                            | 0.926857338  | 1.00000000  |
| Lag 100                                                          | 0.838930228  | 0.89252017  |
| Lag 500                                                          | 0.714319171  | 0.73419722  |
| Lag 1000                                                         | 0.601447081  | 0.60685285  |
| Lag 5000                                                         | -0.005767447 | -0.05722557 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |              |             |
| Lag 0                                                            | -0.2276707   | -0.2507033  |
| Lag 100                                                          | -0.2347923   | -0.2562972  |
| Lag 500                                                          | -0.2196866   | -0.2323026  |
| Lag 1000                                                         | -0.1997778   | -0.2149386  |
| Lag 5000                                                         | -0.0414611   | -0.0282441  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |              |             |
| Lag 0                                                            | 1.00000000   | 0.92347712  |
| Lag 100                                                          | 0.89252017   | 0.81767393  |
| Lag 500                                                          | 0.73419722   | 0.65250913  |

---

|          |             |             |
|----------|-------------|-------------|
| Lag 1000 | 0.60685285  | 0.52632018  |
| Lag 5000 | -0.05722557 | -0.07523754 |

, , traittarsus.1:traitbwt.2.units

|          | traittarsus.1:traittarsus.1.animal | traitbwt.1:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.1843936                          | 0.13991929                      |
| Lag 100  | 0.1916063                          | 0.14051238                      |
| Lag 500  | 0.1997384                          | 0.15350375                      |
| Lag 1000 | 0.1779242                          | 0.14333349                      |
| Lag 5000 | 0.1059309                          | 0.02408511                      |

|          | traittarsus.2:traittarsus.1.animal | traitbwt.2:traittarsus.1.animal |
|----------|------------------------------------|---------------------------------|
| Lag 0    | 0.2159001                          | 0.18941930                      |
| Lag 100  | 0.2292097                          | 0.19756460                      |
| Lag 500  | 0.2470830                          | 0.21294438                      |
| Lag 1000 | 0.2430281                          | 0.21181509                      |
| Lag 5000 | 0.1564420                          | 0.09549646                      |

|          | traittarsus.1:traitbwt.1.animal | traitbwt.1:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | 0.13991929                      | 0.11864383                   |
| Lag 100  | 0.14051238                      | 0.11475981                   |
| Lag 500  | 0.15350375                      | 0.12191537                   |
| Lag 1000 | 0.14333349                      | 0.12719171                   |
| Lag 5000 | 0.02408511                      | -0.00131242                  |

|          | traittarsus.2:traitbwt.1.animal | traitbwt.2:traitbwt.1.animal |
|----------|---------------------------------|------------------------------|
| Lag 0    | 0.1778597                       | 0.1473359                    |
| Lag 100  | 0.1832967                       | 0.1558541                    |
| Lag 500  | 0.2035349                       | 0.1829320                    |
| Lag 1000 | 0.2085231                       | 0.1947599                    |
| Lag 5000 | 0.1096782                       | 0.1076230                    |

|         | traittarsus.1:traittarsus.2.animal | traitbwt.1:traittarsus.2.animal |
|---------|------------------------------------|---------------------------------|
| Lag 0   | 0.2159001                          | 0.1778597                       |
| Lag 100 | 0.2292097                          | 0.1832967                       |
| Lag 500 | 0.2470830                          | 0.2035349                       |



---

|                                                                    |              |            |              |
|--------------------------------------------------------------------|--------------|------------|--------------|
| Lag 1000                                                           | 0.2430281    | 0.2085231  |              |
| Lag 5000                                                           | 0.1564420    | 0.1096782  |              |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |              |            |              |
| Lag 0                                                              | 0.2148002    | 0.2380759  |              |
| Lag 100                                                            | 0.2260582    | 0.2490150  |              |
| Lag 500                                                            | 0.2434250    | 0.2659527  |              |
| Lag 1000                                                           | 0.2396319    | 0.2621966  |              |
| Lag 5000                                                           | 0.2311987    | 0.2321063  |              |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |              |            |              |
| Lag 0                                                              | 0.18941930   | 0.1473359  |              |
| Lag 100                                                            | 0.19756460   | 0.1558541  |              |
| Lag 500                                                            | 0.21294438   | 0.1829320  |              |
| Lag 1000                                                           | 0.21181509   | 0.1947599  |              |
| Lag 5000                                                           | 0.09549646   | 0.1076230  |              |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |              |            |              |
| Lag 0                                                              | 0.2380759    | 0.2116292  |              |
| Lag 100                                                            | 0.2490150    | 0.2231338  |              |
| Lag 500                                                            | 0.2659527    | 0.2429222  |              |
| Lag 1000                                                           | 0.2621966    | 0.2441661  |              |
| Lag 5000                                                           | 0.2321063    | 0.2236298  |              |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |              |            |              |
| Lag 0                                                              | -0.032369970 | 0.05768879 | 0.06054337   |
| Lag 100                                                            | -0.031641858 | 0.05864059 | 0.06580478   |
| Lag 500                                                            | -0.001530924 | 0.04503326 | 0.07399288   |
| Lag 1000                                                           | -0.016005519 | 0.03161375 | 0.08585903   |
| Lag 5000                                                           | 0.010619935  | 0.01730446 | -0.02417570  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |              |            |              |
| Lag 0                                                              | -0.032356483 | 0.1703875  | 0.012376673  |
| Lag 100                                                            | -0.034654911 | 0.1647608  | 0.013581085  |
| Lag 500                                                            | -0.052597308 | 0.1600416  | 0.005553001  |
| Lag 1000                                                           | -0.039591608 | 0.1872703  | -0.009042770 |
| Lag 5000                                                           | 0.004455343  | 0.1972374  | -0.027185781 |
| traittarsus.2.mother traitbwt.2.mother                             |              |            |              |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | -0.03846399 | 0.07575889  |
| Lag 100                                                          | -0.03390447 | 0.06853459  |
| Lag 500                                                          | -0.02862102 | 0.06274714  |
| Lag 1000                                                         | -0.01799649 | 0.06065063  |
| Lag 5000                                                         | -0.05555704 | 0.02202828  |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |             |
| Lag 0                                                            | -0.2645685  | -0.18013988 |
| Lag 100                                                          | -0.2616234  | -0.17607364 |
| Lag 500                                                          | -0.2652457  | -0.18034511 |
| Lag 1000                                                         | -0.2671219  | -0.17702256 |
| Lag 5000                                                         | -0.2093233  | -0.06028968 |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |             |
| Lag 0                                                            | 0.9603400   | 1.0000000   |
| Lag 100                                                          | 0.9432077   | 0.9762869   |
| Lag 500                                                          | 0.9124477   | 0.9252632   |
| Lag 1000                                                         | 0.8838890   | 0.8792611   |
| Lag 5000                                                         | 0.6117319   | 0.6007465   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.18013988 | -0.15026999 |
| Lag 100                                                          | -0.17607364 | -0.14749648 |
| Lag 500                                                          | -0.18034511 | -0.15145709 |
| Lag 1000                                                         | -0.17702256 | -0.15538707 |
| Lag 5000                                                         | -0.06028968 | -0.02621414 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.9303986   | 0.9527160   |
| Lag 100                                                          | 0.9147001   | 0.9321247   |
| Lag 500                                                          | 0.8850543   | 0.8884276   |
| Lag 1000                                                         | 0.8576398   | 0.8431799   |
| Lag 5000                                                         | 0.5876791   | 0.5807055   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.9603400   | 0.9303986   |
| Lag 100                                                          | 0.9432077   | 0.9147001   |
| Lag 500                                                          | 0.9124477   | 0.8850543   |

---

|                                                                    |            |            |
|--------------------------------------------------------------------|------------|------------|
| Lag 1000                                                           | 0.8838890  | 0.8576398  |
| Lag 5000                                                           | 0.6117319  | 0.5876791  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |            |            |
| Lag 0                                                              | -0.2006563 | -0.2276707 |
| Lag 100                                                            | -0.2124295 | -0.2354203 |
| Lag 500                                                            | -0.2379446 | -0.2556157 |
| Lag 1000                                                           | -0.2389714 | -0.2561306 |
| Lag 5000                                                           | -0.2130152 | -0.2137306 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |            |            |
| Lag 0                                                              | 1.0000000  | 0.9527160  |
| Lag 100                                                            | 0.9762869  | 0.9321247  |
| Lag 500                                                            | 0.9252632  | 0.8884276  |
| Lag 1000                                                           | 0.8792611  | 0.8431799  |
| Lag 5000                                                           | 0.6007465  | 0.5807055  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |            |            |
| Lag 0                                                              | -0.2276707 | -0.2128094 |
| Lag 100                                                            | -0.2354203 | -0.2168142 |
| Lag 500                                                            | -0.2556157 | -0.2324758 |
| Lag 1000                                                           | -0.2561306 | -0.2373888 |
| Lag 5000                                                           | -0.2137306 | -0.1974646 |
| , , traitbwt.1:traitbwt.2.units                                    |            |            |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |            |            |
| Lag 0                                                              | 0.1045202  | 0.10675486 |
| Lag 100                                                            | 0.1144301  | 0.10909010 |
| Lag 500                                                            | 0.1244192  | 0.13415361 |
| Lag 1000                                                           | 0.1101292  | 0.11544191 |
| Lag 5000                                                           | 0.1331335  | 0.04367713 |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |            |            |
| Lag 0                                                              | 0.1751169  | 0.18455878 |
| Lag 100                                                            | 0.1936998  | 0.19497627 |
| Lag 500                                                            | 0.2199790  | 0.20989089 |

---

|                                                                    |            |             |
|--------------------------------------------------------------------|------------|-------------|
| Lag 1000                                                           | 0.2292872  | 0.21238609  |
| Lag 5000                                                           | 0.1693953  | 0.06254965  |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |            |             |
| Lag 0                                                              | 0.10675486 | 0.15466547  |
| Lag 100                                                            | 0.10909010 | 0.15753844  |
| Lag 500                                                            | 0.13415361 | 0.17187977  |
| Lag 1000                                                           | 0.11544191 | 0.14889411  |
| Lag 5000                                                           | 0.04367713 | -0.01393939 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |            |             |
| Lag 0                                                              | 0.1618783  | 0.17860886  |
| Lag 100                                                            | 0.1807130  | 0.20406604  |
| Lag 500                                                            | 0.2315550  | 0.24587070  |
| Lag 1000                                                           | 0.2433453  | 0.24000705  |
| Lag 5000                                                           | 0.1313066  | 0.07959284  |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |            |             |
| Lag 0                                                              | 0.1751169  | 0.1618783   |
| Lag 100                                                            | 0.1936998  | 0.1807130   |
| Lag 500                                                            | 0.2199790  | 0.2315550   |
| Lag 1000                                                           | 0.2292872  | 0.2433453   |
| Lag 5000                                                           | 0.1693953  | 0.1313066   |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |            |             |
| Lag 0                                                              | 0.2339355  | 0.2581627   |
| Lag 100                                                            | 0.2476485  | 0.2736295   |
| Lag 500                                                            | 0.2790724  | 0.3078386   |
| Lag 1000                                                           | 0.2842308  | 0.3166841   |
| Lag 5000                                                           | 0.2101542  | 0.1923682   |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |            |             |
| Lag 0                                                              | 0.18455878 | 0.17860886  |
| Lag 100                                                            | 0.19497627 | 0.20406604  |
| Lag 500                                                            | 0.20989089 | 0.24587070  |
| Lag 1000                                                           | 0.21238609 | 0.24000705  |
| Lag 5000                                                           | 0.06254965 | 0.07959284  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |            |             |

---

|          |           |           |
|----------|-----------|-----------|
| Lag 0    | 0.2581627 | 0.2416106 |
| Lag 100  | 0.2736295 | 0.2608676 |
| Lag 500  | 0.3078386 | 0.2963880 |
| Lag 1000 | 0.3166841 | 0.3068141 |
| Lag 5000 | 0.1923682 | 0.1712560 |

traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear

|          |              |            |             |
|----------|--------------|------------|-------------|
| Lag 0    | -0.023701193 | 0.04700881 | 0.05361674  |
| Lag 100  | -0.021944414 | 0.05837761 | 0.06114321  |
| Lag 500  | -0.004019018 | 0.04124133 | 0.05472308  |
| Lag 1000 | -0.014592442 | 0.02897141 | 0.08299624  |
| Lag 5000 | -0.001455363 | 0.01713092 | -0.03330245 |

traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother

|          |             |           |              |
|----------|-------------|-----------|--------------|
| Lag 0    | -0.03458419 | 0.1501420 | -0.030979909 |
| Lag 100  | -0.02882752 | 0.1495980 | -0.034573917 |
| Lag 500  | -0.05044049 | 0.1542703 | -0.037881298 |
| Lag 1000 | -0.04326520 | 0.1753867 | -0.049179995 |
| Lag 5000 | 0.01976572  | 0.1700994 | -0.002621514 |

traittarsus.2.mother traitbwt.2.mother

|          |             |            |
|----------|-------------|------------|
| Lag 0    | -0.05091781 | 0.08248472 |
| Lag 100  | -0.04612708 | 0.07643625 |
| Lag 500  | -0.04592155 | 0.07211789 |
| Lag 1000 | -0.02428388 | 0.04810929 |
| Lag 5000 | -0.05454058 | 0.02230047 |

traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units

|          |            |             |
|----------|------------|-------------|
| Lag 0    | -0.1971901 | -0.17008659 |
| Lag 100  | -0.2006748 | -0.16466175 |
| Lag 500  | -0.2146434 | -0.17764629 |
| Lag 1000 | -0.2152829 | -0.15594317 |
| Lag 5000 | -0.2163189 | -0.07951097 |

traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units

|         |           |           |
|---------|-----------|-----------|
| Lag 0   | 0.9034487 | 0.9527160 |
| Lag 100 | 0.8867034 | 0.9318204 |
| Lag 500 | 0.8558412 | 0.8820862 |

---

|                                                                  |             |            |
|------------------------------------------------------------------|-------------|------------|
| Lag 1000                                                         | 0.8281770   | 0.8332104  |
| Lag 5000                                                         | 0.5777198   | 0.5582022  |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |            |
| Lag 0                                                            | -0.17008659 | -0.1902055 |
| Lag 100                                                          | -0.16466175 | -0.1881759 |
| Lag 500                                                          | -0.17764629 | -0.1918480 |
| Lag 1000                                                         | -0.15594317 | -0.1592987 |
| Lag 5000                                                         | -0.07951097 | -0.0307547 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |            |
| Lag 0                                                            | 0.9457989   | 1.0000000  |
| Lag 100                                                          | 0.9171312   | 0.9591397  |
| Lag 500                                                          | 0.8650556   | 0.8846934  |
| Lag 1000                                                         | 0.8286470   | 0.8252821  |
| Lag 5000                                                         | 0.5520607   | 0.5384118  |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |            |
| Lag 0                                                            | 0.9034487   | 0.9457989  |
| Lag 100                                                          | 0.8867034   | 0.9171312  |
| Lag 500                                                          | 0.8558412   | 0.8650556  |
| Lag 1000                                                         | 0.8281770   | 0.8286470  |
| Lag 5000                                                         | 0.5777198   | 0.5520607  |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |            |
| Lag 0                                                            | -0.2187484  | -0.2507033 |
| Lag 100                                                          | -0.2283894  | -0.2593514 |
| Lag 500                                                          | -0.2686154  | -0.2983604 |
| Lag 1000                                                         | -0.2818961  | -0.3082824 |
| Lag 5000                                                         | -0.1951997  | -0.1732268 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |            |
| Lag 0                                                            | 0.9527160   | 1.0000000  |
| Lag 100                                                          | 0.9318204   | 0.9591397  |
| Lag 500                                                          | 0.8820862   | 0.8846934  |
| Lag 1000                                                         | 0.8332104   | 0.8252821  |
| Lag 5000                                                         | 0.5582022   | 0.5384118  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units       |             |            |

---

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.2507033 | -0.2443613 |
| Lag 100  | -0.2593514 | -0.2551879 |
| Lag 500  | -0.2983604 | -0.2884325 |
| Lag 1000 | -0.3082824 | -0.2942164 |
| Lag 5000 | -0.1732268 | -0.1455134 |

, , traittarsus.2:traitbwt.2.units

    traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.09066377 | -0.12847847 |
| Lag 100  | -0.06791265 | -0.11660390 |
| Lag 500  | -0.02813480 | -0.08793510 |
| Lag 1000 | -0.02464042 | -0.10829057 |
| Lag 5000 | 0.09059977  | 0.04232726  |

    traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal

|          |            |            |
|----------|------------|------------|
| Lag 0    | -0.5208193 | -0.6200400 |
| Lag 100  | -0.4814739 | -0.5760115 |
| Lag 500  | -0.4160646 | -0.4939845 |
| Lag 1000 | -0.3309504 | -0.3961937 |
| Lag 5000 | 0.1719378  | 0.1704323  |

    traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal

|          |             |             |
|----------|-------------|-------------|
| Lag 0    | -0.12847847 | -0.16728073 |
| Lag 100  | -0.11660390 | -0.16431801 |
| Lag 500  | -0.08793510 | -0.14580635 |
| Lag 1000 | -0.10829057 | -0.17627597 |
| Lag 5000 | 0.04232726  | 0.02621324  |

    traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal

|          |            |             |
|----------|------------|-------------|
| Lag 0    | -0.4729141 | -0.49584021 |
| Lag 100  | -0.4400289 | -0.46178772 |
| Lag 500  | -0.3667218 | -0.38855190 |
| Lag 1000 | -0.3075053 | -0.32632756 |
| Lag 5000 | 0.1176403  | 0.08762877  |

    traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal

---

|                                                                    |             |              |             |
|--------------------------------------------------------------------|-------------|--------------|-------------|
| Lag 0                                                              | -0.5208193  |              | -0.4729141  |
| Lag 100                                                            | -0.4814739  |              | -0.4400289  |
| Lag 500                                                            | -0.4160646  |              | -0.3667218  |
| Lag 1000                                                           | -0.3309504  |              | -0.3075053  |
| Lag 5000                                                           | 0.1719378   |              | 0.1176403   |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |              |             |
| Lag 0                                                              | -0.84107771 |              | -0.93853010 |
| Lag 100                                                            | -0.80537663 |              | -0.88900373 |
| Lag 500                                                            | -0.70011206 |              | -0.74957513 |
| Lag 1000                                                           | -0.58965340 |              | -0.62094832 |
| Lag 5000                                                           | -0.01326034 |              | 0.04348338  |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |              |             |
| Lag 0                                                              | -0.6200400  |              | -0.49584021 |
| Lag 100                                                            | -0.5760115  |              | -0.46178772 |
| Lag 500                                                            | -0.4939845  |              | -0.38855190 |
| Lag 1000                                                           | -0.3961937  |              | -0.32632756 |
| Lag 5000                                                           | 0.1704323   |              | 0.08762877  |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal       |             |              |             |
| Lag 0                                                              | -0.93853010 |              | -0.85484247 |
| Lag 100                                                            | -0.88900373 |              | -0.80629890 |
| Lag 500                                                            | -0.74957513 |              | -0.66699739 |
| Lag 1000                                                           | -0.62094832 |              | -0.53588327 |
| Lag 5000                                                           | 0.04348338  |              | 0.06798066  |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear           |             |              |             |
| Lag 0                                                              | -0.04533921 | -0.001638125 | -0.07410803 |
| Lag 100                                                            | -0.04526568 | 0.002752216  | -0.05435921 |
| Lag 500                                                            | -0.02361959 | -0.010266769 | -0.04066778 |
| Lag 1000                                                           | -0.08661480 | -0.055915794 | -0.03404716 |
| Lag 5000                                                           | 0.02028860  | -0.015718402 | 0.05000555  |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother            |             |              |             |
| Lag 0                                                              | 0.09224924  | -0.08507034  | 0.117048388 |
| Lag 100                                                            | 0.09712475  | -0.09090158  | 0.119562799 |
| Lag 500                                                            | 0.06704383  | -0.06950074  | 0.093258667 |



---

|                                                                  |             |             |              |
|------------------------------------------------------------------|-------------|-------------|--------------|
| Lag 1000                                                         | 0.06946025  | -0.09719184 | 0.091569829  |
| Lag 5000                                                         | -0.05957284 | -0.01506617 | -0.004965446 |
| traittarsus.2.mother traitbwt.2.mother                           |             |             |              |
| Lag 0                                                            | 0.11599294  | -0.06384447 |              |
| Lag 100                                                          | 0.12644815  | -0.06061567 |              |
| Lag 500                                                          | 0.12233108  | -0.05933391 |              |
| Lag 1000                                                         | 0.11302925  | -0.05409993 |              |
| Lag 5000                                                         | 0.02623229  | 0.01443371  |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |             |             |              |
| Lag 0                                                            |             | 0.11602826  | 0.13276682   |
| Lag 100                                                          |             | 0.08412251  | 0.10350782   |
| Lag 500                                                          |             | 0.06419072  | 0.10039070   |
| Lag 1000                                                         |             | 0.08359941  | 0.11689682   |
| Lag 5000                                                         |             | -0.10317497 | -0.07583069  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |             |             |              |
| Lag 0                                                            |             | -0.1801544  | -0.2276707   |
| Lag 100                                                          |             | -0.1866591  | -0.2347923   |
| Lag 500                                                          |             | -0.1769783  | -0.2196866   |
| Lag 1000                                                         |             | -0.1617965  | -0.1997778   |
| Lag 5000                                                         |             | -0.0380065  | -0.0414611   |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |              |
| Lag 0                                                            |             | 0.13276682  | 0.13741353   |
| Lag 100                                                          |             | 0.10350782  | 0.11641231   |
| Lag 500                                                          |             | 0.10039070  | 0.12001620   |
| Lag 1000                                                         |             | 0.11689682  | 0.15480317   |
| Lag 5000                                                         |             | -0.07583069 | -0.03093518  |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |              |
| Lag 0                                                            |             | -0.21678569 | -0.2507033   |
| Lag 100                                                          |             | -0.22198433 | -0.2562972   |
| Lag 500                                                          |             | -0.20689452 | -0.2323026   |
| Lag 1000                                                         |             | -0.19261030 | -0.2149386   |
| Lag 5000                                                         |             | -0.02955808 | -0.0282441   |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |              |

---

|                                                                    |              |             |
|--------------------------------------------------------------------|--------------|-------------|
| Lag 0                                                              | -0.1801544   | -0.21678569 |
| Lag 100                                                            | -0.1866591   | -0.22198433 |
| Lag 500                                                            | -0.1769783   | -0.20689452 |
| Lag 1000                                                           | -0.1617965   | -0.19261030 |
| Lag 5000                                                           | -0.0380065   | -0.02955808 |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units   |              |             |
| Lag 0                                                              | 0.926857338  | 1.00000000  |
| Lag 100                                                            | 0.838930228  | 0.89252017  |
| Lag 500                                                            | 0.714319171  | 0.73419722  |
| Lag 1000                                                           | 0.601447081  | 0.60685285  |
| Lag 5000                                                           | -0.005767447 | -0.05722557 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units         |              |             |
| Lag 0                                                              | -0.2276707   | -0.2507033  |
| Lag 100                                                            | -0.2347923   | -0.2562972  |
| Lag 500                                                            | -0.2196866   | -0.2323026  |
| Lag 1000                                                           | -0.1997778   | -0.2149386  |
| Lag 5000                                                           | -0.0414611   | -0.0282441  |
| traittarsus.2:traitbwt.2.units traitbwt.2:traitbwt.2.units         |              |             |
| Lag 0                                                              | 1.00000000   | 0.92347712  |
| Lag 100                                                            | 0.89252017   | 0.81767393  |
| Lag 500                                                            | 0.73419722   | 0.65250913  |
| Lag 1000                                                           | 0.60685285   | 0.52632018  |
| Lag 5000                                                           | -0.05722557  | -0.07523754 |
| , , traitbwt.2:traitbwt.2.units                                    |              |             |
| traittarsus.1:traittarsus.1.animal traitbwt.1:traittarsus.1.animal |              |             |
| Lag 0                                                              | -0.043059890 | -0.10947233 |
| Lag 100                                                            | -0.021006970 | -0.08947374 |
| Lag 500                                                            | -0.001331729 | -0.06046211 |
| Lag 1000                                                           | -0.008677951 | -0.10284705 |
| Lag 5000                                                           | 0.111890868  | 0.03101996  |
| traittarsus.2:traittarsus.1.animal traitbwt.2:traittarsus.1.animal |              |             |

---

|                                                                    |             |              |
|--------------------------------------------------------------------|-------------|--------------|
| Lag 0                                                              | -0.3939711  | -0.5649761   |
| Lag 100                                                            | -0.3674489  | -0.5248935   |
| Lag 500                                                            | -0.3564907  | -0.4716725   |
| Lag 1000                                                           | -0.3026255  | -0.3959870   |
| Lag 5000                                                           | 0.1392553   | 0.1506927    |
| traittarsus.1:traitbwt.1.animal traitbwt.1:traitbwt.1.animal       |             |              |
| Lag 0                                                              | -0.10947233 | -0.187811354 |
| Lag 100                                                            | -0.08947374 | -0.167609258 |
| Lag 500                                                            | -0.06046211 | -0.153207032 |
| Lag 1000                                                           | -0.10284705 | -0.191376283 |
| Lag 5000                                                           | 0.03101996  | -0.009266993 |
| traittarsus.2:traitbwt.1.animal traitbwt.2:traitbwt.1.animal       |             |              |
| Lag 0                                                              | -0.43960419 | -0.56988609  |
| Lag 100                                                            | -0.41008412 | -0.52185804  |
| Lag 500                                                            | -0.34705550 | -0.43385778  |
| Lag 1000                                                           | -0.31152313 | -0.36023626  |
| Lag 5000                                                           | 0.07052962  | 0.07222864   |
| traittarsus.1:traittarsus.2.animal traitbwt.1:traittarsus.2.animal |             |              |
| Lag 0                                                              | -0.3939711  | -0.43960419  |
| Lag 100                                                            | -0.3674489  | -0.41008412  |
| Lag 500                                                            | -0.3564907  | -0.34705550  |
| Lag 1000                                                           | -0.3026255  | -0.31152313  |
| Lag 5000                                                           | 0.1392553   | 0.07052962   |
| traittarsus.2:traittarsus.2.animal traitbwt.2:traittarsus.2.animal |             |              |
| Lag 0                                                              | -0.67910852 | -0.85888343  |
| Lag 100                                                            | -0.65466001 | -0.81238323  |
| Lag 500                                                            | -0.60647885 | -0.70052835  |
| Lag 1000                                                           | -0.54835587 | -0.61884830  |
| Lag 5000                                                           | -0.02772098 | 0.03373215   |
| traittarsus.1:traitbwt.2.animal traitbwt.1:traitbwt.2.animal       |             |              |
| Lag 0                                                              | -0.5649761  | -0.56988609  |
| Lag 100                                                            | -0.5248935  | -0.52185804  |
| Lag 500                                                            | -0.4716725  | -0.43385778  |

---

|                                                                  |              |              |              |
|------------------------------------------------------------------|--------------|--------------|--------------|
| Lag 1000                                                         |              | -0.3959870   | -0.36023626  |
| Lag 5000                                                         |              | 0.1506927    | 0.07222864   |
| traittarsus.2:traitbwt.2.animal traitbwt.2:traitbwt.2.animal     |              |              |              |
| Lag 0                                                            |              | -0.85888343  | -0.90703776  |
| Lag 100                                                          |              | -0.81238323  | -0.83971454  |
| Lag 500                                                          |              | -0.70052835  | -0.68420955  |
| Lag 1000                                                         |              | -0.61884830  | -0.57020879  |
| Lag 5000                                                         |              | 0.03373215   | 0.07361913   |
| traittarsus.1.byear traitbwt.1.byear traittarsus.2.byear         |              |              |              |
| Lag 0                                                            | -0.044815839 | -0.007601342 | -0.09539294  |
| Lag 100                                                          | -0.046632422 | -0.009618828 | -0.07733270  |
| Lag 500                                                          | -0.039749429 | -0.026455434 | -0.05466463  |
| Lag 1000                                                         | -0.096232932 | -0.063248471 | -0.04859295  |
| Lag 5000                                                         | -0.007307801 | -0.011838332 | 0.07303267   |
| traitbwt.2.byear traittarsus.1.mother traitbwt.1.mother          |              |              |              |
| Lag 0                                                            | 0.07926090   | -0.11755479  | 0.1533639724 |
| Lag 100                                                          | 0.09525987   | -0.11193096  | 0.1505628703 |
| Lag 500                                                          | 0.05998165   | -0.06092327  | 0.1143539752 |
| Lag 1000                                                         | 0.06572238   | -0.09798980  | 0.1117926423 |
| Lag 5000                                                         | -0.05632018  | -0.00703712  | 0.0007242137 |
| traittarsus.2.mother traitbwt.2.mother                           |              |              |              |
| Lag 0                                                            | 0.09798770   | -0.07912857  |              |
| Lag 100                                                          | 0.09530597   | -0.05178227  |              |
| Lag 500                                                          | 0.10285009   | -0.03607903  |              |
| Lag 1000                                                         | 0.08664286   | -0.05475142  |              |
| Lag 5000                                                         | 0.03594017   | 0.03608548   |              |
| traittarsus.1:traittarsus.1.units traitbwt.1:traittarsus.1.units |              |              |              |
| Lag 0                                                            |              | 0.10062986   | 0.11344391   |
| Lag 100                                                          |              | 0.07039582   | 0.08507484   |
| Lag 500                                                          |              | 0.04998573   | 0.08084299   |
| Lag 1000                                                         |              | 0.07979248   | 0.11040825   |
| Lag 5000                                                         |              | -0.11441105  | -0.06475361  |
| traittarsus.2:traittarsus.1.units traitbwt.2:traittarsus.1.units |              |              |              |

---

|                                                                  |             |             |
|------------------------------------------------------------------|-------------|-------------|
| Lag 0                                                            | -0.13454205 | -0.21280936 |
| Lag 100                                                          | -0.14136050 | -0.21928247 |
| Lag 500                                                          | -0.12646869 | -0.19382371 |
| Lag 1000                                                         | -0.12162967 | -0.17751558 |
| Lag 5000                                                         | -0.02590274 | -0.02880894 |
| traittarsus.1:traitbwt.1.units traitbwt.1:traitbwt.1.units       |             |             |
| Lag 0                                                            | 0.11344391  | 0.131450832 |
| Lag 100                                                          | 0.08507484  | 0.105602349 |
| Lag 500                                                          | 0.08084299  | 0.125553514 |
| Lag 1000                                                         | 0.11040825  | 0.165398766 |
| Lag 5000                                                         | -0.06475361 | 0.003312387 |
| traittarsus.2:traitbwt.1.units traitbwt.2:traitbwt.1.units       |             |             |
| Lag 0                                                            | -0.18001725 | -0.24436125 |
| Lag 100                                                          | -0.18862947 | -0.25750569 |
| Lag 500                                                          | -0.17962012 | -0.23342183 |
| Lag 1000                                                         | -0.16368716 | -0.20650410 |
| Lag 5000                                                         | -0.02638416 | -0.02742457 |
| traittarsus.1:traittarsus.2.units traitbwt.1:traittarsus.2.units |             |             |
| Lag 0                                                            | -0.13454205 | -0.18001725 |
| Lag 100                                                          | -0.14136050 | -0.18862947 |
| Lag 500                                                          | -0.12646869 | -0.17962012 |
| Lag 1000                                                         | -0.12162967 | -0.16368716 |
| Lag 5000                                                         | -0.02590274 | -0.02638416 |
| traittarsus.2:traittarsus.2.units traitbwt.2:traittarsus.2.units |             |             |
| Lag 0                                                            | 0.75421487  | 0.92347712  |
| Lag 100                                                          | 0.68911416  | 0.81697064  |
| Lag 500                                                          | 0.62438762  | 0.68932883  |
| Lag 1000                                                         | 0.56071503  | 0.60055861  |
| Lag 5000                                                         | 0.01204694  | -0.04603239 |
| traittarsus.1:traitbwt.2.units traitbwt.1:traitbwt.2.units       |             |             |
| Lag 0                                                            | -0.21280936 | -0.24436125 |
| Lag 100                                                          | -0.21928247 | -0.25750569 |
| Lag 500                                                          | -0.19382371 | -0.23342183 |

---

|                                                             |             |             |
|-------------------------------------------------------------|-------------|-------------|
| Lag 1000                                                    | -0.17751558 | -0.20650410 |
| Lag 5000                                                    | -0.02880894 | -0.02742457 |
| traittarsus.2:traitbwt.2.units  traitbwt.2:traitbwt.2.units |             |             |
| Lag 0                                                       | 0.92347712  | 1.00000000  |
| Lag 100                                                     | 0.81697064  | 0.84520133  |
| Lag 500                                                     | 0.68932883  | 0.66116525  |
| Lag 1000                                                    | 0.60055861  | 0.55204450  |
| Lag 5000                                                    | -0.04603239 | -0.08094849 |

# Chapter 7

## brms

First load brms:

```
library(brms)
```

Loading required package: Rcpp

Loading 'brms' package (version 2.21.0). Useful instructions can be found by typing `help('brms')`. A more detailed introduction to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following object is masked from 'package:stats':

ar

```
Amat <- as.matrix(nadiv::makeA(gryphonped))
```

### 7.0.1. Fitting the model

Fitting a multivariate model in brms involves several new consideration above those for fitting univariate models. First, we need to create two models/objects with the function `bf` fitting the desired univariate model structure for each

response variable (here `bwt` and `tarsus`). It is the equivalent of writing `mvbf(bwt, tarsus)`, but the advantage to create two distinct model is to specific different model structure (fixed or random effect) for each response variable.

Then, the two objects/models are added into a third model to quantify all the estimates in addition to their covariance. Contrary to `MCMCglmm` or `asreml-R`, `brms` directly estimate the covariance and the correlation in its outputs. Our most basic model can be specified as:

```
bf_bwt <- bf(bwt ~ 1 + (1 | a | gr(animal, cov = Amat)))
bf_tarsus <- bf(tarsus ~ 1 + (1 | a | gr(animal, cov = Amat)))
brms_m2.1 <- brm(
  bf_bwt + bf_tarsus + set_rescor(TRUE),
  data = gryphon,
  data2 = list(Amat = Amat),
  chains = 2, cores = 2, iter = 1000
)
save(brms_m2.1, file = "r-obj/brms_m2_1.rda")
```

Again we have provided the data from one such run. It can be accessed using the code:

```
load("r-obj/brms_m2_1.rda")
summary(brms_m2.1)
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

```
Family: MV(gaussian, gaussian)
Links: mu = identity; sigma = identity
       mu = identity; sigma = identity
Formula: bwt ~ 1 + (1 | p | gr(animal, cov = Amat))
         tarsus ~ 1 + (1 | p | gr(animal, cov = Amat))
Data: gryphon (Number of observations: 683)
Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;
       total post-warmup draws = 1000
```



---

Multilevel Hyperparameters:

~animal (Number of levels: 683)

|                                     | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat |
|-------------------------------------|----------|-----------|----------|----------|------|
| sd(bwt_Intercept)                   | 1.81     | 0.21      | 1.41     | 2.20     | 1.06 |
| sd(tarsus_Intercept)                | 3.44     | 0.43      | 2.49     | 4.25     | 1.05 |
| cor(bwt_Intercept,tarsus_Intercept) | 0.38     | 0.14      | 0.08     | 0.62     | 1.02 |
|                                     | Bulk_ESS | Tail_ESS  |          |          |      |
| sd(bwt_Intercept)                   | 31       | 192       |          |          |      |
| sd(tarsus_Intercept)                | 61       | 173       |          |          |      |
| cor(bwt_Intercept,tarsus_Intercept) | 101      | 232       |          |          |      |

Regression Coefficients:

|                  | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|------------------|----------|-----------|----------|----------|------|----------|----------|
| bwt_Intercept    | 7.49     | 0.16      | 7.20     | 7.79     | 1.00 | 608      | 839      |
| tarsus_Intercept | 20.47    | 0.30      | 19.92    | 21.03    | 1.00 | 868      | 803      |

Further Distributional Parameters:

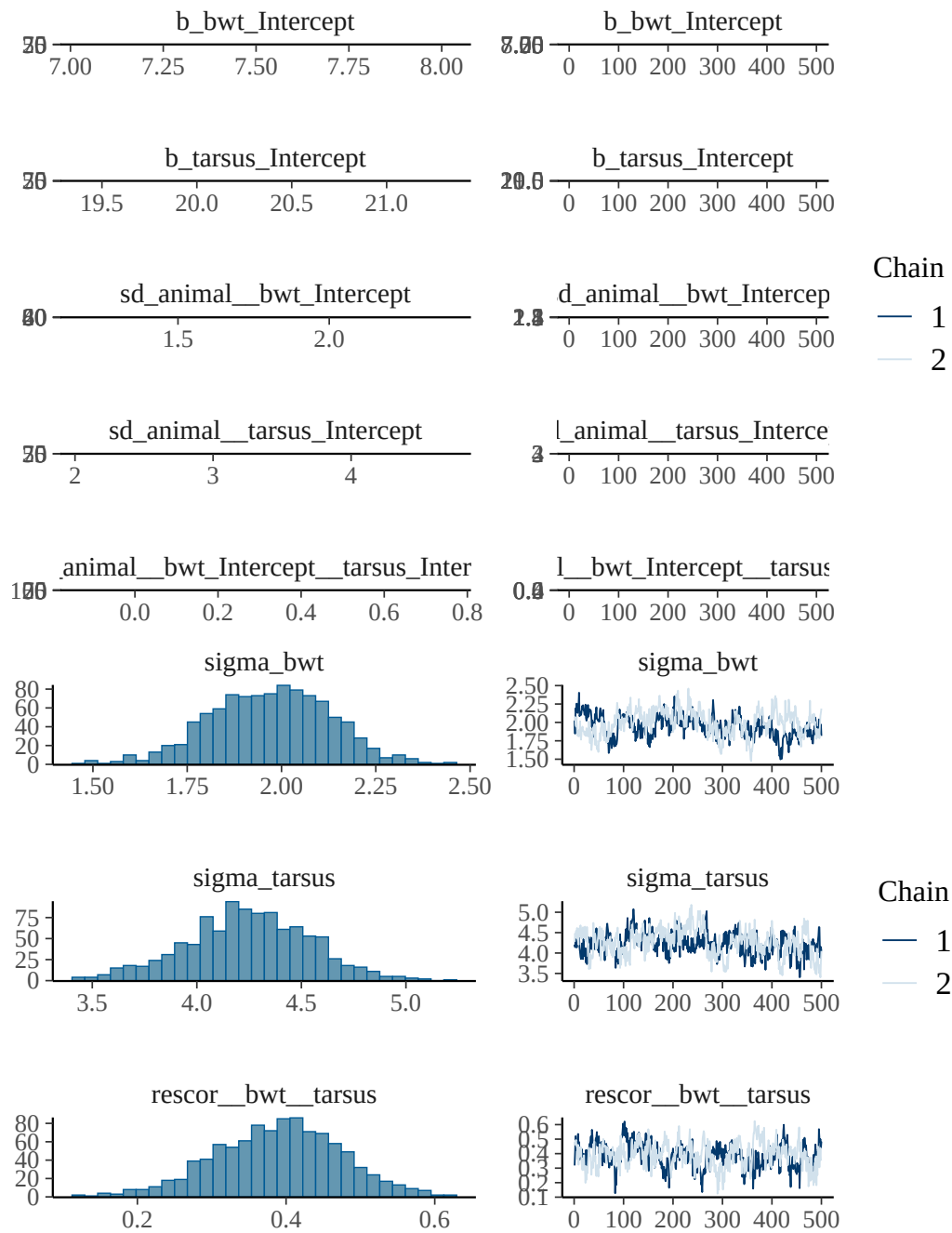
|              | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|--------------|----------|-----------|----------|----------|------|----------|----------|
| sigma_bwt    | 1.97     | 0.16      | 1.66     | 2.28     | 1.06 | 27       | 172      |
| sigma_tarsus | 4.24     | 0.30      | 3.63     | 4.82     | 1.04 | 72       | 162      |

Residual Correlations:

|                    | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|--------------------|----------|-----------|----------|----------|------|----------|----------|
| rescor(bwt,tarsus) | 0.39     | 0.09      | 0.21     | 0.55     | 1.02 | 95       | 179      |

Draws were sampled using `sampling(NUTS)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

```
plot(brms_m2.1, ask = FALSE)
```



```
VarCorr(brms_m2.1)
```

```
$animal
```

```
$animal$sd
```

|                  | Estimate | Est.Error | Q2.5     | Q97.5    |
|------------------|----------|-----------|----------|----------|
| bwt_Intercept    | 1.808171 | 0.2050233 | 1.412824 | 2.204805 |
| tarsus_Intercept | 3.438368 | 0.4283612 | 2.491218 | 4.245264 |

---

\$animal\$cor

, , bwt\_Intercept

|                  | Estimate  | Est.Error | Q2.5       | Q97.5     |
|------------------|-----------|-----------|------------|-----------|
| bwt_Intercept    | 1.0000000 | 0.0000000 | 1.00000000 | 1.0000000 |
| tarsus_Intercept | 0.3814062 | 0.1380014 | 0.07581464 | 0.6209038 |

, , tarsus\_Intercept

|                  | Estimate  | Est.Error | Q2.5       | Q97.5     |
|------------------|-----------|-----------|------------|-----------|
| bwt_Intercept    | 0.3814062 | 0.1380014 | 0.07581464 | 0.6209038 |
| tarsus_Intercept | 1.0000000 | 0.0000000 | 1.00000000 | 1.0000000 |

\$animal\$cov

, , bwt\_Intercept

|                  | Estimate | Est.Error | Q2.5      | Q97.5    |
|------------------|----------|-----------|-----------|----------|
| bwt_Intercept    | 3.311473 | 0.7430185 | 1.9960721 | 4.861167 |
| tarsus_Intercept | 2.440166 | 1.0901689 | 0.3870783 | 4.668720 |

, , tarsus\_Intercept

|                  | Estimate  | Est.Error | Q2.5      | Q97.5    |
|------------------|-----------|-----------|-----------|----------|
| bwt_Intercept    | 2.440166  | 1.090169  | 0.3870783 | 4.66872  |
| tarsus_Intercept | 12.005688 | 2.918741  | 6.2061701 | 18.02226 |

\$residual\_\_

\$residual\_\_\$sd

|     | Estimate | Est.Error | Q2.5     | Q97.5    |
|-----|----------|-----------|----------|----------|
| bwt | 1.970532 | 0.1597581 | 1.658782 | 2.276074 |

```
tarsus 4.244704 0.2984518 3.632824 4.820109
```

```
$residual__$cor
```

```
, , bwt
```

|        | Estimate  | Est.Error  | Q2.5      | Q97.5     |
|--------|-----------|------------|-----------|-----------|
| bwt    | 1.0000000 | 0.0000000  | 1.0000000 | 1.0000000 |
| tarsus | 0.3888754 | 0.08510488 | 0.2127907 | 0.5526631 |

```
, , tarsus
```

|        | Estimate  | Est.Error  | Q2.5      | Q97.5     |
|--------|-----------|------------|-----------|-----------|
| bwt    | 0.3888754 | 0.08510488 | 0.2127907 | 0.5526631 |
| tarsus | 1.0000000 | 0.0000000  | 1.0000000 | 1.0000000 |

```
$residual__$cov
```

```
, , bwt
```

|        | Estimate | Est.Error | Q2.5     | Q97.5    |
|--------|----------|-----------|----------|----------|
| bwt    | 3.908493 | 0.6282892 | 2.751557 | 5.180511 |
| tarsus | 3.289995 | 0.9305960 | 1.572647 | 5.147133 |

```
, , tarsus
```

|        | Estimate  | Est.Error | Q2.5      | Q97.5     |
|--------|-----------|-----------|-----------|-----------|
| bwt    | 3.289995  | 0.930596  | 1.572647  | 5.147133  |
| tarsus | 18.106495 | 2.530138  | 13.197409 | 23.233452 |

It is also possible to calculate the heritability for each trait using the function ‘as.mcmc’

```
v_animal <- (VarCorr(brms_m2.1, summary = FALSE)$animal$sd)^2  
v_r <- (VarCorr(brms_m2.1, summary = FALSE)$residual$sd)^2
```

---

```

h.bwt.2 <- as.mcmc(v_animal[, 1] / (v_animal[, 1] + v_r[, 1]))
h.tarsus.2 <- as.mcmc(v_animal[, 2] / (v_animal[, 2] + v_r[, 2]))

summary(h.bwt.2)

```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.457051 | 0.090878 | 0.002874 | 0.011675       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.2878 | 0.3926 | 0.4596 | 0.5254 | 0.6297 |

```
summary(h.tarsus.2)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

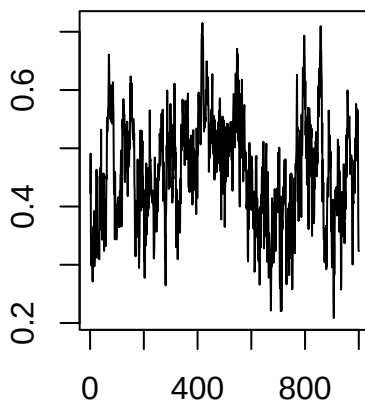
| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.397237 | 0.087350 | 0.002762 | 0.009971       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.2174 | 0.3390 | 0.3982 | 0.4553 | 0.5682 |

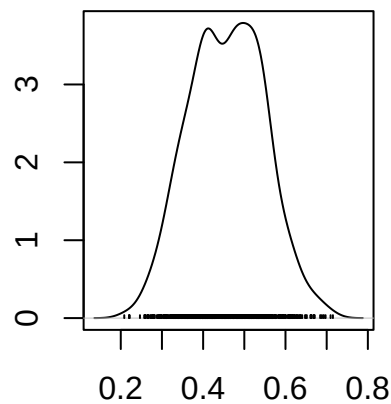
```
plot(h.bwt.2)
```

**Trace of var1**



Iterations

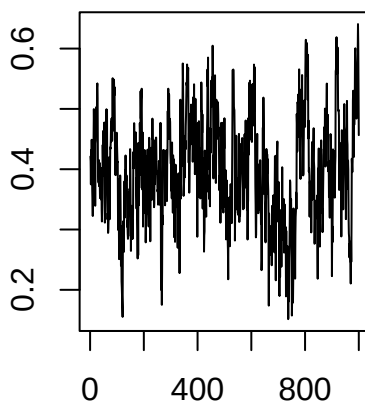
**Density of var1**



N = 1000 Bandwidth = 0.0242

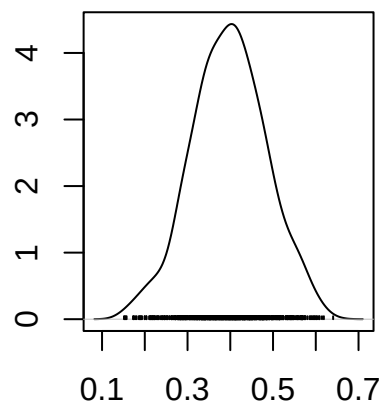
```
plot(h.tarsus.2)
```

**Trace of var1**



Iterations

**Density of var1**



N = 1000 Bandwidth = 0.02312

---

It is also possible to extract the correlation. Just to remember it is an example, the correlation distribution is skewed to 1 due to a weak prior and model parameters. Note, since

```
cor_g <- as.mcmc((VarCorr(brms_m2.1, summary = FALSE)$animal$cor[, 1, 2]))
cor_res <- as.mcmc((VarCorr(brms_m2.1, summary = FALSE)$residual$cor[, 1, 2]))

summary(cor_g)
```

```
Iterations = 1:1000
Thinning interval = 1
Number of chains = 1
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.381406 | 0.138001 | 0.004364 | 0.014946       |

2. Quantiles for each variable:

| 2.5%    | 25%     | 50%     | 75%     | 97.5%   |
|---------|---------|---------|---------|---------|
| 0.07581 | 0.30354 | 0.39041 | 0.47497 | 0.62090 |

```
summary(cor_res)
```

```
Iterations = 1:1000
Thinning interval = 1
Number of chains = 1
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,

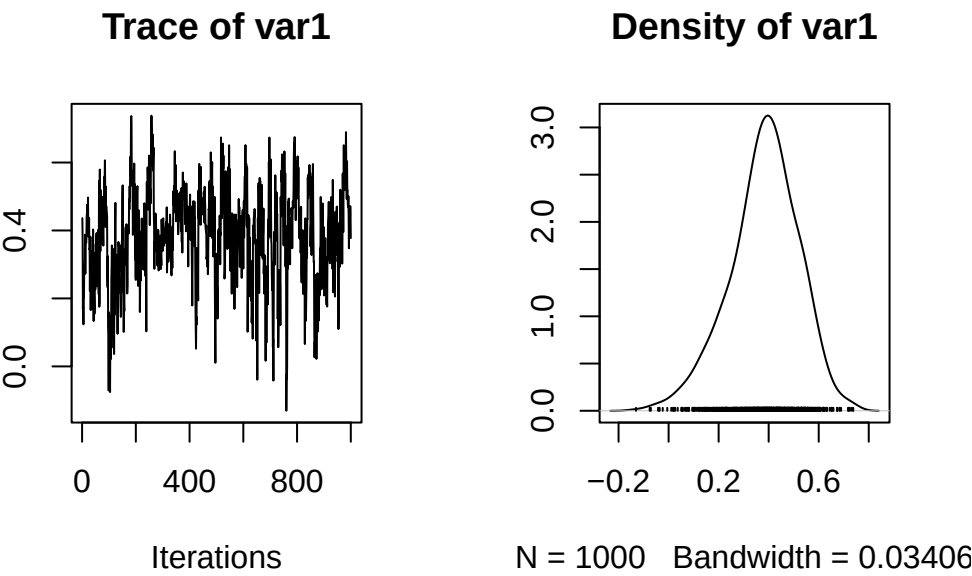
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.388875 | 0.085105 | 0.002691 | 0.009215       |

2. Quantiles for each variable:

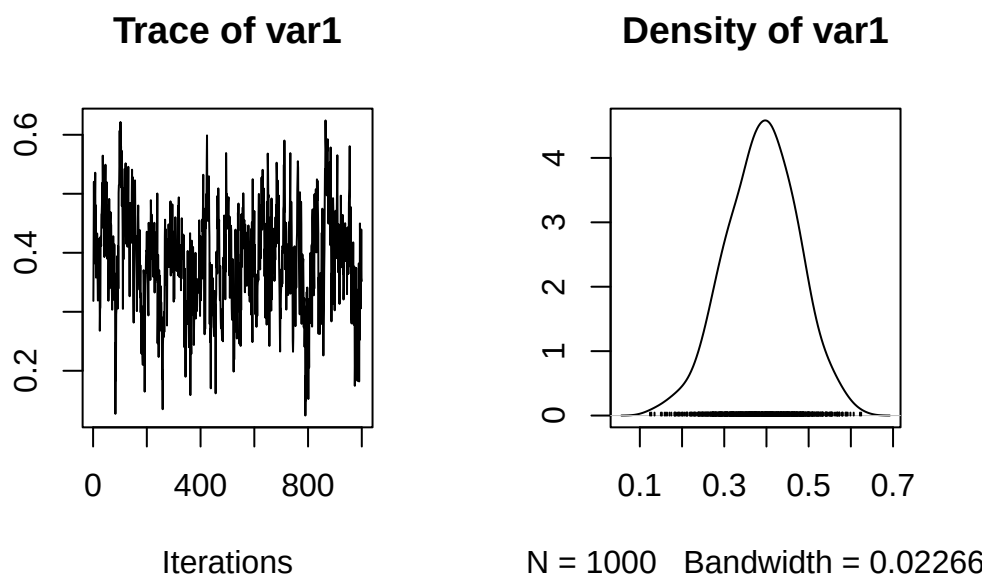
| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.2128 | 0.3319 | 0.3913 | 0.4484 | 0.5527 |

```
plot(cor_g)
```



```
plot(cor_res)
```





Here we can plot the genetic correlation by extraction the breeding values or BLUP.

```
bls_m2.1 <- ranef(brms_m2.1)$animal
bl_m2.1 <- as.data.frame(abind::abind(lapply(1:dim(bls_m2.1)[[3]], function(x) bls_m2.1[, c(1, 3,
colnames(bl_m2.1) <- paste0(rep(dimnames(bls_m2.1)[[3]], each = 3), c("", "_lo", "_up"))
bl_m2.1$id <- rownames(bl_m2.1)
```

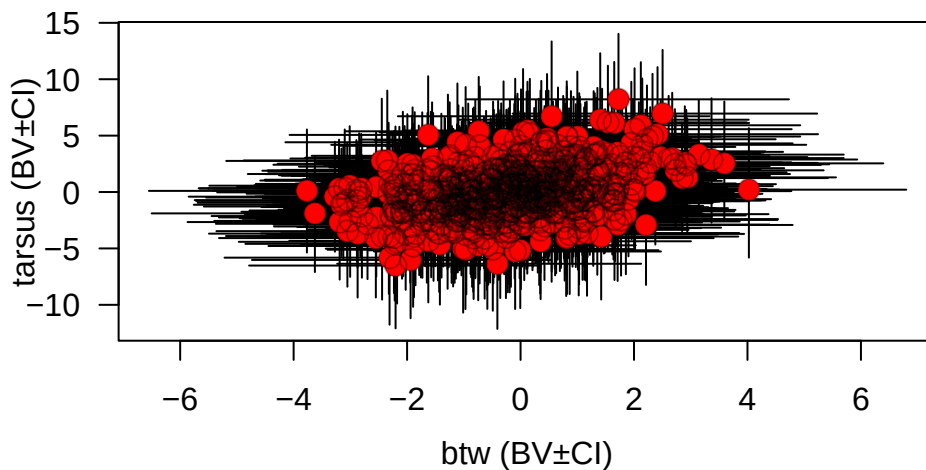
Here, some simple code to plot the genetic correlation.

```
plot(tarsus_Intercept ~ bwt_Intercept, bl_m2.1,
     xlab = "", ylab = "",
     xlim = c(min(bl_m2.1$bwt_Intercept_lo), max(bl_m2.1$bwt_Intercept_up)),
     ylim = c(min(bl_m2.1$tarsus_Intercept_lo), max(bl_m2.1$tarsus_Intercept_up)),
     las = 1.2, type = "n"
)
with(
  bl_m2.1,
  segments(
    x0 = bwt_Intercept, y0 = tarsus_Intercept_lo,
    x1 = bwt_Intercept, y1 = tarsus_Intercept_up,
    col = "black"
  )
)
```

```

with(bl_m2.1, segments(
  x0 = bwt_Intercept_lo, y0 = tarsus_Intercept,
  x1 = bwt_Intercept_up, y1 = tarsus_Intercept,
  col = "black"
))
points(tarsus_Intercept ~ bwt_Intercept, bl_m2.1, pch = 16, col = "red", cex = 1.5)
points(tarsus_Intercept ~ bwt_Intercept, bl_m2.1, pch = 1, col = rgb(0, 0, 0, 0.3), cex = c(1.5))
mtext("btw (BV±CI)", side = 1, line = 2.4)
mtext("tarsus (BV±CI)", side = 2, line = 2, las = 3)

```



### 7.0.2. Adding fixed and random effects

Fixed and random effects can be added just as for the univariate case. Given that our full model of bwt from tutorial 1 had sex as a fixed effect as well as random effects of byear and mother, we could specify a bivariate formulation of this using the following code (including a line to save the output):

```

bf_bwt_2 <- bf(bwt ~ 1 + sex + (1 | a | gr(animal, cov = Amat)) + (1 | b | byear) + (1 | c | moth
bf_tarsus_2 <- bf(tarsus ~ 1 + sex + (1 | a | gr(animal, cov = Amat)) + (1 | b | byear) + (1 | c
brms_m2.2 <- brm(
  bf_bwt_2 + bf_tarsus_2 + set_rescor(TRUE),
  data = gryphon,
  data2 = list(Amat = Amat),
  chains = 2, cores = 2, iter = 1000

```

```
)

save(brms_m2.2, file = "r-obj/brms_m2_2.rda")
```

Again we have provided the data from one such run. It can be accessed using the code:

```
load("r-obj/brms_m2_2.rda")
summary(brms_m2.2)
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

Warning: There were 4 divergent transitions after warmup. Increasing adapt\_delta above 0.8 may help. See <http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup>

```
Family: MV(gaussian, gaussian)
Links: mu = identity; sigma = identity
       mu = identity; sigma = identity
Formula: bwt ~ 1 + sex + (1 | a | gr(animal, cov = Amat)) + (1 | b | byear) + (1 | c | mother)
         tarsus ~ 1 + sex + (1 | a | gr(animal, cov = Amat)) + (1 | b | byear) + (1 | c | mother)
Data: gryphon (Number of observations: 683)
Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;
       total post-warmup draws = 1000
```

Multilevel Hyperparameters:

~animal (Number of levels: 683)

|                                     | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat |
|-------------------------------------|----------|-----------|----------|----------|------|
| sd(bwt_Intercept)                   | 1.31     | 0.21      | 0.86     | 1.69     | 1.06 |
| sd(tarsus_Intercept)                | 2.88     | 0.47      | 1.88     | 3.70     | 1.01 |
| cor(bwt_Intercept,tarsus_Intercept) | 0.60     | 0.17      | 0.18     | 0.89     | 1.09 |
|                                     | Bulk_ESS |           | Tail_ESS |          |      |
| sd(bwt_Intercept)                   | 54       | 58        |          |          |      |

```
sd(tarsus_Intercept)          52      162
cor(bwt_Intercept,tarsus_Intercept)  25      30
```

```
~byear (Number of levels: 34)
```

|                                     | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat |
|-------------------------------------|----------|-----------|----------|----------|------|
| sd(bwt_Intercept)                   | 0.99     | 0.17      | 0.71     | 1.39     | 1.00 |
| sd(tarsus_Intercept)                | 2.02     | 0.34      | 1.45     | 2.78     | 1.00 |
| cor(bwt_Intercept,tarsus_Intercept) | 0.01     | 0.22      | -0.44    | 0.44     | 1.01 |

|                                     | Bulk_ESS | Tail_ESS |
|-------------------------------------|----------|----------|
| sd(bwt_Intercept)                   | 424      | 509      |
| sd(tarsus_Intercept)                | 525      | 699      |
| cor(bwt_Intercept,tarsus_Intercept) | 478      | 492      |

```
~mother (Number of levels: 352)
```

|                                     | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat |
|-------------------------------------|----------|-----------|----------|----------|------|
| sd(bwt_Intercept)                   | 1.14     | 0.12      | 0.90     | 1.36     | 1.01 |
| sd(tarsus_Intercept)                | 2.09     | 0.29      | 1.54     | 2.67     | 1.01 |
| cor(bwt_Intercept,tarsus_Intercept) | -0.64    | 0.20      | -0.97    | -0.24    | 1.02 |

|                                     | Bulk_ESS | Tail_ESS |
|-------------------------------------|----------|----------|
| sd(bwt_Intercept)                   | 370      | 764      |
| sd(tarsus_Intercept)                | 134      | 420      |
| cor(bwt_Intercept,tarsus_Intercept) | 84       | 147      |

Regression Coefficients:

|                  | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|------------------|----------|-----------|----------|----------|------|----------|----------|
| bwt_Intercept    | 6.28     | 0.24      | 5.80     | 6.73     | 1.00 | 453      | 703      |
| tarsus_Intercept | 20.39    | 0.52      | 19.45    | 21.39    | 1.00 | 755      | 760      |
| bwt_sex2         | 2.05     | 0.17      | 1.71     | 2.37     | 1.00 | 1097     | 715      |
| tarsus_sex2      | 0.11     | 0.42      | -0.67    | 0.90     | 1.00 | 780      | 578      |

Further Distributional Parameters:

|           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------|----------|-----------|----------|----------|------|----------|----------|
| sigma_bwt | 1.40     | 0.16      | 1.05     | 1.68     | 1.04 | 59       | 60       |

---

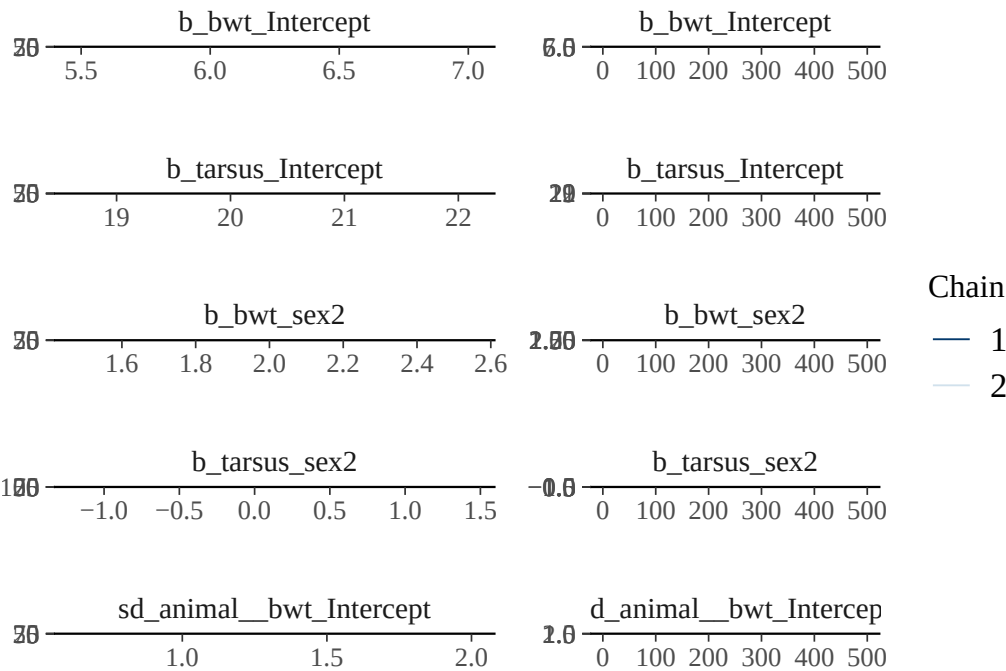
|              |      |      |      |      |      |    |     |
|--------------|------|------|------|------|------|----|-----|
| sigma_tarsus | 3.73 | 0.32 | 3.14 | 4.32 | 1.00 | 55 | 176 |
|--------------|------|------|------|------|------|----|-----|

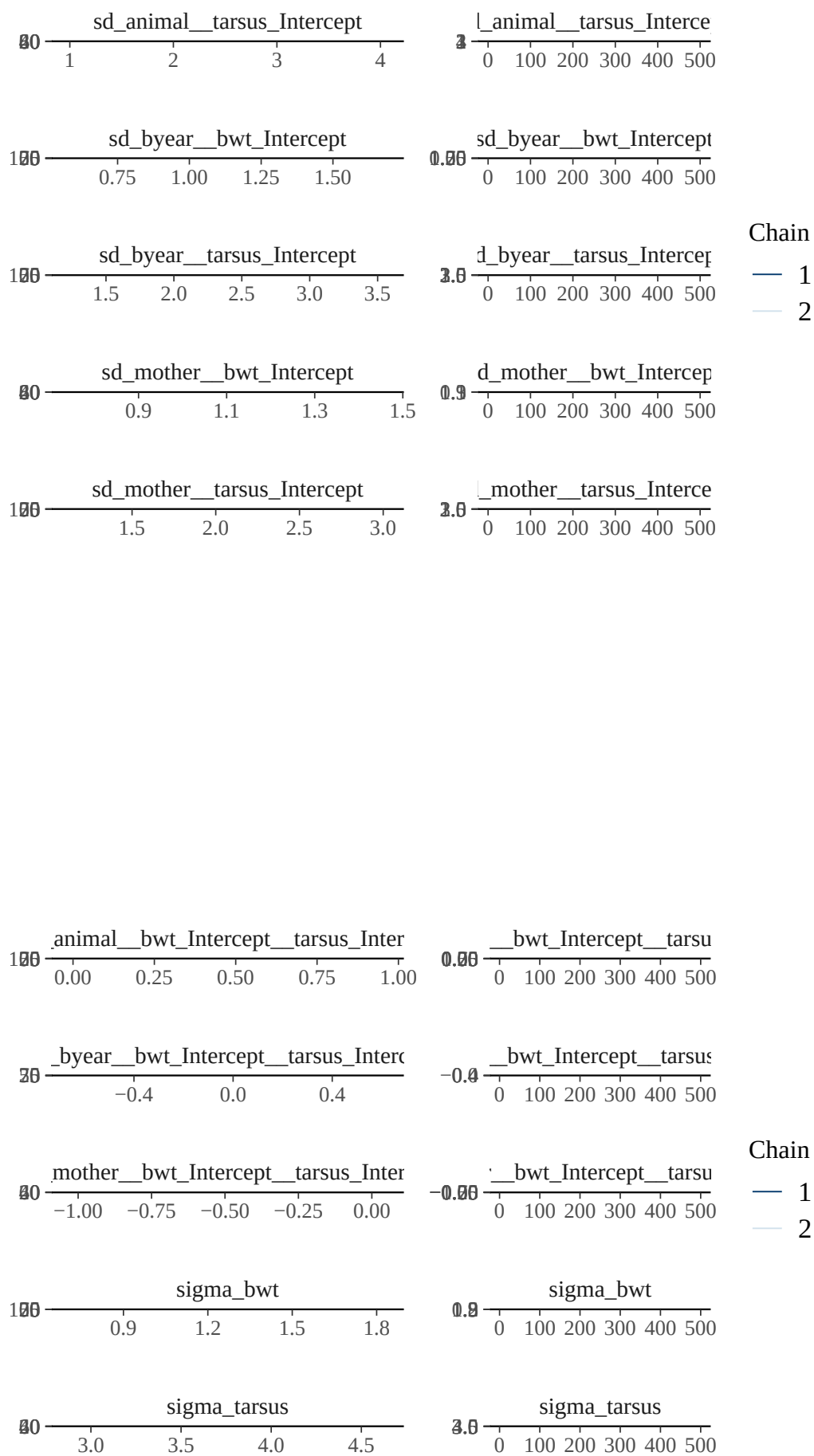
Residual Correlations:

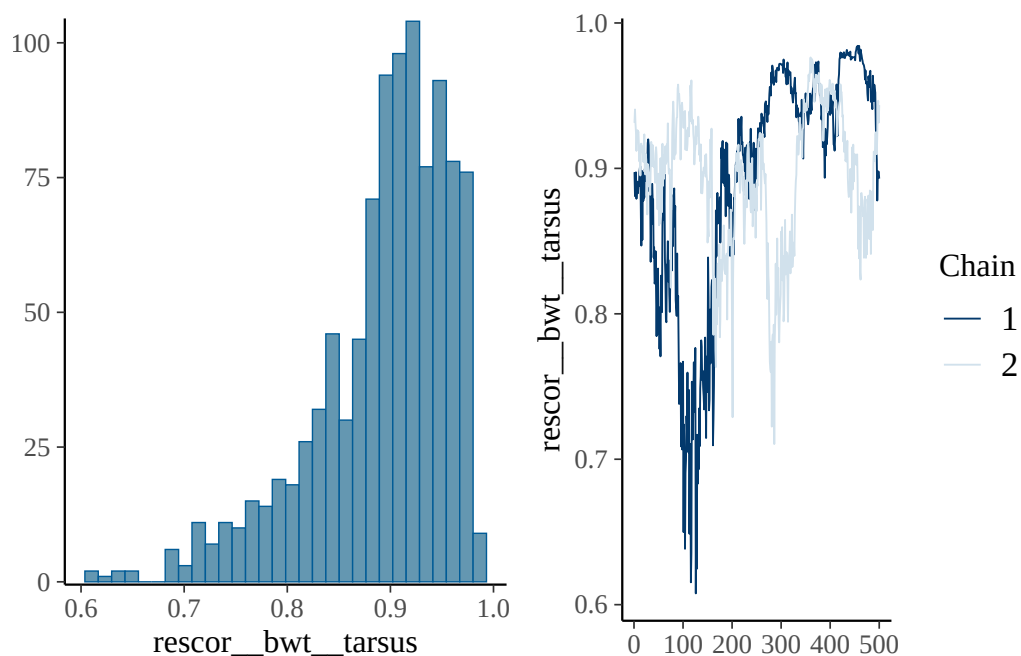
|                    | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|--------------------|----------|-----------|----------|----------|------|----------|----------|
| rescor(bwt,tarsus) | 0.89     | 0.07      | 0.72     | 0.98     | 1.50 | 4        | 24       |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(brms_m2.2, ask = FALSE)
```







```
VarCorr(brms_m2.2)
```

```
$animal
```

```
$animal$sd
```

|                  | Estimate | Est.Error | Q2.5      | Q97.5    |
|------------------|----------|-----------|-----------|----------|
| bwt_Intercept    | 1.306921 | 0.2109780 | 0.8555702 | 1.687254 |
| tarsus_Intercept | 2.876731 | 0.4718524 | 1.8827128 | 3.699873 |

```
$animal$cor
```

```
, , bwt_Intercept
```

|                  | Estimate  | Est.Error | Q2.5      | Q97.5     |
|------------------|-----------|-----------|-----------|-----------|
| bwt_Intercept    | 1.0000000 | 0.0000000 | 1.0000000 | 1.0000000 |
| tarsus_Intercept | 0.5961514 | 0.1727943 | 0.1849725 | 0.8929607 |

```
, , tarsus_Intercept
```

|                  | Estimate  | Est.Error | Q2.5      | Q97.5     |
|------------------|-----------|-----------|-----------|-----------|
| bwt_Intercept    | 0.5961514 | 0.1727943 | 0.1849725 | 0.8929607 |
| tarsus_Intercept | 1.0000000 | 0.0000000 | 1.0000000 | 1.0000000 |

\$animal\$cov

, , bwt\_Intercept

|                  | Estimate | Est.Error | Q2.5      | Q97.5    |
|------------------|----------|-----------|-----------|----------|
| bwt_Intercept    | 1.752510 | 0.5433004 | 0.7320013 | 2.846827 |
| tarsus_Intercept | 2.357852 | 1.0462644 | 0.3960956 | 4.526731 |

, , tarsus\_Intercept

|                  | Estimate | Est.Error | Q2.5      | Q97.5     |
|------------------|----------|-----------|-----------|-----------|
| bwt_Intercept    | 2.357852 | 1.046264  | 0.3960956 | 4.526731  |
| tarsus_Intercept | 8.498003 | 2.672465  | 3.5446115 | 13.689062 |

\$byear

\$byear\$sd

|                  | Estimate  | Est.Error | Q2.5      | Q97.5    |
|------------------|-----------|-----------|-----------|----------|
| bwt_Intercept    | 0.9904858 | 0.1696752 | 0.7102391 | 1.390260 |
| tarsus_Intercept | 2.0166804 | 0.3366954 | 1.4463862 | 2.777005 |

\$byear\$cor

, , bwt\_Intercept

|                  | Estimate   | Est.Error | Q2.5       | Q97.5     |
|------------------|------------|-----------|------------|-----------|
| bwt_Intercept    | 1.00000000 | 0.0000000 | 1.0000000  | 1.0000000 |
| tarsus_Intercept | 0.01351405 | 0.2206186 | -0.4367809 | 0.4412319 |

, , tarsus\_Intercept

|               | Estimate   | Est.Error | Q2.5       | Q97.5     |
|---------------|------------|-----------|------------|-----------|
| bwt_Intercept | 0.01351405 | 0.2206186 | -0.4367809 | 0.4412319 |



---

tarsus\_Intercept 1.00000000 0.0000000 1.0000000 1.0000000

\$byear\$cov

, , bwt\_Intercept

|                  | Estimate   | Est.Error | Q2.5       | Q97.5    |
|------------------|------------|-----------|------------|----------|
| bwt_Intercept    | 1.00982294 | 0.3583697 | 0.5044397  | 1.932823 |
| tarsus_Intercept | 0.06412895 | 0.4880116 | -0.8559486 | 1.092317 |

, , tarsus\_Intercept

|                  | Estimate   | Est.Error | Q2.5       | Q97.5    |
|------------------|------------|-----------|------------|----------|
| bwt_Intercept    | 0.06412895 | 0.4880116 | -0.8559486 | 1.092317 |
| tarsus_Intercept | 4.18025049 | 1.4249595 | 2.0920330  | 7.711755 |

\$mother

\$mother\$sd

|                  | Estimate | Est.Error | Q2.5      | Q97.5    |
|------------------|----------|-----------|-----------|----------|
| bwt_Intercept    | 1.137249 | 0.1175495 | 0.8968349 | 1.358078 |
| tarsus_Intercept | 2.088602 | 0.2865291 | 1.5425683 | 2.673791 |

\$mother\$cor

, , bwt\_Intercept

|                  | Estimate   | Est.Error | Q2.5       | Q97.5      |
|------------------|------------|-----------|------------|------------|
| bwt_Intercept    | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000  |
| tarsus_Intercept | -0.6413329 | 0.1968102 | -0.9745925 | -0.2367393 |

, , tarsus\_Intercept

|                  | Estimate   | Est.Error | Q2.5       | Q97.5      |
|------------------|------------|-----------|------------|------------|
| bwt_Intercept    | -0.6413329 | 0.1968102 | -0.9745925 | -0.2367393 |
| tarsus_Intercept | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000  |

\$mother\$cov

, , bwt\_Intercept

|                  | Estimate  | Est.Error | Q2.5       | Q97.5      |
|------------------|-----------|-----------|------------|------------|
| bwt_Intercept    | 1.307139  | 0.2669748 | 0.8043132  | 1.8443762  |
| tarsus_Intercept | -1.467511 | 0.3650113 | -2.1491979 | -0.6718737 |

, , tarsus\_Intercept

|                  | Estimate  | Est.Error | Q2.5      | Q97.5      |
|------------------|-----------|-----------|-----------|------------|
| bwt_Intercept    | -1.467511 | 0.3650113 | -2.149198 | -0.6718737 |
| tarsus_Intercept | 4.444274  | 1.2018090 | 2.379517  | 7.1491575  |

\$residual\_\_

\$residual\_\_\$sd

|        | Estimate | Est.Error | Q2.5     | Q97.5    |
|--------|----------|-----------|----------|----------|
| bwt    | 1.396815 | 0.1597393 | 1.052042 | 1.684810 |
| tarsus | 3.733614 | 0.3170549 | 3.140204 | 4.319446 |

\$residual\_\_\$cor

, , bwt

|        | Estimate  | Est.Error | Q2.5      | Q97.5     |
|--------|-----------|-----------|-----------|-----------|
| bwt    | 1.0000000 | 0.0000000 | 1.0000000 | 1.0000000 |
| tarsus | 0.8925119 | 0.0672286 | 0.7167862 | 0.9774545 |

---

```
, , tarsus
```

```
      Estimate Est.Error      Q2.5      Q97.5
bwt    0.8925119 0.0672286 0.7167862 0.9774545
tarsus 1.0000000 0.0000000 1.0000000 1.0000000
```

```
$residual__$cov
```

```
, , bwt
```

```
      Estimate Est.Error      Q2.5      Q97.5
bwt    1.976584 0.4347085 1.106793 2.838585
tarsus 4.683451 0.9103808 2.846560 6.440550
```

```
, , tarsus
```

```
      Estimate Est.Error      Q2.5      Q97.5
bwt    4.683451 0.9103808 2.846560 6.44055
tarsus 14.040300 2.3674563 9.860883 18.65761
```

Evaluation of the statistical support for these genetic and maternal correlations is straightforward. Because we imposed no constraint on their estimation, we can evaluate the extent to which the posterior distributions overlap zero:

```
cor_g <- as.mcmc((VarCorr(brms_m2.2, summary = FALSE)$animal$cor[, 1, 2]))
cor_res <- as.mcmc((VarCorr(brms_m2.2, summary = FALSE)$residual$cor[, 1, 2]))
cor_mother <- as.mcmc((VarCorr(brms_m2.2, summary = FALSE)$mother$cor[, 1, 2]))
cor_byear <- as.mcmc((VarCorr(brms_m2.2, summary = FALSE)$byear$cor[, 1, 2]))

summary(cor_g)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

Number of chains = 1

Sample size per chain = 1000

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.596151 | 0.172794 | 0.005464 | 0.028178       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.1850 | 0.5065 | 0.6117 | 0.7064 | 0.8930 |

```
summary(cor_mother)
```

Iterations = 1:1000

Thinning interval = 1

Number of chains = 1

Sample size per chain = 1000

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean      | SD       | Naive SE | Time-series SE |
|-----------|----------|----------|----------------|
| -0.641333 | 0.196810 | 0.006224 | 0.021355       |

2. Quantiles for each variable:

| 2.5%    | 25%     | 50%     | 75%     | 97.5%   |
|---------|---------|---------|---------|---------|
| -0.9746 | -0.7933 | -0.6507 | -0.5027 | -0.2367 |

---

```
summary(cor_byear)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.013514 | 0.220619 | 0.006977 | 0.009633       |

2. Quantiles for each variable:

| 2.5%     | 25%      | 50%     | 75%     | 97.5%   |
|----------|----------|---------|---------|---------|
| -0.43678 | -0.13505 | 0.02408 | 0.16947 | 0.44123 |

```
summary(cor_res)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

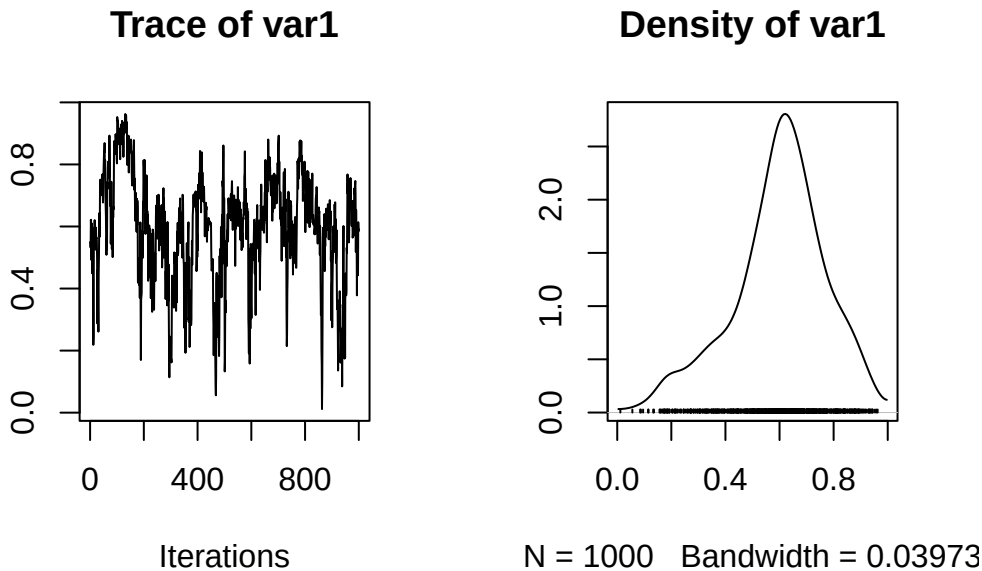
1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.892512 | 0.067229 | 0.002126 | 0.026496       |

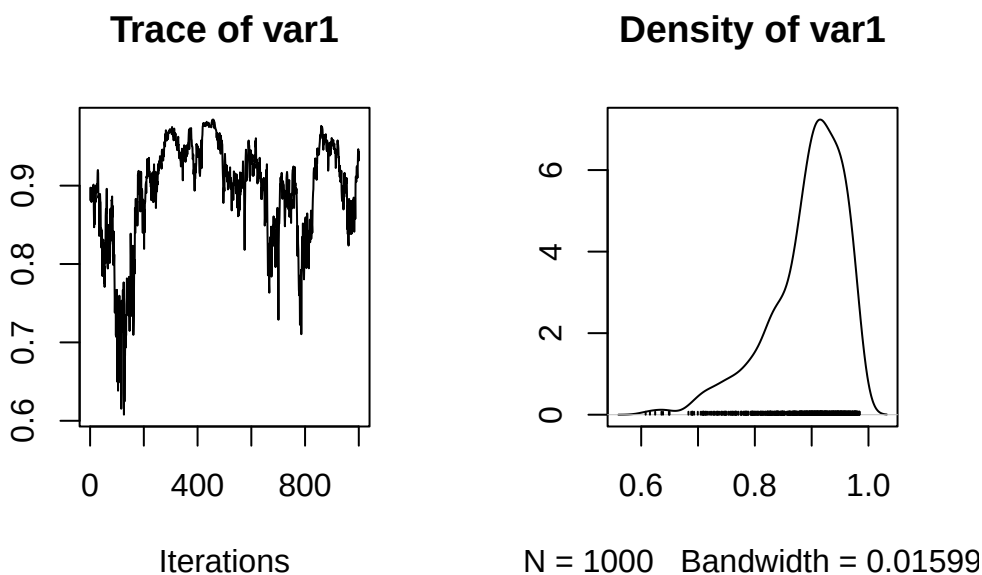
2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.7168 | 0.8620 | 0.9069 | 0.9425 | 0.9775 |

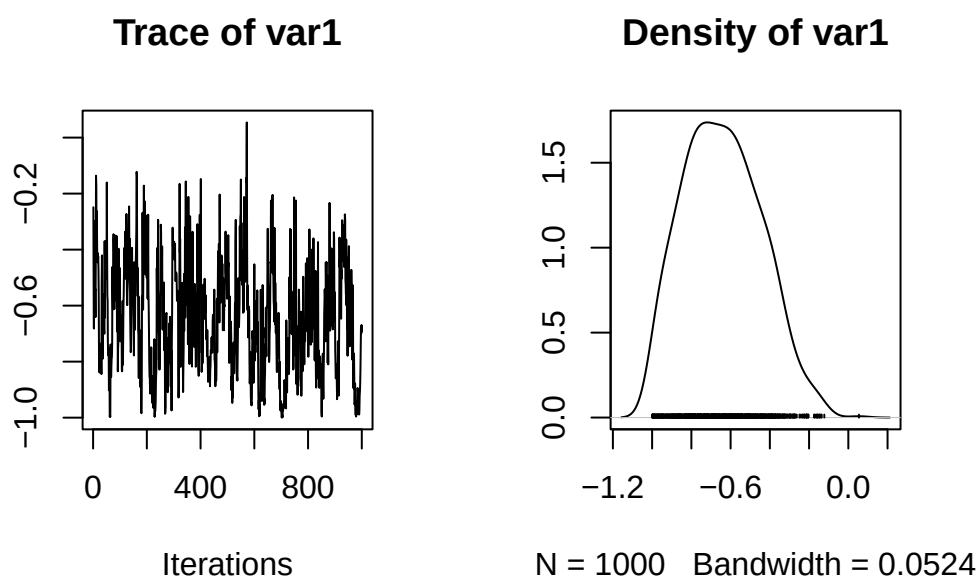
```
plot(cor_g)
```



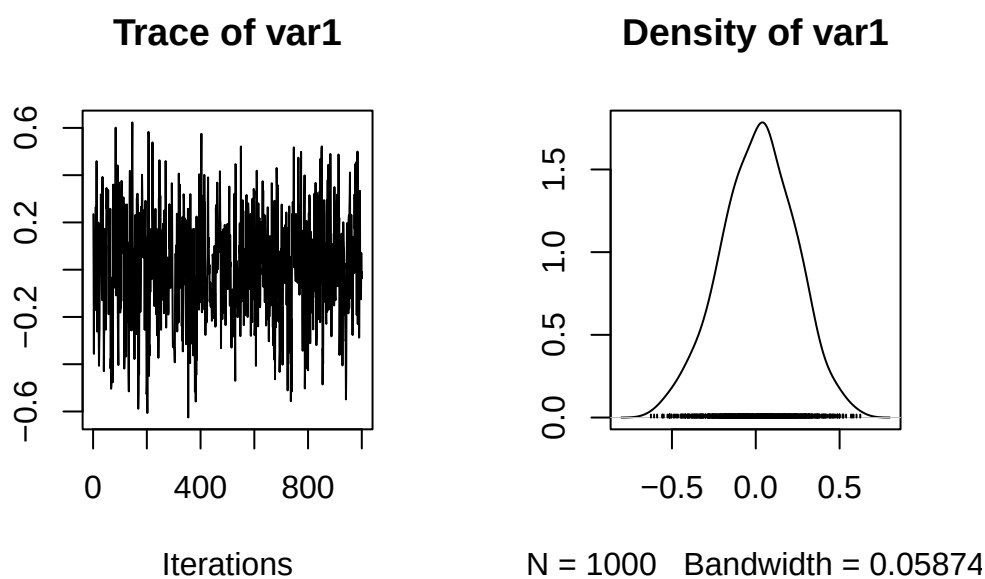
```
plot(cor_res)
```



```
plot(cor_mother)
```



```
plot(cor_byear)
```



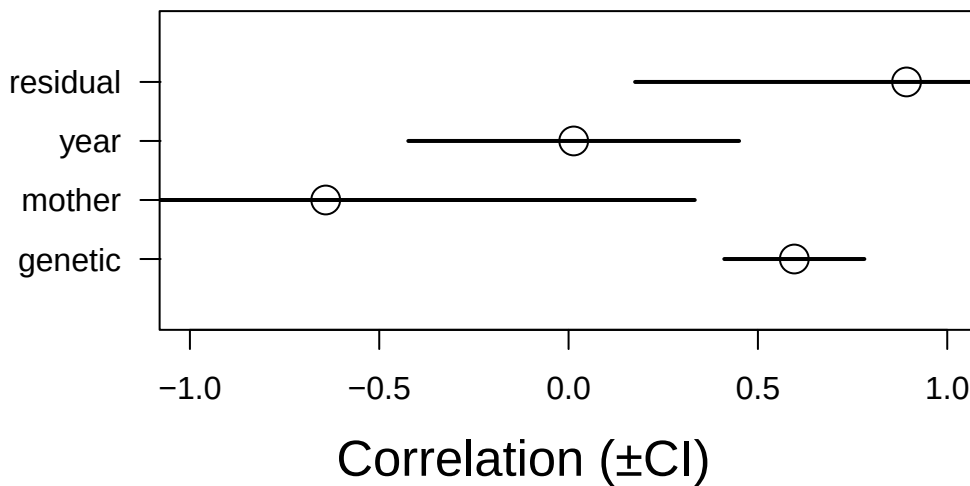
Neither of these posterior distributions overlaps zero, so we can consider them both statistically supported.

```
cor.est <- rbind(
  cbind(summary(cor_g)$statistics[1], summary(cor_g)$quantiles[1], summary(cor_g)$quantiles[5]),
  cbind(summary(cor_mother)$statistics[1], summary(cor_mother)$quantiles[1], summary(cor_mother)$quantiles[5]),
  cbind(summary(cor_byear)$statistics[1], summary(cor_byear)$quantiles[1], summary(cor_byear)$quantiles[5]),
  cbind(summary(cor_res)$statistics[1], summary(cor_res)$quantiles[1], summary(cor_res)$quantiles[5])
)
```

```

plot(c(1, 2, 3, 4) ~ cor.est[, 1], xlim = c(-1, 1), ylim = c(0, 5), xlab = "", ylab = "", cex = 2)
segments(y0 = 1, x0 = cor.est[1, 1] - cor.est[1, 2], y1 = 1, x1 = cor.est[1, 1] + cor.est[1, 2],
segments(y0 = 2, x0 = cor.est[2, 1] - cor.est[2, 2], y1 = 2, x1 = cor.est[2, 1] + cor.est[2, 2],
segments(y0 = 3, x0 = cor.est[3, 1] - cor.est[3, 2], y1 = 3, x1 = cor.est[3, 1] + cor.est[3, 2],
segments(y0 = 4, x0 = cor.est[4, 1] - cor.est[4, 2], y1 = 4, x1 = cor.est[4, 1] + cor.est[4, 2],
mtext("Correlation ( $\pm$ CI)", side = 1, las = 1, adj = 0.4, line = 3, cex = 1.6)
axis(2, at = 1, labels = c("genetic"), las = 2, cex.axis = 1)
axis(2, at = 2, labels = c("mother"), las = 2, cex.axis = 1)
axis(2, at = 3, labels = c("year"), las = 2, cex.axis = 1)
axis(2, at = 4, labels = c("residual"), las = 2, cex.axis = 1)

```



Note, brms estimates the correlation and also the covariance. We can also recalculate the correlation directly from the covariance. To facilitate the extraction of the different parameter, we can use the function `as_draws_df`

```

cov_g <- (VarCorr(brms_m2.2, summary = FALSE)$animal$cov)[, 1, 2]
cov_res <- (VarCorr(brms_m2.2, summary = FALSE)$residual$cov)[, 1, 2]
cov_mother <- (VarCorr(brms_m2.2, summary = FALSE)$mother$cov)[, 1, 2]
cov_byear <- (VarCorr(brms_m2.2, summary = FALSE)$byear$cov)[, 1, 2]

var.est <- as_draws_df(brms_m2.2, variable = c("sd", "sigma"), regex = TRUE)
var.est <- var.est^2

cor_g_2 <- as.mcmc(cov_g / sqrt(var.est[1] * var.est[2]))
cor_byear_2 <- as.mcmc(cov_byear / sqrt(var.est[3] * var.est[4]))

```



---

```
cor_mother_2 <- as.mcmc(cov_g / sqrt(var.est[5] * var.est[6]))
cor_res_2 <- as.mcmc(cov_res / sqrt(var.est[7] * var.est[8]))

summary(cor_g_2)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.596151 | 0.172794 | 0.005464 | 0.028178       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.1850 | 0.5065 | 0.6117 | 0.7064 | 0.8930 |

```
summary(cor_byear_2)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.013514 | 0.220619 | 0.006977 | 0.009633       |

2. Quantiles for each variable:

| 2.5%     | 25%      | 50%     | 75%     | 97.5%   |
|----------|----------|---------|---------|---------|
| -0.43678 | -0.13505 | 0.02408 | 0.16947 | 0.44123 |

```
summary(cor_mother_2)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean    | SD      | Naive SE | Time-series SE |
|---------|---------|----------|----------------|
| 1.01862 | 0.46337 | 0.01465  | 0.06126        |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.1514 | 0.6969 | 1.0310 | 1.3241 | 2.0032 |

```
summary(cor_res_2)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

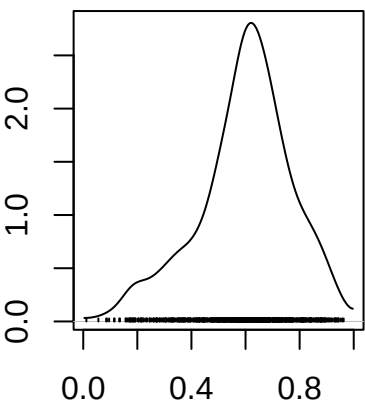
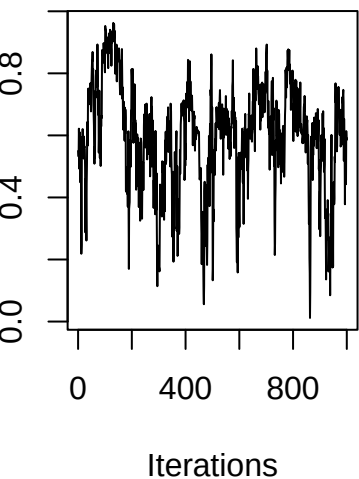
| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.892512 | 0.067229 | 0.002126 | 0.026496       |

2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.7168 | 0.8620 | 0.9069 | 0.9425 | 0.9775 |

```
plot(cor_g_2)
```

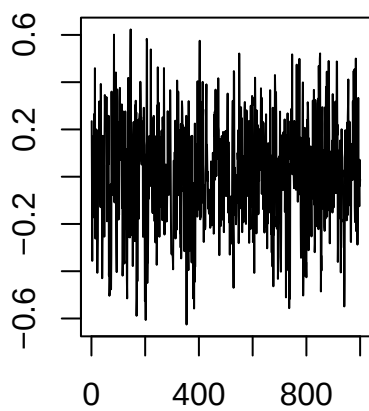
**Trace of sd\_animal\_\_bwt\_Intensity of sd\_animal\_\_bwt\_Intensity**



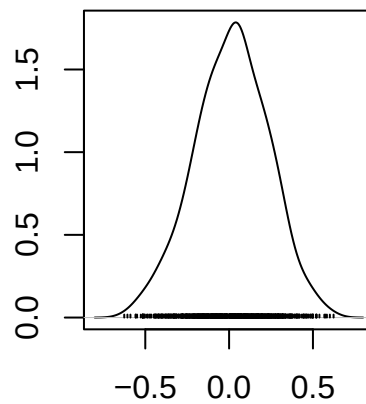
N = 1000 Bandwidth = 0.03973

```
plot(cor_byear_2)
```

## Trace of sd\_byear\_\_bwt\_Inter Density of sd\_byear\_\_bwt\_Inte



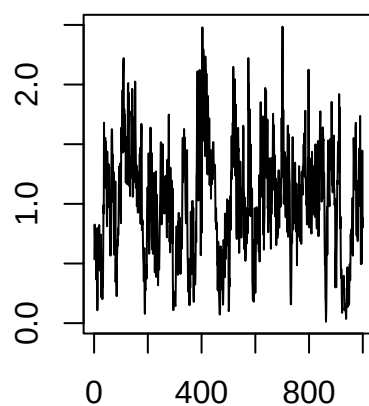
Iterations



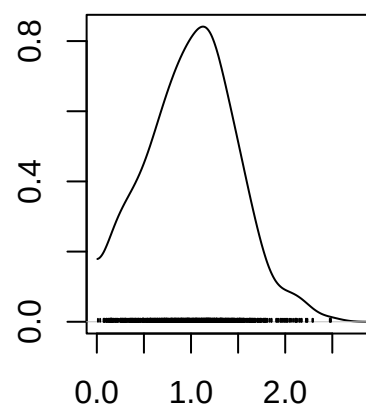
N = 1000 Bandwidth = 0.05874

```
plot(cor_mother_2)
```

## Trace of sd\_mother\_\_bwt\_Intensity of sd\_mother\_\_bwt\_Inte



Iterations

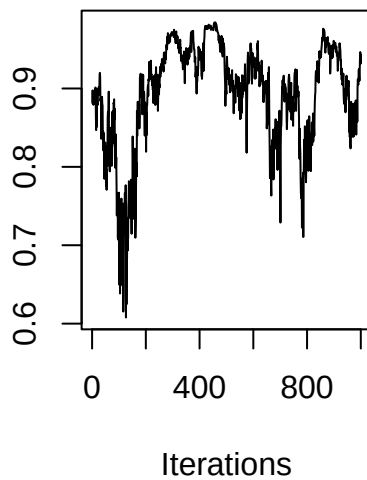


N = 1000 Bandwidth = 0.1234

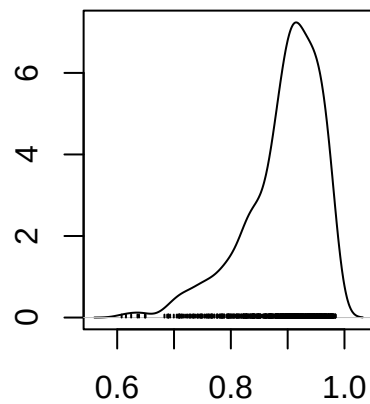
```
plot(cor_res_2)
```

---

Trace of sigma\_bwt



Density of sigma\_bwt



N = 1000 Bandwidth = 0.01599

### 7.0.3. Partitioning (co)variances

As in the tutorial 1, it is possible to partition the variance-covariance matrix between groups (here sex)

```
bf_bwt_3 <- bf(bwt ~ 1 + sex + ((1 | a | gr(animal, cov = Amat, by = sex))) + (1 | b | byear) + (1 | c | sex))
bf_tarsus_3 <- bf(tarsus ~ 1 + sex + (1 | a | gr(animal, cov = Amat, by = sex)) + (1 | b | byear) + (1 | c | sex))

brms_m2.3 <- brm(
  bf_bwt_3 + bf_tarsus_3 + set_rescor(TRUE),
  data = gryphon,
  data2 = list(Amat = Amat),
  chains = 2, cores = 2, iter = 1000
)

save(brms_m2.3, file = "r-obj/brms_m2_3.rda")
```

Again we have provided the data from one such run. It can be accessed using the code:

```
load("r-obj/brms_m2_3.rda")
summary(brms_m2.3)
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be

careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

Warning: There were 6 divergent transitions after warmup. Increasing adapt\_delta above 0.8 may help. See <http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup>

```
Family: MV(gaussian, gaussian)
Links: mu = identity; sigma = identity
       mu = identity; sigma = identity
Formula: bwt ~ 1 + sex + ((1 | a | gr(animal, cov = Amat, by = sex))) + (1 | b | byear) + (1 | c
          tarsus ~ 1 + sex + (1 | a | gr(animal, cov = Amat, by = sex)) + (1 | b | byear) + (1 | c
Data: gryphon (Number of observations: 683)
Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;
       total post-warmup draws = 1000
```

Multilevel Hyperparameters:

~animal (Number of levels: 683)

|                                               | Estimate | Est.Error | 1-95% CI |          |
|-----------------------------------------------|----------|-----------|----------|----------|
| sd(bwt_Intercept:sex1)                        | 1.05     | 0.28      | 0.48     |          |
| sd(tarsus_Intercept:sex1)                     | 1.57     | 0.78      | 0.14     |          |
| sd(bwt_Intercept:sex2)                        | 1.23     | 0.24      | 0.71     |          |
| sd(tarsus_Intercept:sex2)                     | 3.24     | 0.48      | 2.17     |          |
| cor(bwt_Intercept:sex1,tarsus_Intercept:sex1) | 0.32     | 0.43      | -0.78    |          |
| cor(bwt_Intercept:sex2,tarsus_Intercept:sex2) | 0.70     | 0.12      | 0.45     |          |
|                                               | u-95% CI | Rhat      | Bulk_ESS | Tail_ESS |
| sd(bwt_Intercept:sex1)                        | 1.58     | 1.27      | 6        | 39       |
| sd(tarsus_Intercept:sex1)                     | 3.03     | 1.08      | 24       | 117      |
| sd(bwt_Intercept:sex2)                        | 1.65     | 1.24      | 7        | 46       |
| sd(tarsus_Intercept:sex2)                     | 4.13     | 1.05      | 37       | 79       |
| cor(bwt_Intercept:sex1,tarsus_Intercept:sex1) | 0.90     | 1.07      | 25       | 202      |
| cor(bwt_Intercept:sex2,tarsus_Intercept:sex2) | 0.90     | 1.09      | 28       | 44       |

~byear (Number of levels: 34)

---

|                                     | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat |
|-------------------------------------|----------|-----------|----------|----------|------|
| sd(bwt_Intercept)                   | 0.97     | 0.14      | 0.70     | 1.26     | 1.01 |
| sd(tarsus_Intercept)                | 2.03     | 0.34      | 1.47     | 2.80     | 1.00 |
| cor(bwt_Intercept,tarsus_Intercept) | 0.01     | 0.21      | -0.41    | 0.41     | 1.01 |

|                                     | Bulk_ESS | Tail_ESS |
|-------------------------------------|----------|----------|
| sd(bwt_Intercept)                   | 282      | 292      |
| sd(tarsus_Intercept)                | 324      | 361      |
| cor(bwt_Intercept,tarsus_Intercept) | 183      | 393      |

~mother (Number of levels: 352)

|                                     | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat |
|-------------------------------------|----------|-----------|----------|----------|------|
| sd(bwt_Intercept)                   | 1.18     | 0.11      | 0.96     | 1.39     | 1.02 |
| sd(tarsus_Intercept)                | 2.04     | 0.34      | 1.33     | 2.65     | 1.09 |
| cor(bwt_Intercept,tarsus_Intercept) | -0.63    | 0.21      | -0.97    | -0.23    | 1.08 |

|                                     | Bulk_ESS | Tail_ESS |
|-------------------------------------|----------|----------|
| sd(bwt_Intercept)                   | 170      | 352      |
| sd(tarsus_Intercept)                | 24       | 60       |
| cor(bwt_Intercept,tarsus_Intercept) | 22       | 100      |

Regression Coefficients:

|                  | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|------------------|----------|-----------|----------|----------|------|----------|----------|
| bwt_Intercept    | 6.27     | 0.22      | 5.82     | 6.68     | 1.00 | 195      | 123      |
| tarsus_Intercept | 20.35    | 0.48      | 19.35    | 21.21    | 1.00 | 494      | 578      |
| bwt_sex2         | 2.04     | 0.17      | 1.71     | 2.36     | 1.01 | 265      | 384      |
| tarsus_sex2      | 0.14     | 0.39      | -0.60    | 0.88     | 1.00 | 691      | 621      |

Further Distributional Parameters:

|              | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|--------------|----------|-----------|----------|----------|------|----------|----------|
| sigma_bwt    | 1.49     | 0.15      | 1.16     | 1.74     | 1.33 | 5        | 81       |
| sigma_tarsus | 3.88     | 0.27      | 3.27     | 4.35     | 1.04 | 42       | 85       |

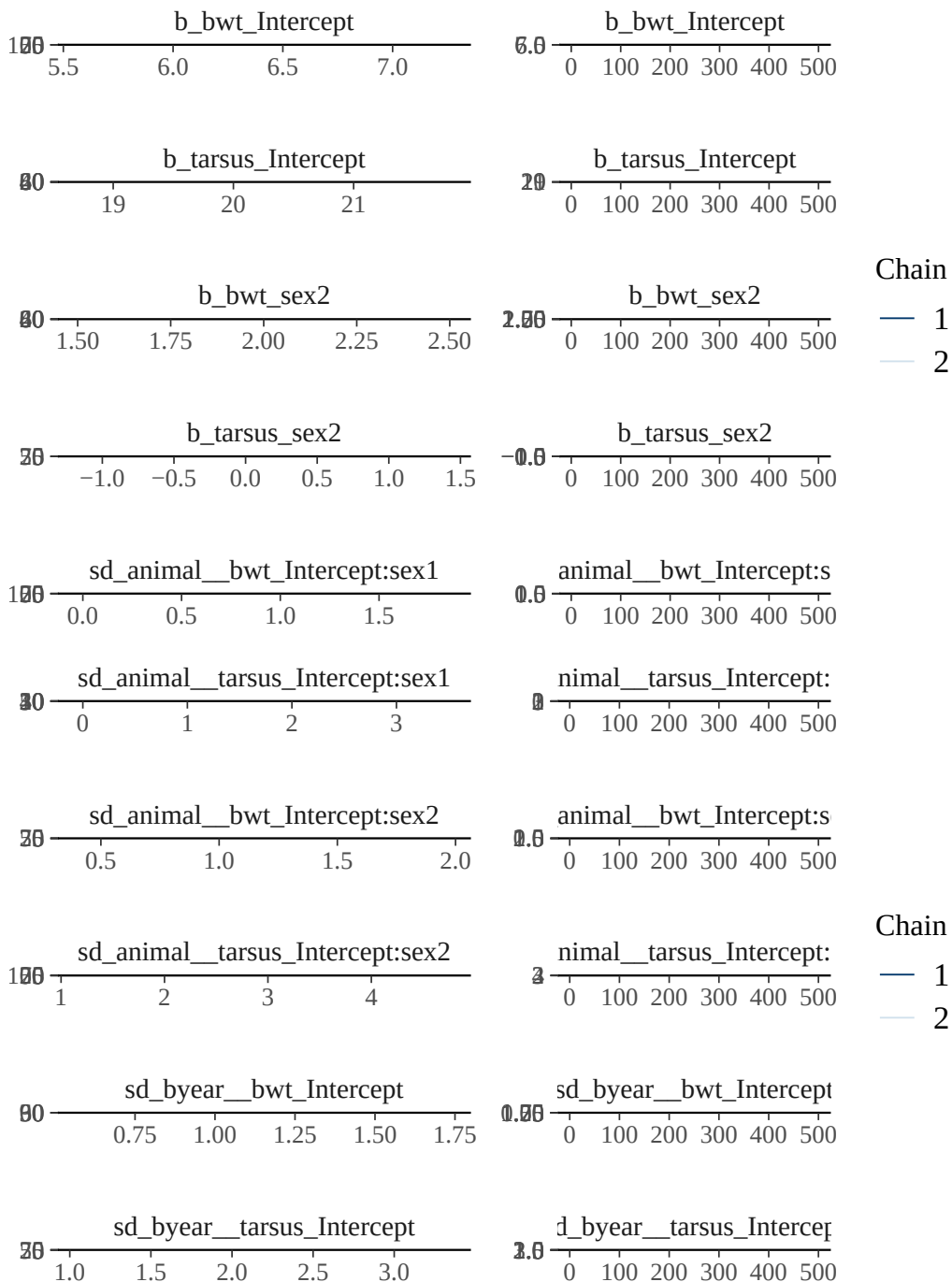
Residual Correlations:

|  | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|--|----------|-----------|----------|----------|------|----------|----------|
|--|----------|-----------|----------|----------|------|----------|----------|

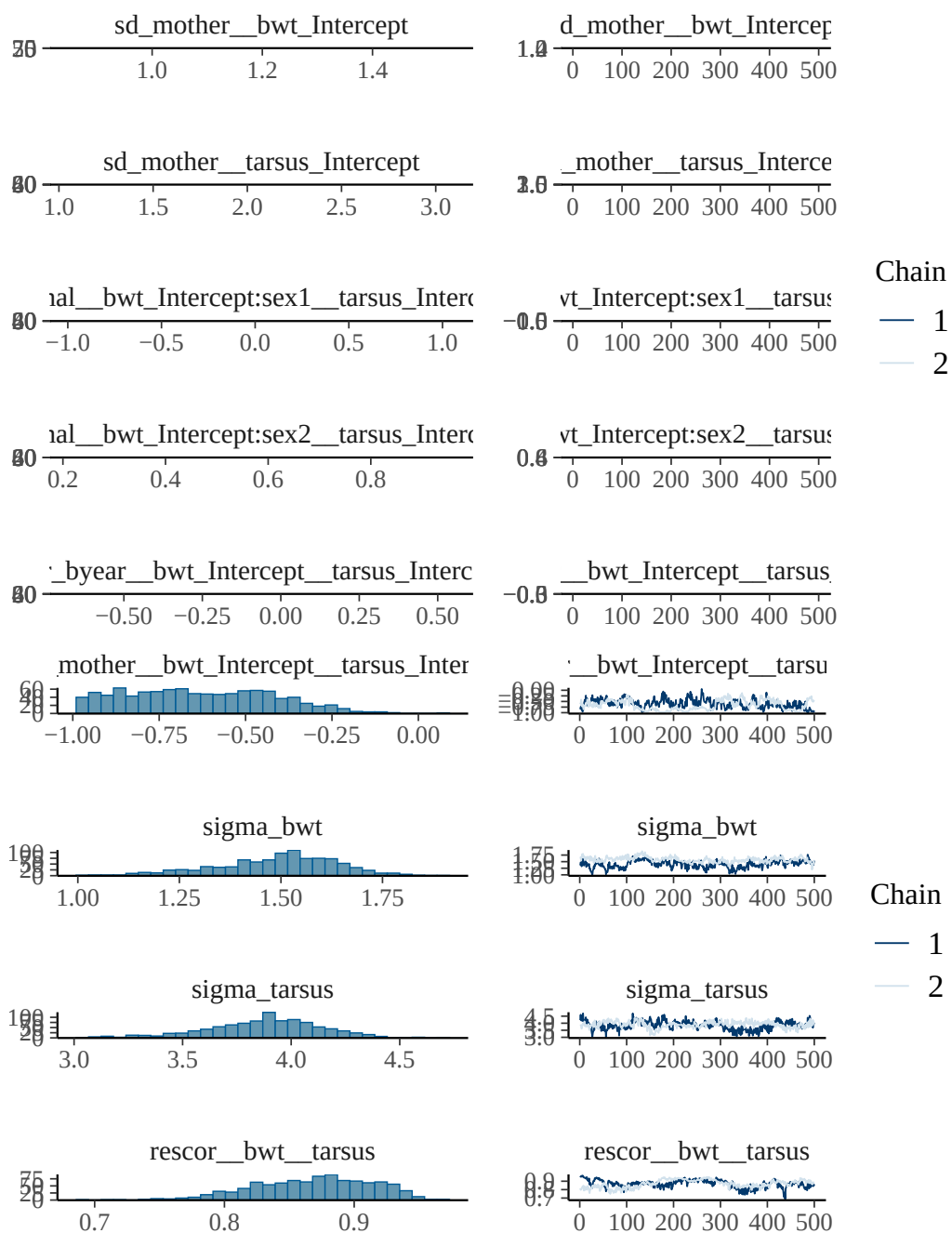
```
rescor(bwt,tarsus)      0.87      0.05      0.78      0.94 1.04      22      94
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(brms_m2.3, ask = FALSE)
```







```
VarCorr(brms_m2.3)
```

```
$animal
```

```
$animal$sd
```

|                       | Estimate | Est.Error | Q2.5      | Q97.5    |
|-----------------------|----------|-----------|-----------|----------|
| bwt_Intercept:sex1    | 1.051791 | 0.2757021 | 0.4836926 | 1.575837 |
| tarsus_Intercept:sex1 | 1.573820 | 0.7777373 | 0.1389929 | 3.025426 |
| bwt_Intercept:sex2    | 1.232722 | 0.2436415 | 0.7075357 | 1.650628 |

tarsus\_Intercept:sex2 3.237363 0.4773725 2.1667846 4.126560

\$animal\$cor

, , bwt\_Intercept:sex1

|                       | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-----------------------|-----------|-----------|------------|-----------|
| bwt_Intercept:sex1    | 1.0000000 | 0.0000000 | 1.0000000  | 1.0000000 |
| tarsus_Intercept:sex1 | 0.3232502 | 0.4324034 | -0.7811983 | 0.9007145 |
| bwt_Intercept:sex2    | 0.0000000 | 0.0000000 | 0.0000000  | 0.0000000 |
| tarsus_Intercept:sex2 | 0.0000000 | 0.0000000 | 0.0000000  | 0.0000000 |

, , tarsus\_Intercept:sex1

|                       | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-----------------------|-----------|-----------|------------|-----------|
| bwt_Intercept:sex1    | 0.3232502 | 0.4324034 | -0.7811983 | 0.9007145 |
| tarsus_Intercept:sex1 | 1.0000000 | 0.0000000 | 1.0000000  | 1.0000000 |
| bwt_Intercept:sex2    | 0.0000000 | 0.0000000 | 0.0000000  | 0.0000000 |
| tarsus_Intercept:sex2 | 0.0000000 | 0.0000000 | 0.0000000  | 0.0000000 |

, , bwt\_Intercept:sex2

|                       | Estimate  | Est.Error | Q2.5      | Q97.5     |
|-----------------------|-----------|-----------|-----------|-----------|
| bwt_Intercept:sex1    | 0.0000000 | 0.0000000 | 0.0000000 | 0.0000000 |
| tarsus_Intercept:sex1 | 0.0000000 | 0.0000000 | 0.0000000 | 0.0000000 |
| bwt_Intercept:sex2    | 1.0000000 | 0.0000000 | 1.0000000 | 1.0000000 |
| tarsus_Intercept:sex2 | 0.6975925 | 0.1213692 | 0.4474181 | 0.9047623 |

, , tarsus\_Intercept:sex2

|                       | Estimate  | Est.Error | Q2.5      | Q97.5     |
|-----------------------|-----------|-----------|-----------|-----------|
| bwt_Intercept:sex1    | 0.0000000 | 0.0000000 | 0.0000000 | 0.0000000 |
| tarsus_Intercept:sex1 | 0.0000000 | 0.0000000 | 0.0000000 | 0.0000000 |
| bwt_Intercept:sex2    | 0.6975925 | 0.1213692 | 0.4474181 | 0.9047623 |

---

tarsus\_Intercept:sex2 1.0000000 0.0000000 1.0000000 1.0000000

\$animal\$cov

, , bwt\_Intercept:sex1

|                       | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-----------------------|-----------|-----------|------------|-----------|
| bwt_Intercept:sex1    | 1.1821990 | 0.5768155 | 0.2339592  | 2.483263  |
| tarsus_Intercept:sex1 | 0.7957577 | 0.9482719 | -0.4879668 | 3.035395  |
| bwt_Intercept:sex2    | 0.0000000 | 0.0000000 | 0.0000000  | 0.0000000 |
| tarsus_Intercept:sex2 | 0.0000000 | 0.0000000 | 0.0000000  | 0.0000000 |

, , tarsus\_Intercept:sex1

|                       | Estimate  | Est.Error | Q2.5        | Q97.5     |
|-----------------------|-----------|-----------|-------------|-----------|
| bwt_Intercept:sex1    | 0.7957577 | 0.9482719 | -0.48796685 | 3.035395  |
| tarsus_Intercept:sex1 | 3.0811806 | 2.5183190 | 0.01931907  | 9.153208  |
| bwt_Intercept:sex2    | 0.0000000 | 0.0000000 | 0.00000000  | 0.0000000 |
| tarsus_Intercept:sex2 | 0.0000000 | 0.0000000 | 0.00000000  | 0.0000000 |

, , bwt\_Intercept:sex2

|                       | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-----------------------|-----------|-----------|------------|-----------|
| bwt_Intercept:sex1    | 0.0000000 | 0.0000000 | 0.00000000 | 0.0000000 |
| tarsus_Intercept:sex1 | 0.0000000 | 0.0000000 | 0.00000000 | 0.0000000 |
| bwt_Intercept:sex2    | 1.578907  | 0.5865683 | 0.5006069  | 2.724572  |
| tarsus_Intercept:sex2 | 2.842372  | 1.0153859 | 1.1593760  | 5.086448  |

, , tarsus\_Intercept:sex2

|                       | Estimate  | Est.Error | Q2.5      | Q97.5     |
|-----------------------|-----------|-----------|-----------|-----------|
| bwt_Intercept:sex1    | 0.0000000 | 0.0000000 | 0.0000000 | 0.0000000 |
| tarsus_Intercept:sex1 | 0.0000000 | 0.0000000 | 0.0000000 | 0.0000000 |

```
bwt_Intercept:sex2      2.842372  1.015386  1.159376  5.086448
tarsus_Intercept:sex2 10.708178  3.017245  4.694964 17.028497
```

```
$byear
```

```
$byear$sd
```

|                  | Estimate  | Est.Error | Q2.5      | Q97.5    |
|------------------|-----------|-----------|-----------|----------|
| bwt_Intercept    | 0.9676965 | 0.1434955 | 0.7018779 | 1.258518 |
| tarsus_Intercept | 2.0290144 | 0.3382466 | 1.4650518 | 2.801932 |

```
$byear$cor
```

```
, , bwt_Intercept
```

|                  | Estimate    | Est.Error | Q2.5       | Q97.5     |
|------------------|-------------|-----------|------------|-----------|
| bwt_Intercept    | 1.000000000 | 0.0000000 | 1.0000000  | 1.0000000 |
| tarsus_Intercept | 0.009103073 | 0.2077977 | -0.4096021 | 0.410902  |

```
, , tarsus_Intercept
```

|                  | Estimate    | Est.Error | Q2.5       | Q97.5     |
|------------------|-------------|-----------|------------|-----------|
| bwt_Intercept    | 0.009103073 | 0.2077977 | -0.4096021 | 0.410902  |
| tarsus_Intercept | 1.000000000 | 0.0000000 | 1.0000000  | 1.0000000 |

```
$byear$cov
```

```
, , bwt_Intercept
```

|                  | Estimate   | Est.Error | Q2.5       | Q97.5    |
|------------------|------------|-----------|------------|----------|
| bwt_Intercept    | 0.95700691 | 0.2875804 | 0.4926327  | 1.583869 |
| tarsus_Intercept | 0.04233908 | 0.4457863 | -0.8475401 | 1.014920 |

```
, , tarsus_Intercept
```

---

|                  | Estimate   | Est.Error | Q2.5       | Q97.5    |
|------------------|------------|-----------|------------|----------|
| bwt_Intercept    | 0.04233908 | 0.4457863 | -0.8475401 | 1.014920 |
| tarsus_Intercept | 4.23119588 | 1.4453420 | 2.1463767  | 7.850826 |

\$mother

\$mother\$sd

|                  | Estimate | Est.Error | Q2.5     | Q97.5    |
|------------------|----------|-----------|----------|----------|
| bwt_Intercept    | 1.175168 | 0.1083817 | 0.958963 | 1.388926 |
| tarsus_Intercept | 2.038471 | 0.3445355 | 1.327549 | 2.648997 |

\$mother\$cor

, , bwt\_Intercept

|                  | Estimate   | Est.Error | Q2.5       | Q97.5      |
|------------------|------------|-----------|------------|------------|
| bwt_Intercept    | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000  |
| tarsus_Intercept | -0.6301934 | 0.2113836 | -0.9652044 | -0.2283133 |

, , tarsus\_Intercept

|                  | Estimate   | Est.Error | Q2.5       | Q97.5      |
|------------------|------------|-----------|------------|------------|
| bwt_Intercept    | -0.6301934 | 0.2113836 | -0.9652044 | -0.2283133 |
| tarsus_Intercept | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000  |

\$mother\$cov

, , bwt\_Intercept

|                  | Estimate  | Est.Error | Q2.5       | Q97.5     |
|------------------|-----------|-----------|------------|-----------|
| bwt_Intercept    | 1.392755  | 0.2562907 | 0.9196105  | 1.929117  |
| tarsus_Intercept | -1.434727 | 0.3664843 | -2.0769067 | -0.674943 |

, , tarsus\_Intercept

|                  | Estimate  | Est.Error | Q2.5      | Q97.5     |
|------------------|-----------|-----------|-----------|-----------|
| bwt_Intercept    | -1.434727 | 0.3664843 | -2.076907 | -0.674943 |
| tarsus_Intercept | 4.273951  | 1.3914013 | 1.762387  | 7.017188  |

\$residual\_\_

\$residual\_\_\$sd

|        | Estimate | Est.Error | Q2.5     | Q97.5    |
|--------|----------|-----------|----------|----------|
| bwt    | 1.489342 | 0.1489641 | 1.162192 | 1.741559 |
| tarsus | 3.875633 | 0.2650604 | 3.268802 | 4.345400 |

\$residual\_\_\$cor

, , bwt

|        | Estimate  | Est.Error  | Q2.5      | Q97.5     |
|--------|-----------|------------|-----------|-----------|
| bwt    | 1.0000000 | 0.0000000  | 1.0000000 | 1.0000000 |
| tarsus | 0.8685184 | 0.04534008 | 0.7772864 | 0.9414488 |

, , tarsus

|        | Estimate  | Est.Error  | Q2.5      | Q97.5     |
|--------|-----------|------------|-----------|-----------|
| bwt    | 0.8685184 | 0.04534008 | 0.7772864 | 0.9414488 |
| tarsus | 1.0000000 | 0.0000000  | 1.0000000 | 1.0000000 |

\$residual\_\_\$cov

, , bwt

|  | Estimate | Est.Error | Q2.5 | Q97.5 |
|--|----------|-----------|------|-------|
|--|----------|-----------|------|-------|

---

```

bwt      2.240307 0.4341334 1.350691 3.033029
tarsus   5.034739 0.7852137 3.272088 6.400539

, , tarsus

```

```

      Estimate Est.Error      Q2.5      Q97.5
bwt      5.034739 0.7852137  3.272088  6.400539
tarsus   15.090721 2.0309529 10.685067 18.882505

```

However, this model is lacking an important and essential group-specific partitioning (we do with the `asreml-R` and `MCMCglmm`). We need to partition the residual variance (or sigma) as well. Doing so, we will use the argument ‘sigma’ to partition the model by sex. To avoid an estimation of the difference between sexes, we need to remove the estimate of the intercept at the sigma level.

```

bf_bwt_4 <- bf(bwt ~ 1 + sex + ((1 | a | gr(animal, cov = Amat, by = sex))) + (1 | b | byear) + (
bf_tarsus_4 <- bf(tarsus ~ 1 + sex + (1 | a | gr(animal, cov = Amat, by = sex)) + (1 | b | byear)

brms_m2.4 <- brm(
  bf_bwt_4 + bf_tarsus_4 + set_rescor(TRUE),
  data = gryphon,
  data2 = list(Amat = Amat),
  chains = 2, cores = 2, iter = 1000
)
save(brms_m2.4, file = "r-obj/brms_m2_4.rda")

```

Again we have provided the data from one such run. It can be accessed using the code:

```

load("r-obj/brms_m2_4.rda")
summary(brms_m2.4)

```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

Warning: There were 6 divergent transitions after warmup. Increasing adapt\_delta above 0.8 may help. See <http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup>

```
Family: MV(gaussian, gaussian)
Links: mu = identity; sigma = log
       mu = identity; sigma = log
Formula: bwt ~ 1 + sex + ((1 | a | gr(animal, cov = Amat, by = sex))) + (1 | b | byear) + (1 | c
          sigma ~ sex - 1
          tarsus ~ 1 + sex + (1 | a | gr(animal, cov = Amat, by = sex)) + (1 | b | byear) + (1 | c
          sigma ~ sex - 1
Data: gryphon (Number of observations: 683)
Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;
       total post-warmup draws = 1000
```

Multilevel Hyperparameters:

~animal (Number of levels: 683)

|                                               | Estimate | Est.Error | 1-95% CI |          |
|-----------------------------------------------|----------|-----------|----------|----------|
| sd(bwt_Intercept:sex1)                        | 0.86     | 0.35      | 0.14     |          |
| sd(tarsus_Intercept:sex1)                     | 1.40     | 0.82      | 0.10     |          |
| sd(bwt_Intercept:sex2)                        | 1.40     | 0.23      | 0.94     |          |
| sd(tarsus_Intercept:sex2)                     | 3.49     | 0.72      | 1.96     |          |
| cor(bwt_Intercept:sex1,tarsus_Intercept:sex1) | 0.28     | 0.51      | -0.85    |          |
| cor(bwt_Intercept:sex2,tarsus_Intercept:sex2) | 0.78     | 0.10      | 0.56     |          |
|                                               | u-95% CI | Rhat      | Bulk_ESS | Tail_ESS |
| sd(bwt_Intercept:sex1)                        | 1.48     | 1.00      | 59       | 101      |
| sd(tarsus_Intercept:sex1)                     | 3.00     | 1.07      | 45       | 111      |
| sd(bwt_Intercept:sex2)                        | 1.83     | 1.04      | 42       | 40       |
| sd(tarsus_Intercept:sex2)                     | 4.62     | 1.22      | 8        | 50       |
| cor(bwt_Intercept:sex1,tarsus_Intercept:sex1) | 0.94     | 1.05      | 60       | 287      |
| cor(bwt_Intercept:sex2,tarsus_Intercept:sex2) | 0.97     | 1.09      | 30       | 94       |

~byear (Number of levels: 34)

Estimate Est.Error 1-95% CI u-95% CI Rhat



---

|                                     |       |      |       |      |      |
|-------------------------------------|-------|------|-------|------|------|
| sd(bwt_Intercept)                   | 0.97  | 0.15 | 0.73  | 1.30 | 1.01 |
| sd(tarsus_Intercept)                | 2.01  | 0.33 | 1.41  | 2.70 | 1.00 |
| cor(bwt_Intercept,tarsus_Intercept) | -0.00 | 0.22 | -0.42 | 0.45 | 1.02 |

Bulk\_ESS Tail\_ESS

|                                     |     |     |
|-------------------------------------|-----|-----|
| sd(bwt_Intercept)                   | 283 | 412 |
| sd(tarsus_Intercept)                | 349 | 554 |
| cor(bwt_Intercept,tarsus_Intercept) | 225 | 256 |

~mother (Number of levels: 352)

Estimate Est.Error l-95% CI u-95% CI Rhat

|                                     |       |      |       |       |      |
|-------------------------------------|-------|------|-------|-------|------|
| sd(bwt_Intercept)                   | 1.19  | 0.11 | 0.98  | 1.42  | 1.00 |
| sd(tarsus_Intercept)                | 2.14  | 0.30 | 1.57  | 2.71  | 1.05 |
| cor(bwt_Intercept,tarsus_Intercept) | -0.55 | 0.21 | -0.96 | -0.16 | 1.04 |

Bulk\_ESS Tail\_ESS

|                                     |     |     |
|-------------------------------------|-----|-----|
| sd(bwt_Intercept)                   | 279 | 434 |
| sd(tarsus_Intercept)                | 46  | 227 |
| cor(bwt_Intercept,tarsus_Intercept) | 46  | 122 |

Regression Coefficients:

Estimate Est.Error l-95% CI u-95% CI Rhat Bulk\_ESS Tail\_ESS

|                   |       |      |       |       |      |     |     |
|-------------------|-------|------|-------|-------|------|-----|-----|
| bwt_Intercept     | 6.27  | 0.23 | 5.82  | 6.73  | 1.01 | 336 | 556 |
| tarsus_Intercept  | 20.39 | 0.49 | 19.42 | 21.37 | 1.00 | 384 | 509 |
| bwt_sex2          | 2.04  | 0.17 | 1.70  | 2.38  | 1.00 | 483 | 507 |
| sigma_bwt_sex1    | 0.45  | 0.12 | 0.18  | 0.63  | 1.00 | 68  | 128 |
| sigma_bwt_sex2    | 0.26  | 0.17 | -0.17 | 0.52  | 1.06 | 38  | 33  |
| tarsus_sex2       | 0.10  | 0.41 | -0.69 | 0.92  | 1.00 | 658 | 659 |
| sigma_tarsus_sex1 | 1.37  | 0.08 | 1.19  | 1.50  | 1.04 | 65  | 215 |
| sigma_tarsus_sex2 | 1.24  | 0.16 | 0.91  | 1.50  | 1.22 | 7   | 67  |

Residual Correlations:

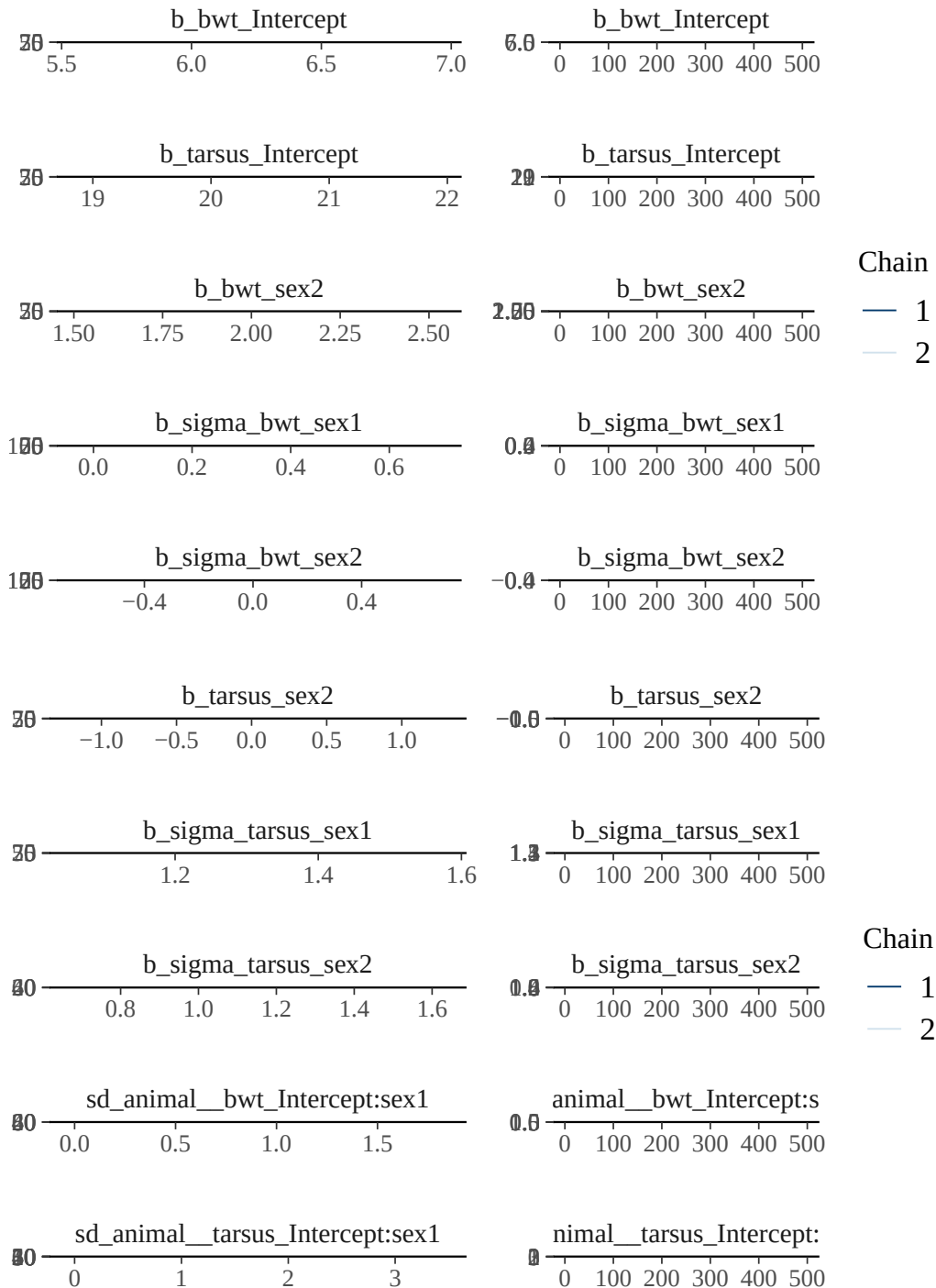
Estimate Est.Error l-95% CI u-95% CI Rhat Bulk\_ESS Tail\_ESS

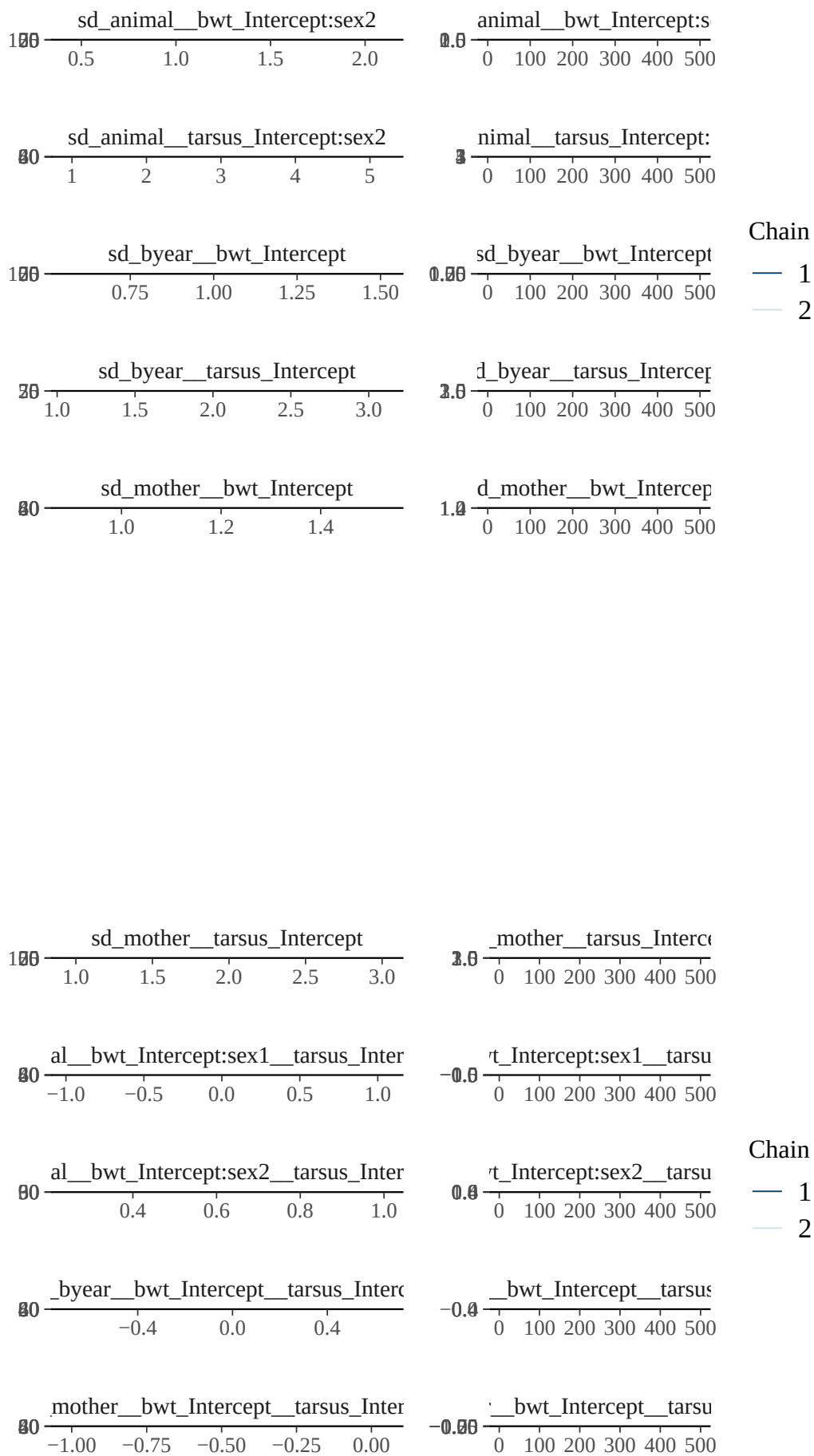
|                    |      |      |      |      |      |    |    |
|--------------------|------|------|------|------|------|----|----|
| rescor(bwt,tarsus) | 0.84 | 0.05 | 0.73 | 0.93 | 1.15 | 11 | 29 |
|--------------------|------|------|------|------|------|----|----|

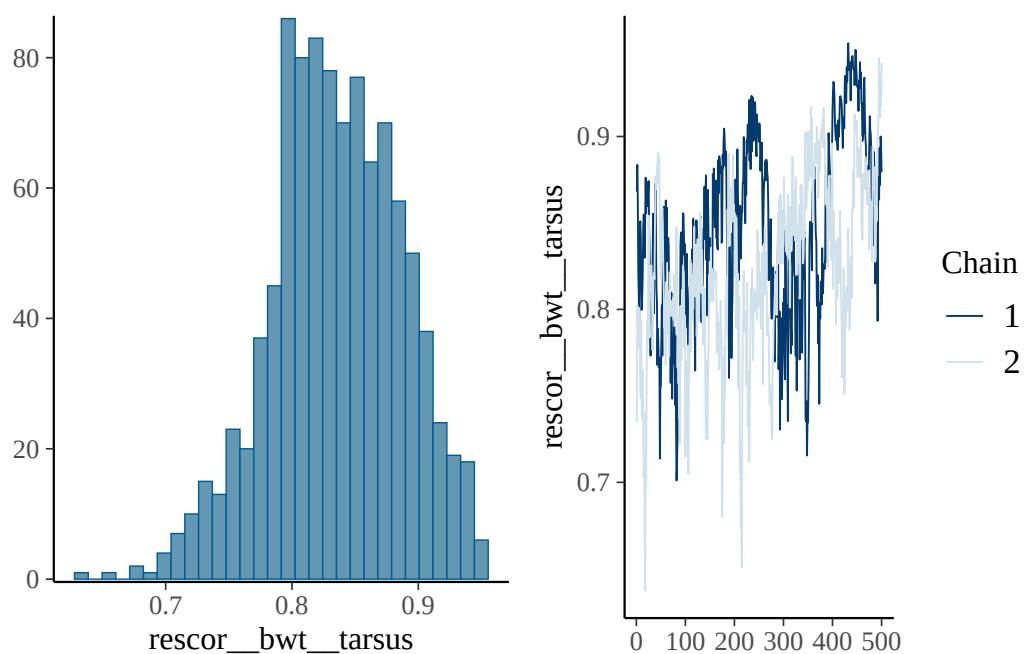
---

Draws were sampled using `sampling(NUTS)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

```
plot(brms_m2.4, ask = FALSE)
```







```
VarCorr(brms_m2.4)
```

```
$animal
```

```
$animal$sd
```

|                       | Estimate  | Est.Error | Q2.5       | Q97.5    |
|-----------------------|-----------|-----------|------------|----------|
| bwt_Intercept:sex1    | 0.8628389 | 0.3531696 | 0.14360929 | 1.481121 |
| tarsus_Intercept:sex1 | 1.3985239 | 0.8163731 | 0.09977045 | 3.002609 |
| bwt_Intercept:sex2    | 1.4023202 | 0.2325863 | 0.93896619 | 1.833688 |
| tarsus_Intercept:sex2 | 3.4858243 | 0.7167230 | 1.96177030 | 4.620105 |

```
$animal$cor
```

```
, , bwt_Intercept:sex1
```

|                       | Estimate | Est.Error | Q2.5       | Q97.5     |
|-----------------------|----------|-----------|------------|-----------|
| bwt_Intercept:sex1    | 1.000000 | 0.000000  | 1.000000   | 1.000000  |
| tarsus_Intercept:sex1 | 0.277338 | 0.5119501 | -0.8479996 | 0.9398158 |
| bwt_Intercept:sex2    | 0.000000 | 0.000000  | 0.000000   | 0.000000  |
| tarsus_Intercept:sex2 | 0.000000 | 0.000000  | 0.000000   | 0.000000  |

```
, , tarsus_Intercept:sex1
```

---

|                       | Estimate | Est.Error | Q2.5       | Q97.5     |
|-----------------------|----------|-----------|------------|-----------|
| bwt_Intercept:sex1    | 0.277338 | 0.5119501 | -0.8479996 | 0.9398158 |
| tarsus_Intercept:sex1 | 1.000000 | 0.0000000 | 1.0000000  | 1.0000000 |
| bwt_Intercept:sex2    | 0.000000 | 0.0000000 | 0.0000000  | 0.0000000 |
| tarsus_Intercept:sex2 | 0.000000 | 0.0000000 | 0.0000000  | 0.0000000 |

, , bwt\_Intercept:sex2

|                       | Estimate  | Est.Error  | Q2.5     | Q97.5     |
|-----------------------|-----------|------------|----------|-----------|
| bwt_Intercept:sex1    | 0.0000000 | 0.00000000 | 0.000000 | 0.0000000 |
| tarsus_Intercept:sex1 | 0.0000000 | 0.00000000 | 0.000000 | 0.0000000 |
| bwt_Intercept:sex2    | 1.0000000 | 0.00000000 | 1.000000 | 1.0000000 |
| tarsus_Intercept:sex2 | 0.7781659 | 0.09850178 | 0.564036 | 0.9699765 |

, , tarsus\_Intercept:sex2

|                       | Estimate  | Est.Error  | Q2.5     | Q97.5     |
|-----------------------|-----------|------------|----------|-----------|
| bwt_Intercept:sex1    | 0.0000000 | 0.00000000 | 0.000000 | 0.0000000 |
| tarsus_Intercept:sex1 | 0.0000000 | 0.00000000 | 0.000000 | 0.0000000 |
| bwt_Intercept:sex2    | 0.7781659 | 0.09850178 | 0.564036 | 0.9699765 |
| tarsus_Intercept:sex2 | 1.0000000 | 0.00000000 | 1.000000 | 1.0000000 |

\$animal\$cov

, , bwt\_Intercept:sex1

|                       | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-----------------------|-----------|-----------|------------|-----------|
| bwt_Intercept:sex1    | 0.8690949 | 0.6002446 | 0.0206250  | 2.193721  |
| tarsus_Intercept:sex1 | 0.6483841 | 0.9143700 | -0.4481203 | 2.989048  |
| bwt_Intercept:sex2    | 0.0000000 | 0.0000000 | 0.0000000  | 0.0000000 |
| tarsus_Intercept:sex2 | 0.0000000 | 0.0000000 | 0.0000000  | 0.0000000 |

, , tarsus\_Intercept:sex1

|                       | Estimate  | Est.Error | Q2.5         | Q97.5    |
|-----------------------|-----------|-----------|--------------|----------|
| bwt_Intercept:sex1    | 0.6483841 | 0.914370  | -0.448120294 | 2.989048 |
| tarsus_Intercept:sex1 | 2.6216677 | 2.500989  | 0.009954235  | 9.015666 |
| bwt_Intercept:sex2    | 0.0000000 | 0.000000  | 0.000000000  | 0.000000 |
| tarsus_Intercept:sex2 | 0.0000000 | 0.000000  | 0.000000000  | 0.000000 |

, , bwt\_Intercept:sex2

|                       | Estimate | Est.Error | Q2.5      | Q97.5    |
|-----------------------|----------|-----------|-----------|----------|
| bwt_Intercept:sex1    | 0.000000 | 0.000000  | 0.0000000 | 0.000000 |
| tarsus_Intercept:sex1 | 0.000000 | 0.000000  | 0.0000000 | 0.000000 |
| bwt_Intercept:sex2    | 2.020544 | 0.639550  | 0.8816577 | 3.362416 |
| tarsus_Intercept:sex2 | 3.875299 | 1.298562  | 1.4247624 | 6.415724 |

, , tarsus\_Intercept:sex2

|                       | Estimate  | Est.Error | Q2.5     | Q97.5     |
|-----------------------|-----------|-----------|----------|-----------|
| bwt_Intercept:sex1    | 0.000000  | 0.000000  | 0.000000 | 0.000000  |
| tarsus_Intercept:sex1 | 0.000000  | 0.000000  | 0.000000 | 0.000000  |
| bwt_Intercept:sex2    | 3.875299  | 1.298562  | 1.424762 | 6.415724  |
| tarsus_Intercept:sex2 | 12.664149 | 4.814957  | 3.848544 | 21.345370 |

\$byear

\$byear\$sd

|                  | Estimate  | Est.Error | Q2.5      | Q97.5    |
|------------------|-----------|-----------|-----------|----------|
| bwt_Intercept    | 0.9707654 | 0.1478361 | 0.7256973 | 1.298668 |
| tarsus_Intercept | 2.0073203 | 0.3290043 | 1.4102530 | 2.699770 |

\$byear\$cor

, , bwt\_Intercept

---

|                  | Estimate     | Est.Error | Q2.5       | Q97.5     |
|------------------|--------------|-----------|------------|-----------|
| bwt_Intercept    | 1.000000000  | 0.0000000 | 1.0000000  | 1.0000000 |
| tarsus_Intercept | -0.001551923 | 0.2236193 | -0.4237849 | 0.4526618 |

, , tarsus\_Intercept

|                  | Estimate     | Est.Error | Q2.5       | Q97.5     |
|------------------|--------------|-----------|------------|-----------|
| bwt_Intercept    | -0.001551923 | 0.2236193 | -0.4237849 | 0.4526618 |
| tarsus_Intercept | 1.000000000  | 0.0000000 | 1.0000000  | 1.0000000 |

\$byear\$cov

, , bwt\_Intercept

|                  | Estimate  | Est.Error | Q2.5       | Q97.5    |
|------------------|-----------|-----------|------------|----------|
| bwt_Intercept    | 0.9642191 | 0.3021042 | 0.5266366  | 1.686538 |
| tarsus_Intercept | 0.0252866 | 0.4713288 | -0.8080314 | 1.069567 |

, , tarsus\_Intercept

|                  | Estimate  | Est.Error | Q2.5       | Q97.5    |
|------------------|-----------|-----------|------------|----------|
| bwt_Intercept    | 0.0252866 | 0.4713288 | -0.8080314 | 1.069567 |
| tarsus_Intercept | 4.1374703 | 1.3676889 | 1.9888135  | 7.288761 |

\$mother

\$mother\$sd

|                  | Estimate | Est.Error | Q2.5      | Q97.5    |
|------------------|----------|-----------|-----------|----------|
| bwt_Intercept    | 1.189902 | 0.1113270 | 0.9791374 | 1.423333 |
| tarsus_Intercept | 2.139290 | 0.2985875 | 1.5714814 | 2.708199 |

```
$mother$cor
```

```
, , bwt_Intercept
```

|                  | Estimate   | Est.Error | Q2.5       | Q97.5      |
|------------------|------------|-----------|------------|------------|
| bwt_Intercept    | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000  |
| tarsus_Intercept | -0.5501934 | 0.2066985 | -0.9589938 | -0.1591737 |

```
, , tarsus_Intercept
```

|                  | Estimate   | Est.Error | Q2.5       | Q97.5      |
|------------------|------------|-----------|------------|------------|
| bwt_Intercept    | -0.5501934 | 0.2066985 | -0.9589938 | -0.1591737 |
| tarsus_Intercept | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000  |

```
$mother$cov
```

```
, , bwt_Intercept
```

|                  | Estimate  | Est.Error | Q2.5      | Q97.5      |
|------------------|-----------|-----------|-----------|------------|
| bwt_Intercept    | 1.428247  | 0.2672833 | 0.958710  | 2.0258758  |
| tarsus_Intercept | -1.335450 | 0.3932707 | -2.051866 | -0.5146954 |

```
, , tarsus_Intercept
```

|                  | Estimate  | Est.Error | Q2.5      | Q97.5      |
|------------------|-----------|-----------|-----------|------------|
| bwt_Intercept    | -1.335450 | 0.3932707 | -2.051866 | -0.5146954 |
| tarsus_Intercept | 4.665626  | 1.2688941 | 2.469554  | 7.3343400  |

Evaluation of the statistical support for these sex-specific correlations is straightforward. Because we imposed no constraint on their estimation, we can evaluate the extent to which the posterior distributions overlap zero or overlap each other:

```
cor_g_F <- as.mcmc((VarCorr(brms_m2.4, summary = FALSE)$animal$cor[, 1, 2]))
cor_g_M <- as.mcmc((VarCorr(brms_m2.4, summary = FALSE)$animal$cor[, 3, 4]))
```



---

```
summary(cor_g_F)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean    | SD      | Naive SE | Time-series SE |
|---------|---------|----------|----------------|
| 0.27734 | 0.51195 | 0.01619  | 0.05981        |

2. Quantiles for each variable:

| 2.5%     | 25%      | 50%     | 75%     | 97.5%   |
|----------|----------|---------|---------|---------|
| -0.84800 | -0.02476 | 0.43226 | 0.67124 | 0.93982 |

```
summary(cor_g_M)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

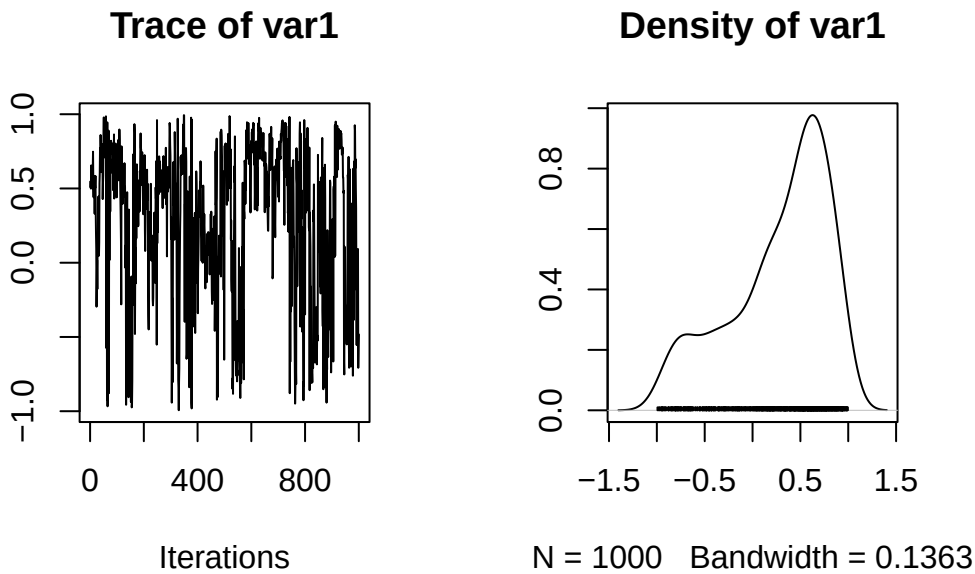
1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.778166 | 0.098502 | 0.003115 | 0.013827       |

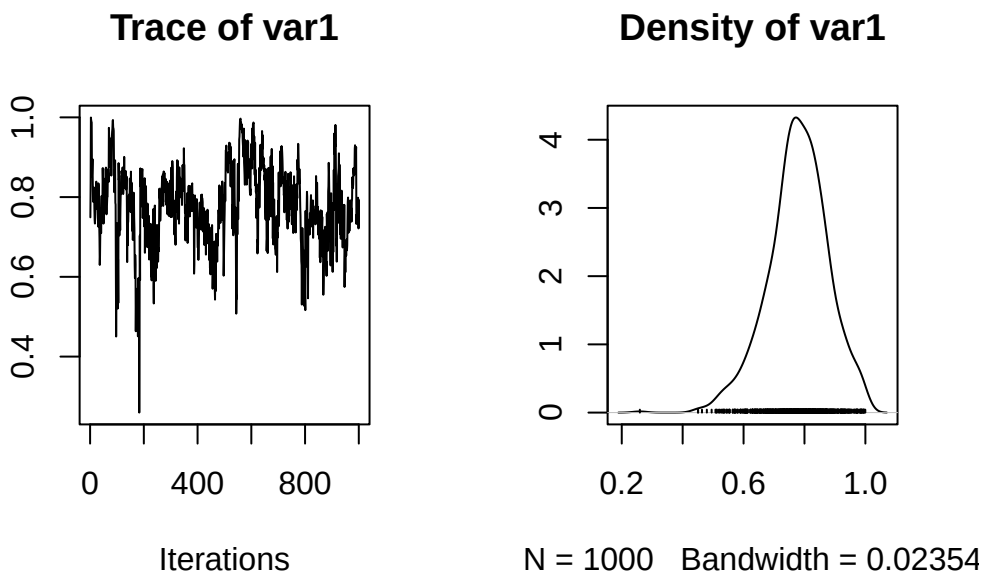
2. Quantiles for each variable:

| 2.5%   | 25%    | 50%    | 75%    | 97.5%  |
|--------|--------|--------|--------|--------|
| 0.5640 | 0.7240 | 0.7805 | 0.8424 | 0.9700 |

```
plot(cor_g_F)
```



```
plot(cor_g_M)
```



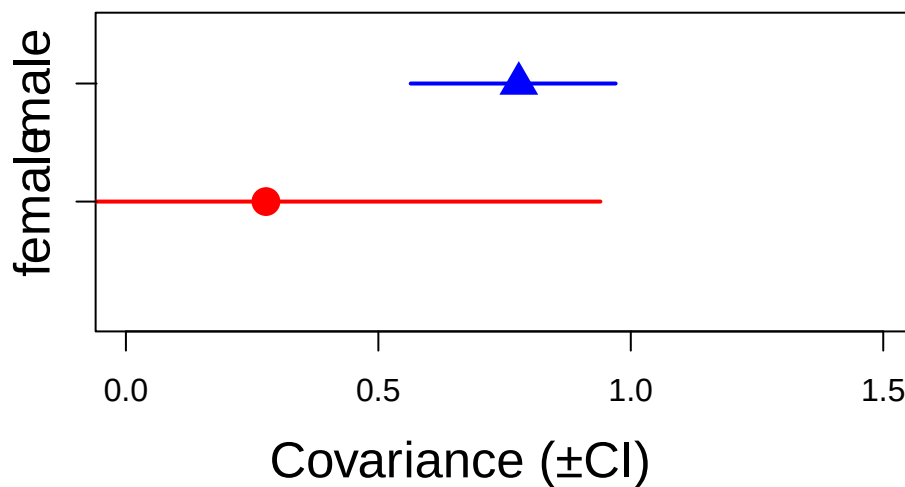
Here a plot to visualize the overlaps of covariances.

```

cor.est <- rbind(
  cbind(summary(cor_g_F)$statistics[1], summary(cor_g_F)$quantiles[1], summary(cor_g_F)$quantiles
  cbind(summary(cor_g_M)$statistics[1], summary(cor_g_M)$quantiles[1], summary(cor_g_M)$quantiles
)

plot(c(1, 2) ~ cor.est[, 1], xlim = c(0, 1.5), ylim = c(0, 2.5), xlab = "", ylab = "", col = c("r", "b"),
      segments(y0 = 1, x0 = cor.est[1, 2], y1 = 1, x1 = cor.est[1, 3], col = c("red"), lwd = 2)
      segments(y0 = 2, x0 = cor.est[2, 2], y1 = 2, x1 = cor.est[2, 3], col = c("blue"), lwd = 2)
      mtext("Covariance ( $\pm$ CI)", side = 1, las = 1, adj = 0.4, line = 3, cex = 1.6)
      axis(2, at = 1, labels = c("female"), las = 3, cex.axis = 1.6)
      axis(2, at = 2, labels = c("male"), las = 3, cex.axis = 1.6)

```



Here a simple plot of the sex-specific genetic correlation using the BLUPs

```

bls_m2.4 <- ranef(brms_m2.4)$animal
bl_m2.4 <- as.data.frame(abind::abind(lapply(1:dim(bls_m2.4)[3], function(x) bls_m2.4[, c(1, 3, 4)
colnames(bl_m2.4) <- paste0(rep(dimnames(bls_m2.4)[[3]], each = 3), c("", "_lo", "_up"))
bl_m2.4$id <- rownames(bl_m2.4)
bl_m2.4$sex <- attr(dimnames(bls_m2.4)[[1]], "by")
FEM <- subset(bl_m2.4, sex == "1")
MAL <- subset(bl_m2.4, sex == "2")

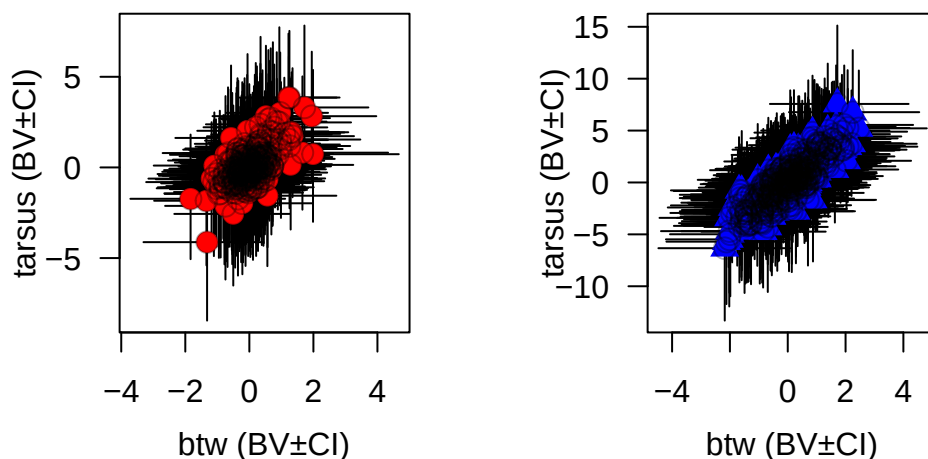
```

```
#
par(mfrow = c(1, 2))
plot(tarsus_Intercept ~ bwt_Intercept, FEM,
     xlab = "", ylab = "",
     xlim = c(min(FEM$bwt_Intercept_lo), max(FEM$bwt_Intercept_up)),
     ylim = c(min(FEM$tarsus_Intercept_lo), max(FEM$tarsus_Intercept_up)),
     las = 1.2, type = "n"
)
segments(
  x0 = FEM$bwt_Intercept, y0 = FEM$tarsus_Intercept_lo,
  x1 = FEM$bwt_Intercept, y1 = FEM$tarsus_Intercept_up,
  col = "black"
)
segments(
  x0 = FEM$bwt_Intercept_lo, y0 = FEM$tarsus_Intercept,
  x1 = FEM$bwt_Intercept_up, y1 = FEM$tarsus_Intercept,
  col = "black"
)
points(tarsus_Intercept ~ bwt_Intercept, FEM, pch = 16, col = "red", cex = 1.5)
points(tarsus_Intercept ~ bwt_Intercept, FEM, pch = 1, col = rgb(0, 0, 0, 0.3), cex = c(1.5))
mtext("btw (BV±CI)", side = 1, line = 2.4)
mtext("tarsus (BV±CI)", side = 2, line = 2, las = 3)
#
plot(tarsus_Intercept ~ bwt_Intercept, MAL,
     xlab = "", ylab = "",
     xlim = c(min(MAL$bwt_Intercept_lo), max(MAL$bwt_Intercept_up)),
     ylim = c(min(MAL$tarsus_Intercept_lo), max(MAL$tarsus_Intercept_up)),
     las = 1.2, type = "n"
)
segments(
  x0 = MAL$bwt_Intercept, y0 = MAL$tarsus_Intercept_lo,
  x1 = MAL$bwt_Intercept, y1 = MAL$tarsus_Intercept_up, col = "black"
)
)
```

```

segments(
  x0 = MAL$bwt_Intercept_lo, y0 = MAL$tarsus_Intercept,
  x1 = MAL$bwt_Intercept_up, y1 = MAL$tarsus_Intercept, col = "black"
)
points(tarsus_Intercept ~ bwt_Intercept, MAL, pch = 17, col = "blue", cex = 1.5)
points(tarsus_Intercept ~ bwt_Intercept, MAL, pch = 1, col = rgb(0, 0, 0, 0.3), cex = c(1.5))
mtext("btw (BV±CI)", side = 1, line = 2.4)
mtext("tarsus (BV±CI)", side = 2, line = 2, las = 3)

```



#### 7.0.4. Between groups (co)variances and the B-matrix

Animal models are amazing model. With different group within a population, it is also possible to estimate how much the different groups shared the same genetic via the cross-group genetic covariance. This covariance is essential to understand ontogenic or sexual conflict, which can constraint or enhanced response to evolution. As an example, we estimate the cross-sex genetic correlation  $r_{fm}$

It is important to keep in mind the covariance matrix at the residual level is zero and it is important to avoid estimating the cross-sex residual covariance because no individual switched sex during the experiment.

Note: the way of partitionning variance per sex is a bit different then the previous code “by=sex”.

This code is faster and also easier to understand. Note, it is possible to play with the | or || to estimate or not covariance between sexes.

```

bf_bwt_5 <- bf(
  bwt ~ 1 + sex + (0 + sex | a | gr(animal, cov = Amat)) + (0 + sex | b | mother) + (0 + sex | c
  sigma ~ sex - 1

```

```
)  
bf_tarsus_5 <- bf(  
  tarsus ~ 1 + sex + (0 + sex | a | gr(animal, cov = Amat)) + (0 + sex | b | mother) + (0 + sex |  
  sigma ~ sex - 1  
)  
  
brms_m2.5 <- brm(  
  bf_bwt_5 + bf_tarsus_5 + set_rescor(TRUE),  
  data = gryphon,  
  data2 = list(Amat = Amat),  
  chains = 2, cores = 2, iter = 1000  
)  
save(brms_m2.5, file = "r-obj/brms_m2_5.rda")
```

Again we have provided the data from one such run. It can be accessed using the code:

```
load("r-obj/brms_m2_5.rda")  
summary(brms_m2.5)
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

Warning: There were 45 divergent transitions after warmup. Increasing adapt\_delta above 0.8 may help. See <http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup>

```
Family: MV(gaussian, gaussian)  
Links: mu = identity; sigma = log  
       mu = identity; sigma = log  
Formula: bwt ~ 1 + sex + (0 + sex | a | gr(animal, cov = Amat)) + (0 + sex | b | mother) + (0 + s  
         sigma ~ sex - 1  
         tarsus ~ 1 + sex + (0 + sex | a | gr(animal, cov = Amat)) + (0 + sex | b | mother) + (0  
         sigma ~ sex - 1
```

---

Data: gryphon (Number of observations: 683)

Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;

total post-warmup draws = 1000

Multilevel Hyperparameters:

~animal (Number of levels: 683)

|                              | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS |
|------------------------------|----------|-----------|----------|----------|------|----------|
| sd(bwt_sex1)                 | 1.26     | 0.30      | 0.63     | 1.73     | 1.06 | 21       |
| sd(bwt_sex2)                 | 1.08     | 0.42      | 0.20     | 1.77     | 1.08 | 18       |
| sd(tarsus_sex1)              | 2.26     | 0.72      | 0.61     | 3.57     | 1.04 | 40       |
| sd(tarsus_sex2)              | 2.74     | 1.05      | 0.61     | 4.47     | 1.13 | 12       |
| cor(bwt_sex1,bwt_sex2)       | 0.48     | 0.29      | -0.24    | 0.87     | 1.02 | 84       |
| cor(bwt_sex1,tarsus_sex1)    | 0.57     | 0.25      | -0.07    | 0.89     | 1.14 | 10       |
| cor(bwt_sex2,tarsus_sex1)    | 0.38     | 0.38      | -0.53    | 0.91     | 1.25 | 7        |
| cor(bwt_sex1,tarsus_sex2)    | 0.17     | 0.31      | -0.49    | 0.75     | 1.04 | 60       |
| cor(bwt_sex2,tarsus_sex2)    | 0.52     | 0.33      | -0.37    | 0.87     | 1.20 | 8        |
| cor(tarsus_sex1,tarsus_sex2) | 0.44     | 0.29      | -0.30    | 0.87     | 1.03 | 47       |

Tail\_ESS

|                              |     |
|------------------------------|-----|
| sd(bwt_sex1)                 | 104 |
| sd(bwt_sex2)                 | 25  |
| sd(tarsus_sex1)              | 99  |
| sd(tarsus_sex2)              | 42  |
| cor(bwt_sex1,bwt_sex2)       | 112 |
| cor(bwt_sex1,tarsus_sex1)    | 145 |
| cor(bwt_sex2,tarsus_sex1)    | 67  |
| cor(bwt_sex1,tarsus_sex2)    | 94  |
| cor(bwt_sex2,tarsus_sex2)    | 50  |
| cor(tarsus_sex1,tarsus_sex2) | 44  |

~byear (Number of levels: 34)

|              | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS |
|--------------|----------|-----------|----------|----------|------|----------|
| sd(bwt_sex1) | 0.80     | 0.16      | 0.53     | 1.16     | 1.00 | 394      |
| sd(bwt_sex2) | 1.14     | 0.19      | 0.81     | 1.55     | 1.01 | 358      |

|                              |       |      |       |      |      |     |
|------------------------------|-------|------|-------|------|------|-----|
| sd(tarsus_sex1)              | 2.23  | 0.46 | 1.50  | 3.18 | 1.01 | 297 |
| sd(tarsus_sex2)              | 2.34  | 0.49 | 1.56  | 3.41 | 1.01 | 229 |
| cor(bwt_sex1,bwt_sex2)       | 0.74  | 0.15 | 0.35  | 0.96 | 1.01 | 266 |
| cor(bwt_sex1,tarsus_sex1)    | -0.11 | 0.24 | -0.55 | 0.35 | 1.01 | 190 |
| cor(bwt_sex2,tarsus_sex1)    | -0.39 | 0.20 | -0.73 | 0.00 | 1.01 | 410 |
| cor(bwt_sex1,tarsus_sex2)    | 0.29  | 0.23 | -0.17 | 0.71 | 1.00 | 256 |
| cor(bwt_sex2,tarsus_sex2)    | 0.29  | 0.21 | -0.16 | 0.66 | 1.01 | 327 |
| cor(tarsus_sex1,tarsus_sex2) | 0.52  | 0.19 | 0.12  | 0.84 | 1.00 | 285 |

## Tail\_ESS

|                              |     |
|------------------------------|-----|
| sd(bwt_sex1)                 | 619 |
| sd(bwt_sex2)                 | 653 |
| sd(tarsus_sex1)              | 559 |
| sd(tarsus_sex2)              | 239 |
| cor(bwt_sex1,bwt_sex2)       | 433 |
| cor(bwt_sex1,tarsus_sex1)    | 603 |
| cor(bwt_sex2,tarsus_sex1)    | 656 |
| cor(bwt_sex1,tarsus_sex2)    | 319 |
| cor(bwt_sex2,tarsus_sex2)    | 474 |
| cor(tarsus_sex1,tarsus_sex2) | 600 |

~mother (Number of levels: 352)

|                              | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat | Bulk_ESS |
|------------------------------|----------|-----------|----------|----------|------|----------|
| sd(bwt_sex1)                 | 1.08     | 0.15      | 0.79     | 1.39     | 1.01 | 281      |
| sd(bwt_sex2)                 | 1.33     | 0.15      | 1.03     | 1.62     | 1.00 | 233      |
| sd(tarsus_sex1)              | 2.21     | 0.40      | 1.36     | 2.95     | 1.01 | 72       |
| sd(tarsus_sex2)              | 2.31     | 0.49      | 1.38     | 3.34     | 1.05 | 50       |
| cor(bwt_sex1,bwt_sex2)       | 0.83     | 0.11      | 0.57     | 0.98     | 1.01 | 68       |
| cor(bwt_sex1,tarsus_sex1)    | -0.50    | 0.24      | -0.91    | -0.07    | 1.01 | 57       |
| cor(bwt_sex2,tarsus_sex1)    | -0.64    | 0.17      | -0.93    | -0.28    | 1.06 | 50       |
| cor(bwt_sex1,tarsus_sex2)    | -0.51    | 0.22      | -0.88    | -0.08    | 1.05 | 67       |
| cor(bwt_sex2,tarsus_sex2)    | -0.36    | 0.26      | -0.84    | 0.11     | 1.05 | 54       |
| cor(tarsus_sex1,tarsus_sex2) | 0.72     | 0.16      | 0.37     | 0.95     | 1.01 | 249      |

## Tail\_ESS



---

```
sd(bwt_sex1)          518
sd(bwt_sex2)          585
sd(tarsus_sex1)       224
sd(tarsus_sex2)       191
cor(bwt_sex1,bwt_sex2) 296
cor(bwt_sex1,tarsus_sex1) 156
cor(bwt_sex2,tarsus_sex1) 268
cor(bwt_sex1,tarsus_sex2) 124
cor(bwt_sex2,tarsus_sex2) 297
cor(tarsus_sex1,tarsus_sex2) 596
```

#### Regression Coefficients:

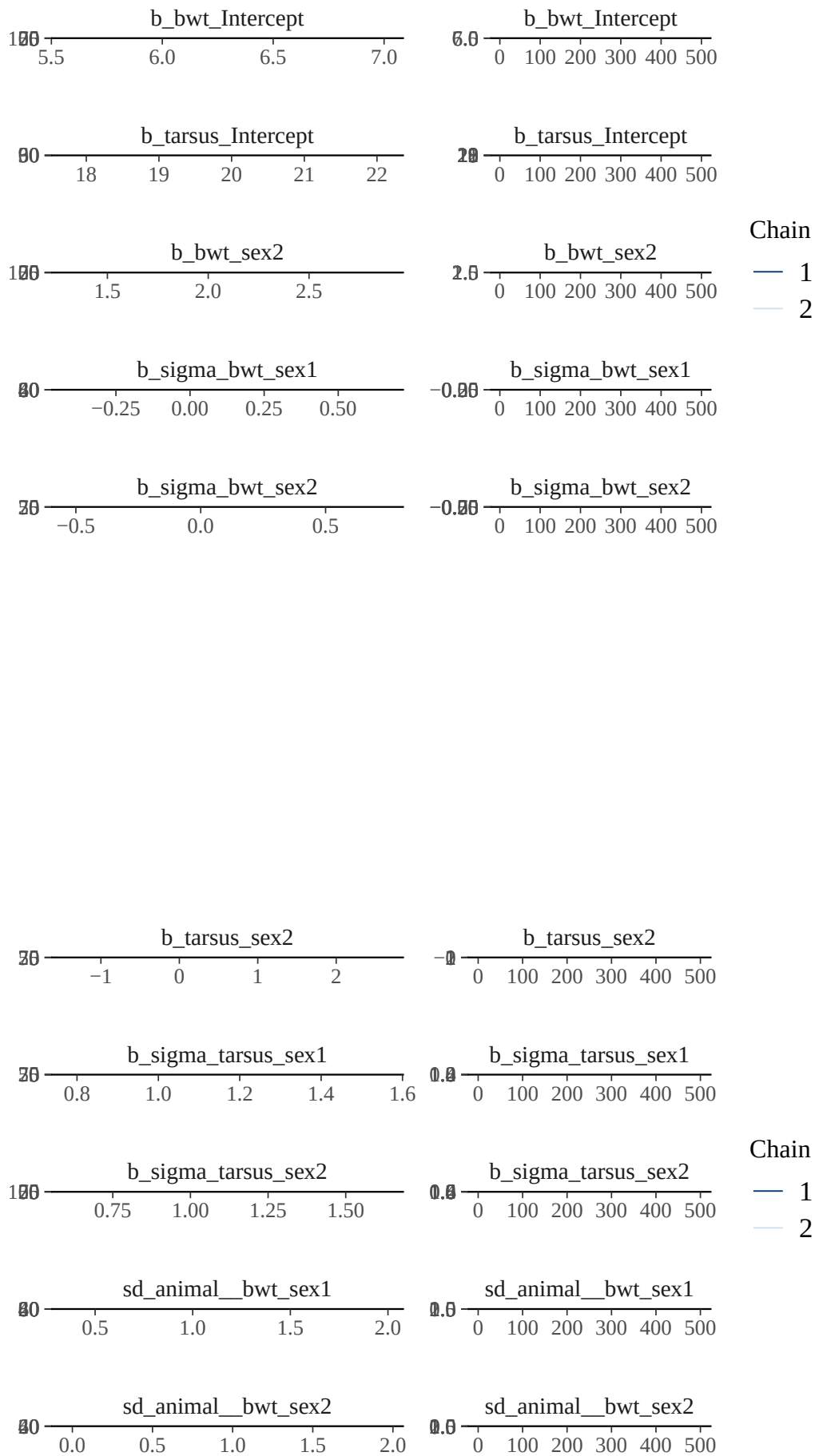
|                   | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------------------|----------|-----------|----------|----------|------|----------|----------|
| bwt_Intercept     | 6.18     | 0.22      | 5.81     | 6.66     | 1.01 | 421      | 413      |
| tarsus_Intercept  | 20.24    | 0.55      | 19.12    | 21.25    | 1.01 | 497      | 527      |
| bwt_sex2          | 2.11     | 0.24      | 1.61     | 2.57     | 1.01 | 541      | 569      |
| sigma_bwt_sex1    | 0.27     | 0.20      | -0.16    | 0.56     | 1.08 | 18       | 66       |
| sigma_bwt_sex2    | 0.31     | 0.23      | -0.35    | 0.59     | 1.11 | 18       | 22       |
| tarsus_sex2       | 0.29     | 0.63      | -0.96    | 1.56     | 1.00 | 490      | 535      |
| sigma_tarsus_sex1 | 1.26     | 0.12      | 1.01     | 1.47     | 1.02 | 50       | 109      |
| sigma_tarsus_sex2 | 1.28     | 0.20      | 0.80     | 1.53     | 1.13 | 12       | 42       |

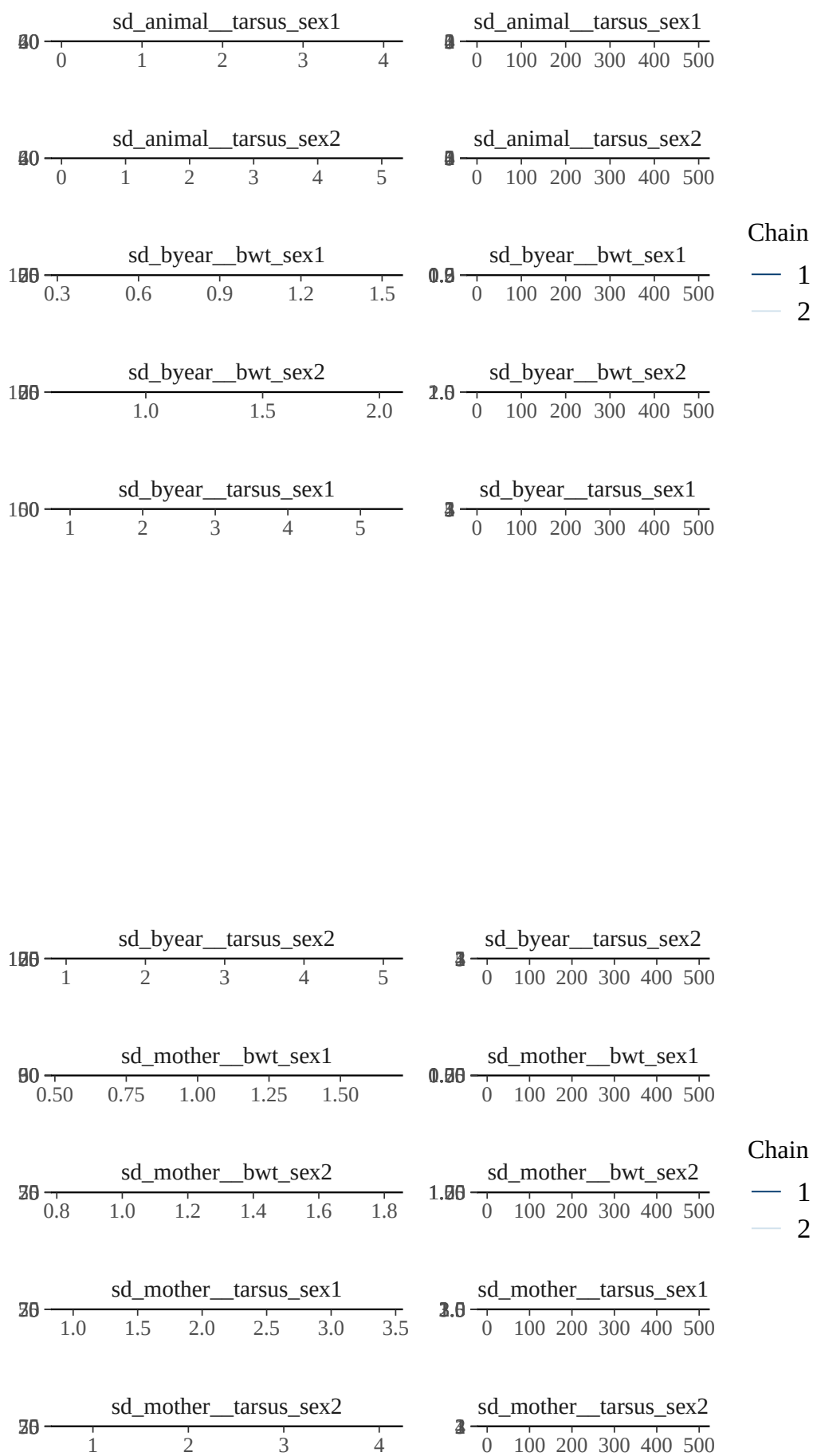
#### Residual Correlations:

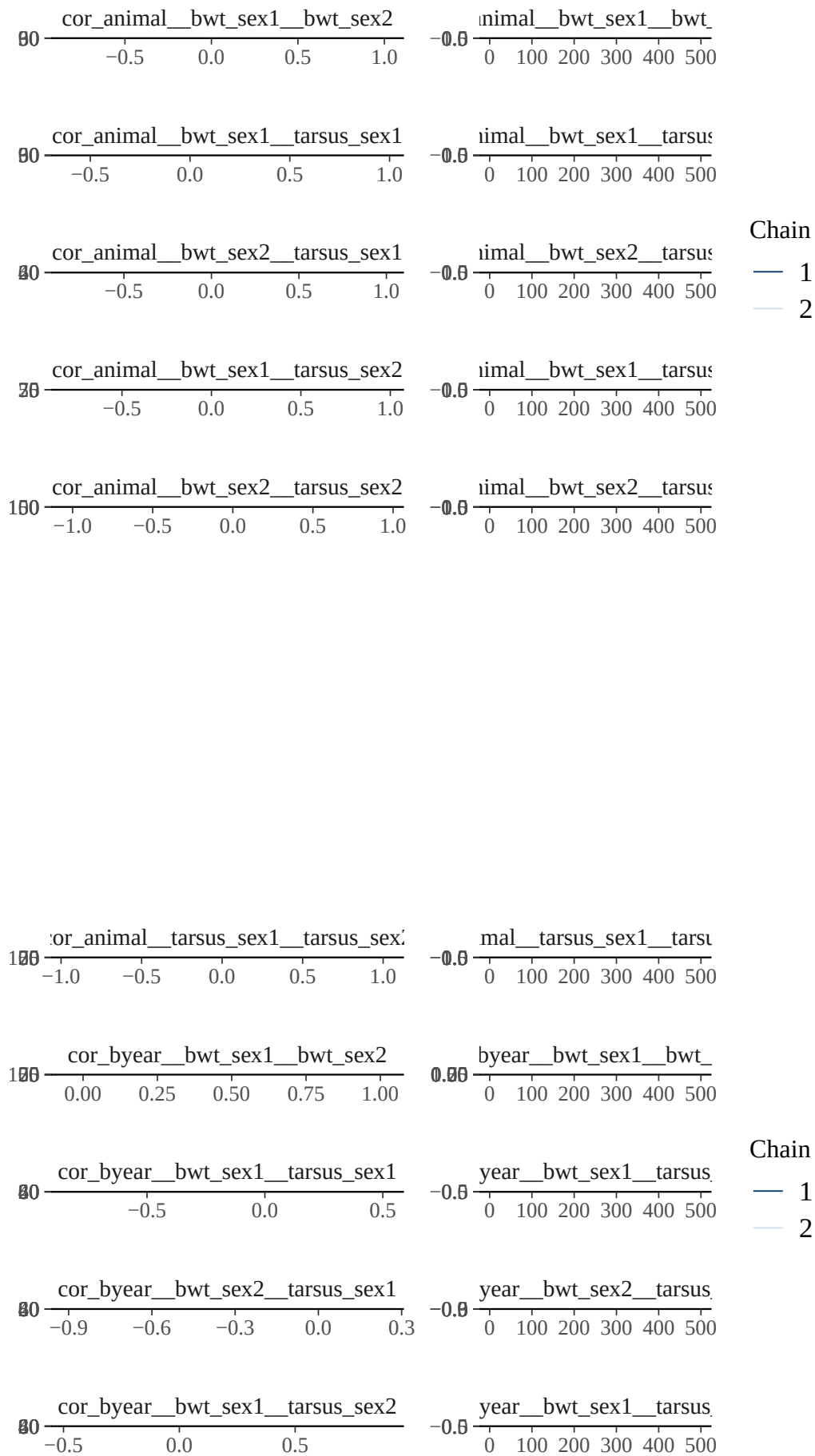
|                    | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|--------------------|----------|-----------|----------|----------|------|----------|----------|
| rescor(bwt,tarsus) | 0.88     | 0.05      | 0.71     | 0.95     | 1.32 | 5        | 33       |

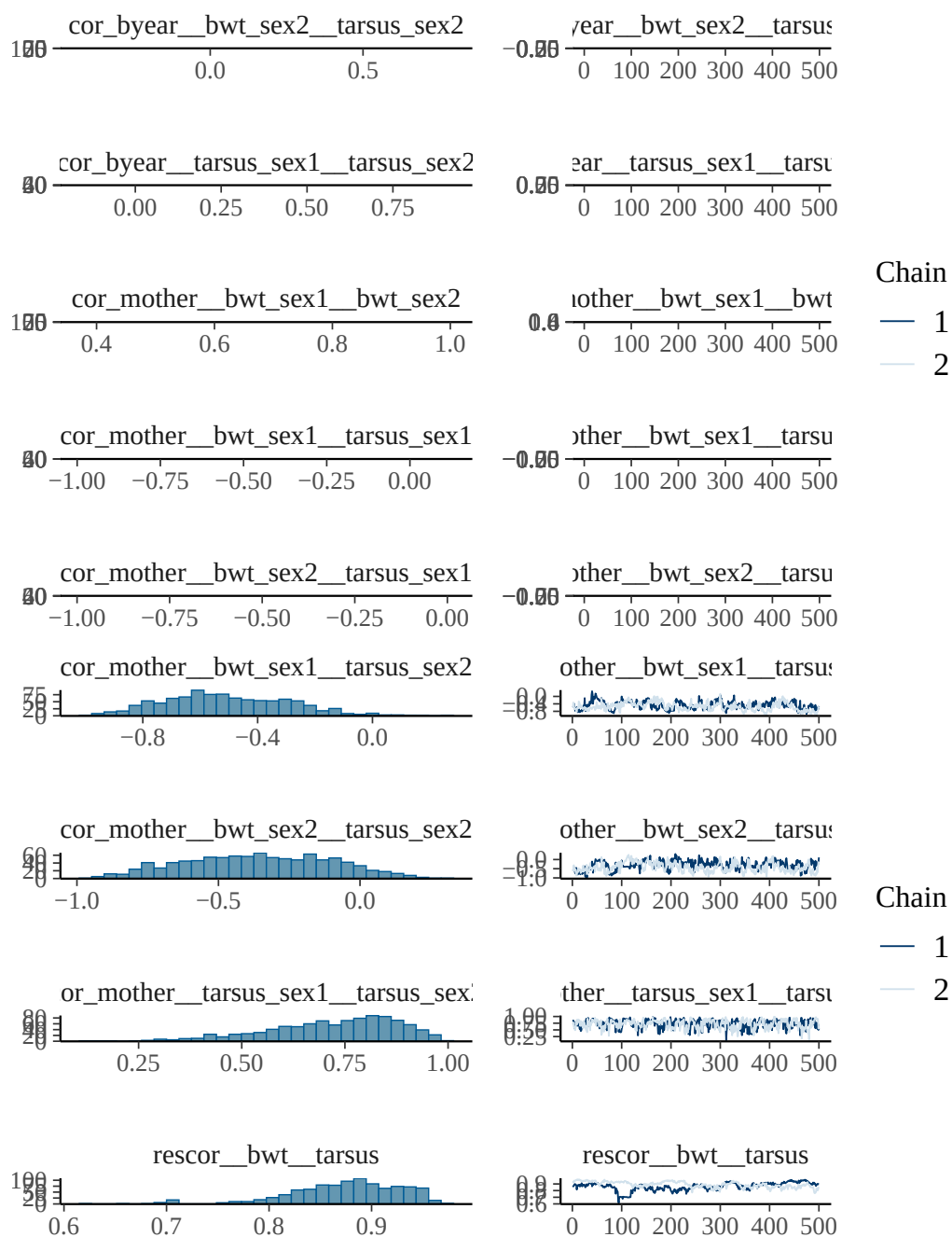
Draws were sampled using `sampling(NUTS)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

```
plot(brms_m2.5, ask = FALSE)
```









```
VarCorr(brms_m2.5)
```

```
$animal
```

```
$animal$sd
```

|             | Estimate | Est.Error | Q2.5      | Q97.5    |
|-------------|----------|-----------|-----------|----------|
| bwt_sex1    | 1.256797 | 0.3027123 | 0.6293789 | 1.733715 |
| bwt_sex2    | 1.077842 | 0.4168427 | 0.1958640 | 1.767136 |
| tarsus_sex1 | 2.259727 | 0.7239967 | 0.6135135 | 3.567762 |

tarsus\_sex2 2.744785 1.0537686 0.6051135 4.468542

\$animal\$cor

, , bwt\_sex1

|             | Estimate  | Est.Error | Q2.5        | Q97.5     |
|-------------|-----------|-----------|-------------|-----------|
| bwt_sex1    | 1.0000000 | 0.0000000 | 1.00000000  | 1.0000000 |
| bwt_sex2    | 0.4823461 | 0.2872049 | -0.23539931 | 0.8699222 |
| tarsus_sex1 | 0.5711397 | 0.2473353 | -0.06616842 | 0.8936581 |
| tarsus_sex2 | 0.1685774 | 0.3102320 | -0.48914110 | 0.7539686 |

, , bwt\_sex2

|             | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-------------|-----------|-----------|------------|-----------|
| bwt_sex1    | 0.4823461 | 0.2872049 | -0.2353993 | 0.8699222 |
| bwt_sex2    | 1.0000000 | 0.0000000 | 1.00000000 | 1.0000000 |
| tarsus_sex1 | 0.3772862 | 0.3765302 | -0.5312556 | 0.9085651 |
| tarsus_sex2 | 0.5246223 | 0.3336679 | -0.3678213 | 0.8692250 |

, , tarsus\_sex1

|             | Estimate  | Est.Error | Q2.5        | Q97.5     |
|-------------|-----------|-----------|-------------|-----------|
| bwt_sex1    | 0.5711397 | 0.2473353 | -0.06616842 | 0.8936581 |
| bwt_sex2    | 0.3772862 | 0.3765302 | -0.53125561 | 0.9085651 |
| tarsus_sex1 | 1.0000000 | 0.0000000 | 1.00000000  | 1.0000000 |
| tarsus_sex2 | 0.4401433 | 0.2929178 | -0.29616405 | 0.8720453 |

, , tarsus\_sex2

|             | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-------------|-----------|-----------|------------|-----------|
| bwt_sex1    | 0.1685774 | 0.3102320 | -0.4891411 | 0.7539686 |
| bwt_sex2    | 0.5246223 | 0.3336679 | -0.3678213 | 0.8692250 |
| tarsus_sex1 | 0.4401433 | 0.2929178 | -0.2961641 | 0.8720453 |

---

tarsus\_sex2 1.0000000 0.0000000 1.0000000 1.0000000

\$animal\$cov

, , bwt\_sex1

|             | Estimate  | Est.Error | Q2.5        | Q97.5    |
|-------------|-----------|-----------|-------------|----------|
| bwt_sex1    | 1.6710810 | 0.7317405 | 0.39611874  | 3.005768 |
| bwt_sex2    | 0.7428820 | 0.5579922 | -0.10600955 | 2.049110 |
| tarsus_sex1 | 1.8733421 | 1.2159805 | -0.05815667 | 4.340959 |
| tarsus_sex2 | 0.6471034 | 1.0646022 | -1.25297349 | 2.930019 |

, , bwt\_sex2

|             | Estimate | Est.Error | Q2.5        | Q97.5    |
|-------------|----------|-----------|-------------|----------|
| bwt_sex1    | 0.742882 | 0.5579922 | -0.10600955 | 2.049110 |
| bwt_sex2    | 1.335327 | 0.8557561 | 0.03836336  | 3.122771 |
| tarsus_sex1 | 1.105102 | 1.0991534 | -0.68104784 | 3.388813 |
| tarsus_sex2 | 2.171388 | 1.8198507 | -0.18718074 | 5.946047 |

, , tarsus\_sex1

|             | Estimate | Est.Error | Q2.5        | Q97.5     |
|-------------|----------|-----------|-------------|-----------|
| bwt_sex1    | 1.873342 | 1.215980  | -0.05815667 | 4.340959  |
| bwt_sex2    | 1.105102 | 1.099153  | -0.68104784 | 3.388813  |
| tarsus_sex1 | 5.630014 | 3.143741  | 0.37639882  | 12.728924 |
| tarsus_sex2 | 3.150235 | 2.476892  | -0.67170548 | 8.755673  |

, , tarsus\_sex2

|          | Estimate  | Est.Error | Q2.5       | Q97.5    |
|----------|-----------|-----------|------------|----------|
| bwt_sex1 | 0.6471034 | 1.064602  | -1.2529735 | 2.930019 |
| bwt_sex2 | 2.1713876 | 1.819851  | -0.1871807 | 5.946047 |

```
tarsus_sex1 3.1502347 2.476892 -0.6717055 8.755673
tarsus_sex2 8.6431609 5.649764 0.3661935 19.967865
```

```
$byear
```

```
$byear$sd
```

|             | Estimate  | Est.Error | Q2.5      | Q97.5    |
|-------------|-----------|-----------|-----------|----------|
| bwt_sex1    | 0.7989572 | 0.1620838 | 0.5318383 | 1.156162 |
| bwt_sex2    | 1.1420876 | 0.1912205 | 0.8090083 | 1.549360 |
| tarsus_sex1 | 2.2286834 | 0.4609182 | 1.4995560 | 3.183107 |
| tarsus_sex2 | 2.3428101 | 0.4941316 | 1.5596304 | 3.405106 |

```
$byear$cor
```

```
, , bwt_sex1
```

|             | Estimate   | Est.Error | Q2.5       | Q97.5     |
|-------------|------------|-----------|------------|-----------|
| bwt_sex1    | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000 |
| bwt_sex2    | 0.7404024  | 0.1542690 | 0.3534945  | 0.9558481 |
| tarsus_sex1 | -0.1137836 | 0.2357386 | -0.5534464 | 0.3543949 |
| tarsus_sex2 | 0.2922708  | 0.2345550 | -0.1749420 | 0.7113421 |

```
, , bwt_sex2
```

|             | Estimate   | Est.Error | Q2.5       | Q97.5       |
|-------------|------------|-----------|------------|-------------|
| bwt_sex1    | 0.7404024  | 0.1542690 | 0.3534945  | 0.955848060 |
| bwt_sex2    | 1.0000000  | 0.0000000 | 1.0000000  | 1.000000000 |
| tarsus_sex1 | -0.3874250 | 0.1987378 | -0.7323699 | 0.004258363 |
| tarsus_sex2 | 0.2937354  | 0.2059346 | -0.1577533 | 0.659531338 |

```
, , tarsus_sex1
```

|  | Estimate | Est.Error | Q2.5 | Q97.5 |
|--|----------|-----------|------|-------|
|--|----------|-----------|------|-------|

---



---

|             |            |           |            |             |
|-------------|------------|-----------|------------|-------------|
| bwt_sex1    | -0.1137836 | 0.2357386 | -0.5534464 | 0.354394929 |
| bwt_sex2    | -0.3874250 | 0.1987378 | -0.7323699 | 0.004258363 |
| tarsus_sex1 | 1.0000000  | 0.0000000 | 1.0000000  | 1.000000000 |
| tarsus_sex2 | 0.5226217  | 0.1897518 | 0.1183176  | 0.839911077 |

, , tarsus\_sex2

|             | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-------------|-----------|-----------|------------|-----------|
| bwt_sex1    | 0.2922708 | 0.2345550 | -0.1749420 | 0.7113421 |
| bwt_sex2    | 0.2937354 | 0.2059346 | -0.1577533 | 0.6595313 |
| tarsus_sex1 | 0.5226217 | 0.1897518 | 0.1183176  | 0.8399111 |
| tarsus_sex2 | 1.0000000 | 0.0000000 | 1.0000000  | 1.0000000 |

\$byear\$cov

, , bwt\_sex1

|             | Estimate   | Est.Error | Q2.5       | Q97.5     |
|-------------|------------|-----------|------------|-----------|
| bwt_sex1    | 0.6645776  | 0.2761781 | 0.2828520  | 1.3367115 |
| bwt_sex2    | 0.6843681  | 0.2597687 | 0.2749460  | 1.2955568 |
| tarsus_sex1 | -0.1581409 | 0.4589758 | -1.0281983 | 0.8339776 |
| tarsus_sex2 | 0.5456796  | 0.5013040 | -0.3608892 | 1.5951028 |

, , bwt\_sex2

|             | Estimate   | Est.Error | Q2.5       | Q97.5       |
|-------------|------------|-----------|------------|-------------|
| bwt_sex1    | 0.6843681  | 0.2597687 | 0.2749460  | 1.295556807 |
| bwt_sex2    | 1.3408929  | 0.4593417 | 0.6544951  | 2.400516888 |
| tarsus_sex1 | -1.0167438 | 0.6693667 | -2.4775184 | 0.009371017 |
| tarsus_sex2 | 0.8646682  | 0.7242669 | -0.3384814 | 2.623722863 |

, , tarsus\_sex1

|             | Estimate   | Est.Error | Q2.5       | Q97.5        |
|-------------|------------|-----------|------------|--------------|
| bwt_sex1    | -0.1581409 | 0.4589758 | -1.0281983 | 0.833977585  |
| bwt_sex2    | -1.0167438 | 0.6693667 | -2.4775184 | 0.009371017  |
| tarsus_sex1 | 5.1792626  | 2.3047474 | 2.2486683  | 10.132170288 |
| tarsus_sex2 | 2.7818157  | 1.5318128 | 0.5297591  | 5.970346660  |

, , tarsus\_sex2

|             | Estimate  | Est.Error | Q2.5       | Q97.5     |
|-------------|-----------|-----------|------------|-----------|
| bwt_sex1    | 0.5456796 | 0.5013040 | -0.3608892 | 1.595103  |
| bwt_sex2    | 0.8646682 | 0.7242669 | -0.3384814 | 2.623723  |
| tarsus_sex1 | 2.7818157 | 1.5318128 | 0.5297591  | 5.970347  |
| tarsus_sex2 | 5.7326811 | 2.5639312 | 2.4324504  | 11.594758 |

\$mother

\$mother\$sd

|             | Estimate | Est.Error | Q2.5      | Q97.5    |
|-------------|----------|-----------|-----------|----------|
| bwt_sex1    | 1.076447 | 0.1526711 | 0.7866154 | 1.390273 |
| bwt_sex2    | 1.325206 | 0.1540539 | 1.0256350 | 1.621175 |
| tarsus_sex1 | 2.214033 | 0.3976449 | 1.3647309 | 2.946910 |
| tarsus_sex2 | 2.310902 | 0.4940802 | 1.3795156 | 3.339175 |

\$mother\$cor

, , bwt\_sex1

|             | Estimate   | Est.Error | Q2.5       | Q97.5       |
|-------------|------------|-----------|------------|-------------|
| bwt_sex1    | 1.0000000  | 0.0000000 | 1.0000000  | 1.00000000  |
| bwt_sex2    | 0.8260360  | 0.1110784 | 0.5676601  | 0.97826511  |
| tarsus_sex1 | -0.5024557 | 0.2385053 | -0.9137423 | -0.06798800 |
| tarsus_sex2 | -0.5073494 | 0.2162931 | -0.8806394 | -0.07677131 |

---

, , bwt\_sex2

|             | Estimate   | Est.Error | Q2.5       | Q97.5      |
|-------------|------------|-----------|------------|------------|
| bwt_sex1    | 0.8260360  | 0.1110784 | 0.5676601  | 0.9782651  |
| bwt_sex2    | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000  |
| tarsus_sex1 | -0.6373889 | 0.1744852 | -0.9272366 | -0.2775698 |
| tarsus_sex2 | -0.3593672 | 0.2577845 | -0.8364419 | 0.1135921  |

, , tarsus\_sex1

|             | Estimate   | Est.Error | Q2.5       | Q97.5      |
|-------------|------------|-----------|------------|------------|
| bwt_sex1    | -0.5024557 | 0.2385053 | -0.9137423 | -0.0679880 |
| bwt_sex2    | -0.6373889 | 0.1744852 | -0.9272366 | -0.2775698 |
| tarsus_sex1 | 1.0000000  | 0.0000000 | 1.0000000  | 1.0000000  |
| tarsus_sex2 | 0.7206209  | 0.1561300 | 0.3688228  | 0.9518425  |

, , tarsus\_sex2

|             | Estimate   | Est.Error | Q2.5       | Q97.5       |
|-------------|------------|-----------|------------|-------------|
| bwt_sex1    | -0.5073494 | 0.2162931 | -0.8806394 | -0.07677131 |
| bwt_sex2    | -0.3593672 | 0.2577845 | -0.8364419 | 0.11359208  |
| tarsus_sex1 | 0.7206209  | 0.1561300 | 0.3688228  | 0.95184250  |
| tarsus_sex2 | 1.0000000  | 0.0000000 | 1.0000000  | 1.00000000  |

\$mother\$cov

, , bwt\_sex1

|             | Estimate  | Est.Error | Q2.5       | Q97.5      |
|-------------|-----------|-----------|------------|------------|
| bwt_sex1    | 1.182023  | 0.3330334 | 0.6187659  | 1.9328578  |
| bwt_sex2    | 1.179468  | 0.2744898 | 0.6821359  | 1.7479435  |
| tarsus_sex1 | -1.110068 | 0.4557029 | -1.9379754 | -0.2103052 |
| tarsus_sex2 | -1.223308 | 0.5582766 | -2.3558658 | -0.1900476 |

```
, , bwt_sex2
```

|             | Estimate   | Est.Error | Q2.5       | Q97.5      |
|-------------|------------|-----------|------------|------------|
| bwt_sex1    | 1.1794683  | 0.2744898 | 0.6821359  | 1.7479435  |
| bwt_sex2    | 1.7798788  | 0.4088241 | 1.0519272  | 2.6282093  |
| tarsus_sex1 | -1.8437371 | 0.5678929 | -2.9129058 | -0.7723013 |
| tarsus_sex2 | -0.9438083 | 0.6393327 | -2.0044262 | 0.4581811  |

```
, , tarsus_sex1
```

|             | Estimate  | Est.Error | Q2.5      | Q97.5      |
|-------------|-----------|-----------|-----------|------------|
| bwt_sex1    | -1.110068 | 0.4557029 | -1.937975 | -0.2103052 |
| bwt_sex2    | -1.843737 | 0.5678929 | -2.912906 | -0.7723013 |
| tarsus_sex1 | 5.059904  | 1.7495725 | 1.862491  | 8.6842757  |
| tarsus_sex2 | 3.692391  | 1.3309441 | 1.432767  | 6.5202798  |

```
, , tarsus_sex2
```

|             | Estimate   | Est.Error | Q2.5      | Q97.5      |
|-------------|------------|-----------|-----------|------------|
| bwt_sex1    | -1.2233079 | 0.5582766 | -2.355866 | -0.1900476 |
| bwt_sex2    | -0.9438083 | 0.6393327 | -2.004426 | 0.4581811  |
| tarsus_sex1 | 3.6923914  | 1.3309441 | 1.432767  | 6.5202798  |
| tarsus_sex2 | 5.5841373  | 2.3245515 | 1.903063  | 11.1500916 |

The cross-sex genetic correlation can estimate from the output of the model. For tarsus length at fledging, sexes shared a lot of genetic variance which is common for a trait with low sexual dimorphism. If the selection is antagonistic between males and females, sexes can not evolve freely from the other sexes and an intralocus sexual conflict can appear.

```
cross_sex.cor.btw <- as.mcmc((VarCorr(brms_m2.5, summary = FALSE)$animal$cor[, 1, 2]))
cross_sex.cor.tarsus <- as.mcmc((VarCorr(brms_m2.5, summary = FALSE)$animal$cor[, 3, 4]))

summary(cross_sex.cor.btw)
```

---

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.482346 | 0.287205 | 0.009082 | 0.032430       |

2. Quantiles for each variable:

| 2.5%    | 25%    | 50%    | 75%    | 97.5%  |
|---------|--------|--------|--------|--------|
| -0.2354 | 0.3433 | 0.5365 | 0.6861 | 0.8699 |

```
summary(cross_sex.cor.tarsus)
```

```
Iterations = 1:1000
```

```
Thinning interval = 1
```

```
Number of chains = 1
```

```
Sample size per chain = 1000
```

1. Empirical mean and standard deviation for each variable,  
plus standard error of the mean:

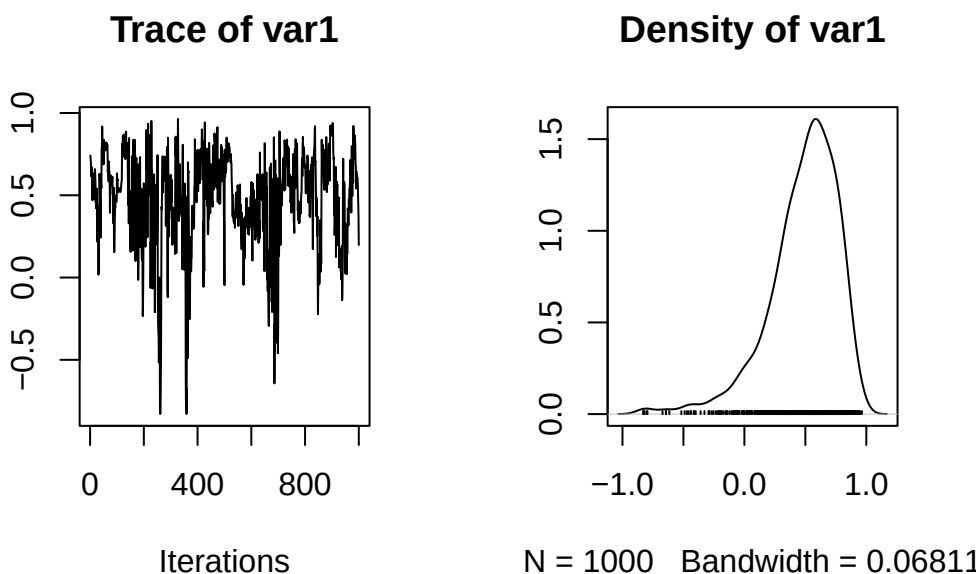
| Mean     | SD       | Naive SE | Time-series SE |
|----------|----------|----------|----------------|
| 0.440143 | 0.292918 | 0.009263 | 0.048863       |

2. Quantiles for each variable:

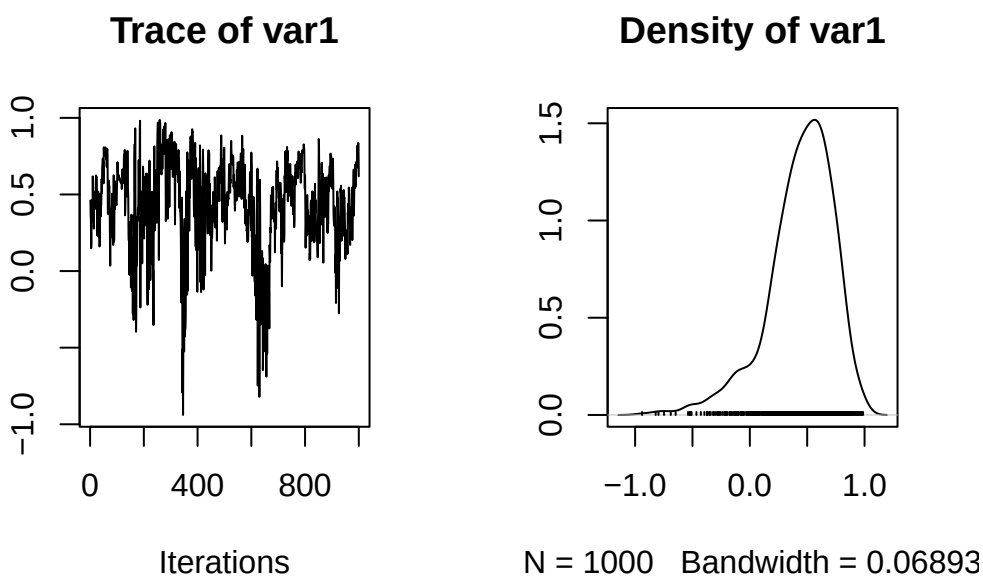
| 2.5% | 25% | 50% | 75% | 97.5% |
|------|-----|-----|-----|-------|
|------|-----|-----|-----|-------|

-0.2962 0.2950 0.4846 0.6419 0.8720

```
plot(cross_sex.cor.btw)
```



```
plot(cross_sex.cor.tarsus)
```



Here, some simple code to extract the BLUP.

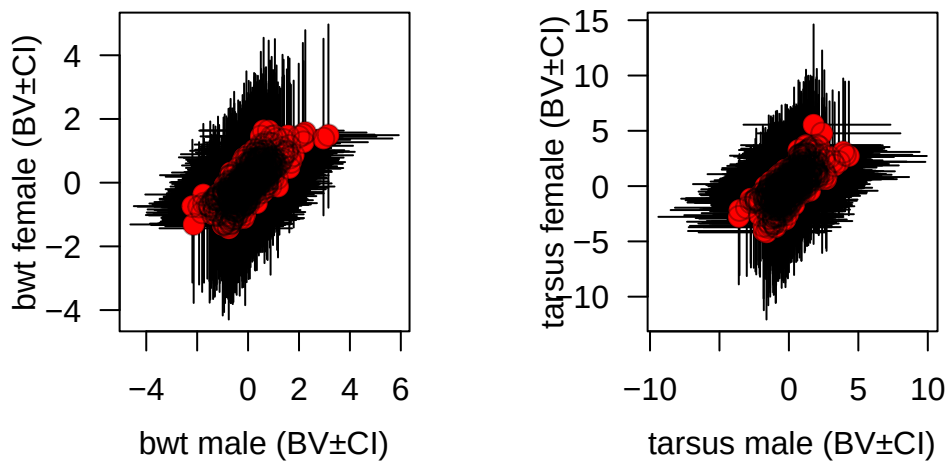
```
bls_m2.5 <- ranef(brms_m2.5)$animal
bl_m2.5 <- as.data.frame(abind::abind(lapply(1:4, function(x) bls_m2.5[, c(1, 3, 4), x])))
colnames(bl_m2.5) <- paste0(rep(dimnames(bls_m2.5)[[3]], each = 3), c("", "_lo", "_up"))
bl_m2.5$id <- rownames(bl_m2.5)
```

---

Here, some simple code to plot the cross-sex genetic correlation.

```
par(mfrow = c(1, 2))
plot(bwt_sex2 ~ bwt_sex1, bl_m2.5,
     xlab = "", ylab = "", las = 1.2, type = "n",
     xlim = c(min(bl_m2.5$bwt_sex1_lo), max(bl_m2.5$bwt_sex1_up)),
     ylim = c(min(bl_m2.5$bwt_sex2_lo), max(bl_m2.5$bwt_sex2_up))
)
with(bl_m2.5, segments(x0 = bwt_sex1, y0 = bwt_sex2_lo, x1 = bwt_sex1, y1 = bwt_sex2_up, col = "b
with(bl_m2.5, segments(x0 = bwt_sex1_lo, y0 = bwt_sex2, x1 = bwt_sex1_up, y1 = bwt_sex2, col = "b
points(bwt_sex2 ~ bwt_sex1, bl_m2.5, pch = 16, col = "red", cex = 1.5)
points(bwt_sex2 ~ bwt_sex1, bl_m2.5, pch = 1, col = rgb(0, 0, 0, 0.3), cex = c(1.5))
mtext("bwt male (BV±CI)", side = 1, line = 2.4)
mtext("bwt female (BV±CI)", side = 2, line = 2, las = 3)

plot(tarsus_sex2 ~ tarsus_sex1, bl_m2.5,
     xlab = "", ylab = "", las = 1.2, type = "n",
     xlim = c(min(bl_m2.5$tarsus_sex1_lo), max(bl_m2.5$tarsus_sex1_up)),
     ylim = c(min(bl_m2.5$tarsus_sex2_lo), max(bl_m2.5$tarsus_sex2_up))
)
with(bl_m2.5, segments(x0 = tarsus_sex1, y0 = tarsus_sex2_lo, x1 = tarsus_sex1, y1 = tarsus_sex2_
with(bl_m2.5, segments(x0 = tarsus_sex1_lo, y0 = tarsus_sex2, x1 = tarsus_sex1_up, y1 = tarsus_se
points(tarsus_sex2 ~ tarsus_sex1, bl_m2.5, pch = 16, col = "red", cex = 1.5)
points(tarsus_sex2 ~ tarsus_sex1, bl_m2.5, pch = 1, col = rgb(0, 0, 0, 0.3), cex = c(1.5))
mtext("tarsus male (BV±CI)", side = 1, line = 2.4)
mtext("tarsus female (BV±CI)", side = 2, line = 2, las = 3)
```



Within this model, we also have access to the rest of the B-matrix. Note, the cross-sex genetic correlation is just the diagonal of the B matrix. For now on, you can explore this matrix and estimate the cross-sex-cross-trait genetic correlation.



## **Part III.**

### **A repeated measures animal model**

---

This tutorial will demonstrate how to run a univariate animal model for a trait with repeated observations using different R packages with an example data files provided.

## Scenario and data

### scenario

Since gryphons are iteroparous, multiple observations of reproductive traits are available for some individuals. Here we have repeated measures of lay date (measured in days after January 1) for individual females varying in age from 2 (age of sexual maturation) up until age 6. Not all females lay every year so the number of observations per female is variable (between 1 to 5). We want to know how repeatable the trait is, and (assuming it is repeatable) how heritable it is.

### Data files

The pedigree file `gryphonped.csv` is that used in the preceding tutorials but we now use a new data file `gryphonRM.csv`. Columns correspond to individual identity (`animal`), birth year (`byear`), age in years (`age`), year of measurement (`year`) and lay date (`laydate`). Each row of the data file corresponds to a single phenotypic observation. Here the data is sorted by identity and then age so that the repeated observations on individuals are apparent. However this is not a requirement for analysis - data could equally be sorted by some other variable (*e.g.*, measurement year) or be in a random order.

```
str(gryphonRM)
```

```
'data.frame':  1607 obs. of  5 variables:
 $ animal : Factor w/ 469 levels "1","2","3","8",...: 1 1 1 1 1 2 2 2 3 3 ...
 $ byear  : Factor w/ 34 levels "968","970","971",...: 22 22 22 22 22 22 22 22 22 22 ...
 $ age    : Factor w/ 5 levels "2","3","4","5",...: 1 2 3 4 5 1 2 3 1 2 ...
 $ year   : Factor w/ 39 levels "970","971","972",...: 23 24 25 26 27 23 24 25 23 24 ...
 $ laydate: num  19 23 24 23 29 21 17 21 20 20 ...
```

```
summary(gryphonRM)
```

|              | animal |   | byear        |   | age |       | year         |   | laydate       |
|--------------|--------|---|--------------|---|-----|-------|--------------|---|---------------|
| 1            | :      | 5 | 1000         | : | 109 | 2:308 | 1004         | : | 79            |
|              |        |   |              |   |     |       |              |   | Min. : 0.00   |
| 3            | :      | 5 | 1001         | : | 98  | 3:322 | 1005         | : | 78            |
|              |        |   |              |   |     |       |              |   | 1st Qu.:20.00 |
| 9            | :      | 5 | 999          | : | 86  | 4:339 | 1003         | : | 69            |
|              |        |   |              |   |     |       |              |   | Median :24.00 |
| 17           | :      | 5 | 1002         | : | 85  | 5:315 | 1006         | : | 64            |
|              |        |   |              |   |     |       |              |   | Mean :23.54   |
| 42           | :      | 5 | 987          | : | 70  | 6:323 | 1002         | : | 60            |
|              |        |   |              |   |     |       |              |   | 3rd Qu.:27.00 |
| 50           | :      | 5 | 989          | : | 66  |       | 988          | : | 54            |
|              |        |   |              |   |     |       |              |   | Max. :41.00   |
| (Other):1577 |        |   | (Other):1093 |   |     |       | (Other):1203 |   |               |

```
head(gryphonRM)
```

| animal | byear | age | year | laydate |
|--------|-------|-----|------|---------|
| 1      | 990   | 2   | 992  | 19      |
| 1      | 990   | 3   | 993  | 23      |
| 1      | 990   | 4   | 994  | 24      |
| 1      | 990   | 5   | 995  | 23      |
| 1      | 990   | 6   | 996  | 29      |
| 2      | 990   | 2   | 992  | 21      |

# Chapter 8

## Asreml-R

First we need to load the `asreml` library:

```
library(asreml)
```

Loading required package: Matrix

Online License checked out Tue Oct 8 11:38:12 2024

Loading ASReml-R version 4.2

### 8.0.1. Estimating repeatability

With repeated measures on individuals it is often of interest to see how repeatable a trait is. We can estimate the repeatability of a trait as the proportion of phenotypic variance  $V_P$  explained by individual variance  $V_{ind}$ ;  $R = V_{ind}/V_P = V_{ind}/(V_{ind} + V_R)$ .

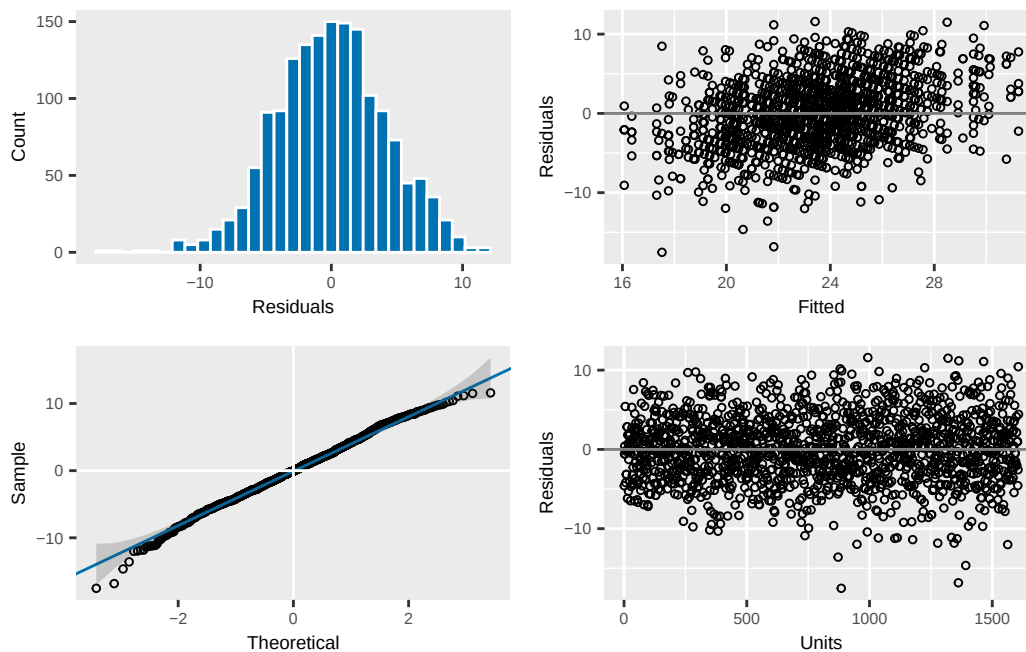
```
modelv <- asreml(  
  fixed = laydate ~ 1,  
  random = ~animal,  
  residual = ~ idv(units),  
  data = gryphonRM,  
  na.action = na.method(x = "omit", y = "omit")  
)
```

---

ASReml Version 4.2 08/10/2024 11:38:12

|    | LogLik    | Sigma2 | DF   | wall     |
|----|-----------|--------|------|----------|
| 1  | -10182.83 | 1.0    | 1606 | 11:38:12 |
| 2  | -8266.097 | 1.0    | 1606 | 11:38:12 |
| 3  | -6145.012 | 1.0    | 1606 | 11:38:12 |
| 4  | -4651.572 | 1.0    | 1606 | 11:38:12 |
| 5  | -3819.309 | 1.0    | 1606 | 11:38:12 |
| 6  | -3554.215 | 1.0    | 1606 | 11:38:12 |
| 7  | -3501.557 | 1.0    | 1606 | 11:38:12 |
| 8  | -3497.576 | 1.0    | 1606 | 11:38:12 |
| 9  | -3497.536 | 1.0    | 1606 | 11:38:12 |
| 10 | -3497.536 | 1.0    | 1606 | 11:38:12 |

```
plot(modelv)
```



The model assumption seems correct, so we can look at the different estimates. Note that since we want to estimate the amount of variance explained by individual identity (rather than by additive genetic effects), we fit `animal` as a normal random effect and we don't associate it with the pedigree. Here, we also ask the model to remove any NA in `laydate`.

This model partitions the phenotypic variance in `laydate` as follows:

```
summary(modelv)$varcomp
```

|             | component | std.error | z.ratio   | bound | %ch |
|-------------|-----------|-----------|-----------|-------|-----|
| animal      | 11.08634  | 1.1794319 | 9.399728  | P     | 0   |
| units!units | 21.29643  | 0.8896196 | 23.938799 | P     | 0   |
| units!R     | 1.00000   | NA        | NA        | F     | 0   |

Between-individual (or among-individual) variance is given by the `animal` component, while the residual component (`units!units`) represents within-individual variance. Here then the repeatability of the trait can be determined by hand as 0.34 (*i.e.*, as  $11.086/(11.086 + 21.296)$ ).

Mean lay date might change with age, so we could ask what the repeatability of lay date is after conditioning on age. This would be done by adding age into the model as a fixed effect.

```
modelw <- asreml(
  fixed = laydate ~ age,
  random = ~animal,
  residual = ~ idv(units),
  data = gryphonRM,
  na.action = na.method(x = "omit", y = "omit")
)
```

ASReml Version 4.2 08/10/2024 11:38:13

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -8402.968 | 1.0    | 1602 | 11:38:14 |
| 2 | -6912.361 | 1.0    | 1602 | 11:38:14 |
| 3 | -5274.379 | 1.0    | 1602 | 11:38:14 |
| 4 | -4143.634 | 1.0    | 1602 | 11:38:14 |
| 5 | -3541.895 | 1.0    | 1602 | 11:38:14 |
| 6 | -3372.909 | 1.0    | 1602 | 11:38:14 |
| 7 | -3347.670 | 1.0    | 1602 | 11:38:14 |
| 8 | -3346.655 | 1.0    | 1602 | 11:38:14 |
| 9 | -3346.652 | 1.0    | 1602 | 11:38:14 |

```
summary(modelw)$varcomp
```

|             | component | std.error | z.ratio  | bound | %ch |
|-------------|-----------|-----------|----------|-------|-----|
| animal      | 12.28982  | 1.156116  | 10.63027 | P     | 0   |
| units!units | 16.37989  | 0.686619  | 23.85586 | P     | 0   |
| units!R     | 1.00000   | NA        | NA       | F     | 0   |

The repeatability of lay date, after accounting for age effects, is now estimated as 0.43 (*i.e.*, as  $12.29/(12.29 + 16.38)$ ). So, just as we saw when estimating  $h^2$  in Tutorial 1, the inclusion of fixed effects will alter the estimated effect size if we determine total phenotypic variance as the sum of the variance components. Thus, proper interpretation is vital.

|             | solution  | std error | z.ratio   |
|-------------|-----------|-----------|-----------|
| (Intercept) | 20.305073 | 0.2899515 | 70.029214 |
| age_2       | 0.000000  | NA        | NA        |
| age_3       | 2.577777  | 0.3355253 | 7.682811  |
| age_4       | 4.247276  | 0.3309028 | 12.835418 |
| age_5       | 6.094490  | 0.3375537 | 18.054872 |
| age_6       | 3.132675  | 0.3371074 | 9.292811  |

ASReml Version 4.2 08/10/2024 11:38:14

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -3346.652 | 1.0    | 1602 | 11:38:14 |
| 2 | -3346.652 | 1.0    | 1602 | 11:38:14 |

|             | Df | denDF  | F.inc    | F.con    | Margin | Pr |
|-------------|----|--------|----------|----------|--------|----|
| (Intercept) | 1  | 460.2  | 14880.00 | 14880.00 |        | 0  |
| age         | 4  | 1225.3 | 88.73    | 88.73    | A      | 0  |

Here age is modeled as a 5-level factor (specified using the function `as.factor()` at the beginning of the analysis). We could equally have fitted it as a continuous variable, in which case, given potential for a late life decline, we would probably also include a quadratic term. In addition, using age as continuous variable can help in saving some degree of freedom in the analysis.

### 8.0.2. Partitioning additive and permanent environment effects

Generally we expect that the repeatability will set the upper limit for heritability since among individual variation can be decomposed in the additive genetic variation and non additive genetic variation. In other word, the additive genetic variation is a subcomponent of the difference between individuals. Non-additive contributions to fixed among-individual differences are normally referred to as *permanent environment effects*. If a trait has repeated measures then it is necessary to model permanent environment effects in an animal model to prevent upward bias in  $V_A$ .

To illustrate it, we first fit the animal model:

```
gryphonped <- read.csv("data/gryphonped.csv")
gryphonped$id <- as.factor(gryphonped$id)
gryphonped$father <- as.factor(gryphonped$father)
gryphonped$mother <- as.factor(gryphonped$mother)

ainv <- ainverse(gryphonped)

modelx <- asreml(
  fixed = laydate ~ age,
  random = ~ vm(animal, ainv),
  residual = ~ idv(units),
  data = gryphonRM,
  na.action = na.method(x = "omit", y = "omit")
)
```

ASReml Version 4.2 08/10/2024 11:38:14

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -8751.390 | 1.0    | 1602 | 11:38:14 |
| 2 | -7169.205 | 1.0    | 1602 | 11:38:14 |
| 3 | -5427.604 | 1.0    | 1602 | 11:38:14 |
| 4 | -4219.598 | 1.0    | 1602 | 11:38:14 |
| 5 | -3569.815 | 1.0    | 1602 | 11:38:14 |
| 6 | -3382.341 | 1.0    | 1602 | 11:38:14 |
| 7 | -3352.867 | 1.0    | 1602 | 11:38:14 |



|   |           |     |      |          |
|---|-----------|-----|------|----------|
| 8 | -3351.565 | 1.0 | 1602 | 11:38:14 |
| 9 | -3351.560 | 1.0 | 1602 | 11:38:14 |

Variance components are almost unchanged if we compare the previous model:

```
summary(modelx)$varcomp
```

|                  | component | std.error | z.ratio   | bound | %ch |
|------------------|-----------|-----------|-----------|-------|-----|
| vm(animal, ainv) | 13.91784  | 1.443968  | 9.638607  | P     | 0   |
| units!units      | 16.84008  | 0.707365  | 23.806768 | P     | 0   |
| units!R          | 1.00000   | NA        | NA        | F     | 0   |

```
summary(modelw)$varcomp
```

|             | component | std.error | z.ratio  | bound | %ch |
|-------------|-----------|-----------|----------|-------|-----|
| animal      | 12.28982  | 1.156116  | 10.63027 | P     | 0   |
| units!units | 16.37989  | 0.686619  | 23.85586 | P     | 0   |
| units!R     | 1.00000   | NA        | NA       | F     | 0   |

This suggests that most of the among-individual variance is – rightly or wrongly – being partitioned as  $V_A$  here. To instead to obtain an unbiased estimate of  $V_A$ , we need to partition for both additive genetic *and* non-genetic sources of individual variation. We do it by fitting `animal` twice, once with a pedigree, and once without a pedigree (using `ide()`). Here, the command `ide` allow to create a second effect using a similar variable.

```
modely <- asreml(
  fixed = laydate ~ age,
  random = ~ vm(animal, ainv) + ide(animal),
  residual = ~ idv(units),
  data = gryphonRM,
  na.action = na.method(x = "omit", y = "omit")
)
```

ASReml Version 4.2 08/10/2024 11:38:14

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -7731.394 | 1.0    | 1602 | 11:38:14 |
| 2 | -6426.548 | 1.0    | 1602 | 11:38:14 |
| 3 | -4997.252 | 1.0    | 1602 | 11:38:14 |
| 4 | -4018.486 | 1.0    | 1602 | 11:38:14 |
| 5 | -3504.988 | 1.0    | 1602 | 11:38:14 |
| 6 | -3363.160 | 1.0    | 1602 | 11:38:14 |
| 7 | -3341.611 | 1.0    | 1602 | 11:38:14 |
| 8 | -3340.682 | 1.0    | 1602 | 11:38:14 |
| 9 | -3340.679 | 1.0    | 1602 | 11:38:14 |

```
summary(modely)$varcomp
```

|                  | component | std.error | z.ratio   | bound | %ch |
|------------------|-----------|-----------|-----------|-------|-----|
| vm(animal, ainv) | 4.876101  | 1.8087709 | 2.695809  | P     | 0   |
| ide(animal)      | 7.400983  | 1.7280113 | 4.282948  | P     | 0   |
| units!units      | 16.380188 | 0.6866189 | 23.856300 | P     | 0   |
| units!R          | 1.000000  | NA        | NA        | F     | 0   |

The estimate of  $V_A$  is now much lower since the additive and permanent environment effects are being properly separated. We can estimate  $h^2$  and the repeatability from this model:

```
vpredict(modely, h2 ~ V1 / (V1 + V2 + V3))
```

|    | Estimate  | SE        |
|----|-----------|-----------|
| h2 | 0.1701523 | 0.0607397 |

```
vpredict(modely, repeatability ~ (V1 + V2) / (V1 + V2 + V3))
```

|               | Estimate  | SE       |
|---------------|-----------|----------|
| repeatability | 0.4284108 | 0.027416 |

---

### 8.0.3. Adding additional effects and testing significance

Models of repeated measures can be extended to include other fixed or random effects. For example try including year of measurement (year) and birth year (byear) as random effects.

```
modelz <- asreml(  
  fixed = laydate ~ age,  
  random = ~ vm(animal, ainv) + ide(animal) +  
    year + byear,  
  residual = ~ idv(units),  
  data = gryphonRM,  
  na.action = na.method(x = "omit", y = "omit")  
)
```

ASReml Version 4.2 08/10/2024 11:38:14

|   | LogLik    | Sigma2 | DF   | wall     |                 |
|---|-----------|--------|------|----------|-----------------|
| 1 | -4650.748 | 1.0    | 1602 | 11:38:14 |                 |
| 2 | -4088.264 | 1.0    | 1602 | 11:38:14 |                 |
| 3 | -3494.147 | 1.0    | 1602 | 11:38:14 |                 |
| 4 | -3127.161 | 1.0    | 1602 | 11:38:14 | ( 1 restrained) |
| 5 | -2976.449 | 1.0    | 1602 | 11:38:14 | ( 1 restrained) |
| 6 | -2955.785 | 1.0    | 1602 | 11:38:14 | ( 1 restrained) |
| 7 | -2955.097 | 1.0    | 1602 | 11:38:14 | ( 1 restrained) |
| 8 | -2955.095 | 1.0    | 1602 | 11:38:14 | ( 1 restrained) |
| 9 | -2955.095 | 1.0    | 1602 | 11:38:14 |                 |

```
summary(modelz)$varcomp
```

|                  | component | std.error | z.ratio   | bound | %ch |
|------------------|-----------|-----------|-----------|-------|-----|
| byear            | 0.0000002 | NA        | NA        | B     | NA  |
| year             | 7.9385763 | 1.9344619 | 4.103765  | P     | 0   |
| vm(animal, ainv) | 4.8151355 | 1.6682351 | 2.886365  | P     | 0   |
| ide(animal)      | 8.4333254 | 1.5495778 | 5.442337  | P     | 0   |
| units!units      | 7.7955597 | 0.3324411 | 23.449443 | P     | 0   |

|         | component | std.error | z.ratio | bound | %ch |
|---------|-----------|-----------|---------|-------|-----|
| units!R | 1.0000000 | NA        | NA      | F     | 0   |

This model will return additional variance components corresponding to variation in lay dates between years of measurement and between birth cohorts of females.  $V_{year}$  is very low and B appeared which tell us that the model had fixed the variance as a boundary. If you compare this model to a reduced model with byear excluded the log-likelihood remains unchanged.

```
modelz_2 <- asreml(
  fixed = laydate ~ age,
  random = ~ vm(animal, ainv) + ide(animal) +
    year,
  residual = ~ idv(units),
  data = gryphonRM,
  na.action = na.method(x = "omit", y = "omit")
)
```

ASReml Version 4.2 08/10/2024 11:38:15

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -4665.606 | 1.0    | 1602 | 11:38:15 |
| 2 | -4097.928 | 1.0    | 1602 | 11:38:15 |
| 3 | -3498.611 | 1.0    | 1602 | 11:38:15 |
| 4 | -3128.789 | 1.0    | 1602 | 11:38:15 |
| 5 | -2976.883 | 1.0    | 1602 | 11:38:15 |
| 6 | -2955.806 | 1.0    | 1602 | 11:38:15 |
| 7 | -2955.096 | 1.0    | 1602 | 11:38:15 |
| 8 | -2955.095 | 1.0    | 1602 | 11:38:15 |

```
summary(modelz_2)$varcomp
```

|      | component | std.error | z.ratio  | bound | %ch |
|------|-----------|-----------|----------|-------|-----|
| year | 7.938576  | 1.9344829 | 4.103720 | P     | 0   |

|                  | component | std.error | z.ratio   | bound | %ch |
|------------------|-----------|-----------|-----------|-------|-----|
| vm(animal, ainv) | 4.815137  | 1.6682366 | 2.886364  | P     | 0   |
| ide(animal)      | 8.433324  | 1.5495828 | 5.442319  | P     | 0   |
| units!units      | 7.795560  | 0.3324384 | 23.449637 | P     | 0   |
| units!R          | 1.000000  | NA        | NA        | F     | 0   |

```
modelz$loglik
```

```
[1] -2955.095
```

```
modelz_2$loglik
```

```
[1] -2955.095
```

```
1 - pchisq(2 * (modelz_2$loglik - modelz$loglik), 1)
```

```
[1] 0.9990453
```

year effects could alternatively be included as fixed effects (try it!). This will reduce  $V_R$  and increase the estimates of heritability and repeatability, which must now be interpreted as proportions of phenotypic variance after conditioning on both age and year of measurement effects.

```
modelz_3 <- asreml(
  fixed = laydate ~ age + byear,
  random = ~ vm(animal, ainv) + ide(animal) +
    year,
  residual = ~ idv(units),
  data = gryphonRM,
  na.action = na.method(x = "omit", y = "omit")
)
```

ASReml Version 4.2 08/10/2024 11:38:15

|        |        |    |      |
|--------|--------|----|------|
| LogLik | Sigma2 | DF | wall |
|--------|--------|----|------|

|   |           |     |      |          |
|---|-----------|-----|------|----------|
| 1 | -4623.985 | 1.0 | 1569 | 11:38:15 |
| 2 | -4063.535 | 1.0 | 1569 | 11:38:15 |
| 3 | -3471.618 | 1.0 | 1569 | 11:38:15 |
| 4 | -3105.972 | 1.0 | 1569 | 11:38:15 |
| 5 | -2955.436 | 1.0 | 1569 | 11:38:15 |
| 6 | -2934.435 | 1.0 | 1569 | 11:38:15 |
| 7 | -2933.721 | 1.0 | 1569 | 11:38:15 |
| 8 | -2933.720 | 1.0 | 1569 | 11:38:15 |

```
summary(modelz_3)$varcomp
```

|                  | component | std.error | z.ratio   | bound | %ch |
|------------------|-----------|-----------|-----------|-------|-----|
| year             | 8.029139  | 1.9920127 | 4.030666  | P     | 0   |
| vm(animal, ainv) | 5.060775  | 1.7855255 | 2.834334  | P     | 0   |
| ide(animal)      | 8.412539  | 1.6494894 | 5.100087  | P     | 0   |
| units!units      | 7.805139  | 0.3331474 | 23.428484 | P     | 0   |
| units!R          | 1.000000  | NA        | NA        | F     | 0   |

|             | solution  | std error | z.ratio   |
|-------------|-----------|-----------|-----------|
| (Intercept) | 20.305073 | 0.2899515 | 70.029214 |
| age_2       | 0.000000  | NA        | NA        |
| age_3       | 2.577777  | 0.3355253 | 7.682811  |
| age_4       | 4.247276  | 0.3309028 | 12.835418 |
| age_5       | 6.094490  | 0.3375537 | 18.054872 |
| age_6       | 3.132675  | 0.3371074 | 9.292811  |

ASReml Version 4.2 08/10/2024 11:38:15

|   | LogLik    | Sigma2 | DF   | wall     |
|---|-----------|--------|------|----------|
| 1 | -2933.720 | 1.0    | 1569 | 11:38:15 |
| 2 | -2933.720 | 1.0    | 1569 | 11:38:15 |

---

|             | Df | denDF | F.inc     | F.con     | Margin | Pr        |
|-------------|----|-------|-----------|-----------|--------|-----------|
| (Intercept) | 1  | 55.3  | 1894.0000 | 1894.0000 |        | 0.0000000 |
| age         | 4  | 845.2 | 152.7000  | 132.9000  | A      | 0.0000000 |
| byear       | 33 | 466.5 | 0.7722    | 0.7722    | A      | 0.8164617 |

---

# Chapter 9

## MCMCglmm

### 9.0.1. Estimating repeatability

With repeated measures on individuals it is often of interest to see how repeatable a trait is. We can estimate the repeatability of a trait as the proportion of phenotypic variance  $V_P$  explained by individual variance  $V_{ind}$ ;  $R = V_{ind}/V_P = V_{ind}/(V_{ind} + V_R)$ . As you already know, bayesian modelisation requires prior. Here, we create a uninformative prior with one estimate for the G matrix and one estimate for the Residual matrix, in addition

```
# p.var <- var(gryphonRM$laydate, na.rm = TRUE)
prior3.1 <- list(G = list(G1 = list(V = 1, nu = 0.002)), R = list(
  V = 1,
  nu = 0.002
))
model3.1 <- MCMCglmm(laydate ~ 1,
  random = ~animal, data = gryphonRM,
  prior = prior3.1, verbose = FALSE
)
posterior.mode(model3.1$VCV)
```

```
    animal    units
10.67985 21.28520
```

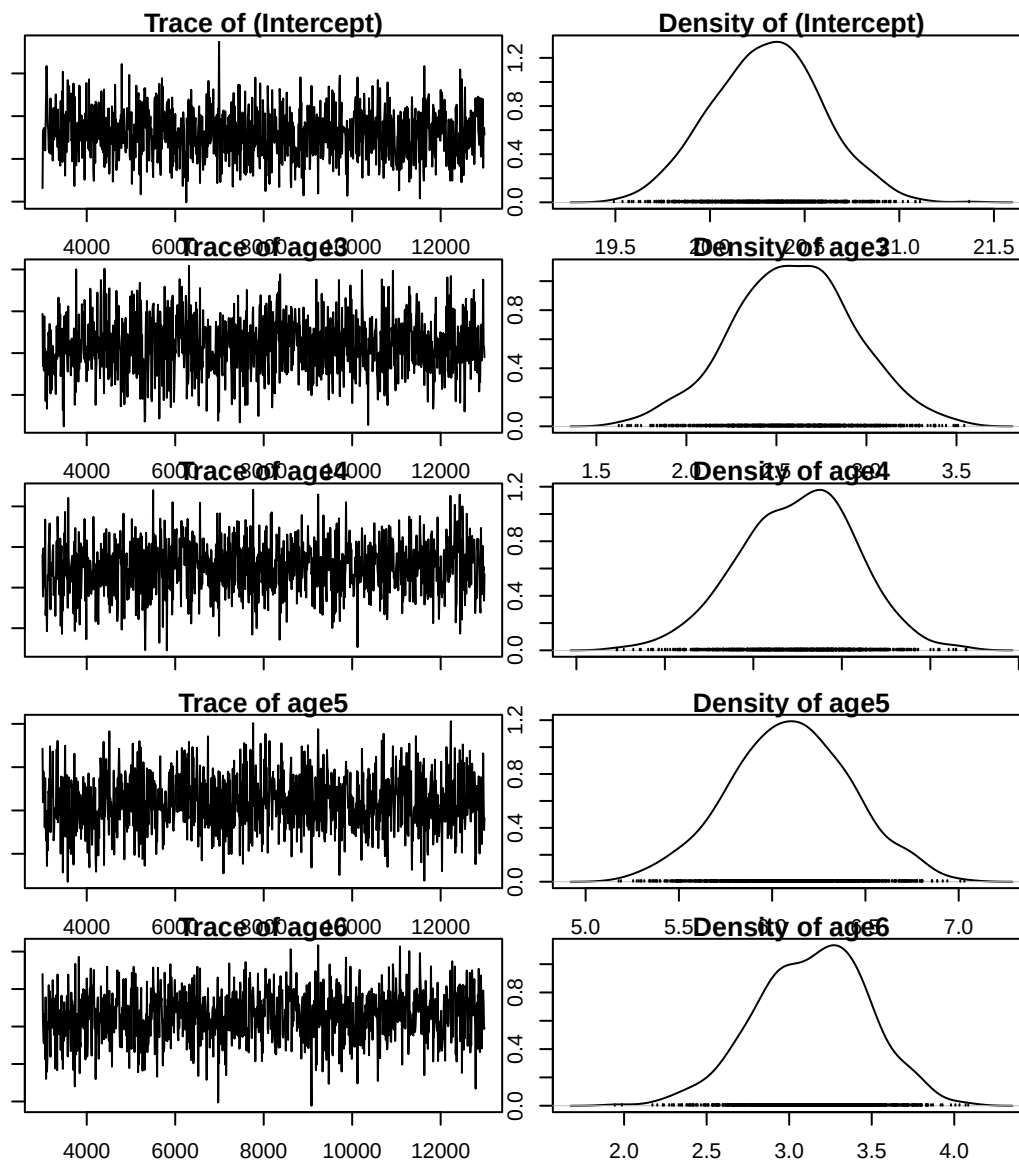
Note the use of the term `animal` as random allowed to partition the phenotypic variance  $V_P$  into among individual variance  $V_{ind}$  associated with `animal` and residual variance  $V_R$  associated with `units`.



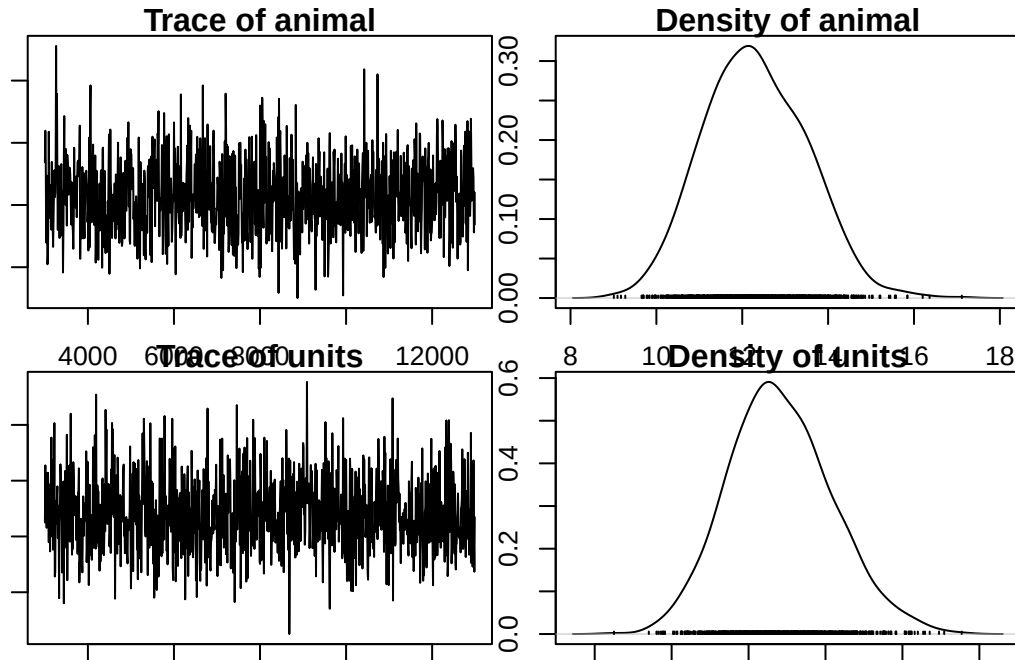
Here then the repeatability of the laydate can be determined as: 22.29 (*i.e.*, as  $10.68/(10.68 + 21.285)$ ). Just a friendly reminder, we work with Monte Carlo chain with model iteration, so the point estimate can be different (but very similar) each time you run the model.

Mean lay date might change with age, so we could ask what the repeatability of lay date is after conditioning on age. This would be done by adding age into the model as a fixed effect.

```
model3.2 <- MCMCglmm(laydate ~ age,
  random = ~animal, data = gryphonRM,
  prior = prior3.1, verbose = FALSE
)
par(mar = c(1, 1, 1, 1))
plot(model3.2$Sol)
```



```
plot(model3.2$VCV)
```



```
posterior.mode(model3.2$VCV)
```

```

  animal    units
12.18011 16.28056

```

The model assumption seems correct, so we can look at the different estimates. Note that the random effect structure has remained unchanged because we did not modified the prior `prior3.1`. The repeatability of `laydate`, after accounting for age effects, is now estimated as 22.29 (*i.e.*, as  $10.68/(10.68 + 21.285)$ ). Just as we saw when estimating  $h_2$  in tutorial 1, the inclusion of fixed effects will alter the estimated effect size if we determine total phenotypic variance as the sum of the variance components. Thus, proper interpretation is vital.

```
posterior.mode(model3.2$Sol)
```

```

(Intercept)    age3    age4    age5    age6
 20.361160    2.743338    4.419162    6.038813    3.290132

```

```
HPDinterval(model3.2$Sol, 0.95)
```

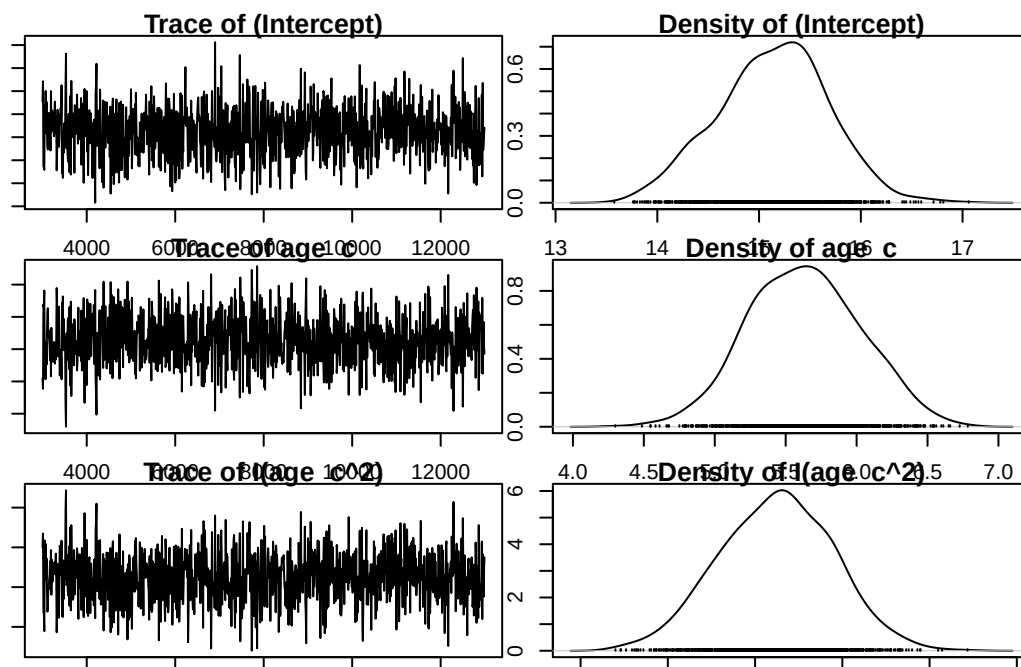
---

```
      lower    upper
(Intercept) 19.775827 20.887996
age3         1.863914  3.209940
age4         3.657741  4.913932
age5         5.480571  6.753545
age6         2.509874  3.803416
attr(,"Probability")
[1] 0.95
```

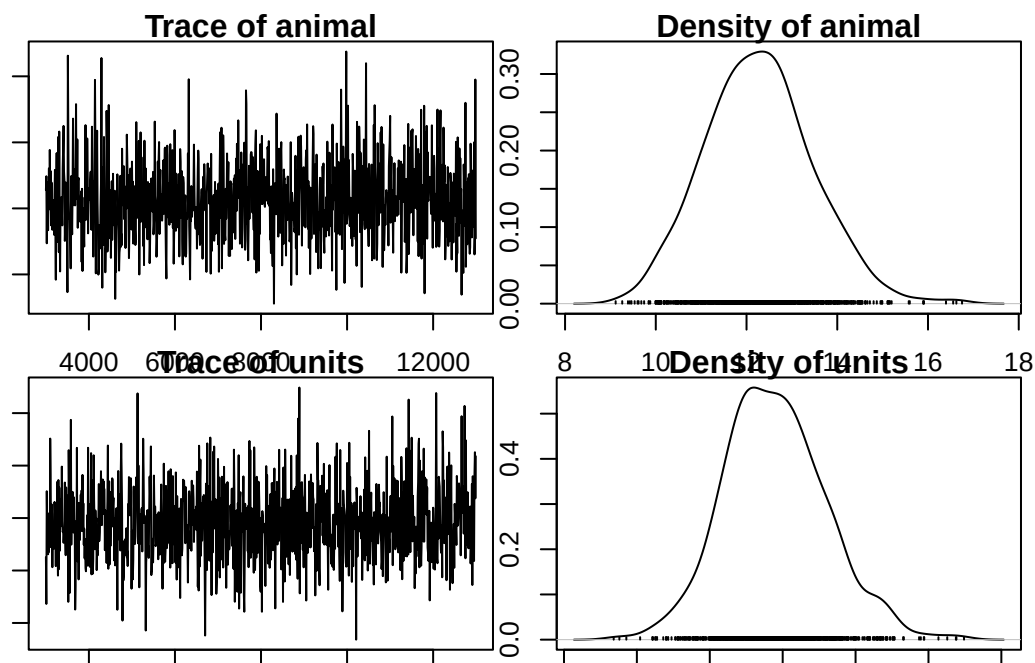
Here age is modeled as a 5-level factor (specified using the function `as.factor()` at the beginning of the analysis). We could equally have fitted it as a continuous variable, in which case, given potential for a late life decline, we would probably also include a quadratic term. In addition, using age as continuous variable can help in saving some degree of freedom in the analysis.

```
gryphonRM$age_c <- as.numeric(gryphonRM$age)

model3.2_2 <- MCMCglmm(laydate ~ age_c + I(age_c^2),
  random = ~animal, data = gryphonRM,
  prior = prior3.1, verbose = FALSE
)
par(mar = c(1, 1, 1, 1))
plot(model3.2_2$Sol)
```



```
plot(model3.2_2$VCV)
```



```
posterior.mode(model3.2_2$VCV)
```

```

  animal    units
12.77266 16.51431

```

---

```
posterior.mode(model3.2_2$Sol)
```

```
(Intercept)      age_c  I(age_c^2)
15.3014700    5.5882993  -0.7634834
```

```
HPDinterval(model3.2_2$Sol, 0.95)
```

```
              lower      upper
(Intercept) 14.0702157 16.1519997
age_c        4.9036170  6.4275478
I(age_c^2)   -0.9048494 -0.6561443
attr(,"Probability")
[1] 0.95
```

### 9.0.2. Partitioning additive and permanent environment effects

Generally we expect that the repeatability will set the upper limit for heritability since among individual variation can be decomposed in the additive genetic variation and non additive genetic variation. In other word, the additive genetic variation is a subcomponent of the difference between individuals. Non-additive contributions to fixed among-individual differences are normally referred to as *permanent environment effects*. If a trait has repeated measures then it is necessary to model permanent environment effects in an animal model to prevent upward bias in  $V_A$ .

To illustrate it, we first fit the animal model:

```
Ainv <- inverseA(gryphonped)$Ainv
model3.3 <- MCMCglmm(laydate ~ 1 + age,
  random = ~animal, ginv = list(animal = Ainv),
  data = gryphonRM, prior = prior3.1, verbose = FALSE
)
```

Variance components are almost unchanged if we compare the previous model:

```
posterior.mode(model3.3$VCV)
```

```
    animal    units  
13.79501 17.19446
```

```
posterior.mode(model3.2$VCV)
```

```
    animal    units  
12.18011 16.28056
```

This suggests that most of the among-individual variance is – rightly or wrongly – being partitioned as  $V_A$  here. In fact here the partition is wrong since the simulation included both additive genetic effects and additional fixed heterogeneity that was not associated with the pedigree structure (i.e. permanent environment effects). In order to obtain an unbiased estimate of  $V_A$ , we need to fit the individual identity twice in the model: once linked to the pedigree (genetic effect) and once not linked to the pedigree (permanent environment effect). To do so, we need to duplicate the variable containing the individual identity `animal` and give it a new name. In addition, the prior needs to be modified to integrate a second random effect. A more appropriate estimate of  $V_A$  is given by the model:

```
gryphonRM$animal_pe <- gryphonRM$animal  
# p.var <- var(gryphonRM$laydate, na.rm = TRUE)  
prior3.4 <- list(G = list(G1 = list(V = 1, nu = 0.002), G2 = list(  
  V = 1,  
  nu = 0.002  
))), R = list(V = 1, nu = 0.002))  
model3.4 <- MCMCglmm(laydate ~ 1 + age,  
  random = ~ animal + animal_pe,  
  ginvs = list(animal = Ainv), data = gryphonRM, prior = prior3.4, verbose = FALSE  
)  
posterior.mode(model3.4$VCV)
```

```
    animal animal_pe    units  
5.000774  6.931818 16.712970
```

---

The estimate of  $V_A$  is now much lower (reduced from 13.6735 to 5.1238) due to a proper separation in the additive and permanent environment effects. We can estimate  $h^2$  and the repeatability from this model:

```
model3.4.VP <- model3.4$VCV[, "animal"] + model3.4$VCV[, "animal_pe"] + model3.4$VCV[, "units"]
model3.4.PE_VA <- model3.4$VCV[, "animal"] + model3.4$VCV[, "animal_pe"]
posterior.mode(model3.4.PE_VA / model3.4.VP)
```

```
var1
0.4304769
```

```
posterior.mode(model3.4$VCV[, "animal"] / model3.4.VP)
```

```
var1
0.1735927
```

### 9.0.3. Adding additional effects and testing significance

Models of repeated measures can be extended to include other fixed or random effects. For example we can try including year of measurement (year) and birth year (byear) as other random effects.

```
# p.var <- var(gryphonRM$laydate, na.rm = TRUE)
prior3.5 <- list(G = list(G1 = list(V = 1, nu = 0.002), G2 = list(
  V = 1,
  nu = 0.002
), G3 = list(V = 1, nu = 0.002), G4 = list(
  V = 1,
  nu = 0.002
)), R = list(V = 1, nu = 0.002))

model3.5 <- MCMCglmm(laydate ~ 1 + age,
  random = ~ animal + animal_pe +
    year + byear, ginvs = list(animal = Ainv), data = gryphonRM, prior = prior3.5,
  verbose = FALSE
)
posterior.mode(model3.5$VCV)
```

```

      animal  animal_pe      year      byear      units
4.955518461 8.396521169 7.673492341 0.002060264 7.747678616

```

```
HPDinterval(model3.5$VCV, 0.95)
```

```

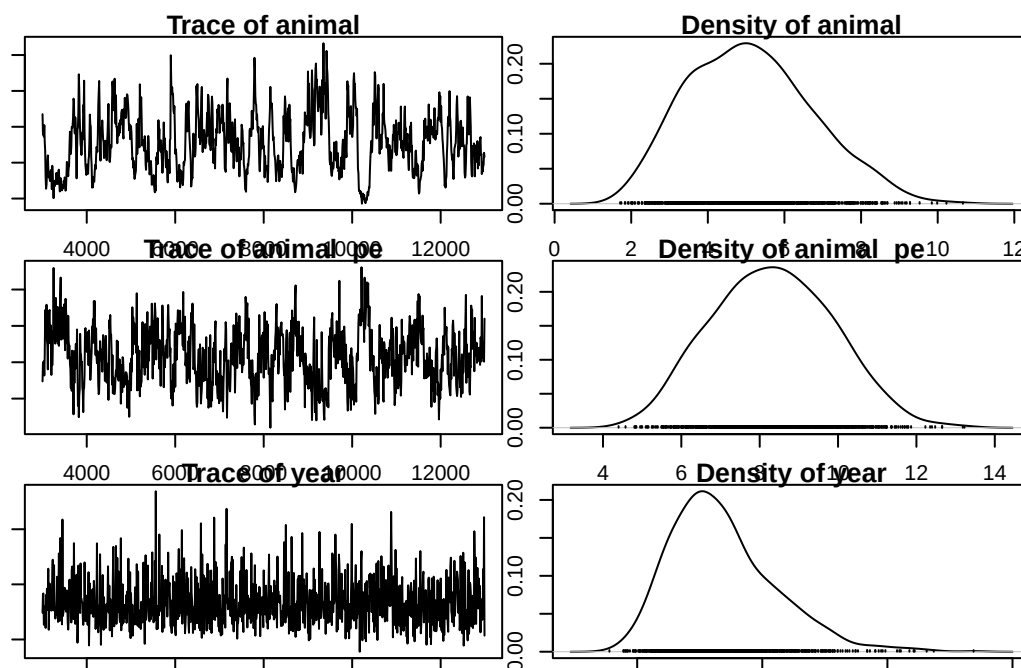
           lower      upper
animal    2.3685070901  8.3886556
animal_pe 5.5459568326 11.2385830
year      5.0750570665 12.8090116
byear     0.0003190255  0.2065312
units     7.1497435613  8.3953730
attr(,"Probability")
[1] 0.95

```

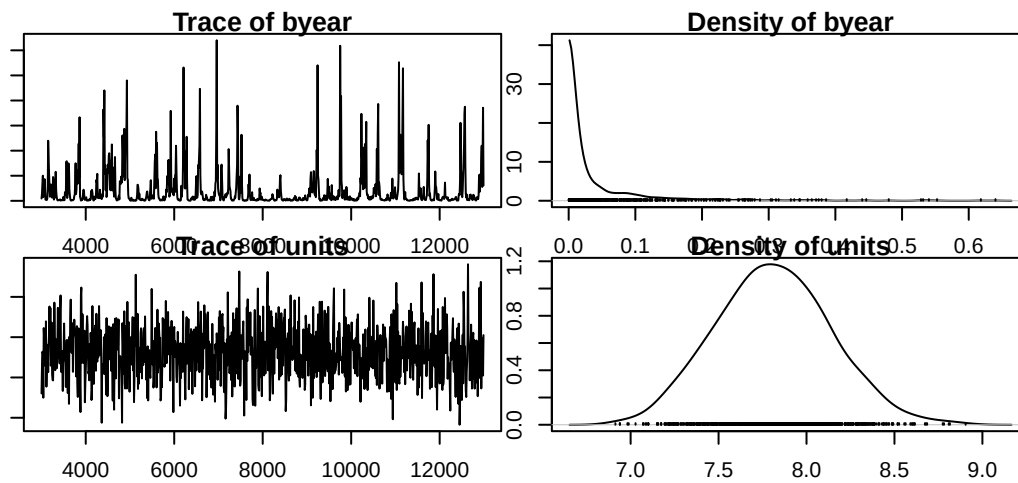
```

par(mar = c(1, 1, 1, 1))
plot(model3.5$VCV)

```







This model will return additional variance components corresponding to year of measurement effects and birth year of the female effects.

$V_{byear}$  is very low and its posterior distribution (via the function `HPDinterval` or `plot`) is very close to zero indicating its not significance. You have to remember bayesian model never estimate variable to 0 or passing zero, so you will never see a credible interval CI crossing zero for a variance. If you compared the DIC of model3.5 to a reduced model without byear, it should be very similar.

```
prior3.5_2 <- list(
  G = list(G1 = list(V = 1, nu = 0.002), G2 = list(
    V = 1,
    nu = 0.002
  )), G3 = list(V = 1, nu = 0.002)),
  R = list(V = 1, nu = 0.002)
)

model3.5_2 <- MCMCglmm(laydate ~ 1 + age,
  random = ~ animal + animal_pe +
    year, ginv = list(animal = Ainv), data = gryphonRM, prior = prior3.5_2,
  verbose = FALSE
)
posterior.mode(model3.5_2$VCV)
```

| animal   | animal_pe | year     | units    |
|----------|-----------|----------|----------|
| 4.725104 | 7.834043  | 8.109265 | 7.781416 |

```
model3.5$DIC
```

```
[1] 8290.793
```

```
model3.5_2$DIC
```

```
[1] 8290.662
```

year effects could alternatively be included as fixed effects (try it!, you should be able to handle the new prior specification at this point). This will reduce  $V_R$  and increase the estimates of heritability and repeatability, which must now be interpreted as proportions of phenotypic variance after conditioning on both age and year of measurement effects.

# Chapter 10

## brms

```
library(brms)

Amat <- as.matrix(nadiv::makeA(gryphonped))
gryphonRM$animal_pe <- gryphonRM$animal

model_simple1.1 <- brm(
  laydate ~ 1 + (1 | gr(animal, cov = Amat)) + (1 | animal_pe),
  data = gryphonRM,
  family = gaussian(),
  data2 = list(Amat = Amat),
  chains = 2, cores = 2, iter = 1000
)

summary(model_simple1.1)
plot(model_simple1.1)
```

# Chapter 11

## Quick comparison of codes

### 11.1. Univariate model with repeated measures

11.1.1. Asreml-R

11.1.2. gremlin

11.1.3. MCMCglmm

11.1.4. brms

### 11.2. bivariate model

11.2.1. Asreml-R

11.2.2. gremlin

11.2.3. MCMCglmm

11.2.4. brms

# References

## R packages

This book was produced using all the following R packages

A large number of files (2811 in total) have been discovered.  
It may take renv a long time to crawl these files for dependencies.  
Consider using `.renvignore` to ignore irrelevant files.  
See `?renv::dependencies` for more information.  
Set `options(renv.config.dependencies.limit = Inf)` to disable this warning.

| Package   | Version   | Citation                                                    |
|-----------|-----------|-------------------------------------------------------------|
| abind     | 1.4.5     | Plate and Heiberger (2016)                                  |
| asreml    | 4.2.0.302 | The VSNi Team (2023)                                        |
| base      | 4.4.1     | R Core Team (2024)                                          |
| brms      | 2.21.0    | Bürkner (2017); Bürkner (2018); Bürkner (2021)              |
| knitr     | 1.48      | Xie (2014); Xie (2015); Xie (2024)                          |
| lme4      | 1.1.35.5  | Bates et al. (2015)                                         |
| MCMCglmm  | 2.36      | Hadfield (2010)                                             |
| nadiv     | 2.18.0    | Wolak (2012)                                                |
| rmarkdown | 2.28      | Xie et al. (2018); Xie et al. (2020); Allaire et al. (2024) |
| rptR      | 0.9.22    | Stoffel et al. (2017)                                       |
| tidyverse | 2.0.0     | Wickham et al. (2019)                                       |
| visreg    | 2.7.0     | Breheny and Burchett (2017)                                 |

## Bibliography

- Allaire, J., Y. Xie, C. Dervieux, J. McPherson, J. Luraschi, K. Ushey, A. Atkins, H. Wickham, J. Cheng, W. Chang, and R. Iannone. 2024. [rmarkdown: Dynamic documents for r](#).
- Bates, D., M. Mächler, B. Bolker, and S. Walker. 2015. [Fitting linear mixed-effects models using lme4](#). *Journal of Statistical Software* 67:1–48.
- Breheny, P., and W. Burchett. 2017. Visualization of regression models using visreg. *The R Journal* 9:56–71.
- Bürkner, P.-C. 2017. [brms: An R package for Bayesian multilevel models using Stan](#). *Journal of Statistical Software* 80:1–28.
- Bürkner, P.-C. 2018. [Advanced Bayesian multilevel modeling with the R package brms](#). *The R Journal* 10:395–411.
- Bürkner, P.-C. 2021. [Bayesian item response modeling in R with brms and Stan](#). *Journal of Statistical Software* 100:1–54.
- Hadfield, J. D. 2010. [MCMC methods for multi-response generalized linear mixed models: The MCMCglmm R package](#). *Journal of Statistical Software* 33:1–22.
- Martin, J., M. Wolak, S. Johnston, and M. Morrissey. 2024. [Pedtricks: Visualize, summarize and simulate data from pedigrees](#).
- Plate, T., and R. Heiberger. 2016. [abind: Combine multidimensional arrays](#).
- R Core Team. 2024. [R: A language and environment for statistical computing](#). R Foundation for Statistical Computing, Vienna, Austria.
- Stoffel, M. A., S. Nakagawa, and H. Schielzeth. 2017. [rptR: Repeatability estimation and variance decomposition by generalized linear mixed-effects models](#). *Methods in Ecology and Evolution* 8:1639–1644.
- The VSNi Team. 2023. [asreml: Fits linear mixed models using REML](#).
- Wickham, H., M. Averick, J. Bryan, W. Chang, L. D. McGowan, R. François, G. Grolemund, A. Hayes, L. Henry, J. Hester, M. Kuhn, T. L. Pedersen, E. Miller, S. M. Bache, K. Müller, J. Ooms, D. Robinson, D. P. Seidel, V. Spinu, K. Takahashi, D. Vaughan, C. Wilke, K. Woo, and H. Yutani. 2019. [Welcome to the tidyverse](#). *Journal of Open Source Software* 4:1686.
- Wilson, A. J., D. Réale, M. N. Clements, M. M. Morrissey, E. Postma, C. A. Walling, L. E. B. Kruuk, and D. H. Nussey. 2010. [An ecologist’s guide to the animal model](#). *Journal of Animal Ecology* 79:13–26.
- Wolak, M. E. 2012. [nadiv: An R package to create relatedness matrices for estimating non-additive genetic variances in animal models](#). *Methods in Ecology and Evolution* 3:792–796.
- Xie, Y. 2014. knitr: A comprehensive tool for reproducible research in R. *in* V. Stodden, F. Leisch, and R. D. Peng, editors. *Implementing reproducible computational research*. Chapman; Hall/CRC.
- Xie, Y. 2015. [Dynamic documents with R and knitr](#). 2nd edition. Chapman; Hall/CRC, Boca Raton, Florida.
- Xie, Y. 2024. [knitr: A general-purpose package for dynamic report generation in r](#).

Xie, Y., J. J. Allaire, and G. Grolemund. 2018. [R markdown: The definitive guide](#). Chapman; Hall/CRC, Boca Raton, Florida.

Xie, Y., C. Dervieux, and E. Riederer. 2020. [R markdown cookbook](#). Chapman; Hall/CRC, Boca Raton, Florida.





# Appendix A

## Data used in this book

### A.1. All in one zip file

All the data and code used in the book in a [zip file](#).

### A.2. Data files

- [gryphon.csv](#)
- [gryphonped.csv](#)
- [gryphonRM.csv](#)
- [gryphonRM.txt](#)

### A.3. R objects

- [brms\\_m1\\_1.rda](#)
- [brms\\_m1\\_2.rda](#)
- [brms\\_m1\\_3.rda](#)
- [brms\\_m1\\_4.rda](#)
- [brms\\_m1\\_5.rda](#)
- [brms\\_m2\\_1.rda](#)
- [brms\\_m2\\_2.rda](#)
- [brms\\_m2\\_3.rda](#)

- [brms\\_m2\\_4.rda](#)
- [brms\\_m2\\_5.rda](#)
- [MCMCglmm\\_model\\_BivSex\\_LongRun.rda](#)
- [MCMCglmm\\_model\\_MultivSex\\_LongRun.rda](#)
- [MCMCglmm\\_model2\\_1\\_LongRun.rda](#)
- [MCMCglmm\\_model2\\_2\\_LongRun.rda](#)
- [MCMCglmm\\_model2\\_3\\_LongRun.rda](#)
- [MCMCglmm\\_model2\\_4\\_LongRun.rda](#)
- [MCMCglmm\\_model2\\_5\\_LongRun.rda](#)

## A.4. R code & functions